



MT6250 GSM/GPRS/EDGE-RX SOC Processor Data Sheet

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Preface

Acronyms for register types

R/W	For both read and write access
RO	Read only
RC	Read only. After the register bank is read, every bit that is HIGH(1) will be cleared to LOW(0) automatically.
WO	Write only
W1S	Write only. When data bits are written to the register bank, every bit that is HIGH(1) will cause the corresponding bit to be set to 1. Data bits that are LOW(0) have no effects on the corresponding bit.
W1C	Write only. When data bits are written to the register bank, every bit that is HIGH(1) will cause the corresponding bit to be cleared to 0. Data bits that are LOW(0) have no effects on the corresponding bit.

1 System Overview

MT6250 is a monolithic chip integrating leading edge power management unit, analog baseband and radio circuitry based on the low-power CMOS process.

MT6250 is a feature-rich and extremely powerful single-chip solution for high-end GSM/GPRS and EDGE-Rx capability. Based on the 32-bit ARM7EJ-S™ RISC processor, MT6250's superb processing power, along with high bandwidth architecture and dedicated hardware support, provides a platform for high-performance GPRS/EDGE-Rx Class 12 MODEM application and leading-edge multimedia applications.

MT6250 also features:

- A highly integrated Bluetooth transceiver which is fully compliant with Bluetooth specification v3.0.
- A FM receiver supporting both audio broadcast de-modulation and RDS/RBDS data decoding.

Typical application diagram is shown in Figure 1.

Platform

MT6250 is capable of running the ARM7EJ-S™ RISC processor at up to 260MHz, which provides the best trade-off between system performance and power consumption.

For large amounts of data transfer, high-performance DMA (Direct Memory Access) with hardware flow control is implemented, which greatly enhances the data movement speed while reducing the MCU processing load.

Targeted as a media-rich platform for mobile applications, MT6250 also provides hardware security digital rights management for copyright protection. For further safeguard and to protect

the manufacturer's development investment, hardware flash content protection is provided to prevent unauthorized porting of the software load.

Memory

MT6250 supports serial flash interface with various operating frequencies.

Multimedia

The MT6250 multimedia subsystem provides conventional parallel interface and 2-bit serial interface for CMOS sensors. The camera resolution is up to 2M pixels. The built-in Hybrid Motion JPEG Encoder hardware enables real-time capture of high-resolution images and recorder of video format such as MPEG-4 or Motion JPEG with smooth quality. The software-based codec can also be used to process various video types. Besides, MT6250 provides fancy UI capabilities through its hardware 2D accelerator. The 2D accelerator performs high-speed linear transformations with filtering. To take advantage of the high MCU performance, GIF and PNG decoders are implemented by the software.

In addition, MT6250 is implemented with a high-performance audio synthesis technology, as well as a high-quality audio amplifier to provide superior audio experiences.

Connectivity and storage

MT6250 supports UART, USB 1.1 FS/LS, SDIO, HIF interface and MMC/SD storage systems. These interfaces provide MT6250 users with the highest level of flexibility in implementing high-end solutions.

To achieve a complete user interface, MT6250 also brings together all the necessary peripheral

blocks for a multimedia GSM/GPRS/EDGE-RX phone. The peripheral blocks include the keypad scanner with the capability to detect multiple key presses, SIM controller, real-time clock, PWM, serial/parallel LCD controller and general-purpose programmable I/Os.

Audio

Using a highly integrated mixed-signal audio front-end, the MT6250 architecture provides easy audio interfacing with direct connection to the audio transducers. The audio interface integrates A/D converters for voice band, as well as high-resolution stereo D/A converters for both audio and voice band.

MT6250 supports AMR codec to adaptively optimize the quality of speech and audio. Moreover, HE-AAC codec is implemented to deliver CD-quality audio at low bit rates.

In addition, a 850mW class-AB amplifier is also embedded to save the BOM cost of adopting external amplifiers.

GSM/GPRS/EDGE-Rx radio

MT6250 integrates a mixed-signal baseband front-end in order to provide a well-organized radio interface with flexibility for efficient customization. The front-end contains gain and offset calibration mechanisms and filters with programmable coefficients for comprehensive compatibility control on RF modules. MT6250 achieves outstanding MODEM performance by utilizing a highly dynamic range ADC in the RF downlink path.

MT6250 embeds a high-performance and completely integrated SAW-less RF transceiver for multi-band GSM cellular system. In this RF transceiver, a quad-band receiving feature with high sensitivity is supported utilizing two RF differential receivers and a fully integrated

channel filter. With ultra-high dynamic range, the SAW filters on the receiving path can be removed for BOM cost reduction. In addition, the minimum component count is guaranteed by realizing a highly integrated transmitter, low-spur frequency synthesizer and a Digitally-Controlled Crystal Oscillator (DCXO).

Bluetooth radio

MT6250 offers a highly integrated Bluetooth radio and baseband processor. Only a minimum of external components are required. MT6250 provides superior sensitivity and class 1 output power and thus ensures the quality of the connection with a wide range of Bluetooth devices.

MT6250 is fully compliant with Bluetooth v3.0 and offers enhanced data rates of up to 3Mbps. It also provides the coexistence protocol with 802.11 system.

MT6250 supports rich Bluetooth profiles, enabling diversified applications that are widely used on the handset with excellent interoperability.

FM radio

The FM radio subsystem provides a completely integrated FM Rx receiver supporting 87.5 ~ 108MHz FM bands with 50kHz tuning step. It also performs fast channel seek/scan algorithm to validate 200 carrier frequencies in 6 seconds. In addition to receiving FM audio broadcasting, the digital RDS/RBDS data system is supported as well. The integrated FM transceiver utilizes state-of-the-art digital demodulation/modulation techniques to achieve excellent performance.

In order to achieve high SINAD, good sensitivity and excellent noise suppression, the FM receiver adopts adaptive demodulation scheme to optimize Rx system performance in all ranges of signal quality by reference of a very

sophisticated channel quality index (CQI). When the received signal quality is poor, the design not only enhances the ACI rejection capability but also uses a very ingenious skill to soft mute annoying noise so as to provide good perception quality.

The FM radio subsystem supports both long antenna, which is usually an earphone, and auto-calibrated short antenna, which is usually a FPC short antenna or shared antenna with GSM for different application scenarios.

Debugging function

The JTAG interface enables in-circuit debugging of the software program with the ARM7EJ-S™ core. With this standardized debugging interface, MT6250 provides developers with a wide set of options in choosing ARM development kits from different third party vendors.

Power management

A power management is embedded in MT6250 to provide rich features a high-end feature phone supports, including Li-ion battery charger, high performance and low quiescent current LDOs, and drivers for LED and backlight.

MT6250 offers various low-power features to help reduce the system power consumption. MT6250 is also fabricated in an advanced low-power CMOS process, hence providing an overall ultra-low leakage solution.

Package

The MT6250 device is offered in a 9.8mm×9.6mm, 233-ball, 0.5mm pitch, TFBGA package.

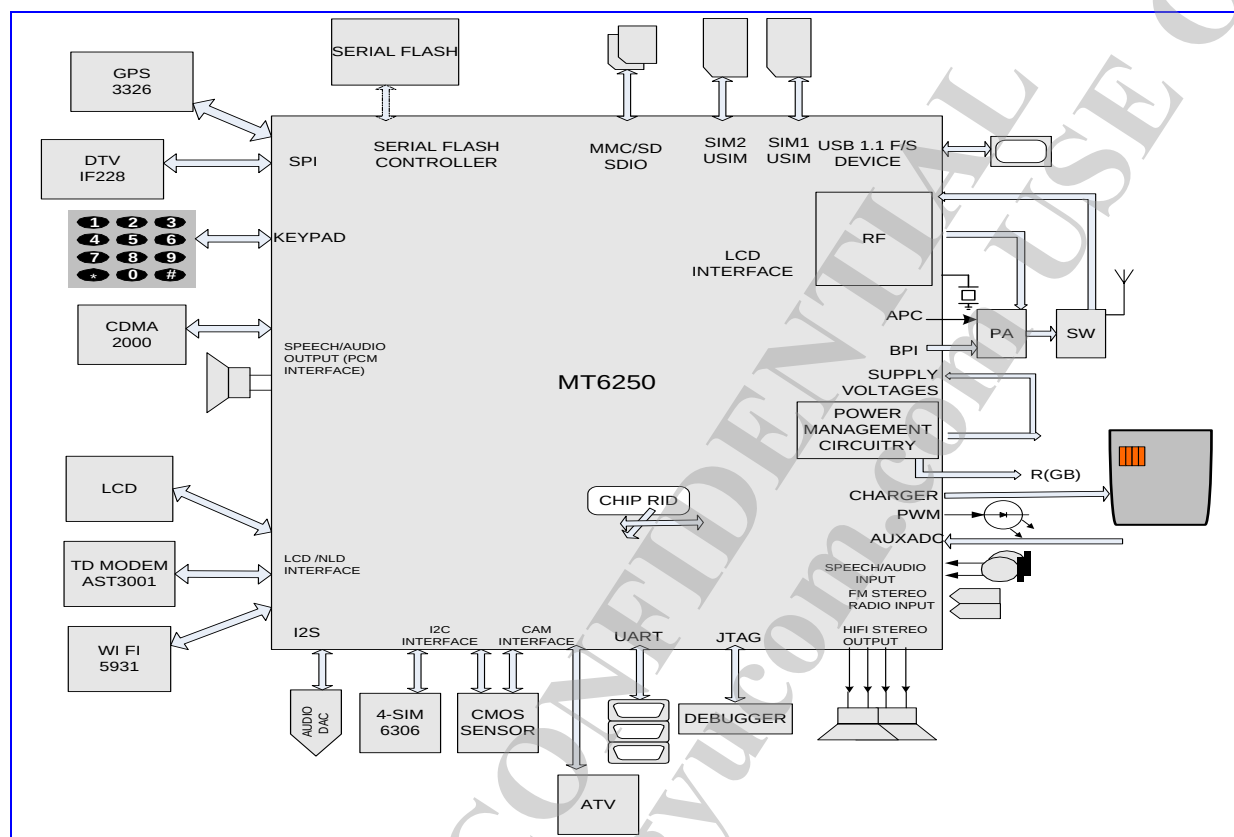


Figure 1. Typical application of MT6250

1.1 Platform Features

General

- Integrated voice-band, audio-band and base-band analog front-end
- Integrated full-featured power management unit

MCU subsystem

- ARM7EJ-S™ 32-bit RISC processor
- Java hardware acceleration for fast Java-based games and applets
- Operating frequency: Max. 260MHz with dynamic clock gating
- High-performance multi-layer AHB bus
- Dedicated DMA bus with 17 DMA channels
- On-chip boot ROM for factory flash programming
- Watchdog timer for system crash recovery
- 4 sets of general-purpose timers
- Circuit switch data coprocessor
- Division coprocessor

Serial flash interfaces

- Supports various operating frequency combinations for serial flash
- Supports QPI and SPI serial flash

User interfaces

- 8-row × 8-column and 5-row × 5-column keypad controller with hardware scanner
- Supports multiple key presses for gaming
- Dual SIM/USIM controller with hardware T = 0/T = 1 protocol control
- Real-time clock (RTC) operating with a low-quiescent-current power supply
- General-purpose I/Os (GPIOs) available for auxiliary applications
- 1 sets of Pulse Width Modulation (PWM) output
- 13 external interrupt lines
- 1 external channel auxiliary 10-bit A/D converter

Security

- Supports security key and chip random ID

Connectivity

- 3 UARTs with hardware flow control and supports baud rate up to 921,600 bps
- FS/LS USB 1.1 device controller
- Multimedia card, secure digital Memory Card, host controller with flexible I/O voltage power
- Supports SDIO interface for SDIO peripherals as well as WIFI connectivity
- DAI/PCM and I2S interface for audio applications
- I2C master interface for peripheral management including image sensors
- SPI master interface for peripheral management including digital TV chips

Power management

- Li-ion battery charger
- 16 LDOs for the power supply of memory card, camera, Bluetooth, RF, SIM card and other diversified usage
- 4 open-drain output switches to supply/control the LED
- LDO type vibrator
- One NMOS switch to control keypad LED
- Thermal overload protection
- Under-voltage lock-out protection
- Over-voltage protection
- Different levels of power-down modes with sophisticated software control enables excellent power saving performance.

Test and debugging

- Built-in digital and analog loop back modes for both audio and baseband front-end
- DAI port complies with GSM Rec.11.10.
- JTAG port for debugging embedded MCU

1.2 MODEM Features

Radio interface and baseband front-end

- Digital PM data path with baseband front-end
- High dynamic range delta-sigma ADC converts the downlink analog I and Q signals to digital baseband.
- 10-bit D/A converter for Automatic Power Control (APC)
- Programmable radio Rx filter with adaptive gain control
- Dedicated Rx filter for FB acquisition
- 4-pin Baseband Parallel Interface (BPI) with programmable driving strength
- Supports multi-band

Voice and modem CODEC

- Dial tone generation
- Voice memo
- Noise reduction
- Echo suppression
- Advanced sidetone oscillation reduction
- Digital sidetone generator with programmable gain
- Two programmable acoustic compensation filters
- GSM quad vocoders for adaptive multirate (AMR), enhanced full rate (EFR), full rate (FR) and half rate (HR)
- GSM channel coding, equalization and A5/1, A5/2 and A5/3 ciphering
- GPRS GEA1, GEA2 and GEA3 ciphering
- Programmable GSM/GPRS/EDGE-Rx modem
- Packet switched data with CS1/CS2/CS3/CS4 coding schemes
- GSM circuit switch data
- GPRS/EDGE-Rx Class 12
- Supports SAIC (single antenna interference cancellation) technology
- Supports VAMOS (Voice services over Adaptive Multi-user channels on One Slot) technology in R9 spec.

Voice interface and voice front-end

- Two microphone inputs share one low-noise amplifier with programmable gain and Automatic Gain Control (AGC) mechanisms
- Voice power amplifier with programmable gain
- 2nd order Sigma-Delta A/D converter for voice uplink path
- Shares D/A converter with audio playback path
- Supports full-duplex hands-free operation
- Compliant with GSM 03.50

1.3 GSM/GPRS/EDGE RF Features

Receiver

- Quad band differential input LNAs
- Quadrature RF mixers
- Fully integrated channel filter
- High dynamic range ADC
- 24dB PGA gain with 6dB gain step

Transmitter

- Transmitter outputs support quad bands.
- Highly precise and low noise RF transmitter for GSM/GPRS applications

Frequency synthesizer

- Programmable fractional-N synthesizer
- Integrated wide range RFVCO
- Integrated loop filter
- Fast settling time suitable for multi-slot GPRS/EDGE-Rx applications

Digitally-Controlled Crystal Oscillator (DCXO)

- Two-pin 26MHz crystal oscillator
- On-chip programmable capacitor array for coarse-tuning
- On-chip programmable capacitor array for fine-tuning
- Low power mode supports 32K crystal removal

1.4 Multimedia Features

LCD/TD-modem/WiFi interface

- Dedicated parallel interface supports 3 external devices with 8-bit for TD-modem/WiFi interface and 8-/9-bit for parallel LCD interface.

LCD controller

- Supports simultaneous connection to 2 parallel and 2 serial LCD modules
- LCM formats supported: RGB332, RGB444, RGB565, RGB666, RGB888
- Supports LCD module with maximum resolution up to 480x320
- Per pixel alpha channel
- True color engine
- Supports hardware display rotation
- Capable of combining display memories with up to 4 blending layers

Camera interface

- YUV422 format image input
- Capable of processing image of size up to 0.3M pixels (Mediatek serial interface) and 2M pixels (w/o compression)

JPEG decoder

- Baseline JPEG decoding
- Supports various YUV formats, DC/AC Huffman tables and quantization tables

JPEG encoder

- Hybrid motion JPEG encoder for video encoding
- ISO/IEC 10918-1 JPEG baseline mode
- ISO/IEC 10918-2 compliance
- Supports YUV420 and grayscale formats
- Supports EXIF/JFIF
- Standard DC and AC Huffman tables
- Provides 5 levels of encode quality
- Supports zeros shutter delay

Image data processing

- Supports 4x digital zoom
- High throughput hardware scaler. Capable of tailoring an image to an arbitrary size.
- Horizontal scaling with bilinear interpolation
- Vertical scaling with bilinear interpolation
- YUV and RGB color space conversion
- RGB/YCbCr format thumbnail output

MPEG-4/H.263 CODEC

- Hybrid MPEG4 encoder
- Software-based MPEG4 decoder
- ISO/IEC 14496-2 simple profile:
 - Decode @ level 0/1/2/3
 - Encode @ level 0
- ISO/IEC 14496-2 advanced simple profile:
 - Decode @ level 0/1/2/3
 - ITU-T H.263 profile 0 @ level 40
- Supports visual tools for decoder: I-VOP, P-VOP, B-VOP, AC/DC prediction, 4-MV, unrestricted MV, error resilience, short header, global motion compensation, method 1/2 quantization, quarter-pel motion compensation.
- Error resilience for decoder: Slice resynchronization, data partitioning, reversible VLC
- Supports visual tools for encoder: I-VOP, P-VOP, Half-Pel, DC prediction, unrestricted MV, short header

H.264

- ISO/IEC 14496-10 baseline profile:
 - Decode @ level 1

2D accelerator

- Supports 32-bpp ARGB8888, 24-bpp RGB888, 16-bpp RGB565, 24-bpp ARGB6666.

- 4 layers overlay with individual color format, window size, source key, constant alpha and rotation
- Rectangle fill with constant
- BitBlt: Capable with 7 rotation types
- Alpha blending with 7 rotation types, per-pixel alpha and pre-multiplied alpha
- Font drawing: Normal font and anti-aliasing font
- Linear transformation: Supports perspective transform, truncate/nearest/bi-linear sample filter.

Audio CODEC

- Supports HE-AAC codec decoding
- Supports AAC codec decoding
- Wavetable synthesis with up to 64 tones
- Advanced wavetable synthesizer capable of generating simulated stereo
- Wavetable including GM full set of 128 instruments and 47 sets of percussions
- PCM playback and record
- Digital audio playback

Audio interface and audio front-end

- Supports I2S interface
- High-resolution D/A converters for stereo audio playback
- Voice band A/D converter with digital MIC input support
- Stereo to mono conversion

1.5 Bluetooth Features

Radio features

- Fully compliant with Bluetooth specification 3.0 + EDR
- Low out-of-band spurious emissions support simultaneous operation with GPS and GSM/GPRS worldwide radio systems
- Low-IF architecture with high degree of linearity and high order channel filter
- Integrated T/R switch and Balun
- Fully integrated PA provides 10dBm output power
- -95dBm sensitivity with excellent interference rejection performance
- Hardware AGC dynamically adjusts receiver performance in changing environments
- Embedded processor for Bluetooth protocol stack with built-in memory system
- Fully verified ROM based system with code patch for feature enhancement

Baseband features

- Up to 4 simultaneous active ACL links
- Up to 1 simultaneous SCO or eSCO link with CVSD coding
- Supports eSCO
- Scatternet support: Up to 4 piconets simultaneously with background inquiry/page scan
- Supports sniff mode
- AFH and PTA collaborative support for WLAN/BT coexistence
- Idle mode and sleep mode enables ultra-low power consumption.
- Supports PCM interface and built-in programmable transcoders for linear voice with re-transmission
- Built-in hardware modem engine for access code correlation, header error correction, forward error correction, CRC, whitening and encryption
- Channel quality driven data rate adaptation
- Channel assessment for AFH

Platform features

1.6 FM Features

- 87.5-108MHz worldwide FM bands with 50kHz tuning step
- Supports RDS/RBDS radio data system
- Supports long/short antenna
- 30ms seek time per channel, and 6sec search time for all channels
- Superior stereo noise reduction
- Supports 32.768kHz clock as reference clock
- Soft mute volume control
- Supports short antenna, auto calibration for different FM channels
- 58dB SINAD with 22.5kHz FM deviation
- 3dBuVemf FM RX sensitivity with superior interference rejection
- 20dBuVemf RDS sensitivity (dev: 2kHz)
- More than 55dBc rejection capability against -200kHz ACI

1.7 General Descriptions

Figure 2 is the block diagram of MT6250. Based on a multi-processor architecture, MT6250 integrates an ARM7EJ-S™ core, the main processor running high-level GSM/EDGE-Rx protocol software as well as multimedia applications, single digital signal processor core, which manages the low-level MODEM and advanced audio functions, an embedded processor running Bluetooth baseband and link control protocol and the Bluetooth radio control.

MT6250 consists of the following subsystems:

- Microcontroller Unit (MCU) subsystem: Includes an ARM7EJ-S™ RISC processor and its accompanying memory management and interrupt handling logics
- Digital Signal Processor (DSP) subsystem: Includes a DSP and its accompanying memory, memory controller and interrupt controller
- MCU/DSP interface: Junction at which the MCU and the DSP exchange hardware and software information
- Microcontroller peripherals: Include all user interface modules and RF control interface modules
- Microcontroller coprocessors: Run computing-intensive processes in place of the microcontroller
- DSP peripherals: Hardware accelerators for GSM/GPRS/EDGE-Rx channel codec
- Multimedia subsystem: Integrates several advanced accelerators to support multimedia applications
- Voice front-end: Data path for converting analog speech to and from digital speech
- Audio front-end: Data path for converting stereo audio from an audio source
- Baseband front-end: Data path for converting a digital signal to and from an analog signal from the RF modules

- Timing generator: Generates the control signals related to the TDMA frame timing
- Power, reset and clock subsystem: Manage the power, reset and clock distribution inside MT6250.
- Bluetooth subsystem: Includes an embedded processor with embedded ROM/RAM system, baseband processor, and a high-performance radio block
- Power management unit: Self-contained power supply source which also controls the charging and system startup circuitry.

Details of the individual subsystems and blocks are described in the following chapters.

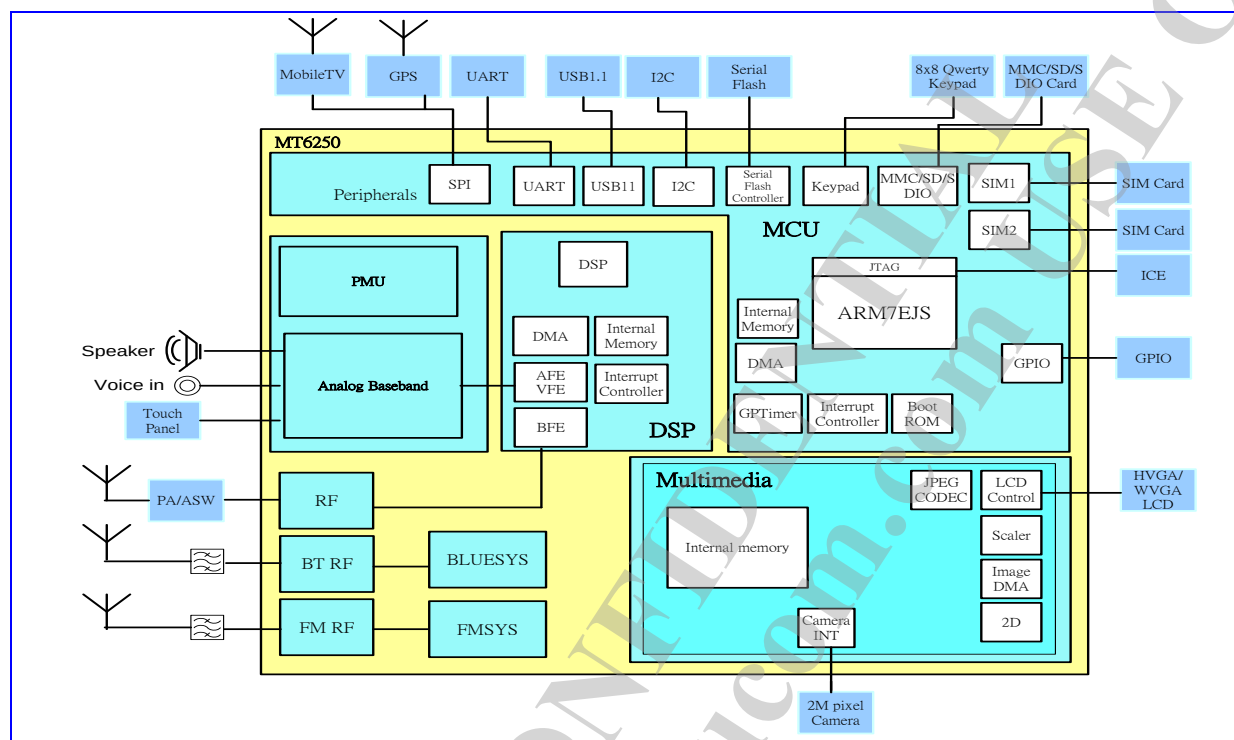


Figure 2. MT6250 block diagram

2 Product Descriptions

2.1 Pin Description

2.1.1 Ball Diagram

For MT6250, an TFBGA 9.8mm*9.6mm, 233-ball, 0.5mm pitch package is offered. Pin-outs and the top view are illustrated in Figure 283 for this package.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19			
A	AVSS_2G	AVSS_2G	TP2	TP4	XTAL1	AVDD28_TDO		BTSPKIN_N	AVSS_BT	DVDD28		UTXD2	URXD2		CMDAT3	CMPCLK		CMPDN	GND	A		
B	RXLB_P	RXLB_N	TP1	TP3	XTAL2	CLK_SEL	AVDD28_BT	AVSS_BT1	AVSS_BT	DVDD28_PERC	JRTCK	URXD1	UTXD1	CMVREF	CMDAT7	CMRST	CMDAT0	CMDAT6	CMMCLK	B		
C			AVSS_2G		AVSS_2G		AVSS_BT		BTREXT	BPL_BUS0	JTCK	JTDO	CMDAT4		CMDAT2	CMDAT5	CMDAT1	NLD2	NLD0	C		
D	RXHB_N	RXHB_P		AVSS_2G	FREF1	FREF2	AVSS_BT	AVDD28_BT	VDDK	BPL_BUS2	BPL_BUS3	JTMS	CMHREF		DVDD18_EMI		WATCHDOG	NLD1	NLD8	D		
E	TXO_LB	TXO_HB	AVSS_2G		AVSS_2G						BPL_BUS1	JTRST_B	JTDI		DVDD18_EMI	DVDD18_EMI		NLD3	NLD4	E		
F	AVSS_2G	AVSS_2G		AVDD28_VRF	AVSS_2G	AVSS_2G									DVDD18_EMI		LSCE1_B	LSRSTB	NLD7	F		
G	VRF	VBAT_RF	VTCXO	VCAMA	VA	VB		AVDD28_VREF								LPA0		NLD5	SCL18	G		
H		VBAT_ANALOG						AVSS43_PMU		VDDK	GND	GND				LPCE1_B	SDA18	LRD_B		H		
J	PWRKEY	ISINK0	VREF	AVSS43_PMU	AVSS43_PMU	AVSS43_PMU	AVSS43_PMU	GND		GND	GND	GND			LPTE	NLD6	LWR_B	SCL28	LPRSTB	J		
K	KPLED	ISINK3		ISINK1				DVDD18		BRCLKENAI	RESETB	EINT12				LPCE0_B		SDA28	MCINS	K		
L	BATSNS	ISENSE	AVSS43_SPK		AGND	TESTMODE	AVSS43_PMU										KROW7	KCOL5	SPL_CS	L		
M	VBAT_SPK	SPK_OUTP		DRV	ISINK2	CHRLDD											KCOL6	KCOL4	KROW6	KROW5	KCOL1	M
N		SPK_OUTN	BATDET	BATON		VCDT	VUSB	VSIM2	VSIM1								SPL_SCK	DVDD28	KCOL0	KCOL2	KROW1	N
P					AVSS28_ABB				AVSS43_PMU	XTAL_SEL												P
R	APC	ACCDET			AVDD28_ABB						SIM1_SCLK		DVDD28_HSD				EINT0	KROW4	KROW2		R	
T	AU_MISO_A50	AU_MISO_A51		HSP	VSF	VCAMD	VRTC	AVDD28_FM			SIM2_SCLK				MCDAA0	DVDD28_SFP	SPL_MISO	KROW3	KCOL7		T	
U	AU_VIN0_P	AU_VIN0_N		HSN	VCORE	VIO18		AVSS_FM		AVSS_FM	USB11_DP	SIM1_SRST		MCCK		GPIOW4	SD_PWRCK	SPL_MISO	SFCS0		U	
V	AU_VIN1_P	AU_VIN1_N	YP	XM	HPL	VBAT_DIGITAL	VIO28	VMC	XIN	RGUAI_VIFA	RGUAI_VIFA	USB11_DM	GND	SIM2_SIO	MCCM0	SFHHOLD	SFOUT	SFWP	SFIN		V	
W	AVSS28_ABB	AUX_IN4	XP	YM	HPR	VBAT_DIGITAL		VIBR	XOUT	RGUAI_VIFA	RGUAI_VIFA			SIM1_SIO	SIM2_SRST		DVDD28_SF	SFCK	GND		W	
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19			

Figure 3. Ball diagram and top view

2.1.2 Pin Coordination

Pin#	Net name	Pin#	Net name	Pin#	Net name
A1	AVSS_2G	F19	NLD7	N7	VUSB
A10	DVDD28	F2	AVSS_2G	N8	VSIM2
A12	UTXD2	F4	AVDD28_VRF	N9	VSIM1
A13	URXD2	F5	AVSS_2G	P17	KROW0
A15	CMDAT3	F6	AVSS_2G	P18	KCOL3
A16	CMPCLK	G1	VRF	P4	AVSS28_ABB
A18	CMPDN	G16	LPA0	P8	AVSS43_PMU
A19	GND	G18	NLD5	P9	XTAL_SEL

Pin#	Net name	Pin#	Net name	Pin#	Net name
A2	AVSS_2G	G19	SCL18	R1	APC
A3	TP2	G2	VBAT_RF	R11	SIM1_SCLK
A4	TP4	G3	VTCXO	R13	DVDD33_MSDC
A5	XTAL1	G4	VCAMA	R17	EINT0
A6	AVDD28_TCXO	G5	VA	R18	KROW4
A8	BTRF2P4G_N	G6	VBT	R19	KROW2
A9	AVSS_BT	G8	AVDD28_2GAFE	R2	ACCDET
B1	RXLB_P	H10	VDDK	R4	AVDD28_ABB
B10	DVDD28_FSRC	H11	GND	T1	AU_MICBIAS0
B11	JRTCK	H12	GND	T12	SIM2_SCLK
B12	URXD1	H16	LPCE1_B	T15	MCDA0
B13	UTXD1	H17	SDA18	T16	DVDD28_SFP
B14	CMVREF	H18	LRD_B	T17	SPI_MOSI
B15	CMDAT7	H2	VBAT_ANALOG	T18	KROW3
B16	CMRST	H7	AVSS43_PMU	T19	KCOL7
B17	CMDAT0	J1	PWRKEY	T2	AU_MICBIAS1
B18	CMDAT6	J10	GND	T4	HSP
B19	CMMCLK	J11	GND	T5	VSF
B2	RXLB_N	J12	GND	T7	VCAMD
B3	TP1	J13	GND	T8	VRTC
B4	TP3	J15	LPTE	T9	AVDD28_FM
B5	XTAL2	J16	NLD6	U1	AU_VIN0_P
B6	CLK_SEL	J17	LWR_B	U11	AVSS_FM
B7	AVDD28_ABT	J18	SCL28	U12	USB11_DP
B8	AVSS_BT1	J19	LPRSTB	U13	SIM1_SRST
B9	AVSS_BT	J2	ISINK0	U14	MCCK
C10	BPI_BUS0	J3	VREF	U16	GPIO74
C11	JTCK	J4	AVSS43_PMU	U17	SD_PWROK
C12	JTDO	J5	AVSS43_PMU	U18	SPI_MISO
C13	CMDAT4	J6	AVSS43_PMU	U19	SFCS0
C15	CMDAT2	J7	AVSS43_PMU	U2	AU_VIN0_N
C16	CMDAT5	J8	GND	U4	HSN
C17	CMDAT1	K1	KPLED	U5	VCORE
C18	NLD2	K10	SRCLKENAI	U7	VIO18
C19	NLD0	K11	RESETB	U9	AVSS_FM
C3	AVSS_2G	K12	EINT12	V1	AU_VIN1_P
C5	AVSS_2G	K16	LPCE0_B	V10	RXLNA_INP_LA
C7	AVSS_BT	K18	SDA28	V11	RXLNA_INN_SA
C9	BTREXT	K19	MCINS	V12	USB11_DM
D1	RXHB_N	K2	ISINK3	V13	GND
D10	BPI_BUS2	K4	ISINK1	V14	SIM2_SIO
D11	BPI_BUS3	K8	DVDD18	V15	MCCM0
D12	JTMS	L1	BATSNS	V16	SFHOLD
D13	CMHREF	L16	KROW7	V17	SFOUT

Pin#	Net name	Pin#	Net name	Pin#	Net name
D15	DVDD18_EMI	L17	KCOL5	V18	SFWP
D17	WATCHDOG	L18	SPI_CS	V19	SFIN
D18	NLD1	L2	ISENSE	V2	AU_VIN1_N
D19	NLD8	L3	AVSS43_SPK	V3	YP
D2	RXHB_P	L5	AGND	V4	XM
D4	AVSS_2G	L6	TESTMODE	V5	HPL
D5	FREF1	L7	AVSS43_PMU	V6	VBAT_DIGITAL
D6	FREF2	M1	VBAT_SPK	V7	VIO28
D7	AVSS_BT	M15	KCOL6	V8	VMC
D8	AVDD28_DBT	M16	KCOL4	V9	XIN
D9	VDDK	M17	KROW6	W1	AVSS28_ABB
E1	TXO_LB	M18	KROW5	W10	RXLNA_INN_LA
E11	BPI_BUS1	M19	KCOL1	W11	RXLNA_INP_SA
E12	JTRST_B	M2	SPK_OUTP	W14	SIM1_SIO
E13	JTDI	M4	DRV	W15	SIM2_SRST
E15	DVDD18_EMI	M5	ISINK2	W17	DVDD28_SF
E16	DVDD18_EMI	M6	CHR_LDO	W18	SFCK
E17	NLD3	N15	SPI_SCK	W19	GND
E18	NLD4	N16	DVDD28	W2	AUX_IN4
E2	TXO_HB	N17	KCOL0	W3	XP
E3	AVSS_2G	N18	KCOL2	W4	YM
E5	AVSS_2G	N19	KROW1	W5	HPR
F1	AVSS_2G	N2	SPK_OUTN	W6	VBAT_DIGITAL
F15	DVDD18_EMI	N3	BATDET	W8	VIBR
F17	LSCE1_B	N4	BATON	W9	XOUT
F18	LSRSTB	N6	VCDT		

Table 1. Pin coordinates

2.1.3 Detailed Pin Description

Abbreviation	Description
AI	Analog input
AO	Analog output
AIO	Analog bi-direction
DI	Digital input
DO	Digital output
DIO	Digital bi-direction
P	Power
G	Ground

Table 2. Acronym for pin types

Pin name	Type	Description	Power domain
System			
GPIO74	DIO	Reserved	DVDD28_SFP
RESETB	DIO	System reset	DVDD18
SRCLKENAI	DIO	26MHz clock request by external devices	DVDD18
RF control circuitry			
BPI_BUS0	DIO	RF hard-wire control bus bit 0	DVDD28
BPI_BUS1	DIO	RF hard-wire control bus bit 1	DVDD28
BPI_BUS2	DIO	RF hard-wire control bus bit 2	DVDD28
BPI_BUS3	DIO	RF hard-wire control bus bit 3	DVDD28
UART interface			
URXD1	DIO	UART1 receive data	DVDD28
UTXD1	DIO	UART1 transmit data	DVDD28
URXD2	DIO	UART2 receive data	DVDD28
UTXD2	DIO	UART2 transmit data	DVDD28
JTAG interface			
JTMS	DIO	JTAG test port mode switch	DVDD28
JTDI	DIO	JTAG test port data input	DVDD28
JTCK	DIO	JTAG test port clock input	DVDD28
JTRST_B	DIO	JTAG test port reset input	DVDD28
JRTCK	DIO	JTAG test port returned clock output	DVDD28
JTDO	DIO	JTAG test port data output	DVDD28
External interrupt			
EINT0	DIO	External interrupt 0	DVDD28
EINT12	DIO	External interrupt 12	DVDD18
Keypad interface			
KCOL0	DIO	Keypad column 0	DVDD28
KCOL1	DIO	Keypad column 1	DVDD28
KCOL2	DIO	Keypad column 2	DVDD28
KCOL3	DIO	Keypad column 3	DVDD28
KCOL4	DIO	Keypad column 4	DVDD28
KCOL5	DIO	Keypad column 5	DVDD28
KCOL6	DIO	Keypad column 6	DVDD28
KCOL7	DIO	Keypad column 7	DVDD28
KROW0	DIO	Keypad row 0	DVDD28
KROW1	DIO	Keypad row 1	DVDD28
KROW2	DIO	Keypad row 2	DVDD28
KROW3	DIO	Keypad row 3	DVDD28

Pin name	Type	Description	Power domain
KROW4	DIO	Keypad row 4	DVDD28
KROW5	DIO	Keypad row 5	DVDD28
KROW6	DIO	Keypad row 6	DVDD28
KROW7	DIO	Keypad row 7	DVDD28
Camera interface			
CMRST	DIO	CMOS sensor reset signal output	DVDD28
CMPDN	DIO	CMOS sensor power down control	DVDD28
CMVREF	DIO	CMOS sensor vertical reference signal input	DVDD28
CMHREF	DIO	CMOS sensor horizontal reference signal input	DVDD28
CMDAT0	DIO	CMOS sensor data input 0	DVDD28
CMDAT1	DIO	CMOS sensor data input 1	DVDD28
CMDAT2	DIO	CMOS sensor data input 2	DVDD28
CMDAT3	DIO	CMOS sensor data input 3	DVDD28
CMDAT4	DIO	CMOS sensor data input 4	DVDD28
CMDAT5	DIO	CMOS sensor data input 5	DVDD28
CMDAT6	DIO	CMOS sensor data input 6	DVDD28
CMDAT7	DIO	CMOS sensor data input 7	DVDD28
CMPCCLK	DIO	CMOS sensor master clock output	DVDD28
CMMCLK	DIO	CMOS sensor master clock output	DVDD28
MS/SD card interface			
MCINS	DIO	SD card detect Input	DVDD18_EMI
MCDA0	DIO	SD serial data IO 0/memory stick serial data IO	DVDD33_MSDC
MCCK	DIO	SD serial clock/memory stick serial clock	DVDD33_MSDC
MCCM0	DIO	SD command output/memory stick bus state output	DVDD33_MSDC
SD_PWROK	DIO	For SD card general-purpose IO	DVDD33_MSDC
SIM card interface			
SIM1_SIO	DIO	SIM1 data input/outputs	VSIM1
SIM1_SRST	DIO	SIM1 card reset output	VSIM1
SIM1_SCLK	DIO	SIM1 card clock output	VSIM1
SIM2_SIO	DIO	SIM2 data input/outputs	VSIM2
SIM2_SRST	DIO	SIM2 card reset output	VSIM2
SIM2_SCLK	DIO	SIM2 card clock output	VSIM2
I2C interface			
SCL28	DIO	I2C clock 2.8v power domain	DVDD28
SDA28	DIO	I2C data 2.8v power domain	DVDD28

Pin name	Type	Description	Power domain
SCL18	DIO	I2C clock 1.8v power domain	DVDD18_EMI
SDA18	DIO	I2C data 1.8v power domain	DVDD18_EMI
LCD interface			
LSRSTB	DIO	Serial display interface reset signal	DVDD18_EMI
LSCE1_B	DIO	Serial display interface chip select 1 output	DVDD18_EMI
LPCE1_B	DIO	Parallel display interface chip select 1 output	DVDD18_EMI
LPCE0_B	DIO	Parallel display interface chip select 0 output	DVDD18_EMI
LPTE	DIO	Parallel display interface tearing effect	DVDD18_EMI
LPRSTB	DIO	Parallel display interface reset signal	DVDD18_EMI
LRD_B	DIO	Parallel display interface read strobe	DVDD18_EMI
LPA0	DIO	Parallel display interface address output	DVDD18_EMI
LWR_B	DIO	Parallel display interface write strobe	DVDD18_EMI
NLD8	DIO	Parallel LCD data 8	DVDD18_EMI
NLD7	DIO	Parallel LCD data 7	DVDD18_EMI
NLD6	DIO	Parallel LCD data 6	DVDD18_EMI
NLD5	DIO	Parallel LCD data 5	DVDD18_EMI
NLD4	DIO	Parallel LCD data 4	DVDD18_EMI
NLD3	DIO	Parallel LCD data 3	DVDD18_EMI
NLD2	DIO	Parallel LCD data 2	DVDD18_EMI
NLD1	DIO	Parallel LCD data 1	DVDD18_EMI
NLD0	DIO	Parallel LCD data 0	DVDD18_EMI
Serial port interface			
SPI_MOSI	DIO	Data signal from master output to slave input	DVDD28
SPI_MISO	DIO	Data signal from slave output to master input	DVDD28
SPI_SCK	DIO	Bit serial clock	DVDD28
SPI_CS	DIO	Low-active chip select signal	DVDD28
Watchdog reset			
WATCHDOG	DIO	Reset external memory device	DVDD18_EMI
Serial flash interface			
SFCS0	DIO	Serial Flash chip select	DVDD28_SFP
SFIN	DIO	Serial Flash data input/output	DVDD28_SFP
SFOUT	DIO	Serial Flash data input/output	DVDD28_SFP
SFSHOLD	DIO	Serial Flash data input/output	DVDD28_SFP
SFWP	DIO	Serial Flash data input/output	DVDD28_SFP
SFCK	DIO	Serial Flash clock	DVDD28_SFP

Pin name	Type	Description	Power domain
FM			
RXLNA_INN_LA	AI	FM input from long antenna	AVDD28_FM
RXLNA_INP_LA	AI	FM input from long antenna	AVDD28_FM
RXLNA_INN_SA	AI	FM input from short antenna	AVDD28_FM
RXLNA_INP_SA	AI	FM input from short antenna	AVDD28_FM
Bluetooth			
BTRF2P4G_N	AIO	Bluetooth RF single-ended input	-
BTREXT	AIO	Bluetooth external reference resistor	-
2G RF			
RXHB_P	AIO	Differential RF input for highband Rx (DCS/PCS)	-
RXHB_N	AIO	Differential RF input for highband Rx (DCS/PCS)	-
RXLB_P	AIO	Differential RF input for lowband Rx (GSM900/GSM850)	-
RXLB_N	AIO	Differential RF input for lowband Rx (GSM900/GSM850)	-
TXO_HB	AIO	RF output for highband Tx (DCS/PCS)	-
TXO_LB	AIO	RF output pin for lowband Tx (GSM900/GSM850)	-
FREF1	AIO	DCXO reference clock output	-
FREF2	AIO	DCXO reference clock output	-
XTAL1	AIO	Input 1 for DCXO crystal	-
XTAL2	AIO	Input 2 for DCXO crystal	-
TP1	AIO	Test pin 1	-
TP2	AIO	Test pin 2	-
TP3	AIO	Test pin 3	-
TP4	AIO	Test pin 4	-
CLK_SEL	AIO	DCXO mode selection	-
USB			
USB11_DM	AIO	D- data Input/Output	-
USB11_DP	AIO	D+ data Input/Output	-
Analog baseband			
HPR	AIO	Audio head phone output (R channel)	AVDD28_ABB
HPL	AIO	Audio head phone output (L channel)	AVDD28_ABB
HSP	AIO	Voice handset output (positive)	AVDD28_ABB
HSN	AIO	Voice handset output (negative)	AVDD28_ABB
AU_VIN0_P	AIO	Microphone 0 input (positive)	AVDD28_ABB
AU_VIN0_N	AIO	Microphone 0 input (negative)	AVDD28_ABB
AU_VIN1_P	AIO	Microphone 1 input (positive)	AVDD28_ABB

Pin name	Type	Description	Power domain
AU_VIN1_N	AIO	Microphone 1 input (negative)	AVDD28_ABB
AUX_IN4	AIO	Auxiliary ADC input	AVDD28_ABB
SPK_OUTP	AIO	Speaker positive output	AVDD28_ABB
SPK_OUTN	AIO	Speaker negative output	AVDD28_ABB
APC	AIO	Automatic power control DAC output	AVDD28_ABB
XP	AIO	Touch panel X-axis positive input	AVDD28_ABB
XM	AIO	Touch panel X-axis negative input	AVDD28_ABB
YP	AIO	Touch panel Y-axis positive input	AVDD28_ABB
YM	AIO	Touch panel Y-axis negative input	AVDD28_ABB
AU_MICBIAS0	AIO	Microphone bias source 0	AVDD28_ABB
AU_MICBIAS1	AIO	Microphone bias source 1	AVDD28_ABB
ACCDDET	AIO	Accessory detection	AVDD28_ABB
Real-time clock			
XIN	AIO	Input pin for 32K crystal	VRTC
XOUT	AIO	Input pin for 32K crystal	VRTC
XTAL_SEL	DIO	Pin option for external 32K crystal	VRTC
Power management unit			
VA	AIO	LDO output for ABB - VA	VBAT_ANALOG
VBAT	AIO	LDO output for BTRF - VBT	VBAT_RF
VCAMA	AIO	LDO output for sensor - VCAMA	VBAT_ANALOG
VCAMD	AIO	LDO output for sensor - VCAMD	VBAT_DIGITAL
VIBR	AIO	LDO output for vibrator - VIBR	VBAT_DIGITAL
VIO18	AIO	LDO output for 1.8V power - VIO18	VBAT_DIGITAL
VIO28	AIO	LDO output for 2.8V power - VIO28	VBAT_DIGITAL
VMC	AIO	LDO output for memory card - VMC	VBAT_DIGITAL
VSF	AIO	LDO output for serial flash - VSF	VBAT_DIGITAL
VRF	AIO	LDO output for GSMRF - VRF	VBAT_DIGITAL
VRTC	AIO	LDO output for RTC - VRTC	VBAT_DIGITAL
VSIM1	AIO	LDO output for 1 st SIM - VSIM	VBAT_DIGITAL
VSIM2	AIO	LDO output for 2 nd SIM - VSIM2	VBAT_DIGITAL
VTCXO	AIO	LDO output for DCXO - VTCXO	VBAT_ANALOG
VUSB	AIO	LDO output for USB - VUSB	VBAT_DIGITAL
VCORE	AIO	LDO output for core circuit - Vcore	VBAT_DIGITAL
VREF	AIO	Band gap reference	BATSNS
VCDT	AIO	Charger-In level sense pin	BATSNS
DRV	AIO	IDAC current output open-drain pin	BATSNS
BATON	AIO	Battery Pack, NTC connected pin	BATSNS
ISENSE	AIO	Top node of current sensing 0.2ohm Rsense resistor	BATSNS

Pin name	Type	Description	Power domain
CHR_LDO	AIO	2.8V shunt-regulator output	BATSNS
BATDET	AIO	Battery detection pin	BATSNS
ISINK0	AIO	Backlight driver channel 0	VBAT_SPK
ISINK1	AIO	Backlight driver channel 1	VBAT_SPK
ISINK2	AIO	Backlight driver channel 2	VBAT_SPK
ISINK3	AIO	Backlight driver channel 3	VBAT_SPK
KPLED	AIO	Keypad led driver	VBAT_SPK
TESTMODE	AIO	Test mode	BATSNS
PWRKEY	AIO	PWR key	BATSNS
Analog power			
AVDD28_FM	P	FM power	-
AVDD28_VRF	P	2.8V power supply for 2G RF	-
AVDD28_TCXO	P	2.8V power supply for 2G TCXO	-
AVDD28_2GAFE	P	2.8V power supply for 2G AFE	-
AVDD28_ABB	P	ABB 2.8V power	-
AVDD28_DBT	P	2.8V power supply for DBT	-
AVDD28_ABT	P	2.8V power supply for ABT	-
VBAT_RF	P	RF LDOs used battery voltage input	-
VBAT_DIGITAL	P	Digital LDOs used battery voltage input	-
VBAT_ANALOG	P	Analog LDOs used battery voltage input	-
VBAT_SPK	P	VBAT input for loud speaker driver	-
BATSNS	P	Battery node of battery pack	-
Analog ground			
AVSS28_ABB	G	ABB 2.8V ground	-
AVSS_BT	G	BT ground	-
AVSS_BT1	G	BT1 ground	-
AVSS_2G	G	2G RF ground	-
AVSS_FM	G	FM ground	-
AVSS43_PMU	G	PMU ground	-
AVSS43_SPK	G	SPK ground	-
AGND	G	GND for VREF	-
Digital power			
DVDD28	P	2.8V power supply for digital macros in transceiver	-
DVDD18	P	1.8V power supply for digital macros in transceiver	-
DVDD28_FSRC	P	E-FUSE blowing power control	-
DVDD33_MSDC	P	3.3V memory card power	-
DVDD18_EMI	P	1.8V EMI IO power	-
DVDD28_SF	P	2.8V serial flash IO power	-

Pin name	Type	Description	Power domain
DVDD28_SFP	P	2.8V serial flash IO power	-
VDDK	P	1.2V core power	-
Digital ground			
GND	G	Ground	-

Table 3. PIN function description and power domain

Abbreviation	Description
I	Input
LO	Low output
HO	High output
XO	Low or high output
PU	Pull-up
PD	Pull-down
-	No PU/PD
0~N	Aux. function number
X	Delicate function pin

Table 4. Acronym for state of pins

Name	Reset			Output drivability	Termination when not used	IO type
	State ¹	Aux ²	PU/PD ³			
System						
GPIO74	I	1	PU	DIOH2/DIOL2	No need	IO Type2
RESETB	O	1	-	DIOH6/DIOL6	No need	IO Type6
SRCLKENAI	I	7	PD	DIOH6/DIOL6	No need	IO Type6
RF control circuitry						
BPI_BUS0	O	1	-	DIOH1/DIOL1	No need	IO Type1
BPI_BUS1	O	1	-	DIOH1/DIOL1	No need	IO Type1
BPI_BUS2	O	1	-	DIOH1/DIOL1	No need	IO Type1
BPI_BUS3	O	1	-	DIOH1/DIOL1	No need	IO Type1
UART interface						
URXD1	I	1	PU	DIOH3/DIOL3	No need	IO Type3
UTXD1	O	1	-	DIOH1/DIOL1	No need	IO Type1
URXD2	I	7	PD	DIOH1/DIOL1	No need	IO Type1

¹ The column "State" of "Reset" shows the pin state during reset. (Input, High Output, Low Output, etc)

² The column "Aux" for "Reset" means the default aux function number, shown in the table "Pin Multiplexing, Capability and Settings".

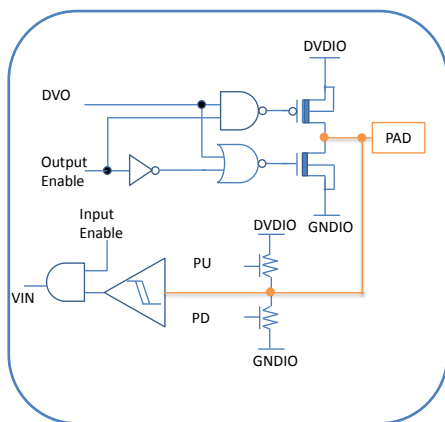
³ The column "PU/PD" for "Reset" means if there is internal pull-up or pull-down when the pin is input in the reset state.

Name	Reset			Output drivability	Termination when not used	IO type
	State ¹	Aux ²	PU/PD ³			
UTXD2	I	7	PD	DIOH1/DIOL1	No need	IO Type1
JTAG interface						
JTMS	I	1	PU	DIOH1/DIOL1	No need	IO Type1
JTDI	I	1	PU	DIOH1/DIOL1	No need	IO Type1
JTCK	I	1	PU	DIOH1/DIOL1	No need	IO Type1
JTRST_B	I	1	PD	DIOH1/DIOL1	No need	IO Type1
JRTCK	O	1	-	DIOH1/DIOL1	No need	IO Type1
JTDO	O	1	-	DIOH1/DIOL1	No need	IO Type1
External interrupt						
EINT0	I	7	PD	DIOH1/DIOL1	No need	IO Type1
EINT12	I	7	PD	DIOH6/DIOL6	No need	IO Type6
Keypad Interface						
KCOL0	I	0	PU	DIOH4/DIOL4	No need	IO Type4
KCOL1	I	7	PD	DIOH4/DIOL4	No need	IO Type4
KCOL2	I	7	PD	DIOH4/DIOL4	No need	IO Type4
KCOL3	I	7	PD	DIOH4/DIOL4	No need	IO Type4
KCOL4	I	7	PD	DIOH4/DIOL4	No need	IO Type4
KCOL5	I	7	PD	DIOH1/DIOL1	No need	IO Type1
KCOL6	I	7	PD	DIOH3/DIOL3	No need	IO Type3
KCOL7	I	7	PD	DIOH3/DIOL3	No need	IO Type3
KROW0	O	0	PU	DIOH5/DIOL5	No need	IO Type5
KROW1	I	7	PD	DIOH5/DIOL5	No need	IO Type5
KROW2	I	7	PD	DIOH5/DIOL5	No need	IO Type5
KROW3	I	7	PD	DIOH5/DIOL5	No need	IO Type5
KROW4	I	7	PD	DIOH5/DIOL5	No need	IO Type5
KROW5	I	7	PD	DIOH1/DIOL1	No need	IO Type1
KROW6	I	7	PD	DIOH3/DIOL3	No need	IO Type3
KROW7	I	7	PD	DIOH1/DIOL1	No need	IO Type1
Camera interface						
CMRST	I	0	PD	DIOH1/DIOL1	No need	IO Type1
CMPDN	I	0	-	DIOH1/DIOL1	No need	IO Type1
CMVREF	I	7	PD	DIOH1/DIOL1	No need	IO Type1
GMHREF	I	7	PD	DIOH1/DIOL1	No need	IO Type1
CMDAT0	I	7	PD	DIOH1/DIOL1	No need	IO Type1
CMDAT1	I	7	PD	DIOH1/DIOL1	No need	IO Type1
CMDAT2	I	7	PD	DIOH1/DIOL1	No need	IO Type1
CMDAT3	I	7	PD	DIOH1/DIOL1	No need	IO Type1
CMDAT4	I	7	PD	DIOH1/DIOL1	No need	IO Type1
CMDAT5	I	7	PD	DIOH1/DIOL1	No need	IO Type1

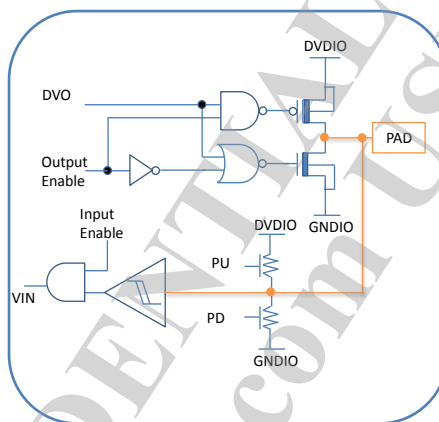
Name	Reset			Output drivability	Termination when not used	IO type
	State ¹	Aux ²	PU/PD ³			
CMDAT6	I	7	PD	DIOH1/DIOL1	No need	IO Type1
CMDAT7	I	7	PD	DIOH1/DIOL1	No need	IO Type1
CMPCLK	I	7	PD	DIOH1/DIOL1	No need	IO Type1
CMMCLK	I	7	PD	DIOH1/DIOL1	No need	IO Type1
MS/SD card interface						
MCINS	I	7	PD	DIOH2/DIOL2	No need	IO Type2
SD_PWROK	I	7	PD	DIOH2/DIOL2	No need	IO Type2
MCDA0	I	0	PD	DIOH3/DIOL3	No need	IO Type3
MCCM	I	0	PD	DIOH3/DIOL3	No need	IO Type3
MCMM0	I	0	PD	DIOH3/DIOL3	No need	IO Type3
I2C interface						
SCL28	I	7	PD	DIOH1/DIOL1	No need	IO Type1
SDA28	I	7	PD	DIOH1/DIOL1	No need	IO Type1
SCL18	I	7	PD	DIOH2/DIOL2	No need	IO Type2
SDA18	I	7	PD	DIOH2/DIOL2	No need	IO Type2
LCD interface						
LSRSTB	I	7	PD	DIOH2/DIOL2	No need	IO Type2
LSCE1B	I	0	PU	DIOH2/DIOL2	No need	IO Type2
LPCE0B	O	1	-	DIOH2/DIOL2	No need	IO Type2
LPCE1B	O	1	-	DIOH2/DIOL2	No need	IO Type2
LPTE	I	7	PD	DIOH2/DIOL2	No need	IO Type2
LPRSTB	I	7	PD	DIOH2/DIOL2	No need	IO Type2
LRD_B	I	7	PD	DIOH2/DIOL2	No need	IO Type2
LPAD0	I	7	PD	DIOH2/DIOL2	No need	IO Type2
LWR_B	I	7	PD	DIOH2/DIOL2	No need	IO Type2
NLD8	I	7	PD	DIOH2/DIOL2	No need	IO Type2
NLD7	I	7	PD	DIOH2/DIOL2	No need	IO Type2
NLD6	I	7	PD	DIOH2/DIOL2	No need	IO Type2
NLD5	I	7	PD	DIOH2/DIOL2	No need	IO Type2
NLD4	I	7	PD	DIOH2/DIOL2	No need	IO Type2
NLD3	I	7	PD	DIOH2/DIOL2	No need	IO Type2
NLD2	I	7	PD	DIOH2/DIOL2	No need	IO Type2
NLD1	I	7	PD	DIOH2/DIOL2	No need	IO Type2
NLD0	I	7	PD	DIOH2/DIOL2	No need	IO Type2
Serial port interface						
SPI_MOSI	I	7	PD	DIOH1/DIOL1	No need	IO Type1
SPI_MISO	I	7	PD	DIOH1/DIOL1	No need	IO Type1
SPI_SCK	I	7	PD	DIOH1/DIOL1	No need	IO Type1
SPI_SCS	I	7	PD	DIOH1/DIOL1	No need	IO Type1

Name	Reset			Output drivability	Termination when not used	IO type
	State ¹	Aux ²	PU/PD ³			
Watchdog reset						
WATCHDOG	O	1	-	DIOH1/DIOL1	No need	IO Type1
Serial flash interface						
SFCS0	O	1	-	DIOH2/DIOL2	No need	IO Type2
SFIN	I	1	-	DIOH2/DIOL2	No need	IO Type2
SFOUT	O	1	-	DIOH2/DIOL2	No need	IO Type2
SFSHOLD	O	1	-	DIOH2/DIOL2	No need	IO Type2
SFWP	O	1	-	DIOH2/DIOL2	No need	IO Type2
SFCK	O	1	-	DIOH2/DIOL2	No need	IO Type2

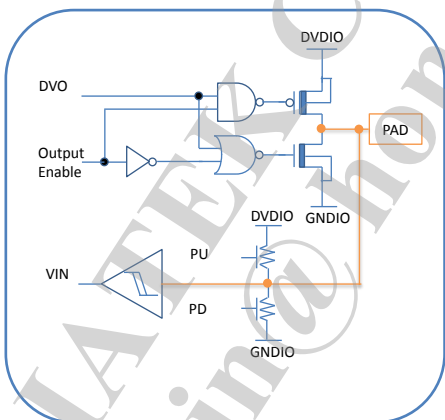
Table 5. State of pins



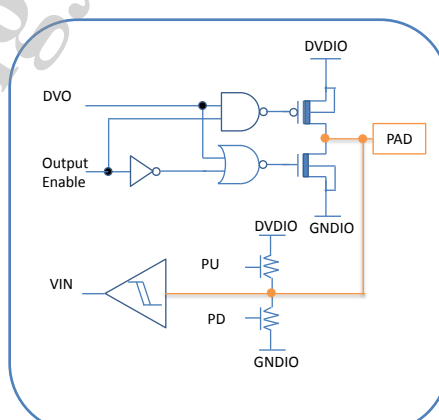
IO type1



IO type2



IO type3



IO type4

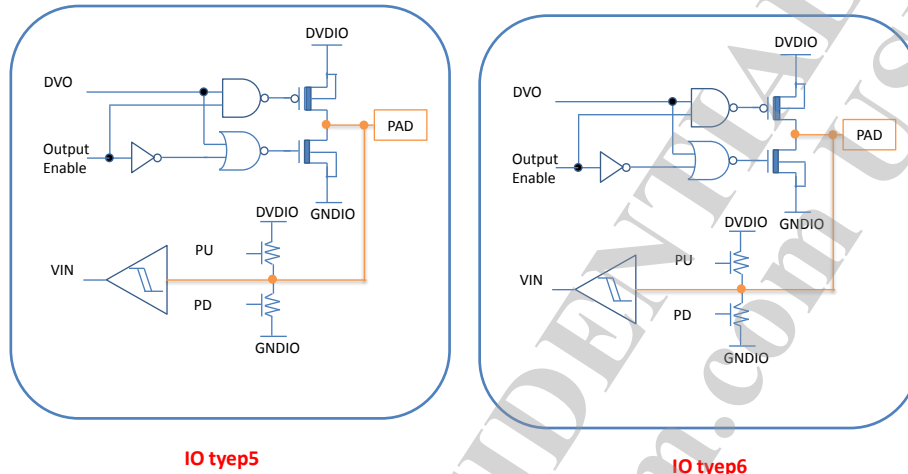


Figure 4. IO types in state of pins

2.1.4 Pin Multiplexing, Capability and Settings

Abbreviation	Description
PU	Pull-up, not controllable
PD	Pull-down, not controllable
CU	Pull-up, controllable
CD	Pull-down, controllable
X	Cannot pull-up or pull-down

Table 6. Acronym for pull-up and pull-down types

Name	Aux. function	Aux. name	Aux. type	PU/PD/CU/CD	Driving	SMT
SPI_MOSI	0	GPIO0	IO	CU, CD	4, 8, 12, 16mA	0
	1	CLKO0	O	-	4, 8, 12, 16mA	0
	3	EDICK	O	-	4, 8, 12, 16mA	0
	4	SPIMOSI0	O	-	4, 8, 12, 16mA	0
	5	U3RXD	I	CU, CD	4, 8, 12, 16mA	0
	6	PWM	O	-	4, 8, 12, 16mA	0
	7	EGND0	I	PD	4, 8, 12, 16mA	0

Name	Aux. function	Aux. name	Aux. type	PU/PD/CU/CD	Driving	SMT
SCL28	0	GPIO1	IO	CU, CD	4, 8, 12, 16mA	0
	1	SCL	IO	CU, CD	4, 8, 12, 16mA	0
	4	SPISCK0	O	-	4, 8, 12, 16mA	0
	7	EGND1	I	PD	4, 8, 12, 16mA	0
SDA28	0	GPIO2	IO	CU, CD	4, 8, 12, 16mA	0
	1	SDA	IO	CU, CD	4, 8, 12, 16mA	0
	4	SPICS0	O	-	4, 8, 12, 16mA	0
	7	EGND2	I	CU, CD	4, 8, 12, 16mA	0
KCOL7	0	GPIO3	IO	CU, CD	4, 8, 12, 16mA	0
	1	KCOL7	IO	CU, CD	4, 8, 12, 16mA	0
	2	EDICK	O	-	4, 8, 12, 16mA	0
	5	MCDA1	IO	CU, CD	4, 8, 12, 16mA	0
	6	LSA0DA10	O	-	4, 8, 12, 16mA	0
	7	EGND3	I	PD	4, 8, 12, 16mA	0
KCOL6	0	GPIO4	IO	CU, CD	4, 8, 12, 16mA	0
	1	KCOL6	IO	CU, CD	4, 8, 12, 16mA	0
	2	U1RXD	I	PU	4, 8, 12, 16mA	0
	3	CLKO5	O	-	4, 8, 12, 16mA	0
	4	WIFITOB	I	CU, CD	4, 8, 12, 16mA	0
	5	MCDA2	IO	CU, CD	4, 8, 12, 16mA	0
	7	EGND4	I	PD	4, 8, 12, 16mA	0
KCOL5	0	GPIO5	IO	CU, CD	4, 8, 12, 16mA	0
	1	KCOL5	IO	CU, CD	4, 8, 12, 16mA	0
	2	EDI2CK	O	-	4, 8, 12, 16mA	0
	5	EINT2	I	CU, CD	4, 8, 12, 16mA	0
	7	EGND5	I	PD	4, 8, 12, 16mA	0
KCLO4	0	GPIO6	IO	CU, CD	4, 8, 12, 16mA	0
	1	KCOL4	IO	CU, CD	4, 8, 12, 16mA	0
	3	U3CTS	I	CU, CD	4, 8, 12, 16mA	0
	7	EGND6	I	PD	4, 8, 12, 16mA	0
KCOL3	0	GPIO7	IO	CU, CD	4, 8, 12, 16mA	0
	1	KCOL3	IO	CU, CD	4, 8, 12, 16mA	0
	2	DAIPCMIN	I	CU, CD	4, 8, 12, 16mA	0
	7	EGND7	I	PD	4, 8, 12, 16mA	0
KCOL2	0	GPIO8	IO	CU, CD	4, 8, 12, 16mA	0

Name	Aux. function	Aux. name	Aux. type	PU/PD/CU/CD	Driving	SMT
	1	KCOL2	IO	CU, CD	4, 8, 12, 16mA	0
	2	DAICLK	O	-	4, 8, 12, 16mA	0
	7	EGND8	I	PD	4, 8, 12, 16mA	0
KCOL1	0	GPIO9	IO	CU, CD	4, 8, 12, 16mA	0
	1	KCOL1	IO	CU, CD	4, 8, 12, 16mA	0
	2	SPIMOSI1	O	-	4, 8, 12, 16mA	0
	7	EGND9	I	PD	4, 8, 12, 16mA	0
KCOL0	0	GPIO10	IO	CU, CD	4, 8, 12, 16mA	0
	1	KCLO0	IO	CU, CD	4, 8, 12, 16mA	0
KROW7	0	GPIO11	IO	CU, CD	4, 8, 12, 16mA	0
	1	KROW7	IO	CU, CD	4, 8, 12, 16mA	0
	2	EINT3	I	CU, CD	4, 8, 12, 16mA	0
	4	EDIDAT	IO	CU, CD	4, 8, 12, 16mA	0
	5	MCDA3	IO	CU, CD	4, 8, 12, 16mA	0
	6	CLKO1	O	-	4, 8, 12, 16mA	0
	7	EGND10	I	PD	4, 8, 12, 16mA	0
KROW6	0	GPIO12	IO	CU, CD	4, 8, 12, 16mA	0
	1	KROW6	IO	CU, CD	4, 8, 12, 16mA	0
	2	EDI2WS	O	CU, CD	4, 8, 12, 16mA	0
	3	WIFITOB	I	CU, CD	4, 8, 12, 16mA	0
	4	SPIMISO0	I	CU, CD	4, 8, 12, 16mA	0
	7	EGND11	I	PD	4, 8, 12, 16mA	0
KROW5	0	GPIO13	IO	CU, CD	4, 8, 12, 16mA	0
	1	KROW5	IO	CU, CD	4, 8, 12, 16mA	0
	2	U1TXD	O	-	4, 8, 12, 16mA	0
	3	EDI2DAT	O	-	4, 8, 12, 16mA	0
	4	SRCLKENA	O	-	4, 8, 12, 16mA	0
	7	EGND12	I	PD	4, 8, 12, 16mA	0
KROW4	0	GPIO14	IO	CU, CD	4, 8, 12, 16mA	0
	1	KROW4	IO	CU, CD	4, 8, 12, 16mA	0
	2	DAIPCMOUT	O	-	4, 8, 12, 16mA	0
	7	EGND13	I	CU, CD	4, 8, 12, 16mA	0
KROW3	0	GPIO15	IO	CU, CD	4, 8, 12, 16mA	0
	1	KROW3	IO	CU, CD	4, 8, 12, 16mA	0
	2	LSA0DA11	O	-	4, 8, 12, 16mA	0

Name	Aux. function	Aux. name	Aux. type	PU/PD/CU/CD	Driving	SMT
	3	DAISYNC	O	-	4, 8, 12, 16mA	0
	7	EGND14	I	PD	4, 8, 12, 16mA	0
KROW2	0	GPIO16	IO	CU, CD	4, 8, 12, 16mA	0
	1	KROW2	IO	CU, CD	4, 8, 12, 16mA	0
	2	LCK0	O	-	4, 8, 12, 16mA	0
	5	LSDI0	I	CU, CD	4, 8, 12, 16mA	0
	7	EGND15	I	PD	4, 8, 12, 16mA	0
KROW1	0	GPIO17	IO	CU, CD	4, 8, 12, 16mA	0
	1	KROW1	IO	CU, CD	4, 8, 12, 16mA	0
	2	LSDA00	IO	PU, CD	4, 8, 12, 16mA	0
	3	U1CTS	I	CU, CD	4, 8, 12, 16mA	0
	4	DAIRST	I	CU, CD	4, 8, 12, 16mA	0
	7	EGND16	I	PD	4, 8, 12, 16mA	0
KROW0	0	GPIO18	IO	CU, CD	4, 8, 12, 16mA	0
	1	KROW0	IO	CU, CD	4, 8, 12, 16mA	0
	2	LSDI0	I	CU, CD	4, 8, 12, 16mA	0
	4	CLKO6	O	-	4, 8, 12, 16mA	0
	5	LCK0	O	-	4, 8, 12, 16mA	0
	7	EGND17	I	PD	4, 8, 12, 16mA	0
SPI_MOSI	0	GPIO19	IO	CU, CD	4, 8, 12, 16mA	0
	1	SPIMISO1	I	CU, CD	4, 8, 12, 16mA	0
	2	GPSFSYNC	O	-	4, 8, 12, 16mA	0
	3	LRSTB	O	-	4, 8, 12, 16mA	0
	4	DAIRST	I	CU, CD	4, 8, 12, 16mA	0
	7	EGND18	I	PD	4, 8, 12, 16mA	0
SPI_SCK	0	GPIO20	IO	CU, CD	4, 8, 12, 16mA	0
	1	SPISCK1	O	-	4, 8, 12, 16mA	0
	2	BPI_BUS5	O	-	4, 8, 12, 16mA	0
	7	EGND19	I	CU, CD	4, 8, 12, 16mA	0
SD_PWROK	0	GPIO21	IO	CU, CD	4, 8, 12, 16mA	0
	1	CLKO2	O	-	4, 8, 12, 16mA	0
	2	EDIWS	O	-	4, 8, 12, 16mA	0
	3	BPIBUS5	O	-	4, 8, 12, 16mA	0
	4	LCK1	O	-	4, 8, 12, 16mA	0
	7	EGND20	I	PD	4, 8, 12, 16mA	0

Name	Aux. function	Aux. name	Aux. type	PU/PD/CU/CD	Driving	SMT
SPI_CS	0	GPIO22	IO	CU, CD	4, 8, 12, 16mA	0
	1	SPICS1	O	-	4, 8, 12, 16mA	0
	2	BPIBUS4	O	-	4, 8, 12, 16mA	0
	7	EGND21	I	CU, CD	4, 8, 12, 16mA	0
URXD2	0	GPIO23	IO	CU, CD	4, 8, 12, 16mA	0
	1	U2RXD	I	CU, CD	4, 8, 12, 16mA	0
	2	BTPRI	IO	CU, CD	4, 8, 12, 16mA	0
	3	LSCE0B0	O	-	4, 8, 12, 16mA	0
	4	U1RTS	O	-	4, 8, 12, 16mA	0
	7	EGND22	I	PD	4, 8, 12, 16mA	0
UTXD2	0	GPIO24	IO	CU, CD	4, 8, 12, 16mA	0
	1	U2TXD	O	-	4, 8, 12, 16mA	0
	2	U3TXD	O	-	4, 8, 12, 16mA	0
	3	BPIBUS4	O	-	4, 8, 12, 16mA	0
	4	CLKO7	O	-	4, 8, 12, 16mA	0
	7	EGND23	I	PD	4, 8, 12, 16mA	0
URXD1	0	GPIO25	IO	CU, CD	4, 8, 12, 16mA	0
	1	U1RXD	I	PU	4, 8, 12, 16mA	0
	2	LSDI1	I	CU, CD	4, 8, 12, 16mA	0
	3	EINT11	I	CU, CD	4, 8, 12, 16mA	0
UTXD1	0	GPIO26	IO	CU, CD	4, 8, 12, 16mA	0
	1	U1TXD	O	-	4, 8, 12, 16mA	0
	2	U3CTS	I	CU, CD	4, 8, 12, 16mA	0
	3	LSDA01	IO	PU, CD	4, 8, 12, 16mA	0
SCL18	0	GPIO27	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	1	EINT4	I	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	2	DAICLK	O	-	2,4,6,8,10,12 ,14,16mA	0
	3	GPT_INT	I	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	4	SCL	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	5	NLD14	IO	PU, CD	2,4,6,8,10,12 ,14,16mA	0
	7	EGND24	I	PD	2,4,6,8,10,12 ,14,16mA	0
SDA18	0	GPIO28	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	1	U3RTS	O	-	2,4,6,8,10,12 ,14,16mA	0
	2	SDA	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	3	NLD13	IO	PU, CD	2,4,6,8,10,12 ,14,16mA	0

Name	Aux. function	Aux. name	Aux. type	PU/PD/CU/CD	Driving	SMT
	4	U3TXD	O	-	2,4,6,8,10,12 ,14,16mA	0
	5	CMCSD1	I	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	6	EINT1	I	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	7	EGND25	I	PD	2,4,6,8,10,12 ,14,16mA	0
JTDO	1	JTDO	O	-	4, 8, 12, 16mA	0
JTMS	1	JTMS	I	PU	4, 8, 12, 16mA	0
JTCK	1	JTCK	I	PU	4, 8, 12, 16mA	0
JTDI	1	JTDI	I	PU	4, 8, 12, 16mA	0
JTRST_B	1	JTRST_B	I	PD	4, 8, 12, 16mA	0
JRTCK	1	JRTCK	O	-	4, 8, 12, 16mA	0
MCINS	0	GPIO29	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	1	EINT5	I	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	2	DAIPCMOUT	O	-	2,4,6,8,10,12 ,14,16mA	0
	3	NLD12	IO	PU, CD	2,4,6,8,10,12 ,14,16mA	0
	4	CMCSK	I	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	5	LSA0DA12	O	-	2,4,6,8,10,12 ,14,16mA	0
	6	LPTE1	I	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	7	EGND26	I	PD	4, 8, 12, 16mA	0
MCCK	0	GPIO30	IO	CU, CD	4, 8, 12, 16mA	0
	1	MCCK	IO	CU, CD	4, 8, 12, 16mA	0
MCDA0	0	GPIO31	IO	CU, CD	4, 8, 12, 16mA	0
	1	MCDA0	IO	CU, CD	4, 8, 12, 16mA	0
MCCM0	0	GPIO32	IO	CU, CD	4, 8, 12, 16mA	0
	1	MCCM0	IO	CU, CD	4, 8, 12, 16mA	0
NLD8	0	GPIO33	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	1	NLD8	IO	PU, CD	2,4,6,8,10,12 ,14,16mA	0
	2	DAIPCMIN	I	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	3	CMMCLK	O	-	2,4,6,8,10,12 ,14,16mA	0
	4	EINT6	I	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	5	CMPDN	O	-	2,4,6,8,10,12 ,14,16mA	0
	6	LSDI2	I	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	7	EGND27	I	PD	2,4,6,8,10,12 ,14,16mA	0
NLD7	0	GPIO34	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	1	NLD7	IO	PU, CD	2,4,6,8,10,12 ,14,16mA	0
	2	EINT7	I	CU, CD	2,4,6,8,10,12 ,14,16mA	0

Name	Aux. function	Aux. name	Aux. type	PU/PD/CU/CD	Driving	SMT
	3	LPTE1	I	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	7	EGND28	I	PD	2,4,6,8,10,12 ,14,16mA	0
NLD6	0	GPIO35	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	1	NLD6	IO	PU, CD	2,4,6,8,10,12 ,14,16mA	0
	7	EGND29	I	PD	2,4,6,8,10,12 ,14,16mA	0
NLD5	0	GPIO36	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	1	NLD5	IO	PU, CD	2,4,6,8,10,12 ,14,16mA	0
	7	EGND30	I	PD	2,4,6,8,10,12 ,14,16mA	0
NLD4	0	GPIO37	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	1	NLD4	IO	PU, CD	2,4,6,8,10,12 ,14,16mA	0
	7	EGND31	I	CU, CD	2,4,6,8,10,12 ,14,16mA	0
NLD3	0	GPIO38	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	1	NLD3	IO	PU, CD	2,4,6,8,10,12 ,14,16mA	0
	2	LSDI3	I	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	7	EGND32	I	PD	2,4,6,8,10,12 ,14,16mA	0
NLD2	0	GPIO39	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	1	NLD2	IO	PU, CD	2,4,6,8,10,12 ,14,16mA	0
	2	LSDA02	IO	PU, CD	2,4,6,8,10,12 ,14,16mA	0
	7	EGND33	I	PD	2,4,6,8,10,12 ,14,16mA	0
NLD1	0	GPIO40	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	1	NLD1	IO	PU, CD	2,4,6,8,10,12 ,14,16mA	0
	2	LSA0DA13	O	-	2,4,6,8,10,12 ,14,16mA	0
	7	EGND34	I	PD	2,4,6,8,10,12 ,14,16mA	0
NLD0	0	GPIO41	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	1	NLD0	IO	PU, CD	2,4,6,8,10,12 ,14,16mA	0
	2	LSCK2	O	-	2,4,6,8,10,12 ,14,16mA	0
	7	EGND35	I	PD	2,4,6,8,10,12 ,14,16mA	0
LWR_B	0	GPIO42	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	1	LWRB	O	-	2,4,6,8,10,12 ,14,16mA	0
	7	EGND36	I	PD	2,4,6,8,10,12 ,14,16mA	0
LRD_B	0	GPIO43	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	1	LRDB	O	-	2,4,6,8,10,12 ,14,16mA	0
	7	EGND37	I	PD	2,4,6,8,10,12 ,14,16mA	0
LPA0	0	GPIO44	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	1	LPA0	O	-	2,4,6,8,10,12 ,14,16mA	0

Name	Aux. function	Aux. name	Aux. type	PU/PD/CU/CD	Driving	SMT
	7	EGND38	I	PD	2,4,6,8,10,12 ,14,16mA	0
LPCE0_B	0	GPIO45	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	1	LPCE0B	O	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	2	LSCE0B1	O	CU, CD	2,4,6,8,10,12 ,14,16mA	0
LSCE1_B	0	GPIO46	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	1	LSCE1B	O	-	2,4,6,8,10,12 ,14,16mA	0
	2	DAISYNC	O	-	2,4,6,8,10,12 ,14,16mA	0
	3	SCL	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	4	NLD9	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	5	LPCE2B	I	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	6	LSA0DA14	O	-	2,4,6,8,10,12 ,14,16mA	0
	7	EGND39	I	PD	2,4,6,8,10,12 ,14,16mA	0
LPCE1_B	0	GPIO47	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	1	LPCE1B	O	-	2,4,6,8,10,12 ,14,16mA	0
	2	CMMCLK	O	-	2,4,6,8,10,12 ,14,16mA	0
	3	SDA	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	4	NLD10	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	5	CMPDN	O	-	2,4,6,8,10,12 ,14,16mA	0
	6	LPTE1	I	CU, CD	2,4,6,8,10,12 ,14,16mA	0
LSRSTB	0	GPIO48	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	1	LSRSTB	O	-	2,4,6,8,10,12 ,14,16mA	0
	2	CLKO3	O	-	2,4,6,8,10,12 ,14,16mA	0
	3	U3RXD	I	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	4	NLD11	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	5	CMCSD0	I	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	6	LSCK3	O	-	2,4,6,8,10,12 ,14,16mA	0
	7	EGND40	I	PD	2,4,6,8,10,12 ,14,16mA	0
WATCHDOG	0	GPIO49	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	1	WATCHDOG	O	-	2,4,6,8,10,12 ,14,16mA	0
	2	NLD15	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	3	CMCSK	I	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	4	CMCSD1	I	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	5	LSDA03	IO	PU, CD	2,4,6,8,10,12 ,14,16mA	0
LPTE	0	GPIO50	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	1	LPTE1	I	CU, CD	2,4,6,8,10,12 ,14,16mA	0

Name	Aux. function	Aux. name	Aux. type	PU/PD/CU/CD	Driving	SMT
	2	CLKO4	O	-	2,4,6,8,10,12 ,14,16mA	0
	7	EGND41	I	CU, CD	2,4,6,8,10,12 ,14,16mA	0
LPRSTB	0	GPIO51	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	1	LPRSTB	O	-	2,4,6,8,10,12 ,14,16mA	0
	2	EINT8	I	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	3	LSCK4	O	-	2,4,6,8,10,12 ,14,16mA	0
	4	LPTE1	I	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	5	LPCE3B	O	-	2,4,6,8,10,12 ,14,16mA	0
	7	EGND42	I	PD	2,4,6,8,10,12 ,14,16mA	0
CMDAT0	0	GPIO52	IO	CU, CD	4, 8, 12, 16mA	0
	1	CMDAT0	I	CU, CD	4, 8, 12, 16mA	0
	2	CMCSD0	I	CU, CD	4, 8, 12, 16mA	0
	7	EGND43	I	PD	4, 8, 12, 16mA	0
CMDAT1	0	GPIO53	IO	CU, CD	4, 8, 12, 16mA	0
	1	CMDAT1	I	CU, CD	4, 8, 12, 16mA	0
	2	CMCSD1	I	CU, CD	4, 8, 12, 16mA	0
	7	EGND44	I	PD	4, 8, 12, 16mA	0
CMDAT2	0	GPIO54	IO	CU, CD	4, 8, 12, 16mA	0
	1	CMDAT2	I	CU, CD	4, 8, 12, 16mA	0
	2	SCL	IO	CU, CD	4, 8, 12, 16mA	0
	7	EGND45	I	PD	4, 8, 12, 16mA	0
CMDAT3	0	GPIO55	IO	CU, CD	4, 8, 12, 16mA	0
	1	CMDAT3	I	CU, CD	4, 8, 12, 16mA	0
	2	LRSTB	O	-	4, 8, 12, 16mA	0
	3	DAIPCMIN	I	CU, CD	4, 8, 12, 16mA	0
	7	EGND46	I	PD	4, 8, 12, 16mA	0
CMDAT4	0	GPIO56	IO	CU, CD	4, 8, 12, 16mA	0
	1	CMDAT4	I	CU, CD	4, 8, 12, 16mA	0
	2	LSA0DA15	O	-	4, 8, 12, 16mA	0
	3	DAICLK	O	-	4, 8, 12, 16mA	0
	4	U3RTS	O	-	4, 8, 12, 16mA	0
	7	EGND47	I	PD	4, 8, 12, 16mA	0
CMDAT5	0	GPIO57	IO	CU, CD	4, 8, 12, 16mA	0
	1	CMDAT5	I	CU, CD	4, 8, 12, 16mA	0
	2	LSCK5	O	-	4, 8, 12, 16mA	0

Name	Aux. function	Aux. name	Aux. type	PU/PD/CU/CD	Driving	SMT
	3	DAIPCMOUT	O	-	4, 8, 12, 16mA	0
	4	U2RXD	I	CU, CD	4, 8, 12, 16mA	0
	7	EGND48	I	PD	4, 8, 12, 16mA	0
CMDAT6	0	GPIO58	IO	CU, CD	4, 8, 12, 16mA	0
	1	CMDAT6	I	CU, CD	4, 8, 12, 16mA	0
	2	LSDA04	IO	PU, CD	4, 8, 12, 16mA	0
	3	DAISYNC	O	-	4, 8, 12, 16mA	0
	4	U2TXD	O	-	4, 8, 12, 16mA	0
	7	EGND49	I	PD	4, 8, 12, 16mA	0
CMDAT7	0	GPIO59	IO	CU, CD	4, 8, 12, 16mA	0
	1	CMDAT7	I	CU, CD	4, 8, 12, 16mA	0
	2	U3CTS	I	CU, CD	4, 8, 12, 16mA	0
	3	PWM	O	-	4, 8, 12, 16mA	0
	7	EGND50	I	PD	4, 8, 12, 16mA	0
CMHREF	0	GPIO60	IO	CU, CD	4, 8, 12, 16mA	0
	1	CMHREF	I	CU, CD	4, 8, 12, 16mA	0
	2	EINT9	I	CU, CD	4, 8, 12, 16mA	0
	3	U3RXD	I	CU, CD	4, 8, 12, 16mA	0
	7	EGND51	I	PD	4, 8, 12, 16mA	0
CMVREF	0	GPIO61	IO	CU, CD	4, 8, 12, 16mA	0
	1	CMVREF	I	CU, CD	4, 8, 12, 16mA	0
	2	SDA	IO	CU, CD	4, 8, 12, 16mA	0
	7	EGND52	I	PD	4, 8, 12, 16mA	0
CMPDN	0	GPIO62	IO	CU, CD	4, 8, 12, 16mA	0
	1	CMPDN	O	-	4, 8, 12, 16mA	0
	2	LSCE0B2	O	-	4, 8, 12, 16mA	0
	3	U3TXD	O	-	4, 8, 12, 16mA	0
CMMCLK	0	GPIO63	IO	CU, CD	4, 8, 12, 16mA	0
	1	CMMCLK	O	-	4, 8, 12, 16mA	0
	7	EGND54	I	CU, CD	4, 8, 12, 16mA	0
CMPCLK	0	GPIO64	IO	CU, CD	4, 8, 12, 16mA	0
	1	CMPCLK	I	CU, CD	4, 8, 12, 16mA	0
	2	CMCSK	I	CU, CD	4, 8, 12, 16mA	0
	7	EGND54	I	PD	4, 8, 12, 16mA	0
CMRST	0	GPIO65	IO	CU, CD	4, 8, 12, 16mA	0

Name	Aux. function	Aux. name	Aux. type	PU/PD/CU/CD	Driving	SMT
	1	CMRST	O	-	4, 8, 12, 16mA	0
	7	EGND55	I	PD	4, 8, 12, 16mA	0
EINT0	0	GPIO66	IO	CU, CD	4, 8, 12, 16mA	0
	1	EINT0	I	CU, CD	4, 8, 12, 16mA	0
	2	EDIDAT	IO	CU, CD	4, 8, 12, 16mA	0
	7	EGND56	I	PD	4, 8, 12, 16mA	0
BPI_BUS3	0	GPIO77	IO	CU, CD	4, 8, 12, 16mA	0
	1	BPIBUS3	O	-	4, 8, 12, 16mA	0
BPI_BUS2	0	GPIO78	IO	CU, CD	4, 8, 12, 16mA	0
	1	BPIBUS2	O	-	4, 8, 12, 16mA	0
BPI_BUS1	0	GPIO79	IO	CU, CD	4, 8, 12, 16mA	0
	1	BPIBUS1	O	-	4, 8, 12, 16mA	0
BPI_BUS0	0	GPIO80	IO	CU, CD	4, 8, 12, 16mA	0
	1	BPIBUS0	O	-	4, 8, 12, 16mA	0
SFSCK	1	SFSCK	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	7	EGND57	I	PD	2,4,6,8,10,12 ,14,16mA	0
SFSWP	1	SFSWP	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	7	EGND58	I	PD	2,4,6,8,10,12 ,14,16mA	0
SFSHOLD	1	SFSHOLD	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	7	EGND59	I	PD	2,4,6,8,10,12 ,14,16mA	0
SFCS0	1	SFCS0	O	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	7	EGND60	I	CU, CD	2,4,6,8,10,12 ,14,16mA	0
SFSIN	1	SFSIN	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	7	EGND61	I	PD	2,4,6,8,10,12 ,14,16mA	0
SFSOUT	1	SFSOUT	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	7	EGND62	I	PD	2,4,6,8,10,12 ,14,16mA	0
SIM1_SIO	0	GPIO67	IO	CU, CD	2, 4, 6, 8mA	0
	1	SIMSIO	IO	CU, CD	2, 4, 6, 8mA	0
	7	EGND63	I	PD	2, 4, 6, 8mA	0
SIM1_SRST	0	GPIO68	IO	CU, CD	2, 4, 6, 8mA	0
	1	SIMSRST	IO	CU, CD	2, 4, 6, 8mA	0
	7	EGND64	I	PD	2, 4, 6, 8mA	0
SIM2_SIO	0	GPIO69	IO	CU, CD	2, 4, 6, 8mA	0
	1	SIM2SIO	IO	CU, CD	2, 4, 6, 8mA	0
	7	EGND65	I	PD	2, 4, 6, 8mA	0

Name	Aux. function	Aux. name	Aux. type	PU/PD/CU/CD	Driving	SMT
SIM1_SCLK	0	GPIO70	IO	CU, CD	2, 4, 6, 8mA	0
	1	SIMCLK	IO	CU, CD	2, 4, 6, 8mA	0
	7	EGND66	I	PD	2, 4, 6, 8mA	0
SIM2_SCLK	0	GPIO71	IO	CU, CD	2, 4, 6, 8mA	0
	1	SIM2SCLK	IO	CU, CD	2, 4, 6, 8mA	0
	7	EGND67	I	PD	2, 4, 6, 8mA	0
SIM2_SRST	0	GPIO72	IO	CU, CD	2, 4, 6, 8mA	0
	1	SIM2SRST	IO	CU, CD	2, 4, 6, 8mA	0
	7	EGND68	I	PD	2, 4, 6, 8mA	0
GPIO74	0	GPIO74	IO	CU, CD	2,4,6,8,10,12 ,14,16mA	0
	1	SFSCS1	O	-	2,4,6,8,10,12 ,14,16mA	0
	2	EINT10	I	CU, CD	2,4,6,8,10,12 ,14,16mA	0
SRCLKENAI	0	GPIO105	IO	CU, CD	4, 8, 12, 16mA	0
	1	SRCLKENAI	I	CU, CD	4, 8, 12, 16mA	0
	7	EGND69	I	PD	4, 8, 12, 16mA	0
RESETB	0	GPIO106	IO	CU, CD	4, 8, 12, 16mA	0
	1	RESETB	IO	CU, CD	4, 8, 12, 16mA	0
	7	EGND70	I	PD	4, 8, 12, 16mA	0
EINT12	0	GPIO107	IO	CU, CD	4, 8, 12, 16mA	0
	1	EINT12	I	CU, CD	4, 8, 12, 16mA	0
	7	EGND71	I	PD	4, 8, 12, 16mA	0

Table 7. Capability of PU/PD, driving and Schmitt trigger

2.2 Electrical Characteristics

2.2.1 Absolute Maximum Ratings

Symbol or pin name	Description	Min.	Max.	Unit
VBAT_DIGITAL	Digital used battery voltage input	-0.3	+4.3	V
VBAT_ANALOG	Analog used battery voltage input	-0.3	+4.3	V
VBAT_SPK	VBAT input for loud speaker driver	-0.3	+4.3	V
VBAT_RF	RF used battery voltage input	-0.3	+4.3	V
VUSB	LDO output for USB-VUSB	+3.0	+3.6	V
AVDD28_FM	2.8V power supply for FM	+2.52	+3.08	V

Symbol or pin name	Description	Min.	Max.	Unit
AVDD28_VRF	2.8V power supply for 2G RF	+2.52	+3.08	V
AVDD28_TCXO	2.8V power supply for 2G TCXO	+2.52	+3.08	V
AVDD28_2GAFF	2.8V power supply for 2G AFE	+2.52	+3.08	V
AVDD28_ABB	2.8V power supply for ABB	+2.52	+3.08	V
AVDD28_DBT	2.8V power supply for DBT	+2.52	+3.08	V
AVDD28_ABT	2.8V power supply for ABT	+2.52	+3.08	V
DVDD28	2.8V power supply for digital macros in transceiver	+2.52	+3.08	V
DVDD18	1.8V power supply for digital macros in transceiver	+1.62	+1.98	V
DVDD28_SF	2.8V serial flash IO power	+2.7	+3.6	V
	1.8V serial flash IO power	+1.7	+1.98	V
DVDD28_SFP	2.8V serial flash IO power	+2.7	+3.6	V
	1.8V serial flash IO power	+1.7	+1.98	V
DVDD33_MSDC	3.3V memory card power	+3.0	+3.6	V
DVDD18_EMI	1.8V EMI IO power	+1.62	+1.98	V
DVDD28_FSRC	E-FUSE blowing power control	+2.52	+3.08	V
VDDK	1.2v core power	+1.08	+1.32	V

Table 8. Absolute maximum ratings for power supply

Symbol or pin name	Description	Min.	Max.	Unit
VIN1	Digital input voltage for IO Type 1	-0.3	3.08	V
VIN2	Digital input voltage for IO Type 2	-0.3	3.08	V
VIN3	Digital input voltage for IO Type 3	-0.3	3.08	V
VIN4	Digital input voltage for IO Type 4	-0.3	3.08	V
VIN5	Digital input voltage for IO Type 5	-0.3	3.08	V
VIN6	Digital input voltage for IO Type 6	-0.3	3.08	V

Table 9. Absolute maximum ratings for voltage input

Symbol or pin name	Description	Min.	Max.	Unit
Tstg	Storage temperature	-55	125	°C

Table 10. Absolute maximum ratings for storage temperature

2.2.2 Recommended Operating Conditions

Symbol or pin name	Description	Min.	Typ.	Max.	Unit
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Symbol or pin name	Description	Min.	Typ.	Max.	Unit
VBAT_DIGITAL	Digital used battery voltage input	3.4	3.8	4.2	V
VBAT_ANALOG	Analog used battery voltage input	3.4	3.8	4.2	V
VBAT_SPK	VBAT input for loud speaker driver	3.4	3.8	4.2	V
VBAT_RF	RF used battery voltage input	3.4	3.8	4.2	V
VUSB	LDO output for USB-VUSB	3.0	3.3	3.6	V
AVDD28_FM	2.8V power supply for FM	2.6	2.8	3.0	V
AVDD28_VRF	2.8V power supply for 2G RF	2.65	2.8	2.95	V
AVDD28_TCXO	2.8V power supply for 2G TCXO	2.65	2.8	2.95	V
AVDD28_2GAFF	2.8V power supply for 2G AFE	2.65	2.8	2.95	V
AVDD28_ABB	2.8V power supply for ABB	2.6	2.8	3.0	V
AVDD28_DBT	2.8V power supply for DBT	2.6	2.8	3.0	V
AVDD28_ABT	2.8V power supply for ABT	2.6	2.8	3.0	V
DVDD28	2.8V power supply for digital macros in transceiver	2.7	2.8	2.9	V
DVDD18	1.8V power supply for digital macros in transceiver	1.62	1.8	1.98	V
DVDD28_SF	2.8V serial flash IO power	2.7	3.3	3.6	V
	1.8V serial flash IO power	1.7	1.8	1.98	
DVDD28_SFP	2.8V serial flash IO power	2.7	3.3	3.6	V
	1.8V serial flash IO power	1.7	1.8	1.98	
DVDD33_MSDC	3.3V memory card power	3.0	3.3	3.6	V
DVDD18_EMI	1.8V EMI IO power	1.62	1.8	1.98	V
DVDD28_FSRC	E-FUSE blowing powr control	2.7	2.8	3.08	V
VDDK	1.2v core power	1.08	1.2	1.32	VDDK

Table 11. Recommended operating conditions for power supply

Symbol or pin name	Description	Min.	Typ.	Max.	Unit
VIN1	Digital input voltage for IO Type 1	-0.3	-	DVDIO+0.3	V
VIN2	Digital input voltage for IO Type 2	-0.3	-	DVDIO+0.3	V
VIN3	Digital input voltage for IO Type 3	-0.3	-	DVDIO+0.3	V
VIN4	Digital input voltage for IO Type 4	-0.3	-	DVDIO+0.3	V
VIN5	Digital input voltage for IO Type 5	-0.3	-	DVDIO+0.3	V
VIN6	Digital input voltage for IO Type 6	-0.3	-	DVDIO+0.3	V

Table 12. Recommended operating conditions for voltage input

Symbol or pin name	Description	Min.	Typ	Max.	Unit
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Symbol or pin name	Description	Min.	Typ	Max.	Unit
Tc	Operating temperature	-20	-	85	°C

Table 13. Recommended operating conditions for operating temperature

2.2.3 Electrical Characteristics under Recommended Operating Conditions

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
DIIH1	Digital high input current for IO Type 1	PU/PD disabled, DVDIO = 2.8V, 2.1 < VIN1 < 3.1	-5	-	5	μA
		PU enabled, DVDIO = 2.8V, 2.1 < VIN1 < 3.1	-22.5	-	12.5	
		PD enabled, DVDIO = 2.8V, 2.1 < VIN1 < 3.1	6.1	-	82.5	
		PU/PD disabled, DVDIO = 1.8V, 1.35 < VIN1 < 2.1	-5	-	5	μA
		PU enabled, DVDIO = 1.8V, 1.35 < VIN1 < 2.1	-11.4	-	9.3	
		PD enabled, DVDIO = 1.8V, 1.35 < VIN1 < 2.1	-0.8	-	35	
DIIL1	Digital low input current for IO Type 1	PU/PD disabled, DVDIO = 2.8V, -0.3 < VIN1 < 0.7	-5	-	5	μA
		PU enabled, DVDIO = 2.8V, -0.3 < VIN1 < 0.7	-82.5	-	-6.1	
		PD enabled, DVDIO = 2.8V, -0.3 < VIN1 < 0.7	-12.5	-	22.5	
		PU/PD disabled, DVDIO = 1.8V, -0.3 < VIN1 < 0.45	-5	-	5	μA
		PU enabled, DVDIO = 1.8V, -0.3 < VIN1 < 0.45	-35	-	0.8	
		PD enabled, DVDIO = 1.8V, -0.3 < VIN1 < 0.45	-9.3	-	11.4	

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
DIOH1	Digital high output current for IO Type 1	DVOH > 2.38V, DVDIO = 2.8V	-16	-	-	mA
		DVOH > 1.53V, DVDIO = 1.8V	-12	-	-	mA
DIOL1	Digital low output current for IO Type 1	DVOL < 0.42V, DVDIO = 2.8V	-	-	16	mA
		DVOL < 0.27V, DVDIO = 1.8V	-	-	12	mA
DRPU1	Digital I/O pull-up resistance for IO Type 1	DVDIO = 2.8V	40	85	190	kΩ
		DVDIO = 1.8V	70	150	320	kΩ
DRPD1	Digital I/O pull-down resistance for IO Type 1	DVDIO = 2.8V	40	85	190	kΩ
		DVDIO = 1.8V	70	150	320	kΩ
DVOH1	Digital output high voltage for IO Type 1	DVDIO = 2.8V	2.38			V
		DVDIO = 1.8V	1.53			V
DVOL1	Digital output low voltage for IO Type 1	DVDIO = 2.8V			0.42	V
		DVDIO = 1.8V			0.27	V
DIIH2	Digital high input current for IO Type 2	PU/PD disabled, DVDIO = 2.8V, 2.1 < VIN2 < 3.1	-5	-	5	μA
		PU enabled, DVDIO = 2.8V, 2.1 < VIN2 < 3.1	-22.5	-	12.5	
		PD enabled, DVDIO = 2.8V, 2.1 < VIN2 < 3.1	6.1	-	82.5	
		PU/PD disabled, DVDIO = 1.8V, 1.35 < VIN2 < 2.1	-5	-	5	μA
		PU enabled, DVDIO = 1.8V, 1.35 < VIN2 < 2.1	-11.4	-	9.3	
		PD enabled, DVDIO = 1.8V, 1.35 < VIN2 < 2.1	-0.8	-	35	
DIIL2	Digital low input current for IO Type 2	PU/PD disabled, DVDIO = 2.8V, -0.3 < VIN2 < 0.7	-5	-	5	μA
		PU enabled, DVDIO = 2.8V, -0.3 < VIN2 < 0.7	-82.5	-	-6.1	

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
		PD enabled, DVDIO = 2.8V, -0.3 < VIN2 < 0.7	-12.5	-	22.5	
		PU/PD disabled, DVDIO = 1.8V, -0.3 < VIN2 < 0.45	-5	-	5	μA
		PU enabled, DVDIO = 1.8V, -0.3 < VIN2 < 0.45	-35	-	0.8	
		PD enabled, DVDIO = 1.8V, -0.3 < VIN2 < 0.45	-9.3	-	11.4	
DIOH2	Digital high output current for IO Type 2	DVOH > 2.38V, DVDIO = 2.8V	-16	-	-	mA
		DVOH > 1.53V, DVDIO = 1.8V	-12	-	-	mA
DIOL2	Digital low output current for IO Type 2	DVOL < 0.42V, DVDIO = 2.8V	-	-	16	mA
		DVOL < 0.27V, DVDIO = 1.8V	-	-	12	mA
DRPU2	Digital I/O pull-up resistance for IO Type 2	DVDIO = 2.8V	40	85	190	kΩ
		DVDIO = 1.8V	70	150	320	kΩ
DRPD2	Digital I/O pull-down resistance for IO Type 2	DVDIO = 2.8V	40	85	190	kΩ
		DVDIO = 1.8V	70	150	320	kΩ
DVOH2	Digital output high voltage for IO Type 2	DVDIO = 2.8V	2.38			V
		DVDIO = 1.8V	1.53			V
DVOL2	Digital output low voltage for IO Type 2	DVDIO = 2.8V			0.42	V
		DVDIO = 1.8V			0.27	V
DIIH3	Digital high input current for IO Type 3	PU/PD disabled, DVDIO = 2.8V, 2.1 < VIN3 < 3.1	-5	-	5	μA
		PU enabled, DVDIO = 2.8V, 2.1 < VIN3 < 3.1	-22.5	-	12.5	
		PD enabled, DVDIO = 2.8V, 2.1 < VIN3 < 3.1	6.1	-	82.5	
		PU/PD disabled, DVDIO = 1.8V, 1.35 < VIN3 < 2.1	-5	-	5	μA

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
		PU enabled, DVDIO = 1.8V, 1.35 < VIN3 < 2.1	-11.4	-	9.3	
		PD enabled, DVDIO = 1.8V, 1.35 < VIN3 < 2.1	-0.8	-	35	
DIIL3	Digital low input current for IO Type 3	PU/PD disabled, DVDIO = 2.8V, -0.3 < VIN3 < 0.7	-5	-	5	μA
		PU enabled, DVDIO = 2.8V, -0.3 < VIN3 < 0.7	-82.5	-	-6.1	
		PD enabled, DVDIO = 2.8V, -0.3 < VIN3 < 0.7	-12.5	-	22.5	
		PU/PD disabled, DVDIO = 1.8V, -0.3 < VIN3 < 0.45	-5	-	5	μA
		PU enabled, DVDIO = 1.8V, -0.3 < VIN3 < 0.45	-35	-	0.8	
		PD enabled, DVDIO = 1.8V, -0.3 < VIN3 < 0.45	-9.3	-	11.4	
DIOH3	Digital high output current for IO Type 3	DVOH > 2.38V, DVDIO = 2.8V	-16	-	-	mA
		DVOH > 1.53V, DVDIO = 1.8V	-12	-	-	mA
DIOL3	Digital low output current for IO Type 3	DVOL < 0.42V, DVDIO = 2.8V	-	-	16	mA
		DVOL < 0.27V, DVDIO = 1.8V	-	-	12	mA
DRPU3	Digital I/O pull-up resistance for IO Type 3	DVDIO = 2.8V	10	47	100	kΩ
		DVDIO = 1.8V	10	47	100	kΩ
DRPD3	Digital I/O pull-down resistance for IO Type 3	DVDIO = 2.8V	10	47	100	kΩ
		DVDIO = 1.8V	10	47	100	kΩ
DVOH3	Digital output high voltage for IO Type 3	DVDIO = 2.8V	2.38			V
		DVDIO = 1.8V	1.53			V
DVOL3	Digital output low voltage for IO Type 3	DVDIO = 2.8V			0.42	V
		DVDIO = 1.8V			0.27	V

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
DIIH4	Digital high input current for IO Type 4	PU/PD disabled, DVDIO = 2.8V, 2.1 < VIN4 < 3.1	-5	-	5	μA
		PU enabled, DVDIO = 2.8V, 2.1 < VIN4 < 3.1	-22.5	-	12.5	
		PD enabled, DVDIO = 2.8V, 2.1 < VIN4 < 3.1	6.1	-	82.5	
		PU/PD disabled, DVDIO = 1.8V, 1.35 < VIN4 < 2.1	-5	-	5	μA
		PU enabled, DVDIO = 1.8V, 1.35 < VIN4 < 2.1	-11.4	-	9.3	
		PD enabled, DVDIO = 1.8V, 1.35 < VIN4 < 2.1	-0.8	-	35	
DIIL4	Digital low input current for IO Type 4	PU/PD disabled, DVDIO = 2.8V, -0.3 < VIN4 < 0.7	-5	-	5	μA
		PU enabled, DVDIO = 2.8V, -0.3 < VIN4 < 0.7	-82.5	-	-6.1	
		PD enabled, DVDIO = 2.8V, -0.3 < VIN4 < 0.7	-12.5	-	22.5	
		PU/PD disabled, DVDIO = 1.8V, -0.3 < VIN4 < 0.45	-5	-	5	μA
		PU enabled, DVDIO = 1.8V, -0.3 < VIN4 < 0.45	-35	-	0.8	
		PD enabled, DVDIO = 1.8V, -0.3 < VIN4 < 0.45	-9.3	-	11.4	
DIOH4	Digital high output current for IO Type 4	DVOH > 2.38V, DVDIO = 2.8V	-16	-	-	mA
		DVOH > 1.53V, DVDIO = 1.8V	-12	-	-	mA
DIOL4	Digital low output current for IO Type 4	DVOL < 0.42V, DVDIO = 2.8V	-	-	16	mA

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
		DVOL < 0.27V, DVDIO = 1.8V	-	-	12	mA
DRPU4	Digital I/O pull-up resistance for IO Type 4 (GPIO mode)	DVDIO = 2.8V	40	75	190	kΩ
		DVDIO = 1.8V	70	150	320	kΩ
DRPD4	Digital I/O pull-down resistance for IO Type 4 (GPIO mode)	DVDIO = 2.8V	40	75	190	kΩ
		DVDIO = 1.8V	70	150	320	kΩ
DRPU4 200K	Digital I/O pull-up resistance for IO Type 4 (Key PAD mode)	DVDIO = 2.8V	200	-	380	kΩ
DRPD4 200K	Digital I/O pull-down resistance for IO Type 4 (Key PAD mode)	DVDIO = 2.8V	200	-	380	kΩ
DVOH4	Digital output high voltage for IO Type 4	DVDIO = 2.8V	2.38			V
		DVDIO = 1.8V	1.53			V
DVOL4	Digital output low voltage for IO Type 4	DVDIO = 2.8V			0.42	V
		DVDIO = 1.8V			0.27	V
DIIH5	Digital high input current for IO Type 5	PU/PD disabled, DVDIO = 2.8V, 2.1 < VIN5 < 3.1	-5	-	5	μA
		PU enabled, DVDIO = 2.8V, 2.1 < VIN5 < 3.1	-22.5	-	12.5	
		PD enabled, DVDIO = 2.8V, 2.1 < VIN5 < 3.1	6.1	-	82.5	
		PU/PD disabled, DVDIO = 1.8V, 1.35 < VIN5 < 2.1	-5	-	5	μA
		PU enabled, DVDIO = 1.8V, 1.35 < VIN5 < 2.1	-11.4	-	9.3	
		PD enabled, DVDIO = 1.8V, 1.35 < VIN5 < 2.1	-0.8	-	35	
DIIL5	Digital low input current for IO Type 5	PU/PD disabled, DVDIO = 2.8V, -0.3 < VIN5 < 0.7	-5	-	5	μA
		PU enabled, DVDIO = 2.8V, -0.3 < VIN5 < 0.7	-82.5	-	-6.1	

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
		PD enabled, DVDIO = 2.8V, -0.3 < VIN5 < 0.7	-12.5	-	22.5	
		PU/PD disabled, DVDIO = 1.8V, -0.3 < VIN5 < 0.45	-5	-	5	μA
		PU enabled, DVDIO = 1.8V, -0.3 < VIN5 < 0.45	-35	-	0.8	
		PD enabled, DVDIO = 1.8V, -0.3 < VIN5 < 0.45	-9.3	-	11.4	
DIOH5	Digital high output current for IO Type 5	DVOH > 2.38V, DVDIO = 2.8V	-16	-	-	mA
		DVOH > 1.53V, DVDIO = 1.8V	-12	-	-	mA
DIOL5	Digital low output current for IO Type 5	DVOL < 0.42V, DVDIO = 2.8V	-	-	16	mA
		DVOL < 0.27V, DVDIO = 1.8V	-	-	12	mA
DRPU5	Digital I/O pull-up resistance for IO Type 5 (GPIO mode)	DVDIO = 2.8V	40	75	190	kΩ
		DVDIO = 1.8V	70	150	320	kΩ
DRPD5	Digital I/O pull-down resistance for IO Type 5 (GPIO mode)	DVDIO = 2.8V	40	75	190	kΩ
		DVDIO = 1.8V	70	150	320	kΩ
DRPU5 2K	Digital I/O pull-up resistance for IO Type 4 (Key PAD mode)	DVDIO = 2.8V	1	-	2	kΩ
DRPD5 2K	Digital I/O pull-down resistance for IO Type 4 (Key PAD mode)	DVDIO = 2.8V	1	-	2	kΩ
DVOH5	Digital output high voltage for IO Type 5	DVDIO = 2.8V	2.38			V
		DVDIO = 1.8V	1.53			V
DVOL5	Digital output low voltage for IO Type 5	DVDIO = 2.8V			0.42	V
		DVDIO = 1.8V			0.27	V
DIIH6	Digital high input current for IO Type 6	PU/PD disabled, DVDIO = 2.8V, 2.1 < VIN6 < 3.1	-5	-	5	μA
		PU enabled, DVDIO = 2.8V, 2.1 < VIN6 < 3.1	-22.5	-	12.5	

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
		PD enabled, DVDIO = 2.8V, 2.1 < VIN6 < 3.1	6.1	-	82.5	
		PU/PD disabled, DVDIO = 1.8V, 1.35 < VIN6 < 2.1	-5	-	5	μA
		PU enabled, DVDIO = 1.8V, 1.35 < VIN6 < 2.1	-11.4	-	9.3	
		PD enabled, DVDIO = 1.8V, 1.35 < VIN6 < 2.1	-0.8	-	35	
DIIL6	Digital low input current for IO Type 6	PU/PD disabled, DVDIO = 2.8V, -0.3 < VIN6 < 0.7	-5	-	5	μA
		PU enabled, DVDIO = 2.8V, -0.3 < VIN6 < 0.7	-82.5	-	-6.1	
		PD enabled, DVDIO = 2.8V, -0.3 < VIN6 < 0.7	-12.5	-	22.5	
		PU/PD disabled, DVDIO = 1.8V, -0.3 < VIN6 < 0.45	-5	-	5	μA
		PU enabled, DVDIO = 1.8V, -0.3 < VIN6 < 0.45	-35	-	0.8	
		PD enabled, DVDIO = 1.8V, -0.3 < VIN6 < 0.45	-9.3	-	11.4	
DIOH6	Digital high output current for IO Type 6	DVOH > 2.38V, DVDIO = 2.8V	-8	-	-	mA
		DVOH > 1.53V, DVDIO = 1.8V	-6	-	-	mA
DIOL6	Digital low output current for IO Type 6	DVOL < 0.42V, DVDIO = 2.8V	-	-	8	mA
		DVOL < 0.27V, DVDIO = 1.8V	-	-	6	mA
DRPU6	Digital I/O pull-up resistance for IO Type 6	DVDIO = 2.8V	40	85	190	kΩ
		DVDIO = 1.8V	70	150	320	kΩ
DRPD6	Digital I/O pull-down resistance for IO Type 6	DVDIO = 2.8V	40	85	190	kΩ
		DVDIO = 1.8V	70	150	320	kΩ

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
DVOH6	Digital output high voltage for IO Type 6	DVDIO = 2.8V	2.38			V
		DVDIO = 1.8V	1.53			V
DVOL6	Digital output low voltage for IO Type 6	DVDIO = 2.8V			0.42	V
		DVDIO = 1.8V			0.27	V

Table 14. Electrical characteristics

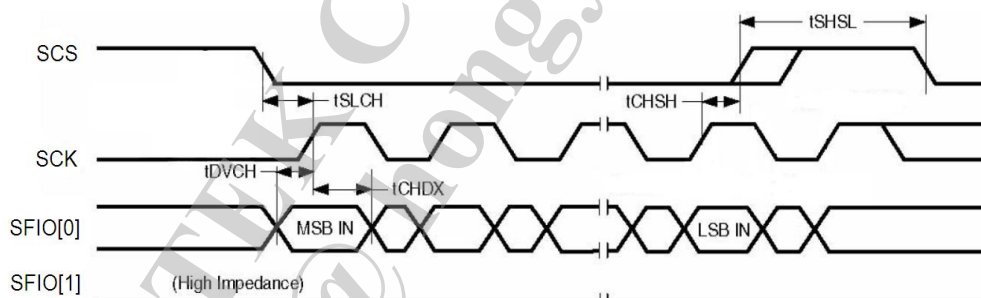
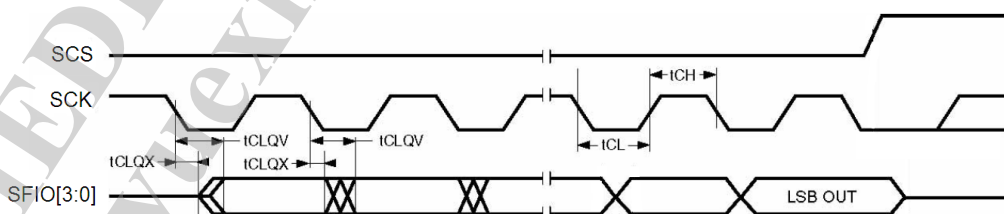
2.2.4 AC Electrical Characteristics and Timing Diagrams

2.2.4.1 Serial Flash Interface

The serial flash interface, as shown in Figure 5 and Figure 6, is used to connect serial flash devices for this chip, including the pins SCK, SCS and SFIO[3:0]. (SFIO[3:0] are SFHOLD, SFWP, SFDIN, SFDOUT) Table 15. Serial flash AC characteristics (temperature = -20 ~ 65°C, Vcc = 1.65 ~ 2.0 V or Vcc = 2.7 ~ 3.6V)

explains the definitions of the symbols and the related timing specifications.

Timing diagram of serial flash:


Figure 5. Write timing for serial flash

Figure 6. Read timing for serial flash

Symbol	Description	Min.	Typ.	Max.	Unit
fSCLK	Clock frequency for supporting serial flash			104	MHz

Symbol	Description		Min.	Typ.	Max.	Unit
tSLCH	SCS active setup time (relative to SCK)			5		ns
tSHSL	SCS deselect time			40		ns
tCHSL	SCS not active hold time (relative to SCK)			7		ns
tDVCH	Data in setup time			4.5		ns
tCHDX	Data in hold time			4.5		ns
tCHSH	SCS active hold time (relative to SCK)			5		ns
tCLQV	Clock low to output valid loading: 30pF/15pF	Loading: 30pF			8	ns
		Loading: 15pF			6	ns
tCLQX	Output hold time			0		ns
tCH	Clock high time			4.5		ns
tCL	Clock low time			4.5		ns

Table 15. Serial flash AC characteristics (temperature = -20 ~ 65°C, Vcc = 1.65 ~2.0 V or Vcc = 2.7 ~ 3.6V)

2.3 System Configuration

2.3.1 Strapping Resistors

Pin name	Description	Trapping condition
CMPDN	Pull-up with 75K register	Power-on reset
CMRST	Pull-up with 75K register	Power-on reset/watchdog reset

Table 16. Strapping table

2.3.2 Mode Selection

Pin name	Description
XTAL_SEL	GND: Uses external 32K crystal as RTC clock source VRTC: Uses internal 32K as RTC clock source
CMPDN	GND: Use 1.8V serial flash device DVDD28: Use 3.0V serial flash device
KCOL0	GND: Boot ROM enter USB download mode DVDD28: Normal boot-up mode
CMRST	GND: Disables USB download sub-feature for supporting virtual 2 USB com port DVDD28: Enables USB download sub-feature for supporting virtual 2 USB com port

Table 17. Mode selection of chip

2.3.3 Constant Tied Pins

Pin name	Description
TESTMODE	Tie to GND

Table 18. Constant tied pin of chip

2.4 Power-on Sequence and Protection Logic

MT6250 provides 32K crystal removal feature. The XOSC32_ENB state tells if MT6250 provides this feature or not. VTCXO will be turned on at the same time with VRTC when XOSC32_ENB = 1. The power-on/off sequence controlled by “Control” and “Reset Generator” is shown as the figure below.

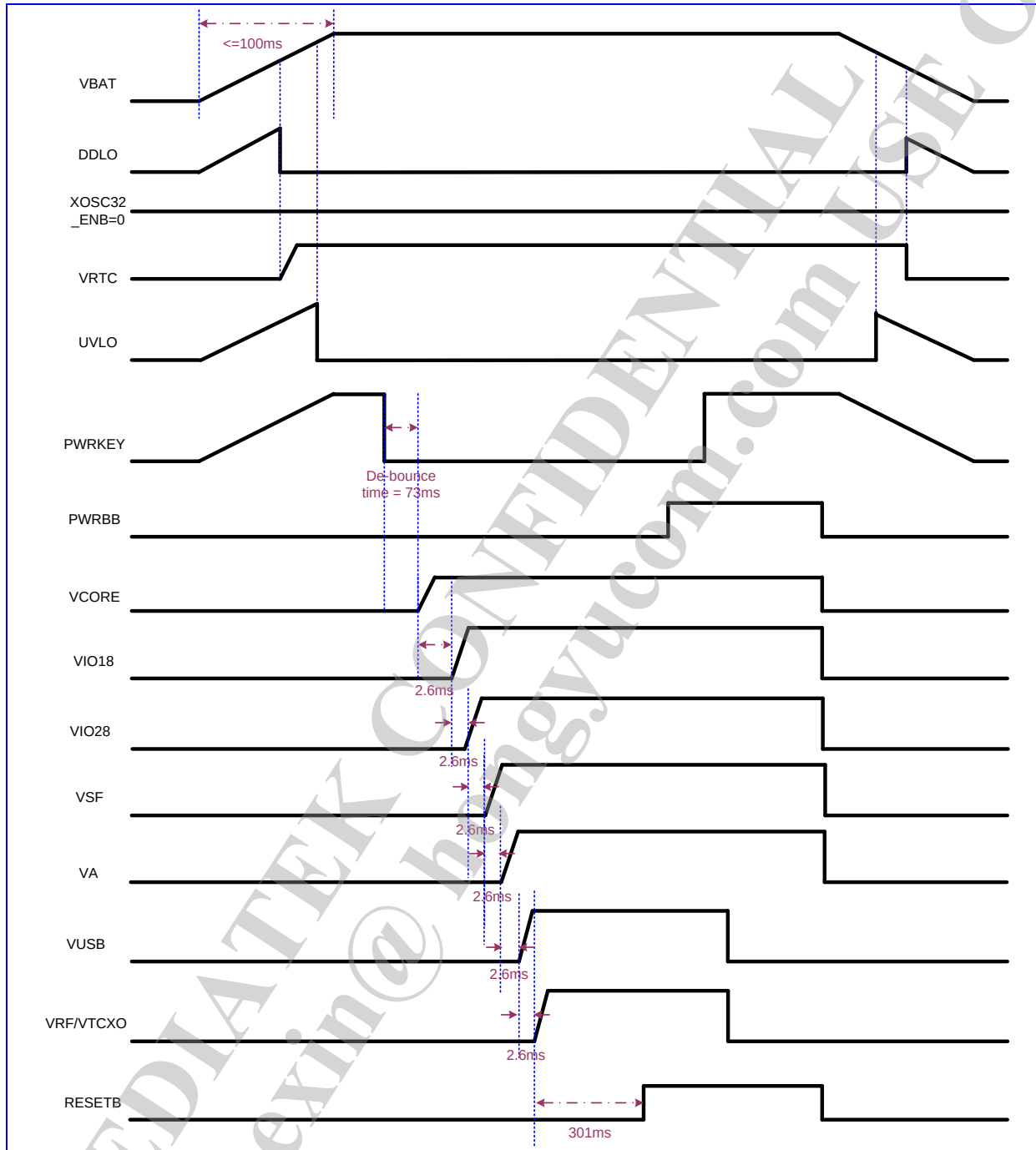


Figure 7. Power-on/off control sequence by pressing PWRKEY and XOSC32_ENB = 0

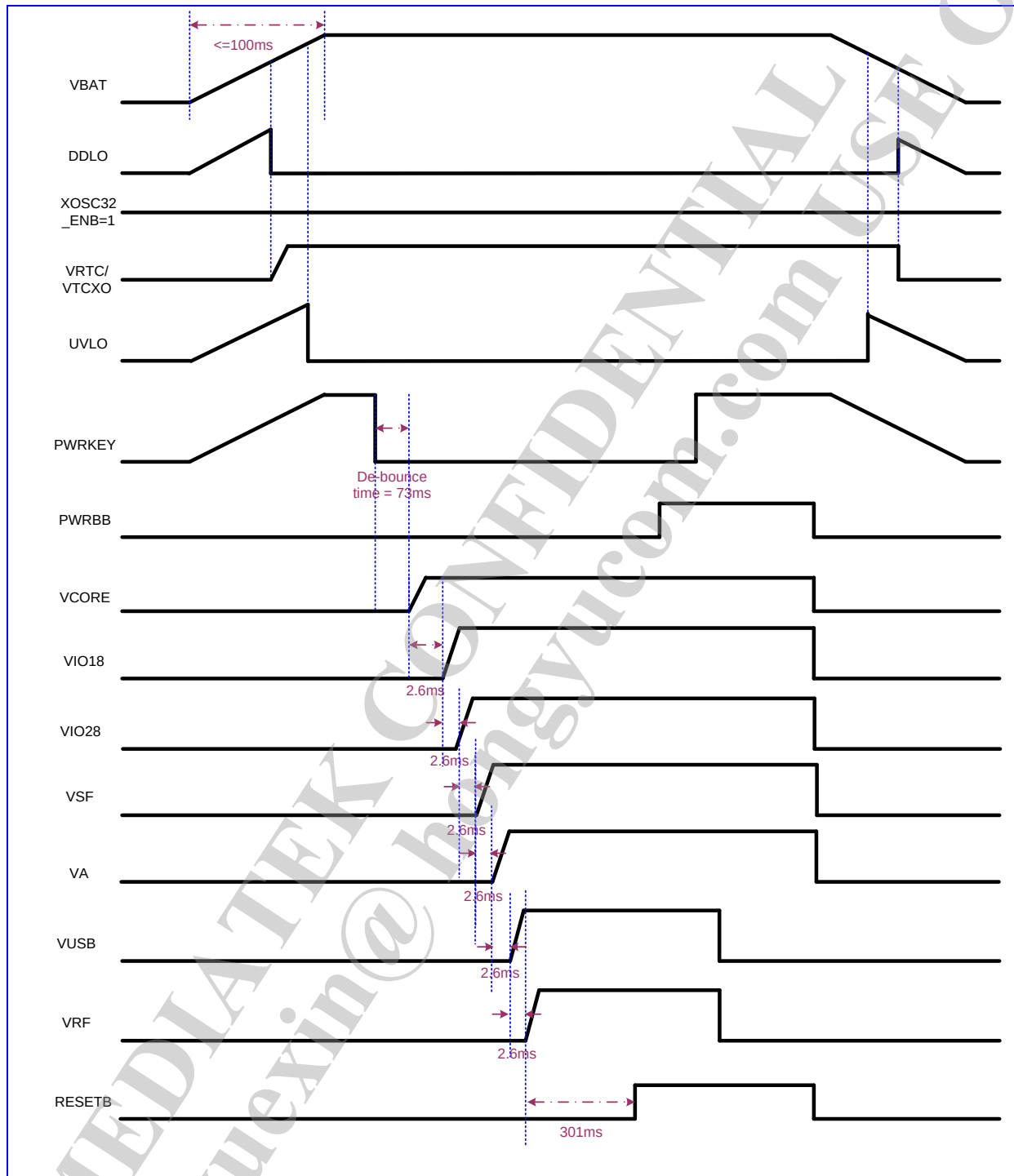


Figure 8. Power-on/off control sequence by pressing PWRKEY and XOSC32_ENB = 1

Note that each of the above figures only shows one power-on/off condition when XOSC32_ENB = 0 or XOSC32_ENB = 1. MT6250 handles the power-on and off of the handset. The following three methods can switch on the handset (when leaving UVLO): XOSC32_ENB = 0

1. Push PWRKEY (Pull the PWRKEY pin to the low level.)

Pulling PWRKEY low is the typical way to turn on the handset. The turn-on sequence is
 VCORE → VIO18 → VIO28 → VSF → VA → VUSB → VRF/VTCXO

The supplies for the baseband are ready, and the system reset ends at the moment when the above LDOs are fully turned on to ensure correct timing and function. After that, the baseband will send the PWRBB signal back to the PMU for acknowledgement. To successfully power on the handset, PWRKEY should be kept low until PMU receives PWRBB from the baseband.

2. RTC module generates PWRBB to wake up the system.

If the RTC module is scheduled to wake up the handset at a certain time, the PWRBB signal will be directly sent to the PMU. In this case, PWRBB will become high at specific moment and allow the PMU to be powered on as the sequence described above. This is called the RTC alarm.

3. Valid charger plug-in (CHVIN voltage is within the valid range.)

The charger plug-in will also turn on the handset if the charger is valid (no OVP takes place). However, if the battery voltage is too low to power on the handset (UVLO state), the system will not be turned on by any of the three methods. In this case, the charger will charge the battery first and the handset will be powered on automatically as long as the battery voltage is high enough.

Under-voltage lockout (UVLO)

The UVLO state in the PMU prevents startup if the initial voltage of the main battery is below the 3.2V threshold. It ensures that the handset is powered on with the battery in good condition. The UVLO function is performed by a hysteretic comparator which ensures a smooth power-on sequence. In addition, when the battery voltage is getting lower, it will enter the UVLO state, and the PMU will be turned off by itself, except for VRTC LDO, to prevent further discharging. Once the PMU enters the UVLO state, it will draw low quiescent current. The RTC LDO will still be working until the DDLO disables it.

Deep discharge lockout (DDLO)

The PMU will enter the deep discharge lockout (DDLO) state when the battery voltage drops below 2.5V. In this state, the VRTC LDO will be shut down. Otherwise, it will draw very low quiescent current to prevent further discharging or damage to the cells.

Reset

The PMU contains a reset control circuit which takes effect at both power-up and power-down. The RESETB pin is held low in the beginning of power-up and returns to high after the pre-determined delay time. The delay time is controlled by a large counter which uses the clock from internal ring-oscillator. At power-off, the RESETB pin will return to low immediately without any delay.

Over-temperature protection

If the die temperature of PMU exceeds 150°C, the PMU will automatically disable all the LDOs except for VRTC. Once the over-temperature state is resolved, a new power-on sequence will be required to enable the LDOs.

2.5 Analog Baseband

To communicate with analog blocks, a common control interface for all analog blocks is implemented. In addition, there are some dedicated interfaces for data transfer. The common control interface translates the APB bus write and read cycle for specific addresses related to analog front-end control. During the writing or reading of any of these control registers, there is a latency associated with the transfer of data to or from the analog front-end. Dedicated data interface of each analog block is implemented in the corresponding digital block. An analog block includes the following analog functions for the complete GSM/GPRS baseband signal processing:

1. RF control: DAC for automatic power control (APC) is included, and its output is provided to external RF power amplifier respectively.
2. Auxiliary ADC: Provides an ADC for the battery and other auxiliary analog functions monitoring
3. Audio mixed-signal block: Provides complete analog voice signal processing including microphone amplification, A/D conversion, D/A conversion, earphone driver, etc. Dedicated stereo D/A conversion and amplification for audio signals are also included.
4. Clock generation: Includes a clock squarer for shaping the system clock, and PLL providing clock signals to DSP, MCU and USB unit
5. XOSC32: A 32-kHz crystal oscillator circuit for RTC applications on analog blocks

2.5.1 APC-DAC

2.5.1.1 Block Description

APC-DAC is a 10-bit DAC with output buffer aiming at automatic power control. See the tables below for its analog pin assignment and functional specifications. It is an event-driven scheme for power saving purpose.

2.5.1.2 Functional Specifications

Symbol	Parameter	Min.	Typ.	Max.	Unit
N	Resolution		10		Bit
FS	Sampling rate			1.0833	MSPS
SINAD	Signal to noise and distortion ratio (10-kHz sine with 1.0V swing & 100-kHz BW)	47			dB
	99% settling time (full swing on maximal capacitance)			5	μS
	Output swing	0		AVDD	V
	Output capacitance		200	2200	pF
	Output resistance	0.47	10		KΩ
DNL	Differential nonlinearity for code 20 to 970		± 1		LSB
INL	Integral nonlinearity for code 20 to 970		± 1		LSB
DVDD	Digital power supply	1.1	1.2	1.3	V
AVDD	Analog power supply	2.6	2.8	3.0	V

Symbol	Parameter	Min.	Typ.	Max.	Unit
T	Operating temperature	-20		80	°C
	Current consumption				
	Power-up		400		μA
	Power-down		1		μA

Table 19. APC-DAC specifications

2.5.2 Auxiliary ADC

2.5.2.1 Block Description

The auxiliary ADC includes the following functional blocks:

1. Analog multiplexer: Selects signal from one of the seven auxiliary input pins. Real-world messages to be monitored, e.g. temperature, should be transferred to the voltage domain.
2. 10-bit A/D converter: Converts the multiplexed input signal to 10-bit digital data.

Channel	Application	Input range [V]
0	BATSNS	3.2 ~ 4.2
1	ISENSE	3.2 ~ 4.2
2	VCDT	Decided by application circuit
3	BATON	0 ~ AVDD28
4	AUXIN4	0 ~ AVDD28
others	Internal use	N/A

2.5.2.2 Functional Specifications

The functional specifications of the auxiliary ADC are listed in the following table.

Symbol	Parameter	Min.	Typ.	Max.	Unit
N	Resolution		10		Bit
FC	Clock rate		1.08		MHz
FS	Sampling rate @ N-Bit		1.08/(N+1)		MSPS
	Input swing	0		AVDD	V
CIN	Input capacitance				
	Unselected channel			50	fF
	Selected channel			4	pF
RIN	Input resistance				
	Unselected channel	400			MΩ
	Selected channel	1			MΩ
	Clock latency		N+1		1/FC
DNL	Differential nonlinearity		± 1		LSB
INL	Integral nonlinearity		± 1		LSB

Symbol	Parameter	Min.	Typ.	Max.	Unit
OE	Offset error		± 10		mV
FSE	Full swing error		± 10		mV
SINAD	Signal to noise and distortion ratio (10-kHz full swing input & 1.0833-MHz clock rate)		50		dB
DVDD	Digital power supply	1.1	1.2	1.3	V
AVDD	Analog power supply	2.6	2.8	3.0	V
T	Operating temperature	-20		80	°C
	Current consumption				
	Power-up		280		μ A
	Power-down		1		μ A

Table 20. Functional specifications of auxiliary ADC

2.5.3 Audio Mixed-Signal Blocks

2.5.3.1 Block Description

Audio mixed-signal blocks (AMB) integrate complete voice uplink/downlink and audio playback functions. As shown in the figure below, it includes three parts. The first consists of stereo audio DACs and audio amplifiers for audio playback. The second part is the voice downlink path, including voice-band DACs (right channel audio DAC) and voice amplifier, which produces voice signals to earphones or other auxiliary output devices. Amplifiers in the two blocks are equipped with multiplexers to accept signals from the internal audio/voice. The last part is the voice uplink path, which is the interface between the microphone (or other auxiliary input device) input and MT6250 DSP. A set of bias voltage is provided for the external electric microphone.

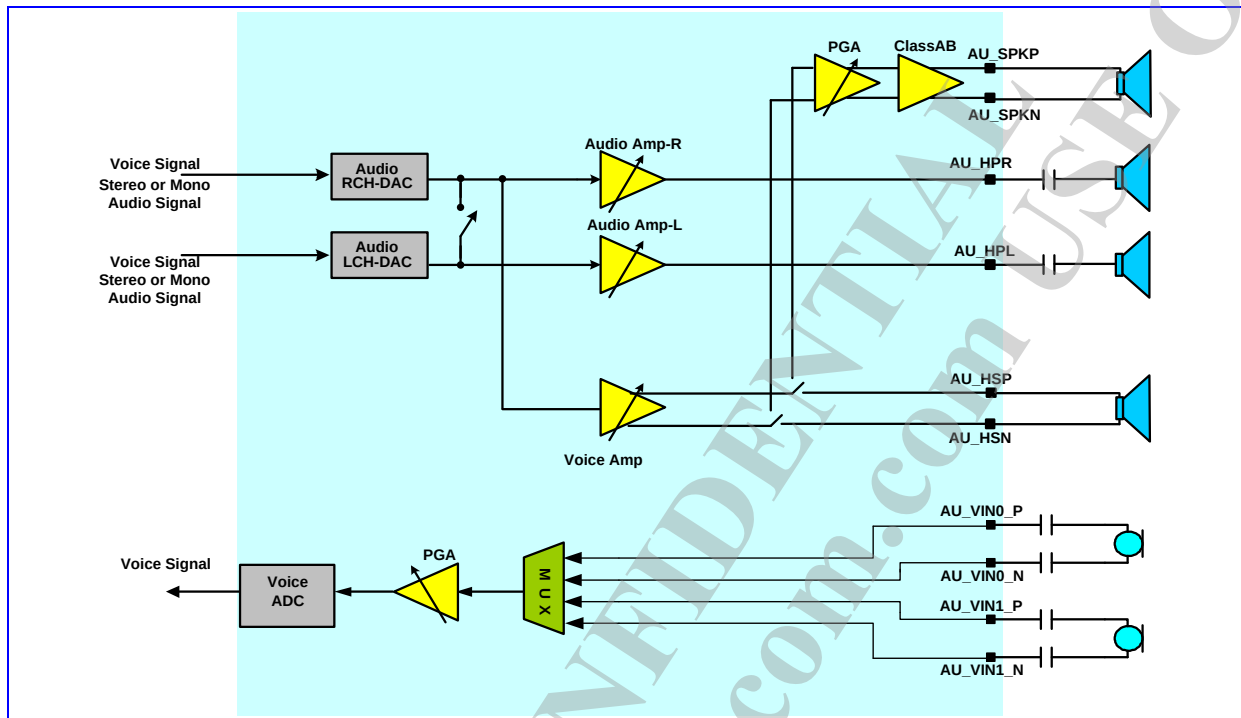


Figure 9. Block diagram of audio mixed-signal blocks

2.5.3.2 Functional Specifications

See the table below for the functional specifications of voice-band uplink/downlink blocks.

Symbol	Parameter	Min.	Typ.	Max.	Unit
FS	Sampling rate		6,500		kHz
DVDD	Digital power supply	1.1	1.2	1.3	V
AVDD	Analog power supply	2.6	2.8	3.0	V
T	Operating temperature	-20		80	°C
VMIC	Microphone biasing voltage		1.9	2.2	V
IMIC	Current draw from microphone bias pins			2	mA
Uplink path⁴					
IDC	Current consumption for one channel		1.5		mA
SINAD	Signal to noise and distortion ratio				
	Input level: -40 dbm0	29	69		dB
	Input level: 0 dbm0				dB
RIN	Input impedance (differential)	13	20	27	KΩ
ICN	Idle channel noise			-67	dBm0
Downlink path					
IDC	Current consumption		4		mA

⁴ For uplink-path, not all gain settings of VUPG meet the specifications listed in the table, especially for several the lowest gains. The minimum gain that meets the specifications is to be determined.

Symbol	Parameter	Min.	Typ.	Max.	Unit
SINAD	Signal to noise and distortion ratio Input level: -40 dBm0 Input level: 0 dBm0	29	69		dB dB
RLOAD	Output resistor load (differential)	16	32		Ω
CLOAD	Output capacitor load			250	pF
ICN	Idle channel noise of transmit path			-67	dBm0
XT	Crosstalk level on transmit path			-66	dBm0

Table 21. Functional specifications of analog voice blocks

See the table below for the functional specifications of audio blocks.

Symbol	Parameter	Min.	Typ.	Max.	Unit
FCK	Clock frequency		6.5		MHz
Fs	Sampling rate	32	44.1	48	kHz
AVDD	Power supply	2.6	2.8	3.0	V
T	Operating temperature	-20		80	°C
IDC	Current consumption		4		mA
PSNR	Peak signal to noise ratio		88		dB
DR	Dynamic range		88		dB
VOUT	Output swing for 0dBFS input level @ -1dB headphone gain		0.78		Vrms
VOUT _{MAX}	Maximum output swing		2.4		Vpp
THD	Total harmonic distortion 10mW at 64 Ω load			-70	dB
RLOAD	Output resistor load (single-ended)	64			Ω
CLOAD	Output capacitor load			250	pF
XT	L-R channel cross talk	70			dB

Table 22. Functional specifications of analog audio blocks

2.5.4 32-kHz Crystal Oscillator

2.5.4.1 Block Description

The low-power 32-kHz crystal oscillator XOSC32 is designed to work with an external piezoelectric 32.768 kHz crystal and a load composed of two functional capacitors. See the figure below.

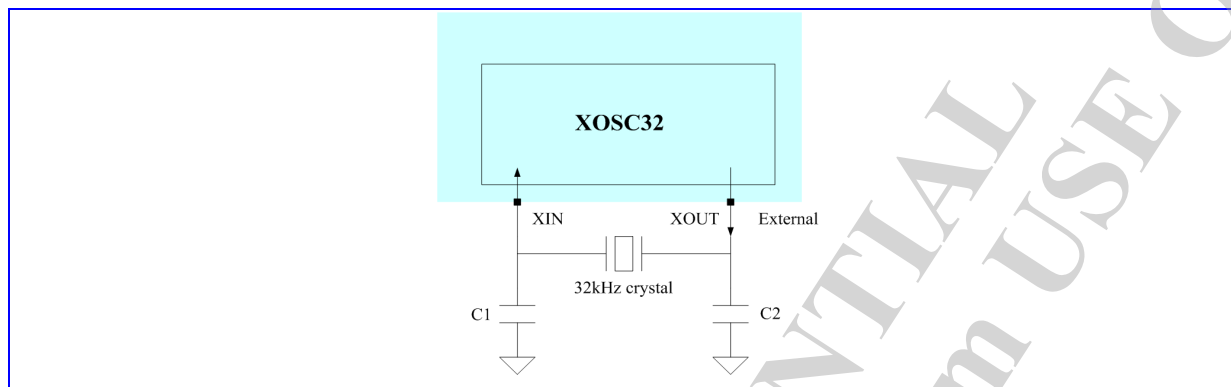


Figure 10. Block diagram of XOSC32

2.5.4.2 Functional Specifications

See the table below for the functional specifications of XOSC32.

Symbol	Parameter	Min.	Typ.	Max.	Unit
AVDDRTC	Analog power supply	1	2.8		V
Tosc	Start-up time			1	sec
Dcyc	Duty cycle	30	50	70	%
	Current consumption			5	μA
T	Operating temperature	-25		70	°C

Table 23. Functional specifications of XOSC32

See the table below for recommendations on crystal parameters to use with XOSC32.

Symbol	Parameter	Min.	Typ.	Max.	Unit
F	Frequency range		32,768		Hz
GL	Drive level			1	uW
$\Delta f/f$	Frequency tolerance		± 20		ppm
ESR	Series resistance		50	70	KΩ
C0	Static capacitance		0.9		pF
CL	Load capacitance		12.5		pF

Table 24. Recommended parameters of 32kHz crystal

2.6 Power Management Unit Blocks

The power management unit (PMU) manages the power supply of the entire chip, such as baseband, processor, memory, SIM cards, camera, vibrator, etc. The digital part of PMU is integrated into the analog part (see the figure below). PMU includes the following analog functions for signal processing:

- LDO: Regulates battery voltage to lower voltage level
- Keypad LED driver (KPLED) and current sink (ISINK) switches: Sink current for keypad LED and LCM module.
- Start-up (STRUP): Generates power-on/off control sequence of start-up circuits
- Pulse charger (PCHR): Controls battery charging
- Class-AB audio amplifier: Supports high-power audio amplifier

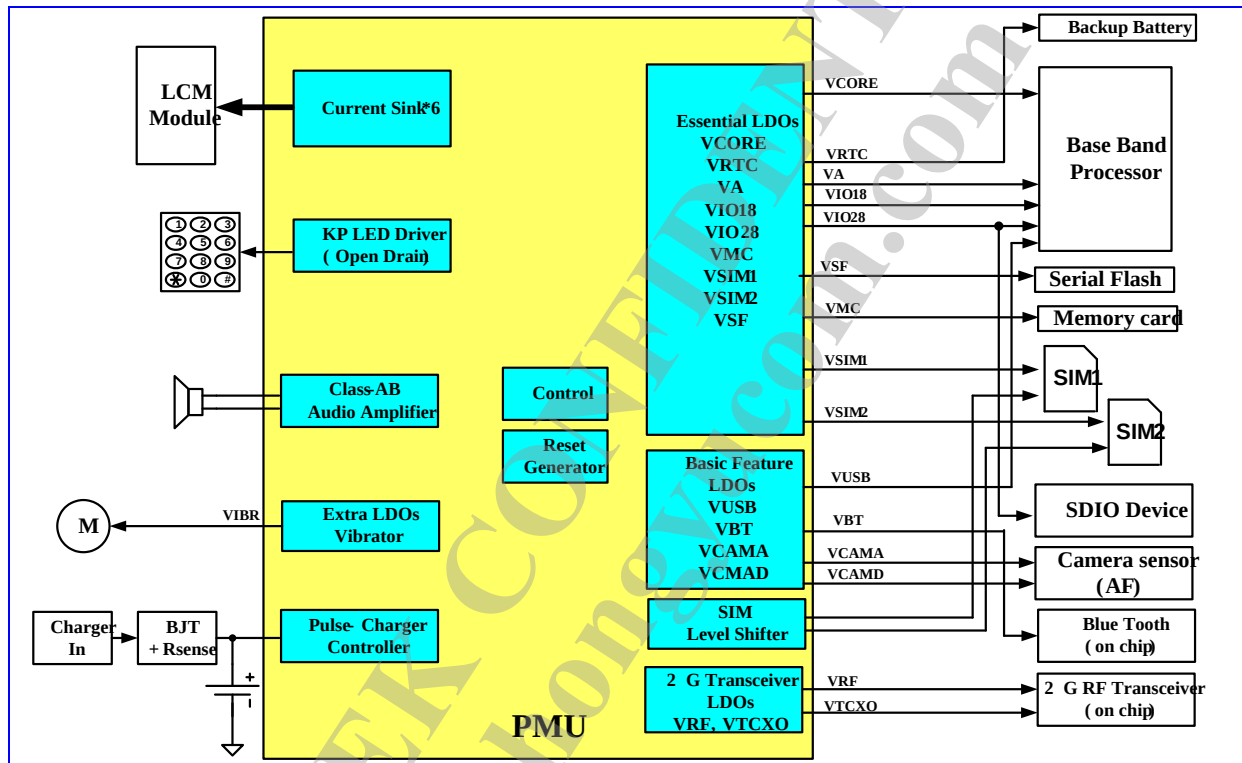


Figure 11. PMU system block diagram

2.6.1 LDO

PMU integrates 16 general low dropout regulators (LDO) optimized for their given functions by balancing the quiescent current, dropout voltage, line/load regulation, ripple rejection and output noise.

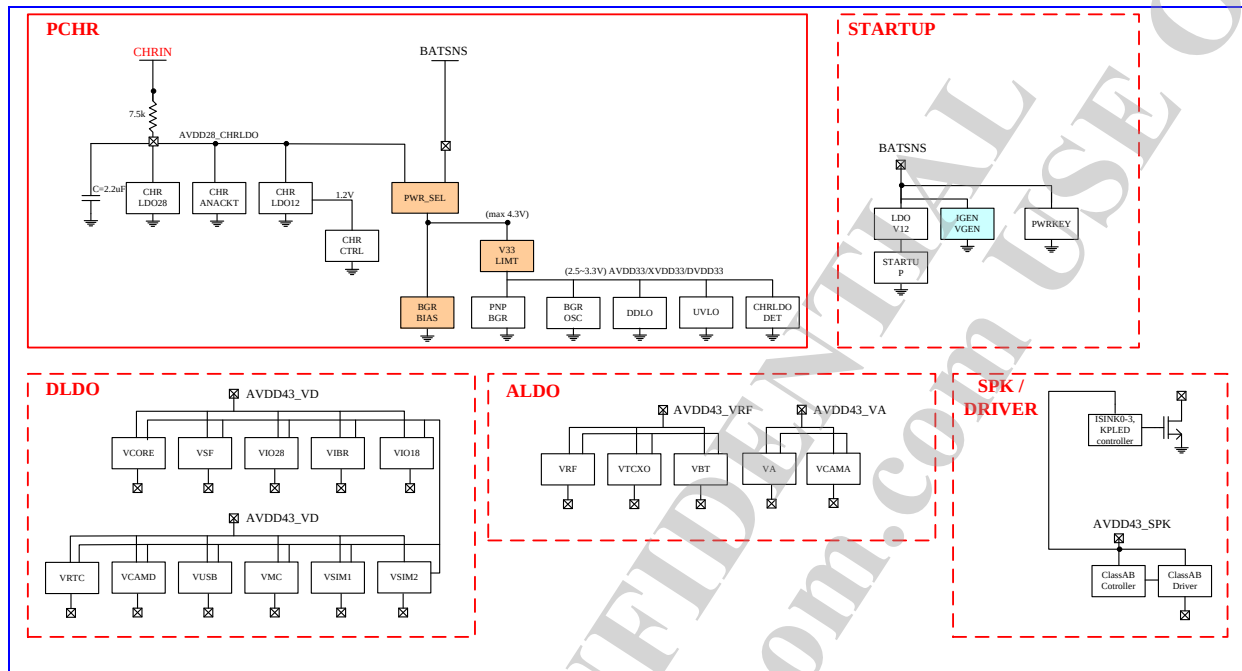


Figure 12. Power domain

2.6.1.1 LDO

A low-dropout regulator (LDO) is capable of maintaining its specified output voltage over a wide range of load current and input voltage, down to a very small difference between input and output voltages.

There are several features in the design of LDO, including discharge control, soft start and current limit. Before LDO is enabled, the output pin of LDO should be discharged first to avoid voltage accumulation on the capacitance. The soft-start limits inrush current and controls output-voltage rising time during the power-up. The current limit is the current protection to limit the LDO's output current and power dissipation.

There are three types of LDOs in PMU of MT6250 PMU. The analog LDO is optimized for low-frequency ripple rejection in order to reject the ripples coming from the burst of RF power amplifier. The digital IO LDO is a linear regulator optimized for very low quiescent current. The single-step RTC LDO is a linear regulator that can charge up a capacitor-type backup coin cell, which also supplies the RTC module even at the absence of the main battery. The single-step LDO features reverse current protection and is optimized for ultra-low quiescent current while sustaining the RTC function as long as possible.

2.6.1.1.1 Block Description

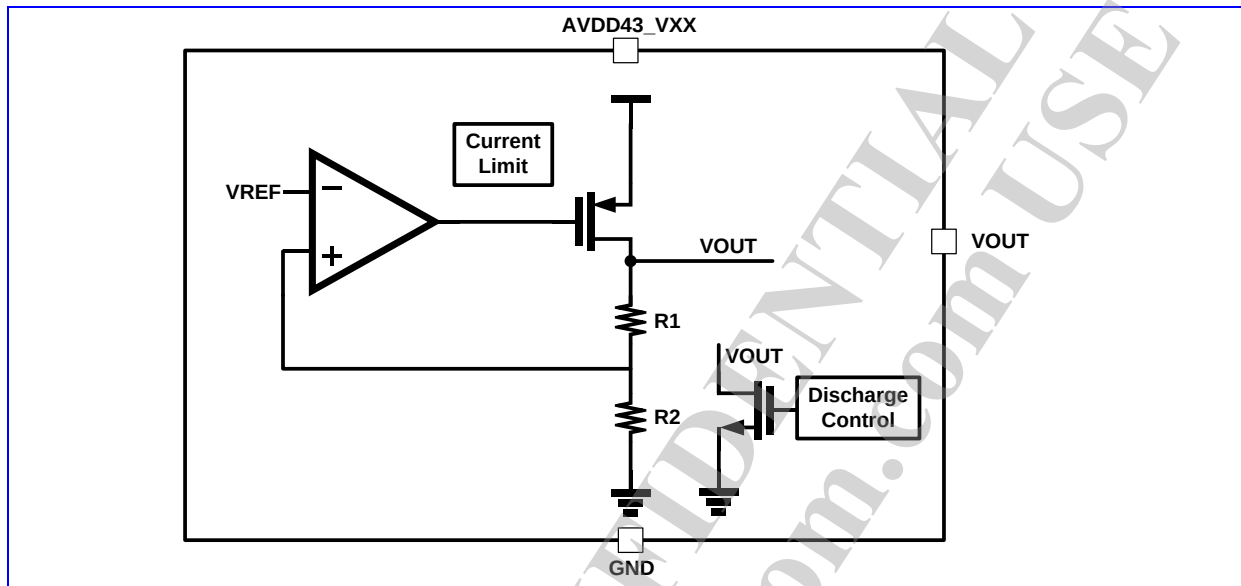


Figure 13. LDO block diagram

2.6.1.1.2 LDO Types

Type	LDO name	Vout (Volt)	I _{max} (mA)	Description
ALDO	VRF	2.8	150	RF chip
ALDO	VA	2.8	150	Analog baseband
ALDO	VTCXO	2.8	50	13/26MHz reference clock
ALDO	VCAMA	1.5/1.8/2.5/2.8	150	Camera sensor
ALDO	VBT	2.8	50	Blue tooth
DLDO	VIO28	2.8	200	Digital IO
DLDO	VSIM	1.8/3.0	30	SIM card
DLDO	VSIM2	1.3/1.5/1.8/2.5/2.8/3.0/3.3	30	SIM card
DLDO	VUSB	3.3	50	USB
DLDO	VIO18	1.8	200	Digital IO
DLDO	VCORE	0.85~1.35	200	Digital baseband
DLDO	VCAMD	1.3/1.5/1.8/2.5/2.8/3.0/3.3	100	Camera Sensor
DLDO	VIBR	1.3/1.5/1.8/2.5/2.8/3.0/3.3	150	Vibrator
DLDO	VMC	1.6/1.8/2.8/3.0	200	Memory card
DLDO	VSF	1.86/3.3	100	Serial flash
RTCLDO	VRTC	2.8	2	Real-time clock

Table 25. LDO types and brief specifications

2.6.1.1.3 Functional Specifications

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
	Load capacitor			1 ⁵		μF
	Current limit		1.2*I _{max}		5*I _{max}	mA
	V _{out}	Includes load regulation, line regulation, and temperature coefficient	Max. (-5%, -0.1V)		Max. (+5%, +0.1V)	V
	Transient response	Slew: 15mA/us	Max. (-5%, -0.1V)		Max. (+5%, +0.1V)	V
	Temperature coefficient				100	ppm/C
	PSRR	I _{out} < 0.5*I _{max} 10 < f < 3 kHz	65			dB
		I _{out} < 0.5*I _{max} 3K < f < 30 kHz	45			dB
	Output noise	With A-weighted filter			90	uVrms
	Quiescent current	I _{out} = 0		55		μA
	Turn-on overshoot	I _{out} = 0			Max. (+10%, +0.1V)	V
	Turn-on settling time	I _{out} = 0			240	μsec

Table 26. Analog LDO specifications

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
	Load capacitor			1		μF
	Current limit		1.2*I _{max}		5*I _{max}	mA
	V _{out}	Includes load regulation, line regulation, and temperature coefficient	Max. (-5%, -0.1V)		Max. (+5%, +0.1V)	V
	Transient response	Slew: 15mA/us	Max. (-5%, -0.1V)		Max. (+5%, +0.1V)	V
	Temperature coefficient				100	ppm/C
	Quiescent current	I _{out} = 0		15		μA
	Turn-on overshoot	I _{out} = 0			Max. (+10%, +0.1V)	V
	Turn-on settling	I _{out} = 0			240	μs

⁵ VRF loading capacitor typical value is 2.2uF. Other analog LDOs are 1uF.

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
	time					

Table 27. Digital LDO specifications

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
	Load capacitor			1		μF
	Vout	Includes load regulation, line regulation, and temperature coefficient	2	2.8	3	V
	Temperature coefficient				100	ppm/C
	Quiescent current	Iout = 0		15		μA

Table 28. RTC LDO specification

2.6.2 ISINK and KPLED Switches

One built-in open-drain output switch drives the keypad LED (KPLED) in the handset. The switch is controlled by baseband with enabling registers. The switch of keypad LED can sink as much as 60mA current, and the output is high impedance when disabled. The value of the sink current decides the brightness of the LED.

Four current controlled open drain drivers (ISINK0 ~ 3) are also implemented to drive the LCM backlight module, and each channel provides 6-current-level steps up to 24mA.

2.6.2.1 Block Description

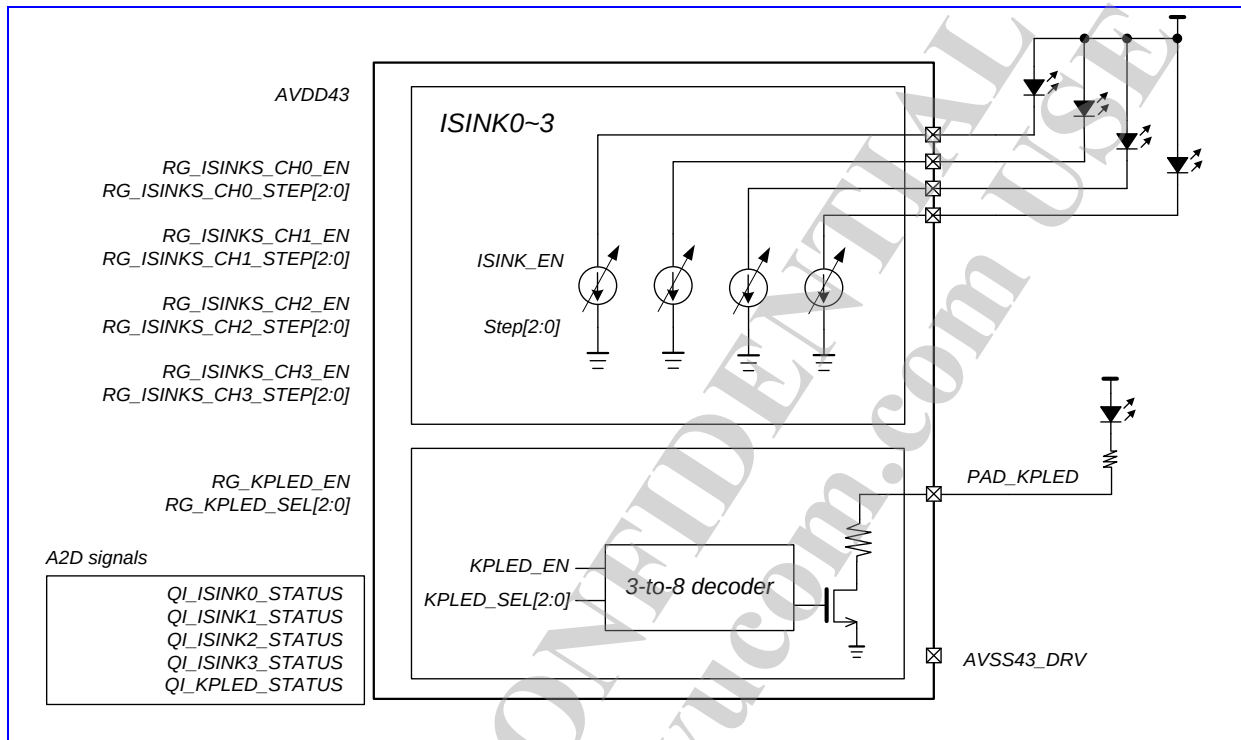


Figure 14. ISINKs and KPLED switches block diagram

2.6.2.2 Functional Specifications

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
	Sink current of keypad LED driver	Von > 0.5V, 100% dimming duty	60			mA
	Sink current of ISINK0 ~ 3	Von > 0.15V, 100% dimming duty, ISINKS_CHx_STEP = 000		4		mA
	Sink current of ISINK0 ~ 3	Von > 0.15V, 100% dimming duty, ISINKS_CHx_STEP = 001		8		mA
	Sink current of ISINK0 ~ 3	Von > 0.15V, 100% dimming duty, ISINKS_CHx_STEP = 010		12		mA
	Sink current of ISINK0 ~ 3	Von > 0.15V, 100% dimming duty, ISINKS_CHx_STEP = 011		16		mA
	Sink current of ISINK0 ~ 3	Von > 0.15V, 100% dimming duty,		20		mA

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
		ISINKS_CHx_STEP = 100				
	Sink current of ISINK0 ~ 3	Von > 0.15V, 100% dimming duty, ISINKS_CHx_STEP = 101		24		mA
	Current mismatch between the 4 channels	Von > 0.15V, 100% dimming duty	-5		5	%

Table 29. ISINKs and KPLED switches specifications

2.6.3 STRUP

PMU handles the power-on and off of the handset. If the battery voltage is neither in the UVLO state (VBAT ≥ 3.4V) nor in the thermal condition, there are three methods to power on the handset system: pulling PWRKEY low (the user pushes PWRKEY), pulling PWRBB high (baseband BB_WakeUp) or valid charger plug-in.

According to different battery voltage (VBAT) and phone states, control signals and regulators will have different responses.

2.6.4 PCHR

The charger controller senses the charger input voltage from either a standard AC-DC adaptor or an USB charger. When the charger input voltage is within a pre-determined range, the charging process will be activated. This detector can resist higher input voltage than other parts of the PMU.

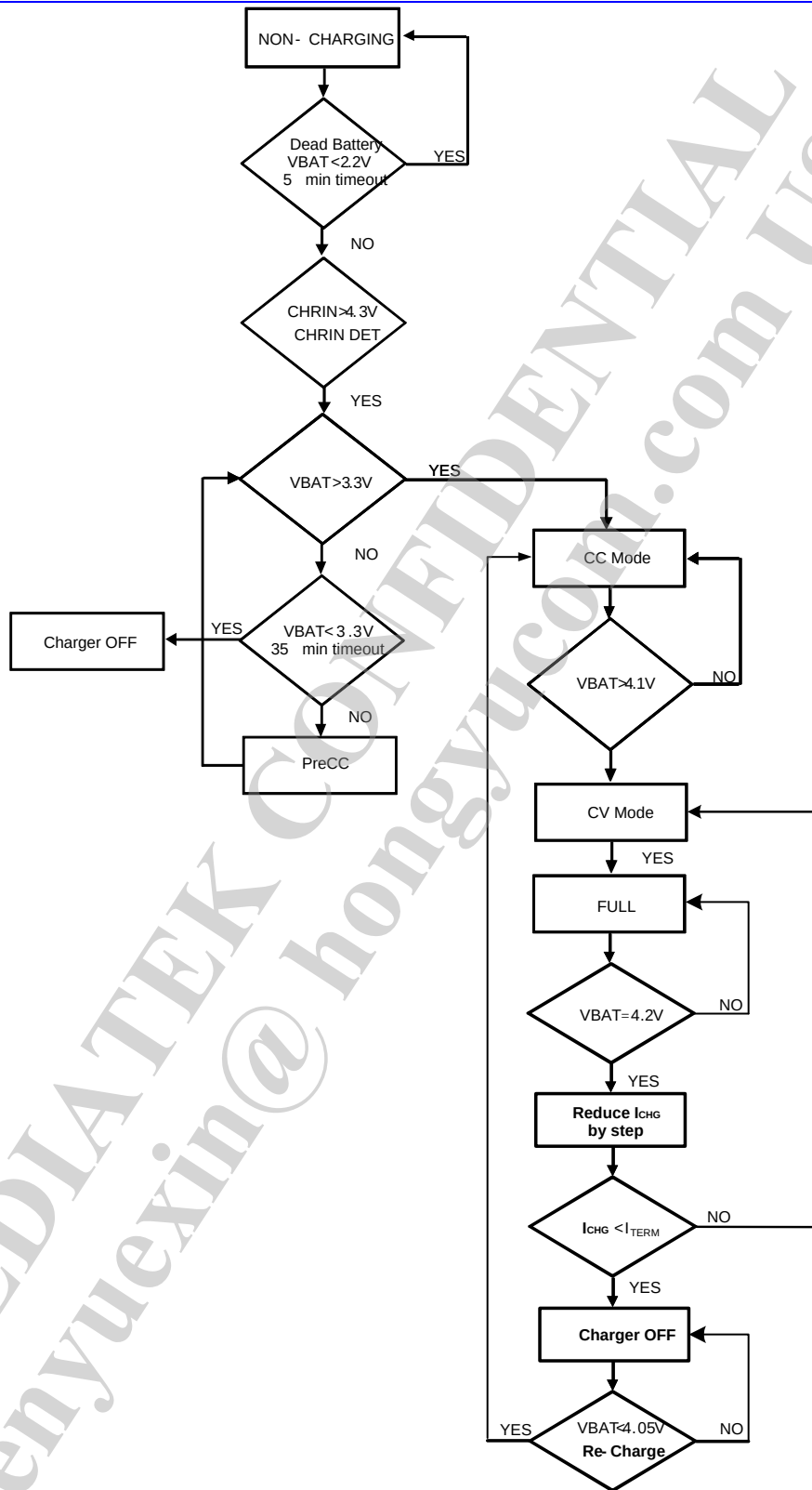


Figure 16. Charging states diagram

Pre-charge mode

When the battery voltage is in the UVLO state, the charger operates in the pre-charge mode. There are two steps in this mode. When the battery voltage is deeply discharged below 2.2V, PRECC0 trickle charging current will be applied to the battery.

The PRECC1 trickle charging current is about 550ms pulse 70mA current when VBAT is under 2.2V.

When the battery voltage exceeds 2.2V, called the PRECC2 stage, the closed-loop pre-charge is enabled. The voltage drop across the external RSENSE is kept around 40mV (AC charger) or 14mV (USB host). The closed-loop pre-charge current can be calculated:

$$I_{\text{PRECC2, AC adapter}} = \frac{V_{\text{SENSE}}}{R_{\text{sense}}} = \frac{40\text{mV}}{R_{\text{sense}}}$$

$$I_{\text{PRECC2, USBHOST}} = \frac{V_{\text{SENSE}}}{R_{\text{sense}}} = \frac{14\text{mV}}{R_{\text{sense}}}$$

Constant current mode

As the battery is charged up and over 3.4V, it can switch to the CC mode. (CHR_EN should be high) In the CC mode, several charging currents can be set by programming registers or the external RSENSE resistor. The charging current can be determined by CS_VTH/RSENSE, where CS_VTH is programmed by registers. For example, if RSENSE is selected as 0.2ohm, the CC mode charging current can be set from 70 to 800mA. It can accommodate the battery charger to various charger inputs with different current capabilities.

Constant voltage mode and over-voltage protection (OV)

While the battery voltage reaches about 4.2V, a constant voltage is used for charging. This is called the full-voltage charging mode or constant-voltage charging mode in correspondence to a linear charger. While the battery voltage actually reaches 4.2V, the charging current is gradually decreased step-by-step, the end-of-charging process starts. It may prolong the charging and detecting period for acquiring optimized full charging volume. The charging process is completed once the current reaches zero automatically and this mechanism is optimized for different battery

BC1.1 Dead-Battery Support of China Standard

MT6250 supports dead-battery condition from China standard (called BC1.1). The specification protects dead-battery charging by timer and trickle current. Once the battery voltage is below 2.2V, a period (TUNIT) of trickle current (IUNIT) will be applied to the battery.

If the battery voltage is still below 2.2V after applying trickle current, the charger will be disabled. On the other hand, if the battery voltage is raised to above 2.2V, the charger will enter the PRECC1 stage, and the charging current will be 70mA or 200mA depending on the type of charging port.

Under the condition of battery voltage from 2.2V to 3.3V, the charger will charge the battery with the PRECC1 current.

A dedicated 5mins (T1) timer will be timed out and disable the charger if the battery voltage is always below 2.7V under charging. Another 35mins (T2) timer will also be timed out and disable the charger if the battery voltage is always kept between 2.7V and 3.3V under charging.

The trickle current (IUNIT) and two dedicated timers protect the charging action if the battery is dead.

2.6.4.2 Functional Specifications

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
	Charger detect-on range		4.3		7	V

Table 30. Charger detection specifications

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
	IUNIT with 500ms pulse	VBAT < 2.2V	28	56	84	mA
	Pre-charging current	VBAT < 2.2V (500ms pulse)	28	56	84	mA
		VBAT ≥ 2.2V (USB host)	7/R _{sense}	14/R _{sense}	20/R _{sense}	mA
		VBAT ≥ 2.2V (AC adapter < 7V)	30/R _{sense}	40/R _{sense}	50/R _{sense}	mA
		VBAT ≥ 2.2V (AC adapter > 7V)	7/R _{sense}	14/R _{sense}	20/R _{sense}	mA
	Pre-charging off threshold	CHR_EN = L		3.3		V
	Pre-charging off hysteresis			0.4		V

Table 31. Pre-charge specifications

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
	CC mode charging current (CS_VTH)	CS_VTH [2:0] = 000		160/R _{sense}		mA
		CS_VTH [2:0] = 001		140/R _{sense}		mA
		CS_VTH [2:0] = 010		130/R _{sense}		mA
		CS_VTH [2:0] = 011		110/R _{sense}		mA
		CS_VTH [2:0] = 100		90/R _{sense}		mA
		CS_VTH [2:0] = 101		60/R _{sense}		mA

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
		CS_VTH [2:0] = 110		40/R _{sense}		mA
		CS_VTH [2:0] = 111		14/R _{sense}		mA
	Current sensing resistor	RSENSE		0.2		ohm

Table 32. Constant current specifications

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
	Charging complete threshold		4.15	4.2	4.25	V
	Battery over-voltage protection threshold (OV)			4.35		V

Table 33. Constant voltage and over-voltage protection specifications

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
IUNIT	BC1.1 trickle current	VBAT < 2.2V		56	100	mA
IPRECC1 (USB host)	PRECC1 current	2.2 < VBAT < 3.3V		70	100	mA
IPRECC1 (AC adapter)	PRECC1 current	2.2 < VBAT < 3.3V		200	250	mA
T1	5 minute dedicated timer	2.2 < VBAT < 2.7V		5	7	min.
T2	35 minute dedicated timer	2.7 < VBAT < 3.3V		35	36	min.
TUNIT	BC1.1 trickle current period				1	sec.

Table 34. BC1.1 specifications

2.6.5 Class-AB Audio Amplifier

2.6.5.1 Block Description

MT6250 has built-in high efficiency class AB audio power amplifier, capable of delivering 1 watts of power to an 8 ohm BTL load with less than 10% distortion (THD+N) from a 4.2V battery supply. The speaker amplifier in MT6250 is integrated in PMU SOC. MT6250 class AB also supports 2-in-1 receivers.

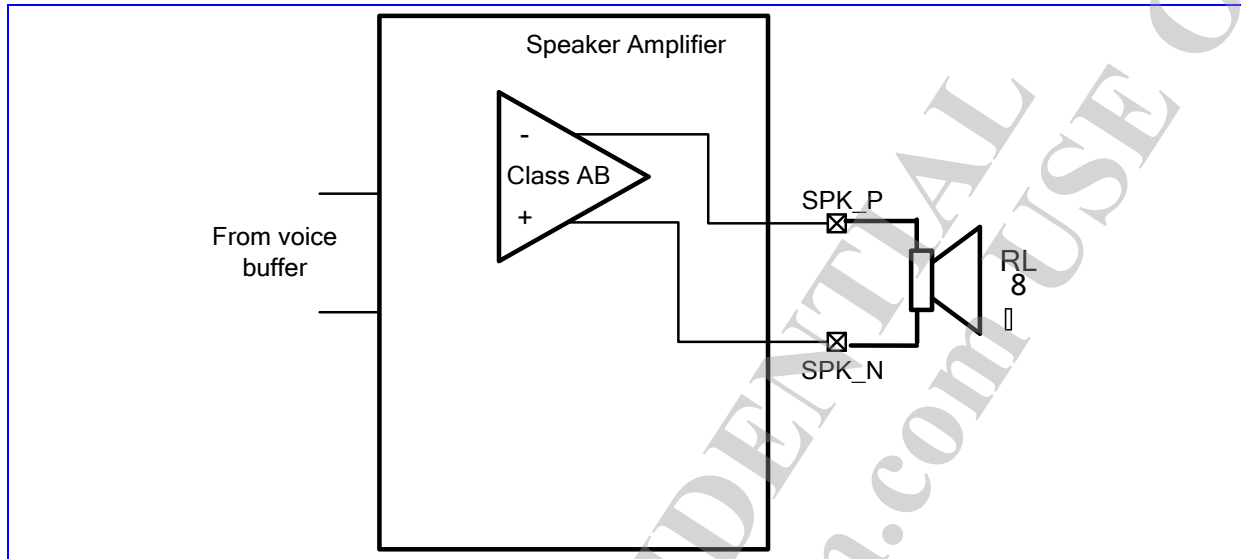


Figure 17. Class-AB audio amplifier block diagram

2.6.5.2 Functional Specifications

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
	RMS power@1% THD	8Ω load, VBAT = 4.2V		850		mW
	THD + N	1kHz, Po = 600mWrms, 4.2V		0.1		%
	PSRR	20Hz ~ 1kHz, diff. mode	60	85		dB
	Shutdown current	SPK_EN = SPK_OUTSTG_EN = 0		1		μA
	Quiescent power supply current	VBAT = 4.2V, no input		4		mA
	Gain adjustment	Audio speaker mode		6/12		dB
	Gain adjustment	Voice receiver mode		-6		dB
	Gain adjustment steps	Audio speaker mode		6		dB

Table 35. Class-AB audio amplifier specifications

2.7 GSM/GPRS/EDGE-Rx RF

2.7.1 General Description

2G RFSYS which is built in MT6250 SOC is a highly integrated RF transceiver for multi-band GSM, GPRS and EDGE-Rx cellular systems.

The features include:

Receiver

- Saw-less Rx
- Quadrature RF mixers
- Fully integrated channel filter
- High dynamic range ADC
- 24dB PGA gain with 6dB gain step

Transmitter

- High accurate transmitter modulator for GSM/GPRS application
- Built-in calibration of SX loop filter and loop gain

Frequency synthesizer

- Programmable fractional-N synthesizer
- Integrated wide range RFVCO
- Integrated loop filter
- Fast settling time suitable for multi-slot GSM/GPRS/EDGE-Rx applications

Digitally-Controlled Crystal Oscillator (DCXO)

- Two-pin 26 MHz crystal oscillator
- On-chip programmable capacitor array for coarse-tuning
- On-chip programmable capacitor array for fine-tuning
- Supports 32K XTAL-less operation

2.7.2 Functional Block Diagram

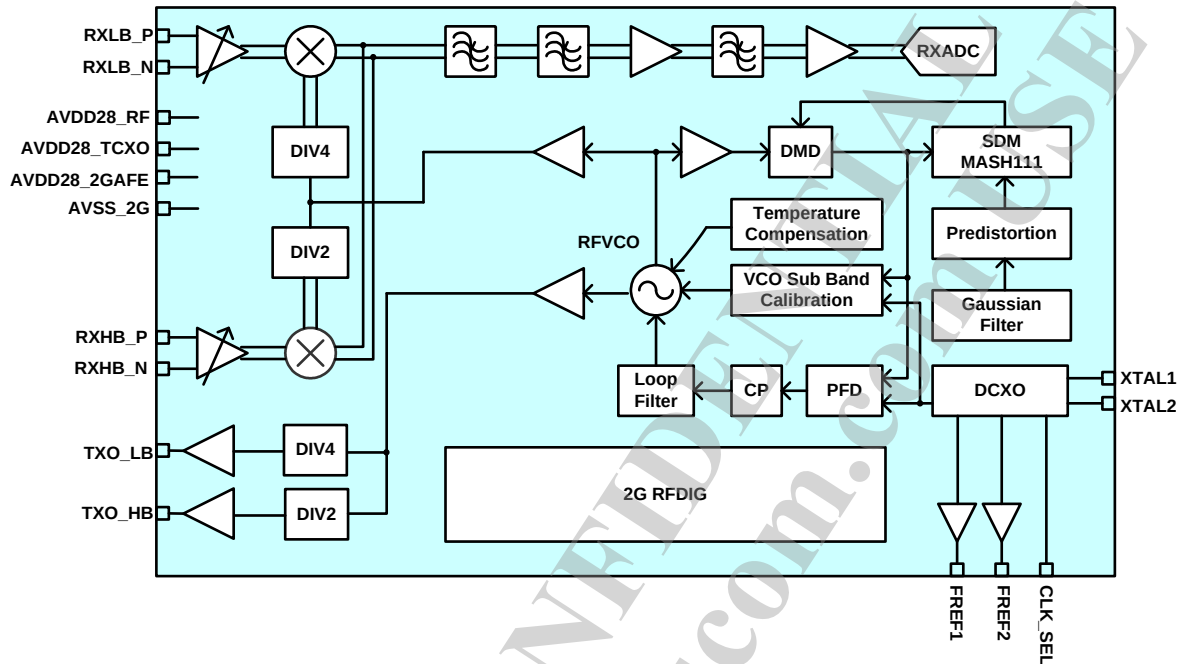


Figure 18. Diagram of MT6250 2G RFSYS

2.7.3 Electrical Characteristics

RFSYS Mode	AVDD28_RF	AVDD28_TCXO	AVDD28_2GAFF	RFSYS Total	Unit
BCM_Deep Sleep (DCXO is off)	3	14	1	18	uA
BCM_Sleep (DCXO is on)	0.003	2.8	0.26	3.1	mA
Low power mode	3	88	1	92	uA
Full power mode	0.003	2.8	0.26	3.1	mA
RX (GSM850/EGSM)	53	8	5	66	mA
RX (DCS/PCS)	60	8	5	73	mA
TX (GSM850/EGSM)	46	10	2	58	mA
TX (DCS/PCS)	40	10	2	52	mA

Table 36. DC characteristics (TA = 25oC, VDD = 2.8V unless otherwise stated)

Item	Symbol	Band	Test condition	Min.	Typ.	Max.	Unit
Input frequency	F _{RX}	GSM850		869		894	MHz
		GSM900		925		960	MHz
		DCS1800		1,805		1,880	MHz

Item	Symbol	Band	Test condition	Min.	Typ.	Max.	Unit
		PCS1900		1,930		1,990	MHz
Differential voltage gain 1	G ₁	GSM850	LNA = High gain	54 ¹	57		dB
		GSM900	PGA = Middle high gain	54 ²	57		dB
		DCS1800	LNA = High gain	53.5 ³	56.5		dB
		PCS1900	PGA = Middle high gain	53.5 ⁴	56.5		dB
Differential voltage gain 2	G ₂	GSM850	LNA = Middle high gain		55.5		dB
		GSM900	PGA = Middle high gain		55.5		dB
		DCS1800	LNA = Middle high gain		53.5		dB
		PCS1900	PGA = Middle high gain		53.5		dB
Differential voltage gain 3	G ₃	GSM850	LNA = Middle gain		51		dB
		GSM900	PGA = Middle high gain		51		dB
		DCS1800	LNA = Middle gain		50		dB
		PCS1900	PGA = Middle high gain		50		dB
Differential voltage gain 4	G ₄	GSM850	LNA = Low gain		24		dB
		GSM900	PGA = Middle high gain		24		dB
		DCS1800	LNA = Low gain		24		dB
		PCS1900	PGA = Middle high gain		24		dB
Noise figure at 25°C	NF ₂₅	GSM850	G ₁		2.5	4.5 ¹	dB
		GSM900			2.5	4.5 ²	dB
		DCS1800			2.5	4.5 ³	dB
		PCS1900			2.5	4.5 ⁴	dB
Noise figure at 85°C	NF ₈₅	GSM850	G ₁		3.5		dB
		GSM900			3.5		dB
		DCS1800			3.5		dB
		PCS1900			3.5		dB
2 nd -order input intercept point	IIP2	GSM850	G2	31 ¹	43		dBm
		GSM900		31 ²	43		dBm
		DCS1800		31 ³	43		dBm
		PCS1900		31 ⁴	43		dBm
3 rd -order input intercept point	IIP3	GSM850	G2	-14 ¹	-3		dBm
		GSM900		-14 ²	-3		dBm
		DCS1800		-14 ³	-3		dBm
		PCS1900		-14 ⁴	-3		dBm

Item	Symbol	Band	Test condition	Min.	Typ.	Max.	Unit
3 rd -order input intercept point @ -20°C	IIP _{3.20}	GSM850	G2		-5		dBm
		GSM900			-5		dBm
		DCS1800			-5		dBm
		PCS1900			-5		dBm
Receiver S/N with 3MHz blocker	SN _{3M}	GSM850	G2	8 ¹	14		dB
		GSM900	Blocker = rN wdBm	8 ²	14		dB
		DCS1800	G2	8 ³	14		dB
		PCS1900	Blocker = rN wdBm	8 ⁴	14		dB
Receiver S/N with OBB	SN _{OBB}	GSM850	G3	6 ⁵	8		dB
		GSM900	Blocker = 3dBm, offset 20MHz	6 ⁵	8		dB
		DCS1800	G3	6 ⁵	8		dB
		PCS1900	Blocker = 3dBm, offset 80MHz	6 ⁵	8		dB
Image rejection ratio	IRR	ALL	G1	32 ^{1,2,3,4}	40		dB
Receiver channel response attenuation		ALL	@3MHz offset		20		dB
			@6MHz offset		35		dB
Receiver filtering 3-dB bandwidth		ALL	For all gain settings		900		kHz
PGA gain linearity		ALL	INL		0.2	1 ⁵	dBΩ
			DNL		0.1	0.5 ⁵	dBΩ
PGA gain step		ALL			6		dBΩ
PGA dynamic range		ALL			24		dBΩ
I/Q common-mode output voltage		ALL	G1	1.1 ⁵	1.2	1.3 ⁵	V
Output static dc offset		ALL	G1		100	200	mV

Table 37. Rx AC characteristics (TA = 25oC, VDD = 2.8V unless otherwise stated)

Item	Symbol	Band	Test condition	Min.	Typ.	Max.	Unit
Frequency	F _{TX}	GSM850		824		849	MHz
		GSM900		880		915	MHz
		DCS1800		1,710		1,785	MHz
		PCS1900		1,850		1,910	MHz
RMS phase error	PE _{rms}	GSM850 GSM900			1	2.5 ^{1,2}	degree

Item	Symbol	Band	Test condition	Min.	Typ.	Max.	Unit
		DCS1800 PCS1900			1	2.5 ^{3,4}	degree
Output modulation spectrum	ORFS	GSM850 GSM900	400kHz offset (RBW = 30kHz bandwidth)		-66	-64 ^{1,2}	dBc
		DCS1800 PCS1900			-66	-64 ^{3,4}	dBc
		GSM850 GSM900	1.8MHz offset (RBW = 30kHz bandwidth)			-75 ⁵	dBc
		DCS1800 PCS1900				-75 ⁵	dBc
Tx noise in Rx band		GSM850	20MHz offset		-165	-164 ⁵	dBc/Hz
			35MHz offset		-168	-167 ⁵	dBc/Hz
		GSM900	20MHz offset		-165	-164 ⁵	dBc/Hz
			35MHz offset		-168	-167 ⁵	dBc/Hz
		DCS1800	20MHz offset		-160	-156 ⁵	dBc/Hz
		PCS1900	20MHz offset		-160	-156 ⁵	dBc/Hz
Output power level	P _{out}	GSM850 GSM900	PA driver amplifier R _{load} = 50Ω	1 ^{1,2}	3	6 ^{1,2}	dBm
		DCS1800 PCS1900		1 ^{3,4}	3	6 ^{3,4}	dBm
Output 3 rd harmonics		ALL	PA driver amplifier		-10		dBc

Table 38. Tx GMSK AC characteristics (TA = 25oC, VDD = 2.8V unless otherwise stated)

Item	Symbol	Test condition	Min.	Typ.	Max.	Unit
Frequency range	F _{range}		3,296		3,980	MHz
Reference frequency	F _{ref}			26		MHz
Frequency step resolution	F _{res}			3		Hz
Phase noise	PN _{10k}	@ 10kHz offset		-83		dBc/Hz
	PN _{400k}	@ 400kHz offset		-116		dBc/Hz
	PN _{3M}	@ 3MHz offset		-136		dBc/Hz
Lock time of Rx burst	T _{lock_rx}	Frequency error < ± 0.1ppm		150	200 ⁵	us
Lock time of Tx burst	T _{lock_tx}	Frequency error < ± 0.1ppm		200	300 ⁵	us
Pushing figure		With internal RFVCO LDO		400		kHz/V

Table 39. SX AC characteristics (TA = 25oC, VDD = 2.8V unless otherwise stated)

Item	Symbol	Test condition	Min.	Typ.	Max.	Unit
Operating frequency	F _{ref}			26		MHz
Crystal C load	C _L			7.5		pF
Crystal tuning sensitivity	T _s		27.5	32.3		ppm/pF
Static range	SR	CDAC from 0 to 255	± 22	± 50		ppm
Dynamic range	DR	CAFC from 0 to 8191	36	50		ppm
AFC tuning step	F _{res-AFC}			0.006		ppm/DAC
AFC settling time	T _{AFC}	CAFC from 0 to 8191 CAFC from 8191 to 0 Frequency error < 0.1ppm		100	200 ⁵	us
Start-up time	T _{DCXO}	Frequency error < 1ppm Amplitude > 90 %			2 ⁵	ms
Pushing figure				0.2		ppm/V
Fref buffer output level	V _{Fref}	Max. loading = 19pF	0.8 ⁵			V _{p-p}
Fref buffer output phase noise		10kHz offset Jitter noise		-140		dBc/Hz

Table 40. DCXO AC characteristics (TA = 25oC, VDD = 2.8V unless otherwise stated)

^{1, 2}: Tested at E-GSM Tx channel 0 and GSM850 Rx channel 190.

^{3, 4}: Tested at PCS Tx channel 601 and DCS Rx channel 636.

⁵: Not subject to production test – verified by characterization and design.

2.8 Bluetooth

2.8.1 Block Description

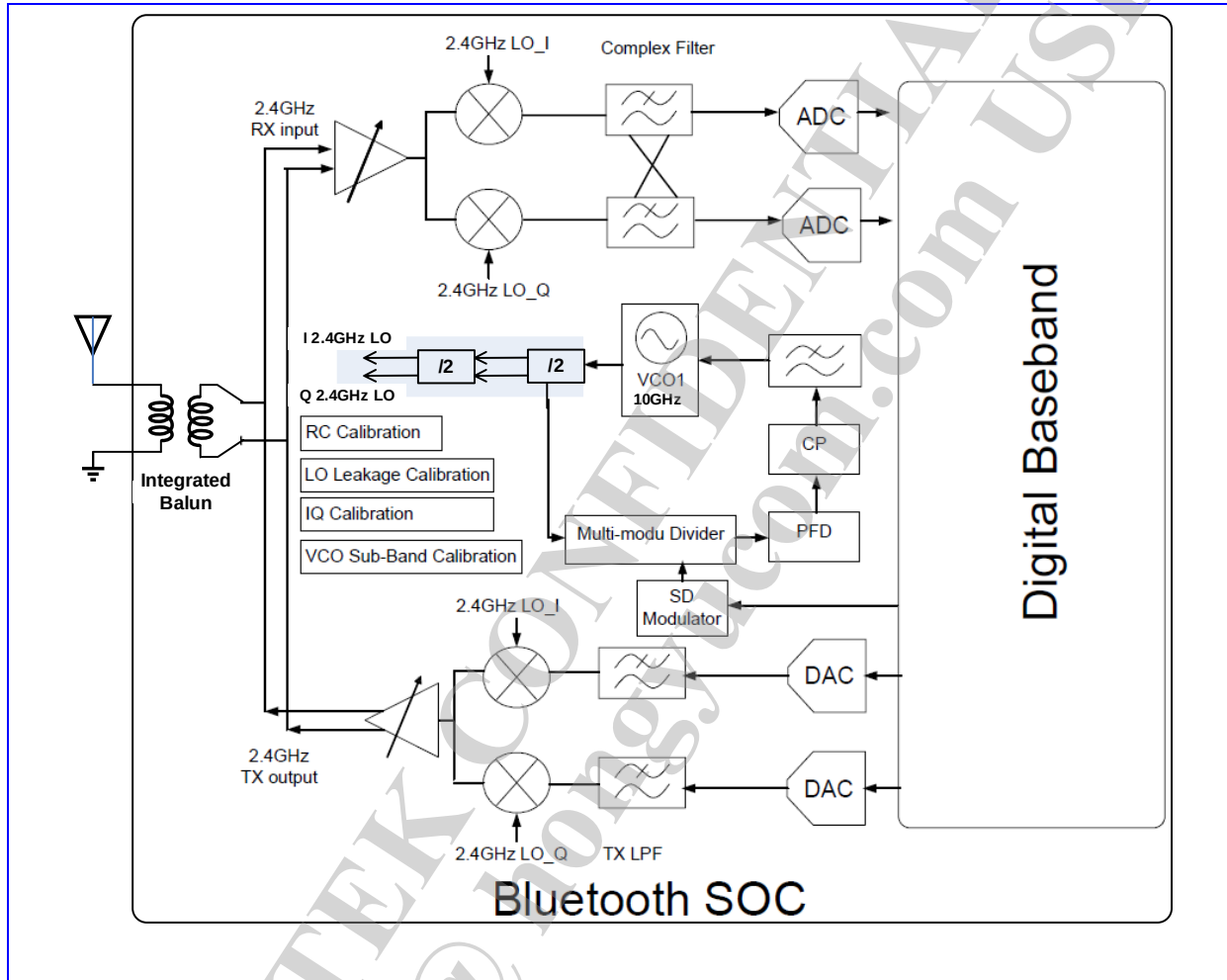


Figure 19. System diagram of Bluetooth RF transceiver

The Bluetooth RF subsystem contains a fully integrated transceiver.

For the Tx path, the baseband transmit data are digitally modulated in the baseband processor then up-converted to 2.4GHz RF channels through the DA converter, filter, IQ up-converter and power amplifier. The power amplifier is capable of transmitting 10dBm power for class-1.5 operation.

For the Rx path, MT6250 is a low IF receiver architecture. An image-reject mixer down-converts the RF signal to the IF with the LO from the synthesizer, which supports different clock frequencies as the reference clock. The mixer output is then converted to digital signal, down-converted to baseband for demodulation. A fast AGC enables the effective discovery of device within the dynamic range of the receiver.

MT6250 features self calibration schemes to compensate the process and temperature variation to maintain high performance. Those calibrations are performed automatically right after the system boot-up.

2.8.2 Functional Specifications

2.8.2.1 Basic Data Rate – Receiver Specifications

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
	Frequency range		2,402	-	2,480	MHz
	Receiver sensitivity	BER < 0.1%	-	-95	-	dBm
	Max. detectable input power	BER < 0.1%	-	0	-	dBm
	C/I co-channel selectivity	BER < 0.1%	-	4	-	dB
	C/I 1MHz adj. channel selectivity	BER < 0.1%	-	-15	-	dB
	C/I 2MHz adj. channel selectivity	BER < 0.1%	-	-47.5	-	dB
	C/I ≥ 3MHz adj. channel selectivity	BER < 0.1%	-	-47.5	-	dB
	C/I image channel selectivity	BER < 0.1%	-	-27.5	-	dB
	C/I image 1MHz adj. channel selectivity	BER < 0.1%	-	-47.5	-	dB
	Out-of-band blocking	30 to 2,000MHz	-	10	-	dBm
		2,000 to 2,399MHz	-	-2	-	dBm
		2,498 to 3,000MHz	-	-3	-	dBm
		3,000MHz to 12.75GHz	-	10	-	dBm
	Intermodulation		-	-20	-	dBm

Table 41. Basic data rate – receiver specifications

2.8.2.2 Basic Data Rate – Transmitter Specification

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
	Frequency range		2,402	-	2,480	MHz
	Max. transmit power		-	10	-	dBm
	Gain step		-	4	-	dB
	Δf1avg (00001111)		140	158	175	kHz
	Δf2max (10101010)		115	145	-	kHz
	Δf1avg/Δf2avg		0.8	0.98	-	kHz
	Initial carrier frequency drift		-75	5	75	kHz
	Frequency drift	DH1	-25	9	25	kHz
		DH3	-40	10	40	kHz

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
		DH5	-40	10	40	kHz
	Max. drift rate		-	100	400	Hz/μs
	BW _{20dB} of Tx output spectrum		-	920	1,000	kHz
	In-band spurious emission	±2 MHz offset	-	-44	-	dBm
		±3 MHz offset	-	-48	-	dBm
		> ±3 MHz offset	-	-48	-	dBm
	Out-of-band spurious emission	30 MHz to 1 GHz	-	-75	-	dBm
		1 to 12.75 GHz	-	-41	-	dBm
		1.8 to 1.9 GHz	-	-84	-	dBm
		5.15 to 5.3 GHz	-	-88	-	dBm

Table 42. Basic data rate – transmitter specifications

2.8.2.3 Enhanced Data Rate – Receiver Specifications

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
	Frequency range		2,402	-	2,480	MHz
	Receiver sensitivity	π/4 DQPSK, BER < 0.01%	-	-95	-	dBm
		8PSK, BER < 0.01%	-	-88	-	dBm
	Max. detectable input power	π/4 DQPSK, BER < 0.01%	-	0	-	dBm
		8PSK, BER < 0.01%	-	0	-	dBm
	C/I co-channel selectivity	π/4 DQPSK, BER < 0.01%	-	8	-	dB
		8PSK, BER < 0.01%	-	14.5	-	dB
	C/I 1MHz adj. channel selectivity	π/4 DQPSK, BER < 0.01%	-	-13.5	-	dB
		8PSK, BER < 0.01%	-	-7.5	-	dB
	C/I 2MHz adj. channel selectivity	π/4 DQPSK, BER < 0.01%	-	-47.5	-	dB
		8PSK, BER < 0.01%	-	-41.5	-	dB
	C/I ≥ 3MHz adj. channel selectivity	π/4 DQPSK, BER < 0.01%	-	-50.5	-	dB
		8PSK, BER < 0.01%	-	-44.5	-	dB
	C/I image channel selectivity	π/4 DQPSK, BER < 0.01%	-	-30	-	dB
		8PSK, BER < 0.01%	-	-25	-	dB
	C/I image 1 MHz adj. channel selectivity	π/4 DQPSK, BER < 0.01%	-	-48.5	-	dB
		8PSK, BER < 0.01%	-	-44.5	-	dB

Table 43. Enhanced data rate – receiver specifications

2.8.2.4 Enhanced Data Rate – Transmitter Specifications

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
	Frequency range		2,402	-	2,480	MHz
	Max. transmit power	$\pi/4$ DQPSK	-	8	-	dBm
		8PSK	-	8	-	dBm
	Relative transmit power	$\pi/4$ DQPSK	-	-1.7	-	dB
		8PSK	-	-1.7	-	dB
	Freq. stability ω_0	$\pi/4$ DQPSK	-	1.5	-	kHz
		8PSK	-	1.5	-	kHz
	Freq. stability ω_1	$\pi/4$ DQPSK	-	3	-	kHz
		8PSK	-	3	-	kHz
	$ \omega_0 + \omega_1 $	$\pi/4$ DQPSK	-	2.8	-	kHz
		8PSK	-	2.8	-	kHz
	RMS DEVM	$\pi/4$ DQPSK	-	5.4	-	%
		8PSK	-	5.7	-	%
	99% DEVM	$\pi/4$ DQPSK	-	10	-	%
		8PSK	-	11	-	%
	Peak DEVM	$\pi/4$ DQPSK	-	18	-	%
		8PSK	-	18	-	%
	In-band spurious emission	$\pi/4$ DQPSK, ± 1 MHz offset	-	-40	-	dBm
		8PSK, ± 1 MHz offset	-	-40	-	dBm
		$\pi/4$ DQPSK, ± 2 MHz offset	-	-34	-	dBm
		8PSK, ± 2 MHz offset	-	-34	-	dBm
		$\pi/4$ DQPSK, ± 3 MHz offset	-	-42	-	dBm
		8PSK, ± 3 MHz offset	-	-42	-	dBm

Table 44. Enhanced data rate – transmitter specifications

Note: To meet the specifications, use a front-end band-pass filter.

2.9 FM RF

2.9.1 Block Descriptions

The connection between internal modules, as well as, external interfaces can be found in Figure 20. The FM receiver section incorporates the complete receiving path with wide tuning range. The FM baseband signal processor incorporates the digital demodulator and audio processing function which provides superior audio quality.

FM contains completely integrated FM audio receiver functions (RDS/RBDS may also be supported depending on the model number). The integrated receiver enables superior sensitivity, ACI performance and FM audio performances with minimum external BOM.

The FM subsystem supports either high-performance stereo analog line out or digital audio output.

For models supporting RDS/RBDS, large dedicated internal data buffers are allocated to reduce the frequency of the interrupt to the host, so that the receiving host can enter low-power states efficiently.

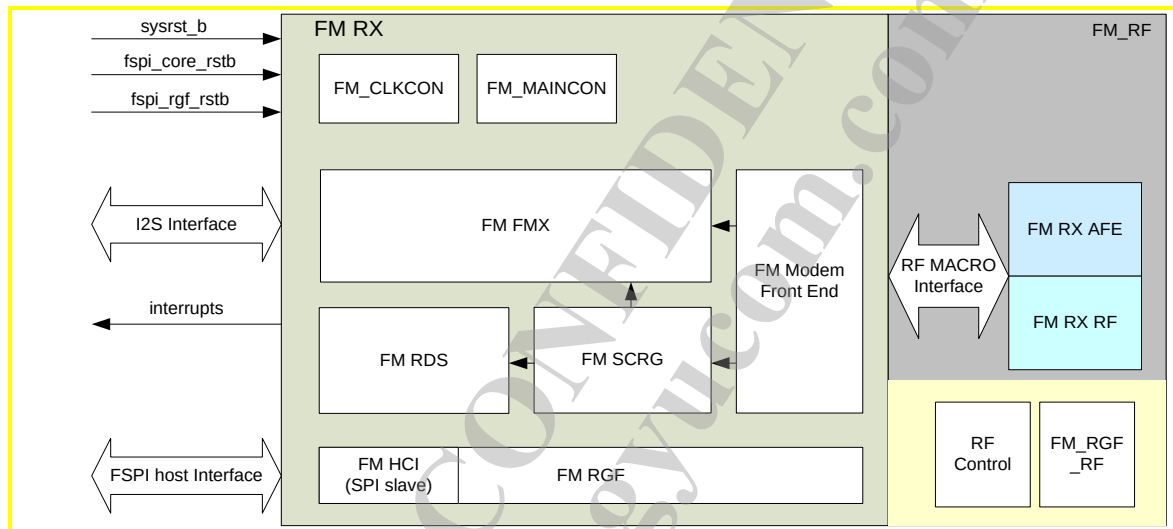


Figure 20. Block diagram of hardware top-level architecture

2.9.2 Functional Specifications

Operating mode	Current consumption	Unit
Idle	5	μA
FM receiver	10.5	mA

Table 45. FM receiver DC characteristics (TA = 25°C, VDD = 2.8V unless otherwise stated)

Unless otherwise stated, all receiver characteristics are applicable to both long and short antenna ports when operated under the recommended operating conditions. Typical specifications are for channel 98MHz, default register settings and under recommended operating conditions. The minimum and maximum specifications are for extreme operating voltage and temperature conditions, unless otherwise stated.

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
	Input frequency range		87.5		108	MHz

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
	Sensitivity (long antenna) ^{1,3}	(S+N)/N = 26dB, unmatched		3		dB μ Vemf
		(S+N)/N = 26dB, matched		2		dB μ Vemf
	RDS sensitivity (long antenna)	Δf = 2kHz, BLER < 5%, unmatched		19		dB μ Vemf
	Sensitivity (short antenna) ^{1,3}	(S+N)/N = 26dB, unmatched		4		dB μ Vemf
	RDS sensitivity (short antenna)	Δf = 2kHz, BLER < 5%, unmatched		20		dB μ Vemf
	LNA input resistance ⁴			2.4k		Ohm
	LNA input capacitance ⁴			8		pF
	AM suppression ^{1,4}	M = 0.3		60		dB
	Adjacent channel selectivity ^{1,4}	± 200 kHz		53		dB
	Alternate channel selectivity ^{1,4}	± 400 kHz		66		dB
	Spurious response rejection ⁴	In-band		55		dB
	Maximum input level				117	dB μ Vemf
	Audio mono (S+N)/N ^{1,3,4}			58		dB
	Audio stereo (S+N)/N ^{2,3,4}			55		dB
	Audio stereo separation ⁴	Δf = 75 kHz		35		dB

¹ Δf = 22.5kHz, fm = 1kHz, 75 μ s de-emphasis, mono, L = R

² Δf = 22.5kHz, fm = 1kHz, 75s de-emphasis, stereo

³ A-weighting, BW = 300Hz to 15kHz

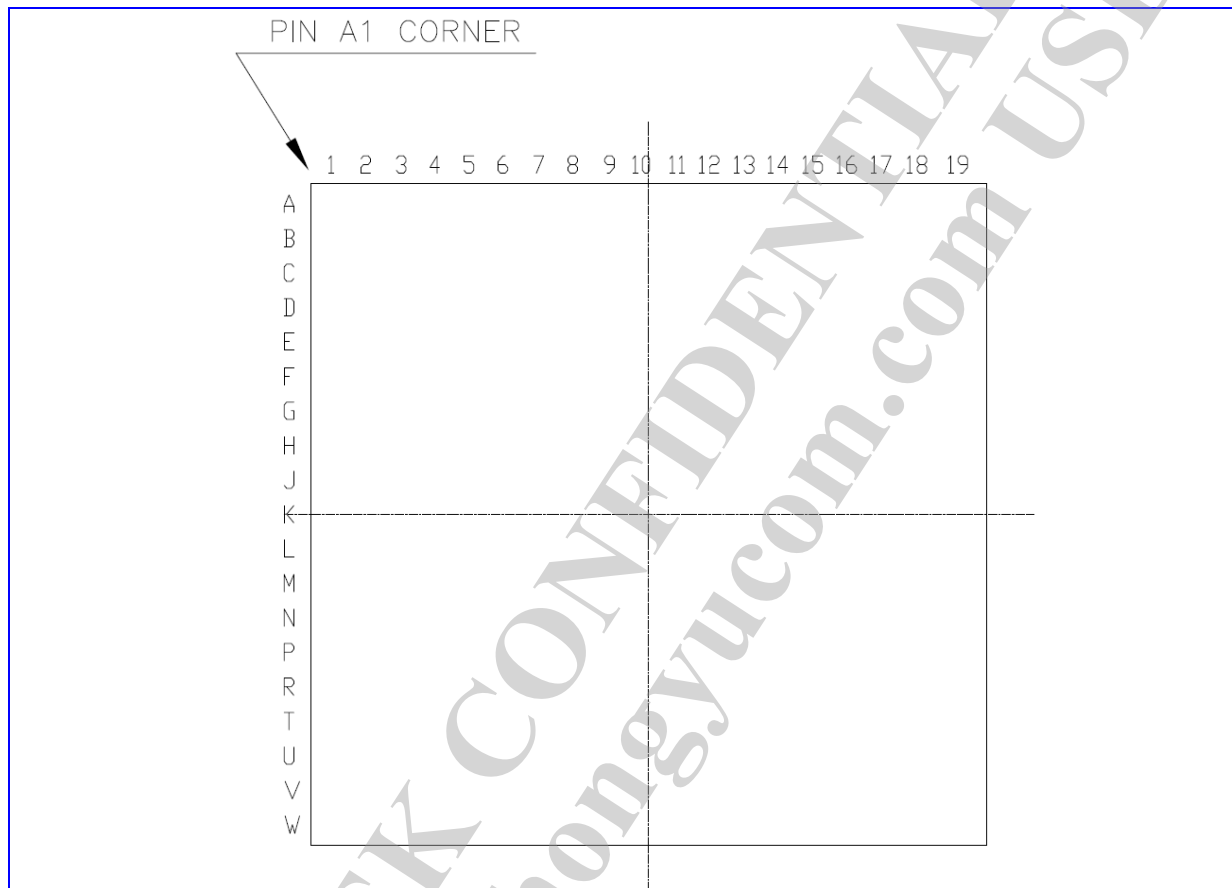
⁴ Vin = 60dB μ Vemf

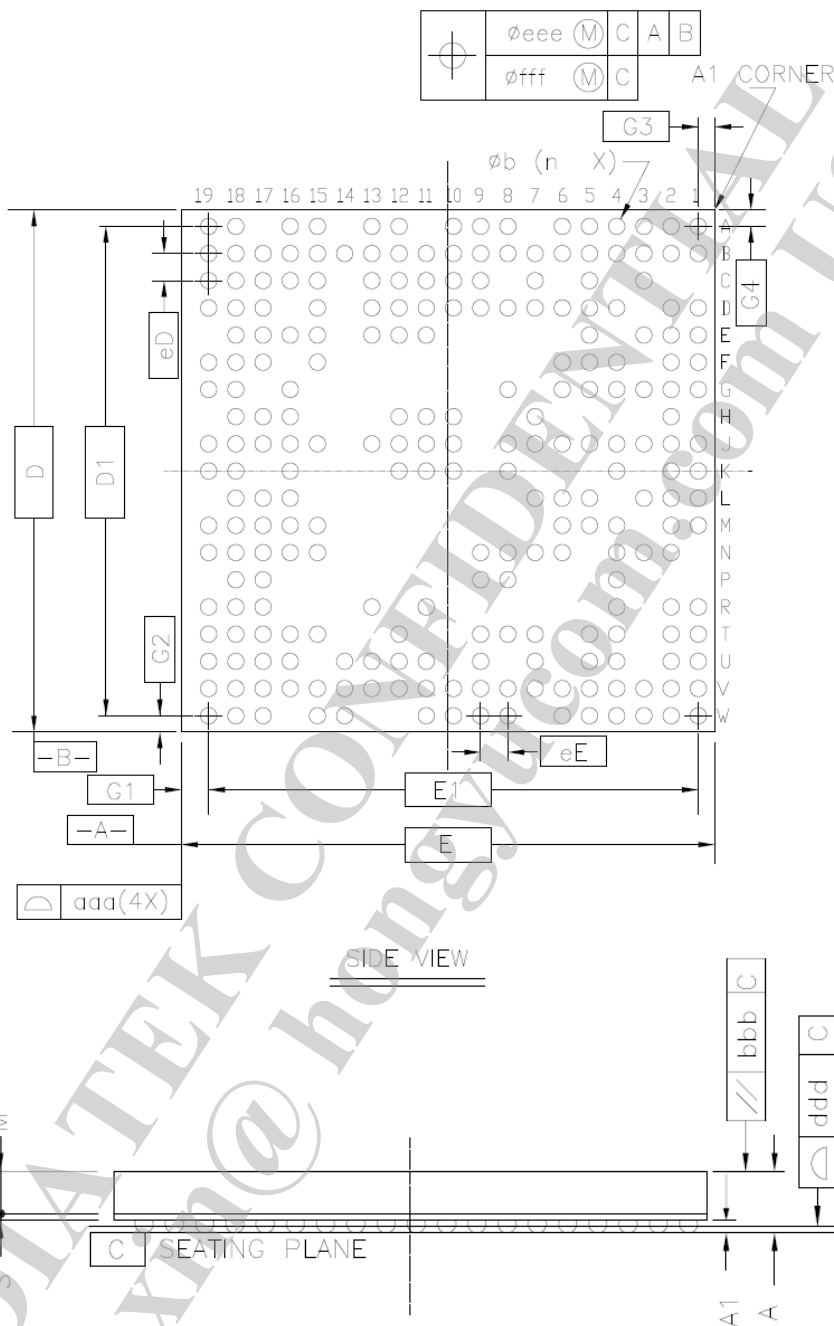
⁵ Reference clock accuracy assumes ideal FM source. If the input FM source has less frequency error, then it is recommended to use a reference clock of accuracy within ± 100 ppm so as not to affect the channel scan quality.

Table 46. FM receiver AC characteristics

2.10 Package Information

2.10.1 Package Outlines





		Symbol	Common Dimensions
Package :			TFBGA
Body Size:	X	E	9.800
	Y	D	9.600
Ball Pitch :	X	eE	0.500
	Y	eD	0.500
Total Thickness :		A	1.100 MAX.
Mold Thickness :		M	0.700 Ref.
Substrate Thickness :		S	0.110 Ref.
Ball Diameter :			0.300
Stand Off :		A1	0.160 ~ 0.260
Ball Width :		b	0.250 ~ 0.350
Package Edge Tolerance :		aaa	0.100
Mold Flatness :		bbb	0.100
Coplanarity:		ddd	0.080
Ball Offset (Package) :		eee	0.150
Ball Offset (Ball) :		fff	0.050
Ball Center TO Package Edge :		G1	0.500
		G2	0.300
		G3	0.300
		G4	0.300
Ball Count :		n	233
Edge Ball Center to Center :	X	E1	9.000
	Y	D1	9.000

Figure 21. Outlines and dimension of TFBGA 9.8mm*9.6mm, 233-ball, 0.5 mm pitch package

2.10.2 Thermal Operating Specifications

Symbol	Description	Value	Unit	Notes
	Thermal resistance from device junction to package case	45.3	C/W	
	Maximum package temperature	65	Deg C	
	Maximum power dissipation	1.328	W	

2.10.3 Lead-free Packaging

MT6250 is provided in a lead-free package and meets RoHS requirements

2.11 Ordering Information

2.11.1 Top Marking Definition

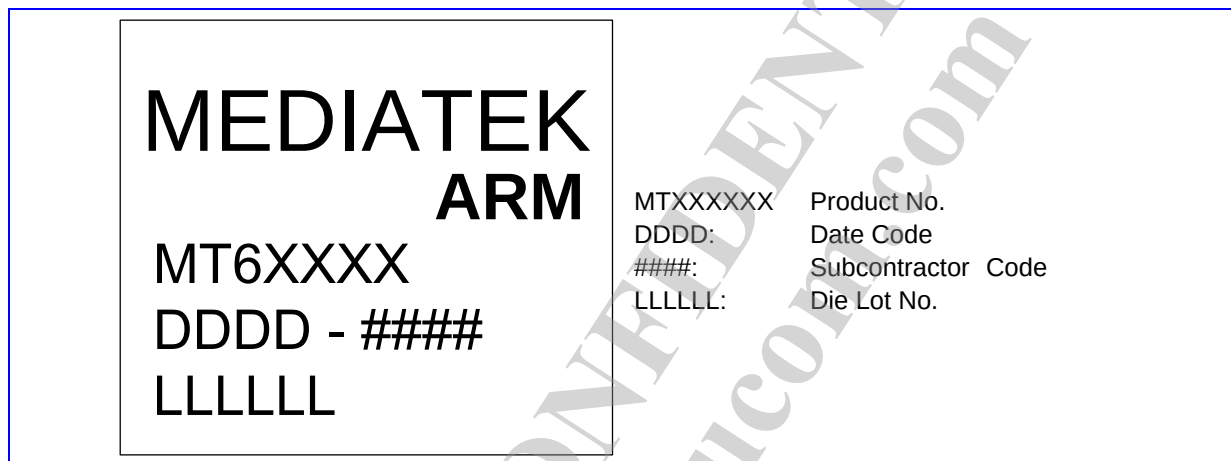


Figure 22. Mass production top marking of MT6250

Part number	Package	Description
-------------	---------	-------------

	MT6250 GSM/GPRS/EDGE-RX SOC Processor Data Sheet v1.0 Confidential A
---	--

Part number	Package	Description
3 MT6250A/BMB-PCU-H	TFBGA	9.8mm*9.6mm, 233-ball, 0.5 mm pitch package, non-security version

4 Micro-Controller Unit Peripherals

4.1 Pulse-Width Modulation Outputs

4.1.1 General Description

1 generic pulse-width modulator is implemented to generate pulse sequences with programmable frequency and duty cycles for LCD backlight. As long as the internal counter value is bigger than or equal to the threshold value, the duration of the PWM output signal is LOW. The waveform is shown in Figure 23.

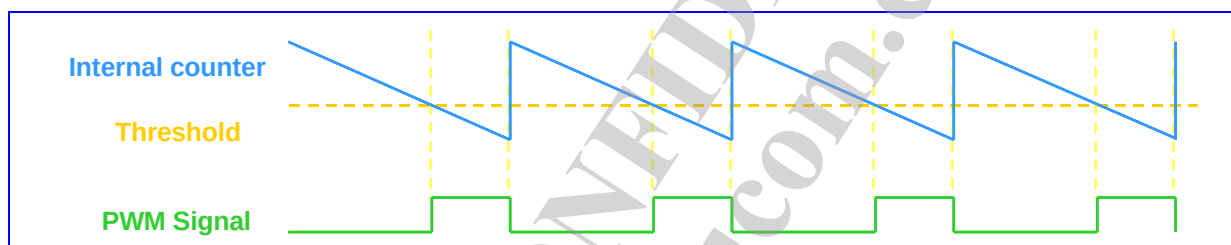


Figure 23. PWM waveform

The frequency and volume of the PWM output signal are determined by PWM1_COUNT, PWM1_THRES and PWM1_CON registers. The POWERDOWN (pdn1_pwm) signal is applied to power-down the PWM_1ch module. When PWM_1ch is deactivated (pwm1_pdn = 1), the output is in the LOW state.

The output PWM frequency is determined by:

$$\frac{CLK}{CLOCK_DIV \times (PWM_COUNT + 1)} \quad \text{CLK} = 13000000 \text{ when CLKSEL} = 0, \text{CLK} = 32000 \text{ when CLKSEL} = 1$$

CLOCK_DIV = 1, when CLK[1:0] = 00b

CLOCK_DIV = 2, when CLK[1:0] = 01b

CLOCK_DIV = 4, when CLK[1:0] = 10b

CLOCK_DIV = 8, when CLK[1:0] = 11b

The output PWM duty cycle is determined by: $\frac{PWM_THRES}{PWM_COUNT + 1}$

Note that PWM_THRES should be less than the PWM_COUNT. If this condition is not satisfied, the output pulse of the PWM will always be HIGH.

Figure 24 shows the PWM waveform with the indicated register values.

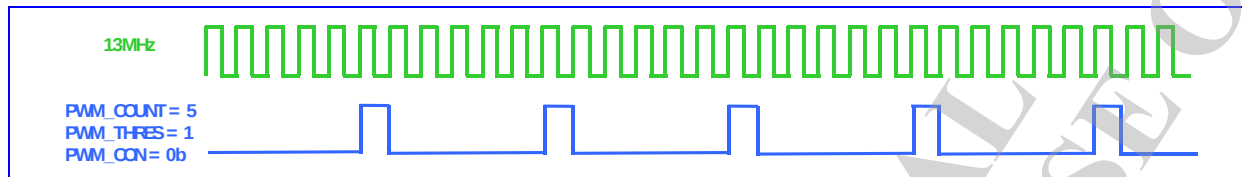


Figure 24. PWM waveform with register values

4.1.2 Register Definition

Module name: Pulse Width Modulation Base address: (+A00E0000h)

Address	Name	Width	Register Function
A00E0000	PWM1_CTRL_ADDR	16	PWM1 control register
A00E0004	PWM1_COUNT_ADDR	16	PWM1 max, counter value register
A00E0008	PWM1_THRESH_ADDR	16	PWM1 threshold value register

A00E0000 PWM1_CTRL_ADDR PWM1 control register

0000

D

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name														PWM1_CLK_SEL	PWM1_CLK_DIV	
Type														RW	RW	
Reset														0	0	0

PWM1_CLK_DIV Selects clock prescale of PWM1

- 00** f=fclk
- 01** f=fclk/2
- 10** f=fclk/4
- 11** f=fclk/8

PWM1_CLK_SEL Selects source clock frequency of PWM1

- 0** CLK = 13MHz
- 1** CLK = 32kHz

A00E0004 PWM1_COUNT_ADDR PWM1 max. counter value register

1FFF

A

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name				PWM1_COUNT												
Type				RW												
Reset				1	1	1	1	1	1	1	1	1	1	1	1	1

PWM1_COUNT

PWM1 maximum counter value. This value is the initial value of the internal counter. Regardless of the operation mode, if PWM1_COUNT is written while the internal counter is counting backwards, the new initial value will not take effect until the internal counter counts down to zero, i.e. a complete period.

A00E0008 PWM1_THRES ADDR PWM1 threshold value register 0000 H

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name				PWM1_THRES												
Type				RW												
Reset				0	0	0	0	0	0	0	0	0	0	0	0	0

PWM1_THRESH

PWM1 threshold value. When the internal counter value is bigger than or equals to PWM1_THRESH, the PWM1 output signal will be "0". When the internal counter is less than PWM1_THRESH, the PWM1 output signal will be "1".

4.2 SIM Interface

MT6250 contains two dedicated smart card interfaces that allow the MCU to access two SIM cards. Each interface can operate via 4 terminals. See Figure 25, SIMVCC, SIMRST, SIMCLK and SIMDATA are for one SIM interface, and SIM2VCC, SIM2RST, SIM2CLK and SIM2DATA are for the other.

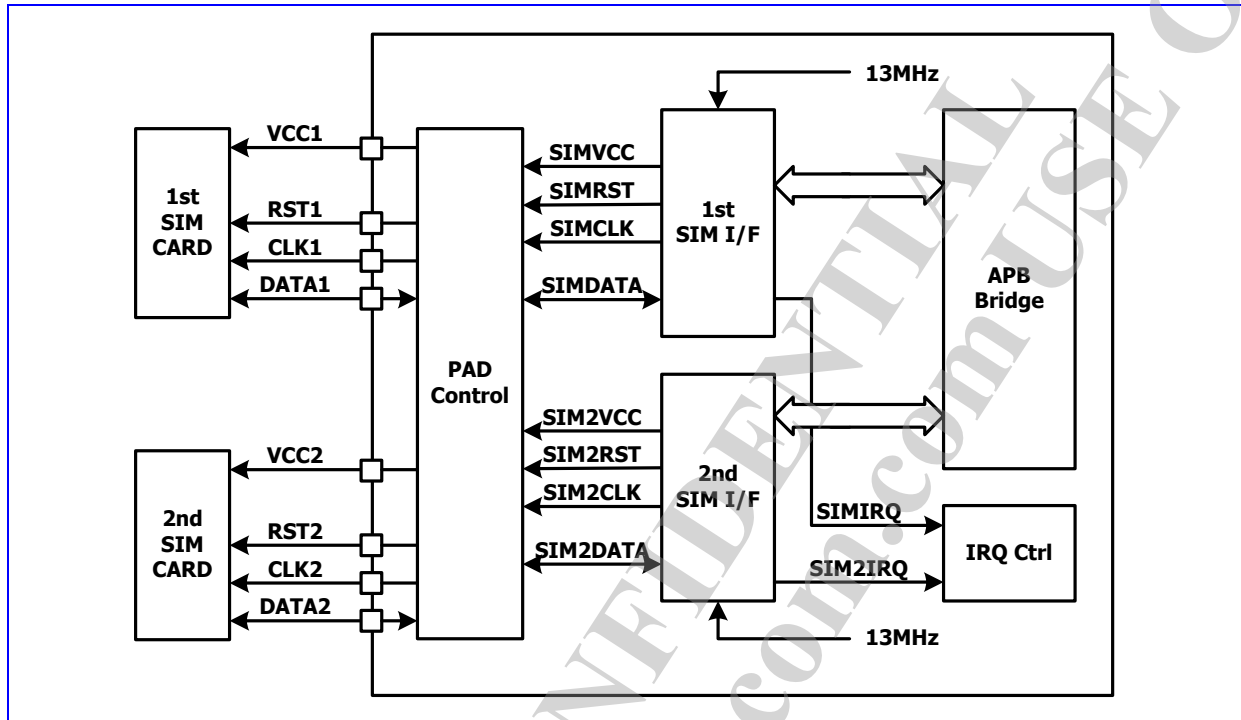


Figure 25. Block diagram of SIM interface

The functions of the two SIM interfaces are identical; therefore, only the first SIM interface will be described in this document. SIMVCC is used to control the external voltage supply to the SIM card. SIMRST is used as the SIM card reset signal. SIMDATA and SIMCLK are used for data exchange. (the control of the regulated smart card supply voltage moves from SIM interface to PMU)

The SIM interface is a half duplex asynchronous communication port, and its data format is composed of ten consecutive bits. The bits are a start bit in state “low”, eight information bits and a tenth bit used for parity checking. The data format can be divided into two modes as follows:

- Direct convention mode (ODD = SDIR = SINV = 0)

SB D0 D1 D2 D3 D4 D5 D6 D7 PB

SB: Start bit (in state “low”)

Dx: Data byte (LSB is the first and logic level ONE is in state “high”)

PB: Even parity check bit

- Inverse convention mode (ODD = SDIR = SINV = 1)

SB N7 N6 N5 N4 N3 N2 N1 N0 PB

SB: Start bit (in state “low”)

Nx: Data byte (MSB is the first and logic level ONE is in state “low”)

PB: Odd parity check bit

If the receiver obtains a wrong parity bit, it will respond by pulling the SIMDATA “low” to inform the transmitter, The transmitter will then retransmit the character.

If the receiver is an SIM card, the error response will start 0.5 bit after the PB and may last for 1 ~ 2-bit period. If the receiver is an SIM interface, the error response will start 0.5 bit after the PB and last for 1.5-bit period.

If the SIM interface is a transmitter, a total of 14 bits guard period will occur wherever the error response appears. If the receiver shows the error response, the SIM interface will retransmit the previous character again, or it will transmit the next character.

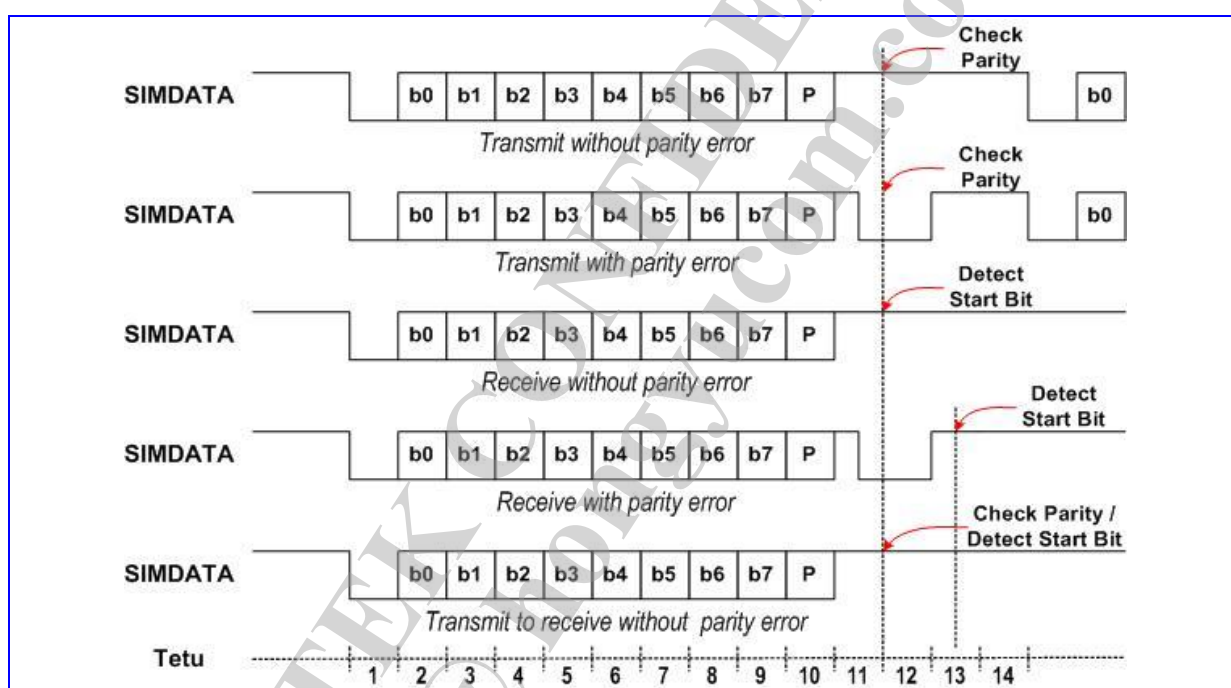


Figure 26. Timing diagram of SIM interface

4.2.1 Register Definition

When the MCU controls two SIM card interfaces, two copies of all registers are made and each copy has a different base address. n = “0” is for the first SIM card interface; n=1 is for the second SIM card interface. For example, address SIMIF0+0000h is mapped to the SIMIF0_SIM_CTRL register while address SIMIF1+0000h is mapped to the SIMIF1_SIM_CTRL register.

4.2.1.1 Register Overview

MCU register address (hex)	Acronym	Description
1 st SIM card interface		

MCU register address (hex)	Acronym	Description
SIMIF0+0000h	SIMIF0_SIM_CTRL	Control register
SIMIF0+0004h	SIMIF0_SIM_CONF	Configuration register
SIMIF0+0008h	SIMIF0_SIM_BRR	Baudrate register
SIMIF0+0010h	SIMIF0_SIM_IRQEN	Interrupt enabling register
SIMIF0+0014h	SIMIF0_SIM_STS	Status register
SIMIF0+0020h	SIMIF0_SIM_RETRY	Retry limit register
SIMIF0+0024h	SIMIF0_SIM_TIDE	FIFO tide mark register
SIMIF0+0030h	SIMIF0_SIM_DATA	TX/RX data register
SIMIF0+0034h	SIMIF0_SIM_COUNT	FIFO count register
SIMIF0+0040h	SIMIF0_SIM_ETIME	Activation time register
SIMIF0+0044h	SIMIF0_SIM_DTIME	Deactivation time register
SIMIF0+0048h	SIMIF0_SIM_TOUT	Character to character waiting time register
SIMIF0+004Ch	SIMIF0_SIM_GTIME	Block to block guard time register
SIMIF0+0050h	SIMIF0_SIM_ETIME	Block to error signal time register
SIMIF0+0054h	SIMIF0_SIM_EXT_TIME	Extend data I/O state switch time register
SIMIF0+0058h	SIMIF0_SIM_CGTIME	Character to character guard time register
SIMIF0+0060h	SIMIF0_SIM_INS	Command header register: INS
SIMIF0+0064h	SIMIF0_SIM_IMP3	Command header register: P3
SIMIF0+0068h	SIMIF0_SIM_SW1	Procedure byte register: SW1
SIMIF0+006Ch	SIMIF0_SIM_SW2	Procedure byte register: SW2
SIMIF0+0070h	SIMIF0_SIM_ATRSTA	ATR state register
SIMIF0+0074h	SIMIF0_SIM_STATUS	Protocol state register
SIMIF0+0080h	SIMIF0_SIM_DMADATA	TX/RX data register for DMA
SIMIF0+0090h	SIMIF0_SIM_DBG	Debug register
SIMIF0+0094h	SIMIF0_SIM_DBGDATA	FIFO data debug register
SIMIF0+0098h	SIMIF0_SIM_TOUT2	Character to character waiting time 2 register
SIMIF0+00A0h	SIMIF0_SIM_SCLK	SCLK PAD control register
SIMIF0+00A4h	SIMIF0_SIM_SRST	SRST PAD control register
SIMIF0+00A8h	SIMIF0_SIM_SIO	SIO PAD control register
SIMIF0+00ACh	SIMIF0_SIM_MON	PAD monitor register
2nd SIM card interface		
SIMIF1+0000h	SIMIF1_SIM_CTRL	Control register
SIMIF1+0004h	SIMIF1_SIM_CONF	Configuration register
SIMIF1+0008h	SIMIF1_SIM_BRR	Baudrate register
SIMIF1+0010h	SIMIF1_SIM_IRQEN	Interrupt enabling register
SIMIF1+0014h	SIMIF1_SIM_STS	Status register
SIMIF1+0020h	SIMIF1_SIM_RETRY	Retry limit register

MCU register address (hex)	Acronym	Description
SIMIF1+0024h	SIMIF1_SIM_TIDE	FIFO tide mark register
SIMIF1+0030h	SIMIF1_SIM_DATA	TX/RX data register
SIMIF1+0034h	SIMIF1_SIM_COUNT	FIFO count register
SIMIF1+0040h	SIMIF1_SIM_ETIME	Activation time register
SIMIF1+0044h	SIMIF1_SIM_DTIME	Deactivation time register
SIMIF1+0048h	SIMIF1_SIM_TOUT	Character to character waiting time register
SIMIF1+004Ch	SIMIF1_SIM_GTIME	Block to block guard time register
SIMIF1+0050h	SIMIF1_SIM_ETIME	Block to error signal time register
SIMIF1+0054h	SIMIF1_SIM_EXT_TIME	Extend data I/O state switch time register
SIMIF1+0058h	SIMIF1_SIM_CGTIME	Character to character guard time register
SIMIF1+0060h	SIMIF1_SIM_INS	Command header register: INS
SIMIF1+0064h	SIMIF1_SIM_IMP3	Command header register: P3
SIMIF1+0068h	SIMIF1_SIM_SW1	Procedure byte register: SW1
SIMIF1+006Ch	SIMIF1_SIM_SW2	Procedure byte register: SW2
SIMIF1+0070h	SIMIF1_SIM_ATRSTA	ATR state register
SIMIF1+0074h	SIMIF1_SIM_STATUS	Protocol state register
SIMIF1+0080h	SIMIF1_SIM_DMADATA	TX/RX data register for DMA
SIMIF1+0090h	SIMIF1_SIM_DBGD	Debug register
SIMIF1+0094h	SIMIF1_SIM_DBGDATA	FIFO data debug register
SIMIF1+0098h	SIMIF1_SIM_TOUT2	Character to character waiting time 2 register
SIMIF1+00A0h	SIMIF1_SIM_SCLK	SCLK PAD control register
SIMIF1+00A4h	SIMIF1_SIM_SRST	SRST PAD control register
SIMIF1+00A8h	SIMIF1_SIM_SIO	SIO PAD control register
SIMIF1+00ACh	SIMIF1_SIM_MON	PAD monitor register

4.2.1.2 Register Description

SIMn+0000h SIM module control register

SIMIFN_SIM_CTRL

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name										VCC CTRL	VCCL V	RSTC TRL	RSTL V	WRS T	CSTO P	SIMO N
Type										R/W	R/W	R/W	R/W	W	R/W	R/W
Reset										0	0	0	0	0	0	0

SIMON SIM card power-up/power-down control

- 0** A 1-to-0 change will start the card deactivation sequence.
- 1** A 0-to-1 change will start the card activation sequence.

- CSTOP** Enables clock stop mode. Together with CPOL in the SIM_CONF register, it determines the polarity of SIMCLK in this mode.
0 Enable SIMCLK output
1 Disable SIMCLK output
- WRST** Controls SIM card warm reset
- RSTLV** Controls SIMRST parking level in SIMRST direct control mode
- RSTCTRL** Enables SIMRST direct control mode
- VCCLV** Controls SIMVCC parking level in SIMVCC direct control mode
- VCCCTRL** Enables SIMVCC direct control mode

SIMn+0004h SIM module configuration register

SIMIFN_SIM_CONF

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name		TOUT 2		TXRD IS	RXR DIS	HFEN	T0EN	T1EN	TOUT		ODD	SDIR	SINV	CPOL	TXAC K	RXA CK
Type		R/W		R/W	R/W	R/W	R/W	R/W	R/W		R/W	R/W	R/W	R/W	R/W	R/W
Reset		0		0	0	0	0	0	0		0	0	0	0	0	0

- RXACK** Handshaking control of SIM card reception error
0 Disable character receipt handshaking
1 Enable character receipt handshaking
- TXACK** Handshaking control of SIM card transmission error
0 Disable character transmission handshaking
1 Enable character transmission handshaking
- CPOL** SIMCLK polarity control in clock stop mode
0 Make SIMCLK stop in "low" level
1 Make SIMCLK stop in "high" level
- SINV** Data invert mode
0 Does not invert the transmitted and received data. Data logic ONE is in "high" state
1 Invert the transmitted and received data. Data logic ONE is in "low" state
- SDIR** Direction of data transfer
0 LSB is transmitted and received first
1 MSB is transmitted and received first
- ODD** Selects odd or even parity
0 Even parity
1 Odd parity

- TOUT** Controls SIM work waiting time counter
0 Disable time-out counter
1 Enable time-out counter
- T1EN** Controls T = 1 protocol controller
0 Disable T = 1 protocol controller
1 Enable T = 1 protocol controller
- T0EN** Controls T = 0 protocol controller
0 Disable T = 0 protocol controller
1 Enable T = 0 protocol controller
- HFEN** Controls hardware flow
0 Disable hardware flow control
1 Enable hardware flow control
- RXRDIS** Disables Rx DMA request
0 Enable Rx DMA request (default). RXRDIS must be set to 0 for protocol T = 1
1 Disable Rx DMA request. During Tx transmission and not protocol T = 1, the recommended setting of RXRDIS is 1
- TXRDIS** Disables Tx DMA request
0 Enable Tx DMA request (default). TXRDIS must be set to 0 for protocol T = 1
1 Disable Tx DMA request. During Rx transmission and not protocol T = 1, the recommended setting of TXRDIS is 1
- TOUT2** Controls SIM work waiting time counter 2
0 Disable time-out counter 2
1 Enable time-out counter 2

SIMn+0008h SIM baudrate register
SIMIFN_SIM_BRR

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name										ETU[8:0]					SIMCLK[1:0]	
Type										R/W					R/W	
Reset										372d					01	

- SIMCLK** Sets up SIMCLK frequency
00 Reserved
01 13/4 MHz
10 13/8 MHz
11 13/12 MHz

ETU Determines the duration of elementary time unit in SIMCLK. The minimum valid setting of ETU is 8

SIMn+0010h SIM interrupt enable register

**SIMIFN_SIM_IRQ
EN**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name				TOUT 2	UDR UN	EDCE RR	T1EN D	RXER R	T0EN D	SIMO FF	ATRE RR	TXER R	TOU T	OVR UN	RXTI DE	TXTI DE
Type				R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset				0	0	0	0	0	0	0	0	0	0	0	0	0

For all the bits

- 0** Interrupt is disabled.
- 1** Interrupt is enabled.

SIMn+0014h SIM module status register

SIMIFN_SIM_STS

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name				TOUT 2	UDR UN	EDCE RR	T1EN D	RXER R	T0EN D	SIMO FF	ATRE RR	TXER R	TOU T	OVR UN	RXTI DE	TXTI DE
Type				R/C	R/C	R/C	R/C	R/C	R/C	R/C	R/C	R/C	R/C	R/C	R	R
Reset				-	-	-	-	-	-	-	-	-	-	-	-	-

- TXTIDE** The interrupt occurs when the number of transmitted data in the FIFO is less than the transmitted tide.
- RXTIDE** The interrupt occurs when the number of received data in the FIFO is less than the received tide.
- OVRUN** Receiving FIFO overflow interrupt occurs
- TOUT** Between characters time-out interrupt occurs
- TXERR** Character transmission error interrupt occurs
- ATRERR** ATR start time-out interrupt occurs
- SIMOFF** Card deactivation completed interrupt occurs.
- T0END** Data transfer handled by T = 0 controller completed interrupt occurs.
- RXERR** Character reception error interrupt occurs.
- T1END** Data transfer handled by T = 1 controller completed interrupt occurs.
- EDCERR** T = 1 controller CRC error occurs.
- UDRUN** FIFO underflow interrupt occurs (still reading FIFO when FIFO is empty).
- TOUT2** Between characters 2nd time-out interrupt occurs.

SIMn+0020h SIM retry limit register

SIMIFN_SIM_RETRY

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name						TXRETRY[2:0]								RXRETRY[2:0]		
Type						R/W								R/W		
Reset						3h								3h		

RXRETRY Specifies the maximum number of receive retries allowed when parity error occurs.

TXRETRY Specifies the maximum number of transmit retries allowed when parity error occurs.

SIMn+0024h SIM FIFO tide mark register

SIMIFN_SIM_TIDE

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name						TXTIDE[3:0]								RXTIDE[3:0]		
Type						R/W								R/W		
Reset						0h								0h		

RXTIDE Trigger point of RXTIDE interrupt

TXIDE Trigger point of TXTIDE interrupt

SIMn+0030h Data register used as Tx/Rx data register

SIMIFN_SIM_DATA

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name												DATA[7:0]				
Type												R/W				
Reset												-				

DATA Eight data digits, corresponding to the character being read or written

SIMn+0034h SIM FIFO count register

SIMIFN_SIM_COUNT

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name													COUNT[4:0]			
Type													R/W			
Reset													0h			

COUNT Number of characters in the SIM FIFO when read and flushes when written.

SIMn+0040h SIM activation time register

**SIMIFN_SIM_ATI
ME**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name							ATIME[9:0]									
Type							R/W									
Reset							2BEh									

ATIME Defines the duration, in 64 SIM clock cycles, of the time taken for each of the three stages of the card activation process, from SIMON transiting to “high” to turning on VCC, from turning on VCC to pull data “high” and then from pulling data “high” to turning on CLK.

SIMn+0044h SIM deactivation time register

**SIMIFN_SIM_DTI
ME**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name											DTIME[5:0]					
Type											R/W					
Reset											Fh					

DTIME Defines the duration, in 64 13 MHz clock cycles, of the time taken for each of the three stages of the card deactivation sequence, from pulling RST “low” to turning off CLK, from turning off CLK to pulling data “low”, from pulling data “low” to turning off VCC.

SIMn+0048h Character to character waiting time register

**SIMIFN_SIM_TOU
T**

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name											WTIME[21:0]					
Type											R/W					
Reset											260h					
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	WTIME[21:0]															
Type	R/W															
Reset	260h															

WTIME Maximum interval between the leading edge of two consecutive characters in 16 ETU units

SIMn+004Ch Block to block guard time register

SIMIFN_SIM_GTIME

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name													GTIME[3:0]			
Type													R/W			
Reset													10d			

GTIME Minimum interval between the leading edge of two consecutive characters sent in opposite directions in ETU unit

SIMn+0050h Block to error signal time register

SIMN_SIM_ETIME

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name													ETIME[5:0]			
Type													R/W			
Reset													15d			

ETIME Defines the interval, in 1/16 ETU unit, between the end of the transmitted parity bit and the time to check the parity error signal sent from SIM card.

SIMn+0054h Active high period control register

SIMIFN_SIM_EXT_TIME

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name													EXT_TIME[3:0]			
Type													R/W			
Reset													1d			

EXT_TIME Defines the interval, in 1/16 ETU unit, between the end of the transmitted parity bit and the time to switch SIO to input mode. This value should be smaller than ETIME.

SIMn+0058h Character to character guard time register

SIMIFN_SIM_CGTIME

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name													CGTIME[7:0]			
Type													R/W			
Reset													2h			

CGTIME Defines the minimum interval between the leading edges of two consecutive characters in ETU unit. In the same transmission direction, the minimum interval is (12 + CGTIME) ETU. In opposite transmission direction, the minimum interval is (12 + CGTIME + GTIME) ETU.

SIMn+0060h SIM command header register: INS

SIMIFN_SIM_INS

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name								INS								
Type								R/W								
Reset								0h								

SIMINS This field is identical to the INS instruction code. When writing to this register, the T = 0 controller will be activated and data transfer initiated.

INS Instruction direction
0 T = 0 controller receives data from the SIM card.
1 T = 0 controller sends data to the SIM card.

SIMn+0064h SIM command header register: P3

**SIMIFN_SIM_IMP3
(ICC_LEN)**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name								SIMP3[8]								
Type								R								
Reset								0h								

SIMP3 This field is identical to the P3 instruction code. It should be written prior to the SIM_INS register. When the data transfer is being conducted, this field will show the number of the remaining data to be sent or to be received.

SIMn+0068h SIM procedure byte register: SW1

**SIMIFN_SIM_SW1
(ICC_LEN)**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name																
Type																
Reset																

SIMSW1 Holds the last received procedure byte for debugging. When the T0END interrupt occurs, it will keep the SW1 procedure byte.

SIMn+006Ch SIM procedure byte register: SW2

**SIMIFN_SIM_SW2
(ICC_EDC)**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
-----	----	----	----	----	----	----	---	---	---	---	---	---	---	---	---	---

Name								SIMSW2[7:0]
Type								R
Reset								0h

SIMSW2 Holds the SW2 procedure byte

SIMn+0070h SIM ATR state register

SIMIFN_SIM_ATTRS

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name								AL	IR							OFF
Type								R	R							R
Reset								0h	0h							1h

The SIM card is initially turned off. After configuring SIMON of SIMn_SIM_CTRL and ATR procedure, SIMn_SIM_ATRSTA will set IR or AL to 1 to indicate the card's feature.

OFF Indicates On/Off of the SIM card

IR SIM card is IR (internal reset) card

AL SIM card is AL (active low reset) card

SIMn+0074h SIM protocol state register

SIMIFN_SIM_STA
TUS

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name											ALL	ONE				IDLE
Type											R	R				R
Reset											0h	0h				1h

T0 or T1 protocol of the SIM card is initially turned off. When T0 or T1 protocol is turned on, SIMn SIM T0STA will transit between ONE or ALL according to the procedure byte of the SIM card.

IDLE SIM card's T0 or T1 protocol is active or idle.

ONE SIM card will send the next byte

ALL SIM card will send all the remaining bytes.

SIMn+0080h Data register used as Tx/Rx data register

SIMIFN_SIM_DMADA
TA[illegible]

DATA Eight data digits, corresponding to the character being read or written

SIMn+0090h SIM module debug register

SIMIFN_SIM_DBG

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name					DBG7	DBG6	DBG5	DBG4								DBG3[4:0]
Type					R	R	R	R								R
Reset					0h	0h	0h	0h								0h
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name																DBG1[4:0]
Type																R
Reset																0h

DBG1 Debugging register 1
DBG2 Debugging register 2
DBG3 Debugging register 3
DBG4 Debugging register 4
DBG5 Debugging register 5
DBG6 Debugging register 6
DBG7 Debugging register 7

SIMn+0094h SIM FIFO data debug register

SIMIFN_SIM_DBGDATA

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name																DBGDATA[7:0]
Type																R
Reset																0h

DBGDATA FIFO data debug register. There is no influence on data transmission when reading this register

DBGRPTR FIFO read pointer related to DBGDATA. Automatically increased by 1 after this register is read.

SIMn+0098h Character to character waiting time 2 register

SIMIFN_SIM_TOUT2

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																WTIME2[21:0]
Type																R/W

Reset																260h
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	WTIME2[21:0]															
Type	R/W															
Reset	260h															

WTIME2 Maximum interval between the leading edge of two consecutive characters in 16 ETU units

SIMn+00Ah **SIM SCLK PAD control register** **SIMIFN_SIM_SCLK**

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name															DEBU G	ACD_FU NC
Type															R/W	R/W
Reset															0h	0h
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	IES_CTR L	IES_LV			TDSEL[1:0]		RDSEL[1:0]	R1	R0	PUPD	SMT	E4	E2		SR[1:0]	
Type	R/W	R/W			R/W		R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Reset	0h	0h			0h		0h	0h	0h	0h	1h	0h	1h		3h	

SR Output slew rate control. High asserted. SR = 1, slower slew. SR = 0, no slew rate control. For SIM card mode, SR[1:0] = [1 1] is the recommended setting to eliminate overshooting/undershooting. For non-SIM card mode, SR[1:0] = [0 0] is set for the best speed.

E2 Tx driving strength control. For SIM card mode, E2 = [1] is the recommended setting for SCLK/SRST/SIO. (SIO/SRST can use [0])

E4 Tx driving strength control. For SIM card mode, E4 = [0] is the recommended setting for SCLK/SRST/SIO.

SMT Enables Rx input buffer Schmit trigger hysteresis control. High asserted. SMT = 1, Schmit trigger enable. For SIM card mode, SMT = [1] is the recommended setting.

PUPD Weak pull-up/pull-down control
0 Pull-up
1 Pull-down

R0 Selects weak pull-up/pull-down resistance. Refer to the table in register "R1".

R1 Selects weak pull-up/pull-down resistance. Refer to the following table.

E	PUPD	R1	R0	R Value
0	0	0	0	High – Z
0	0	0	1	PU – 20k
0	0	1	0	PU – 5k
0	0	1	1	PU – 4k
0	1	0	0	High – Z
0	1	0	1	PD – 75k
0	1	1	0	PD – 75k
0	1	1	1	PD – 37.5k
1	x	x	x	High - Z

RDSEL Selects Rx duty

RDSEL[0]: Input buffer duty high when asserted (high pulse width adjustment)

RDSEL[1]: Input buffer duty low when asserted (low pulse width adjustment)

For SIM card mode, RDSEL = [0 0] is the recommended setting.

TDSEL Selects Tx duty

TDSEL[0]: Output level shifter duty high when asserted (high pulse width adjustment)

TDSEL[1]: Output level shifter duty low when asserted (low pulse width adjustment)

For SIM card mode, TDSEL = [0 0] is the recommended setting.

IES_LV Controls IES (Rx input buffer enabling) parking level in IES direct control mode

High asserted. Datapath: From IO to O. IES = 0, O = 0.

In the quiescent mode, IES = 0 is suggested for power saving.

IES_CTRL Enables IES direct control mode

ACD_FUNC ACD function mode for analog designer

DEBUG Output PAD related signals for monitoring

0 Disable

1 Enable

SIMn+00A4h **SIM SRST PAD control register**

SIMIFN_SIM_SRST

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	IES_CTRL	IES_LV			TDSEL[1:0]		RDSEL[1:0]		R1	R0	PUPD	SMT	E4	E2	SR[1:0]	
Type	R/W	R/W			R/W		R/W		R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Reset	0h	0h			0h		0h		0h	0h	0h	1h	0h	1h	3h	

SR Output slew rate control. High asserted. SR = 1, slower slew. SR = 0, no slew rate control.

For SIM card mode, SR[1:0] = [1 1] is the recommended setting to eliminate

overshooting/undershooting. For non-SIM card mode, SR[1:0] = [0 0] is set for the best speed.

- E2** Tx driving strength control. For SIM card mode, E2 = [1] is the recommended setting for SCLK/SRST/SIO. (SIO/SRST can use [0])
- E4** Tx driving strength control. For SIM card mode, E4 = [0] is the recommended setting for SCLK/SRST/SIO.
- SMT** Enables Rx input buffer Schmit trigger hysteresis control. High asserted. SMT = 1, Schmit trigger enable. For SIM card mode, SMT = [1] is the recommended setting.
- PUPD** Weak pull-up/pull-down control
0 Pull-up
1 Pull-down
- R0** Selects weak pull-up/pull-down resistance. Refer to the table in register "R1".
- R1** Selects weak pull-up/pull-down resistance. Refer to the following table.

E	PUPD	R1	R0	R Value
0	0	0	0	High – Z
0	0	0	1	PU – 20k
0	0	1	0	PU – 5k
0	0	1	1	PU – 4k
0	1	0	0	High – Z
0	1	0	1	PD – 75k
0	1	1	0	PD – 75k
0	1	1	1	PD – 37.5k
1	x	x	x	High - Z

- RDSEL** Selects Rx duty
RDSEL[0]: Input buffer duty high when asserted (high pulse width adjustment)
RDSEL[1]: Input buffer duty low when asserted (low pulse width adjustment)
For SIM card mode, RDSEL = [0 0] is the recommended setting.
- TDSEL** Selects Tx duty
TDSEL[0]: Output level shifter duty high when asserted (high pulse width adjustment)
TDSEL[1]: Output level shifter duty low when asserted (low pulse width adjustment)
For SIM card mode, TDSEL = [0 0] is the recommended setting.
- IES_LV** Controls IES (Rx input buffer enabling) parking level in IES direct control mode
High asserted. Datapath: From IO to O. IES = 0, O = 0.
In the quiescent mode, IES = 0 is suggested for power saving.

IES_CTRL Enables IES direct control mode

SIMn+00A8h SIM SIO PAD control register

SIMIFN_SIM_SIO

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	IES_CTRL	IES_LV			TDSEL[1:0]		RDSEL[1:0]		R1	R0	PUPD	SMT	E4	E2	SR[1:0]	
Type	R/W	R/W			R/W		R/W		R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0h	0h			0h		0h		1h	0h	0h	1h	0h	1h		3h

SR Output slew rate control. High asserted. SR = 1, slower slew. SR = 0, no slew rate control. For SIM card mode, SR[1:0] = [1 1] is the recommended setting to eliminate overshooting/undershooting. For non-SIM card mode, SR[1:0] = [0 0] is set for the best speed.

E2 Tx driving strength control. For SIM card mode, E2 = [1] is the recommended setting for SCLK/SRST/SIO. (SIO/SRST can use [0])

E4 Tx driving strength control. For SIM card mode, E4 = [0] is the recommended setting for SCLK/SRST/SIO.

SMT Enables Rx input buffer Schmit trigger hysteresis control. High asserted. SMT = 1, Schmit trigger enable. For SIM card mode, SMT = [1] is the recommended setting.

PUPD Weak pull-up/pull-down control
0 Pull-up
1 Pull-down

R0 Selects weak pull-up/pull-down resistance. Refer to the table in register "R1".

R1 Selects weak pull-up/pull-down resistance. Refer to the following table. For SIO, [R1 R0] = [1 0] is the recommended setting for 5k weak pull-up. In 4 SIM application and SIO is connected to the external SIM switch, please disable the pull-up resistance. ([R1 R0] = [0 0])

E	PUPD	R1	R0	R Value
0	0	0	0	High – Z
0	0	0	1	PU – 20k
0	0	1	0	PU – 5k
0	0	1	1	PU – 4k
0	1	0	0	High – Z
0	1	0	1	PD – 75k
0	1	1	0	PD – 75k

E	PUPD	R1	R0	R Value
0	1	1	1	PD – 37.5k
1	x	x	x	High - Z

RDSEL Selects Rx duty

RDSEL[0]: Input buffer duty high when asserted (high pulse width adjustment)

RDSEL[1]: Input buffer duty low when asserted (low pulse width adjustment)

For SIM card mode, RDSEL = [0 0] is the recommended setting.

TDSEL Selects Tx duty

TDSEL[0]: Output level shifter duty high when asserted (high pulse width adjustment)

TDSEL[1]: Output level shifter duty low when asserted (low pulse width adjustment)

For SIM card mode, TDSEL = [0 0] is the recommended setting.

IES_LV Controls IES (Rx input buffer enabling) parking level in IES direct control mode

High asserted. Datapath: From IO to O. IES = 0, O = 0.

In the quiescent mode, IES = 0 is suggested for power saving.

IES_CTRL Enables IES direct control mode

SIMn **SIM monitor register** **SIMIFN_SIM_MON**
+00ACH **N**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name		MON 12	MON 11	MON 10		MON 9	MON 8	MON 7		MON 6	MON 5	MON 4		MON 3	MON 2	MON 1
Type		R	R	R		R	R	R		R	R	R		R	R	R
Reset		0h	0h	0h		0h	0h	0h		0h	0h	0h		0h	0h	0h

MON1 Monitor signal 1

MON2 Monitor signal 2

MON3 Monitor signal 3

MON4 Monitor signal 4

MON5 Monitor signal 5

MON6 Monitor signal 6

MON7 Monitor signal 7

MON8 Monitor signal 8

MON9 Monitor signal 9

MON10 Monitor signal 10

MON11 Monitor signal 11

MON12 Monitor signal 12

4.2.2 SIM Card Insertion and Removal

The detection of physical connection to the SIM card and card removal can be done by the external interrupt controller or by GPIO.

4.2.3 Card Activation and Deactivation

The card activation and deactivation sequence are both controlled by hardware. The MCU initiates the activation sequence by writing "1" to bit 0 of the SIM_CTRL register, and then the interface performs the following activation sequence:

- Assert SIMRST "low".
- Set SIMVCC at "high" level and SIMDATA in the reception mode.
- Enable SIMCLK clock.
- De-assert SIMRST "high" (required if it belongs to active low reset SIM card).

The final step in a typical card session is contacting deactivation in case the card will be electrically damaged. The deactivation sequence is initiated by writing "0" to bit 0 of the SIM_CTRL register, and the interface will perform the following deactivation sequence:

- Assert SIMRST "low".
- Set SCIMCLK at "low" level.
- Set SIMDATA at "low" level.
- Set SIMVCC at "low" level.

4.2.4 Answering to Reset Sequence

After the card is activated, a reset operation will result in an answer from the card consisting of the initial character TS, followed by maximum 32 characters. The initial character TS provides a bit synchronization sequence and defines the conventions to interpret data bytes in all subsequent characters.

On reception of the first character, TS, the MCU should read this character, establish the respective required convention and re-program the related registers. These processes should be completed prior to the completion of reception of the next character. Next, the remainder of the ATR sequence will be received, read via the SIM_DATA in the selected convention and interpreted by the software.

The timing requirement and procedures for ATR sequence are handled by hardware and shall meet the requirement of ISO 7816-3, as shown in Figure 27.

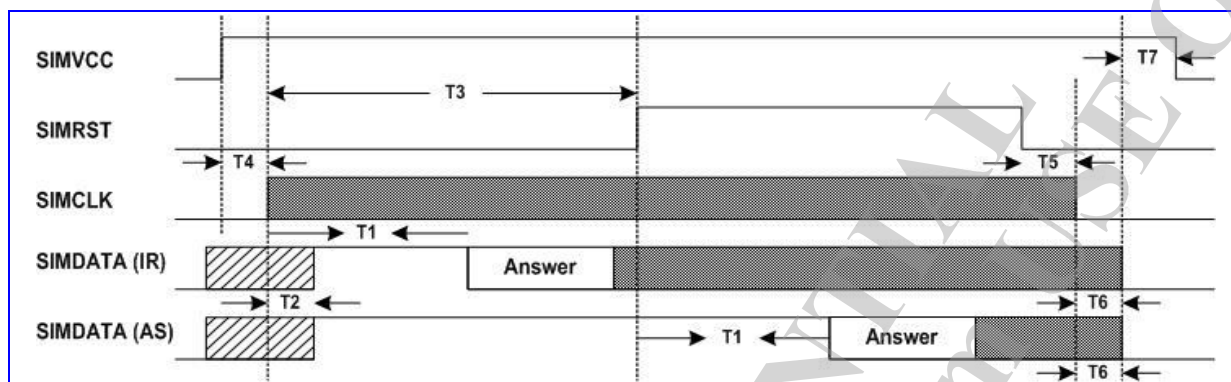


Figure 27. Answering to reset sequence

Time	Value	Comment
T1	> 400 SIMCLK	SIMCLK start to ATR appears
T2	< 200 SIMCLK	SIMCLK start to SIMDATA in reception mode
T3	> 40,000 SIMCLK	SIMCLK start to SIMRST "high"
T4	-	SIMVCC "high" to SIMCLK start
T5	-	SIMRST "low" to SIMCLK stop
T6	-	SIMCLK stop to SIMDATA "low"
T7	-	SIMDATA "low" to SIMVCC "low"

Table 47. Time-out condition for answering to reset sequence

4.2.5 SIM Data Transfer

There are two transfer modes provided, in software controlled byte by byte fashion or in a block fashion using T=0 controller and DMA controller. In both modes, the time-out counter can be enabled to monitor the elapsed time between two consecutive bytes.

4.2.5.1 Byte Transfer Mode

This mode is used during the ATR and PPS procedure. In this mode, the SIM interface only ensures error free character transmission and reception.

Receiving characters

Upon detection of the start-bit sent by the SIM card, the interface transforms into the reception mode, and the following bits are shifted into an internal register. If no parity error is detected or the character-received handshaking is disabled, the received-character will be written into the SIM FIFO and the SIM_COUNT register increased by one. Otherwise, the SIMDATA line will be held "low" at 0.5 etu after detecting the parity error for 1.5 etus, and the character will be re-received. If a character fails to be received correctly for the RXRETRY times, the receive-handshaking will be aborted, the

last-received character written into the SIM FIFO, the SIM_COUNT increased by one and the RXERR interrupt generated.

When the number of characters held in the received FIFO exceeds the level defined in the SIM_TIDE register, a RXTIDE interrupt will be generated. The number of characters held in the SIM FIFO can be determined by reading the SIM_COUNT register, and writing to this register will flush the SIM FIFO.

Sending characters

Characters that are to be sent to the card are first written into the SIM FIFO and then automatically transmitted to the card at timed intervals. If character-transmitted handshaking is enabled, the SIMDATA line will be sampled at 1 etu after the parity bit. If the card indicates that it does not receive the character correctly, the character will be re-transmitted for maximum TXRETRY times before a TXERR interrupt is generated and the transmission is aborted. Otherwise, the succeeding byte in the SIM FIFO will be transmitted.

If a character fails to be transmitted and a TXERR interrupt is generated, the interface will need to be reset by flushing the SIM FIFO before any subsequent transmission or reception operation.

When the number of characters held in the SIM FIFO falls below the level defined in the SIM_TIDE register, a TXTIDE interrupt will be generated. The number of characters held in the SIM FIFO can be determined by reading the SIM_COUNT register, and writing to this register will flush the SIM FIFO.

4.2.5.2 Block Transfer Mode

Basically, the SIM interface is designed to work in conjunction with the T=0 protocol controller and the DMA controller during non-ATR and non-PPS phase, though it is still possible for software to service the data transfer manually as in the byte transfer mode if necessary. Thus the T=0 protocol should be controlled by software.

The T=0 controller can be accessed via four registers representing the instruction header bytes INS and P3, and the procedure bytes SW1 and SW2. The registers are:

- SIM_INS, SIM_P3
- SIM_SW1, SIM_SW2

During the character transfer, SIM_P3 holds the number of characters to be sent or to be received, and SIM_SW1 holds the last received procedure byte including NULL, ACK, NACK and SW1 for debugging.

Data receiving instruction

Data receiving instructions receive data from the SIM card. See the following instantiated procedure.

1. Enable the T=0 protocol controller by setting the T0EN bit to 1 in the SIM_CONF register.
2. Program the SIM_TIDE register to 0x0000 (TXTIDE = 0, RXTIDE = 0).

3. Program the SIM_IRQEN to 0x019C (enable RXERR, TXERR, T0END, TOUT and OVRUN interrupts).
4. Write CLA, INS, P1, P2 and P3 into SIM FIFO.
5. Program the DMA controller:
 - DMA_n_MSBSRC and DMA_n_LSBSRC: Address of the SIM_DATA register
 - DMA_n_MSBDST and DMA_n_LSBDST: Memory address reserved to store the received characters
 - DMA_n_COUNT: Identical to P3 or 256 (if P3 = 0)
 - DMA_n_CON: 0x0078
6. Write P3 into the SIM_P3 register and then INS into SIM_INS register. (Data transfer is initiated now.)
7. Enable the time-out counter by setting the TOUT bit to 1 in the SIM_CONF register.
8. Start the DMA controller by writing 0x8000 into the DMA_n_START register.

Upon completion of the data receiving instruction, the T0END interrupt will be generated, and the time-out counter should be disabled by setting the TOUT bit to 0 in the SIM_CONF register.

If error occurs during data transfer (RXERR, TXERR, OVRUN or TOUT interrupt is generated), the SIM card should be deactivated first and then activate the prior subsequent operation.

Data sending instruction

Data sending instructions send data to the SIM card. See the following instantiated procedure.

1. Enable the T=0 protocol controller by setting the T0EN bit to 1 in the SIM_CONF register.
2. Program the SIM_TIDE register to 0x0100 (TXTIDE = 1, RXTIDE = 0)
3. Program the SIM_IRQEN to 0x019C (enable RXERR, TXERR, T0END, TOUT and OVRUN interrupts)
4. Write CLA, INS, P1, P2 and P3 into SIM FIFO
5. Program the DMA controller:
 - DMA_n_MSBSRC and DMA_n_LSBSRC: Memory address reserved to store the transmitted characters
 - DMA_n_MSBDST and DMA_n_LSBDST: Address of the SIM_DATA register
 - DMA_n_COUNT: Identical to P3
 - DMA_n_CON: 0x0074
6. Write P3 into the SIM_P3 register and then (0x0100 | INS) into SIM_INS register. (Data transfer is initiated now.)
7. Enable the time-out counter by setting the TOUT bit to 1 in the SIM_CONF register.
8. Start the DMA controller by writing 0x8000 into the DMA_n_START register.

Upon completion of the data sending instruction, the T0END interrupt will be generated, and the time-out counter should be disabled by setting the TOUT bit back to 0 in the SIM_CONF register.

If error occurs during the data transfer (RXERR, TXERR, OVRUN or TOUT interrupt is generated), the SIM card should be deactivated first and then activate the prior subsequent operation.

4.2.6 SIM IO PAD Electrical Characteristics

SIM card interface (SIM interface): Provides level shifting needs for low-voltage GSM controller

There are two SIM card interface modules to support two SIM cards simultaneously. The SIM card interface circuitry of PMU meets all ETSI and IMT-2000 SIM interface requirements. It provides level shifting for low-voltage GSM controller to communicate with either 1.8V or 3V SIM cards. All SIM cards contain a clock input, a reset input and a bi-directional data input/output. The clock and reset inputs to SIM cards are level shifted from the supply of digital IO (Vio) of baseband to the SIM supply (Vsim). The bi-directional data bus is internally pulled high to Vsim via the 5KΩ resistor.

The second SIM card interface can be used for supporting another SIM card or mobile TV. The interface pins such as SIO2, SRST2 and SCLK2 can be configured as GPIO when there is no need to use the second SIM card interface.

All pins that are connected to the SIM card (Vsim, SRST, SCLK, SIO) withstand over 2kV HBM (human body mode) ESD. In order to ensure proper ESD protection, careful board layout is required.

4.2.6.1 Block Descriptions

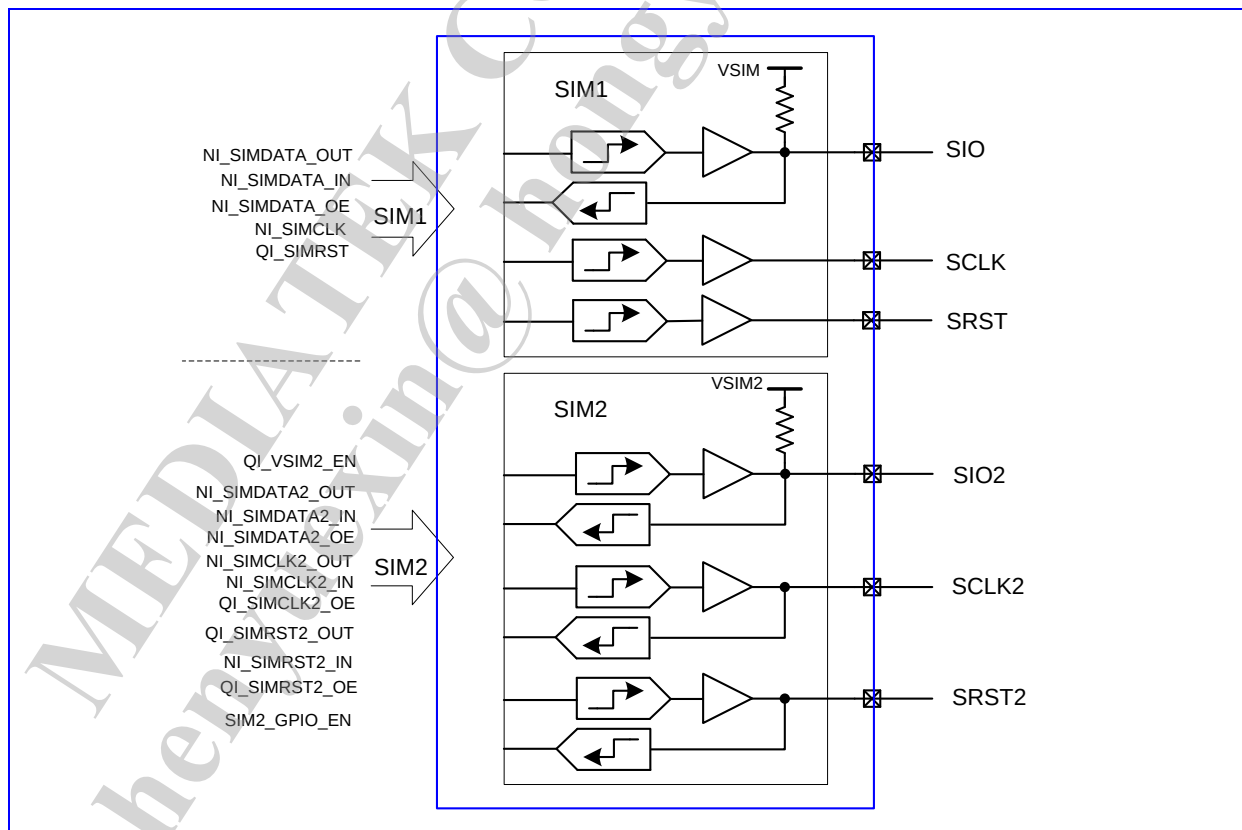


Figure 28. SIM interface block diagram

4.2.6.2 Functional Specifications

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
Interface to 3V SIM card						
	Output low of SRST	I = -200μA			0.36	V
	Output high of SRST	I = 200μA	0.9*VSIM			V
	Output low of SCLK	I = -100μA			0.4	V
	Output high of SCLK	I = 200μA	0.9*VSIM			V
	Input/Output low of SIO	I = -1mA			0.4	V
	Input/Output high of SIO	I = +20μA	VSIM-0.4			V
	(lii) Pull high current of SIO	Vil = 0V			+1	mA
Interface to 1.8V SIM card						
	Output low of SRST	I = -200μA			0.12*VSIM	V
	Output high of SRST	I = +200μA	0.9*VSIM			V
	Output low of SCLK	I = -100μA			0.2*VSIM	V
	Output high of SCLK	I = 200μA	0.9*VSIM			V
	Input/Output low of SIO	I = -1mA			0.15*VSIM	V
	Input/Output high of SIO	I = +20μA	VSIM-0.4			V
	(lii) Pull high current of SIO	Vil = 0V			+1	mA
	SIO pull-up resistance to VSIM		4	5	6	kΩ
SIM card interface timing						
	SRST, SIO rise/fall times	VSIM = 3, 1.8V, load with 30pF			1	μs
	SCLK rise/fall times	VSIM = 3V, CLK load with 30pF			18	ns
		VSIM = 1.8V, CLK load with 30pF			50	ns
	SCLK frequency	CLK load with 30pF			5	MHz
	SCLK duty cycle	SIMCLK Duty = 50%, fsmclk = 5MHz	47		53	%

Table 48. SIM interface specifications

4.3 Keypad Scanner

4.3.1 General Description

The keypad supports two keypad modes: 8*8 normal keypad mode and 5*5 extended keypad mode.

The 8*8 normal keypad mode can be divided into two parts: One is the keypad interface including 8 columns and 8 rows with one dedicated power-key (see Figure 29); the other is the key detection block which provides key pressed, key released and de-bounce mechanisms.

Each time the key is pressed or released, i.e. something different in the 8x8 matrix or power-key, the key detection block senses the change and identifies if a key has been pressed or released. Whenever the key status changes and is stable, a KEYPAD IRQ is issued. The MCU can then read the key(s) pressed directly in the KP_MEM1, KP_MEM2, KP_MEM3, KP_MEM4 and KP_MEM5 (only 8*8 is needed) registers. To ensure that the key pressed information is not missed, the status register in the keypad is not read-cleared by the APB read command. The status register can only be changed by the key-pressed detection FSM.

This keypad detects one or two key-pressed simultaneously with any combination. Figure 31 shows the scan waveform of the 8*8 normal keypad mode. Figure 33 shows one key pressed condition. Figure 34(a) and (b) illustrate two keys pressed cases. Since the key press detection depends on the HIGH or LOW level of the external keypad interface, if keys are pressed at the same time and there exists a key that is on the same column and the same row with the other keys, the pressed key cannot be correctly decoded. For example, if there are three keys pressed: key1 = (x1, y1), key2 = (x2, y2), and key3 = (x1, y2), then both key3 and key4 = (x2, y1) will be detected, and therefore they cannot be distinguished correctly. Hence, the keypad can detect only one or two keys pressed simultaneously at any combination. More than two keys pressed simultaneously in a specific pattern will retrieve the wrong information.

The 5*5 extended keypad mode supports $5*5*2 = 50$ keys matrix and one dedicated power key. The 50 keys are divided into 25 sub groups, and each group consists of 2 keys and a 20Kohm resistor. The power key is connected with PMIC, and PMIC is connected with the PWR_KEY pad of the Baseband chip. Figure 32 shows the scan waveform of the 5*5 extended keypad mode. Besides the limitation of the 8*8 normal keypad mode, the 5*5 extended keypad mode has another limitation which is it cannot detect two keys pressed simultaneously when the two keys are in one group, i.e. the 5*5 extended keypad mode cannot detect key0 and key1 pressed simultaneously or key16 and key17 pressed simultaneously.

The key order number of the two modes in COL/ROW matrix is showed in Table 49 and Table 50.

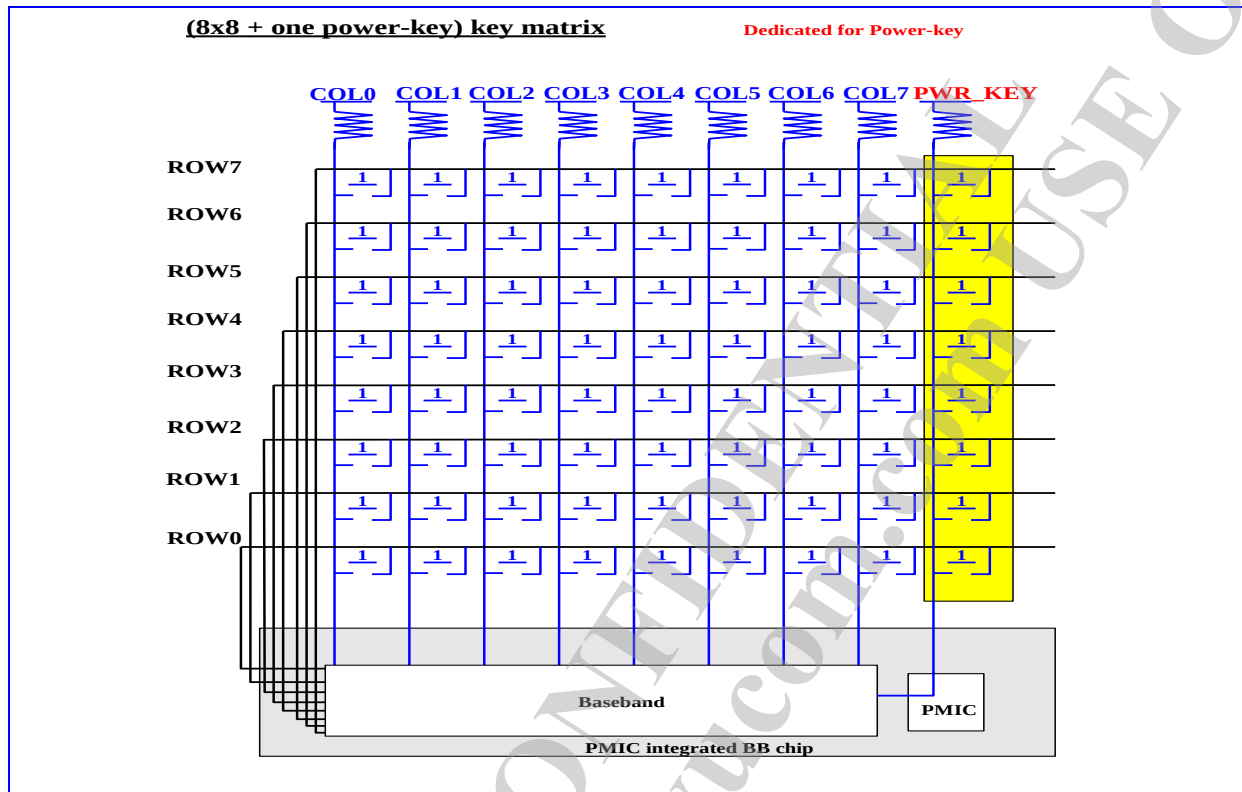


Figure 29. 8x8 normal keypad mode matrix (64 keys and one dedicated power-key)

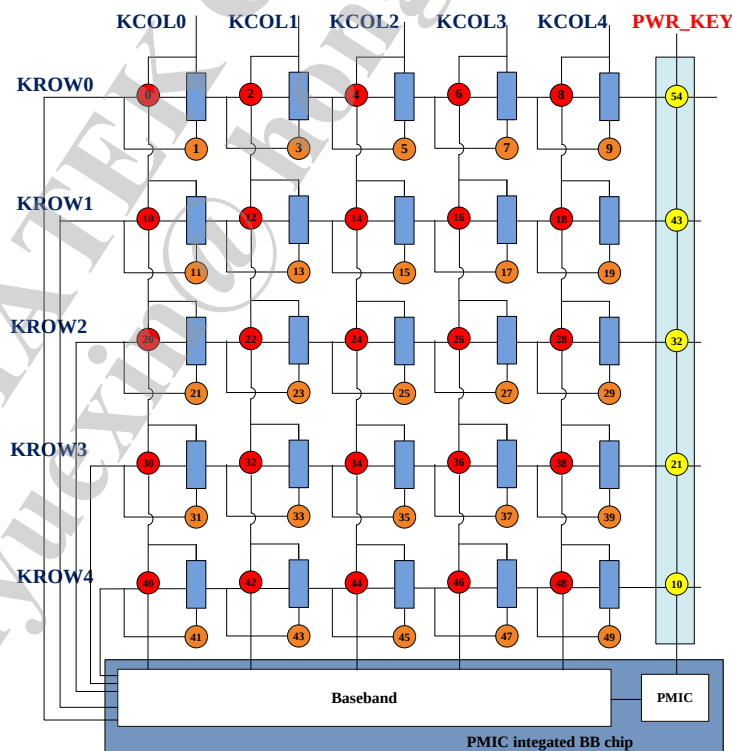


Figure 30. Extended keypad mode matrix (50 keys and one dedicated power key)

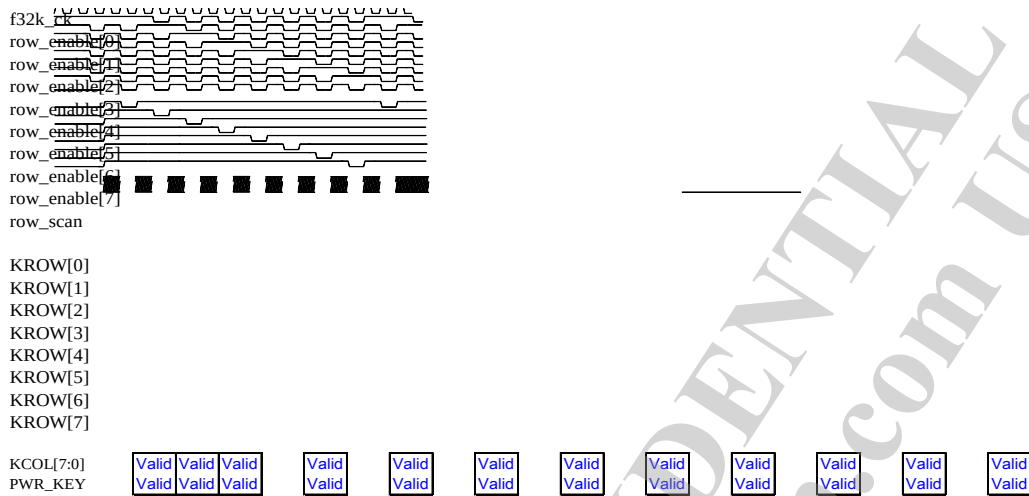


Figure 31. 8x8 normal keypad mode scan waveform

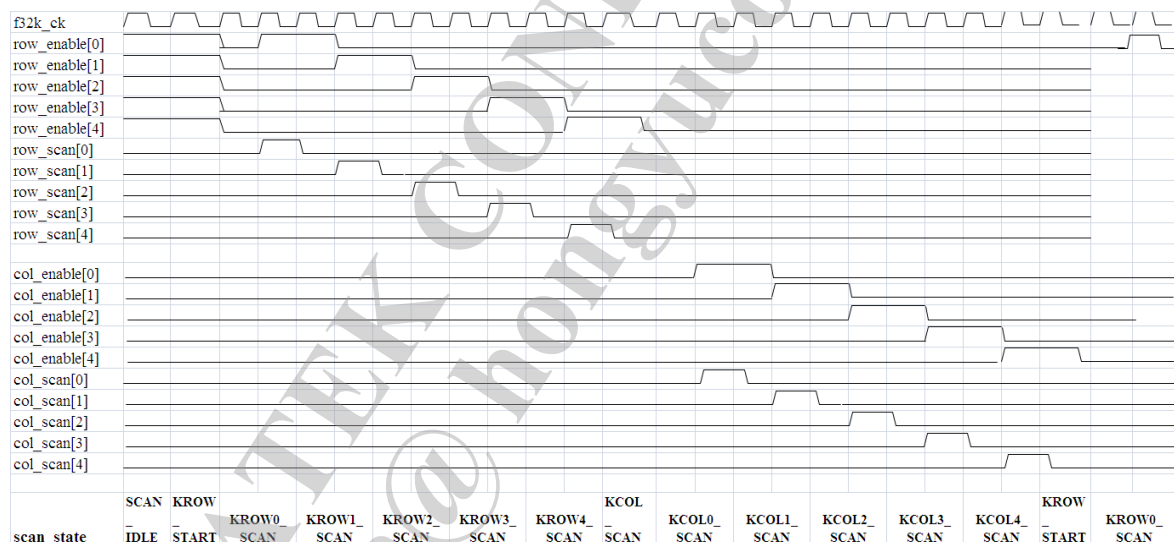


Figure 32. 5*5 extended keypad mode scan waveform

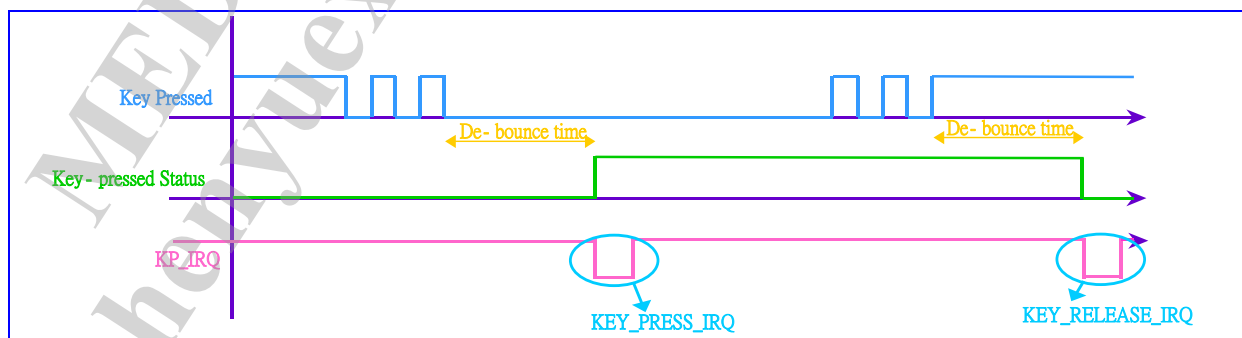


Figure 33. One key pressed with de-bounce mechanism denoted

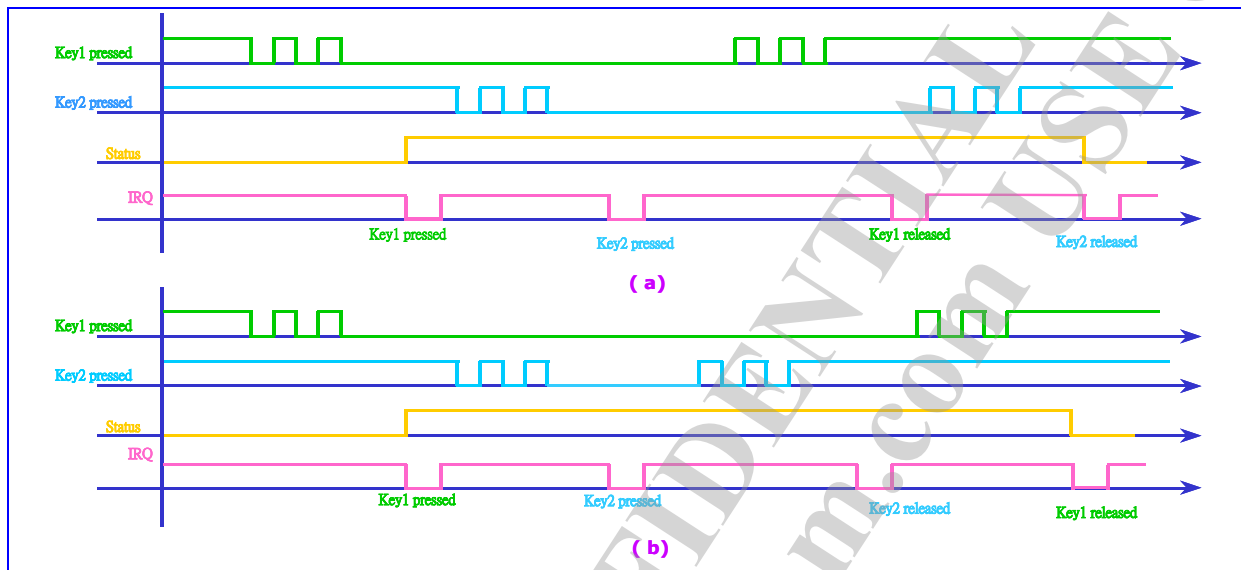


Figure 34. (a) Two keys pressed, case 1 (b): Two keys pressed, case 2

4.3.2 Register Definition

KP+0000h Keypad status															KP_STA	
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name																STA
Type																RO
Reset																0

STA Indicates the keypad status. The register is not cleared by the read operation.

- 0 No key pressed
- 1 Key pressed

KP+0004h Keypad scanning output register															KP_MEM1	
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	KEY 15	KEY 14	KEY 13	KEY 12	KEY 11	KEY 10	KEY 9	KEY 8	KEY 7	KEY 6	KEY 5	KEY 4	KEY 3	KEY 2	KEY 1	KEY 0
Type	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

The register shows the key-press status of key0 (LSB) ~ key15. Refer to Table 49 and Table 50.

KP+0008h Keypad scanning output register

KP_MEM2

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	KEY 31	KEY 30	KEY 29	KEY 28	KEY 27	KEY 26	KEY 25	KEY 24	KEY 23	KEY 22	KEY 21	KEY 20	KEY 19	KEY 18	KEY 17	KEY 16
Type	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

The register shows the key-press status of key16 (LSB) ~ key31. Refer to Table 49 and Table 50.

KP+000Ch Keypad scanning output register

KP_MEM3

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	KEY 47	KEY 46	KEY 45	KEY 44	KEY 43	KEY 42	KEY 41	KEY 40	KEY 39	KEY 38	KEY 37	KEY 36	KEY 35	KEY 34	KEY 33	KEY 32
Type	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

The register shows the key-press status of key32 (LSB) ~ key47. Refer to Table 49 and Table 50.

KP+0010h Keypad scanning output register

KP_MEM4

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	KEY 63	KEY 62	KEY 61	KEY 60	KEY 59	KEY 58	KEY 57	KEY 56	KEY 55	KEY 54	KEY 53	KEY 52	KEY 51	KEY 50	KEY 49	KEY 48
Type	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

The register shows the key-press status of key48 (LSB) ~ key63. Refer to Table 49 and Table 50.

KP+0014h Keypad scanning output register

KP_MEM5

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name									KEY 71	KEY 70	KEY 69	KEY 68	KEY 67	KEY 66	KEY 65	KEY 64
Type									RO	RO	RO	RO	RO	RO	RO	RO
Reset									1	1	1	1	1	1	1	1

The register shows the key-press status of key64 (LSB) ~ key71. Refer to Table 49.

For the 8*8 normal keypad mode, the five registers list the status of the 72 keys on the keypad but KEY[8], KEY[17], KEY[26], KEY[35], KEY[44], KEY[53], KEY[62] and KEY[71] are dedicated for the power key. For the 5*5 extended keypad mode, KP_MEM1 ~ 4 registers list the status of the 55 keys

on the keypad. When the MCU receives KEYPAD IRQ, both KP_STA and KP_MEM1~5 registers must be read. If any key is pressed, the relative bit will be set to 0.

In order to work normally, all COLs must be pulled up. If some keys cannot be used because their COL or ROW is used as GPIO, the corresponding COL/ROW must be tied to high.

KEYS Status list of the 72 keys.

KP+00018h De-bounce period setting

KP_DEBOUNCE

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name			DEBOUNCE [13:0]													
Type			R/W													
Reset			400h													

This register defines the waiting period before key press or release events are considered stale. If the de-bounce setting is too small, the keypad will be too sensitive and detect too many unexpected key-presses. The suitable debounce time setting must be adjustable for the user's demand.

DEBOUNCE De-bounce time = KP_DEBOUNCE/32 ms.

KP+001CH Keypad scan timing adjustment register

KP_SCAN_TIMING

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	COL_HIGH_PULSE				ROW_HIGH_PULSE				COL_SCAN_DIV				ROW_SCAN_DIV			
Type	R/W				R/W				R/W				R/W			
Reset	4'h0				4'h0				4'h1				4'h1			

This register is used to set up the scan timing for the 5*5 extended keypad mode.

As Figure 35 shows, ROW_HIGH_PULSE/COL_HIGH_PULSE are used to set up the scan high cycle, default 0 means there is 1 cycle when row/col output high.

ROW_SCAN_DIV/COL_SCAN_DIV are used to set up the whole scan cycle which includes ROW_HIGH_PULSE/COL_HIGH_PULSE and scan interval cycles, default 1 means there is 2 cycles for each scan, including 1 cycle high pulse and 1 cycle interval.

Note: ROW_SCAN_DIV > ROW_HIGH_PULSE and COL_SCAN_DIV > COL_HIGH_PULSE.

ROW_SCAN_DIV/COL_SCAN_DIV are used to lower the power consumption for it decreases the actual scan number in the de-bounce time. It is recommend that

ROW_HIGH_PULSE/COL_HIGH_PULSE set to 0, and you can adjust ROW_SCAN_DIV/COL_SCAN_DIV when debugging.

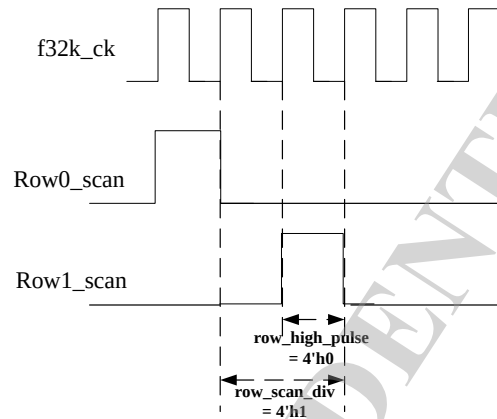


Figure 35. 5*5 extended keypad mode scan timing

KP+0020h Keypad selection register

KP_SEL

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name																KPSEL
Type																R/W
Reset																0

KPSEL Selects the 8*8 normal keypad mode or the 5*5 extended keypad mode

- 0** Use the 8*8 normal keypad mode
- 1** Use the 5*5 extended keypad mode

KP+0024h Keypad enable register

KP_EN

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name																KPEN
Type																R/W
Reset																1

KPEN Enables/Disables keypad

- 0** Disable keypad (both of the two keypad modes will not work)
- 1** Enable keypad (only one of the two keypad modes can work)

	COL0	COL1	COL2	COL3	COL4	COL5	COL6	COL7	PWRKEY
ROW7	63	64	65	66	67	68	69	70	71
ROW6	54	55	56	57	58	59	60	61	62
ROW5	45	46	47	48	49	50	51	52	53

	COL0	COL1	COL2	COL3	COL4	COL5	COL6	COL7	PWRKEY
ROW4	36	37	38	39	40	41	42	43	44
ROW3	27	28	29	30	31	32	33	34	35
ROW2	18	19	20	21	22	23	24	25	26
ROW1	9	10	11	12	13	14	15	16	17
ROW0	0	1	2	3	4	5	6	7	8

Table 49. 8*8 normal keypad mode KEY's order number in COL/ROW matrix

	COL0	COL1	COL2	COL3	COL4	PWRKEY
ROW4	44/45	46/47	48/49	50/51	52/53	54
ROW3	33/34	35/36	37/38	39/40	41/42	43
ROW2	22/23	24/25	26/27	28/29	30/31	32
ROW1	11/12	13/14	15/16	17/18	19/20	21
ROW0	0/1	2/3	4/5	6/7	8/9	10

Table 50. 5*5 extended keypad mode KEY's order number in COL/ROW matrix

4.4 General Purpose Inputs/Outputs

MT6250 offers 107 general purpose I/O pins. By setting up the control registers, the MCU software can control the direction, the output value, and read the input values on these pins. These GPIOs and GPOs are multiplexed with other functions to reduce the pin count. In addition, all GPO pins are removed. To facilitate application use, the software can configure which clock to send outside the chip. There are 8 clock-out ports embedded in 107 GPIO pins, and each clock-out can be programmed to output appropriate clock source. GPIO73-104 are for debugging mode only. Besides, when 2 GPIO function for the same peripheral IP, the smaller GPIO serial numbers have higher priority than larger numbers.

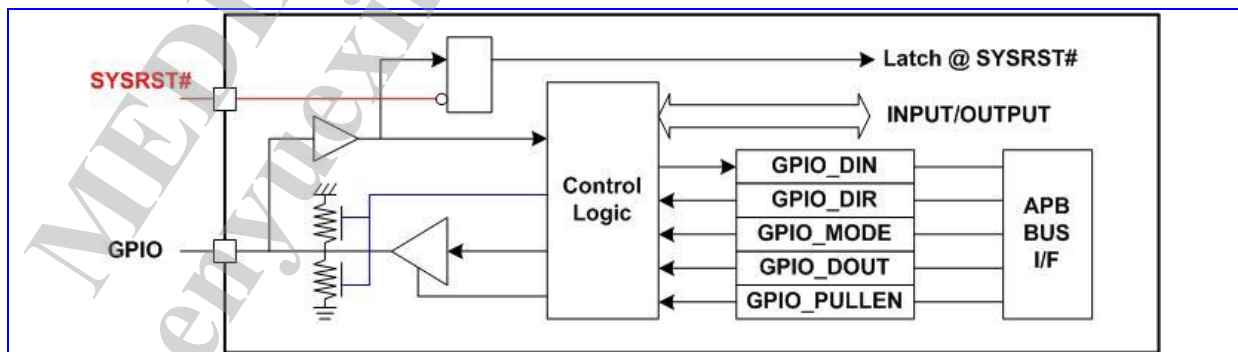


Figure 36. GPIO block diagram

Register address	Register name	Synonym
0xA002_0000h	GPIO direction control register 0	GPIO_DIR0
0xA002_0010h	GPIO direction control register 1	GPIO_DIR1
0xA002_0020h	GPIO direction control register 2	GPIO_DIR2
0xA002_0030h	GPIO direction control register 3	GPIO_DIR3
0xA002_0040h	GPIO direction control register 4	GPIO_DIR4
0xA002_0050h	GPIO direction control register 5	GPIO_DIR5
0xA002_0060h	GPIO direction control register 6	GPIO_DIR6
0xA002_0100h	GPIO pull-up/down enable 0 register	GPIO_PULLEN0
0xA002_0110h	GPIO pull-up/down enable 1 register	GPIO_PULLEN1
0xA002_0120h	GPIO pull-up/down enable 2 register	GPIO_PULLEN2
0xA002_0130h	GPIO pull-up/down enable 3 register	GPIO_PULLEN3
0xA002_0140h	GPIO pull-up/down enable 4 register	GPIO_PULLEN4
0xA002_0150h	GPIO pull-up/down enable 5 register	GPIO_PULLEN5
0xA002_0160h	GPIO pull-up/down enable 6 register	GPIO_PULLEN6
0xA002_0200h	GPIO data inversion control register 0	GPIO_DINV0
0xA002_0210h	GPIO data inversion control register 1	GPIO_DINV1
0xA002_0220h	GPIO data inversion control register 2	GPIO_DINV2
0xA002_0230h	GPIO data inversion control register 3	GPIO_DINV3
0xA002_0240h	GPIO data inversion control register 4	GPIO_DINV4
0xA002_0250h	GPIO data inversion control register 5	GPIO_DINV5
0xA002_0260h	GPIO data inversion control register 6	GPIO_DINV6
0xA002_0300h	GPIO data output register 0	GPIO_DOUT0
0xA002_0310h	GPIO data output register 1	GPIO_DOUT1
0xA002_0320h	GPIO data output register 2	GPIO_DOUT2
0xA002_0330h	GPIO data output register 3	GPIO_DOUT3
0xA002_0340h	GPIO data output register 4	GPIO_DOUT4
0xA002_0350h	GPIO data output register 5	GPIO_DOUT5
0xA002_0360h	GPIO data output register 6	GPIO_DOUT6
0xA002_0400h	GPIO data input register 0	GPIO_DIN0
0xA002_0410h	GPIO data input register 1	GPIO_DIN1
0xA002_0420h	GPIO data input register 2	GPIO_DIN2
0xA002_0430h	GPIO data input register 3	GPIO_DIN3
0xA002_0440h	GPIO data input register 4	GPIO_DIN4
0xA002_0450h	GPIO data input register 5	GPIO_DIN5
0xA002_0460h	GPIO data input register 6	GPIO_DIN6
0xA002_0500h	GPIO pull-up/down select 0	GPIO_PULLSEL0
0xA002_0510h	GPIO pull-up/down select 1	GPIO_PULLSEL1
0xA002_0520h	GPIO pull-up/down select 2	GPIO_PULLSEL2
0xA002_0530h	GPIO pull-up/down select 3	GPIO_PULLSEL3
0xA002_0540h	GPIO pull-up/down select 4	GPIO_PULLSEL4
0xA002_0550h	GPIO pull-up/down select 5	GPIO_PULLSEL5
0xA002_0560h	GPIO pull-up/down select 6	GPIO_PULLSEL6

Register address	Register name	Synonym
0xA002_0600h	GPIO Schmit trigger control register 0	GPIO_SMT0
0xA002_0610h	GPIO Schmit trigger control register 1	GPIO_SMT1
0xA002_0620h	GPIO Schmit trigger control register 2	GPIO_SMT2
0xA002_0630h	GPIO Schmit trigger control register 3	GPIO_SMT3
0xA002_0700h	GPIO slew rate control register 0	GPIO_SR0
0xA002_0710h	GPIO slew rate control register 0	GPIO_SR1
0xA002_0720h	GPIO slew rate control register 0	GPIO_SR2
0xA002_0730h	GPIO slew rate control register 0	GPIO_SR3
0xA002_0800h	GPIO driving control register 0	GPIO_DRV0
0xA002_0810h	GPIO driving control register 1	GPIO_DRV1
0xA002_0820h	GPIO driving control register 2	GPIO_DRV2
0xA002_0830h	GPIO driving control register 3	GPIO_DRV3
0xA002_0900h	GPIO Rx input buffer control register 0	GPIO_IES0
0xA002_0910h	GPIO Rx input buffer control register 1	GPIO_IES1
0xA002_0920h	GPIO Rx input buffer control register 2	GPIO_IES2
0xA002_0930h	GPIO Rx input buffer control register 3	GPIO_IES3
0xA002_0a00h	GPIO PUPD enable control register 0	GPIO_PUPDEN0
0xA002_0a10h	GPIO PUPD enable control register 1	GPIO_PUPDEN1
0xA002_0b00h	GPIO resistance enable control register 0	GPIO_RESEN0
0xA002_0b10h	GPIO resistance enable control register 1	GPIO_RESEN1
0xA002_0b20h	GPIO resistance enable control register 2	GPIO_RESEN2
0xA002_0c00h	GPIO mode control register 0	GPIO_MODE0
0xA002_0c10h	GPIO mode control register 1	GPIO_MODE1
0xA002_0c20h	GPIO mode control register 2	GPIO_MODE2
0xA002_0c30h	GPIO mode control register 3	GPIO_MODE3
0xA002_0c40h	GPIO mode control register 4	GPIO_MODE4
0xA002_0c50h	GPIO mode control register 5	GPIO_MODE5
0xA002_0c60h	GPIO mode control register 6	GPIO_MODE6
0xA002_0c70h	GPIO mode control register 7	GPIO_MODE7
0xA002_0c80h	GPIO code control register 8	GPIO_MODE8
0xA002_0c90h	GPIO mode control register 9	GPIO_MODE9
0xA002_0d00h	GPIO Rx duty select control register	GPIO_RDSEL
0xA002_0d10h	GPIO Tx duty select control register	GPIO_TDSEL
0xA002_0e00h	CLK OUT0 setting	CLK_OUT0
0xA002_0e10h	CLK OUT1 setting	CLK_OUT1
0xA002_0e20h	CLK OUT2 setting	CLK_OUT2
0xA002_0e30h	CLK OUT3 setting	CLK_OUT3
0xA002_0e40h	CLK OUT4 setting	CLK_OUT4
0xA002_0e50h	CLK OUT5 setting	CLK_OUT5
0xA002_0e60h	CLK OUT6 setting	CLK_OUT6
0xA002_0e70h	CLK OUT7 setting	CLK_OUT7

0xA002_0000 GPIO direction control register 0

GPIO_DIR0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO15	GPIO14	GPIO13	GPIO12	GPIO11	GPIO10	GPIO9	GPIO8	GPIO7	GPIO6	GPIO5	GPIO4	GPIO3	GPIO2	GPIO1	GPIO0
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
15	GPIO15	GPIO15_DIR	GPIO 15 direction control 0: Input 1: Output
14	GPIO14	GPIO14_DIR	GPIO 14 direction control 0: Input 1: Output
13	GPIO13	GPIO13_DIR	GPIO 13 direction control 0: Input 1: Output
12	GPIO12	GPIO12_DIR	GPIO 12 direction control 0: Input 1: Output
11	GPIO11	GPIO11_DIR	GPIO 11 direction control 0: Input 1: Output
10	GPIO10	GPIO10_DIR	GPIO 10 direction control 0: Input 1: Output
9	GPIO9	GPIO9_DIR	GPIO 9 direction control 0: Input 1: Output
8	GPIO8	GPIO8_DIR	GPIO 8 direction control 0: Input 1: Output
7	GPIO7	GPIO7_DIR	GPIO 7 direction control 0: Input 1: Output
6	GPIO6	GPIO6_DIR	GPIO 6 direction control 0: Input 1: Output
5	GPIO5	GPIO5_DIR	GPIO 5 direction control 0: Input 1: Output
4	GPIO4	GPIO4_DIR	GPIO 4 direction control 0: Input 1: Output
3	GPIO3	GPIO3_DIR	GPIO 3 direction control 0: Input

Bit(s)	Mnemonic	Name	Description
2	GPIO2	GPIO2_DIR	GPIO 2 direction control 1: Output 0: Input
1	GPIO1	GPIO1_DIR	GPIO 1 direction control 1: Output 0: Input
0	GPIO0	GPIO0_DIR	GPIO 0 direction control 1: Output 0: Input

0xA002_0010 GPIO direction control register 1

GPIO_DIR1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO31	GPIO30	GPIO29	GPIO28	GPIO27	GPIO26	GPIO25	GPIO24	GPIO23	GPIO22	GPIO21	GPIO20	GPIO19	GPIO18	GPIO17	GPIO16
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0

Bit(s)	Mnemonic	Name	Description
15	GPIO31	GPIO31_DIR	GPIO 31 direction control 1: Output 0: Input
14	GPIO30	GPIO30_DIR	GPIO 30 direction control 1: Output 0: Input
13	GPIO29	GPIO29_DIR	GPIO 29 direction control 1: Output 0: Input
12	GPIO28	GPIO28_DIR	GPIO 28 direction control 1: Output 0: Input
11	GPIO27	GPIO27_DIR	GPIO 27 direction control 1: Output 0: Input
10	GPIO26	GPIO26_DIR	GPIO 26 direction control 1: Output 0: Input
9	GPIO25	GPIO25_DIR	GPIO 25 direction control 1: Output 0: Input
8	GPIO24	GPIO24_DIR	GPIO 24 direction control 1: Output 0: Input
7	GPIO23	GPIO23_DIR	GPIO 23 direction control

Bit(s)	Mnemonic	Name	Description
6	GPIO22	GPIO22_DIR	GPIO 22 direction control 0: Input 1: Output
5	GPIO21	GPIO21_DIR	GPIO 21 direction control 0: Input 1: Output
4	GPIO20	GPIO20_DIR	GPIO 20 direction control 0: Input 1: Output
3	GPIO19	GPIO19_DIR	GPIO 19 direction control 0: Input 1: Output
2	GPIO18	GPIO18_DIR	GPIO 18 direction control 0: Input 1: Output
1	GPIO17	GPIO17_DIR	GPIO 17 direction control 0: Input 1: Output
0	GPIO16	GPIO16_DIR	GPIO 16 direction control 0: Input 1: Output

0xA002_0020 GPIO direction control register 2 GPIO_DIR2

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO 47	GPIO 46	GPIO 45	GPIO 44	GPIO 43	GPIO 42	GPIO 41	GPIO 40	GPIO 39	GPIO 38	GPIO 37	GPIO 36	GPIO 35	GPIO 34	GPIO 33	GPIO 32
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
15	GPIO47	GPIO47_DIR	GPIO 47 direction control 0: Input 1: Output
14	GPIO46	GPIO46_DIR	GPIO 46 direction control 0: Input 1: Output
13	GPIO45	GPIO45_DIR	GPIO 45 direction control 0: Input 1: Output
12	GPIO44	GPIO44_DIR	GPIO 44 direction control 0: Input

Bit(s)	Mnemonic	Name	Description
11	GPIO43	GPIO43_DIR	1: Output GPIO 43 direction control 0: Input
10	GPIO42	GPIO42_DIR	1: Output GPIO 42 direction control 0: Input
9	GPIO41	GPIO41_DIR	1: Output GPIO 41 direction control 0: Input
8	GPIO40	GPIO40_DIR	1: Output GPIO 40 direction control 0: Input
7	GPIO39	GPIO39_DIR	1: Output GPIO 39 direction control 0: Input
6	GPIO38	GPIO38_DIR	1: Output GPIO 38 direction control 0: Input
5	GPIO37	GPIO37_DIR	1: Output GPIO 37 direction control 0: Input
4	GPIO36	GPIO36_DIR	1: Output GPIO 36 direction control 0: Input
3	GPIO35	GPIO35_DIR	1: Output GPIO 35 direction control 0: Input
2	GPIO34	GPIO34_DIR	1: Output GPIO 34 direction control 0: Input
1	GPIO33	GPIO33_DIR	1: Output GPIO 33 direction control 0: Input
0	GPIO32	GPIO32_DIR	1: Output GPIO 32 direction control 0: Input

0xA002_0030 GPIO direction control register 3

GPIO_DIR3

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO 63	GPIO 62	GPIO 61	GPIO 60	GPIO 59	GPIO 58	GPIO 57	GPIO 56	GPIO 55	GPIO 54	GPIO 53	GPIO 52	GPIO 51	GPIO 50	GPIO 49	GPIO 48
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
15	GPIO63	GPIO63_DIR	GPIO 63 direction control 0: Input 1: Output
14	GPIO62	GPIO62_DIR	GPIO 62 direction control 0: Input 1: Output
13	GPIO61	GPIO61_DIR	GPIO 61 direction control 0: Input 1: Output
12	GPIO60	GPIO60_DIR	GPIO 60 direction control 0: Input 1: Output
11	GPIO59	GPIO59_DIR	GPIO 59 direction control 0: Input 1: Output
10	GPIO58	GPIO58_DIR	GPIO 58 direction control 0: Input 1: Output
9	GPIO57	GPIO57_DIR	GPIO 57 direction control 0: Input 1: Output
8	GPIO56	GPIO56_DIR	GPIO 56 direction control 0: Input 1: Output
7	GPIO55	GPIO55_DIR	GPIO 55 direction control 0: Input 1: Output
6	GPIO54	GPIO54_DIR	GPIO 54 direction control 0: Input 1: Output
5	GPIO53	GPIO53_DIR	GPIO 53 direction control 0: Input 1: Output
4	GPIO52	GPIO52_DIR	GPIO 52 direction control 0: Input 1: Output
3	GPIO51	GPIO51_DIR	GPIO 51 direction control 0: Input 1: Output
2	GPIO50	GPIO50_DIR	GPIO 50 direction control 0: Input 1: Output
1	GPIO49	GPIO49_DIR	GPIO 49 direction control 0: Input 1: Output

Bit(s)	Mnemonic	Name	Description
0	GPIO48	GPIO48_DIR	GPIO 48 direction control 0: Input 1: Output

0xA002_0040 GPIO direction control register 4 GPIO_DIR4

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO 79	GPIO 78	GPIO 77	GPIO 76	GPIO 75	GPIO 74	GPIO 73	GPIO 72	GPIO 71	GPIO 70	GPIO 69	GPIO 68	GPIO 67	GPIO 66	GPIO 65	GPIO 64
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
15	GPIO79	GPIO79_DIR	GPIO 79 direction control 0: Input 1: Output
14	GPIO78	GPIO78_DIR	GPIO 78 direction control 0: Input 1: Output
13	GPIO77	GPIO77_DIR	GPIO 77 direction control 0: Input 1: Output
12	GPIO76	GPIO76_DIR	GPIO 76 direction control 0: Input 1: Output
11	GPIO75	GPIO75_DIR	GPIO 75 direction control 0: Input 1: Output
10	GPIO74	GPIO74_DIR	GPIO 74 direction control 0: Input 1: Output
9	GPIO73	GPIO73_DIR	GPIO 73 direction control 0: Input 1: Output
8	GPIO72	GPIO72_DIR	GPIO 72 direction control 0: Input 1: Output
7	GPIO71	GPIO71_DIR	GPIO 71 direction control 0: Input 1: Output
6	GPIO70	GPIO70_DIR	GPIO 70 direction control 0: Input 1: Output
5	GPIO69	GPIO69_DIR	GPIO 69 direction control

Bit(s)	Mnemonic	Name	Description
			0: Input 1: Output
4	GPIO68	GPIO68_DIR	GPIO 68 direction control 0: Input 1: Output
3	GPIO67	GPIO67_DIR	GPIO 67 direction control 0: Input 1: Output
2	GPIO66	GPIO66_DIR	GPIO 66 direction control 0: Input 1: Output
1	GPIO65	GPIO65_DIR	GPIO 65 direction control 0: Input 1: Output
0	GPIO64	GPIO64_DIR	GPIO 64 direction control 0: Input 1: Output

0xA002_0050 GPIO direction control register 5 GPIO_DIR5

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO 95	GPIO 94	GPIO 93	GPIO 92	GPIO 91	GPIO 90	GPIO 89	GPIO 88	GPIO 87	GPIO 86	GPIO 85	GPIO 84	GPIO 83	GPIO 82	GPIO 81	GPIO 80
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
15	GPIO95	GPIO95_DIR	GPIO 95 direction control 0: Input 1: Output
14	GPIO94	GPIO94_DIR	GPIO 94 direction control 0: Input 1: Output
13	GPIO93	GPIO93_DIR	GPIO 93 direction control 0: Input 1: Output
12	GPIO92	GPIO92_DIR	GPIO 92 direction control 0: Input 1: Output
11	GPIO91	GPIO91_DIR	GPIO 91 direction control 0: Input 1: Output
10	GPIO90	GPIO90_DIR	GPIO 90 direction control 0: Input

Bit(s)	Mnemonic	Name	Description
9	GPIO89	GPIO89_DIR	GPIO 89 direction control 1: Output 0: Input
8	GPIO88	GPIO88_DIR	GPIO 88 direction control 1: Output 0: Input
7	GPIO87	GPIO87_DIR	GPIO 87 direction control 1: Output 0: Input
6	GPIO86	GPIO86_DIR	GPIO 86 direction control 1: Output 0: Input
5	GPIO85	GPIO85_DIR	GPIO 85 direction control 1: Output 0: Input
4	GPIO84	GPIO84_DIR	GPIO 84 direction control 1: Output 0: Input
3	GPIO83	GPIO83_DIR	GPIO 83 direction control 1: Output 0: Input
2	GPIO82	GPIO82_DIR	GPIO 82 direction control 1: Output 0: Input
1	GPIO81	GPIO81_DIR	GPIO 81 direction control 1: Output 0: Input
0	GPIO80	GPIO80_DIR	GPIO 80 direction control 1: Output 0: Input

0xA002_0060 GPIO direction control register 6 GPIO_DIR6

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name								GPIO 104	GPIO 103	GPIO 102	GPIO 101	GPIO 100	GPIO 99	GPIO 98	GPIO 97	GPIO 96
Type								R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset								0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
8	GPIO104	GPIO104_DIR	GPIO 104 direction control 0: Input 1: Output

Bit(s)	Mnemonic	Name	Description
7	GPIO103	GPIO103_DIR	GPIO 103 direction control 0: Input 1: Output
6	GPIO102	GPIO102_DIR	GPIO 102 direction control 0: Input 1: Output
5	GPIO101	GPIO101_DIR	GPIO 101 direction control 0: Input 1: Output
4	GPIO100	GPIO100_DIR	GPIO 100 direction control 0: Input 1: Output
3	GPIO99	GPIO99_DIR	GPIO 99 direction control 0: Input 1: Output
2	GPIO98	GPIO98_DIR	GPIO 98 direction control 0: Input 1: Output
1	GPIO97	GPIO97_DIR	GPIO 97 direction control 0: Input 1: Output
0	GPIO96	GPIO96_DIR	GPIO 96 direction control 0: Input 1: Output

0xA002_0100 GPIO pull-up/down enable 0 **GPIO_PULLEN0**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO 15	GPIO 14	GPIO 13	GPIO 12	GPIO 11	GPIO 10	GPIO 9	GPIO 8	GPIO 7	GPIO 6	GPIO 5	GPIO 4	GPIO 3	GPIO 2	GPIO 1	GPIO 0
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
15	GPIO15	GPIO15_PULLEN	Enables GPIO 15 pull-up/down 0: Disable 1: Enable
14	GPIO14	GPIO14_PULLEN	Enables GPIO 14 pull-up/down 0: Disable 1: Enable
13	GPIO13	GPIO13_PULLEN	Enables GPIO 13 pull-up/down 0: Disable 1: Enable

Bit(s)	Mnemonic	Name	Description
12	GPIO12	GPIO12_PULLEN	Enables GPIO 12 pull-up/down 0: Disable 1: Enable
11	GPIO11	GPIO11_PULLEN	Enables GPIO 11 pull-up/down 0: Disable 1: Enable
10	GPIO10	GPIO10_PULLEN	Enables GPIO 10 pull-up/down 0: Disable 1: Enable
9	GPIO9	GPIO9_PULLEN	Enables GPIO 9 pull-up/down 0: Disable 1: Enable
8	GPIO8	GPIO8_PULLEN	Enables GPIO 8 pull-up/down 0: Disable 1: Enable
7	GPIO7	GPIO7_PULLEN	Enables GPIO 7 pull-up/down 0: Disable 1: Enable
6	GPIO6	GPIO6_PULLEN	Enables GPIO 6 pull-up/down 0: Disable 1: Enable
5	GPIO5	GPIO5_PULLEN	Enables GPIO 5 pull-up/down 0: Disable 1: Enable
4	GPIO4	GPIO4_PULLEN	Enables GPIO 4 pull-up/down 0: Disable 1: Enable
3	GPIO3	GPIO3_PULLEN	Enables GPIO 3 pull-up/down 0: Disable 1: Enable
2	GPIO2	GPIO2_PULLEN	Enables GPIO 2 pull-up/down 0: Disable 1: Enable
1	GPIO1	GPIO1_PULLEN	Enables GPIO 1 pull-up/down 0: Disable 1: Enable
0	GPIO0	GPIO0_PULLEN	Enables GPIO 0 pull-up/down 0: Disable 1: Enable

0xA002_0110 GPIO pull-up/down enable 1

**GPIO_PULLE
N1**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
-----	----	----	----	----	----	----	---	---	---	---	---	---	---	---	---	---

Name	GPIO 31	GPIO 30	GPIO 29	GPIO 28	GPIO 27	GPIO 26	GPIO 25	GPIO 24	GPIO 23	GPIO 22	GPIO 21	PGIO 20	GPIO 19	GPIO 18	GPIO 17	GPIO 16
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0

Bit(s)	Mnemonic	Name	Description
15	GPIO31	GPIO31_PULLEN	Enables GPIO 31 pull-up/down 0: Disable 1: Enable
14	GPIO30	GPIO30_PULLEN	Enables GPIO 30 pull-up/down 0: Disable 1: Enable
13	GPIO29	GPIO29_PULLEN	Enables GPIO 29 pull-up/down 0: Disable 1: Enable
12	GPIO28	GPIO28_PULLEN	Enables GPIO 28 pull-up/down 0: Disable 1: Enable
11	GPIO27	GPIO27_PULLEN	Enables GPIO 27 pull-up/down 0: Disable 1: Enable
10	GPIO26	GPIO26_PULLEN	Enables GPIO 26 pull-up/down 0: Disable 1: Enable
9	GPIO25	GPIO25_PULLEN	Enables GPIO 25 pull-up/down 0: Disable 1: Enable
8	GPIO24	GPIO24_PULLEN	Enables GPIO 24 pull-up/down 0: Disable 1: Enable
7	GPIO23	GPIO23_PULLEN	Enables GPIO 23 pull-up/down 0: Disable 1: Enable
6	GPIO22	GPIO22_PULLEN	Enables GPIO 22 pull-up/down 0: Disable 1: Enable
5	GPIO21	GPIO21_PULLEN	Enables GPIO 21 pull-up/down 0: Disable 1: Enable
4	GPIO20	GPIO20_PULLEN	Enables GPIO 20 pull-up/down 0: Disable 1: Enable
3	GPIO19	GPIO19_PULLEN	Enables GPIO 19 pull-up/down 0: Disable 1: Enable
2	GPIO18	GPIO18_PULLEN	Enables GPIO 18 pull-up/down 0: Disable

Bit(s)	Mnemonic	Name	Description
1	GPIO17	GPIO17_PULLEN	Enables GPIO 17 pull-up/down 1: Enable 0: Disable
0	GPIO16	GPIO16_PULLEN	Enables GPIO 16 pull-up/down 1: Enable 0: Disable

0xA002_0120 GPIO pull-up/down enable 2

GPIO_PULLEN2

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO47	GPIO46	GPIO45	GPIO44	GPIO43	GPIO42	GPIO41	GPIO40	GPIO39	GPIO38	GPIO37	GPIO36	GPIO35	GPIO34	GPIO33	GPIO32
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
15	GPIO47	GPIO47_PULLEN	Enables GPIO 47 pull-up/down 1: Enable 0: Disable
14	GPIO46	GPIO46_PULLEN	Enables GPIO 46 pull-up/down 1: Enable 0: Disable
13	GPIO45	GPIO45_PULLEN	Enables GPIO 45 pull-up/down 1: Enable 0: Disable
12	GPIO44	GPIO44_PULLEN	Enables GPIO 44 pull-up/down 1: Enable 0: Disable
11	GPIO43	GPIO43_PULLEN	Enables GPIO 43 pull-up/down 1: Enable 0: Disable
10	GPIO42	GPIO42_PULLEN	Enables GPIO 42 pull-up/down 1: Enable 0: Disable
9	GPIO41	GPIO41_PULLEN	Enables GPIO 41 pull-up/down 1: Enable 0: Disable
8	GPIO40	GPIO40_PULLEN	Enables GPIO 40 pull-up/down 1: Enable 0: Disable
7	GPIO39	GPIO39_PULLEN	Enables GPIO 39 pull-up/down 0: Disable

Bit(s)	Mnemonic	Name	Description
			1: Enable
6	GPIO38	GPIO38_PULLEN	Enables GPIO 38 pull-up/down 0: Disable 1: Enable
5	GPIO37	GPIO37_PULLEN	Enables GPIO 37 pull-up/down 0: Disable 1: Enable
4	GPIO36	GPIO36_PULLEN	Enables GPIO 36 pull-up/down 0: Disable 1: Enable
3	GPIO35	GPIO35_PULLEN	Enables GPIO 35 pull-up/down 0: Disable 1: Enable
2	GPIO34	GPIO34_PULLEN	Enables GPIO 34 pull-up/down 0: Disable 1: Enable
1	GPIO33	GPIO33_PULLEN	Enables GPIO 33 pull-up/down 0: Disable 1: Enable
0	GPIO32	GPIO32_PULLEN	Enables GPIO 32 pull-up/down 0: Disable 1: Enable

0xA002_0130 GPIO pull-up/down enable 3 **GPIO_PULLEN**
3

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO 63	GPIO 62	GPIO 61	GPIO 60	GPIO 59	GPIO 58	GPIO 57	GPIO 56	GPIO 55	GPIO 54	GPIO 53	GPIO 52	GPIO 51	GPIO 50	GPIO 49	GPIO 48
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
15	GPIO63	GPIO63_PULLEN	Enables GPIO 63 pull-up/down 0: Disable 1: Enable
14	GPIO62	GPIO62_PULLEN	Enables GPIO 62 pull-up/down 0: Disable 1: Enable
13	GPIO61	GPIO61_PULLEN	Enables GPIO 61 pull-up/down 0: Disable 1: Enable
12	GPIO60	GPIO60_PULLEN	Enables GPIO 60 pull-up/down 0: Disable

Bit(s)	Mnemonic	Name	Description
11	GPIO59	GPIO59_PULLEN	1: Enable Enables GPIO 59 pull-up/down 0: Disable 1: Enable
10	GPIO58	GPIO58_PULLEN	1: Enable Enables GPIO 58 pull-up/down 0: Disable 1: Enable
9	GPIO57	GPIO57_PULLEN	1: Enable Enables GPIO 57 pull-up/down 0: Disable 1: Enable
8	GPIO56	GPIO56_PULLEN	1: Enable Enables GPIO 56 pull-up/down 0: Disable 1: Enable
7	GPIO55	GPIO55_PULLEN	1: Enable Enables GPIO 55 pull-up/down 0: Disable 1: Enable
6	GPIO54	GPIO54_PULLEN	1: Enable Enables GPIO 54 pull-up/down 0: Disable 1: Enable
5	GPIO53	GPIO53_PULLEN	1: Enable Enables GPIO 53 pull-up/down 0: Disable 1: Enable
4	GPIO52	GPIO52_PULLEN	1: Enable Enables GPIO 52 pull-up/down 0: Disable 1: Enable
3	GPIO51	GPIO51_PULLEN	1: Enable Enables GPIO 51 pull-up/down 0: Disable 1: Enable
2	GPIO50	GPIO50_PULLEN	1: Enable Enables GPIO 50 pull-up/down 0: Disable 1: Enable
1	GPIO49	GPIO49_PULLEN	1: Enable Enables GPIO 49 pull-up/down 0: Disable 1: Enable
0	GPIO48	GPIO48_PULLEN	1: Enable Enables GPIO 48 pull-up/down 0: Disable 1: Enable

GPIO_PULLEN
0xA002_0140 GPIO pull-up/down enable 4
4

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO 79	GPIO 78	GPIO 77	GPIO 76	GPIO 75	GPIO 74	GPIO 73	GPIO 72	GPIO 71	GPIO 70	GPIO 69	GPIO 68	GPIO 67	GPIO 66	GPIO 65	GPIO 64
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
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Bit(s)	Mnemonic	Name	Description
15	GPIO79	GPIO79_PULLEN	Enables GPIO 79 pull-up/down 0: Disable 1: Enable
14	GPIO78	GPIO78_PULLEN	Enables GPIO 78 pull-up/down 0: Disable 1: Enable
13	GPIO77	GPIO77_PULLEN	Enables GPIO 77 pull-up/down 0: Disable 1: Enable
12	GPIO76	GPIO76_PULLEN	Enables GPIO 76 pull-up/down 0: Disable 1: Enable
11	GPIO75	GPIO75_PULLEN	Enables GPIO 75 pull-up/down 0: Disable 1: Enable
10	GPIO74	GPIO74_PULLEN	Enables GPIO 74 pull-up/down 0: Disable 1: Enable
9	GPIO73	GPIO73_PULLEN	Enables GPIO 73 pull-up/down 0: Disable 1: Enable
8	GPIO72	GPIO72_PULLEN	Enables GPIO 72 pull-up/down 0: Disable 1: Enable
7	GPIO71	GPIO71_PULLEN	Enables GPIO 71 pull-up/down 0: Disable 1: Enable
6	GPIO70	GPIO70_PULLEN	Enables GPIO 70 pull-up/down 0: Disable 1: Enable
5	GPIO69	GPIO69_PULLEN	Enables GPIO 69 pull-up/down 0: Disable 1: Enable
4	GPIO68	GPIO68_PULLEN	Enables GPIO 68 pull-up/down 0: Disable 1: Enable
3	GPIO67	GPIO67_PULLEN	Enables GPIO 67 pull-up/down 0: Disable 1: Enable
2	GPIO66	GPIO66_PULLEN	Enables GPIO 66 pull-up/down 0: Disable 1: Enable
1	GPIO65	GPIO65_PULLEN	Enables GPIO 65 pull-up/down 0: Disable

Bit(s)	Mnemonic Name	Description
0	GPIO64 GPIO64_PULLEN	1: Enable Enables GPIO 64 pull-up/down 0: Disable 1: Enable

0xA002_0150 GPIO pull-up/down enable 5 **GPIO_PULLEN 5**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO 95	GPIO 94	GPIO 93	GPIO 92	GPIO 91	GPIO 90	GPIO 89	GPIO 88	GPIO 87	GPIO 86	GPIO 85	GPIO 84	GPIO 83	GPIO 82	GPIO 81	GPIO 80
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic Name	Description
15	GPIO95 GPIO95_PULLEN	Enables GPIO 95 pull-up/down 0: Disable 1: Enable
14	GPIO94 GPIO94_PULLEN	Enables GPIO 94 pull-up/down 0: Disable 1: Enable
13	GPIO93 GPIO93_PULLEN	Enables GPIO 93 pull-up/down 0: Disable 1: Enable
12	GPIO92 GPIO92_PULLEN	Enables GPIO 92 pull-up/down 0: Disable 1: Enable
11	GPIO91 GPIO91_PULLEN	Enables GPIO 91 pull-up/down 0: Disable 1: Enable
10	GPIO90 GPIO90_PULLEN	Enables GPIO 90 pull-up/down 0: Disable 1: Enable
9	GPIO89 GPIO89_PULLEN	Enables GPIO 89 pull-up/down 0: Disable 1: Enable
8	GPIO88 GPIO88_PULLEN	Enables GPIO 88 pull-up/down 0: Disable 1: Enable
7	GPIO87 GPIO87_PULLEN	Enables GPIO 87 pull-up/down 0: Disable 1: Enable
6	GPIO86 GPIO86_PULLEN	Enables GPIO 86 pull-up/down 0: Disable

Bit(s)	Mnemonic	Name	Description
5	GPIO85	GPIO85_PULLEN	Enables GPIO 85 pull-up/down 0: Disable 1: Enable
4	GPIO84	GPIO84_PULLEN	Enables GPIO 84 pull-up/down 0: Disable 1: Enable
3	GPIO83	GPIO83_PULLEN	Enables GPIO 83 pull-up/down 0: Disable 1: Enable
2	GPIO82	GPIO82_PULLEN	Enables GPIO 82 pull-up/down 0: Disable 1: Enable
1	GPIO81	GPIO81_PULLEN	Enables GPIO 81 pull-up/down 0: Disable 1: Enable
0	GPIO80	GPIO80_PULLEN	Enables GPIO 80 pull-up/down 0: Disable 1: Enable

0xA002_0160 GPIO pull-up/down enable 6 GPIO_PULLEN 6

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name								GPIO 104	GPIO 103	GPIO 102	GPIO 101	GPIO 100	GPIO 99	GPIO 98	GPIO 97	GPIO 96
Type								R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset								0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
8	GPIO104	GPIO104_PULLEN	Enables GPIO 104 pull-up/down 0: Disable 1: Enable
7	GPIO103	GPIO103_PULLEN	Enables GPIO 103 pull-up/down 0: Disable 1: Enable
6	GPIO102	GPIO102_PULLEN	Enables GPIO 102 pull-up/down 0: Disable 1: Enable
5	GPIO101	GPIO101_PULLEN	Enables GPIO 101 pull-up/down 0: Disable 1: Enable
4	GPIO100	GPIO100_PULLEN	Enables GPIO 100 pull-up/down 0: Disable

Bit(s)	Mnemonic	Name	Description
3	GPIO99	GPIO99_PULLEN	1: Enable Enables GPIO 99 pull-up/down 0: Disable 1: Enable
2	GPIO98	GPIO98_PULLEN	1: Enable Enables GPIO 98 pull-up/down 0: Disable 1: Enable
1	GPIO97	GPIO97_PULLEN	1: Enable Enables GPIO 97 pull-up/down 0: Disable 1: Enable
0	GPIO96	GPIO96_PULLEN	1: Enable Enables GPIO 96 pull-up/down 0: Disable 1: Enable

0xA002_0200 GPIO data inversion control register 0 GPIO_DINV0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	INV15	INV14	INV13	INV12	INV11	INV10	INV9	INV8	INV7	INV6	INV5	INV4	INV3	INV2	INV1	INV0
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
15	INV15	GPIO15_DINV	GPIO 15 inversion control 0: Disable 1: Enable
14	INV14	GPIO14_DINV	GPIO 14 inversion control 0: Disable 1: Enable
13	INV13	GPIO13_DINV	GPIO 13 inversion control 0: Disable 1: Enable
12	INV12	GPIO12_DINV	GPIO 12 inversion control 0: Disable 1: Enable
11	INV11	GPIO11_DINV	GPIO 11 inversion control 0: Disable 1: Enable
10	INV10	GPIO10_DINV	GPIO 10 inversion control 0: Disable 1: Enable
9	INV9	GPIO9_DINV	GPIO 9 inversion control 0: Disable 1: Enable
8	INV8	GPIO8_DINV	GPIO 8 inversion control

Bit(s)	Mnemonic	Name	Description
7	INV7	GPIO7_DINV	GPIO 7 inversion control 0: Disable 1: Enable
6	INV6	GPIO6_DINV	GPIO 6 inversion control 0: Disable 1: Enable
5	INV5	GPIO5_DINV	GPIO 5 inversion control 0: Disable 1: Enable
4	INV4	GPIO4_DINV	GPIO 4 inversion control 0: Disable 1: Enable
3	INV3	GPIO3_DINV	GPIO 3 inversion control 0: Disable 1: Enable
2	INV2	GPIO2_DINV	GPIO 2 inversion control 0: Disable 1: Enable
1	INV1	GPIO1_DINV	GPIO 1 inversion control 0: Disable 1: Enable
0	INV0	GPIO0_DINV	GPIO 0 Inversion Control 0: Disable 1: Enable

0xA002_0210 GPIO data inversion control register 1 **GPIO_DINV1**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	INV31	INV30	INV29	INV28	INV27	INV26	INV25	INV24	INV23	INV22	INV21	INV20	INV19	INV18	INV17	INV16
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
15	INV31	GPIO31_DINV	GPIO 31 inversion control 0: Disable 1: Enable
14	INV30	GPIO30_DINV	GPIO 30 inversion control 0: Disable 1: Enable
13	INV29	GPIO29_DINV	GPIO 29 inversion control 0: Disable

Bit(s)	Mnemonic	Name	Description
12	INV28	GPIO28_DINV	GPIO 28 inversion control 1: Enable 0: Disable
11	INV27	GPIO27_DINV	GPIO 27 inversion control 1: Enable 0: Disable
10	INV26	GPIO26_DINV	GPIO 26 inversion control 1: Enable 0: Disable
9	INV25	GPIO25_DINV	GPIO 25 inversion control 1: Enable 0: Disable
8	INV24	GPIO24_DINV	GPIO 24 inversion control 1: Enable 0: Disable
7	INV23	GPIO23_DINV	GPIO 23 inversion control 1: Enable 0: Disable
6	INV22	GPIO22_DINV	GPIO 22 inversion control 1: Enable 0: Disable
5	INV21	GPIO21_DINV	GPIO 21 inversion control 1: Enable 0: Disable
4	INV20	GPIO20_DINV	GPIO 20 inversion control 1: Enable 0: Disable
3	INV19	GPIO19_DINV	GPIO 19 inversion control 1: Enable 0: Disable
2	INV18	GPIO18_DINV	GPIO 18 inversion control 1: Enable 0: Disable
1	INV17	GPIO17_DINV	GPIO 17 inversion control 1: Enable 0: Disable
0	INV16	GPIO16_DINV	GPIO 16 inversion control 1: Enable 0: Disable

0xA002_0220 GPIO data inversion control register 2

GPIO_DINV2

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
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Name	INV47	INV46	INV45	INV44	INV43	INV42	INV41	INV40	INV39	INV38	INV37	INV36	INV35	INV34	INV33	INV32
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
15	INV47	GPIO47_DINV	GPIO 47 inversion control 0: Disable 1: Enable
14	INV46	GPIO46_DINV	GPIO 46 inversion control 0: Disable 1: Enable
13	INV45	GPIO45_DINV	GPIO 45 inversion control 0: Disable 1: Enable
12	INV44	GPIO44_DINV	GPIO 44 inversion control 0: Disable 1: Enable
11	INV43	GPIO43_DINV	GPIO 43 inversion control 0: Disable 1: Enable
10	INV42	GPIO42_DINV	GPIO 42 inversion control 0: Disable 1: Enable
9	INV41	GPIO41_DINV	GPIO 41 inversion control 0: Disable 1: Enable
8	INV40	GPIO40_DINV	GPIO 40 inversion control 0: Disable 1: Enable
7	INV39	GPIO39_DINV	GPIO 39 inversion control 0: Disable 1: Enable
6	INV38	GPIO38_DINV	GPIO 38 inversion control 0: Disable 1: Enable
5	INV37	GPIO37_DINV	GPIO 37 inversion control 0: Disable 1: Enable
4	INV36	GPIO36_DINV	GPIO 36 inversion control 0: Disable 1: Enable
3	INV35	GPIO35_DINV	GPIO 35 inversion control 0: Disable 1: Enable
2	INV34	GPIO34_DINV	GPIO 34 inversion control 0: Disable

Bit(s)	Mnemonic	Name	Description
1	INV33	GPIO33_DINV	GPIO 33 inversion control 1: Enable 0: Disable
0	INV32	GPIO32_DINV	GPIO 32 inversion control 1: Enable 0: Disable

0xA002_0230 GPIO data inversion control register 3 GPIO_DINV3

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	INV63	INV62	INV61	INV60	INV59	INV58	INV57	INV56	INV55	INV54	INV53	INV52	INV51	INV50	INV49	INV48
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
15	INV63	GPIO63_DINV	GPIO 63 inversion control 1: Enable 0: Disable
14	INV62	GPIO62_DINV	GPIO 62 inversion control 1: Enable 0: Disable
13	INV61	GPIO61_DINV	GPIO 61 inversion control 1: Enable 0: Disable
12	INV60	GPIO60_DINV	GPIO 60 inversion control 1: Enable 0: Disable
11	INV59	GPIO59_DINV	GPIO 59 inversion control 1: Enable 0: Disable
10	INV58	GPIO58_DINV	GPIO 58 inversion control 1: Enable 0: Disable
9	INV57	GPIO57_DINV	GPIO 57 inversion control 1: Enable 0: Disable
8	INV56	GPIO56_DINV	GPIO 56 inversion control 1: Enable 0: Disable
7	INV55	GPIO55_DINV	GPIO 55 inversion control 1: Enable 0: Disable

Bit(s)	Mnemonic	Name	Description
6	INV54	GPIO54_DINV	GPIO 54 inversion control 0: Disable 1: Enable
5	INV53	GPIO53_DINV	GPIO 53 inversion control 0: Disable 1: Enable
4	INV52	GPIO52_DINV	GPIO 52 inversion control 0: Disable 1: Enable
3	INV51	GPIO51_DINV	GPIO 51 inversion control 0: Disable 1: Enable
2	INV50	GPIO50_DINV	GPIO 50 inversion control 0: Disable 1: Enable
1	INV49	GPIO49_DINV	GPIO 49 inversion control 0: Disable 1: Enable
0	INV48	GPIO48_DINV	GPIO 48 inversion control 0: Disable 1: Enable

0xA002_0240 GPIO data inversion control register 4 GPIO_DINV4

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	INV79	INV78	INV77	INV76	INV75	INV74	INV73	INV72	INV71	INV70	INV69	INV68	INV67	INV66	INV65	INV64
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
15	INV79	GPIO79_DINV	GPIO 79 inversion control 0: Disable 1: Enable
14	INV78	GPIO78_DINV	GPIO 78 inversion control 0: Disable 1: Enable
13	INV77	GPIO77_DINV	GPIO 77 inversion control 0: Disable 1: Enable
12	INV76	GPIO76_DINV	GPIO 76 inversion control 0: Disable 1: Enable
11	INV75	GPIO75_DINV	GPIO 75 inversion control

Bit(s)	Mnemonic	Name	Description
10	INV74	GPIO74_DINV	GPIO 74 inversion control 0: Disable 1: Enable
9	INV73	GPIO73_DINV	GPIO 73 inversion control 0: Disable 1: Enable
8	INV72	GPIO72_DINV	GPIO 72 inversion control 0: Disable 1: Enable
7	INV71	GPIO71_DINV	GPIO 71 inversion control 0: Disable 1: Enable
6	INV70	GPIO70_DINV	GPIO 70 inversion control 0: Disable 1: Enable
5	INV69	GPIO69_DINV	GPIO 69 inversion control 0: Disable 1: Enable
4	INV68	GPIO68_DINV	GPIO 68 inversion control 0: Disable 1: Enable
3	INV67	GPIO67_DINV	GPIO 67 inversion control 0: Disable 1: Enable
2	INV66	GPIO66_DINV	GPIO 66 inversion control 0: Disable 1: Enable
1	INV65	GPIO65_DINV	GPIO 65 inversion control 0: Disable 1: Enable
0	INV64	GPIO64_DINV	GPIO 64 inversion control 0: Disable 1: Enable

0xA002_0250 GPIO data inversion control register 5 GPIO_DINV5

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	INV95	INV94	INV93	INV92	INV91	INV90	INV89	INV88	INV87	INV86	INV85	INV84	INV83	INV82	INV81	INV80
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Bit(s)	Mnemonic	Name	Description
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Bit(s)	Mnemonic	Name	Description
15	INV95	GPIO95_DINV	GPIO 95 inversion control 0: Disable 1: Enable
14	INV94	GPIO94_DINV	GPIO 94 inversion control 0: Disable 1: Enable
13	INV93	GPIO93_DINV	GPIO 93 inversion control 0: Disable 1: Enable
12	INV92	GPIO92_DINV	GPIO 92 inversion control 0: Disable 1: Enable
11	INV91	GPIO91_DINV	GPIO 91 inversion control 0: Disable 1: Enable
10	INV90	GPIO90_DINV	GPIO 90 inversion control 0: Disable 1: Enable
9	INV89	GPIO89_DINV	GPIO 89 inversion control 0: Disable 1: Enable
8	INV88	GPIO88_DINV	GPIO 88 inversion control 0: Disable 1: Enable
7	INV87	GPIO87_DINV	GPIO 87 inversion control 0: Disable 1: Enable
6	INV86	GPIO86_DINV	GPIO 86 inversion control 0: Disable 1: Enable
5	INV85	GPIO85_DINV	GPIO 85 inversion control 0: Disable 1: Enable
4	INV84	GPIO84_DINV	GPIO 84 inversion control 0: Disable 1: Enable
3	INV83	GPIO83_DINV	GPIO 83 inversion control 0: Disable 1: Enable
2	INV82	GPIO82_DINV	GPIO 82 inversion control 0: Disable 1: Enable
1	INV81	GPIO81_DINV	GPIO 81 inversion control 0: Disable 1: Enable
0	INV80	GPIO80_DINV	GPIO 80 inversion control

Bit(s)	Mnemonic Name	Description
		0: Disable 1: Enable

0xA002_0260 GPIO data inversion control register 6 GPIO_DINV6

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name								INV10 4	INV10 3	INV10 2	INV10 1	INV10 0	INV99	INV98	INV97	INV9 6
Type								R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset								0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic Name	Description
8	INV104	GPIO104_DINV GPIO 104 inversion control 0: Disable 1: Enable
7	INV103	GPIO103_DINV GPIO 103 inversion control 0: Disable 1: Enable
6	INV102	GPIO102_DINV GPIO 102 inversion control 0: Disable 1: Enable
5	INV101	GPIO101_DINV GPIO 101 inversion control 0: Disable 1: Enable
4	INV100	GPIO100_DINV GPIO 100 inversion control 0: Disable 1: Enable
3	INV99	GPIO99_DINV GPIO 99 inversion control 0: Disable 1: Enable
2	INV98	GPIO98_DINV GPIO 98 inversion control 0: Disable 1: Enable
1	INV97	GPIO97_DINV GPIO 97 inversion control 0: Disable 1: Enable
0	INV96	GPIO96_DINV GPIO 96 inversion control 0: Disable 1: Enable

0xA002_0300 GPIO data output register 0 GPIO_DOUT0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
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Name	GPIO 15	GPIO 14	GPIO 13	GPIO 12	GPIO 11	GPIO 10	GPIO 9	GPIO 8	GPIO 7	GPIO 6	GPIO 5	GPIO 4	GPIO 3	GPIO 2	GPIO 1	GPIO 0
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
15	GPO15	GPIO15_DOUT	GPIO 15 data output value 0: Output 0 1: Output 1
14	GPO14	GPIO14_DOUT	GPIO 14 data output value 0: Output 0 1: Output 1
13	GPO13	GPIO13_DOUT	GPIO 13 data output value 0: Output 0 1: Output 1
12	GPO12	GPIO12_DOUT	GPIO 12 data output value 0: Output 0 1: Output 1
11	GPO11	GPIO11_DOUT	GPIO 11 data output value 0: Output 0 1: Output 1
10	GPO10	GPIO10_DOUT	GPIO 10 data output value 0: Output 0 1: Output 1
9	GPO9	GPIO9_DOUT	GPIO 9 data output value 0: Output 0 1: Output 1
8	GPO8	GPIO8_DOUT	GPIO 8 data output value 0: Output 0 1: Output 1
7	GPO7	GPIO7_DOUT	GPIO 7 data output value 0: Output 0 1: Output 1
6	GPO6	GPIO6_DOUT	GPIO 6 data output value 0: Output 0 1: Output 1
5	GPO5	GPIO5_DOUT	GPIO 5 data output value 0: Output 0 1: Output 1
4	GPO4	GPIO4_DOUT	GPIO 4 data output value 0: Output 0 1: Output 1
3	GPO3	GPIO3_DOUT	GPIO 3 data output value 0: Output 0 1: Output 1
2	GPO2	GPIO2_DOUT	GPIO 2 data output value 0: Output 0

Bit(s)	Mnemonic	Name	Description
1	GPO1	GPIO1_DOUT	GPIO 1 data output value 1: Output 1 0: Output 0
0	GPO0	GPIO0_DOUT	GPIO 0 data output value 1: Output 1 0: Output 0

0xA002_0310 GPIO data output register 1 GPIO_DOUT1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO 31	GPIO 30	GPIO 29	GPIO 28	GPIO 27	GPIO 26	GPIO 25	GPIO 24	GPIO 23	GPIO 22	GPIO 21	GPIO 20	GPIO 19	GPIO 18	GPIO 17	GPIO 16
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
15	GPO31	GPIO31_DOUT	GPIO 31 data output value 1: Output 1 0: Output 0
14	GPO30	GPIO30_DOUT	GPIO 30 data output value 1: Output 1 0: Output 0
13	GPO29	GPIO29_DOUT	GPIO 29 data output value 1: Output 1 0: Output 0
12	GPO28	GPIO28_DOUT	GPIO 28 data output value 1: Output 1 0: Output 0
11	GPO27	GPIO27_DOUT	GPIO 27 data output value 1: Output 1 0: Output 0
10	GPO26	GPIO26_DOUT	GPIO 26 data output value 1: Output 1 0: Output 0
9	GPO25	GPIO25_DOUT	GPIO 25 data output value 1: Output 1 0: Output 0
8	GPO24	GPIO24_DOUT	GPIO 24 data output value 1: Output 1 0: Output 0
7	GPO23	GPIO23_DOUT	GPIO 23 data output value 1: Output 1 0: Output 0
6	GPO22	GPIO22_DOUT	GPIO 22 data output value

Bit(s)	Mnemonic	Name	Description
5	GPO21	GPIO21_DOUT	GPIO 21 data output value 0: Output 0 1: Output 1
4	GPO20	GPIO20_DOUT	GPIO 20 data output value 0: Output 0 1: Output 1
3	GPO19	GPIO19_DOUT	GPIO 19 data output value 0: Output 0 1: Output 1
2	GPO18	GPIO18_DOUT	GPIO 18 data output value 0: Output 0 1: Output 1
1	GPO17	GPIO17_DOUT	GPIO 17 data output value 0: Output 0 1: Output 1
0	GPO16	GPIO16_DOUT	GPIO 16 data output value 0: Output 0 1: Output 1

0xA002_0320 GPIO data output register 2 GPIO_DOUT2

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO 47	GPIO 46	GPIO 45	GPIO 44	GPIO 43	GPIO 42	GPIO 41	GPIO 40	GPIO 39	GPIO 38	GPIO 37	GPIO 36	GPIO 35	GPIO 34	GPIO 33	GPIO 32
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
15	GPO47	GPIO47_DOUT	GPIO 47 data output value 0: Output 0 1: Output 1
14	GPO46	GPIO46_DOUT	GPIO 46 data output value 0: Output 0 1: Output 1
13	GPO45	GPIO45_DOUT	GPIO 45 data output value 0: Output 0 1: Output 1
12	GPO44	GPIO44_DOUT	GPIO 44 data output value 0: Output 0 1: Output 1
11	GPO43	GPIO43_DOUT	GPIO 43 data output value 0: Output 0 1: Output 1

Bit(s)	Mnemonic	Name	Description
10	GPO42	GPIO42_DOUT	GPIO 42 data output value 0: Output 0 1: Output 1
9	GPO41	GPIO41_DOUT	GPIO 41 data output value 0: Output 0 1: Output 1
8	GPO40	GPIO40_DOUT	GPIO 40 data output value 0: Output 0 1: Output 1
7	GPO39	GPIO39_DOUT	GPIO 39 data output value 0: Output 0 1: Output 1
6	GPO38	GPIO38_DOUT	GPIO 38 data output value 0: Output 0 1: Output 1
5	GPO37	GPIO37_DOUT	GPIO 37 data output value 0: Output 0 1: Output 1
4	GPO36	GPIO36_DOUT	GPIO 36 data output value 0: Output 0 1: Output 1
3	GPO35	GPIO35_DOUT	GPIO 35 data output value 0: Output 0 1: Output 1
2	GPO34	GPIO34_DOUT	GPIO 34 data output value 0: Output 0 1: Output 1
1	GPO33	GPIO33_DOUT	GPIO 33 data output value 0: Output 0 1: Output 1
0	GPO32	GPIO32_DOUT	GPIO 32 data output value 0: Output 0 1: Output 1

0xA002_0330 GPIO data output register 3

GPIO_DOUT3

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO 63	GPIO 62	GPIO 61	GPIO 60	GPIO 59	GPIO 58	GPIO 57	GPIO 56	GPIO 55	GPIO 54	GPIO 53	GPIO 52	GPIO 51	GPIO 50	GPIO 49	GPIO 48
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
15	GPO63	GPIO63_DOUT	GPIO 63 data output value

Bit(s)	Mnemonic	Name	Description
			0: Output 0 1: Output 1
14	GPO62	GPIO62_DOUT	GPIO 62 data output value 0: Output 0 1: Output 1
13	GPO61	GPIO61_DOUT	GPIO 61 data output value 0: Output 0 1: Output 1
12	GPO60	GPIO60_DOUT	GPIO 60 data output value 0: Output 0 1: Output 1
11	GPO59	GPIO59_DOUT	GPIO 59 data output value 0: Output 0 1: Output 1
10	GPO58	GPIO58_DOUT	GPIO 58 data output value 0: Output 0 1: Output 1
9	GPO57	GPIO57_DOUT	GPIO 57 data output value 0: Output 0 1: Output 1
8	GPO56	GPIO56_DOUT	GPIO 56 data output value 0: Output 0 1: Output 1
7	GPO55	GPIO55_DOUT	GPIO 55 data output value 0: Output 0 1: Output 1
6	GPO54	GPIO54_DOUT	GPIO 54 data output value 0: Output 0 1: Output 1
5	GPO53	GPIO53_DOUT	GPIO 53 data output value 0: Output 0 1: Output 1
4	GPO52	GPIO52_DOUT	GPIO 52 data output value 0: Output 0 1: Output 1
3	GPO51	GPIO51_DOUT	GPIO 51 data output value 0: Output 0 1: Output 1
2	GPO50	GPIO50_DOUT	GPIO 50 data output value 0: Output 0 1: Output 1
1	GPO49	GPIO49_DOUT	GPIO 49 data output value 0: Output 0 1: Output 1
0	GPO48	GPIO48_DOUT	GPIO 48 data output value 0: Output 0

Bit(s)	Mnemonic Name	Description
		1: Output 1

0xA002_0340 GPIO data output register 4

GPIO DOUT4

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO 79	GPIO 78	GPIO 77	GPIO 76	GPIO 75	GPIO 74	GPIO 73	GPIO 72	GPIO 71	GPIO 70	GPIO 69	GPIO 68	GPIO 67	GPIO 66	GPIO 65	GPIO 64
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic Name	Description
15	GPO79 GPIO79_DOUT	GPIO 79 data output value 0: Output 0 1: Output 1
14	GPO78 GPIO78_DOUT	GPIO 78 data output value 0: Output 0 1: Output 1
13	GPO77 GPIO77_DOUT	GPIO 77 data output value 0: Output 0 1: Output 1
12	GPO76 GPIO76_DOUT	GPIO 76 data output value 0: Output 0 1: Output 1
11	GPO75 GPIO75_DOUT	GPIO 75 data output value 0: Output 0 1: Output 1
10	GPO74 GPIO74_DOUT	GPIO 74 data output value 0: Output 0 1: Output 1
9	GPO73 GPIO73_DOUT	GPIO 73 data output value 0: Output 0 1: Output 1
8	GPO72 GPIO72_DOUT	GPIO 72 data output value 0: Output 0 1: Output 1
7	GPO71 GPIO71_DOUT	GPIO 71 data output value 0: Output 0 1: Output 1
6	GPO70 GPIO70_DOUT	GPIO 70 data output value 0: Output 0 1: Output 1
5	GPO69 GPIO69_DOUT	GPIO 69 data output value 0: Output 0 1: Output 1

Bit(s)	Mnemonic	Name	Description
4	GPO68	GPIO68_DOUT	GPIO 68 data output value 0: Output 0 1: Output 1
3	GPO67	GPIO67_DOUT	GPIO 67 data output value 0: Output 0 1: Output 1
2	GPO66	GPIO66_DOUT	GPIO 66 data output value 0: Output 0 1: Output 1
1	GPO65	GPIO65_DOUT	GPIO 65 data output value 0: Output 0 1: Output 1
0	GPO64	GPIO64_DOUT	GPIO 64 data output value 0: Output 0 1: Output 1

0xA002_0350 GPIO data output register 5

GPIO_DOUT5

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO 95	GPIO 94	GPIO 93	GPIO 92	GPIO 91	GPIO 90	GPIO 89	GPIO 88	GPIO 87	GPIO 86	GPIO 85	GPIO 84	GPIO 83	GPIO 82	GPIO 81	GPIO 80
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
15	GPO95	GPIO95_DOUT	GPIO 95 data output value 0: Output 0 1: Output 1
14	GPO94	GPIO94_DOUT	GPIO 94 data output value 0: Output 0 1: Output 1
13	GPO93	GPIO93_DOUT	GPIO 93 data output value 0: Output 0 1: Output 1
12	GPO92	GPIO92_DOUT	GPIO 92 data output value 0: Output 0 1: Output 1
11	GPO91	GPIO91_DOUT	GPIO 91 data output value 0: Output 0 1: Output 1
10	GPO90	GPIO90_DOUT	GPIO 90 data output value 0: Output 0 1: Output 1
9	GPO89	GPIO89_DOUT	GPIO 89 data output value

Bit(s)	Mnemonic	Name	Description
			0: Output 0 1: Output 1
8	GPO88	GPIO88_DOUT	GPIO 88 data output value 0: Output 0 1: Output 1
7	GPO87	GPIO87_DOUT	GPIO 87 data output value 0: Output 0 1: Output 1
6	GPO86	GPIO86_DOUT	GPIO 86 data output value 0: Output 0 1: Output 1
5	GPO85	GPIO85_DOUT	GPIO 85 data output value 0: Output 0 1: Output 1
4	GPO84	GPIO84_DOUT	GPIO 84 data output value 0: Output 0 1: Output 1
3	GPO83	GPIO83_DOUT	GPIO 83 data output value 0: Output 0 1: Output 1
2	GPO82	GPIO82_DOUT	GPIO 82 data output value 0: Output 0 1: Output 1
1	GPO81	GPIO81_DOUT	GPIO 81 data output value 0: Output 0 1: Output 1
0	GPO80	GPIO80_DOUT	GPIO 80 data output value 0: Output 0 1: Output 1

0xA002_0360 GPIO data output register 6 GPIO_DOUT6

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name								GPIO 104	GPIO 103	GPIO 102	GPIO 101	GPIO 100	GPIO 99	GPIO 98	GPIO 97	GPIO 96
Type								R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset								0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
8	GPO104	GPIO104_DOUT	GPIO 104 data output value 0: Output 0 1: Output 1
7	GPO103	GPIO103_DOUT	GPIO 103 data output value 0: Output 0

Bit(s)	Mnemonic	Name	Description
6	GPO102	GPIO102_DOUT	GPIO 102 data output value 1: Output 1 0: Output 0
5	GPO101	GPIO101_DOUT	GPIO 101 data output value 1: Output 1 0: Output 0
4	GPO100	GPIO100_DOUT	GPIO 100 data output value 1: Output 1 0: Output 0
3	GPO99	GPIO99_DOUT	GPIO 99 data output value 1: Output 1 0: Output 0
2	GPO98	GPIO98_DOUT	GPIO 98 data output value 1: Output 1 0: Output 0
1	GPO97	GPIO97_DOUT	GPIO 97 data output value 1: Output 1 0: Output 0
0	GPO96	GPIO96_DOUT	GPIO 96 data output value 1: Output 1 0: Output 0

0xA002_0400 GPIO data input register 0

GPIO_DIN0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO15	GPIO14	GPIO13	GPIO12	GPIO11	GPIO10	GPIO9	GPIO8	GPIO7	GPIO6	GPIO5	GPIO4	GPIO3	GPIO2	GPIO1	GPIO0
Type	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO
Reset	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

Bit(s)	Mnemonic	Name	Description
15	GPIO15	GPIO15_DIN	GPIO 15 data input value
14	GPIO14	GPIO14_DIN	GPIO 14 data input value
13	GPIO13	GPIO13_DIN	GPIO 13 data input value
12	GPIO12	GPIO12_DIN	GPIO 12 data input value
11	GPIO11	GPIO11_DIN	GPIO 11 data input value
10	GPIO10	GPIO10_DIN	GPIO 10 data input value
9	GPIO9	GPIO9_DIN	GPIO 9 data input value
8	GPIO8	GPIO8_DIN	GPIO 8 data input value
7	GPIO7	GPIO7_DIN	GPIO 7 data input value
6	GPIO6	GPIO6_DIN	GPIO 6 data input value
5	GPIO5	GPIO5_DIN	GPIO 5 data input value

4	GPIO4	GPIO4_DIN	GPIO 4 data input value
3	GPIO3	GPIO3_DIN	GPIO 3 data input value
2	GPIO2	GPIO2_DIN	GPIO 2 data input value
1	GPIO1	GPIO1_DIN	GPIO 1 data input value
0	GPIO0	GPIO0_DIN	GPIO 0 data input value

0xA002_0410 GPIO data input register 1

GPIO_DIN1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO31	GPIO30	GPIO29	GPIO28	GPIO27	GPIO26	GPIO25	GPIO24	GPIO23	GPIO22	GPIO21	GPIO20	GPIO19	GPIO18	GPIO17	GPIO16
Type	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO
Reset	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

Bit(s)	Mnemonic	Name	Description
15	GPIO31	GPIO31_DIN	GPIO 31 data input value
14	GPIO30	GPIO30_DIN	GPIO 30 data input value
13	GPIO29	GPIO29_DIN	GPIO 29 data input value
12	GPIO28	GPIO28_DIN	GPIO 28 data input value
11	GPIO27	GPIO27_DIN	GPIO 27 data input value
10	GPIO26	GPIO26_DIN	GPIO 26 data input value
9	GPIO25	GPIO25_DIN	GPIO 25 data input value
8	GPIO24	GPIO24_DIN	GPIO 24 data input value
7	GPIO23	GPIO23_DIN	GPIO 23 data input value
6	GPIO22	GPIO22_DIN	GPIO 22 data input value
5	GPIO21	GPIO21_DIN	GPIO 21 data input value
4	GPIO20	GPIO20_DIN	GPIO 20 data input value
3	GPIO19	GPIO19_DIN	GPIO 19 data input value
2	GPIO18	GPIO18_DIN	GPIO 18 data input value
1	GPIO17	GPIO17_DIN	GPIO 17 data input value
0	GPIO16	GPIO16_DIN	GPIO 16 data input value

0xA002_0420 GPIO data input register 2

GPIO_DIN2

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO47	GPIO46	GPIO45	GPIO44	GPIO43	GPIO42	GPIO41	GPIO40	GPIO39	GPIO38	GPIO37	GPIO36	GPIO35	GPIO34	GPIO33	GPIO32
Type	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO
Reset	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

Bit(s)	Mnemonic	Name	Description
15	GPIO47	GPIO47_DIN	GPIO 47 data input value
14	GPIO46	GPIO46_DIN	GPIO 46 data input value
13	GPIO45	GPIO45_DIN	GPIO 45 data input value
12	GPIO44	GPIO44_DIN	GPIO 44 data input value
11	GPIO43	GPIO43_DIN	GPIO 43 data input value
10	GPIO42	GPIO42_DIN	GPIO 42 data input value
9	GPIO41	GPIO41_DIN	GPIO 41 data input value
8	GPIO40	GPIO40_DIN	GPIO 40 data input value
7	GPIO39	GPIO39_DIN	GPIO 39 data input value
6	GPIO38	GPIO38_DIN	GPIO 38 data input value
5	GPIO37	GPIO37_DIN	GPIO 37 data input value
4	GPIO36	GPIO36_DIN	GPIO 36 data input value
3	GPIO35	GPIO35_DIN	GPIO 35 data input value
2	GPIO34	GPIO34_DIN	GPIO 34 data input value
1	GPIO33	GPIO33_DIN	GPIO 33 data input value
0	GPIO32	GPIO32_DIN	GPIO 32 data input value

0xA002_0430 GPIO data input register 3 GPIO_DIN3

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO 63	GPIO 62	GPIO 61	GPIO 60	GPIO 59	GPIO 58	GPIO 57	GPIO 56	GPIO 55	GPIO 54	GPIO 53	GPIO 52	GPIO 51	GPIO 50	GPIO 49	GPIO 48
Type	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO
Reset	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

Bit(s)	Mnemonic	Name	Description
15	GPIO63	GPIO63_DIN	GPIO 63 data input value
14	GPIO62	GPIO62_DIN	GPIO 62 data input value
13	GPIO61	GPIO61_DIN	GPIO 61 data input value
12	GPIO60	GPIO60_DIN	GPIO 60 data input value
11	GPIO59	GPIO59_DIN	GPIO 59 data input value
10	GPIO58	GPIO58_DIN	GPIO 58 data input value
9	GPIO57	GPIO57_DIN	GPIO 57 data input value
8	GPIO56	GPIO56_DIN	GPIO 56 data input value
7	GPIO55	GPIO55_DIN	GPIO 55 data input value
6	GPIO54	GPIO54_DIN	GPIO 54 data input value
5	GPIO53	GPIO53_DIN	GPIO 53 data input value
4	GPIO52	GPIO52_DIN	GPIO 52 data input value
3	GPIO51	GPIO51_DIN	GPIO 51 data input value
2	GPIO50	GPIO50_DIN	GPIO 50 data input value

Bit(s)	Mnemonic	Name	Description
1	GPIO49	GPIO49_DIN	GPIO 49 data input value
0	GPIO48	GPIO48_DIN	GPIO 48 data input value

0xA002_0440 **GPIO data input register 4**

GPIO_DIN4

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO79	GPIO78	GPIO77	GPIO76	GPIO75	GPIO74	GPIO73	GPIO72	GPIO71	GPIO70	GPIO69	GPIO68	GPIO67	GPIO66	GPIO65	GPIO64
Type	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO
Reset	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

Bit(s)	Mnemonic	Name	Description
15	GPIO79	GPIO79_DIN	GPIO 79 data input value
14	GPIO78	GPIO78_DIN	GPIO 78 data input value
13	GPIO77	GPIO77_DIN	GPIO 77 data input value
12	GPIO76	GPIO76_DIN	GPIO 76 data input value
11	GPIO75	GPIO75_DIN	GPIO 75 data input value
10	GPIO74	GPIO74_DIN	GPIO 74 data input value
9	GPIO73	GPIO73_DIN	GPIO 73 data input value
8	GPIO72	GPIO72_DIN	GPIO 72 data input value
7	GPIO71	GPIO71_DIN	GPIO 71 data input value
6	GPIO70	GPIO70_DIN	GPIO 70 data input value
5	GPIO69	GPIO69_DIN	GPIO 69 data input value
4	GPIO68	GPIO68_DIN	GPIO 68 data input value
3	GPIO67	GPIO67_DIN	GPIO 67 data input value
2	GPIO66	GPIO66_DIN	GPIO 66 data input value
1	GPIO65	GPIO65_DIN	GPIO 65 data input value
0	GPIO64	GPIO64_DIN	GPIO 64 data input value

0xA002_0450 **GPIO data input register 5**

GPIO_DIN5

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO95	GPIO94	GPIO93	GPIO92	GPIO91	GPIO90	GPIO89	GPIO88	GPIO87	GPIO86	GPIO85	GPIO84	GPIO83	GPIO82	GPIO81	GPIO80
Type	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO
Reset	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

Bit(s)	Mnemonic	Name	Description
15	GPIO95	GPIO95_DIN	GPIO 95 data input value

Bit(s)	Mnemonic	Name	Description
14	GPIO94	GPIO94_DIN	GPIO 94 data input value
13	GPIO93	GPIO93_DIN	GPIO 93 data input value
12	GPIO92	GPIO92_DIN	GPIO 92 data input value
11	GPIO91	GPIO91_DIN	GPIO 91 data input value
10	GPIO90	GPIO90_DIN	GPIO 90 data input value
9	GPIO89	GPIO89_DIN	GPIO 89 data input value
8	GPIO88	GPIO88_DIN	GPIO 88 data input value
7	GPIO87	GPIO87_DIN	GPIO 87 data input value
6	GPIO86	GPIO86_DIN	GPIO 86 data input value
5	GPIO85	GPIO85_DIN	GPIO 85 data input value
4	GPIO84	GPIO84_DIN	GPIO 84 data input value
3	GPIO83	GPIO83_DIN	GPIO 83 data input value
2	GPIO82	GPIO82_DIN	GPIO 82 data input value
1	GPIO81	GPIO81_DIN	GPIO 81 data input value
0	GPIO80	GPIO80_DIN	GPIO 80 data input value

0xA002_0460 GPIO data input register 6 **GPIO_DIN6**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name								GPIO 104	GPIO 103	GPIO 102	GPIO 101	GPIO 100	GPIO 99	GPIO 98	GPIO 97	GPIO 96
Type								RO	RO	RO	RO	RO	RO	RO	RO	RO
Reset								X	X	X	X	X	X	X	X	X

Bit(s)	Mnemonic	Name	Description
8	GPIO104	GPIO104_DIN	GPIO 104 data input value
7	GPIO103	GPIO103_DIN	GPIO 103 data input value
6	GPIO102	GPIO102_DIN	GPIO 102 data input value
5	GPIO101	GPIO101_DIN	GPIO 101 data input value
4	GPIO100	GPIO100_DIN	GPIO 100 data input value
3	GPIO99	GPIO99_DIN	GPIO 99 data input value
2	GPIO98	GPIO98_DIN	GPIO 98 data input value
1	GPIO97	GPIO97_DIN	GPIO 97 data input value
0	GPIO96	GPIO96_DIN	GPIO 96 data input value

0xA002_0500 GPIO pull-up/down select 0 **GPIO_PULLSEL0**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
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Name	GPIO 15	GPIO 14	GPIO 13	GPIO 12	GPIO 11	GPIO 10	GPIO 9	GPIO 8	GPIO 7	GPIO 6	GPIO 5	GPIO 4	GPIO 3	GPIO 2	GPIO 1	GPIO 0
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
15	GPIO15	GPIO15_PULLSEL	GPIO 15 pull-up/down selection 0: Pull-down 1: Pull-up
14	GPIO14	GPIO14_PULLSEL	GPIO 14 pull-up/down selection 0: Pull-down 1: Pull-up
13	GPIO13	GPIO13_PULLSEL	GPIO 13 pull-up/down selection 0: Pull-down 1: Pull-up
12	GPIO12	GPIO12_PULLSEL	GPIO 12 pull-up/down selection 0: Pull-down 1: Pull-up
11	GPIO11	GPIO11_PULLSEL	GPIO 11 pull-up/down selection 0: Pull-down 1: Pull-up
10	GPIO10	GPIO10_PULLSEL	GPIO 10 pull-up/down selection 0: Pull-down 1: Pull-up
9	GPIO9	GPIO9_PULLSEL	GPIO 9 pull-up/down selection 0: Pull-down 1: Pull-up
8	GPIO8	GPIO8_PULLSEL	GPIO 8 pull-up/down selection 0: Pull-down 1: Pull-up
7	GPIO7	GPIO7_PULLSEL	GPIO 7 pull-up/down selection 0: Pull-down 1: Pull-up
6	GPIO6	GPIO6_PULLSEL	GPIO 6 pull-up/down selection 0: Pull-down 1: Pull-up
5	GPIO5	GPIO5_PULLSEL	GPIO 5 pull-up/down selection 0: Pull-down 1: Pull-up
4	GPIO4	GPIO4_PULLSEL	GPIO 4 pull-up/down selection 0: Pull-down 1: Pull-up
3	GPIO3	GPIO3_PULLSEL	GPIO 3 pull-up/down selection 0: Pull-down 1: Pull-up
2	GPIO2	GPIO2_PULLSEL	GPIO 2 pull-up/down selection 0: Pull-down

Bit(s)	Mnemonic	Name	Description
1	GPIO1	GPIO1_PULLSEL	GPIO 1 pull-up/down selection 1: Pull-up 0: Pull-down
0	GPIO0	GPIO0_PULLSEL	GPIO 0 pull-up/down selection 1: Pull-up 0: Pull-down

0xA002_0510 GPIO pull-up/down select 1

GPIO_PULLSEL1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO31	GPIO30	GPIO29	GPIO28	GPIO27	GPIO26	GPIO25	GPIO24	GPIO23	GPIO22	GPIO21	GPIO20	GPIO19	GPIO18	GPIO17	GPIO16
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0

Bit(s)	Mnemonic	Name	Description
15	GPIO31	GPIO31_PULLSEL	GPIO 31 pull-up/down selection 0: Pull-down 1: Pull-up
14	GPIO30	GPIO30_PULLSEL	GPIO 30 pull-up/down selection 0: Pull-down 1: Pull-up
13	GPIO29	GPIO29_PULLSEL	GPIO 29 pull-up/down selection 0: Pull-down 1: Pull-up
12	GPIO28	GPIO28_PULLSEL	GPIO 28 pull-up/down selection 0: Pull-down 1: Pull-up
11	GPIO27	GPIO27_PULLSEL	GPIO 27 pull-up/down selection 0: Pull-down 1: Pull-up
10	GPIO26	GPIO26_PULLSEL	GPIO 26 pull-up/down selection 0: Pull-down 1: Pull-up
9	GPIO25	GPIO25_PULLSEL	GPIO 25 pull-up/down selection 0: Pull-down 1: Pull-up
8	GPIO24	GPIO24_PULLSEL	GPIO 24 pull-up/down selection 0: Pull-down 1: Pull-up
7	GPIO23	GPIO23_PULLSEL	GPIO 23 pull-up/down selection 0: Pull-down 1: Pull-up

Bit(s)	Mnemonic	Name	Description
6	GPIO22	GPIO22_PULLSEL	GPIO 22 pull-up/down selection 0: Pull-down 1: Pull-up
5	GPIO21	GPIO21_PULLSEL	GPIO 21 pull-up/down selection 0: Pull-down 1: Pull-up
4	GPIO20	GPIO20_PULLSEL	GPIO 20 pull-up/down selection 0: Pull-down 1: Pull-up
3	GPIO19	GPIO19_PULLSEL	GPIO 19 pull-up/down selection 0: Pull-down 1: Pull-up
2	GPIO18	GPIO18_PULLSEL	GPIO 18 pull-up/down selection 0: Pull-down 1: Pull-up
1	GPIO17	GPIO17_PULLSEL	GPIO 17 pull-up/down selection 0: Pull-down 1: Pull-up
0	GPIO16	GPIO16_PULLSEL	GPIO 16 pull-up/down selection 0: Pull-down 1: Pull-up

0xA002_0520 GPIO pull-up/down select 2 GPIO_PULLSEL2

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO 47	GPIO 46	GPIO 45	GPIO 44	GPIO 43	GPIO 42	GPIO 41	GPIO 40	GPIO 39	GPIO 38	GPIO 37	GPIO 36	GPIO 35	GPIO 34	GPIO 33	GPIO 32
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
15	GPIO47	GPIO47_PULLSEL	GPIO 47 pull-up/down selection 0: Pull-down 1: Pull-up
14	GPIO46	GPIO46_PULLSEL	GPIO 46 pull-up/down selection 0: Pull-down 1: Pull-up
13	GPIO45	GPIO45_PULLSEL	GPIO 45 pull-up/down selection 0: Pull-down 1: Pull-up
12	GPIO44	GPIO44_PULLSEL	GPIO 44 pull-up/down selection 0: Pull-down 1: Pull-up

Bit(s)	Mnemonic	Name	Description
11	GPIO43	GPIO43_PULLSEL	GPIO 43 pull-up/down selection 0: Pull-down 1: Pull-up
10	GPIO42	GPIO42_PULLSEL	GPIO 42 pull-up/down selection 0: Pull-down 1: Pull-up
9	GPIO41	GPIO41_PULLSEL	GPIO 41 pull-up/down selection 0: Pull-down 1: Pull-up
8	GPIO40	GPIO40_PULLSEL	GPIO 40 pull-up/down selection 0: Pull-down 1: Pull-up
7	GPIO39	GPIO39_PULLSEL	GPIO 39 pull-up/down selection 0: Pull-down 1: Pull-up
6	GPIO38	GPIO38_PULLSEL	GPIO 38 pull-up/down selection 0: Pull-down 1: Pull-up
5	GPIO37	GPIO37_PULLSEL	GPIO 37 pull-up/down selection 0: Pull-down 1: Pull-up
4	GPIO36	GPIO36_PULLSEL	GPIO 36 pull-up/down selection 0: Pull-down 1: Pull-up
3	GPIO35	GPIO35_PULLSEL	GPIO 35 pull-up/down selection 0: Pull-down 1: Pull-up
2	GPIO34	GPIO34_PULLSEL	GPIO 34 pull-up/down selection 0: Pull-down 1: Pull-up
1	GPIO33	GPIO33_PULLSEL	GPIO 33 pull-up/down selection 0: Pull-down 1: Pull-up
0	GPIO32	GPIO32_PULLSEL	GPIO 32 pull-up/down selection 0: Pull-down 1: Pull-up

0xA002_0530 GPIO pull-up/down select 3 **GPIO_PULLSEL3**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO 63	GPIO 62	GPIO 61	GPIO 60	GPIO 59	GPIO 58	GPIO 57	GPIO 56	GPIO 55	GPIO 54	GPIO 53	GPIO 52	GPIO 51	GPIO 50	GPIO 49	GPIO 48
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
15	GPIO63	GPIO63_PULLSEL	GPIO 63 pull-up/down selection 0: Pull-down 1: Pull-up
14	GPIO62	GPIO62_PULLSEL	GPIO 62 pull-up/down selection 0: Pull-down 1: Pull-up
13	GPIO61	GPIO61_PULLSEL	GPIO 61 pull-up/down selection 0: Pull-down 1: Pull-up
12	GPIO60	GPIO60_PULLSEL	GPIO 60 pull-up/down selection 0: Pull-down 1: Pull-up
11	GPIO59	GPIO59_PULLSEL	GPIO 59 pull-up/down selection 0: Pull-down 1: Pull-up
10	GPIO58	GPIO58_PULLSEL	GPIO 58 pull-up/down selection 0: Pull-down 1: Pull-up
9	GPIO57	GPIO57_PULLSEL	GPIO 57 pull-up/down selection 0: Pull-down 1: Pull-up
8	GPIO56	GPIO56_PULLSEL	GPIO 56 pull-up/down selection 0: Pull-down 1: Pull-up
7	GPIO55	GPIO55_PULLSEL	GPIO 55 pull-up/down selection 0: Pull-down 1: Pull-up
6	GPIO54	GPIO54_PULLSEL	GPIO 54 pull-up/down selection 0: Pull-down 1: Pull-up
5	GPIO53	GPIO53_PULLSEL	GPIO 53 pull-up/down selection 0: Pull-down 1: Pull-up
4	GPIO52	GPIO52_PULLSEL	GPIO 52 pull-up/down selection 0: Pull-down 1: Pull-up
3	GPIO51	GPIO51_PULLSEL	GPIO 51 pull-up/down selection 0: Pull-down 1: Pull-up
2	GPIO50	GPIO50_PULLSEL	GPIO 50 pull-up/down selection 0: Pull-down 1: Pull-up
1	GPIO49	GPIO49_PULLSEL	GPIO 49 pull-up/down selection 0: Pull-down 1: Pull-up

Bit(s)	Mnemonic	Name	Description
0	GPIO48	GPIO48_PULLSEL	GPIO 48 pull-up/down selection 0: Pull-down 1: Pull-up

0xA002_0540 GPIO pull-up/down select 4

GPIO_PULLSEL4

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO79	GPIO78	GPIO77	GPIO76	GPIO75	GPIO74	GPIO73	GPIO72	GPIO71	GPIO70	GPIO69	GPIO68	GPIO67	GPIO66	GPIO65	GPIO64
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
15	GPIO79	GPIO79_PULLSEL	GPIO 79 pull-up/down selection 0: Pull-down 1: Pull-up
14	GPIO78	GPIO78_PULLSEL	GPIO 78 pull-up/down selection 0: Pull-down 1: Pull-up
13	GPIO77	GPIO77_PULLSEL	GPIO 77 pull-up/down selection 0: Pull-down 1: Pull-up
12	GPIO76	GPIO76_PULLSEL	GPIO 76 pull-up/down selection 0: Pull-down 1: Pull-up
11	GPIO75	GPIO75_PULLSEL	GPIO 75 pull-up/down selection 0: Pull-down 1: Pull-up
10	GPIO74	GPIO74_PULLSEL	GPIO 74 pull-up/down selection 0: Pull-down 1: Pull-up
9	GPIO73	GPIO73_PULLSEL	GPIO 73 pull-up/down selection 0: Pull-down 1: Pull-up
8	GPIO72	GPIO72_PULLSEL	GPIO 72 pull-up/down selection 0: Pull-down 1: Pull-up
7	GPIO71	GPIO71_PULLSEL	GPIO 71 pull-up/down selection 0: Pull-down 1: Pull-up
6	GPIO70	GPIO70_PULLSEL	GPIO 70 pull-up/down selection 0: Pull-down 1: Pull-up

Bit(s)	Mnemonic	Name	Description
5	GPIO69	GPIO69_PULLSEL	GPIO 69 pull-up/down selection 0: Pull-down 1: Pull-up
4	GPIO68	GPIO68_PULLSEL	GPIO 68 pull-up/down selection 0: Pull-down 1: Pull-up
3	GPIO67	GPIO67_PULLSEL	GPIO 67 pull-up/down selection 0: Pull-down 1: Pull-up
2	GPIO66	GPIO66_PULLSEL	GPIO 66 pull-up/down selection 0: Pull-down 1: Pull-up
1	GPIO65	GPIO65_PULLSEL	GPIO 65 pull-up/down selection 0: Pull-down 1: Pull-up
0	GPIO64	GPIO64_PULLSEL	GPIO 64 pull-up/down selection 0: Pull-down 1: Pull-up

0xA002_0550 GPIO pull-up/down select 5

GPIO_PULLSEL5

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO95	GPIO94	GPIO93	GPIO92	GPIO91	GPIO90	GPIO89	GPIO88	GPIO87	GPIO86	GPIO85	GPIO84	GPIO83	GPIO82	GPIO81	GPIO80
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
15	GPIO95	GPIO95_PULLSEL	GPIO 95 pull-up/down selection 0: Pull-down 1: Pull-up
14	GPIO94	GPIO94_PULLSEL	GPIO 94 pull-up/down selection 0: Pull-down 1: Pull-up
13	GPIO93	GPIO93_PULLSEL	GPIO 93 pull-up/down selection 0: Pull-down 1: Pull-up
12	GPIO92	GPIO92_PULLSEL	GPIO 92 pull-up/down selection 0: Pull-down 1: Pull-up
11	GPIO91	GPIO91_PULLSEL	GPIO 91 pull-up/down selection 0: Pull-down 1: Pull-up

Bit(s)	Mnemonic	Name	Description
10	GPIO90	GPIO90_PULLSEL	GPIO 90 pull-up/down selection 0: Pull-down 1: Pull-up
9	GPIO89	GPIO89_PULLSEL	GPIO 89 pull-up/down selection 0: Pull-down 1: Pull-up
8	GPIO88	GPIO88_PULLSEL	GPIO 88 pull-up/down selection 0: Pull-down 1: Pull-up
7	GPIO87	GPIO87_PULLSEL	GPIO 87 pull-up/down selection 0: Pull-down 1: Pull-up
6	GPIO86	GPIO86_PULLSEL	GPIO 86 pull-up/down selection 0: Pull-down 1: Pull-up
5	GPIO85	GPIO85_PULLSEL	GPIO 85 pull-up/down selection 0: Pull-down 1: Pull-up
4	GPIO84	GPIO84_PULLSEL	GPIO 84 pull-up/down selection 0: Pull-down 1: Pull-up
3	GPIO83	GPIO83_PULLSEL	GPIO 83 pull-up/down selection 0: Pull-down 1: Pull-up
2	GPIO82	GPIO82_PULLSEL	GPIO 82 pull-up/down selection 0: Pull-down 1: Pull-up
1	GPIO81	GPIO81_PULLSEL	GPIO 81 pull-up/down selection 0: Pull-down 1: Pull-up
0	GPIO80	GPIO80_PULLSEL	GPIO 80 pull-up/down selection 0: Pull-down 1: Pull-up

0xA002_0560 **GPIO pull-up/down select 6**

GPIO_PULLSE
L6

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name								GPIO 104	GPIO 103	GPIO 102	GPIO 101	GPIO 100	GPIO 99	GPIO 98	GPIO 97	GPIO 96
Type								R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset								0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
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Bit(s)	Mnemonic	Name	Description
8	GPIO104	GPIO104_PULLSEL	GPIO 104 pull-up/down selection 0: Pull-down 1: Pull-up
7	GPIO103	GPIO103_PULLSEL	GPIO 103 pull-up/down selection 0: Pull-down 1: Pull-up
6	GPIO102	GPIO102_PULLSEL	GPIO 102 pull-up/down selection 0: Pull-down 1: Pull-up
5	GPIO101	GPIO101_PULLSEL	GPIO 101 pull-up/down selection 0: Pull-down 1: Pull-up
4	GPIO100	GPIO100_PULLSEL	GPIO 100 pull-up/down selection 0: Pull-down 1: Pull-up
3	GPIO99	GPIO99_PULLSEL	GPIO 99 pull-up/down selection 0: Pull-down 1: Pull-up
2	GPIO98	GPIO98_PULLSEL	GPIO 98 pull-up/down selection 0: Pull-down 1: Pull-up
1	GPIO97	GPIO97_PULLSEL	GPIO 97 pull-up/down selection 0: Pull-down 1: Pull-up
0	GPIO96	GPIO96_PULLSEL	GPIO 96 pull-up/down selection 0: Pull-down 1: Pull-up

0xA002_0600 GPIO Schmit trigger control 0

GPIO_SMT0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO_SMT[15:0]															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
15	GPIO_SMT	GPIO_SMTEN	Enables GPIO 102 Schmitt trigger 0: Disable 1: Enable
14	GPIO_SMT	GPIO_SMTEN	Enables GPIO 101 Schmitt trigger 0: Disable 1: Enable
13	GPIO_SMT	GPIO_SMTEN	Enables GPIO 100 Schmitt trigger 0: Disable

Bit(s)	Mnemonic Name	Description
12	GPIO_SMT GPIO_SMTEN	Enables GPIO 99 Schmitt trigger 1: Enable 0: Disable
11	GPIO_SMT D2D_SMTEN	Enables D2D_ANA_CON_SCK Schmitt trigger 1: Enable 0: Disable
10	GPIO_SMT D2D_SMTEN	Enables D2D_ANA_CON_SDA0 Schmitt trigger 1: Enable 0: Disable
9	GPIO_SMT D2D_SMTEN	Enables D2D_ANA_CON_SDA1 Schmitt trigger 1: Enable 0: Disable
8	GPIO_SMT GPIO_SMTEN	Enables GPIO 98 Schmitt trigger 1: Enable 0: Disable
7	GPIO_SMT GPIO_SMTEN	Enables GPIO 97 Schmitt trigger 1: Enable 0: Disable
6	GPIO_SMT GPIO_SMTEN	Enables GPIO 96 Schmitt trigger 1: Enable 0: Disable
5	GPIO_SMT GPIO_SMTEN	Enables GPIO 95 Schmitt trigger 1: Enable 0: Disable
4	GPIO_SMT GPIO_SMTEN	Enables GPIO 94 Schmitt trigger 1: Enable 0: Disable
3	GPIO_SMT GPIO_SMTEN	Enables GPIO 93 Schmitt trigger 1: Enable 0: Disable
2	GPIO_SMT GPIO_SMTEN	Enables GPIO 76 Schmitt trigger 1: Enable 0: Disable
1	GPIO_SMT GPIO_SMTEN	Enables GPIO 75 Schmitt trigger 1: Enable 0: Disable
0	GPIO_SMT GPIO_SMTEN	Enables GPIO 73 Schmitt trigger 1: Enable 0: Disable

0xA002_0610 GPIO Schmitt trigger control 1

GPIO_SMT1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO_SMT[31:16]															

Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic Name	Description
31	GPIO_SMT GPIO_SMTEN	Enables GPIO 72 Schmitt trigger 0: Disable 1: Enable
30	GPIO_SMT GPIO_SMTEN	Enables GPIO 71 Schmitt trigger 0: Disable 1: Enable
29	GPIO_SMT GPIO_SMTEN	Enables GPIO 67 Schmitt trigger 0: Disable 1: Enable
28	GPIO_SMT GPIO_SMTEN	Enables GPIO 68 Schmitt trigger 0: Disable 1: Enable
27	GPIO_SMT GPIO_SMTEN	Enables GPIO 70 Schmitt trigger 0: Disable 1: Enable
26	GPIO_SMT Reserved	Reserved
25	GPIO_SMT Reserved	Reserved
24	GPIO_SMT Reserved	Reserved
23	GPIO_SMT Reserved	Reserved
22	GPIO_SMT GPIO_SMTEN	Enables GPIO 103 Schmitt trigger 0: Disable 1: Enable
21	GPIO_SMT GPIO_SMTEN	Enables GPIO 104 Schmitt trigger 0: Disable 1: Enable
20	GPIO_SMT D2D_SMTEN	Enables D2D_BB_SRCLKENA Schmitt trigger 0: Disable 1: Enable
19	GPIO_SMT D2D_SMTEN	Enables D2D_RGU_RESETB Schmitt trigger 0: Disable 1: Enable
18	GPIO_SMT D2D_SMTEN	Enables D2D_SYS_RESETB Schmitt trigger 0: Disable 1: Enable
17	GPIO_SMT D2D_SMTEN	Enables D2D_CLK_128K Schmitt trigger 0: Disable 1: Enable
16	GPIO_SMT D2D_SMTEN	Enables D2D_XOSC_32K_CK Schmitt trigger 0: Disable 1: Enable

0xA002_0620 GPIO Schmit trigger control 2

GPIO_SMT2

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO_SMT[47:32]															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic Name	Description
47	GPIO_SMT GPIO_SMTEN	Enables GPIO 27 Schmitt trigger 0: Disable 1: Enable
46	GPIO_SMT GPIO_SMTEN	Enables GPIO 77/78/79/80 Schmitt trigger 0: Disable 1: Enable
45	GPIO_SMT GPIO_SMTEN	Enables GPIO 25/26 Schmitt trigger 0: Disable 1: Enable
44	GPIO_SMT GPIO_SMTEN	Enables GPIO 23/24 Schmitt trigger 0: Disable 1: Enable
43	GPIO_SMT GPIO_SMTEN	Enables GPIO 52/53/54/55/56/57/58/59/60/61/62/63/64 Schmitt trigger 0: Disable 1: Enable
42	GPIO_SMT GPIO_SMTEN	Enables GPIO 33/34/35/36/37/38/39/40/41/42/44/46/47/48/50/51 Schmitt trigger 0: Disable 1: Enable
41	GPIO_SMT GPIO_SMTEN	Enables GPIO 1/2/28 Schmitt trigger 0: Disable 1: Enable
40	GPIO_SMT GPIO_SMTEN	Enables GPIO 3/4/5/9/11/12/13/16/17 Schmitt trigger 0: Disable 1: Enable
39	GPIO_SMT GPIO_SMTEN	Enables GPIO 22/66 Schmitt trigger 0: Disable 1: Enable
38	GPIO_SMT GPIO_SMTEN	Enables GPIO 0/19/20 Schmitt trigger 0: Disable 1: Enable
37	GPIO_SMT GPIO_SMTEN	Enables GPIO 74/81/82/83/84/85/86 Schmitt trigger 0: Disable 1: Enable
36	GPIO_SMT D2D_SMTEN	Enables D2D_SFSOUT/D2D_SFSCCK/D2D_SFSIN D2D_SFSWP/D2D_SFSCS/D2D_SFSHOLD Schmitt trigger 0: Disable 1: Enable

Bit(s)	Mnemonic Name	Description
35	GPIO_SMT GPIO_SMTEN	Enables GPIO 31 Schmitt trigger 0: Disable 1: Enable
34	GPIO_SMT GPIO_SMTEN	Enables GPIO 32 Schmitt trigger 0: Disable 1: Enable
33	GPIO_SMT GPIO_SMTEN	Enables GPIO 30 Schmitt trigger 0: Disable 1: Enable
32	GPIO_SMT GPIO_SMTEN	Enables GPIO 69 Schmitt trigger 0: Disable 1: Enable

0xA002_0630 GPIO Schmitt trigger control 3

GPIO_SMT3

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO_SMT[63:48]															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic Name	Description
63	GPIO_SMT Reserved	Reserved
62	GPIO_SMT Reserved	Reserved
61	GPIO_SMT Reserved	Reserved
60	GPIO_SMT Reserved	Reserved
59	GPIO_SMT Reserved	Reserved
58	GPIO_SMT Reserved	Reserved
57	GPIO_SMT Reserved	Reserved
56	GPIO_SMT Reserved	Reserved
55	GPIO_SMT Reserved	Reserved
54	GPIO_SMT Reserved	Reserved
53	GPIO_SMT Reserved	Reserved
52	GPIO_SMT Reserved	Reserved
51	GPIO_SMT GPIO_SMTEN	Enables GPIO 87/88/89/90/91/92 Schmitt trigger 0: Disable 1: Enable
50	GPIO_SMT GPIO_SMTEN	Enables GPIO 49 Schmitt trigger 0: Disable 1: Enable
49	GPIO_SMT GPIO_SMTEN	Enables GPIO 29 Schmitt trigger 0: Disable 1: Enable

Bit(s)	Mnemonic Name	Description
48	GPIO_SMT GPIO_SMTEN	Enables GPIO 21 Schmitt trigger 0: Disable 1: Enable

GPIO_n Enables Rx input buffer Schmitt trigger hysteresis control. High asserted. SMT = 1, Schmitt trigger enabled
0 Disable
1 Enable

0xA002_0700 GPIO slew rate control 0 **GPIO_SR0**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO_SR[15:0]															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit(s)	Mnemonic Name	Description
15	GPIO_SR GPIO_SREN	Enables GPIO 102 slew rate 0: Disable 1: Enable
14	GPIO_SR GPIO_SREN	Enables GPIO 101 slew rate 0: Disable 1: Enable
13	GPIO_SR GPIO_SREN	Enables GPIO 100 slew rate 0: Disable 1: Enable
12	GPIO_SR GPIO_SREN	Enables GPIO 99 slew rate 0: Disable 1: Enable
11	GPIO_SR D2D_SREN	Enables D2D_ANA_CON_SCK slew rate 0: Disable 1: Enable
10	GPIO_SR D2D_SREN	Enables D2D_ANA_CON_SDA0 slew rate 0: Disable 1: Enable
9	GPIO_SR D2D_SREN	Enables D2D_ANA_CON_SDA1 slew rate 0: Disable 1: Enable
8	GPIO_SR GPIO_SREN	Enables GPIO 98 slew rate 0: Disable 1: Enable
7	GPIO_SR GPIO_SREN	Enables GPIO 97 slew rate 0: Disable

Bit(s)	Mnemonic Name	Description
6	GPIO_SR GPIO_SREN	Enables GPIO 96 slew rate 1: Enable 0: Disable
5	GPIO_SR GPIO_SREN	Enables GPIO 95 slew rate 1: Enable 0: Disable
4	GPIO_SR GPIO_SREN	Enables GPIO 94 slew rate 1: Enable 0: Disable
3	GPIO_SR GPIO_SREN	Enables GPIO 93 slew rate 1: Enable 0: Disable
2	GPIO_SR GPIO_SREN	Enables GPIO 76 slew rate 1: Enable 0: Disable
1	GPIO_SR GPIO_SREN	Enables GPIO 75 slew rate 1: Enable 0: Disable
0	GPIO_SR GPIO_SREN	Enables GPIO 73 slew rate 1: Enable 0: Disable

0xA002_0710 GPIO slew rate control 1

GPIO_SR1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO_SR[15:0]															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit(s)	Mnemonic Name	Description
31	GPIO_SR GPIO_SREN	Enables GPIO 72 slew rate 1: Enable 0: Disable
30	GPIO_SR GPIO_SREN	Enables GPIO 71 slew rate 1: Enable 0: Disable
29	GPIO_SR GPIO_SREN	Enables GPIO 67 slew rate 1: Enable 0: Disable
28	GPIO_SR GPIO_SREN	Enables GPIO 68 slew rate 1: Enable 0: Disable
27	GPIO_SR GPIO_SREN	Enables GPIO 70 slew rate 0: Disable

Bit(s)	Mnemonic Name	Description
		1: Enable
26	GPIO_SR Reserved	Reserved
25	GPIO_SR Reserved	Reserved
24	GPIO_SR Reserved	Reserved
23	GPIO_SR Reserved	Reserved
22	GPIO_SR GPIO_SREN	Enables GPIO 103 slew rate 0: Disable 1: Enable
21	GPIO_SR GPIO_SREN	Enables GPIO 104 slew rate 0: Disable 1: Enable
20	GPIO_SR D2D_SREN	Enables D2D_BB_SRCLKENA slew rate 0: Disable 1: Enable
19	GPIO_SR D2D_SREN	Enables D2D_RGU_RESETB slew rate 0: Disable 1: Enable
18	GPIO_SR D2D_SREN	Enables D2D_SYS_RESETB slew rate 0: Disable 1: Enable
17	GPIO_SR D2D_SREN	Enables D2D_CLK_128K slew rate 0: Disable 1: Enable
16	GPIO_SR D2D_SREN	Enables D2D_XOSC_32K_CK slew rate 0: Disable 1: Enable

0xA002_0720 GPIO slew rate control 2

GPIO_SR2

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO_SR[15:0]															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit(s)	Mnemonic Name	Description
47	GPIO_SR GPIO_SREN	Enables GPIO 27 slew rate 0: Disable 1: Enable
46	GPIO_SR GPIO_SREN	Enables GPIO 77/78/79/80 slew rate 0: Disable 1: Enable

Bit(s)	Mnemonic	Name	Description
45	GPIO_SR	GPIO_SREN	Enables GPIO 25/26 slew rate 0: Disable 1: Enable
44	GPIO_SR	GPIO_SREN	Enables GPIO 23/24 slew rate 0: Disable 1: Enable
43	GPIO_SR	GPIO_SREN	Enables GPIO 52/53/54/55/56/57/58/59/60/61/62/63/64 slew rate 0: Disable 1: Enable
42	GPIO_SR	GPIO_SREN	Enables GPIO 33/34/35/36/37/38/39/40/41/42/44/46/47/48/50/51 slew rate 0: Disable 1: Enable
41	GPIO_SR	GPIO_SREN	Enables GPIO 1/2/28 slew rate 0: Disable 1: Enable
40	GPIO_SR	GPIO_SREN	Enables GPIO 3/4/5/9/11/12/13/16/17 slew rate 0: Disable 1: Enable
39	GPIO_SR	GPIO_SREN	Enables GPIO 22/66 slew rate 0: Disable 1: Enable
38	GPIO_SR	GPIO_SREN	Enables GPIO 0/19/20 slew rate 0: Disable 1: Enable
37	GPIO_SR	GPIO_SREN	Enables GPIO 74/81/82/83/84/85/86 slew rate 0: Disable 1: Enable
36	GPIO_SR	D2D_SREN	Enables D2D_SFSOUT/D2D_SFSCCK/D2D_SFSIN D2D_SFSWP/D2D_SFSCS/D2D_SFSHOLD slew rate 0: Disable 1: Enable
35	GPIO_SR	GPIO_SREN	Enables GPIO 31 slew rate 0: Disable 1: Enable
34	GPIO_SR	GPIO_SREN	Enables GPIO 32 slew rate 0: Disable 1: Enable
33	GPIO_SR	GPIO_SREN	Enables GPIO 30 slew rate 0: Disable 1: Enable
32	GPIO_SR	GPIO_SREN	Enables GPIO 69 slew rate 0: Disable 1: Enable

0xA002_0730 GPIO slew rate control 3

GPIO_SR3

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO_SR[15:0]															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit(s)	Mnemonic	Name	Description
63	GPIO_SR	Reserved	Reserved
62	GPIO_SR	Reserved	Reserved
61	GPIO_SR	Reserved	Reserved
60	GPIO_SR	Reserved	Reserved
59	GPIO_SR	Reserved	Reserved
58	GPIO_SR	Reserved	Reserved
57	GPIO_SR	Reserved	Reserved
56	GPIO_SR	Reserved	Reserved
55	GPIO_SR	Reserved	Reserved
54	GPIO_SR	Reserved	Reserved
53	GPIO_SR	Reserved	Reserved
52	GPIO_SR	Reserved	Reserved
51	GPIO_SR	GPIO_SREN	Enables GPIO 87/88/89/90/91/92 slew rate 0: Disable 1: Enable
50	GPIO_SR	GPIO_SREN	Enables GPIO 49 slew rate 0: Disable 1: Enable
49	GPIO_SR	GPIO_SREN	Enables GPIO 29 slew rate 0: Disable 1: Enable
48	GPIO_SR	GPIO_SREN	Enables GPIO 21 slew rate 0: Disable 1: Enable

GPIO_n Output slew rate control. High asserted. SR = 1, slower slew. SR = 0, no slew rate controlled.
0 Disable
1 Enable

0xA002_0800 GPIO driving control 0

GPIO_DRV0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO_DRV[15:0]															

Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	1	1	1	0	0	0	0	0	0	1	1

Bit(s)	Mnemonic Name	Description
14:13	GPIO_DRV IO_CFG_DRV	IO driving strength configuration For the IO list: GPIO3/4/5/6/7/8/9/10/11/12/13/14/15/16/17/18 0: 4mA 1: 8mA 2: 12mA 3: 16mA
12:11	GPIO_DRV IO_CFG_DRV	IO driving strength configuration For the IO list: GPIO0/19/20/22 0: 4mA 1: 8mA 2: 12mA 3: 16mA
10:8	GPIO_DRV IO_CFG_DRV	IO driving strength configuration For the IO list: GPIO74/81/82/83/84/85/86 0: 2mA 1: 4mA 2: 6mA 3: 8mA 4: 10mA 5: 12mA 6: 14mA 7: 16mA
7:6	GPIO_DRV IO_CFG_DRV	IO driving strength configuration For the IO list: GPIO30/31/32 0: 4mA 1: 8mA 2: 12mA 3: 16mA
5:4	GPIO_DRV IO_CFG_DRV	IO driving strength configuration For the IO list: GPIO69/72/71 0: 2mA 1: 4mA 2: 6mA 3: 8mA
3:2	GPIO_DRV IO_CFG_DRV	IO driving strength configuration For the IO list: GPIO67/68/70 0: 2mA 1: 4mA 2: 6mA 3: 8mA
1:0	GPIO_DRV IO_CFG_DRV	IO driving strength configuration For the IO list: GPIO73/75/76/93/94/95/96/97/98/99/100/101/102/103/104 0: 4mA 1: 8mA

Bit(s)	Mnemonic Name	Description
		2: 12mA
		3: 16mA

0xA002_0810 GPIO driving control 1

GPIO_DRV1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO_DRV[31:16]															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic Name	Description
30:28	GPIO_DRV Reserved	Reserved
27:25	GPIO_DRV IO_CFG_DRV	IO driving strength configuration For the IO list: GPIO27/28 1: 4mA 2: 6mA 3: 8mA 4: 10mA 5: 12mA 6: 14mA 7: 16mA
24:22	GPIO_DRV IO_CFG_DRV	IO driving strength configuration For the IO list: GPIO29 0: 2mA 1: 4mA 2: 6mA 3: 8mA 4: 10mA 5: 12mA 6: 14mA 7: 16mA
21:20	GPIO_DRV IO_CFG_DRV	IO driving strength configuration For the IO list: GPIO1/2 0: 4mA 1: 8mA 2: 12mA 3: 16mA
19:17	GPIO_DRV IO_CFG_DRV	IO driving strength configuration For the IO list: GPIO33/34/35/36/37/38/39/40/41/42/43/44/45/46/47/48/50/51 0: 2mA 1: 4mA 2: 6mA 3: 8mA 4: 10mA 5: 12mA

Bit(s)	Mnemonic Name	Description
		6: 14mA 7: 16mA
16:15	GPIO_DRV IO_CFG_DRV	IO driving strength configuration For the IO list: GPIO21/66 0: 4mA 1: 8mA 2: 12mA 3: 16mA

0xA002_0820 GPIO driving control 2

GPIO_DRV2

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO_DRV[47:32]															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic Name	Description
48:46	GPIO_DRV Reserved	IO driving strength configuration For the IO list: GPIO49 0: 2mA 1: 4mA 2: 6mA 3: 8mA 4: 10mA 5: 12mA 6: 14mA 7: 16mA
45:43	GPIO_DRV IO_CFG_DRV	IO driving strength configuration For the IO list: GPIO27/28 0: 2mA 1: 4mA 2: 6mA 3: 8mA 4: 10mA 5: 12mA 6: 14mA 7: 16mA
42:40	GPIO_DRV IO_CFG_DRV	IO driving strength configuration For the IO list: GPIO29 0: 2mA 1: 4mA 2: 6mA 3: 8mA 4: 10mA 5: 12mA 6: 14mA

Bit(s)	Mnemonic Name	Description
39:37	GPIO_DRV IO_CFG_DRV	IO driving strength configuration For the IO list: GPIO1/2 0: 4mA 1: 8mA 2: 12mA 3: 16mA
36:34	GPIO_DRV IO_CFG_DRV	IO driving strength configuration For the IO list: GPIO33/34/35/36/37/38/39/40/41/42/43/44/45/46/47/48/50/51 0: 2mA 1: 4mA 2: 6mA 3: 8mA 4: 10mA 5: 12mA 6: 14mA 7: 16mA
33:31	GPIO_DRV IO_CFG_DRV	IO driving strength configuration For the IO list: GPIO21/66 0: 4mA 1: 8mA 2: 12mA 3: 16mA

0xA002_0830 GPIO driving control 3 **GPIO_DRV3**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO_DRV[63:48]															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic Name	Description
56:55	GPIO_DRV IO_CFG_DRV	IO driving strength configuration For the IO list: GPIO77/78/79/80 0: 4mA 1: 8mA 2: 12mA 3: 16mA
54:53	GPIO_DRV IO_CFG_DRV	IO driving strength configuration For the IO list: GPIO87/88/89/90/91/92 0: 4mA 1: 8mA 2: 12mA 3: 16mA
52:51	GPIO_DRV IO_CFG_DRV	IO driving strength configuration

Bit(s)	Mnemonic Name	Description
		For the IO list: GPIO23/24/25/26 0: 4mA 1: 8mA 2: 12mA 3: 16mA
50:49	GPIO_DRV IO_CFG_DRV	IO driving strength configuration For the IO list: GPIO52/53/54/55/56/57/58/59/60/61/62/63/64 0: 4mA 1: 8mA 2: 12mA 3: 16mA

GPIO_n GPIO driving current control
2-bit driving as the following

00 2mA
01 4mA
10 6mA
11 8mA

3-bit driving as the following

000 2mA
001 4mA
010 6mA
011 8mA
100 10mA
101 12mA
110 14mA
111 16mA

0xA002_0900 **GPIO Rx input buffer control 0**

GPIO_IES0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO_IES[15:0]															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit(s)	Mnemonic Name	Description
15	GPIO_IES GPIO_IEN	Enables GPIO 102 Rx input buffer 0: Disable 1: Enable
14	GPIO_IES GPIO_IEN	Enables GPIO 101 Rx input buffer 0: Disable 1: Enable
13	GPIO_IES GPIO_IEN	Enables GPIO 100 Rx input buffer 0: Disable

Bit(s)	Mnemonic	Name	Description
12	GPIO_IES	GPIO_IENSEN	Enables GPIO 99 Rx input buffer 0: Disable 1: Enable
11	GPIO_IES	D2D_IENSEN	Enables D2D_ANA_CON_SCK Rx input buffer 0: Disable 1: Enable
10	GPIO_IES	D2D_IENSEN	Enables D2D_ANA_CON_SDA0 Rx input buffer 0: Disable 1: Enable
9	GPIO_IES	D2D_IENSEN	Enables D2D_ANA_CON_SDA1 Rx input buffer 0: Disable 1: Enable
8	GPIO_IES	GPIO_IENSEN	Enables GPIO 98 Rx input buffer 0: Disable 1: Enable
7	GPIO_IES	GPIO_IENSEN	Enables GPIO 97 Rx input buffer 0: Disable 1: Enable
6	GPIO_IES	GPIO_IENSEN	Enables GPIO 96 Rx input buffer 0: Disable 1: Enable
5	GPIO_IES	GPIO_IENSEN	Enables GPIO 95 Rx input buffer 0: Disable 1: Enable
4	GPIO_IES	GPIO_IENSEN	Enables GPIO 94 Rx input buffer 0: Disable 1: Enable
3	GPIO_IES	GPIO_IENSEN	Enables GPIO 93 Rx input buffer 0: Disable 1: Enable
2	GPIO_IES	GPIO_IENSEN	Enables GPIO 76 Rx input buffer 0: Disable 1: Enable
1	GPIO_IES	GPIO_IENSEN	Enables GPIO 75 Rx input buffer 0: Disable 1: Enable
0	GPIO_IES	GPIO_IENSEN	Enables GPIO 73 Rx input buffer 0: Disable 1: Enable

0xA002_0910 GPIO Rx input buffer control 1

GPIO_IES1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO_IES[31:16]															

Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit(s)	Mnemonic	Name	Description
31	GPIO_IES	GPIO_IENSEN	Enables GPIO 72 Rx input buffer 0: Disable 1: Enable
30	GPIO_IES	GPIO_IENSEN	Enables GPIO 71 Rx input buffer 0: Disable 1: Enable
29	GPIO_IES	GPIO_IENSEN	Enables GPIO 67 Rx input buffer 0: Disable 1: Enable
28	GPIO_IES	GPIO_IENSEN	Enables GPIO 68 Rx input buffer 0: Disable 1: Enable
27	GPIO_IES	GPIO_IENSEN	Enables GPIO 70 Rx input buffer 0: Disable 1: Enable
26	GPIO_IES	Reserved	Reserved
25	GPIO_IES	Reserved	Reserved
24	GPIO_IES	Reserved	Reserved
23	GPIO_IES	Reserved	Reserved
22	GPIO_IES	GPIO_IENSEN	Enables GPIO 103 Rx input buffer 0: Disable 1: Enable
21	GPIO_IES	GPIO_IENSEN	Enables GPIO 104 Rx input buffer 0: Disable 1: Enable
20	GPIO_IES	D2D_IENSEN	Enables D2D_BB_SRCLKENA Rx input buffer 0: Disable 1: Enable
19	GPIO_IES	D2D_IENSEN	Enables D2D_RGU_RESETB Rx input buffer 0: Disable 1: Enable
18	GPIO_IES	D2D_IENSEN	Enables D2D_SYS_RESETB Rx input buffer 0: Disable 1: Enable
17	GPIO_IES	D2D_IENSEN	Enables D2D_CLK_128K Rx input buffer 0: Disable 1: Enable
16	GPIO_IES	D2D_IENSEN	Enables D2D_XOSC_32K_CK Rx input buffer 0: Disable 1: Enable

0xA002_0920 GPIO Rx input buffer control 2

GPIO_IES2

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO_IES[47:32]															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit(s)	Mnemonic Name	Description
47	GPIO_IES GPIO_IENEN	Enables GPIO 27 Rx input buffer 0: Disable 1: Enable
46	GPIO_IES GPIO_IENEN	Enables GPIO 77/78/79/80 Rx input buffer 0: Disable 1: Enable
45	GPIO_IES GPIO_IENEN	Enables GPIO 25/26 Rx input buffer 0: Disable 1: Enable
44	GPIO_IES GPIO_IENEN	Enables GPIO 23/24 Rx input buffer 0: Disable 1: Enable
43	GPIO_IES GPIO_IENEN	Enables GPIO 52/53/54/55/56/57/58/59/60/61/62/63/64 Rx input buffer 0: Disable 1: Enable
42	GPIO_IES GPIO_IENEN	Enables GPIO 33/34/35/36/37/38/39/40/41/42/44/46/47/48/50/51 Rx input buffer 0: Disable 1: Enable
41	GPIO_IES GPIO_IENEN	Enables GPIO 1/2/28 Rx input buffer 0: Disable 1: Enable
40	GPIO_IES GPIO_IENEN	Enables GPIO 3/4/5/9/11/12/13/16/17 Rx input buffer 0: Disable 1: Enable
39	GPIO_IES GPIO_IENEN	Enables GPIO 22/66 Rx input buffer 0: Disable 1: Enable
38	GPIO_IES GPIO_IENEN	Enables GPIO 0/19/20 Rx input buffer 0: Disable 1: Enable
37	GPIO_IES GPIO_IENEN	Enables GPIO 74/81/82/83/84/85/86 Rx input buffer 0: Disable 1: Enable
36	GPIO_IES D2D_IENEN	Enables D2D_SFSOUT/D2D_SFSCCK/ D2D_SFSIN D2D_SFSWP/D2D_SFSCS/D2D_SFSHOLD Rx input buffer 0: Disable 1: Enable

Bit(s)	Mnemonic	Name	Description
35	GPIO_IES	GPIO_IENSEN	Enables GPIO 31 Rx input buffer 0: Disable 1: Enable
34	GPIO_IES	GPIO_IENSEN	Enables GPIO 32 Rx input buffer 0: Disable 1: Enable
33	GPIO_IES	GPIO_IENSEN	Enables GPIO 30 Rx input buffer 0: Disable 1: Enable
32	GPIO_IES	GPIO_IENSEN	Enables GPIO 69 Rx input buffer 0: Disable 1: Enable

0xA002_0930 GPIO Rx input buffer control 3

GPIO_IES3

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO_IES[63:48]															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit(s)	Mnemonic	Name	Description
63	GPIO_IES	Reserved	Reserved
62	GPIO_IES	Reserved	Reserved
61	GPIO_IES	Reserved	Reserved
60	GPIO_IES	Reserved	Reserved
59	GPIO_IES	Reserved	Reserved
58	GPIO_IES	Reserved	Reserved
57	GPIO_IES	Reserved	Reserved
56	GPIO_IES	Reserved	Reserved
55	GPIO_IES	Reserved	Reserved
54	GPIO_IES	Reserved	Reserved
53	GPIO_IES	Reserved	Reserved
52	GPIO_IES	Reserved	Reserved
51	GPIO_IES	GPIO_IENSEN	Enables GPIO 87/88/89/90/91/92 Rx input buffer 0: Disable 1: Enable
50	GPIO_IES	GPIO_IENSEN	Enables GPIO 49 Rx input buffer 0: Disable 1: Enable
49	GPIO_IES	GPIO_IENSEN	Enables GPIO 29 Rx input buffer 0: Disable 1: Enable
48	GPIO_IES	GPIO_IENSEN	Enables GPIO 21 Rx input buffer

Bit(s)	Mnemonic Name	Description
		0: Disable 1: Enable

GPIO_n Enables GPIO Rx input buffer. High asserted. Data path: From IO to O. IES = 0, O = 0
0 Disable
1 Enable

0xA002_0a00 GPIO PUPD enable control registers 0 GPIO_PUPDEN0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO_PUPD[15:0]															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	1	1	1	0	0	0	0	0	0	0

Bit(s)	Mnemonic Name	Description
15	PUPD GPIO7_PUPDEN	Enables GPIO 7 PUPD 0: Disable 1: Enable
14	PUPD GPIO8_PUPDEN	Enables GPIO 8 PUPD 0: Disable 1: Enable
13	PUPD GPIO18_PUPDEN	Enables GPIO 18 PUPD 0: Disable 1: Enable
12	PUPD GPIO6_PUPDEN	Enables GPIO 6 PUPD 0: Disable 1: Enable
11	PUPD GPIO14_PUPDEN	Enables GPIO 14 PUPD 0: Disable 1: Enable
10	PUPD GPIO15_PUPDEN	Enables GPIO 15 PUPD 0: Disable 1: Enable
9	PUPD GPIO10_PUPDEN	Enables GPIO 10 PUPD 0: Disable 1: Enable
8	PUPD GPIO31_PUPDEN	Enables GPIO 31 PUPD 0: Disable 1: Enable
7	PUPD GPIO32_PUPDEN	Enables GPIO 32 PUPD 0: Disable 1: Enable
6	PUPD GPIO30_PUPDEN	Enables GPIO 30 PUPD 0: Disable

Bit(s)	Mnemonic	Name	Description
5	PUPD	GPIO79_PUPDEN	1: Enable Enables GPIO 69 PUPD 0: Disable
4	PUPD	GPIO72_PUPDEN	1: Enable Enables GPIO 72 PUPD 0: Disable
3	PUPD	GPIO71_PUPDEN	1: Enable Enables GPIO 71 PUPD 0: Disable
2	PUPD	GPIO67_PUPDEN	1: Enable Enables GPIO 67 PUPD 0: Disable
1	PUPD	GPIO68_PUPDEN	1: Enable Enables GPIO 68 PUPD 0: Disable
0	PUPD	GPIO70_PUPDEN	1: Enable Enables GPIO 70 PUPD 0: Disable

0xA002_0a10 GPIO PUPD enable control register 1 GPIO_PUPDEN1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO_PUPD[31:16]															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
31	PUPD	Reserved	Reserved
30	PUPD	Reserved	Reserved
29	PUPD	Reserved	Reserved
28	PUPD	Reserved	Reserved
27	PUPD	Reserved	Reserved
26	PUPD	Reserved	Reserved
25	PUPD	Reserved	Reserved
24	PUPD	Reserved	Reserved
23	PUPD	Reserved	Reserved
22	PUPD	GPIO25_PUPDEN	Enables GPIO 25 PUPD 0: Disable 1: Enable
21	PUPD	GPIO9_PUPDEN	Enables GPIO 9 PUPD 0: Disable 1: Enable

Bit(s)	Mnemonic	Name	Description
20	PUPD	GPIO17_PUPDEN	Enables GPIO 17 PUPD 0: Disable 1: Enable
19	PUPD	GPIO11_PUPDEN	Enables GPIO 11 PUPD 0: Disable 1: Enable
18	PUPD	GPIO16_PUPDEN	Enables GPIO 16 PUPD 0: Disable 1: Enable
17	PUPD	GPIO3_PUPDEN	Enables GPIO 3 PUPD 0: Disable 1: Enable
16	PUPD	GPIO4_PUPDEN	Enables GPIO 4 PUPD 0: Disable 1: Enable

0xA002_0b00 GPIO resistance enable control register 0

GPIO_RESEN0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO_R0[15:0]															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
15	R0	GPIO7_R0EN	Enables GPIO 7 R0 0: Disable 1: Enable
14	R0	GPIO8_R0EN	Enables GPIO 8 R0 0: Disable 1: Enable
13	R0	GPIO18_R0EN	Enables GPIO 18 R0 0: Disable 1: Enable
12	R0	GPIO6_R0EN	Enables GPIO 6 R0 0: Disable 1: Enable
11	R0	GPIO14_R0EN	Enables GPIO 14 R0 0: Disable 1: Enable
10	R0	GPIO15_R0EN	Enables GPIO 15 R0 0: Disable 1: Enable
9	R0	GPIO10_R0EN	Enables GPIO 10 R0 0: Disable 1: Enable

Bit(s)	Mnemonic	Name	Description
8	R0	GPIO31_R0EN	Enables GPIO 31 R0 0: Disable 1: Enable
7	R0	GPIO32_R0EN	Enables GPIO 32 R0 0: Disable 1: Enable
6	R0	GPIO30_R0EN	Enables GPIO 30 R0 0: Disable 1: Enable
5	R0	GPIO79_R0EN	Enables GPIO 69 R0 0: Disable 1: Enable
4	R0	GPIO72_R0EN	Enables GPIO 72 R0 0: Disable 1: Enable
3	R0	GPIO71_R0EN	Enables GPIO 71 R0 0: Disable 1: Enable
2	R0	GPIO67_R0EN	Enables GPIO 67 R0 0: Disable 1: Enable
1	R0	GPIO68_R0EN	Enables GPIO 68 R0 0: Disable 1: Enable
0	R0	GPIO70_R0EN	Enables GPIO 70 R0 0: Disable 1: Enable

0xA002_0b10 GPIO resistance enable control register 1 GPIO_RESEN1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO_R1[8:0]								GPIO_R0[22:16]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0

Bit(s)	Mnemonic	Name	Description
8	R1	GPIO31_R1EN	Enables GPIO 31 R1 0: Disable 1: Enable
7	R1	GPIO32_R1EN	Enables GPIO 32 R1 0: Disable 1: Enable
6	R1	GPIO30_R1EN	Enables GPIO 30 R1 0: Disable

Bit(s)	Mnemonic	Name	Description
5	R1	GPIO69_R1EN	1: Enable Enables GPIO 69 R1 0: Disable
4	R1	GPIO72_R1EN	1: Enable Enables GPIO 72 R1 0: Disable
3	R1	GPIO71_R1EN	1: Enable Enables GPIO 71 R1 0: Disable
2	R1	GPIO67_R1EN	1: Enable Enables GPIO 67 R1 0: Disable
1	R1	GPIO68_R1EN	1: Enable Enables GPIO 68 R1 0: Disable
0	R1	GPIO70_R1EN	1: Enable Enables GPIO 70 R1 0: Disable
22	R0	GPIO25_R0EN	1: Enable Enables GPIO 25 R0 0: Disable
21	R0	GPIO9_R0EN	1: Enable Enables GPIO 9 R0 0: Disable
20	R0	GPIO17_R0EN	1: Enable Enables GPIO 17 R0 0: Disable
19	R0	GPIO11_R0EN	1: Enable Enables GPIO 11 R0 0: Disable
18	R0	GPIO16_R0EN	1: Enable Enables GPIO 16 R0 0: Disable
17	R0	GPIO3_R0EN	1: Enable Enables GPIO 3 R0 0: Disable
16	R0	GPIO4_R0EN	1: Enable Enables GPIO 4 R0 0: Disable

0xA002_0b20 GPIO resistance enable control register 2 GPIO_RESEN2

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO_R1[24:9]															

Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
24	R1	Reserved	Reserved
23	R1	Reserved	Reserved
22	R1	GPIO25_R1EN	Enables GPIO 25 R1 0: Disable 1: Enable
21	R1	GPIO9_R1EN	Enables GPIO 9 R1 0: Disable 1: Enable
20	R1	GPIO17_R1EN	Enables GPIO 17 R1 0: Disable 1: Enable
19	R1	GPIO11_R1EN	Enables GPIO 11 R1 0: Disable 1: Enable
18	R1	GPIO16_R1EN	Enables GPIO 16 R1 0: Disable 1: Enable
17	R1	GPI34_R1EN	Enables GPIO 3 R1 0: Disable 1: Enable
16	R1	GPIO4_R1EN	Enables GPIO 4 R1 0: Disable 1: Enable
15	R1	GPIO7_R1EN	Enables GPIO 7 R1 0: Disable 1: Enable
14	R1	GPIO8_R1EN	Enables GPIO 8 R1 0: Disable 1: Enable
13	R1	GPIO18_R1EN	Enables GPIO 18 R1 0: Disable 1: Enable
12	R1	GPIO6_R1EN	Enables GPIO 6 R1 0: Disable 1: Enable
11	R1	GPIO14_R1EN	Enables GPIO 14 R1 0: Disable 1: Enable
10	R1	GPIO15_R1EN	Enables GPIO 15 R1 0: Disable 1: Enable
9	R1	GPIO10_R1EN	Enables GPIO 10 R1

Bit(s) Mnemonic Name	Description
	0: Disable
	1: Enable

GPIO_n PUPD weak pull-up/pull-down control register. See the tables below for details. E means GPIO DRV control.

KeyPad

KCOL0-4

E	PUPD	R1	R0	weak Pull up/down state
0	0	0	0	0 Hi-Z
0	0	0	0	1 PU-75K ohms
0	0	0	1	0 PU-200K ohms
0	0	0	1	1 PU- 75K//200K ohms
0	1	1	0	0 Hi-Z
0	1	1	0	1 PD-75K ohms
0	1	1	1	0 PD-200K ohms
0	1	1	1	1 PD- 75K//200K ohms
1	1	X	X	X Hi-Z

KROW0-4

E	PUPD	R1	R0	weak Pull up/down state
0	0	0	0	0 Hi-Z
0	0	0	0	1 PU-75K ohms
0	0	0	1	0 PU-2K ohms
0	0	0	1	1 PU- 75K//2K ohms
0	1	1	0	0 Hi-Z
0	1	1	0	1 PD-75K ohms
0	1	1	1	0 PD-2K ohms
0	1	1	1	1 PD- 75K//2K ohms
1	1	X	X	X Hi-Z

MSDC/KCOL6/KCOL7/KROW7

MSDC PAD Control

* No keeper function is required

pu_pd	ctrl_r0	ctrl_r1	Description
0	0	0	both resistors disabled
0	0	1	enable pull up 47K
0	1	0	enable pull up 47K
0	1	1	enable pull up 23.5K
1	0	0	both resistors disabled
1	0	1	enable pull down 47K
1	1	0	enable pull down 47K
1	1	1	enable pull down 23.5K

Others :Using GPIO_PULLEN0 ~ 6 &GPIO_PULLSEL0 ~ 6 control

GPIO_PULLSEL_n GPIO pull-up/down selection. Only valid if the corresponding GPIO_PULLEN is high.

0 GPIO_PULLSELs pull-down is selected.

1 GPIO_PULLSELs pull-up is selected.

0xA002_0c00 GPIO mode control register 0

GPIO_MODE0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name	GPIO7_M				GPIO6_M				GPIO5_M				GPIO4_M			
Type	R/W				R/W				R/W				R/W			
Reset	7				7				7				7			
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO3_M				GPIO2_M				GPIO1_M				GPIO0_M			
Type	R/W				R/W				R/W				R/W			
Reset	7				7				7				7			

Bit(s)	Mnemonic Name	Description
15:12	GPIO3_M GPIO3_MODE	GPIO 3 mode selection 0: GPIO3 (IO) 1: KCOL7 (IO) 2: EDICK (O) 3: Reserved 4: Reserved 5: MCDA1 (IO) 6: LSA0DA10 (O) 7: EGND3 (I)
11:8	GPIO2_M GPIO2_MODE	GPIO 2 mode selection 0: GPIO2 (IO) 1: SDA (IO) 2: Reserved 3: Reserved 4: SPICS0 (O) 5: Reserved 6: 7: EGND2 (I)
7:4	GPIO1_M GPIO1_MODE	GPIO 1 mode selection 0: GPIO1 (IO) 1: SCL (IO) 2: Reserved 3: Reserved 4: SPISCK0 (O) 5: Reserved 6: Reserved 7: EGND1 (I)
3:0	GPIO0_M GPIO0_MODE	GPIO 0 mode selection 0: GPIO0 (IO) 1: CLK00 (O) 2: BTDBGIN (I) 3: EDICK (O)

Bit(s)	Mnemonic Name	Description
		4: SPIMOSI0 (O) 5: U3RXD (I) 6: PWM (O) 7: EGND0 (I)
15:12	GPIO7_M GPIO7_MODE	GPIO 7 mode selection 0: GPIO7 (IO) 1: KCOL3 (IO) 2: DAIPCMIN (I) 3: Reserved 4: Reserved 5: Reserved 6: DEBUG0MON1 (O) 7: EGND7 (I)
11:8	GPIO6_M GPIO6_MODE	GPIO 6 mode selection 0: GPIO6 (IO) 1: KCOL4 (IO) 2: U3CTS (I) 3: Reserved 4: Reserved 5: Reserved 6: Reserved 7: EGND6 (I)
7:4	GPIO5_M GPIO5_MODE	GPIO 5 mode selection 0: GPIO5 (IO) 1: KCOL5 (IO) 2: EDI2CK (O) 3: Reserved 4: Reserved 5: EINT2 (I) 6: Reserved 7: EGND5 (I)
3:0	GPIO4_M GPIO4_MODE	GPIO 4 mode selection 0: GPIO4 (IO) 1: KCOL6 (IO) 2: U1RXD (I) 3: CLK05 (O) 4: WIFITOB (I) 5: MCDA2 (IO) 6: Reserved 7: EGND4 (I)

0xA002_0c10 GPIO mode control register 1 **GPIO_MODE1**

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name			GPIO15_M				GPIO14_M				GPIO13_M				GPIO12_M	
Type			R/W				R/W				R/W				R/W	
Reset			7				7				7				7	
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Name		GPIO11_M		GPIO10_M		GPIO9_M		GPIO8_M
Type		R/W		R/W		R/W		R/W
Reset		7		0		7		7

Bit(s)	Mnemonic Name	Description
15:12	GPIO11_M GPIO11_MODE	GPIO 11 mode selection 0: GPIO11 (IO) 1: KROW7 (IO) 2: EINT3 (I) 3: Reserved 4: EDIDAT (IO) 5: MCDA3 (IO) 6: CLK01 (O) 7: EGND10 (I)
11:8	GPIO10_M GPIO10_MODE	GPIO 10 mode selection 0: GPIO10 (IO) 1: KCOL0 (IO) 2: Reserved 3: Reserved 4: Reserved 5: Reserved 6: DEBUG0MON3 (O) 7: Reserved
7:4	GPIO9_M GPIO9_MODE	GPIO 9 mode selection 0: GPIO9 (IO) 1: KCOL1 (IO) 2: SPIMOSI1 (O) 3: Reserved 4: Reserved 5: Reserved 6: Reserved 7: EGND9 (I)
3:0	GPIO8_M GPIO8_MODE	GPIO 8 mode selection 0: GPIO8 (IO) 1: KCOL2 (IO) 2: DAICLK (O) 3: Reserved 4: Reserved 5: Reserved 6: DEBUG0MON2 (O) 7: EGND8 (I)
15:12	GPIO15_M GPIO15_MODE	GPIO 15 Mode Selection 0: GPIO15 (IO) 1: KROW3 (IO) 2: LSA0DA11 (O) 3: DAISYNC (O) 4: Reserved 5: Reserved 6: DEBUG0MON7 (O) 7: EGND14 (I)
11:8	GPIO14_M GPIO14_MODE	GPIO 14 mode selection

Bit(s)	Mnemonic Name	Description
		0: GPIO14 (IO) 1: KROW4 (IO) 2: DAIPCMOUT (O) 3: Reserved 4: Reserved 5: Reserved 6: DEBUG0MON6 (O) 7: EGND13 (I)
7:4	GPIO13_M GPIO13_MODE	GPIO 13 mode selection 0: GPIO13 (IO) 1: KROW5 (IO) 2: U1TXD (O) 3: EDI2DAT (O) 4: SRCLKENA (O) 5: Reserved 6: Reserved 7: EGND12 (I)
3:0	GPIO12_M GPIO12_MODE	GPIO 12 mode selection 0: GPIO12 (IO) 1: KROW6 (IO) 2: EDI2WS (O) 3: WIFITQBT (I) 4: SPIMISO0 (I) 5: Reserved 6: Reserved 7: EGND11 (I)

0xA002_0c20 GPIO mode control register 2 GPIO_MODE2

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16				
Name		GPIO23_M					GPIO22_M					GPIO21_M					GPIO20_M			
Type		R/W					R/W					R/W					R/W			
Reset		7					7					7					7			
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0				
Name		GPIO19_M					GPIO18_M					GPIO17_M					GPIO16_M			
Type		R/W					R/W					R/W					R/W			
Reset		7					0					7					7			

Bit(s)	Mnemonic Name	Description
15:12	GPIO19_M GPIO19_MODE	GPIO 19 mode selection 0: GPIO19 (IO) 1: SPIMISO1 (I) 2: GPSFSYNC (O) 3: LRSTB (O) 4: DAIRST (I) 5: Reserved 6: Reserved

Bit(s)	Mnemonic Name	Description
		7: EGND18 (I)
11:8	GPIO18_M GPIO18_MODE	GPIO 18 mode selection 0: GPIO18 (IO) 1: KROW0 (IO) 2: LSDI0 (I) 3: Reserved 4: CLKO6 (O) 5: LSCK0 (O) 6: Reserved 7: EGND17 (I)
7:4	GPIO17_M GPIO17_MODE	GPIO 17 mode selection 0: GPIO17 (IO) 1: KROW1 (IO) 2: LSDA00 (IO) 3: U1CTS (I) 4: DAIRST (I) 5: Reserved 6: Reserved 7: EGND16 (I)
3:0	GPIO16_M GPIO16_MODE	GPIO 16 mode selection 0: GPIO16 (IO) 1: KROW2 (IO) 2: LSCK0 (O) 3: Reserved 4: Reserved 5: LSDI0 (I) 6: Reserved 7: EGND15 (I)
15:12	GPIO23_M GPIO23_MODE	GPIO 23 mode selection 0: GPIO23 (IO) 1: U2RXD (I) 2: BTPRI (IO) 3: LSCE0B0 (O) 4: U1RTS (O) 5: Reserved 6: Reserved 7: EGND22 (I)
11:8	GPIO22_M GPIO22_MODE	GPIO 22 mode selection 0: GPIO22 (IO) 1: SPICS1 (O) 2: BPIBUS4 (O) 3: Reserved 4: Reserved 5: Reserved 6: Reserved 7: EGND21 (I)
7:4	GPIO21_M GPIO21_MODE	GPIO 21 mode selection 0: GPIO21 (IO) 1: CLKO2 (O) 2: EDIWS (O) 3: BPIBUS5 (O)

Bit(s)	Mnemonic Name	Description
		4: LSCK1 (O) 5: Reserved 6: Reserved 7: EGND20 (I)
3:0	GPIO20_M GPIO20_MODE	GPIO 20 mode selection 0: GPIO20 (IO) 1: SPISCK1 (O) 2: BPIBUS5 (O) 3: Reserved 4: Reserved 5: Reserved 6: Reserved 7: EGND19 (I)

0xA002_0c30 GPIO mode control register 3 **GPIO_MODE3**

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name			GPIO31_M				GPIO30_M				GPIO29_M				GPIO28_M	
Type			R/W				R/W				R/W				R/W	
Reset			0				0				7				7	
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name			GPIO27_M				GPIO26_M				GPIO25_M				GPIO24_M	
Type			R/W				R/W				R/W				R/W	
Reset			7				1				1				7	

Bit(s)	Mnemonic Name	Description
15:12	GPIO27_M GPIO27_MODE	GPIO 27 mode selection 0: GPIO27 (IO) 1: EINT4 (I) 2: DAICLK (O) 3: GPT_INT (I) 4: SCL (IO) 5: NLD14 (IO) 6: PWM (O) 7: EGND24 (I)
11:8	GPIO26_M GPIO26_MODE	GPIO 26 mode selection 0: GPIO26 (IO) 1: U1TXD (O) 2: U3CTS (I) 3: LSDA01 (IO) 4: Reserved 5: Reserved 6: Reserved 7: Reserved
7:4	GPIO25_M GPIO25_MODE	GPIO 25 mode selection 0: GPIO25 (IO) 1: U1RXD (I)

Bit(s)	Mnemonic Name	Description
		2: LSDI1 (I) 3: EINT11 (I) 4: Reserved 5: Reserved 6: Reserved 7: Reserved
3:0	GPIO24_M GPIO24_MODE	GPIO 24 mode selection 0: GPIO24 (IO) 1: U2TXD (O) 2: U3TXD (O) 3: BPIBUS4 (O) 4: CLK07 (O) 5: Reserved 6: Reserved 7: EGND23 (I)

0xA002_0c40 GPIO mode control register 4

GPIO_MODE4

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name	GPIO39_M					GPIO38_M				GPIO37_M				GPIO36_M		
Type	R/W					R/W				R/W				R/W		
Reset	7					7				7				7		
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO35_M					GPIO34_M				GPIO33_M				GPIO32_M		
Type	R/W					R/W				R/W				R/W		
Reset	7					7				7				0		

Bit(s)	Mnemonic Name	Description
15:12	GPIO31_M GPIO31_MODE	GPIO 31 mode selection 0: GPIO31 (IO) 1: MCDA0 (IO) 2: Reserved 3: Reserved 4: Reserved 5: Reserved 6: Reserved 7: Reserved
11:8	GPIO30_M GPIO30_MODE	GPIO 30 mode selection 0: GPIO30 (IO) 1: MCCK (IO) 2: Reserved 3: Reserved 4: Reserved 5: Reserved 6: Reserved 7: Reserved
7:4	GPIO29_M GPIO29_MODE	GPIO 29 mode selection

Bit(s)	Mnemonic Name	Description
		0: GPIO29 (IO) 1: EINT5 (I) 2: DAIPCMOUT (O) 3: NLD12 (IO) 4: CMCSK (I) 5: LSA0DA12 (O) 6: LPTE1 (I) 7: EGND26 (I)
3:0	GPIO28_M GPIO28_MODE	GPIO 28 mode selection 0: GPIO28 (IO) 1: U3RTS (O) 2: SDA (IO) 3: NLD13 (IO) 4: U3TXD (O) 5: CMCS1 (I) 6: EINT1 (I) 7: EGND25 (I)
15:12	GPIO35_M GPIO35_MODE	GPIO 35 mode selection 0: GPIO35 (IO) 1: NLD6 (IO) 2: Reserved 3: Reserved 4: Reserved 5: DEBUG1IN4 (I) 6: DEBUG1MON4 (O) 7: EGND29 (I)
11:8	GPIO34_M GPIO34_MODE	GPIO 34 mode selection 0: GPIO34 (IO) 1: NLD7 (IO) 2: EINT7 (I) 3: LPTE1 (I) 4: Reserved 5: DEBUG1IN3 (I) 6: DEBUG1MON3 (O) 7: EGND28 (I)
7:4	GPIO33_M GPIO33_MODE	GPIO 33 mode selection 0: GPIO33 (IO) 1: NLD8 (IO) 2: DAIPCMIN (I) 3: CMMCLK (O) 4: EINT6 (I) 5: CMPDN (O) 6: LSDI2 (I) 7: EGND27 (I)
3:0	GPIO32_M GPIO32_MODE	GPIO 32 mode selection 0: GPIO32 (IO) 1: MCCM0 (IO) 2: Reserved 3: Reserved 4: Reserved 5: Reserved 6: Reserved

Bit(s)	Mnemonic Name	Description
		7: Reserved
15:12	GPIO39_M GPIO39_MODE	GPIO 39 Mode Selection 0: GPIO39 (IO) 1: NLD2 (IO) 2: LSDA02 (IO) 3: Reserved 4: Reserved 5: BT_RGPIO2 (IO) 6: Reserved 7: EGND33 (I)
11:8	GPIO38_M GPIO38_MODE	GPIO 38 Mode Selection 0: GPIO38 (IO) 1: NLD3 (IO) 2: LSDI3 (I) 3: Reserved 4: Reserved 5: BT_RGPIO1 (IO) 6: Reserved 7: EGND32 (I)
7:4	GPIO37_M GPIO37_MODE	GPIO 37 Mode Selection 0: GPIO37 (IO) 1: NLD4 (IO) 2: Reserved 3: Reserved 4: Reserved 5: BT_RGPIO0 (IO) 6: Reserved 7: EGND31 (I)
3:0	GPIO36_M GPIO36_MODE	GPIO 36 mode selection 0: GPIO36 (IO) 1: NLD5 (IO) 2: Reserved 3: Reserved 4: Reserved 5: Reserved 6: Reserved 7: EGND30 (I)

0xA002_0c50 GPIO mode control register 5

GPIO_MODE5

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16			
Name	GPIO47_M					GPIO46_M					GPIO45_M					GPIO44_M			
Type	R/W					R/W					R/W					R/W			
Reset	1					0					1					7			
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0			
Name	GPIO43_M					GPIO42_M					GPIO41_M					GPIO40_M			
Type	R/W					R/W					R/W					R/W			
Reset	7					7					7					7			

Bit(s)	Mnemonic Name	Description
15:12	GPIO43_M GPIO43_MODE	GPIO 43 mode selection 0: GPIO43 (IO) 1: LRDB (O) 2: Reserved 3: Reserved 4: Reserved 5: Reserved 6: Reserved 7: EGND37 (I)
11:8	GPIO42_M GPIO42_MODE	GPIO 42 mode selection 0: GPIO42 (IO) 1: LWRB (O) 2: Reserved 3: Reserved 4: Reserved 5: BT_RGPI05 (IO) 6: Reserved 7: EGND36 (I)
7:4	GPIO41_M GPIO41_MODE	GPIO 41 mode selection 0: GPIO41 (IO) 1: NLD0 (IO) 2: LSCK2 (O) 3: Reserved 4: Reserved 5: BT_RGPI04 (IO) 6: DEBUG1MON10 (O) 7: EGND35 (I)
3:0	GPIO40_M GPIO40_MODE	GPIO 40 mode selection 0: GPIO40 (IO) 1: NLD1 (IO) 2: LSA0DA13 (O) 3: Reserved 4: Reserved 5: BT_RGPI03 (IO) 6: DEBUG1MON9 (O) 7: EGND34 (I)
15:12	GPIO47_M GPIO47_MODE	GPIO 47 mode selection 0: GPIO47 (IO) 1: LPCE1B (O) 2: CMMCLK (O) 3: SDA (IO) 4: NLD10 (IO) 5: CMPDN (O) 6: LPTE1 (I) 7: Reserved
11:8	GPIO46_M GPIO46_MODE	GPIO 46 mode selection 0: GPIO46 (IO) 1: LSCE1B (O) 2: DAISYNC (O) 3: SCL (IO)

Bit(s)	Mnemonic Name	Description
		4: NLD9 (IO) 5: LPCE2B (O) 6: LSA0DA14 (O) 7: EGND39 (I)
7:4	GPIO45_M GPIO45_MODE	GPIO 45 mode selection 0: GPIO45 (IO) 1: LPCE0B (O) 2: LSCE0B1 (O) 3: Reserved 4: Reserved 5: Reserved 6: Reserved 7: Reserved
3:0	GPIO44_M GPIO44_MODE	GPIO 44 mode selection 0: GPIO44 (IO) 1: LPA0 (O) 2: Reserved 3: Reserved 4: Reserved 5: Reserved 6: Reserved 7: EGND38 (I)

0xA002_0c60 GPIO mode control register 6 **GPIO_MODE6**

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name			GPIO55_M				GPIO54_M				GPIO53_M				GPIO52_M	
Type			R/W				R/W				R/W				R/W	
Reset			7				7				7				7	
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name			GPIO51_M				GPIO50_M				GPIO49_M				GPIO48_M	
Type			R/W				R/W				R/W				R/W	
Reset			7				7				1				7	

Bit(s)	Mnemonic Name	Description
15:12	GPIO51_M GPIO51_MODE	GPIO 51 mode selection 0: GPIO51 (IO) 1: LPRSTB (O) 2: EINT8 (I) 3: LSCK4 (O) 4: LPTE1 (I) 5: LPCE3B (O) 6: Reserved 7: EGND42 (I)
11:8	GPIO50_M GPIO50_MODE	GPIO 50 mode selection 0: GPIO50 (IO) 1: LPTE1 (I)

Bit(s)	Mnemonic Name	Description
		2: CLK04 (O) 3: Reserved 4: Reserved 5: Reserved 6: DEBUG1MON15 (O) 7: EGND41 (I)
7:4	GPIO49_M GPIO49_MODE	GPIO 49 mode selection 0: GPIO49 (IO) 1: WATCHDOG (O) 2: NLD15 (IO) 3: CMCSK (I) 4: CMCS1 (I) 5: LS1A03 (IO) 6: Reserved 7: Reserved
3:0	GPIO48_M GPIO48_MODE	GPIO 48 mode selection 0: GPIO48 (IO) 1: LSRSTB (O) 2: CLK03 (O) 3: U3RXD (I) 4: NLD11 (IO) 5: CMCS0 (I) 6: LSCK3 (O) 7: EGND40 (I)
15:12	GPIO55_M GPIO55_MODE	GPIO 55 mode selection 0: GPIO55 (IO) 1: CMDAT3 (I) 2: LRSTB (O) 3: DAIPCMIN (I) 4: Reserved 5: DEBUG2IN3 (I) 6: DEBUG2MON3 (O) 7: EGND46 (I)
11:8	GPIO54_M GPIO54_MODE	GPIO 54 mode selection 0: GPIO54 (IO) 1: CMDAT2 (I) 2: SCL (IO) 3: Reserved 4: Reserved 5: DEBUG2IN2 (I) 6: DEBUG2MON2 (O) 7: EGND45 (I)
7:4	GPIO53_M GPIO53_MODE	GPIO 53 mode selection 0: GPIO53 (IO) 1: CMDAT1 (I) 2: CMCS1 (I) 3: Reserved 4: Reserved 5: Reserved 6: Reserved 7: EGND44 (I)

Bit(s)	Mnemonic Name	Description
3:0	GPIO52_M GPIO52_MODE	GPIO 52 mode selection 0: GPIO52 (IO) 1: CMDAT0 (I) 2: CMCS0 (I) 3: Reserved 4: Reserved 5: Reserved 6: Reserved 7: EGND43 (I)

0xA002_0c70 GPIO mode control register 7

GPIO_MODE7

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name			GPIO63_M					GPIO62_M				GPIO61_M				GPIO60_M
Type			R/W					R/W				R/W				R/W
Reset			7					0				7				7
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name			GPIO59_M					GPIO58_M				GPIO57_M				GPIO56_M
Type			R/W					R/W				R/W				R/W
Reset			7					7				7				7

Bit(s)	Mnemonic Name	Description
15:12	GPIO59_M GPIO59_MODE	GPIO 59 mode selection 0: GPIO59 (IO) 1: CMDAT7 (I) 2: U3CTS (I) 3: PWM (O) 4: Reserved 5: Reserved 6: Reserved 7: EGND50 (I)
11:8	GPIO58_M GPIO58_MODE	GPIO 58 mode selection 0: GPIO58 (IO) 1: CMDAT6 (I) 2: LSDA04 (IO) 3: DAISYNC (O) 4: U2TXD (O) 5: Reserved 6: Reserved 7: EGND49 (I)
7:4	GPIO57_M GPIO57_MODE	GPIO 57 mode selection 0: GPIO57 (IO) 1: CMDAT5 (I) 2: LSCK5 (O) 3: DAIPCMOUT (O) 4: U2RXD (I) 5: DEBUG2IN5 (I)

Bit(s)	Mnemonic Name	Description
		6: DEBUG2MON5 (O) 7: EGND48 (I)
3:0	GPIO56_M GPIO56_MODE	GPIO 56 mode selection 0: GPIO56 (IO) 1: CMDAT4 (I) 2: LSA0DA15 (O) 3: DAICLK (O) 4: U3RTS (O) 5: DEBUG2IN4 (I) 6: DEBUG2MON4 (O) 7: EGND47 (I)
15:12	GPIO63_M GPIO63_MODE	GPIO 63 mode selection 0: GPIO63 (IO) 1: CMMCLK (O) 2: Reserved 3: Reserved 4: Reserved 5: Reserved 6: Reserved 7: EGND53 (I)
11:8	GPIO62_M GPIO62_MODE	GPIO 62 mode selection 0: GPIO62 (IO) 1: CMPDN (O) 2: LSCE0B2 (O) 3: U3TXD (O) 4: Reserved 5: Reserved 6: Reserved 7: Reserved
7:4	GPIO61_M GPIO61_MODE	GPIO 61 mode selection 0: GPIO61 (IO) 1: CMVREF (I) 2: SDA (IO) 3: Reserved 4: Reserved 5: Reserved 6: Reserved 7: EGND52 (I)
3:0	GPIO60_M GPIO60_MODE	GPIO 60 mode selection 0: GPIO60 (IO) 1: CMHREF (I) 2: EINT9 (I) 3: U3RXD (I) 4: Reserved 5: Reserved 6: Reserved 7: EGND51 (I)

0xA002_0c80 GPIO mode control register 8

GPIO_MODE8

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name			GPIO71_M				GPIO70_M				GPIO69_M				GPIO68_M	
Type			R/W				R/W				R/W				R/W	
Reset			7				7				7				7	
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name			GPIO67_M				GPIO66_M				GPIO65_M				GPIO64_M	
Type			R/W				R/W				R/W				R/W	
Reset			7				7				0				7	

Bit(s)	Mnemonic Name	Description
15:12	GPIO67_M GPIO67_MODE	GPIO 67 mode selection 0: GPIO67 (IO) 1: SIMSIO (IO) 2: Reserved 3: Reserved 4: Reserved 5: Reserved 6: Reserved 7: EGND63 (I)
11:8	GPIO66_M GPIO66_MODE	GPIO 66 mode selection 0: GPIO66 (IO) 1: EINT0 (I) 2: EDIDAT (IO) 3: LSDI4 (I) 4: Reserved 5: Reserved 6: Reserved 7: EGND56 (I)
7:4	GPIO65_M GPIO65_MODE	GPIO 65 mode selection 0: GPIO65 (IO) 1: CMRST (O) 2: Reserved 3: Reserved 4: Reserved 5: Reserved 6: Reserved 7: EGND55 (I)
3:0	GPIO64_M GPIO64_MODE	GPIO 64 mode selection 0: GPIO64 (IO) 1: CMPCLK (I) 2: CMCSK (I) 3: Reserved 4: Reserved 5: Reserved 6: Reserved 7: EGND54 (I)
15:12	GPIO71_M GPIO71_MODE	GPIO 71 mode selection 0: GPIO71 (IO)

Bit(s)	Mnemonic Name	Description
		1: SIM2SCLK (IO) 2: Reserved 3: Reserved 4: Reserved 5: Reserved 6: Reserved 7: EGND67 (I)
11:8	GPIO70_M GPIO70_MODE	GPIO 70 mode selection 0: GPIO70 (IO) 1: SIMSCLK (IO) 2: Reserved 3: Reserved 4: Reserved 5: Reserved 6: Reserved 7: EGND66 (I)
7:4	GPIO69_M GPIO69_MODE	GPIO 69 mode selection 0: GPIO69 (IO) 1: SIM2SIO (IO) 2: Reserved 3: Reserved 4: Reserved 5: Reserved 6: Reserved 7: EGND65 (I)
3:0	GPIO68_M GPIO68_MODE	GPIO 68 mode selection 0: GPIO68 (IO) 1: SIMSRST (IO) 2: Reserved 3: Reserved 4: Reserved 5: Reserved 6: Reserved 7: EGND64 (I)

0xA002_0c90 GPIO mode control register 9 GPIO_MODE9

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name			GPIO79_M				GPIO78_M				GPIO77_M				GPIO76_M	
Type			R/W				R/W				R/W				R/W	
Reset			0				0				1				1	
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name			GPIO75_M				GPIO74_M				GPIO73_M				GPIO72_M	
Type			R/W				R/W				R/W				R/W	
Reset			1				0				1				7	

Bit(s)	Mnemonic Name	Description
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Bit(s)	Mnemonic Name	Description
15:12	GPIO75_M GPIO75_MODE	GPIO 75 mode selection 0: Reserved 1: Reserved 2: Reserved 3: Reserved 4: Reserved 5: Reserved 6: Reserved 7: Reserved
11:8	GPIO74_M GPIO74_MODE	GPIO 74 mode selection 0: GPIO74 (IO) 1: SFSCS1 (O) 2: EINT10 (I) 3: CMPDN (O) 4: Reserved 5: Reserved 6: Reserved 7: Reserved
7:4	GPIO73_M GPIO73_MODE	GPIO 73 mode selection 0: Reserved 1: Reserved 2: Reserved 3: Reserved 4: Reserved 5: Reserved 6: Reserved 7: Reserved
3:0	GPIO72_M GPIO72_MODE	GPIO 72 mode selection 0: GPIO72 (IO) 1: SIM2SRST (IO) 2: Reserved 3: Reserved 4: Reserved 5: Reserved 6: Reserved 7: EGND68 (I)
15:12	GPIO79_M GPIO79_MODE	GPIO 79 mode selection 0: GPIO79 (IO) 1: BPIBUS1 (O) 2: Reserved 3: Reserved 4: Reserved 5: Reserved 6: Reserved 7: Reserved
11:8	GPIO78_M GPIO78_MODE	GPIO 78 mode selection 0: GPIO78 (IO) 1: BPIBUS2 (O) 2: Reserved 3: Reserved 4: Reserved

Bit(s)	Mnemonic Name	Description
		5: Reserved 6: Reserved 7: Reserved
7:4	GPIO77_M GPIO77_MODE	GPIO 77 mode selection 0: GPIO77 (IO) 1: BPIBUS3 (O) 2: Reserved 3: Reserved 4: Reserved 5: Reserved 6: Reserved 7: Reserved
3:0	GPIO76_M GPIO76_MODE	GPIO 76 mode selection 0: GPIO76 (IO) 1: Reserved 2: Reserved 3: Reserved 4: Reserved 5: Reserved 6: Reserved 7: Reserved

0xA002_0d00 GPIO Rx duty select control register

GPIO_RDSEL

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO_RDSEL[15:0]															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
15:13	GPIO_RDSEL	Reserved	Reserved
12:11	GPIO_RDSEL	Reserved	Reserved
10:8	GPIO_RDSEL	Reserved	Reserved
7:6	GPIO_RDSEL	Reserved	Reserved
5:4	GPIO_RDSEL	Reserved	Reserved
3:2	GPIO_RDSEL	D2D_RDSEL	Rx duty configuration For the IO list: GPIO74/81/82/83/84/85/86 0: Input buffer duty high when asserted 1: Input buffer duty low when asserted
1:0	GPIO_RDSEL	GPIO_RDSEL	Rx duty configuration For the IO list: D2D_SFSOUT/D2D_SFSCCK/ D2D_SFSIN D2D_SFSWP/D2D_SFSCS/D2D_SFSHOLD 0: Input buffer duty high when asserted 1: Input buffer duty low when asserted

RDSEL[1:0] Selects Rx duty
Bit 0 Input buffer duty high when asserted (high pulse width adjustment)
Bit 1 Input buffer duty low when asserted (low pulse width adjustment)

0xA002_0d10 GPIO Tx duty select control register **GPIO_TDSEL**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO_TDSEL[15:0]															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Mnemonic	Name	Description
15:13	GPIO_TDSEL	Reserved	Reserved
12:11	GPIO_TDSEL	Reserved	Reserved
10:8	GPIO_TDSEL	Reserved	Reserved
7:6	GPIO_TDSEL	Reserved	Reserved
5:4	GPIO_TDSEL	Reserved	Reserved
3:2	GPIO_TDSEL	D2D_RDSEL	Tx duty configuration For the IO list: GPIO74/81/82/83/84/85/86 0: Output buffer duty high when asserted 1: Output buffer duty low when asserted
1:0	GPIO_RDSEL	GPIO_RDSEL	Tx duty configuration For the IO list: D2D_SFSOUT/D2D_SFSCCK/D2D_SFSIN D2D_SFSWP/D2D_SFSCS/D2D_SFSHOLD 0: Output buffer duty high when asserted 1: Output buffer duty low when asserted

TDSEL[1:0] Selects Tx duty
Bit 0 Output level shifter duty high when asserted (high pulse width adjustment)
Bit 1 Output level shifter duty low when asserted (low pulse width adjustment)

0xA002_0e00 CLK_OUT0 setting **CLKO_MODE0**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name														CLK_MODE0		
Type														R/W		
Reset														0		

CLK_MODE0 Selects the CLK00 output source
0 sfc_ck_div 2 (65MHz, 52MHz, 39MHz, etc.)
1 f26m_ck (26MHz)
2 emi1x_ck_div2 (65MHz, 52MHz, etc.)
3 command d2d_ck_div2 (37.5MHz, 26MHz, etc.)

- 4 f32k_ck (32kHz)
- 5 command d2d_ck (65MHz, 52MHz, 26MHz)
- 6 BT ahb_bus_clk_div2 (65MHz, 52MHz, etc.)
- 7 MMsyst_emi1x_ck_div2 (65MHz, 52MHz)
- 8 MCUSYS ahb bus clk (65MHz, 52MHz)
- 9 MSDC clk_div2 (~49MHz)
- A ARM clk_div4 (65MHz, 52MHz, etc.)
- B FM 128Khz clk (128kHz)
- C USB 48Mhz clk (48MHz)
- D x4g md2g clk (52MHz)
- E BSI_clk_div2 (65MHz)
- F DSP_clk_div2 (78MHz)

0xA002_0e10 CLK_OUT1 setting CLKO_MODE1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name																CLK_MODE1
Type																R/W
Reset																0

- CLK_MODE1** Selects the CLKO1 output source
- 0 sfc_ck_div 2 (65MHz, 52MHz, 39MHz, etc.)
 - 1 f26m_ck (26MHz)
 - 2 emi1x_ck_div2 (65MHz, 52MHz, etc.)
 - 3 command d2d_ck_div2 (37.5MHz, 26MHz, etc.)
 - 4 f32k_ck (32kHz)
 - 5 command d2d_ck (65MHz, 52MHz, 26MHz)
 - 6 BT ahb_bus_clk_div2 (65MHz, 52MHz, etc.)
 - 7 MMsyst_emi1x_ck_div2 (65MHz, 52MHz)
 - 8 MCUSYS ahb bus clk (65MHz, 52MHz)
 - 9 MSDC clk_div2 (~49MHz)
 - A ARM clk_div4 (65MHz, 52MHz, etc.)
 - B FM 128Khz clk (128kHz)
 - C USB 48Mhz clk (48MHz)
 - D x4g md2g clk (52MHz)
 - E BSI_clk_div2 (65MHz)
 - F DSP_clk_div2 (78MHz)

0xA002_0e20 CLK_OUT2 setting CLKO_MODE2

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name																CLK_MODE2
Type																R/W

[illegible]

CLK_MODE2 Selects the CLKO2 output source

0	sfc_ck_div 2 (65MHz, 52MHz, 39MHz, etc.)
1	f26m_ck (26MHz)
2	emi1x_ck_div2 (65MHz, 52MHz, etc.)
3	command d2d_ck_div2 (37.5MHz, 26MHz, etc.)
4	f32k_ck (32kHz)
5	command d2d_ck (65MHz, 52MHz, 26MHz)
6	BT ahb_bus_clk_div2 (65MHz, 52MHz, etc.)
7	MMsys_emi1x_ck_div2 (65MHz, 52MHz)
8	MCUSYS ahb bus clk (65MHz, 52MHz)
9	MSDC clk_div2 (~49MHz)
A	ARM clk_div4 (65MHz, 52MHz, etc.)
B	FM 128Khz clk (128kHz)
C	USB 48Mhz clk (48MHz)
D	x4g md2g clk (52MHz)
E	BSI_clk_div2 (65MHz)
F	DSP clk div2 (78MHz)

0xA002 0e30

CLK OUT3 setting

CLKO MODE3

[illegible]

CLK_MODE3 Selects the CLK03 output source

- 0** sfc_ck_div 2 (65MHz, 52MHz, 39MHz, etc.)
- 1** f26m_ck (26MHz)
- 2** emi1x_ck_div2 (65MHz, 52MHz, etc.)
- 3** command d2d_ck_div2 (37.5MHz, 26MHz, etc.)
- 4** f32k_ck (32kHz)
- 5** command d2d_ck (65MHz, 52MHz, 26MHz)
- 6** BT ahb_bus_clk_div2 (65MHz, 52MHz, etc.)
- 7** MMSys_emi1x_ck_div2 (65MHz, 52MHz)
- 8** MCUSYS ahb bus clk (65MHz, 52MHz)
- 9** MSDC clk_div2 (~49MHz)
- A** ARM clk_div4 (65MHz, 52MHz, etc.)
- B** FM 128Khz clk (128kHz)
- C** USB 48Mhz clk (48MHz)
- D** x4g md2g clk (52MHz)
- E** BSI_clk_div2 (65MHz)
- F** DSP clk_div2 (78MHz)

0xA002_0e40

CLK_OUT4 setting

CLKO_MODE4

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name														CLK_MODE4		
Type														R/W		
Reset														0		

CLK_MODE4

Selects the CLKO4 output source

- 0 sfc_ck_div 2 (65MHz, 52MHz, 39MHz, etc.)
- 1 f26m_ck (26MHz)
- 2 emi1x_ck_div2 (65MHz, 52MHz, etc.)
- 3 command d2d_ck_div2 (37.5MHz, 26MHz, etc.)
- 4 f32k_ck (32kHz)
- 5 command d2d_ck (65MHz, 52MHz, 26MHz)
- 6 BT ahb_bus_clk_div2 (65MHz, 52MHz, etc.)
- 7 MMsys_emi1x_ck_div2 (65MHz, 52MHz)
- 8 MCUSYS ahb bus clk (65MHz, 52MHz)
- 9 MSDC clk_div2 (~49MHz)
- A ARM clk_div4 (65MHz, 52MHz, etc.)
- B FM 128Khz clk (128kHz)
- C USB 48Mhz clk (48MHz)
- D x4g md2g clk (52MHz)
- E BSI_clk_div2 (65MHz)
- F DSP_clk_div2 (78MHz)

0xA002_0e50

CLK_OUT5 setting

CLKO_MODE5

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name														CLK_MODE5		
Type														R/W		
Reset														0		

CLK_MODE5

Selects the CLKO5 output source

- 0 sfc_ck_div 2 (65MHz, 52MHz, 39MHz, etc.)
- 1 f26m_ck (26MHz)
- 2 emi1x_ck_div2 (65MHz, 52MHz, etc.)
- 3 command d2d_ck_div2 (37.5MHz, 26MHz, etc.)
- 4 f32k_ck (32kHz)
- 5 command d2d_ck (65MHz, 52MHz, 26MHz)
- 6 BT ahb_bus_clk_div2 (65MHz, 52MHz, etc.)
- 7 MMsys_emi1x_ck_div2 (65MHz, 52MHz)
- 8 MCUSYS ahb bus clk (65MHz, 52MHz)

- 9 MSDC clk_div2 (~49MHz)
- A ARM clk_div4 (65MHz, 52MHz, etc.)
- B FM 128Khz clk (128kHz)
- C USB 48Mhz clk (48MHz)
- D x4g md2g clk (52MHz)
- E BSI_clk_div2 (65MHz)
- F DSP_clk_div2 (78MHz)

0xA002_0e60

CLK_OUT6 setting

CLKO_MODE6

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name																CLK_MODE6
Type																R/W
Reset																0

CLK_MODE6

Selects the CLKO6 output source

- 0 sfc_ck_div 2 (65MHz, 52MHz, 39MHz, etc.)
- 1 f26m_ck (26MHz)
- 2 emi1x_ck_div2 (65MHz, 52MHz, etc.)
- 3 command d2d_ck_div2 (37.5MHz, 26MHz, etc.)
- 4 f32k_ck (32kHz)
- 5 command d2d_ck (65MHz, 52MHz, 26MHz)
- 6 BT ahb_bus_clk_div2 (65MHz, 52MHz, etc.)
- 7 MMsys_emi1x_ck_div2 (65MHz, 52MHz)
- 8 MCUSYS ahb bus clk (65MHz, 52MHz)
- 9 MSDC clk_div2 (~49MHz)
- A ARM clk_div4 (65MHz, 52MHz, etc.)
- B FM 128Khz clk (128kHz)
- C USB 48Mhz clk (48MHz)
- D x4g md2g clk (52MHz)
- E BSI_clk_div2 (65MHz)
- F DSP_clk_div2 (78MHz)

0xA002_0e70

CLK_OUT7 setting

CLKO_MODE7

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name																CLK_MODE7
Type																R/W
Reset																0

CLK_MODE7

Selects the CLKO7 output source

- 0 sfc_ck_div 2 (65MHz, 52MHz, 39MHz, etc.)
- 1 f26m_ck (26MHz)

- 2 emi1x_ck_div2 (65MHz, 52MHz, etc.)
- 3 command d2d_ck_div2 (37.5MHz, 26MHz, etc.)
- 4 f32k_ck (32kHz)
- 5 command d2d_ck (65MHz, 52MHz, 26MHz)
- 6 BT_ahb_bus_clk_div2 (65MHz, 52MHz, etc.)
- 7 MMsys_emi1x_ck_div2 (65MHz, 52MHz)
- 8 MCUSYS_ahb_bus_clk (65MHz, 52MHz)
- 9 MSDC_clk_div2 (~49MHz)
- A ARM_clk_div4 (65MHz, 52MHz, etc.)
- B FM_128Khz_clk (128kHz)
- C USB_48Mhz_clk (48MHz)
- D x4g_md2g_clk (52MHz)
- E BSI_clk_div2 (65MHz)
- F DSP_clk_div2 (78MHz)

GPIO+0x4h GPIO xxx register SET GPIO_XXX_SET

For all registers addresses listed above, writing to the +4h address offset will perform a bit-wise **OR** function between the 16-bit written value and the 16-bit register value already existing in the corresponding GPIO_XXX registers.

Example:

If GPIO_DIR0 (GPIO+0000h) = 16'h0F0F,
 writing GPIO_DIR0_SET (GPIO+0004h) = 16'F0F0 will result in GPIO_DIR0 = 16'hFFFF.

GPIO+0x8h GPIO xxx register CLR GPIO_XXX_CLR

For all registers addresses listed above, writing to the +8h address offset will perform a bit-wise **AND-NOT** function between the 16-bit written value and the 16-bit register value already existing in the corresponding GPIO_XXX registers.

Example:

If GPIO_DIR0 (GPIO+0000h) = 16'h0F0F,
 writing GPIO_DIR0_CLR (GPIO+0008h) = 16'0F0F will result in GPIO_DIR0 = 16'h0000.

MT6250 GPIO PUPD/R0/R1 Map Relationship

1. GPIO_PUPD

Address: 0xA002_0a00(0-15)

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
GPIO 7	GPIO 8	GPIO 18	GPIO 6	GPIO 14	GPIO 15	GPIO 10	GPIO 31	GPIO 32	GPIO 30	GPIO 69	GPIO 72	GPIO 71	GPIO 67	GPIO 68	GPIO 70

Address: 0xA002_0a10(16-31)

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
----	----	----	----	----	----	---	---	---	---	---	---	---	---	---	---

NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	GPIO 25	GPIO 9	GPIO 17	GPIO 11	GPIO 16	GPIO 3	GPIO 4
------	------	------	------	------	------	------	------	------	---------	--------	---------	---------	---------	--------	--------

2. GPIO_R0

Address:0xA002_0b00(0-15)

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
GPIO 7	GPIO 8	GPIO 18	GPIO 6	GPIO 14	GPIO 15	GPIO 10	GPIO 31	GPIO 32	GPIO 30	GPIO 69	GPIO 72	GPIO 71	GPIO 67	GPIO 68	GPIO 70

Address:0xA002_0b10(16-31)

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	GPIO 25	GPIO 9	GPIO 17	GPIO 11	GPIO 16	GPIO 3	GPIO 4

3. GPIO_R1

Address:0xA002_0b20(0-15)

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
GPIO 7	GPIO 8	GPIO 18	GPIO 6	GPIO 14	GPIO 15	GPIO 10	GPIO 31	GPIO 32	GPIO 30	GPIO 69	GPIO 72	GPIO 71	GPIO 67	GPIO 68	GPIO 70

Address:0xA002_0b30(16-31)

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	GPIO 25	GPIO 9	GPIO 17	GPIO 11	GPIO 16	GPIO 3	GPIO 4

4. GPIO105~107

A0730168 **ACFG_GPIO_MODE** **GPIO mode 0**

C141

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO107_M				GPIO106_M				GPIO105_M				D2D_RSV			
Type	RW				RW				RW				RW			
Reset	1	1	0	0	0	0	0	1	0	1	0	0	0	0	0	1

Bit(s)	Name	Description
15:12	GPIO107_M	GPIO107 mode selection 0x0: GPIO107 0x1: EINT12 0x2~0xB: N/A 0xC: EGND 0xD~0xF: N/A
11:8	GPIO106_M	GPIO106 mode selection 0x0: GPIO106 0x1: RESETB 0x2~0xF: N/A
7:4	GPIO105_M	GPIO105 mode selection 0x0: GPIO105

Bit(s)	Name	Description
		0x1: SRCLKENAI 0x2~0x3: N/A 0x4: EGND 0x5~0xF: N/A
3:0	D2D_RSV	Reserved for testing

A0730160 ACFG_GPIO_CON0 ACFG_GPIO_DIR_DOUT 0000

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name						GPIO107	GPIO106	GPIO105						GPIO107	GPIO106	GPIO105
Type						RW	RW	RW						RW	RW	RW
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Name	Description
10	GPIO107_DIR	GPIO107 direction control 0: Input 1: Output
9	GPIO106_DIR	GPIO106 direction control 0: Input 1: Output
8	GPIO105_DIR	GPIO105 direction control 0: Input 1: Output
2	GPIO107_DOUT	GPIO107 data-out value 0: Data-out is 0 1: Data-out is 1
1	GPIO106_DOUT	GPIO106 data-out value 0: Data-out is 0 1: Data-out is 1
0	GPIO105_DOUT	GPIO105 data-out value 0: Data-out is 0 1: Data-out is 1

A0730164 ACFG_GPIO_CON1 ACFG_GPIO_DI 0000

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name									DLKY_EVT					GPIO107	GPIO106	GPIO105
Type									RO					RO	RO	RO
Reset									0					X	X	X

Bit(s)	Name	Description
7	DLKY_EVENT	USB download: download key event 0: Download key not pressed 1: Download key pressed
2	GPIO107_DIN	GPIO 107 data-in value
1	GPIO106_DIN	GPIO 106 data-in value
0	GPIO105_DIN	GPIO 105 data-in value

A0730104 ACFG_PIN_PULLEN1 ACFG_GPIO_PULLEN 0000

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name							GPIO 107	GPIO 106	GPIO 105							
Type							RW	RW	RW							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Name	Description
9	GPIO107_PULLEN	GPIO107 pull-up/down enabling bit 0: Pull-up/down disabled 1: Pull-up/down enabled
8	GPIO106_PULLEN	GPIO106 pull-up/down enabling bit 0: Pull-up/down disabled 1: Pull-up/down enabled
7	GPIO105_PULLEN	GPIO105 pull-up/down enabling bit 0: Pull-up/down disabled 1: Pull-up/down enabled

A073010C ACFG_PIN_PULLSEL1 ACFG_GPIO_PULLSEL 0000

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name							GPIO 107	GPIO 106	GPIO 105							
Type							RW	RW	RW							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Name	Description
9	GPIO107_PULLSEL	GPIO107 pull-up/down selection 0: Pull-down 1: Pull-up
8	GPIO106_PULLSEL	GPIO106 pull-up/down selection 0: Pull-down 1: Pull-up
7	GPIO105_PULLSEL	GPIO105 pull-up/down selection 0: Pull-down

Bit(s)	Name	Description
		1: Pull-up

A0730118 **ACFG_PIN_DRV2** **ACFG_GPIO_DRV2** **FFFF**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO105															
Type	RW															
Reset	11		1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit(s)	Name	Description
15:14	GPIO105_DRV	GPIO105 driving strength configuration 2'b00: 4mA 2'b01: 8mA 2'b10: 12mA 1'b11: 16mA

A073011C **ACFG_PIN_DRV3** **ACFG_GPIO_DRV3** **5555**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name													GPIO107		GPIO106	
Type													RW		RW	
Reset	0	1	0	1	0	1	0	1	0	1	0	1	01		01	

Bit(s)	Name	Description
3:2	GPIO107_DRV	GPIO107 driving strength configuration 2'b00: 4mA 2'b01: 8mA 2'b10: 12mA 1'b11: 16mA
1:0	GPIO106_DRV	GPIO106 driving strength configuration 2'b00: 4mA 2'b01: 8mA 2'b10: 12mA 1'b11: 16mA

A0730124 **ACFG_PIN_SMT1** **ACFG_GPIO_SMT** **0000**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name							GPIO 107		GPIO 106	GPIO 105						
Type							RW		RW	RW						
Reset	0	0	0	0	0	0	0		0	0	0	0	0	0	0	0

Bit(s)	Name	Description
9	GPIO107_SMT	GPIO107 Schmitt trigger control 0: Schmitt trigger off 1: Schmitt trigger on
8	GPIO106_SMT	GPIO106 Schmitt trigger control 0: Schmitt trigger off 1: Schmitt trigger on
7	GPIO105_SMT	GPIO105 Schmitt trigger control 0: Schmitt trigger off 1: Schmitt trigger on

A073012C	ACFG_PIN_IES1			ACFG_GPIO_IES							FFFF					
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name							GPIO 107	GPIO 106	GPIO 105							
Type							RW	RW	RW							
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit(s)	Name	Description
9	GPIO107_IES	Enables GPIO107 Rx input buffer 0: Disabled 1: Enabled
8	GPIO106_IES	Enables GPIO106 Rx input buffer 0: Disabled 1: Enabled
7	GPIO105_IES	Enables GPIO105 Rx input buffer 0: Disabled 1: Enabled

A0730130	ACFG_PIN_DINV0			ACFG_GPIO_DINV0							0000					
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	GPIO 105															
Type	RW															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Name	Description
15	GPIO105_DINV	GPIO105 inversion control 0: Disabled 1: Enabled

A0730134 ACFG_PIN_DINV1 ACFG_GPIO_DINV1

0000

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name															GPIO107	GPIO106
Type															RW	RW
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Name	Description
1	GPIO107_DINV	GPIO107 inversion control 0: Disabled 1: Enabled
0	GPIO106_DINV	GPIO106 inversion control 0: Disabled 1: Enabled

Power (SiP)	Pin Name	default Func.0	Aux Func.0	Aux Func.1	Aux Func.2	Aux Func.3	Aux Func.4	Aux Func.5	Aux Func.6	Aux Func.7
DVDD33	SPI_MOSI	I:EGND0	B:GPIO0	O:CLK00	I1:BTDBGIN	O:EDICK	O:SPIMOSI0	I1:U3RXD	O:PWM	I:EGND0
DVDD33	SCL28	I:EGND1	B:GPIO1	B1:SCL			O:SPISCK0		I0:D1_JCK	I:EGND1
DVDD33	SDA28	I:EGND2	B:GPIO2	B1:SDA			O:SPIC80		I0:D1_IMS	I:EGND2
DVDD33	KCOL7	I:EGND3	B:GPIO3	B1:KCOL7	O:EDICK	O:BTJTD0	O:FMJTD0	B1:MCD1	O:LSA0DA10	I:EGND3
DVDD33	KCOL6	I:EGND4	B:GPIO4	B1:KCOL6	I1:U1RXD	O:CLK05	I0:WIFIT0BT	B1:MCD2	O:DEBUG0MON3	I:EGND4
DVDD33	KCOL5	I:EGND5	B:GPIO5	B1:KCOL5	O:ED12CK	I1:BTJTMS	I1:FMJTMS	I1:EINT2		I:EGND5
DVDD33	KCOL4	I:EGND6	B:GPIO6	B1:KCOL4	I1:U3CTS	I1:BTJTCK	I0:FMJTCK		B:D1_IDA	I:EGND6
DVDD33	KCOL3	I:EGND7	B:GPIO7	B1:KCOL3	I0:DAIPCMIN				O:DEBUG0MON1	I:EGND7
DVDD33	KCOL2	I:EGND8	B:GPIO8	B1:KCOL2	O:DAICLK				O:DEBUG0MON2	I:EGND8
DVDD33	KCOL1	I:EGND9	B:GPIO9	B1:KCOL1	O:SPIMOSI1	I1:BTJTD1	I1:FMJTD1			I:EGND9
DVDD33	KCOL0	B:GPIO10	B:GPIO10	B1:KCOL0					O:DEBUG0MON3	
DVDD33	KROW7	I:EGND10	B:GPIO11	B1:KROW7	I1:EINT3	O:BTDBGACKN	B0:EDIDAT	B1:MCD3	O:CLK01	I:EGND10
DVDD33	KROW6	I:EGND11	B:GPIO12	B1:KROW6	O:ED12WS	I0:WIFIT0BT	I0:SPIMISO0		O:DEBUG0MON4	I:EGND11
DVDD33	KROW5	I:EGND12	B:GPIO13	B1:KROW5	O:U1TXD	O:ED12DAT	O:SRCLKENA		O:DEBUG0MON5	I:EGND12
DVDD33	KROW4	I:EGND13	B:GPIO14	B1:KROW4	O:DAIPCMOUT				O:DEBUG0MON6	I:EGND13
DVDD33	KROW3	I:EGND14	B:GPIO15	B1:KROW3	O:LSA0DA11	O:DAISYN			O:DEBUG0MON7	I:EGND14
DVDD33	KROW2	I:EGND15	B:GPIO16	B1:KROW2	O:LSCK0	I0:BTJTRSTB	I1:FMJTRST	I0:LSDI0		I:EGND15
DVDD33	KROW1	I:EGND16	B:GPIO17	B1:KROW1	B1:LSDA00	I1:U1CTS	I1:DAIRST		O:DEBUG0MON8	I:EGND16
DVDD33	KROW0	B:GPIO18	B:GPIO18	B1:KROW0	I0:LSDI0	I1:BTDBGIN	O:CLK06	O:LSCK0	O:DEBUG0MON9	I:EGND17
DVDD33	SPI_MISO	I:EGND18	B:GPIO19	I0:SPIMISO1	O:GPSFSYN	O:LRSTB	I1:DAIRST		O:DEBUG0MON10	I:EGND18
DVDD33	SPI_SCK	I:EGND19	B:GPIO20	O:SPISCK1	O:BPBUS5				O:DEBUG0MON11	I:EGND19
DVDD33	SD_PWROK	I:EGND20	B:GPIO21	O:CLK02	O:EDIWS	O:BPBUS5	O:LSCK1		O:DEBUG0MON12	I:EGND20
DVDD33	SPI_CS	I:EGND21	B:GPIO22	O:SPIC81	O:BPBUS4				O:DEBUG0MON13	I:EGND21
DVDD33	URXD2	I:EGND22	B:GPIO23	I1:U2RXD	B:BTPRI	O:LSCE0B0	O:U1RTS		O:DEBUG0MON14	I:EGND22
DVDD33	UTXD2	I:EGND23	B:GPIO24	O:U2TXD	O:U3TXD	O:BPBUS4	O:CLK07		O:DEBUG0MON15	I:EGND23
DVDD33	URXD1	I1:U1RXD	B:GPIO25	I1:U1RXD	I0:LSDI1	I1:EINT11				
DVDD33	UTXD1	O:U1TXD	B:GPIO26	O:U1TXD	I1:U3CTS	B1:LSDA01				
DVDD_EMI	SCL18	I:EGND24	B:GPIO27	I1:EINT4	O:DAICLK	I1:GPT_JNT	B1:SCL	B:NLD14	O:PWM	I:EGND24
DVDD_EMI	SDA18	I:EGND25	B:GPIO28	O:U3RTS	B1:SDA	B:NLD13	O:U3TXD	I0:CMCSD1	I1:EINT1	I:EGND25
DVDD33	JTD0	O:JTD0	B:GPIO29	O:JTD0	O:BTJTD0	O:FMJTD0				
DVDD33	JTMS	I1:JTMS	B:GPIO30	I1:JTMS	I1:BTJTMS	I1:FMJTMS				
DVDD33	JTCK	I1:JTCK	B:GPIO31	I1:JTCK	I1:BTJTCK	I0:FMJTCK				
DVDD33	JTDI	I1:JTDI	B:GPIO32	I1:JTDI	I1:BTJTDI	I1:FMJTDI				
DVDD33	JTRST_B	I0:JTRST_B	B:GPIO33	I0:JTRST_B	I0:BTJTRSTB	I1:FMJTRST				
DVDD33	JTRCK	O:JTRCK	B:GPIO34	O:JTRCK	O:BTDBGACKN					
DVDD_EMI	MCINS	I:EGND26	B:GPIO29	I1:EINT5	O:DAIPCMOUT	B:NLD12	I0:CMCSK	O:LSA0DA12	I0:LPT1	I:EGND26
DVDD_MSDC	MOCK	B:GPIO30	B:GPIO30	B1:MOCK				O:DEBUG1IN3	O:DEBUG1MON3	

Power (SiP)	Pin Name	default Func.0	Aux Func.0	Aux Func.1	Aux Func.2	Aux Func.3	Aux Func.4	Aux Func.5	Aux Func.6	Aux Func.7
DVDD_MSDC	MCD0A0	B:GPIO31	B:GPIO31	B1:MCD0A0				I0:DEBUG1IN1	O:DEBUG1MON1	
DVDD_MSDC	MCCM0	B:GPIO32	B:GPIO32	B1:MCCM0				I0:DEBUG1IN2	O:DEBUG1MON2	
DVDD_EMI	NLD8	I:EGND27	B:GPIO33	B:NLD8	I0:DAIPCMIN	O:CMCLK	I1:EINT6	O:CMPCN	I0:LSD2	I:EGND27
DVDD_EMI	NLD7	I:EGND28	B:GPIO34	B:NLD7	I1:EINT7	I0:LPTE1		I0:DEBUG1IN3	O:DEBUG1MON3	I:EGND28
DVDD_EMI	NLD6	I:EGND29	B:GPIO35	B:NLD6				I0:DEBUG1IN4	O:DEBUG1MON4	I:EGND29
DVDD_EMI	NLD5	I:EGND30	B:GPIO36	B:NLD5				I0:DEBUG1IN5	O:DEBUG1MON5	I:EGND30
DVDD_EMI	NLD4	I:EGND31	B:GPIO37	B:NLD4				B0:BT_RGPIO0	O:DEBUG1MON6	I:EGND31
DVDD_EMI	NLD3	I:EGND32	B:GPIO38	B:NLD3	I0:LSD3			B0:BT_RGPIO1	O:DEBUG1MON7	I:EGND32
DVDD_EMI	NLD2	I:EGND33	B:GPIO39	B:NLD2	B1:LSDA02			B0:BT_RGPIO2	O:DEBUG1MON8	I:EGND33
DVDD_EMI	NLD1	I:EGND34	B:GPIO40	B:NLD1	O:LSA0DA13			B0:BT_RGPIO3	O:DEBUG1MON9	I:EGND34
DVDD_EMI	NLD0	I:EGND35	B:GPIO41	B:NLD0	O:LSCK2			B0:BT_RGPIO4	O:DEBUG1MON10	I:EGND35
DVDD_EMI	LWR_B	I:EGND36	B:GPIO42	O:LWRB				B0:BT_RGPIO5	O:DEBUG1MON11	I:EGND36
DVDD_EMI	LRD_B	I:EGND37	B:GPIO43	O:LRDB					O:DEBUG1MON12	I:EGND37
DVDD_EMI	LP0	I:EGND38	B:GPIO44	O:LP0					O:DEBUG1MON13	I:EGND38
DVDD_EMI	LPCE0_B	O:LPCE0B	B:GPIO45	O:LPCE0B	O:LSCE0B1				O:DEBUG1MON14	
DVDD_EMI	LSCE1_B	B:GPIO46	B:GPIO46	O:LSCE1B	O:DAISYNC	B1:SCL	B:NLD9	O:LPCE2B	O:LSA0DA14	I:EGND39
DVDD_EMI	LPCE1_B	O:LPCE1B	B:GPIO47	O:LPCE1B	O:CMCLK	B1:SDA	B:NLD10	O:CMPCN	I0:LPTE1	
DVDD_EMI	LSRSTB	I:EGND40	B:GPIO48	O:LSRSTB	O:CLK03	I1:U3RXD	B:NLD11	I0:CMCSD0	O:LSCK3	I:EGND40
DVDD_EMI	WATCHDOG	O:WATCHDOG	B:GPIO49	O:WATCHDOG	B:NLD15	I0:CMC8K	I0:CMCSD1	B1:LSDA03		
DVDD_EMI	LPTE	I:EGND41	B:GPIO50	I0:LPTE1	O:CLK04				O:DEBUG1MON15	I:EGND41
DVDD33	LPRSTB	I:EGND42	B:GPIO51	O:LPRSTB	I1:EINT8	O:LSCK4	I0:LPTE1	O:LPCE3B		I:EGND42
DVDD33	CMDAT0	I:EGND43	B:GPIO52	I0:CMDAT0	I0:CMCSD0			I0:DEBUG2IN0	O:DEBUG2MON0	I:EGND43
DVDD33	CMDAT1	I:EGND44	B:GPIO53	I0:CMDAT1	I0:CMCSD1			I0:DEBUG2IN1	O:DEBUG2MON1	I:EGND44
DVDD33	CMDAT2	I:EGND45	B:GPIO54	I0:CMDAT2	B1:SCL			I0:DEBUG2IN2	O:DEBUG2MON2	I:EGND45
DVDD33	CMDAT3	I:EGND46	B:GPIO55	I0:CMDAT3	O:LSRSTB	I0:DAIPCMIN		I0:DEBUG2IN3	O:DEBUG2MON3	I:EGND46
DVDD33	CMDAT4	I:EGND47	B:GPIO56	I0:CMDAT4	O:LSA0DA15	O:DAICLK	O:U3RTS	I0:DEBUG2IN4	O:DEBUG2MON4	I:EGND47
DVDD33	CMDAT5	I:EGND48	B:GPIO57	I0:CMDAT5	O:LSCK5	O:DAIPCMOUT	I1:U2RXD	I0:DEBUG2IN5	O:DEBUG2MON5	I:EGND48
DVDD33	CMDAT6	I:EGND49	B:GPIO58	I0:CMDAT6	B1:LSDA04	O:DAISYNC	O:U2TXD	I0:DEBUG2IN6	O:DEBUG2MON6	I:EGND49
DVDD33	CMDAT7	I:EGND50	B:GPIO59	I0:CMDAT7	I1:U3CTS	O:PWM		I0:DEBUG2IN7	O:DEBUG2MON7	I:EGND50
DVDD33	CMHREF	I:EGND51	B:GPIO60	I0:CMHREF	I1:EINT9	I1:U3RXD		I0:DEBUG2IN8	O:DEBUG2MON8	I:EGND51
DVDD33	CMVREF	I:EGND52	B:GPIO61	I0:CMVREF	B1:SDA			I0:DEBUG2IN9	O:DEBUG2MON9	I:EGND52
DVDD33	CMPCN	B:GPIO62	B:GPIO62	O:CMPCN	O:LSCE0B2	O:U3TXD		I0:DEBUG2IN10	O:DEBUG2MON10	
DVDD33	CMCLK	I:EGND53	B:GPIO63	O:CMCLK				I0:DEBUG2IN11	O:DEBUG2MON11	I:EGND53
DVDD33	CMPLK	I:EGND54	B:GPIO64	I0:CMPLK	I0:CMC8K			I0:DEBUG2IN12	O:DEBUG2MON12	I:EGND54
DVDD33	CMRST	B:GPIO65	B:GPIO65	O:CMRST				I0:DEBUG2IN13	O:DEBUG2MON13	I:EGND55
DVDD33	EINT0	I:EGND56	B:GPIO66	I1:EINT0	B0:EDIDAT	I0:SD4		I0:DEBUG2IN14	O:DEBUG2MON14	I:EGND56
DVDD33	BPIBUS3	O:BPIBUS3	B:GPIO77	O:BPIBUS3						

Power (SiP)	Pin Name	default Func.0	Aux Func.0	Aux Func.1	Aux Func.2	Aux Func.3	Aux Func.4	Aux Func.5	Aux Func.6	Aux Func.7
DVDD33	BPI_BUS2	O:BPIBUS2	B:GPIO78	O:BPIBUS2						
DVDD33	BPI_BUS1	O:BPIBUS1	B:GPIO79	O:BPIBUS1						
DVDD33	BPI_BUS0	O:BPIBUS0	B:GPIO80	O:BPIBUS0						
DVDD_SF	SFSCCK	I:EGND57	B:GPIO81	B:SFSCK						I:EGND57
DVDD_SF	SFSWP	I:EGND58	B:GPIO82	B:SFSWP						I:EGND58
DVDD_SF	SFSHOLD	I:EGND59	B:GPIO83	B:SFSHOLD						I:EGND59
DVDD_SF	SFSCS0	I:EGND60	B:GPIO84	O:SFSCK0						I:EGND60
DVDD_SF	SFSIN	I:EGND61	B:GPIO85	B:SFSIN						I:EGND61
DVDD_SF	SFSOUT	I:EGND62	B:GPIO86	B:SFSOUT						I:EGND62
DVDD33_VSIM	SIM1_SIO	I:EGND63	B:GPIO67	B:SIMSIO						I:EGND63
DVDD33_VSIM	SIM1_SRST	I:EGND64	B:GPIO68	B:SIMSRST						I:EGND64
DVDD33_VSIM	AM2_SIO	I:EGND65	B:GPIO69	B:SIM2SIO						I:EGND65
DVDD33_VSIM	SIM1_SCLK	I:EGND66	B:GPIO70	B:SIMSCLK						I:EGND66
DVDD33_VSIM	SIM2_SCLK	I:EGND67	B:GPIO71	B:SIM2SCLK						I:EGND67
DVDD33_VSIM	AM2_SRST	I:EGND68	B:GPIO72	B:SIM2SRST						I:EGND68
DVDD_SF	SFSCS1	B:GPIO74	B:GPIO74	O:SFSCK1	I1:EINT10	O:CMPCDN				
DVDD33	PRTC_B	I:PRTC_B		I:PRTC_B						
DVDD33	BONDOPT1	I:BONDOPT1		I:BONDOPT1						
	D2D_BT_ACC_IN	O:D2D_BT_ACC	B:GPIO73	O:D2D_BT_ACC_IND						O:D2D_DEBUG2IN I0 D2D_DEBUG2MON0
	D2D_BT_ACC_C0	B0:D2D_BT_ACC	B:GPIO75	B0:D2D_BT_ACC_CON_3						O:D2D_DEBUG2IN I0 D2D_DEBUG2MON1
	D2D_BT_ACC_C1	O:D2D_BT_ACC	B:GPIO76	O:D2D_BT_ACC_CON2						O:D2D_DEBUG2IN I0 D2D_DEBUG2MON2
	D2D_BT_ACC_C2	O:D2D_BT_ACC	B:GPIO93	O:D2D_BT_ACC_CON1						O:D2D_DEBUG2IN I0 D2D_DEBUG2MON3
	D2D_BT_ACC_C3	O:D2D_BT_ACC	B:GPIO94	O:D2D_BT_ACC_CON0						O:D2D_DEBUG2IN I0 D2D_DEBUG2MON4
	D2D_BSI_EN	O:D2D_BSI_EN	B:GPIO95	O:D2D_BSI_EN			O:D2D_FMTTRST			O:D2D_DEBUG2IN I0 D2D_DEBUG2MON5
	D2D_BSI_DATAI1	B0:D2D_BSI_DAT	B:GPIO96	B0:D2D_BSI_DATAI1			O:D2D_FMTTMS			O:D2D_DEBUG2IN I0 D2D_DEBUG2MON6
	D2D_BSI_DATAI0	B0:D2D_BSI_DAT	B:GPIO97	B0:D2D_BSI_DATAI0			O:D2D_FMTDI			O:D2D_DEBUG2IN I0 D2D_DEBUG2MON7
	D2D_BSI_CLK	O:D2D_BSI_CLK	B:GPIO98	O:D2D_BSI_CLK			O:D2D_FMTDO			O:D2D_DEBUG2IN I0 D2D_DEBUG2MON7
	D2D_BSI_CLK	O:D2D_BSI_CLK	B:GPIO98	O:D2D_BSI_CLK			O:D2D_FMTJCK			O:D2D_DEBUG2IN I0 D2D_DEBUG2MON8
	D2D_ANA_CON	B:D2D_ANA_CON_SDA1		B:D2D_ANA_CON_SDA1						
	D2D_ANA_CON	B:D2D_ANA_CON_SDA0		B:D2D_ANA_CON_SDA0						
	D2D_ANA_CON	O:D2D_ANA_CON_SCK		O:D2D_ANA_CON_SCK						
	D2D_ABB_DPH	O:D2D_ABB_DPH	B:GPIO99	O:D2D_ABB_DPH_SCK						O:D2D_DEBUG2IN I0 D2D_DEBUG2MON9
	D2D_ABB_DPH	B0:D2D_ABB_DP	B:GPIO100	B0:D2D_ABB_DPH_SDA2						O:D2D_DEBUG2IN I0 D2D_DEBUG2MON10
	D2D_ABB_DPH	B0:D2D_ABB_DP	B:GPIO101	B0:D2D_ABB_DPH_SDA1						O:D2D_DEBUG2IN I0 D2D_DEBUG2MON11
	D2D_ABB_DPH	B0:D2D_ABB_DP	B:GPIO102	B0:D2D_ABB_DPH_SDA0						O:D2D_DEBUG2IN I0 D2D_DEBUG2MON12
	D2D_XOSC_32K	I:D2D_XOSC_32K_CK		I:D2D_XOSC_32K_CK						
	D2D_CLK_128K	O:D2D_CLK_128K		O:D2D_CLK_128K						
	D2D_SYS_RESE	I:D2D_SYS_RESETB		I:D2D_SYS_RESETB						

Power (SiP)	Pin Name	default Func.0	Aux Func.0	Aux Func.1	Aux Func.2	Aux Func.3	Aux Func.4	Aux Func.5	Aux Func.6	Aux Func.7
	D2D_RGU_RESE	O:D2D_RGU_RESETB		O:D2D_RGU_RESETB						
	D2D_TESTMODE	I:D2D_TESTMODE		I:D2D_TESTMODE						
	D2D_BB_SRCLK	O:D2D_BB_SRCLKENA		O:D2D_BB_SRCLKENA						
	D2D_ANA_WAKE	I1:D2D_ANA_WAKE	B:GPIO103	I1:D2D_ANA_WAKEUP						O:D2D_DEBUG2IN I0 D2D_DEBUG2MON13
	D2D_ANA_INT	I1:D2D_ANA_INT	B:GPIO104	I1:D2D_ANA_INT						O:D2D_DEBUG2IN I0 D2D_DEBUG2MON14

4.5 General-purpose Timer

4.5.1 General Description

Four general-purpose timers are provided. Three timers are 16 bits long and one timer is 32 bits long. Each runs independently. GPT1 ~ 3 use 32k clock source to count, whereas GPT4 uses 26M clock source. The 26M clock source can be gated when the system enters the sleep mode, and this will cause GPT4 to stop counting, whereas 32k clock source is always toggling. GPT1 and GPT2 can operate in one of the two modes: one-shot mode and auto-repeat mode. GPT3 and GPT4 are free running timer. In the one-shot mode, when the timer counts down and reaches 0, it will be halted. In the auto-repeat mode, when the timer reaches 0, it will simply be reset to counting down the initial value and repeating the count-down to 0. This loop keeps repeating until the disabling signal is set to 1. Regardless of the timer's mode, if the countdown initial value (i.e. GPTIMER1_DAT for GPT1 or GPTIMER_DAT2 for GPT2) is written when the timer is running, the new initial value will not take effect until the next time the timer is restarted. In the auto-repeat mode, the new countdown start value is

used on the next countdown iteration. Therefore, before enabling the timer, the desired values for GPTIMER_DAT and the GPTIMER_PRESCALER registers must first be set.

4.5.2 Register Definition

Register address	Register name	Synonym
0xA00C_0000	General-purpose timer 1 control register	GPTIMER1_CON
0xA00C_0004	General-purpose timer 1 time-out interval register	GPTIMER1_DAT
0xA00C_0008	General-purpose timer 2 control register	GPTIMER2_CON
0xA00C_000C	General-purpose timer 2 time-out interval register	GPTIMER2_DAT
0xA00C_0010	General-purpose timer status register	GPTIMER_STA
0xA00C_0014	General-purpose timer 1 prescaler register	GPTIMER1_PRESCALAR
0xA00C_0018	General-purpose timer 2 prescaler register	GPTIMER2_PRESCALAR
0xA00C_001C	General-purpose timer 3 control register	GPTIMER3_CON
0xA00C_0020	General-purpose timer 3 time-out interval register	GPTIMER3_DAT
0xA00C_0024	General-purpose timer 3 prescaler register	GPTIMER3_PRESCALAR
0xA00C_0028	General-purpose timer 4 control register	GPTIMER4_CON
0xA00C_002C	General-purpose timer 4 data register	GPTIMER4_DAT

Table 51. General-purpose timer register map

0xA00C_0000 **GPT1 control register** **GPTIMER1_CON**

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	EN	MODE														
Type	R/W	R/W														
Reset	0	0														

MODE Controls GPT1 to count repeatedly (in a loop) or just for one-shot

- 0** One-shot mode is selected.
- 1** Auto-repeat mode is selected.

EN Controls GPT1 to start counting or to stop

- 0** Disable GPT1
- 1** Enable GPT1

0xA00C_0004 **GPT1 time-out interval register**

**GPTIMER1_D
AT**

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CNT [15:0]															
Type	R/W															
Reset	FFFFh															

CNT [15:0] Initial counting value. GPT1 counts down from GPTIMER1_DAT. When GPT1 counts down to 0, a GPT1 interrupt will be generated.

0xA00C_0008 **GPT2 control register**

**GPTIMER2_C
ON**

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	EN	MOD E														
Type	R/W	R/W														
Reset	0	0														

MODE Controls GPT2 to count repeatedly (in a loop) or just for one-shot

- 0** One-shot mode is selected.
- 1** Auto-repeat mode is selected.

EN Controls GPT2 to start counting or to stop

- 0** Disable GPT2
- 1** Enable GPT2

0xA00C_000 **GPT2 time-out interval register**

**GPTIMER2_D
AT**

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Name	CNT [15:0]
Type	R/W
Reset	FFFFh

CNT [15:0] Initial counting value. GPT2 counts down from GPTIMER2_DAT. When GPT2 counts down to 0, a GPT2 interrupt will be generated.

0xA00C_0010
h **GPT status register**

GPTIMER_ST
A

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name															GPT2	GPT1
Type															RC	RC
Reset															0	0

This register illustrates the GP timer time-out status. Each flag is set when the corresponding timer count-down is completed. Can be cleared when the CPU reads the status register.

0xA00C_0014
h **GPT1 prescaler register**

GPTIMER1_PRESCALER

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name															PRESCALER [2:0]	
Type															R/W	
Reset															100b	

PRESCALER Controls the counting clock for GP timer 1

000	16,384 Hz
001	8,192 Hz
010	4,096 Hz
011	2,048 Hz
100	1,024 Hz
101	512 Hz
110	256 Hz
111	128 Hz

0xA00C_0018h **GPT2 prescaler register**
GPTIMER2_PRESCALER

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name														PRESCALER [2:0]		
Type														R/W		
Reset														100b		

PRESCALER Controls the counting clock for GP timer 2

000	16,384 Hz
001	8,192 Hz
010	4,096 Hz
011	2,048 Hz
100	1,024 Hz
101	512 Hz
110	256 Hz
111	128 Hz

0xA00C_001Ch **GPT3 control register**
GPTIMER3_CON

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name																EN
Type																R/W
Reset																0

EN Controls GPT3 to start counting or to stop

0	Disable GPT3
1	Enable GPT3

0xA00C_0020h **GPT3 time-out interval register**
GPTIMER3_DAT

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
-----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----

Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CNT [15:0]															
Type	RO															
Reset	0															

CNT [15:0] If EN=1, GPT3 will be a free running timer. This register records the GPT3 value. If EN = 0, this register will be cleared to 0.

0xA00C_0024h GPT3 prescaler register GPTIMER3_PRESCALER

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name														PRESCALER [2:0]		
Type														R/W		
Reset														100b		

PRESCALER Controls the counting clock for GP timer 3

- 000** 16,384 Hz
- 001** 8,192 Hz
- 010** 4,096 Hz
- 011** 2,048 Hz
- 100** 1,024 Hz
- 101** 512 Hz
- 110** 256 Hz
- 111** 128 Hz

0xA00C_0028h GPT4 control register GPTIMER4_CON

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name															LOCK	EN

Type																	R/W	R/W
Reset																	0	0

EN Controls GPT4 to start counting or to stop

0 Disable GPT4

1 Enable GPT4

LOCK Controls GPT4 EN bit can be modified or not. If LOCK = 0, EN can be modified. If LOCK = 1, the EN value will be fixed, and the LOCK bit will always be 1 and cannot be modified until hardware reset.

0 Unlock

1 Lock

0xA00C_002 **GPT4 data register**
Ch

GPTIMER4_D
AT

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name	CNT[31:16]															
Type	RO															
Reset	0															
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CNT [15:0]															
Type	RO															
Reset	0															

CNT [31:0] If EN=1, GPT4 will be a free running timer. This register records the GPT4 value. If EN = 0, this register will be cleared to 0. This register does not allow continuous read. It requires at least one 26M clock cycle between two APB reads.

4.6 MCU OSTIMER

4.6.1 Overview

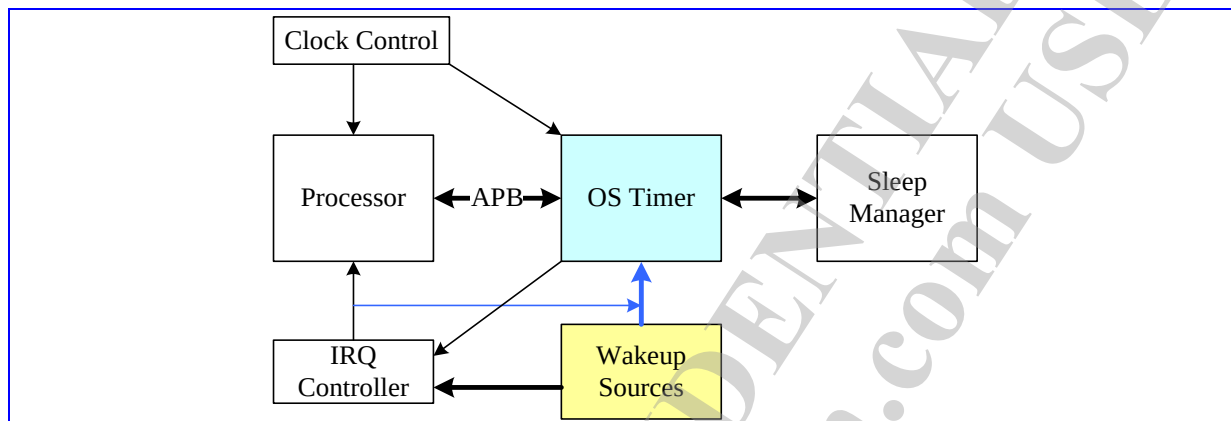


Figure 37. OS timer system view

The OS timer is a hardware timer which specifies the OS time frame duration and generates time-out interrupts by programming the frame counter number. The OS timer has the pause mode. The user can specify pause duration period before the pause mode, and the timer will resume from the pause mode by the external wakeup sources or when the pause duration is timed out.

4.6.2 Terminology

Abbreviation	
R/W	Read/Write
RO	Read only
WO	Write only
W1C	Write 1 to clear
R/W1C	Read/Write 1 to clear
FRC	Free running counter in the system
OST	OS timer

Table 52. Abbreviations

4.6.3 Introduction to Wakeup Source

The OS timer only accepts level trigger wakeup source. The wakeup sources are all treated as asynchronous input (will be changed) and will be synchronized by OST clock in OST wakeup source controller.

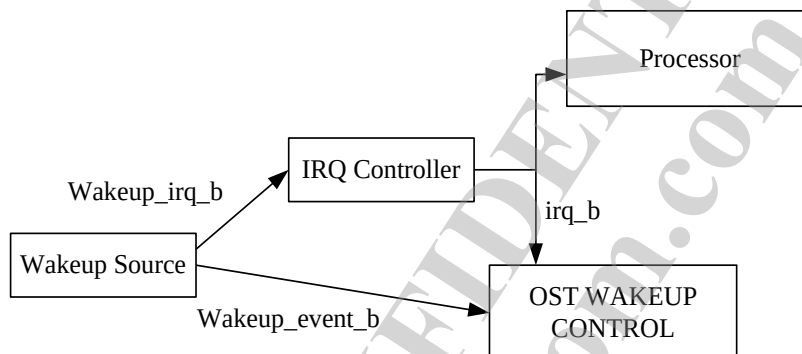
The possible wakeup sources are listed in the table below.

No	Wakeup source
0	GPT
1	EINT
2	Timer trigger
3	KP
4	MSDC
5	ANALOG
6	DSP
7	GPI
8	Reserved
9	Reserved
10	Reserved
11	Reserved
12	DSP_ASYNC
13	GPI_ASYNC
14	Reserved
15	Reserved
16	Reserved
17	Reserved
18	Reserved

Table 53. Wakeup sources

No	Wakeup source	Edge/Level	SW clear	Clock domain
0	GPT	Edge		F32k_CK
1	EINT	Level		F32k_CK
2	Timer trigger	Level		F32k_CK
3	KP	Level		
4	MSDC	Level		
5	ANALOG	Level		
6	DSP	Level		DSP_CK
7	GPI	Level		
8	Reserved			
9	Reserved			
10	Reserved			
11	Reserved			
12	DSP_ASYNC	Level		DSP_CK
13	GPI_ASYNC	Level		MCLK_CK
14	Reserved			
15	Reserved			

No	Wakeup source	Edge/Level	SW clear	Clock domain
16	Reserved			
17	Reserved			
18	Reserved			

Table 54. Characteristics of wakeup sources

Figure 38. Wakeup event and Irq_b integration diagram

wakeup_irq_b may be asserted before *wakeup_event_b* is asserted from the wakeup source peripheral to ensure the software will enter wakeup ISR after the pause command is set.

Recommended software programming sequence:

1. I-BIT is set when a pause command is executed.
2. Set up IRQ mask registers to select IRQ wakeup sources
3. Pause criteria are met.
4. Set up pause command.
5. Confirm the pause command is executed at OST, and check if the pause command is pending or not.
6. Set up processor in request for interrupt state.
7. The processor clock will be off if there is no interrupt.
8. Check if the pause request command is completed (after the processor clock is active or resumes).
9. Clear I-BIT.

4.6.4 Register Definition

Address	Name	Width	Register function
0XA01F0000	OST_CON	32	OS timer control register Only updated when OST_CMD.OST_WR is set and OST_STA.WR_RDY is high.
0XA01F0004	OST_CMD	32	OS timer command Only valid when the write data BIT31 to BIT16 is 0x6276.

Address	Name	Width	Register function
0XA01F0008	OST_STA	32	OS timer status
0XA01F000C	OST_FRM	32	OS timer frame duration Specifies OS timer frame duration by micron second resolution. The maximum duration is 8ms and minimum duration is 1ms. This register should be set before OS timer is enabled. The hardware will set up OST_ISR[0] periodically at frame duration period when the OS timer is enabled.
0XA01F0010	OST_FRM_F32K	32	OS timer frame duration by 32K clock Specifies OS timer frame duration with 30.5176 micron second (32kHz) resolution. The maximum duration is 8ms and minimum duration is 1ms. This register should be set before the OS timer is enabled. OST_FRM_F32K*30.5176us should be less than OST_FRM_NUM*OST_FRM*1us - 30.5176us. Set up OST_FRM_NUM if the frame duration is shorter than system settling time.
0XA01F0014	OST_UFN	32	OS timer un-alignment frame number Specifies OS timer un-alignment event frame number count. This register value is updated to OS timer only when OST_CMD.OST_WR is set. The read value of this register may not be the current OS time Un-alignment frame number. The hardware will count down the OST_UFN counter at frame time-out and stop the count-down when the current value is 0. When the OS timer is in the pause mode, this counter is still active and wakes up the OS timer when OST_UFN becomes 0 at frame time-out boundary. The software should enable the OS timer pause mode when OST_UFN is larger than 1, or the pause command will be ignored by the hardware.
0XA01F0018	OST_AFN	32	OS timer alignment frame number Specifies OS timer alignment event frame number count. This register value is updated to the OS timer only when OST_CMD.OST_WR is set. The read value of this register may not be the current OS time alignment frame number. The hardware will count down the OST_AFN counter at frame time-out and stop the count-down when the current value is 0. The hardware will set OST_ISR[1] when OST_AFN is 1 or 0 at frame time-out, and the OS timer is in normal mode. OST_AFN is current hardware AFN down counter value, and the software cannot read this register directly. The software can only read the current frame number through the OST_AFN_R register, and there is synchronization latency from OST_AFN to OST_AFN_R.

Address	Name	Width	Register function
0XA01F001C	OST_AFN_DLY	32	OS timer alignment frame delay number count Specifies the OS timer alignment event frame delay count due to OS timer pause mode. The software has to use the OST_CMD.OST_RD command to update this register value, or the value may be out of date.
0XA01F0020	OST_UFN_R	32	Current OS timer un-alignment frame number Specifies the OS timer current un-alignment frame number. The software has to use the OST_CMD.OST_RD command to update this register value, or the value may be out of date.
0XA01F0024	OST_AFN_R	32	Current OS timer alignment frame number Specifies the OS timer current alignment frame number. The software has to use the OST_CMD.OST_RD command to update this register value, or the value may be out of date.
0XA01F0030	OST_INT_MASK	32	OS timer interrupt mask Specifies the OS timer interrupt mask control.
0XA01F0040	OST_ISR	32	OS timer interrupt status Specifies the OS timer interrupt status. The software has to write 1 at the corresponding bit to clear the interrupt status bit. OSR_ISR[2-0] are also cleared when the OSR_WR command is executed and AFN, UFN are updated.
0XA01F0050	OST_EVENT_MASK	32	OS timer event mask Specifies which wakeup event will be masked to wake up the OS timer in the OS timer pause mode.
0XA01F0054	OST_WAKEUP_STA	32	OS timer event wakeup status
0XA01F0060	OST_DBG_WAKEUP	32	OS timer debug wakeup

0XA01F0000				OS timer control register										OST_CON		
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name														OST_DBG	UFN_DOWN	EN
Type														RW	RW	RW
Reset														0	1	0

Overview This registers is only updated when OST_CMD.OST_WR is set and OST_STA.WR_RDY is high.

Bit(s)	Name	Description
2	OST_DBG	Enables OST wakeup debugging function 0: Disable 1: Enable
1	UFN_DOWN	Enables OST_UFN count-down 0: Disable OST_UFN count-down 1: Enable OST_UFN count-down
0	EN	Enables OS timer 0: OS timer is disabled. Then all internal timers are stopped. 1: OS Timer is enabled. The software has to ensure OST_AFN, OST_UFN, OST_FRAM are configured before enabling the OS timer.

0XA01F0004				OS timer command								OST_CMD				
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name	OST_KEY															
Type	WO															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OST_CON_WR	OST_AFN_WR	OST_UFN_WR											OST_WR	OST_RD	PAUSE_STR
Type	RW	RW	RW											WO	WO	WO
Reset	0	0	0											0	0	0

Overview The command is only valid when the write data BIT31 to BIT16 is 0x6276.

Bit(s)	Name	Description
31:16	OST_KEY	
15	OST_CON_WR	Updates OST_CON when the OST_WR command is active. .
14	OST_AFN_WR	Updates OST_AFN when the OST_WR command is active.
13	OST_UFN_WR	Updates OST_UFN when the OST_WR command is active.
2	OST_WR	Write 1 to this bit to update the bus clock domain OS timer configuration into the OST SYSCLK domain
1	OST_RD	Write 1 to this bit to update the current OS timer status to the bus clock domain
0	PAUSE_STR	Write 1 to this bit to enable the OS timer pause function. The command will be ignored if the current OST UFN is less than 2. The software has to ensure OST_CMD.OST_WR is completed before the next software pause sequence.

0XA01F0008				OS timer status								OST_STA				
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16

Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CPU_SLEEP									AFN_DLY_OVER		PAUSE_REQ			CMD_CPL	READY
Type	RO									RO		RO			RO	RO
Reset	0									0		0	0		1	0

Overview

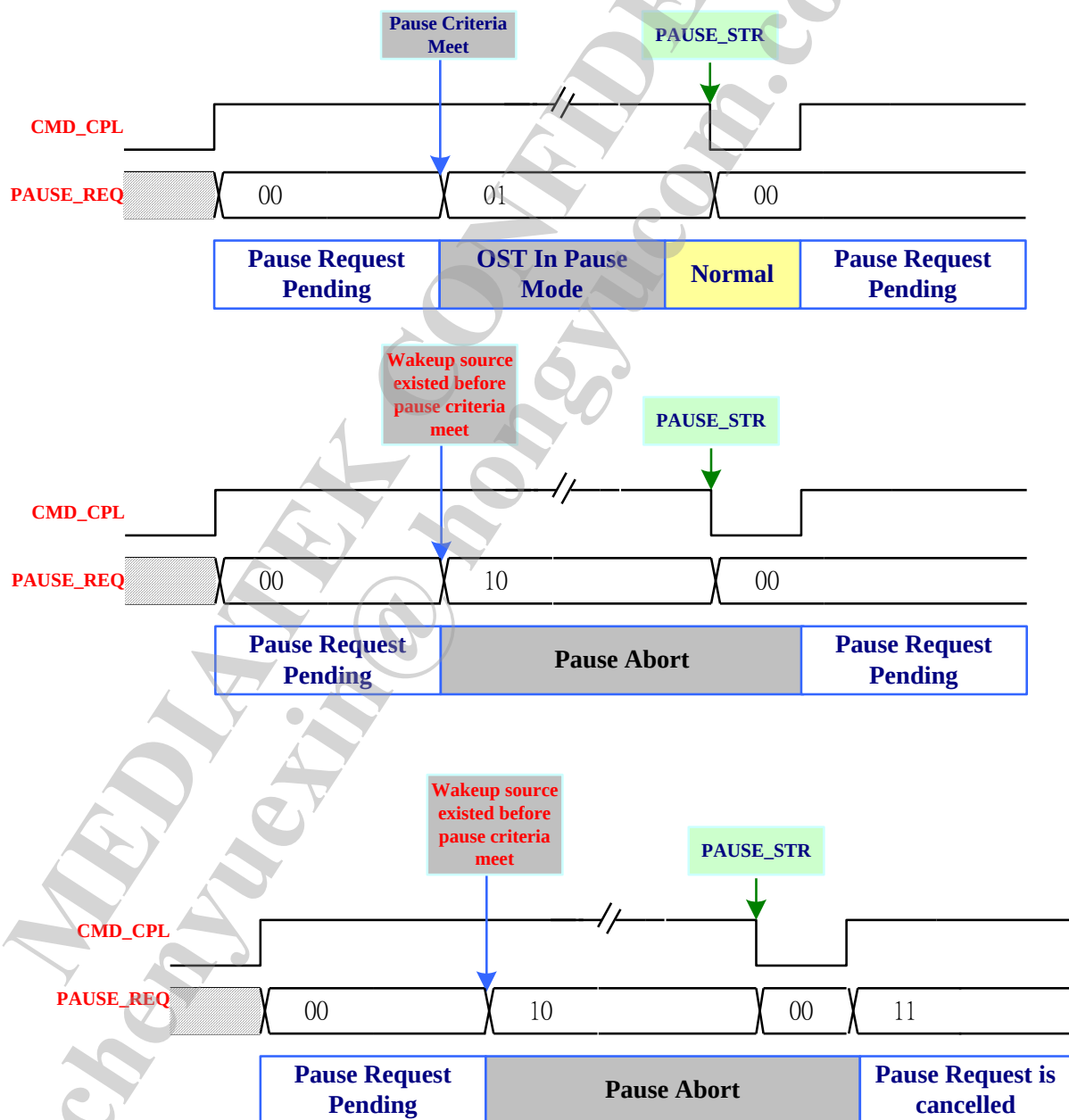


Figure 39. Pause command complete and pause request state

Bit(s)	Name	Description
15	CPU_SLEEP	The processor is in the sleep mode. (for debugging) 0: Active 1: Sleep
6	AFN_DLY_OVER	AFN_DLY counter overflows. This bit is cleared when AFN is updated. 0: Does not overflow 1: Overflow
4:3	PAUSE_REQ	An OS timer pause request is pending. A pause command will be completed when the pause command is set. 1. Processor is in the sleep mode (processor clock is off), UFN ≥ 2 and no wakeup sources 2. Any wakeup sources are sensed after the pause command is set. 3. UFN < 2 00: The last pause command request is not completed yet (CP15 is not enable and no wakeup source). 01: The last pause command request is completed with OST pause mode being active. 10: The last pause command request is completed with wakeup sources. 11: The last pause command request is completed with UFN < 2 .
1	CMD_CPL	OST command is completed. It takes several clocks from OST_CMD being updated to command being active. 0: OST command is not completed. 1: OST command is completed.
0	READY	OS timer status 0: OST is in pause mode. 1: OST is in normal mode.

0XA01F000C				OS timer frame duration												OST_FRM	
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
Name																	
Type																	
Reset																	
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
Name				OST_FRM													
Type				RW													
Reset				1	0	0	0	0	0	1	0	0	1	0	0	1	

Overview This register specifies the OS timer frame duration by micron second resolution. The maximum duration is 8ms and minimum duration is 1ms. This register should be set before the OS timer is enabled. The hardware will set up OST_ISR[0] periodically at

frame duration period when the OS timer is enabled.

Bit(s)	Name	Description
12:0	OST_FRM	

0XA01F0010 OS timer frame duration by 32K clock OST_FRM_F32K

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OST_UFN								OST_FRM_F32K							
Type	RW								RW							
Reset	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0

Overview This register specifies the OS timer frame duration with 30.5176 micron second (32kHz) resolution. The maximum duration is 8ms and minimum duration is 1ms. This register should be set up before the OS timer is enabled. OST_FRAM_F32K*30.5176us should be less than OST_FRM_NUM*OST_FRM*1us - 30.5176us. Set up OST_FRM_NUM if the frame duration is shorter than the system settling time.

Bit(s)	Name	Description
15:12	OST_UFN	
11:0	OST_FRM_F32K	

0XA01F0014 OS timer un-alignment frame number OST_UFN

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name	OST_UFN[31:16]															
Type	RW															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OST_UFN[15:0]															
Type	RW															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Overview This register specifies the OS timer un-alignment event frame number count. This register value is updated to the OS timer only when OST_CMD.OST_WR is set. The read value of this register may not be the current OS time un-alignment frame number. The hardware will count down the OST_UFN counter at frame time-out and stop the count-down when the current value is 0. When the OS timer is in the pause mode, this

counter is still active and wakes up the OS timer when OST_UFN becomes 0 at frame time-out boundary. The software should enable the OS timer pause mode when OST_UFN is larger than 1, or the pause command will be ignored by hardware.

Bit(s)	Name	Description
31:0	OST_UFN	

0XA01F0018				OS timer alignment frame number												OST_AFN	
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
Name	OST_AFN[31:16]																
Type	RW																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
Name	OST_AFN[15:0]																
Type	RW																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

Overview This register specifies the OS timer alignment event frame number count. This register value is updated to the OS timer only when OST_CMD.OST_WR is set. The read value of this register may not be the current OS time alignment frame number. The hardware will count down the OST_AFN counter at frame time-out and stop the count-down when the current value is 0. The hardware will set up OST_ISR[1] when OST_AFN is 1 or 0 at frame time-out and the OS timer is in the normal mode. OST_AFN is the current hardware AFN down counter value, and the software cannot read this register directly. The software can only read the current frame number through the OST_AFN_R register, and there is synchronization latency from OST_AFN to OST_AFN_R.

Bit(s)	Name	Description
31:0	OST_AFN	

0XA01F001C																OS timer alignment frame delay number count																OST_AFN_DLY																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																						
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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Overview This register specifies the OS Timer Alignment event frame delay count due to OS timer pause mode. Software has to use OST_CMD.OST_RD command to update this register value, or the value maybe out of date.

Bit(s)	Name	Description
31:0	OST_AFN_DLY	

0XA01F0020 **Current OS timer un-alignment frame number** **OST_UFN_R**

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name	OST_UFN_R[31:16]															
Type	RO															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OST_UFN_R[15:0]															
Type	RO															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Overview This register specifies the OS timer current un-alignment frame number. The software has to use the OST_CMD.OST_RD command to update this register value, or the value may be out of date.

Bit(s)	Name	Description
31:0	OST_UFN_R	

0XA01F0024 **Current OS timer alignment frame number** **OST_AFN_R**

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name	OST_AFN_R[31:16]															
Type	RO															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OST_AFN_R[15:0]															
Type	RO															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Overview This register specifies the OS timer current alignment frame number. The software has to use the OST_CMD.OST_RD command to update this register value, or the value may be out of date.

Bit(s)	Name	Description
31:0	OST_AFN_R	

0XA01F0030
OS Timer Interrupt Mask
OST_INT_MASK

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name												OST_INT_MASK				
Type												RW				
Reset												1	1	1	1	1

Overview This register specifies the OS timer interrupt mask control.

Bit(s)	Name	Description
4:0	OST_INT_MASK	0: Mask OS timer frame time-out interrupt 1: Mask OS timer alignment frame time-out interrupt 2: Mask OS timer un-alignment frame time-out interrupt 3: Mask OS timer pause abort interrupt 4: Mask OS timer pause interrupt

0XA01F0040
OS timer interrupt status
OST_ISR

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name												OST_ISR				
Type												W1C				
Reset												0	0	0	0	0

Overview This register specifies the OS timer interrupt status. The software has to write 1 to the corresponding bit to clear the interrupt status bit. OSR_ISR[2-0] are also cleared when the OSR_WR command is executed and AFN, UFN are updated.

Bit(s)	Name	Description
4:0	OST_ISR	0: OS timer frame time-out interrupt status 1: OS timer alignment frame time-out interrupt status 2: OS timer un-alignment frame time-out interrupt status 3: OS timer pause abort interrupt status 4: OS timer pause interrupt status

0XA01F0050
OS timer event mask
OST_EVENT_MASK

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name														OST_EVENT_MASK[18:16]		
Type														RW		
Reset														0	1	1
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OST_EVENT_MASK[15:0]															
Type	RW															
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Overview This register specifies which wakeup event will be masked to wake up the OS timer in the OS timer pause mode.

Bit(s)	Name	Description
18:0	OST_EVENT_MASK	

0XA01F0054
OS timer event wakeup status
OST_WAKEUP_STA

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name														OST_WAKEUP_STA[18:16]		
Type														RO		
Reset														0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OST_WAKEUP_STA[15:0]															
Type	RO															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Name	Description
18:0	OST_WAKEUP_STA	

0XA01F0060
OS timer debug wakeup
OST_DBG_WAKEUP

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name	CIRQ_MASK_EN													OST_DBG_WAKEUP[18:16]		
Type	RW															
Reset	0													0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Name	OST_DBG_WAKEUP[15:0]															
Type																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit(s)	Name	Description
31	CIRQ_MASK_EN	0: Disable cirq mask function 1: Enable cirq mask function
18:0	OST_DBG_WAKEUP	Controls wakeup status in debug timing event1

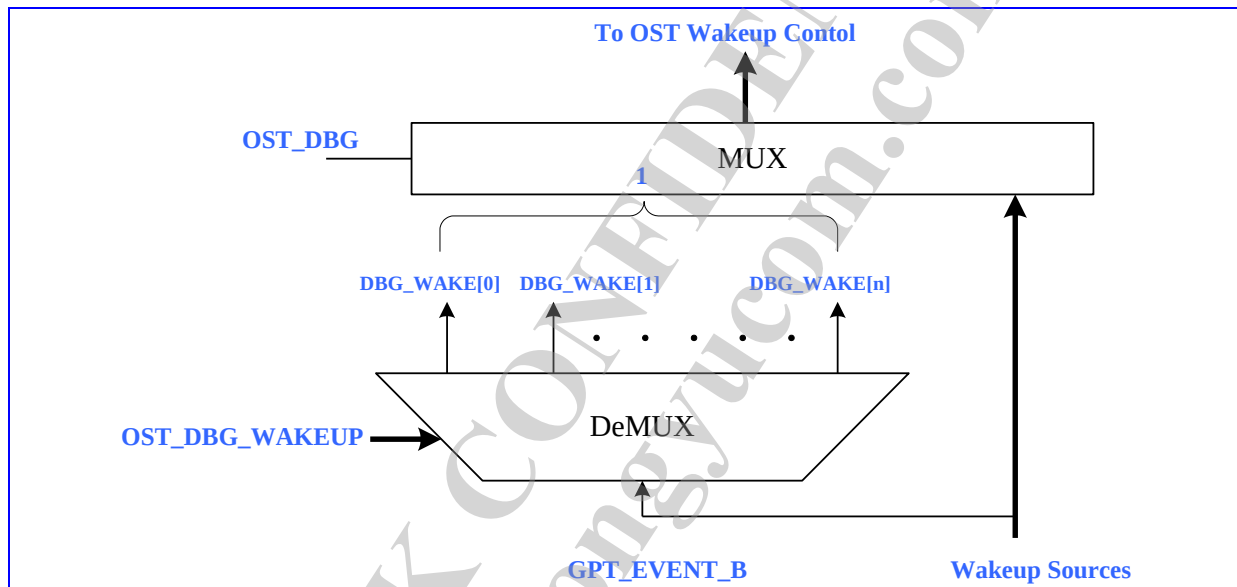


Figure 40. Debug wakeup events

4.7 UART

4.7.1 General Description

The baseband chipset houses three UARTs. The UARTs provide full duplex serial communication channels between baseband chipset and external devices.

UART supports word length from 5 to 8 bits, an optional parity bit and one or two stop bits, and is fully programmable. A 16-bit programmable baud rate generator is included, together with separate transmit and receive FIFOs. A diagnostic loopback mode is provided. UART also includes two pairs of DMA handshake lines used to indicate when the FIFOs are ready to transfer data to or receive data from the CPU. Interrupts can be generated from any of the 10 sources.

After hardware reset, the UART will be in the M16C450 mode. Its FIFOs can be enabled and the UART can then enter the M16550A mode. The UART has further additional functions beyond the M16550A

mode. Each of the extended functions can be selected individually under software control.

The UART provides more powerful enhancements than the industry-standard 16550:

- Hardware flow control: This feature is very useful when the ISR latency is hard to predict and control in the embedded applications. The MCU is relieved of having to fetch the received data within a fixed amount of time.
- High-speed mode: For high-speed external devices, the UART supports up to 3.25M baud rate.
- Low power: To save power, the UART supports dynamic clock change while keeping the same baud rate.

Note: In order to enable any of the enhancements, the Enhanced Mode bit, EFR[4] must be set. If EFR[4] is not set, IER[7:5], FCR[5:4], IIR[5:4] and MCR[7:6] cannot be written. The Enhanced Mode bit ensures that the UART is backward compatible with the software that has been written for 16C450 and 16550A devices.

Figure 41 shows the block diagram of the UART device.

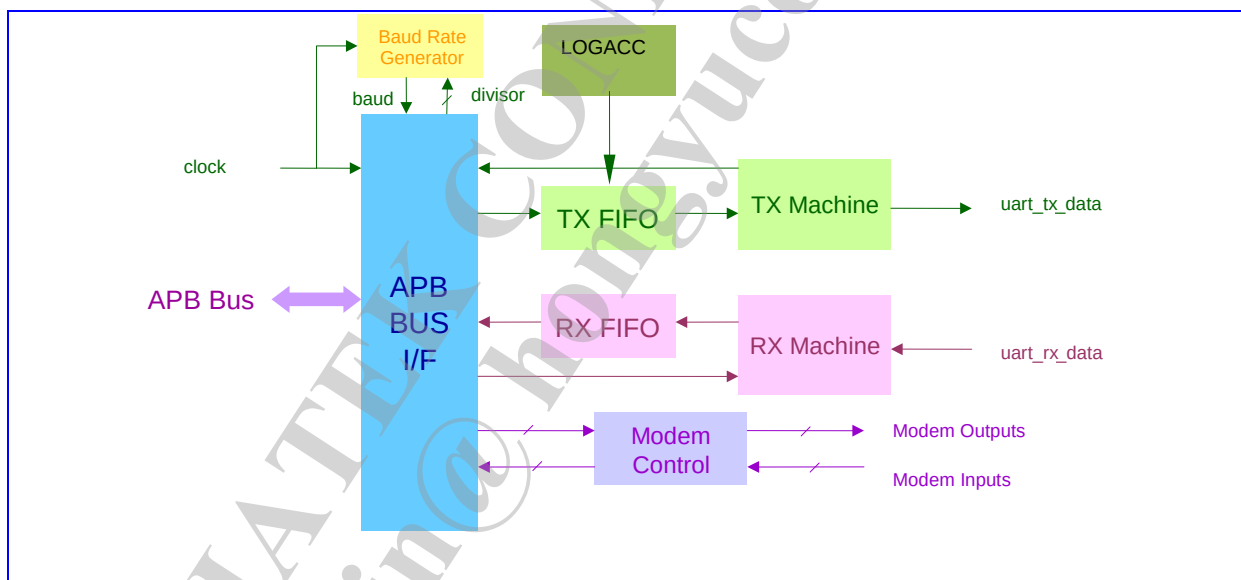


Figure 41. Block diagram of UART

4.7.2 Log Accelerator Description

In order to download more information from the network and to reduce the software loading, the software only writes all information at the specified external memory and triggers the hardware to transmit it through UART to the PC client.

Before transmitting data through the UART, the software needs to specify some settings. At first, the address of layer 1 and PS data are required. The amount of data is so huge that the data are usually located in the external memory. The software needs to specify the starting address and the size of the

ring buffer. However, it is acceptable if the address is located in the internal RAM (in the same layer of the AHB bus).

Besides the location of the original data, the software also needs to specify the target address. There are two methods to transmit data to the UART.

1. The log_acc module writes all encoded data to the specified memory and issues an interrupt to the MCU after the entire block is transmitted. Next, the software uses the DMA to move the data to the UART or USB.
2. The log_acc module writes the UART port directly.

With method 1, the software needs to set up a specified memory to store all encoded data. When the ARM receives the IRQ signal from LOG_ACC, the software can start to use DMA to move the data to the UART or USB. This method has limitation: If the encoded data are not word-aligned, the software should separate the encoded data into two parts. One part is sent by "DMA size=word", and the remaining not-word-aligned data are sent by "DMA size=byte". If the software uses "DMA size=byte" to send all data, the bus performance will be degraded, and the transmission rate will be too slow.

With method 2, LOG_ACC will write the UART port directly by the AHB single mode. The software needs to enable this function in the register and set the target address to UART port. The software should also set up the appreciated threshold value in "UARTn_FCR"; otherwise log_acc will write too many data to UART FIFO. Although log_acc can detect the status of the UART, it may not be the first master to access the UART while the UART FIFO is safe to send. If too many data occur, the software can try to use the DMA approach since this approach will hold the transmission until the FIFO is not full. If the software can guarantee that one UART is dedicated to the LOG_ACC module, there will not be such problem.

The log_acc module writes data to the UART TX FIFO directly.

At a specific timing, only one user of VFIFO/LOGACC can access the UART Tx channel. When the Tx channel is switched from LOGACC to VFIFO, the following conditions should be met:

- FIFO enable
- Tx DMA enable
- Reset Tx FIFO

When the Tx channel is switched from VFIFO to LOGACC, the following conditions should be met:

- FIFO enable
- Tx DMA disable
- Reset Tx FIFO

4.7.3 Register Definition

n = 1, 2, 3; for UART1, UART2 and UART3 respectively.

UARTn+0000h Rx buffer register

UARTn_RBR

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name																RBR[7:0]
Type																RO

RBR Rx buffer register. A read-only register. The received data can be read by accessing this register.

Modified when LCR[7] = 0.

UARTn+0000h Tx holding register

UARTn_T HR

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name																THR[7:0]
Type																WO

THR Tx holding register. A write-only register. The data to be transmitted are written to this register and sent to the PC via serial communication. Modified when LCR[7] = 0.

UARTn+0004h Interrupt enable register

UARTn_IER

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name									CTSI	RTSI	XOFF I	VFF FC_E N	EDSS I	ELSI	ETBE I	ERBF I
Type																R/W
Reset																0

By storing 1 to a specific bit position, the interrupt associated with that bit will be enabled. Otherwise, the interrupt will be disabled. IER[3:0] are modified when LCR[7] = 0. IER[7:4] are modified when LCR[7] = 0 & EFR[4] = 1.

CTSI Masks an interrupt generated when a rising edge is detected on the CTS modem control line. *Note that this interrupt is only enabled when the hardware flow control is enabled.*

0 Mask an interrupt generated when a rising edge is detected on the CTS modem control line.

1 Unmask an interrupt generated when a rising edge is detected on the CTS modem control line.

RTSI Masks an interrupt generated when a rising edge is detected on the RTS modem control line. *Note that this interrupt is only enabled when the hardware flow control is enabled.*

- 0** Mask an interrupt generated when a rising edge is detected on the RTS modem control line.
- 1** Unmask an interrupt generated when a rising edge is detected on the RTS modem control line.

XOFFI Masks an interrupt generated when an XOFF character is received. *Note that this interrupt is only enabled when the software flow control is enabled.*

- 0** Mask an interrupt generated when an XOFF character is received.
- 1** Unmask an interrupt generated when an XOFF character is received.

VFF_FC_EN Enables flow control triggered by Rx FIFO full when VFIFO_EN is set

- 0** Disable flow control triggered by Rx FIFO full when VFIFO_EN is set.
- 1** Enable flow control triggered by Rx FIFO full when VFIFO_EN is set.

EDSSI When set to 1, an interrupt will be generated if DDCD, TERI, DDSR or DCTS (MSR[4:1]) is set.

- 0** No interrupt is generated when DDCD, TERI, DDSR or DCTS (MSR[4:1]) is set.
- 1** An interrupt is generated when DDCD, TERI, DDSR or DCTS (MSR[4:1]) is set.

ELSI When set to 1, an interrupt will be generated if BI, FE, PE or OE (LSR[4:1]) is set.

- 0** No interrupt is generated when BI, FE, PE or OE (LSR[4:1]) is set.
- 1** An interrupt is generated when BI, FE, PE or OE (LSR[4:1]) is set.

ETBEI When set to 1, an interrupt will be generated if the Tx holding register is empty or the contents of Tx FIFO are reduced to its trigger level.

- 0** No interrupt will be generated if the Tx holding register is empty or the contents of Tx FIFO are reduced to its trigger level.
- 1** An interrupt will be generated if the Tx holding register is empty or the contents of Tx FIFO are reduced to its trigger level.

ERBFI When set to 1, an interrupt will be generated if the Rx buffer contains data.

- 0** No interrupt is generated if the Rx buffer contains data.
- 1** An interrupt is generated if the Rx buffer contains data.

UARTn+0008h Interrupt identification register

UARTn_IIR

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name									FIFOE		ID4	ID3	ID2	ID1	ID0	NINT
Type									RO							
Reset									0	0	0	0	0	0	0	1

Identifying if there are pending interrupts. ID4 and ID3 are presented only when EFR[4] = 1. The table below lists the IIR[5:0] codes associated with the possible interrupts:

IIR[5:0]	Priority level	Interrupt	Source
000001	-	No pending interrupt	
000110	1	Line status interrupt	BI, FE, PE or OE set in LSR (when IER[2] = 1)
000100	2	Rx data received	Rx data received or Rx trigger level reached (when IER[0] = 1)
001100	2	Rx data time-out	Time-out on character in Rx FIFO (when IER[0] = 1)
000010	3	Tx holding register empty	Tx holding register empty or Tx FIFO trigger level reached. (when IER[1] = 1)
000000	4	Modem status change	DDCD, TERI, DDSR or DCTS set in MSR (when IER[3]=1)
010000	5	Software flow control	XOFF character received (when IER[5] = 1)
100000	6	Hardware flow control	CTS or RTS rising edge (when IER[6] = 1, EDR[6] = 1)

Table 55. IIR[5:0] codes associated with the possible interrupts

Line status interrupt

A Rx line status Interrupt (IIR[5:0] = 000110b) is generated if ELSI (IER[2]) is set and any of BI, FE, PE or OE (LSR[4:1]) is set. The interrupt is cleared by reading the line status register.

Rx data received interrupt

A Rx received interrupt (IER[5:0] = 000100b) is generated if EFRBI (IER[0]) is set and either Rx data are placed in the Rx buffer register or the Rx trigger level is reached. The interrupt is cleared by reading the Rx buffer register or the Rx FIFO (if enabled).

Rx data time-out interrupt

When virtual FIFO mode is disabled, the Rx data time-out interrupt will be generated if all of the following conditions are applied:

1. FIFO contains at least one character.
2. The most recent character is received longer than four character periods ago (including all start, parity and stop bits).
3. The most recent CPU read of the FIFO was longer than four character periods ago.

The time-out timer is restarted upon receipt of a new byte from the Rx shift register or on a CPU read from the Rx FIFO.

The Rx data time-out interrupt is enabled by setting EFRBI (IER[0]) to 1 and is cleared by reading Rx FIFO.

When virtual FIFO mode is enabled, the Rx data time-out interrupt will be generated if all of the following conditions are applied:

1. FIFO is empty.
2. The most recent character is received longer than four character periods ago (including all start, parity and stop bits);
3. The most recent CPU read of the FIFO was longer than four character periods ago.

The time-out timer is restarted on receipt of a new byte from the Rx shift register.

Rx holding register empty interrupt

A Tx holding register empty interrupt (IIR[5:0] = 000010b) is generated if ETRBI (IER[1]) is set and either the Tx holding register or, if FIFOs are enabled, the Tx FIFO becomes empty. The interrupt is cleared by writing to the Tx holding register or Tx FIFO if FIFO is enabled.

Modem status change interrupt

A modem status change interrupt (IIR[5:0] = 000000b) is generated if EDSSI (IER[3]) is set and either DDCD, TERI, DDSR or DCTS (MSR[3:0]) is set. The interrupt is cleared by reading the modem status register.

Software flow control interrupt

A software flow control interrupt (IIR[5:0] = 010000b) is generated if the software flow control is enabled and XOFFI (IER[5]) is set, indicating that an XOFF character has been received. The interrupt is cleared by reading the interrupt identification register.

Hardware flow control interrupt

A hardware flow control interrupt (IER[5:0] = 100000b) is generated if the hardware flow control is enabled and either RTSI (IER[6]) or CTSI (IER[7]) is set, indicating that a rising edge has been detected on either of RTS/CTS modem control line. The interrupt is cleared by reading the interrupt identification register.

UARTn+0008h FIFO control register

UARTn_FCR

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name									RFTL 1	RFTL 0	TFTL 1	TFTL 0		CLRT	CLRR	FIFO E
Type									WO	WO	WO	WO		WO	WO	WO

FCR is used to control the trigger levels of the FIFOs or flush them. FCR[7:6] is modified when LCR != 0xBF. FCR[5:4] is modified when LCR != BFh & EFR[4] = 1. FCR[4:0] is modified when LCR != BFh.

FCR[7:6] Rx FIFO trigger threshold. (Rx FIFO contains total 32 bytes.)

- 0
- 1 6
- 2 12
- 3 RXTRIG(0~31)

FCR[5:4] Tx FIFO trigger threshold (Tx FIFO contains total 32 bytes.)

- 0 1
- 1 4
- 2 8
- 3 14

CLRT Clears transmit FIFO. This bit is self-clearing.

- 0** Leave Tx FIFO intact.
- 1** Clear all the bytes in Tx FIFO.

CLRR Clears receive FIFO. This bit is self-clearing.

- 0** Leave Rx FIFO intact.
- 1** Clear all the bytes in the Rx FIFO.

FIFOE Enables FIFO. This bit must be set to 1 for any of the other bits in the registers to have any effect.

- 0** Disable both Rx and Tx FIFOs.
- 1** Enable both Rx and Tx FIFOs.

UARTn+000Ch Line control register

UARTn_LC
R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name									DLAB	SB	SP	EPS	PEN	STB	WLS1	WLS0
Type																
Reset									0	0	0	0	0	0	0	0

Line control register. Determining the characteristics of serial communication signals. Modified when LCR[7] = 0.

DLAB Divisor latch access bit

- 0** The Rx and Tx registers are read/written at address 0, and the IER register is read/written at address 4.
- 1** The divisor latch LS is read/written at address 0, and the Divisor Latch MS is read/written at address 4.

SB Sets up break

- 0** No effect
- 1** SOUT signal is forced into the "0" state.

SP Stick parity

- 0** No effect
- 1** The parity bit is forced into a defined state, depending on the states of EPS and PEN:
If EPS = 1 & PEN = 1, the parity bit will be set and checked = 0.
If EPS = 0 & PEN = 1, the parity bit will be set and checked = 1.

EPS Selects even parity

- 0** When EPS = 0, an odd number of ones is sent and checked.
- 1** When EPS = 1, an even number of ones is sent and checked.

- PEN** Enables parity
0 The parity is neither transmitted nor checked.
1 The parity is transmitted and checked.
- STB** Number of STOP bits
0 One STOP bit is always added.
1 Two STOP bits are added after each character is sent; unless the character length is 5 when 1 STOP bit is added.
- WLS1, 0** Selects word length
0 5 bits
1 6 bits
2 7 bits
3 8 bits

UARTn+0010h Modem control register

UARTn_MCR

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name									XOFF STAT US			LOO P			RTS	
Type									R/W			R/W			R/W	
Reset									0			0			0	

Modem control register. Controlling interface signals of the UART. MCR[4:0] are modified when LCR[7] = 0, MCR[7:6] are modified when LCR[7] = 0 & EFR[4] = 1.

- XOFF STATUS** A read-only bit
0 When an XON character is received.
1 When an XOFF character is received.
- LOOP** Loop-back control bit
0 No loop-back is enabled.
1 Loop-back mode is enabled.
- RTS** Controls the state of the output NRTS, even in the loop mode.
0 NRTS = 1
1 NRTS = 0

UARTn+0014h Line status register

UARTn_LS
R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name									FIFO ERR	TEMT	THRE	BI	FE	PE	OE	DR
Type									R/W							
Reset									0	1	1	0	0	0	0	0

Line status register. Modified when LCR[7] = 0.

FIFOERR Rx FIFO error indicator

0 No PE, FE, BI set in Rx FIFO

1 Set to 1 when there is at least one PE, FE or BI in Rx FIFO.

TEMT Tx holding register (or Tx FIFO) and the Tx shift register are empty.

0 Empty conditions below are not met.

1 If FIFOs are enabled, the bit is set whenever Tx FIFO and the Tx shift register are empty. If FIFOs are disabled, the bit will be set whenever the Tx holding register and Tx shift register are empty.

THRE Indicates if there is room for Tx holding register or Tx FIFO is reduced to its trigger level.

0 Reset whenever the contents of Tx FIFO are more than its trigger level (FIFOs are enabled), or whenever the Tx holding register is not empty (FIFOs are disabled).

1 Set whenever the contents of Tx FIFO are reduced to its trigger level (FIFOs are enabled), or whenever the Tx holding register is empty and ready to accept new data (FIFOs are disabled).

BI Break interrupt

0 Reset by the CPU reading this register

1 If the FIFOs are disabled, this bit will be set whenever SIN is held in the 0 state for more than one transmission time (START bit + DATA bits + PARITY + STOP bits). If the FIFOs are enabled, this error will be associated with a corresponding character in FIFO and is flagged when this byte is at the top of the FIFO. When a break occurs, only one zero character will be loaded into the FIFO. The next character transfer is enabled when SIN goes into the marking state and receives the next valid start bit.

FE Framing error

0 Reset by the CPU reading this register

1 If the FIFOs are disabled, this bit will be set if the received data do not have a valid STOP bit. If the FIFOs are enabled, the state of this bit will be revealed when the byte it refers to is the next to be read.

PE Parity error

0 Reset by the CPU reading this register

- 1** If the FIFOs are disabled, this bit will be set if the received data do not have a valid parity bit. If the FIFOs are enabled, the state of this bit will be revealed when the referred byte is the next to be read.

OE Overrun error

- 0** Reset by the CPU reading this register.
- 1** If the FIFOs are disabled, this bit will be set if the Rx buffer is not read by the CPU before the new data from the Rx shift register overwrite the previous contents. If the FIFOs are enabled, an overrun error will occur when Rx FIFO is full and the Rx shift register becomes full. OE is set as soon as this happens. The character in the shift register is then overwritten but not transferred to the FIFO.

DR Data ready

- 0** Cleared by the CPU reading the Rx buffer or by reading all the FIFO bytes.
- 1** Set by the Rx buffer becoming full or by a byte being transferred into the FIFO.

UARTn+0018h Modem status register

UARTn_MS
R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name												CTS				DCTS
Type												R/W				R/W
Reset												Input				0

Note: After a reset, D4-D7 are inputs. A modem status interrupt can be cleared by writing '0' or set by writing '1' to this register. D0-D3 can be written to. Modified when LCR[7] = 0.

CTS Clear to send

When Loop = "0", this value will be the complement of the NCTS input signal.
 When Loop = "1", this value will be equal to the RTS bit in the modem control register.

DCTS Delta clear to send

- 0** Cleared if the state of CTS has not changed since this register was last read.
- 1** Set if the state of CTS has changed since this register was last read.

UARTn+0000h Divisor latch (LS)

UARTn_DL
L

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name												DLL[7:0]				
Type												R/W				
Reset												1				

UARTn+0004h Divisor latch (MS)

UARTn_DL
M

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name												DLM[7:0]				
Type												R/W				
Reset												0				

Note: DLL & DLM can only be updated if DLAB is set to 1. Division by 1 will generate a BAUD signal that is constantly high. Modified when LCR[7] = 1.

UARTn+0008h Enhanced feature register

UARTn_EFR

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name									AUTO CTS	AUTO RTS	LOWER_RATEFIX	ENABLE-E	SW FLOW CONT[3:0]			
Type									R/W	R/W	RW	R/W	R/W			
Reset									0	0	0	0	0			

Note: Only when LCR = BF'h

AUTO CTS Enables hardware transmission flow control
0 Disable
1 Enable

AUTO RTS Enables hardware reception flow control
0 Disable
1 Enable

LOWER_RATEFIX Enables ratefix function for lower clock
0 Disable
1 Enable

ENABLE-E Enables enhancement features
0 Disable
1 Enable

CONT[3:0] Software flow control bits
00xx No Tx flow control
10xx Transmit XON1/XOFF1 as flow control bytes

- 01xx** Transmit XON2/XOFF2 as flow control bytes
- 11xx** Transmit XON1 & XON2 and XOFF1 & XOFF2 as flow control words
- xx00** No Rx flow control
- xx10** Receive XON1/XOFF1 as flow control bytes
- xx01** Receive XON2/XOFF2 as flow control bytes
- xx11** Receive XON1 & XON2 and XOFF1 & XOFF2 as flow control words

Note: When 11xx or xx11, ESC character is not supported and ESC_EN must be 0.

In MT6250, UART only supports 52MHz, 26MHz and 13MHz.

61.44MHz clock domain

	Freq_set (0x34[2]) = 0	Freq_set (0x34[2]) = 1
Lower_ratefix(EFR[5]) = 0	Rate fix 30.72MHz	Rate fix 15.36MHz
Lower_ratefix(EFR[5]) = 1	Rate fix 7.68MHz	Rate fix 3.84MHz

52MHz clock domain

	Freq_set (0x34[2]) = 0	Freq_set (0x34[2]) = 1
Lower_ratefix(EFR[5]) = 0	Rate fix 26MHz	Rate fix 13MHz
Lower_ratefix(EFR[5]) = 1	Rate fix 6.5MHz	Rate fix 3.25MHz

UARTn+0010h XON1

UARTn_XON
1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name																
Type																
Reset																

UARTn+0014h XON2

UARTn_XON2

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name																
Type																
Reset																

UARTn+0018h XOFF1

UARTn_XOFF
1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name																
Type																

Reset									0
-------	--	--	--	--	--	--	--	--	---

UARTn+001Ch XOFF2

UARTn_XOFF 2

[illegible]

XON1, XON2, XOFF1, XOFF2 are valid only when LCR = BFh.

UARTn+0020

AUTOBAUD_EN

UARTn_AUTOBAUD_EN

[illegible]

AUTOBAUD EN

Auto-baud enable signal

- 0 Disable auto-baud function
1 Enable auto-baud function enable (UARTn+0024h SPEED should be set to 0.)

UARTn+0024

HIGH SPEED UART

UARTn HIGHSPEED

[illegible]

SPEED

UART sample counter base

- 0 Based on 16*baud_pulse, baud_rate = system clock frequency/16/{DLH, DLL}
- 1 Based on 8*baud_pulse, baud_rate = system clock frequency/8/{DLH, DLL}
- 2 Based on 4*baud_pulse, baud_rate = system clock frequency/4/{DLH, DLL}
- 3 Based on sampe_count * baud_pulse, baud_rate = system clock frequency /
sampe count

The table below shows the divisors needed to generate given baud rates from CLK inputs of 13MHz based on different HIGHSPEED values.

BAUD	HIGHSPEED = 0	HIGHSPEED = 1	HIGHSPEED = 2
110	7386	14773	29545
300	2708	7386	14773
1,200	677	2708	7386
2,400	338	677	2708
4,800	169	338	677
9,600	85	169	338
19,200	42	85	169
38,400	21	42	85
57,600	14	21	42
115,200	7	14	21
230,400	*	7	14
460,800	*	*	7
921,600	*	*	*

Table 56. Divisors needed to generate given baud rates from 13MHz based on different HIGHSPEED values

The table below shows the divisors needed to generate given baud rates from CLK inputs of 26MHz based on different HIGHSPEED values.

BAUD	HIGHSPEED = 0	HIGHSPEED = 1	HIGHSPEED = 2
110	14773	29545	59091
300	5417	14773	29545
1,200	1354	5417	14773
2,400	677	1354	5417
4,800	339	677	1354
9,600	169	339	667
19,200	85	169	339
38,400	42	85	169
57,600	28	42	85
115,200	14	28	42
230,400	7	14	28
460,800	*	7	14
921,600	*	*	7

**Table 57. Divisors needed to generate given baud rates from 26MHz based on different
HIGH SPEED values**

The table below shows the divisors needed to generate given baud rates from CLK inputs of 52MHz based on different HIGH SPEED values.

BAUD	HIGH SPEED = 0	HIGH SPEED = 1	HIGH SPEED = 2
110	29545	59091	118182
300	10833	29545	59091
1,200	2708	10833	29545
2,400	1354	2708	10833
4,800	677	1354	2708
9,600	339	677	1354
19,200	169	339	677
38,400	85	169	339
57,600	56	85	169
115,200	28	56	85
230,400	14	28	56
460,800	7	14	28
921600	*	7	14

**Table 58. Divisors needed to generate given baud rates from 52MHz based on different
HIGH SPEED values**

The table below shows the divisors needed to generate given baud rates from CLK inputs of 61.44MHz based on different HIGH SPEED values.

BAUD	HIGH SPEED = 0	HIGH SPEED = 1	HIGH SPEED = 2
110	34909	69818	139636
300	12800	25600	51200
1,200	3200	6400	12800
2,400	1600	3200	6400
4,800	800	1600	3200
9,600	400	800	1600
19,200	200	400	800
38,400	100	200	400
57,600	67	133	266
115,200	33	67	133
230,400	17	33	67

BAUD	HIGHSPEED = 0	HIGHSPEED = 1	HIGHSPEED = 2
460,800	8	17	33
921,600	4	8	17

Table 59. Divisors needed to generate given baud rates from 61.44MHz based on different HIGHSPEED values

The table below shows the divisors needed to generate given baud rates from CLK inputs of 30.72MHz based on different HIGHSPEED values.

BAUD	HIGHSPEED = 0	HIGHSPEED = 1	HIGHSPEED = 2
110	17455	34909	69818
300	6400	12800	25600
1,200	1600	3200	6400
2,400	800	1600	3200
4,800	400	800	1600
9,600	200	400	800
19,200	100	200	400
38,400	50	100	200
57,600	33	67	133
115,200	17	33	67
230,400	8	17	33
460,800	4	8	17
921,600	2	4	8

Table 60. Divisors needed to generate given baud rates from 30.72MHz based on different HIGHSPEED values

The table below shows the divisors needed to generate given baud rates from CLK inputs of 15.36MHz based on different HIGHSPEED values.

BAUD	HIGHSPEED = 0	HIGHSPEED = 1	HIGHSPEED = 2
110	8727	17455	34909
300	3200	6400	12800
1,200	800	1600	3200
2,400	400	800	1600
4,800	200	400	800
9,600	100	200	400
19,200	50	100	200

BAUD	HIGHSPEED = 0	HIGHSPEED = 1	HIGHSPEED = 2
38,400	25	50	100
57,600	17	33	67
115,200	8	17	33
230,400	4	8	17
460,800	2	4	8
921,600	1	2	4

Table 61. Divisors needed to generate given baud rates from 15.36MHz based on different HIGHSPEED values

UARTn+0028h **SAMPLE_COUNT** **UARTn_SAMPLE_COUNT**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name																
Type																
Reset																

When HIGHSPEED = 3, sample_count will be the threshold value for UART sample counter (sample_num). Counting from 0 to sample_count.

UARTn+002Ch **SAMPLE_POINT** **UARTn_SAMPLE_POINT**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name																
Type																
Reset																

When HIGHSPEED = 3, UART will obtain the input data when sample_count= sample_num.

For example:

System clock = 13MHz, 921600 = 13000000/14

Sample_count = 14 and sample point = 6 (sample the central point to decrease the inaccuracy)

SAMPLE_POINT is usually (SAMPLE_COUNT-1)/2 without decimal.

UARTn+0034h **Rate fix address** **UARTn_RATEFIX_AD**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
-----	----	----	----	----	----	----	---	---	---	---	---	---	---	---	---	---

Name											AUTO OBA U_61 M		FREQ _SEL	AUTO OBA UD_R ATE_ FIX	RXTX _FIX
Type											RW		R/W	R/W	R/W
Reset											0		0	0	0

AUTOBAU_61M

Selects frequency

- 0 When in 52MHz clock domain
- 1 When in 61.44MHz clock domain

RATE_FIX

When "rate_fix"(34H[0]) is set, you can transmit and receive data only if f13m_en is enable and freq_sel (34H[2]) is set to 1, or f26m_en is enabled and freq_sel (34H[2]) is set to 0.

AUTOBAUD_RATE_FIX

When "autobaud_rate_fix"(34H[1]) is set, Tx/Rx the autobaud packet only if

- | | |
|---|--|
| 0 | f13m_en is enabled and freq_sel (34H[2]) is set to 1, or |
| 1 | f26m_en is enabled and freq_sel (34H[2]) is set to 0. |

FREQ_SEL

Selects frequency

- ```
0 Select f26m_en for rate_fix and autobaud_rate_fix
1 Select f13m_en for rate_fix and autobaud_rate_fix
```

## UARTn+0038

# AUTOBAUDSAMPLE

## UARTn\_AUTOBAUDSAM PLE

[illegible]

Since the system clock may change, the autobaud sample duration should change as system clock changes. When the system clock = 13MHz, autobaudsample = 6. When the system clock = 26MHz, autobaudsample = 13.

UARTn+003C  
h

## Guard time added register

## UARTn GUARD

|      |    |    |    |    |    |    |   |   |   |   |   |         |                |   |   |   |
|------|----|----|----|----|----|----|---|---|---|---|---|---------|----------------|---|---|---|
| Bit  | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4       | 3              | 2 | 1 | 0 |
| Name |    |    |    |    |    |    |   |   |   |   |   | GUARD_N | GUARD_CNT[3:0] |   |   |   |

|       |  |  |  |  |  |  |  |  |  |  |  |     |     |     |     |     |
|-------|--|--|--|--|--|--|--|--|--|--|--|-----|-----|-----|-----|-----|
| Type  |  |  |  |  |  |  |  |  |  |  |  | R/W | R/W | R/W | R/W | R/W |
| Reset |  |  |  |  |  |  |  |  |  |  |  | 0   | 1   | 1   | 1   | 1   |

**GUARD\_CNT** Guard interval count value. Guard interval =  $[1/(\text{system clock}/\text{div\_step}/\text{div})]*\text{GUARD\_CNT}$ .

**GUARD\_EN** Guard interval adds enable signal.  
**0** No guard interval added.  
**1** Add guard interval after stop bit.

**UARTn+0040h**

**Escape character register**

**UARTn\_ESCAPE\_DAT**

|       |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |                        |
|-------|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|------------------------|
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0                      |
| Name  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | <b>ESCAPE_DAT[7:0]</b> |
| Type  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | R/W                    |
| Reset |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | FFh                    |

**ESCAPE\_DAT** Escape character added before software flow control data and escape character, i.e. if Tx data are xon (31h), with esc\_en = 1, UART will transmit data as esc + CEh (~xon).

**UARTn+0044h**

**Escape enable register**

**UARTn\_ESCAPE\_EN**

|       |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |               |
|-------|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---------------|
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0             |
| Name  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | <b>ESC_EN</b> |
| Type  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | R/W           |
| Reset |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | 0             |

**ESC\_EN** Adds escape character in transmitter and remove escape character in receiver by UART  
**0** Does not deal with the escape character.  
**1** Add escape character in transmitter and remove escape character in receiver.

**UARTn+0048h**

**Sleep enable register**

**UARTn\_SLEEP\_EN**

|       |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |                 |
|-------|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|-----------------|
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0               |
| Name  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | <b>SLEEP_EN</b> |
| Type  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | R/W             |
| Reset |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | 0               |

**SLEEP\_EN** For sleep mode issue

- 0** Does not deal with sleep mode indication signal
- 1** Activate hardware flow control or software control according to software initial settings when the chip enters the sleep mode. Release hardware flow when the chip wakes up. However, for software control, UART sends xon when awoken and when FIFO does not reach threshold level.

**UARTn+004C  
h**

**Rx virtual FIFO enable register**

**UARTn\_RXVFF\_EN**

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2                  | 1             | 0             |
|-------|----|----|----|----|----|----|---|---|---|---|---|---|---|--------------------|---------------|---------------|
| Name  |    |    |    |    |    |    |   |   |   |   |   |   |   | TO CNT_A<br>UTORST | TX DMA_E<br>N | RX DMA_E<br>N |
| Type  |    |    |    |    |    |    |   |   |   |   |   |   |   | RW                 | RW            | RW            |
| Reset |    |    |    |    |    |    |   |   |   |   |   |   |   | 0                  | 0             | 0             |

**TO\_CNT\_AUTORST** Time-out counter auto reset register

- 0** After Rx time-out takes place, the software shall reset the interrupt by reading UART 0x4C.
- 1** The time-out counter will be auto reset. Set this register when Rain's new DMA is used.

**TX\_DMA\_EN**

TX\_DMA mechanism enable signal

- 0** Does not use DMA in Tx.
- 1** Use DMA in Tx. When this register is enabled, the flow control is based on the DMA threshold and generates a time-out interrupt for DMA.

**RX\_DMA\_EN**

RX\_DMA mechanism enable signal

- 0** Does not use DMA in Rx
- 1** Use DMA in Rx. When this register is enabled, the flow control is based on the DMA threshold and generates a time-out interrupt.

**UARTn+0050  
h**

**Rx trigger address**

**UARTn\_RXTRI\_  
AD**

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2           | 1 | 0 |
|-------|----|----|----|----|----|----|---|---|---|---|---|---|---|-------------|---|---|
| Name  |    |    |    |    |    |    |   |   |   |   |   |   |   | RXTRIG[3:0] |   |   |
| Type  |    |    |    |    |    |    |   |   |   |   |   |   |   | R/W         |   |   |
| Reset |    |    |    |    |    |    |   |   |   |   |   |   |   | 0           |   |   |

**RXTRIG** When {rtm,rtl} = 2'b11, the Rx FIFO threshold will be RXTRIG. The suggested value is smaller than half of the Rx FIFO size, which is 32 bytes.

## UARTn+0054h Fractional divider LSB address

### UARTn\_FRACDIV\_L

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| Name  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| Type  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| Reset |    |    |    |    |    |    |   |   | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

**FRACDIV\_L** Adds sampling count (+1) from state data7 to data0, in order to contribute fractional divisor.

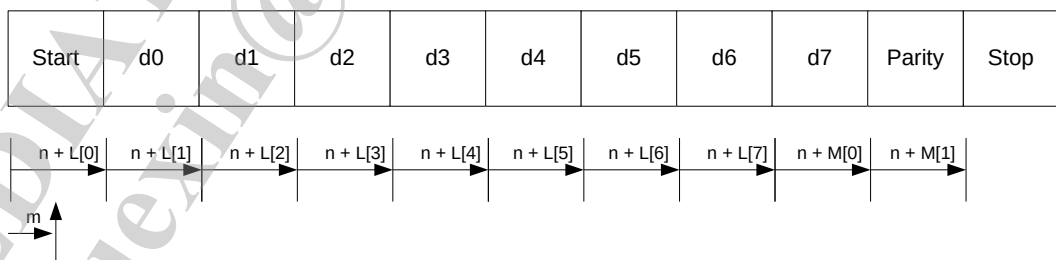
## UARTn+0058h Fractional divider MSB address

### UARTn\_FRACDIV\_M

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| Name  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| Type  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| Reset |    |    |    |    |    |    |   |   |   |   |   |   |   |   | 0 | 0 |

**FRACDIV\_M** Adds sampling count when in state stop to parity, in order to contribute fractional divisor.

bit\_extend register = FRACDIV\_L[7:0]  
FRACDIV\_M[1:0]



## UARTn+005Ch FIFO control register

### UARTn\_FCR\_R D

| Bit  | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|------|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| Name |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| Type |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |



Refer to register UARTn\_FCR.

### UARTn+0060h Debug register

### UARTn\_DEBUG

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0       |
|-------|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---------|
| Name  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | ACC_SEL |
| Type  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | RO      |
| Reset |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | 0       |

**ACC\_SEL** LOG\_ACC select or not

## 4.8 I2C/SCCB Controller

### 4.8.1 General Description

I2C (Inter-IC)/SCCB (Serial Camera Control Bus) is a two-wire serial interface. The two signals are SCL and SDA. SCL is a clock signal driven by the master. SDA is a bi-directional data signal that can be driven by either the master or the slave. This generic controller supports the master role and conforms to the I2C specification.

#### 4.8.1.1 Features

- I2C compliant master mode operation
- Adjustable clock speed for LS/FS mode operation
- Supports 7-bit/10-bit addressing
- Supports high-speed mode
- Supports slave clock extension
- START/STOP/REPEATED START condition
- Manual transfer mode
- Multi-write per transfer (up to 8 data bytes)
- Multi-read per transfer(up to 8 data bytes )
- Multi-transfer per transaction Combined format transfer with length change capability
- Active drive/wired-and I/O configuration

#### 4.8.1.2 Manual Transfer Mode

The controller offers manualtransfer modes

In addition to the slave address register, the controller has a built-in 8-byte deep FIFO which allows the MCU to prepare for up to 8 bytes of data for a write transfer, or read up to 8 bytes of data for a read transfer.

## 4.8.1.3 Transfer Format Support

This controller is designed to be as generic as possible in order to support a wide range of devices that may utilize different combinations of transfer formats. Here are the transfer formats that can be supported through different software configurations:

### Wording convention notes

Transfer = Anything encapsulated within a Start and Stop or Repeated Start

Transfer length = Number of bytes within the transfer

Transaction = This is the top unit. Everything combined equals 1 transaction.

Transaction length = Number of transfers to be conducted

 Master to slave dir

 Slave to master dir

### Single-byte access

Single Byte Write

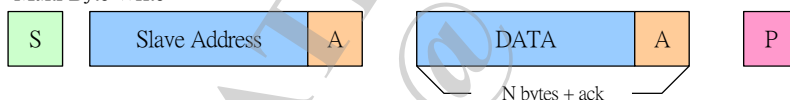


Single Byte Read

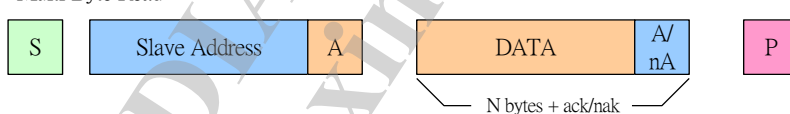


### Multi-byte access

Multi Byte Write

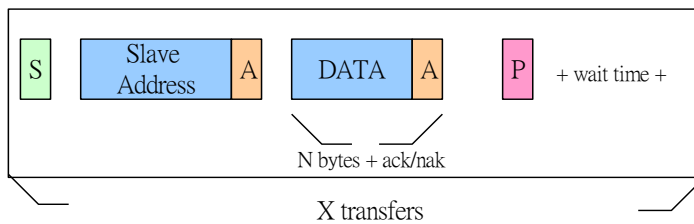


Multi Byte Read

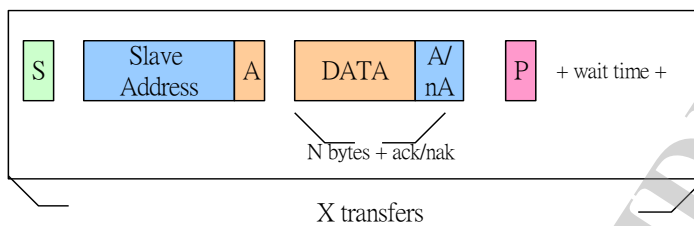


### Multi-byte transfer + multi-transfer (same direction)

## Multi Byte Write + Multi Transfer

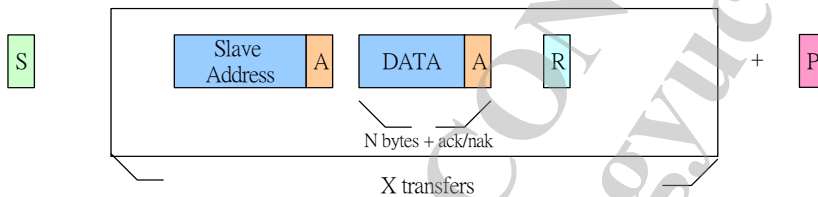


## Multi Byte Read + Multi Transfer

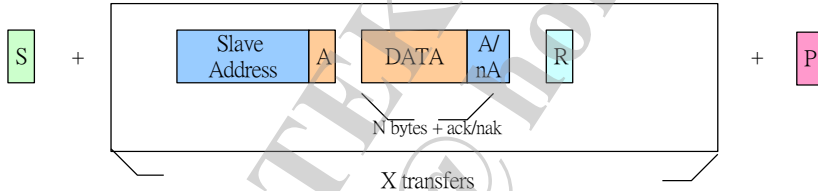


## Multi-byte transfer + multi-transfer w RS (same direction)

### Multi Byte Write + Multi Transfer + Repeated Start



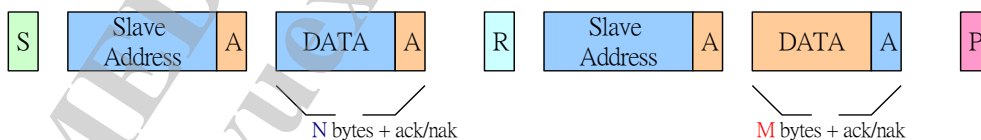
### Multi Byte Read + Multi Transfer + Repeated Start



## Combined write/read with repeated Start (direction change)

*Note: Only supports Write and then Read sequence. Read and then Write is not supported.*

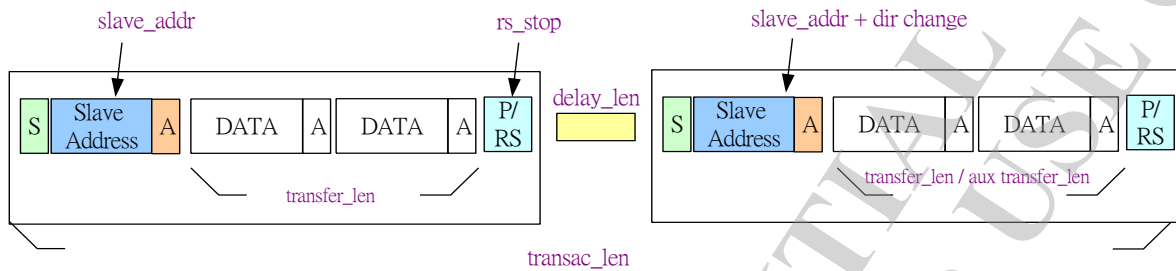
### Combined Multi Byte Write + Multi Byte Read



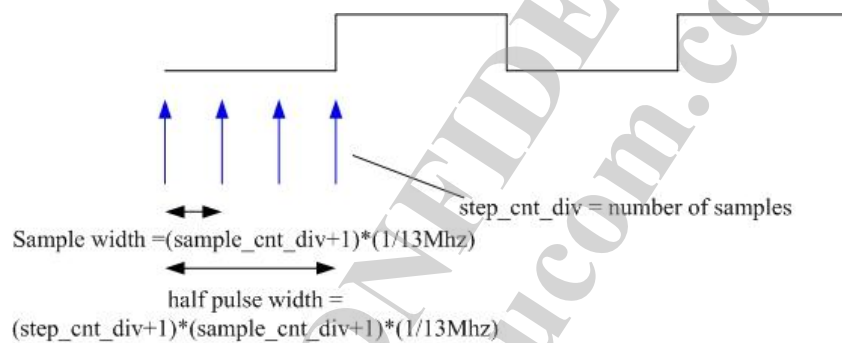
## 4.8.2 Programming Examples

### Common transfer programmable parameters

## Programmable Parameters



## Output waveform timing programmable parameters



## 4.8.3 Register Definition

I2CREG+000  
0h

Data port register

DATA\_PORT

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4         | 3 | 2 | 1 | 0 |
|-------|----|----|----|----|----|----|---|---|---|---|---|-----------|---|---|---|---|
| Name  |    |    |    |    |    |    |   |   |   |   |   | FIFO DATA |   |   |   |   |
| Type  |    |    |    |    |    |    |   |   |   |   |   | R/W       |   |   |   |   |
| Reset |    |    |    |    |    |    |   |   |   |   |   | X         |   |   |   |   |

DATA\_PORT[7:0]

FIFO access port. During master write sequences (slave\_addr[0] = 0), this port can be written by APB. During master read sequences (slave\_addr[0] = 1), this port can be read by APB. *Note: Slave\_addr must be set correctly before accessing the FIFO.*

(For debugging only) If the fifo\_apb\_debug bit is set, the FIFO can be read and written by the APB

I2CREG+000  
4h

Slave address register

SLAVE\_ADDR

| Bit  | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4          | 3 | 2 | 1 | 0 |
|------|----|----|----|----|----|----|---|---|---|---|---|------------|---|---|---|---|
| Name |    |    |    |    |    |    |   |   |   |   |   | SLAVE_ADDR |   |   |   |   |

|       |  |  |  |  |  |  |  |     |
|-------|--|--|--|--|--|--|--|-----|
| Type  |  |  |  |  |  |  |  | R/W |
| Reset |  |  |  |  |  |  |  | BF  |

**SLAVE\_ADDR [7:0]** Specifies the slave address of the device to be accessed. Bit 0 is defined by the I2C protocol as a bit that indicates the direction of transfer. 1 = master read, 0 = master write.

## I2CREG+000

## Interrupt mask register

## INTR MASK

8h

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4           | 3 | 2                  | 1          | 0                     |
|-------|----|----|----|----|----|----|---|---|---|---|---|-------------|---|--------------------|------------|-----------------------|
| Name  |    |    |    |    |    |    |   |   |   |   |   | TIME<br>OUT |   | HS_N<br>ACK<br>ERR | ACK<br>ERR | TRAN<br>SA_C<br>C_OMP |
| Type  |    |    |    |    |    |    |   |   |   |   |   | R/W         |   | R/W                | R/W        | R/W                   |
| Reset |    |    |    |    |    |    |   |   |   |   |   | 0           |   | 0                  | 0          | 0                     |

This register provides masks for the corresponding interrupt sources as indicated in the intr\_stat register.

1 = Allow interrupt

0 = Disable interrupt

*Note: While disabled, the corresponding interrupt will not be asserted. However, intr\_stat will still be updated with the status, i.e. the mask does not affect intr\_stat register values.*

## I2CREG+000

## Interrupt status register

## INTR STAT

## Ch

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4           | 3 | 2                  | 1          | 0                        |
|-------|----|----|----|----|----|----|---|---|---|---|---|-------------|---|--------------------|------------|--------------------------|
| Name  |    |    |    |    |    |    |   |   |   |   |   | TIME<br>OUT |   | HS_N<br>ACK<br>ERR | ACK<br>ERR | TRA<br>NSA<br>C_C<br>OMP |
| Type  |    |    |    |    |    |    |   |   |   |   |   | W1C         |   | W1C                | W1C        | W1C                      |
| Reset |    |    |    |    |    |    |   |   |   |   |   | 0           |   | 0                  | 0          | 0                        |

When an interrupt is issued by the I2C controller, this register will need to be read by the MCU to determine the cause for the interrupt. After this status is read and appropriate actions are taken, the corresponding interrupt source will need to be written to 1 to clear.

## TIMEOUT

This status is asserted if time-out is enabled and the timer expires. The internal master state machine will stop, and the MCU will need to manually clear the

state machine by either issuing software reset by disabling the transact\_en bit. The time-out can be used to detect PCB or I2C malfunction.

**HS\_NACKERR** This status is asserted if the HS master code NACK error detection is enabled. If enabled, the HS master code NACK ERR will cause transaction to end, and stop will be issued.

**ACKERR** This status is asserted if ACK error detection is enabled. If enabled, ACKERR will cause the transaction to end, and stop will be issued.

**TRANSAC\_COMP** This status is asserted when a transaction has completed successfully.

### I2CREG+001 0h

#### Control register

#### CONTROL

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8                  | 7                                 | 6                                  | 5                        | 4                 | 3                | 2          | 1           | 0   |
|-------|----|----|----|----|----|----|---|--------------------|-----------------------------------|------------------------------------|--------------------------|-------------------|------------------|------------|-------------|-----|
| Name  |    |    |    |    |    |    |   | TIME<br>OUT<br>_EN | RESE<br>T_BU<br>S_BU<br>SY_E<br>N | TRAN<br>SFER<br>LEN<br>_CHA<br>NGE | ACK<br>ERR<br>DET<br>_EN | DIR<br>CHA<br>NGE | CLK<br>EXT<br>EN | DMA<br>_EN | RS_S<br>TOP |     |
| Type  |    |    |    |    |    |    |   | R/W                | R/W                               | R/W                                | R/W                      | RW                | RW               | RW         | RW          | R/W |
| Reset |    |    |    |    |    |    |   | 0                  | 0                                 | 0                                  | 0                        | 0                 | 0                | 0          | 0           | 0   |

**TIMEOUT\_EN** Enables time-out mechanism. When enabled, if SCL stays 0 for too long period of time due to the I2C slave holds SCL at 0 for too long or SCL sticks at 0 due to PCB issue, the master shall terminate the transaction, stop the internal state machine and assert the time-out interrupt. The MCU shall handle this case appropriately and reset the master and FIFO address before reissuing the transaction again. If this option is disabled, the hardware timer will not count nor expire forever.

**0** Disable

**1** Enable

**RESET\_BUS\_BUSY\_EN** Enables that software reset can reset bus\_busy signal. When enabled, if the master receives software reset, it will reset the bus\_busy signal to 0. When disabled, the software reset will have no impact on the bus\_busy signal. Normally, the software reset should not reset the bus\_busy signal. Used only for debugging.

**0** Disable

**1** Enable

**TRANSFER\_LEN\_CHANGE** Specifies whether or not to change the transfer length after the first transfer is completed. If enabled, the transfers after the first transfer will use the transfer\_len\_aux parameter.

**ACKERR\_DET\_EN** Enables slave ACK error detection. When enabled, if the slave ACK error is detected, the master shall terminate the transaction by issuing a STOP condition and asserting the ACKERR interrupt. The MCU shall handle this case appropriately and reset the FIFO address before reissuing the transaction. If this option is disabled, the controller will ignore slave ACK error and keep on scheduled transaction.

0 Disable  
 1 Enable

**DIR\_CHANGE** Used for combined transfer format, where the direction of transfer is to be changed from write to read after the FIRST RS condition. *Note that when set to 1, the transfers after the direction change will be based on the transfer\_len\_aux parameter.*

0 Disable  
 1 Enable

**CLK\_EXT\_EN** I2C spec allows slaves to hold the SCL line low if it is not yet ready for further processing. Therefore, if this bit is set to 1, the master controller will enter a high wait state until the slave releases the SCL line.

**RS\_STOP** In the LS/FS mode, this bit affects the multi-transfer transaction only. It controls whether or not the REPEATED-START condition is used between transfers. The last ending transfer always ends with a STOP. In the HS mode, this bit must be set to 1.

0 Use STOP  
 1 Use REPEATED-START

**I2CREG+001 Transfer length register (number of bytes per transfer)** **TRANSFER\_LEN**

| Bit   | 15           | 14  | 13  | 12  | 11  | 10  | 9   | 8   | 7   | 6   | 5   | 4   | 3   | 2   | 1   | 0   |
|-------|--------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Name  | TRANSFER_LEN |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Type  | R/W          | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W |
| Reset | 'h0          | 'h0 | 'h0 | 'h0 | 'h0 | 'h0 | 'h0 | 'h0 | 'h0 | 'h0 | 'h0 | 'h0 | 'h0 | 'h0 | 'h0 | 'h1 |



**TRANSFER\_LEN[15:0]**

Indicates the number of data bytes to be transferred in 1 transfer unit (excluding slave address byte). *Note: The value must be set to bigger than 1; otherwise no transfer will take place.*

**I2CREG+006 Transfer length register (number of bytes per transfer)** **TRANSFER\_LEN**  
**Ch** **AUX**

| Bit   | 15                      | 14  | 13  | 12  | 11  | 10  | 9   | 8   | 7   | 6   | 5   | 4   | 3   | 2   | 1   | 0   |
|-------|-------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Name  | <b>TRANSFER_LEN_AUX</b> |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Type  | R/W                     | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W |
| Reset | 'h0                     | 'h0 | 'h0 | 'h0 | 'h0 | 'h0 | 'h0 | 'h0 | 'h0 | 'h0 | 'h0 | 'h0 | 'h0 | 'h0 | 'h0 | 'h1 |

**TRANSFER\_LEN\_AUX[15:0]**

This field is valid only when dir\_change or transfer\_len\_change is set to 1. This indicates the number of data bytes to be transferred in 1 transfer unit (excluding slave address byte) for the transfers following the direction change or transfer\_len\_change, i.e. if dir\_change = 1, the first write transfer length will depend on transfer\_len, while the second read transfer length will depend on transfer\_len\_aux. Dir change is always after the first transfer. Similarly, the transfer length change is always after the first transfer. *Note: The value must be set to bigger than 1; otherwise no transfer will take place.*

**I2CREG+001 Transaction length register (number of transfers per transaction)** **TRANSAC\_LEN**  
**8h** **N**

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0   |
|-------|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|-----|
| Name  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |     |
| Type  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | R/W |
| Reset |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | 'h1 |

**TRANSAC\_LEN[7:0]**

Indicates the number of transfers to be transferred in 1 transaction. *Note: The value must be set to bigger than 1; otherwise no transfer will take place.*

**I2CREG+001 Inter delay length register** **DELAY\_LEN**  
**Ch**

| Bit  | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0   |
|------|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|-----|
| Name |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |     |
| Type |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | R/W |

|       |  |  |  |  |  |  |  |  |     |
|-------|--|--|--|--|--|--|--|--|-----|
| Reset |  |  |  |  |  |  |  |  | 'h2 |
|-------|--|--|--|--|--|--|--|--|-----|

**DELAY\_LEN[7:0]**

Sets up the wait delay between consecutive transfers when the RS\_STOP bit is set to 0. The unit is same as the half pulse width.

**I2CREG+002**

# Oh

## Timing control register

## TIMING

| Bit   | 15            | 14             | 13 | 12 | 11 | 10             | 9 | 8 | 7 | 6 | 5            | 4 | 3 | 2 | 1 | 0 |
|-------|---------------|----------------|----|----|----|----------------|---|---|---|---|--------------|---|---|---|---|---|
| Name  | DATA_READ_ADJ | DATA_READ_TIME |    |    |    | SAMPLE_CNT_DIV |   |   |   |   | STEP_CNT_DIV |   |   |   |   |   |
| Type  | R/W           | R/W            |    |    |    | R/W            |   |   |   |   | R/W          |   |   |   |   |   |
| Reset | 'h0           | 'h1            |    |    |    | 'h3            |   |   |   |   | 'h3          |   |   |   |   |   |

LS/FS only. This register is used to control the output waveform timing. Each half pulse width (i.e. each high or low pulse) is equal to  $(\text{step\_cnt\_div}+1) * (\text{sample\_cnt\_div}+1) * 1/13\text{MHz}$

|                            |                                                                                                                          |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>SAMPLE_CNT_DIV[2:0]</b> | Used for LS/FS only. This adjusts the width of each sample. Sample width = (sample_cnt_div+1)*1/13MHz. Minimum value: 0. |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------|

|                          |                                                                                                       |
|--------------------------|-------------------------------------------------------------------------------------------------------|
| <b>STEP_CNT_DIV[5:0]</b> | Specifies the number of samples per half pulse width (i.e. each high or low pulse). Minimum value: 1. |
|--------------------------|-------------------------------------------------------------------------------------------------------|

|                      |                                                                                                                                                                                                              |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DATA_READ_ADJ</b> | When set to 1, the data latch in sampling time during master reads are adjusted according to the DATA_READ_TIME value. Otherwise, by default, the data are latched in at half of the high pulse width point. |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                            |                                                                                                                                                                                                                                                                  |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DATA_READ_TIME[2:0]</b> | This value is valid only when DATA_READ_ADJ is set to 1. Can be used to adjust so that the data are latched in at earlier sampling points (assuming the data are settled by then). This value must be set to smaller than or equal to half the high pulse width. |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## I2CREG+002

4h

## Start register

**START**

[illegible]

**START** Starts the transaction on the bus. It is auto de-asserted at the end of the transaction.

### I2CREG+003

0h

### FIFO status register

### FIFO\_STAT

| Bit   | 15      | 14 | 13 | 12 | 11      | 10 | 9 | 8 | 7           | 6 | 5 | 4 | 3 | 2 | 1       | 0        |
|-------|---------|----|----|----|---------|----|---|---|-------------|---|---|---|---|---|---------|----------|
| Name  | RD_ADDR |    |    |    | WR_ADDR |    |   |   | FIFO_OFFSET |   |   |   |   |   | WR_FULL | RD_EMPTY |
| Type  | RO      |    |    |    | RO      |    |   |   | RO          |   |   |   |   |   | RO      | RO       |
| Reset | 0       |    |    |    | 0       |    |   |   | 0           |   |   |   | 0 | 0 | 0       | 0        |

**RD\_ADDR[3:0]** Current RD address pointer (only bit [2:0] have physical meaning)

**WR\_ADDR[3:0]** Current WR address pointer (only bit [2:0] have physical meaning)

**FIFO\_OFFSET[3:0]** wr\_addr[3:0] – rd\_addr[3:0]

**WR\_FULL** Indicates FIFO is full.

**RD\_EMPTY** Indicates FIFO is empty.

### I2CREG+003

8h

### FIFO address clear register

### FIFO\_ADDR\_CLR

LR

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0             |
|-------|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---------------|
| Name  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | FIFO_ADDR_CLR |
| Type  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | WO            |
| Reset |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | 0             |

**FIFO\_ADDR\_CLR** When written 1'b1, one pulse fifo\_addr\_clr will be generated to clear the FIFO address to 0.

### I2CREG+004

0h

### IO config register

### IO\_CONFIG

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3          | 2          | 1             | 0             |
|-------|----|----|----|----|----|----|---|---|---|---|---|---|------------|------------|---------------|---------------|
| Name  |    |    |    |    |    |    |   |   |   |   |   |   | IDLE_OE_EN | IO_SYNC_EN | SDA_IO_CONFIG | SCL_IO_CONFIG |
| Type  |    |    |    |    |    |    |   |   |   |   |   |   | R/W        | R/W        | R/W           | R/W           |
| Reset |    |    |    |    |    |    |   |   |   |   |   |   | 0          | 0          | 0             | 0             |

This register is used to configure the I/O for the SDA and SCL lines to select between normal I/O mode or open-drain mode to support wired-and bus.

- IDLE\_OE\_EN**      **0**    Does not drive bus in idle state  
                          **1**    Drive bus in idle state
- IO\_SYNC\_EN**      For debugging only. When set to 1, SCL and SDA inputs will be first dual synced by bclk\_ck. This should not be needed. Only reserved for debugging.
- SDA\_IO\_CONFIG**   **0**    Normal tristate I/O mode  
                          **1**    Open-drain mode
- SCL\_IO\_CONFIG**   **0**    Normal tristate I/O mode  
                          **1**    Open-drain mode

#### I2CREG+004

4h

Reserved debug register

DEBUG

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2   | 1   | 0   |
|-------|----|----|----|----|----|----|---|---|---|---|---|---|---|-----|-----|-----|
| Name  |    |    |    |    |    |    |   |   |   |   |   |   |   |     |     |     |
| Type  |    |    |    |    |    |    |   |   |   |   |   |   |   | R/W | R/W | R/W |
| Reset |    |    |    |    |    |    |   |   |   |   |   |   |   | 0   | 0   | 0   |

This register is for debugging only. The bits are R/W. Do not change the values from the default value.

#### I2CREG+004

8h

High-speed mode register

HS

| Bit   | 15 | 14                | 13 | 12 | 11 | 10              | 9 | 8 | 7 | 6           | 5 | 4 | 3 | 2 | 1                  | 0     |
|-------|----|-------------------|----|----|----|-----------------|---|---|---|-------------|---|---|---|---|--------------------|-------|
| Name  |    | HS_SAMPLE_CNT_DIV |    |    |    | HS_STEP_CNT_DIV |   |   |   | MASTER_CODE |   |   |   |   | HS_NACK_ERR_DET_EN | HS_EN |
| Type  |    | R/W               |    |    |    | R/W             |   |   |   | R/W         |   |   |   |   | R/W                | R/W   |
| Reset |    | 0                 |    |    |    | 1               |   |   |   | 0           |   |   |   |   | 1                  | 0     |

This register contains options to support high-speed operation features. Each HS half pulse width (i.e. each high or low pulse) = (hs\_step\_cnt\_div+1)\*[(hs\_sample\_cnt\_div+1)\*1/13MHz]

- HS\_SAMPLE\_CNT\_DIV[2:0]**    When the high-speed mode is entered after the master code transfer is completed, the sample width will become dependent on this parameter. Minimum value: 0

- HS\_STEP\_CNT\_DIV[2:0]** When the high-speed mode is entered after the master code transfer is completed, the number of samples per half pulse width will become dependent on this value. Minimum value: 1
- MASTER\_CODE[2:0]** The 3-bit programmable value for the master code to be transmitted.
- HS\_NACKERR\_DET\_EN** Enables NACKERR detection during the master code transmission. When enabled, if NACK is not received after the master code is transmitted, the transaction will be terminated with a STOP condition.
- HS\_EN** Enables the high-speed transaction. *Note: rs\_stop must be set to 1 as well.*

**I2CREG+005**  
**0h**

**Soft reset register**

**SOFTRESET**

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0                 |
|-------|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|-------------------|
| Name  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | <b>SOFT_RESET</b> |
| Type  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | WO                |
| Reset |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | 0                 |

- SOFT\_RESET** When written with 1'b1, one pulse soft reset is used as synchronous reset to reset the I2C internal hardware circuits.

**I2CREG+006**  
**4h**

**Debug status register**

**DEBUGSTAT**

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7               | 6                   | 5                  | 4                   | 3 | 2 | 1 | 0 |
|-------|----|----|----|----|----|----|---|---|-----------------|---------------------|--------------------|---------------------|---|---|---|---|
| Name  |    |    |    |    |    |    |   |   | <b>BUS_BUSY</b> | <b>MASTER_WRITE</b> | <b>MASTER_READ</b> | <b>MASTER_STATE</b> |   |   |   |   |
| Type  |    |    |    |    |    |    |   |   | RO              | RO                  | RO                 | RO                  |   |   |   |   |
| Reset |    |    |    |    |    |    |   |   | 0               | 1                   | 0                  | 0                   |   |   |   |   |

- BUS\_BUSY** For debugging only. Valid when bus\_detect\_en is 1. bus\_busy = 1 indicates a start transaction has been detected and no stop condition has been detected yet.

- MASTER\_WRITE** For debugging only. 1 = Current transfer is in the master write dir

**MASTER\_READ** For debugging only. 1 = Current transfer is in the master read dir

**MASTER\_STATE[4:0]** For debugging only. Reads back the current master\_state.

- 0** Idle state
- 1** I2C master is preparing for sending out the start bit, SCL = 1, SDA = 1
- 2** I2C master is sending out the start bit, SCL = 1, SDA = 0
- 3** I2C master/slave is preparing for transmitting data bit, SCL = 0, SDA = data bit (data bit can be changed when SCL = 0)
- 4** I2C master/slave is transmitting data bit, SCL = 1, SDA = data bit (data bit is stable when SCL = 1)
- 5** I2C master/slave is preparing for transmitting the ACK bit, SCL = 0, SDA = ack (the ACK bit can be changed when SCL = 0)
- 6** I2C master/slave is transmitting the ACK bit, SCL = 1, SDA = 0 (the ACK bit is stable when SCL = 1)
- 7** I2C master is preparing for sending out stop bit or repeated-start bit, SCL = 0, SDA = 0/1 (0 means stop bit; 1 means repeated-start bit)
- 8** I2C master is sending out stop bit or repeated-start bit, SCL = 1, SDA = 1/0 (1 means stop bit; 0 means repeated-start bit)
- 9** I2C master is in delay start between two transfers, SCL = 1, SDA = 1
- 10** I2C master is in FIFO wait state. For writing transaction, it means FIFO is empty and I2C master is waiting for DMA controller to write data into FIFO. For reading transaction, it means FIFO is full and I2C master is waiting for DMA controller to read data from FIFO, SCL = 0, SDA = don't care;
- 12** I2C master is preparing for sending out data bit of master code. This state is used only in high-speed transaction, SCL = 0, SDA = data bit of master code (data bit of master code can be changed when SCL = 0)
- 13** I2C master is sending out data bit of master code. This state is used only in high-speed transaction, SCL = 1, SDA = data bit of master code (data bit of master code is stable when SCL = 1)
- 14** I2C master/slave is preparing for transmitting the NACK bit, SCL = 0, SDA = NACK bit (the NACK bit can be changed when SCL = 0). This state is used only in high-speed transaction.
- 15** I2C master/slave is transmitting the NACK bit, SCL = 1, SDA = 1. This state is used only in high-speed transaction.

**I2CREG+006**  
**8h** **Debug control register**

**DEBUGCTRL**

| Bit  | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1                 | 0               |
|------|----|----|----|----|----|----|---|---|---|---|---|---|---|---|-------------------|-----------------|
| Name |    |    |    |    |    |    |   |   |   |   |   |   |   |   | APB_DEB_UG_R<br>D | FIFO_APB_DEB_UG |

|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |    |     |
|-------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|----|-----|
| Type  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | WO | R/W |
| Reset |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 0  | 0   |

**APB\_DEBUG\_RD** Valid when fifo\_apb\_debug is set to 1. Writing to this register will generate a 1 pulsed FIFO APB RD signal for reading the FIFO data.

**FIFO\_APB\_DEBUG** Used for trace32 debugging purposes. When using trace32, and the memory map is shown, turning this bit on will block the normal APB read access. The APB read access to the FIFO is then enabled by writing to apb\_debug\_rd.

- 0** DISABLE  
**1** ENABLE

## I2CREG+007 Time-out timing register

**TIMEOUT**

|       |         |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-------|---------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Bit   | 15      | 14  | 13  | 12  | 11  | 10  | 9   | 8   | 7   | 6   | 5   | 4   | 3   | 2   | 1   | 0   |
| Name  | TIMEOUT |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Type  | R/W     | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W |
| Reset | 'h1     | 'h1 | 'h1 | 'h1 | 'h1 | 'h1 | 'h1 | 'h1 | 'h1 | 'h1 | 'h1 | 'h1 | 'h1 | 'h1 | 'h1 | 'h1 |

**TIMEOUT[15:0]** Number of steps to count before time-out. The time-out counter will only count when the time-out mechanism is enabled and has started a transaction.

## I2CREG+007 Version code register

**VERSION\_CODE**

|       |              |    |    |    |    |    |   |   |   |   |   |   |   |   |   |      |
|-------|--------------|----|----|----|----|----|---|---|---|---|---|---|---|---|---|------|
| Bit   | 15           | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0    |
| Name  | VERSION_CODE |    |    |    |    |    |   |   |   |   |   |   |   |   |   |      |
| Type  |              |    |    |    |    |    |   |   |   |   |   |   |   |   |   | RO   |
| Reset |              |    |    |    |    |    |   |   |   |   |   |   |   |   |   | 'h10 |

**VERSION\_CODE[7:0]** Hardware version code

## 4.9 Real Time Clock

### 4.9.1 General Description

The Real-Time Clock (RTC) module provides time and data information, as well as 32.768 kHz clock. By configuring pin XOSC32\_ENB, the use of the 32k crystal can be determined, i.e. to use a 32k crystal or not. The RTC block has an independent power supply. When the mobile handset is powered off, a dedicated regulator supplies the RTC block. If the main battery is not present, a backup supply



such as a small mercury cell battery or a large capacitor is used. In addition to providing timing data, an alarm interrupt is generated and can be used to power up the baseband core. Regulator interrupts corresponding to seconds, minutes, hours and days can be generated whenever the time counter value reaches a maximum value (e.g. 59 for seconds and minutes, 23 for hours, etc.). The year span is supported up to 2,127. The maximum day-of-month values, which depend on the leap year condition, are stored in the RTC block.

#### 4.9.2 Register Definition

Module name: RTC Base address: (+A0710000h)

| Address   | Name                     | Width | Register function                                                                                                                                                                                                                                                                                            |
|-----------|--------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A0710000  | <u>RTC_BBPU</u>          | 16    | Baseband power up                                                                                                                                                                                                                                                                                            |
| A071 0004 | <u>RTC_IRQ_STA</u>       | 16    | <b>RTC IRQ status</b><br>This register is fixed to 0 when RTC_POWERKEY1 & RTC_POWERKEY2 unmatch the correct values.                                                                                                                                                                                          |
| A071 0008 | <u>RTC_IRQ_EN</u>        | 16    | <b>RTC IRQ enable</b><br>This register is fixed to 0 when RTC_POWERKEY1 & RTC_POWERKEY2 unmatch the correct values.                                                                                                                                                                                          |
| A071 000C | <u>RTC_CII_EN</u>        | 16    | <b>Counter increment IRQ enable</b><br>This register activates or de-activates the IRQ generation when the TC counter reaches its maximum value.                                                                                                                                                             |
| A071 0010 | <u>RTC_AL_MAS<br/>K</u>  | 16    | <b>RTC alarm mask</b><br>The alarm condition for alarm IRQ generation depends on whether or not the corresponding bit in this register is masked. Warning: If you set all bits to 1 in RTC_AL_MASK (i.e. RTC_AL_MASK=0x7f) and PWREN=1 in RTC_BBPU, it means the alarm will come every second, not disabled. |
| A071 0014 | <u>RTC_TC_SEC</u>        | 16    | <b>RTC seconds time counter register</b>                                                                                                                                                                                                                                                                     |
| A071 0018 | <u>RTC_TC_MIN</u>        | 16    | <b>RTC minutes time counter register</b>                                                                                                                                                                                                                                                                     |
| A071001C  | <u>RTC_TC_HOU</u>        | 16    | <b>RTC hours time counter register</b>                                                                                                                                                                                                                                                                       |
| A0710020  | <u>RTC_TC_DOM</u>        | 16    | <b>RTC day-of-month time counter register</b>                                                                                                                                                                                                                                                                |
| A0710024  | <u>RTC_TC_DOW</u>        | 16    | <b>RTC day-of-week time counter register</b>                                                                                                                                                                                                                                                                 |
| A0710028  | <u>RTC_TC_MTH</u>        | 16    | <b>RTC month time counter register</b>                                                                                                                                                                                                                                                                       |
| A071002C  | <u>RTC_TC_YEA</u>        | 16    | <b>RTC year time counter register</b>                                                                                                                                                                                                                                                                        |
| A0710030  | <u>RTC_AL_SEC</u>        | 16    | <b>RTC second alarm setting register</b>                                                                                                                                                                                                                                                                     |
| A0710034  | <u>RTC_AL_MIN</u>        | 16    | <b>RTC minute alarm setting register</b>                                                                                                                                                                                                                                                                     |
| A0710038  | <u>RTC_AL_HOU</u>        | 16    | <b>RTC hour alarm setting register</b>                                                                                                                                                                                                                                                                       |
| A071003C  | <u>RTC_AL_DOM</u>        | 16    | <b>RTC day-of-month alarm setting register</b>                                                                                                                                                                                                                                                               |
| A0710040  | <u>RTC_AL_DOW</u>        | 16    | <b>RTC day-of-week alarm setting register</b>                                                                                                                                                                                                                                                                |
| A0710044  | <u>RTC_AL_MTH</u>        | 16    | <b>RTC month alarm setting register</b>                                                                                                                                                                                                                                                                      |
| A0710048  | <u>RTC_AL_YEA</u>        | 16    | <b>RTC year alarm setting register</b>                                                                                                                                                                                                                                                                       |
| A071004C  | <u>RTC_OSC32C<br/>ON</u> | 16    | <b>OSC32 control</b><br>The function is only active when RTC_POWERKEY1 &                                                                                                                                                                                                                                     |

| Address  | Name                 | Width | Register function                                                                                                                                                                                                               |
|----------|----------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                      |       | RTC_POWERKEY2 match the correct values.                                                                                                                                                                                         |
| A0710050 | <u>RTC_POWERKEY1</u> | 16    | RTC_POWERKEY1 register                                                                                                                                                                                                          |
| A0710054 | <u>RTC_POWERKEY2</u> | 16    | RTC_POWERKEY2 register                                                                                                                                                                                                          |
| A0710058 | <u>RTC_PDN1</u>      | 16    | PDN1                                                                                                                                                                                                                            |
| A071005C | <u>RTC_PDN2</u>      | 16    | PDN2                                                                                                                                                                                                                            |
| A0710060 | <u>RTC_SPAR0</u>     | 16    | Spare register for specific purpose                                                                                                                                                                                             |
| A0710064 | <u>RTC_SPAR1</u>     | 16    | Spare register for specific purpose                                                                                                                                                                                             |
| A0710068 | <u>RTC_PROT</u>      | 16    | Lock/unlock scheme to prevent RTC miswriting                                                                                                                                                                                    |
| A071006C | <u>RTC_DIFF</u>      | 16    | One-time calibration offset<br>This register is fixed to 0 when RTC_POWERKEY1 & RTC_POWERKEY2 unmatch the correct values.                                                                                                       |
| A0710070 | <u>RTC_CALI</u>      | 16    | Repeat calibration offset<br>This register is fixed to 0 when RTC_POWERKEY1 & RTC_POWERKEY2 unmatch the correct values.                                                                                                         |
| A0710074 | <u>RTC_WRTGR</u>     | 16    | Enable the transfers from core to RTC in the queue                                                                                                                                                                              |
| A0710078 | <u>RTC_CON</u>       | 16    | Other RTC control register<br>Note: GPU,GOE,LPRST,LPEN are tie0 and GPEN,F32KOB are tie1 internally when RTC_POWERKEY1 & RTC_POWERKEY2 do not match the correct values. After changing RTC_CON, write WRTGR = 1 to take effect. |

**Table 62. RTC register map**

| A0710000 |          | RTC_BBPU |    |    | Baseband power up |    |   |   |   |       |        |         |      |      | 0000 |       |
|----------|----------|----------|----|----|-------------------|----|---|---|---|-------|--------|---------|------|------|------|-------|
| Bit      | 15       | 14       | 13 | 12 | 11                | 10 | 9 | 8 | 7 | 6     | 5      | 4       | 3    | 2    | 1    | 0     |
| Mne      | KEY_BBPU |          |    |    |                   |    |   |   |   | CBUSY | RELOAD | CLR PKY | AUTO | BBPU |      | PWREN |
| Type     | WO       |          |    |    |                   |    |   |   |   | RO    | WO     | WO      | RW   | RW   |      | RW    |
| Reset    | 0        | 0        | 0  | 0  | 0                 | 0  | 0 | 0 |   | 0     | 0      | 0       | 0    | 0    |      | 0     |

| Bit (s) | Mnemonic | Name     | Description                                                                                                                                                                                                                                                                  |
|---------|----------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8    | KEY_BBPU | KEY_BBPU | A bus write is acceptable only when KEY_BBPU = 0x43.                                                                                                                                                                                                                         |
| 6       | CBUSY    | CBUSY    | The read/write channels between RTC/Core is busy. This bit indicates high after the software program sequence to anyone of RTC data registers and enables the transfer by RTC_WRTGR = 1. In addition, it is high after the reset from low to high due to RTC reload process. |
| 5       | RELOAD   | RELOAD   | <b>Reloads the values from RTC domain to core domain</b><br>Generally the RTC will reload and synchronize the data from RTC to core when being reset from 0 to 1. This bit can be treated as a debugging bit.                                                                |

| Bit (s) | Mnemonic | Name   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------|----------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4       | CLRPKY   | CLRPKY | <b>Clears powerkey1 and powerkey2 at the same time</b><br>In some cases, the software may clear powerkey1 & powerkey2. BBWAKEUP depends on the matching specific patterns of powerkey1 and powerkey2. If any one of powerkey1 or powerkey2 or BBPU is cleared, BBWAKEUP will go low immediately. The software cannot program other control bits without power. By programming RTC_BBPU with CLRPKY = 1 and BBPU = 0 condition, RTC can clear powerkey1, powerkey2 and BBPU at the same time. |
| 3       | AUTO     | AUTO   | <b>Controls if BBWAKEUP is automatically in the low state when SYSRST transitions from high to low.</b><br>0: BBWAKEUP is not automatically in the low state when SYSRST# transitions are from high to low.<br>1: BBWAKEUP is automatically in the low state when SYSRST# transitions are from high to low. The function is only active when RTC_POWERKEY1 & RTC_POWERKEY2 match the correct values.                                                                                         |
| 2       | BBPU     | BBPU   | <b>Controls the power of PMU</b><br>If powerkey1 = A357h and powerkey2 = 67D2h, PMU takes on the value programmed by software; otherwise PMU is low.<br>0: Power down<br>1: Power on                                                                                                                                                                                                                                                                                                         |
| 0       | PWREN    | PWREN  | 0: RTC alarm has no action on power switch.<br>1: When an RTC alarm occurs, BBPU is set to 1 and the system is powered on by RTC alarm wakeup.                                                                                                                                                                                                                                                                                                                                               |

| A0710004 | RTC_IRQ_STA |    |    |    | RTC_IRQ status |    |   |   |   |   |   |   | 0000      |   |           |           |
|----------|-------------|----|----|----|----------------|----|---|---|---|---|---|---|-----------|---|-----------|-----------|
| Bit      | 15          | 14 | 13 | 12 | 11             | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3         | 2 | 1         | 0         |
| Mne      |             |    |    |    |                |    |   |   |   |   |   |   | LPST<br>A |   | TCS<br>TA | ALS<br>TA |
| Type     |             |    |    |    |                |    |   |   |   |   |   |   | RC        |   | RC        | RC        |
| Reset    |             |    |    |    |                |    |   |   |   |   |   |   | 0         |   | 0         | 0         |

**Overview** This register is read-clear and is fixed to 0 when RTC\_POWERKEY1 and RTC\_POWERKEY2 unmatch the correct values.

| Bit (s) | Mnemonic | Name  | Description                                                                                                                                                                                                                                                                                                          |
|---------|----------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3       | LPSTA    | LPSTA | <b>Indicates the IRQ status and whether or not the LPD is asserted</b><br>(LPD function is either provided by XOSC32 or EOSC32, depending on XOSC32_ENB)<br>0: No IRQ occurred; the 32K clock is good.<br>1: IRQ occurred; the 32K clock stopped or stops. This can be masked by LP_EN or cleared by initialize LPD. |

| Bit (s) | Mnemonic | Name  | Description                                                                                                                                                                                             |
|---------|----------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | TCSTA    | TCSTA | <b>Indicates the IRQ status and whether or not the tick condition has been met.</b><br>0: No IRQ occurred; the tick condition has not been met.<br>1: IRQ occurred; the tick condition has been met.    |
| 0       | ALSTA    | ALSTA | <b>Indicates the IRQ status and whether or not the alarm condition has been met.</b><br>0: No IRQ occurred; the alarm condition has not been met.<br>1: IRQ occurred; the alarm condition has been met. |

| A0710008 | RTC_IRQ_EN |    |    |    | RTC IRQ enable |    |   |   |   |   |   |   | 0000  |          |       |       |
|----------|------------|----|----|----|----------------|----|---|---|---|---|---|---|-------|----------|-------|-------|
| Bit      | 15         | 14 | 13 | 12 | 11             | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3     | 2        | 1     | 0     |
| Mne      |            |    |    |    |                |    |   |   |   |   |   |   | LP_EN | ONE SHOT | TC_EN | AL_EN |
| Type     |            |    |    |    |                |    |   |   |   |   |   |   | RW    | RW       | RW    | RW    |
| Reset    |            |    |    |    |                |    |   |   |   |   |   |   | 0     | 0        | 0     | 0     |

**Overview** This register is fixed to 0 when RTC\_POWERKEY1 and RTC\_POWERKEY2 unmatch the correct values.

| Bit (s) | Mnemonic | Name    | Description                                                                                                                                                                                                                                         |
|---------|----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3       | LP_EN    | LP_EN   | <b>Enables the control bit for IRQ generation if low power detected (32k clock off)</b><br>0: Disable IRQ generations<br>1: Enable LPD                                                                                                              |
| 2       | ONESHOT  | ONESHOT | <b>Controls automatic reset of AL_EN and TC_EN</b>                                                                                                                                                                                                  |
| 1       | TC_EN    | TC_EN   | <b>Enables the control bit for IRQ generation if the tick condition has been met.</b><br>0: Disable IRQ generations<br>1: Enable the tick time match interrupt. Clear the interrupt when ONESHOT is high upon generation of the corresponding IRQ.  |
| 0       | AL_EN    | AL_EN   | <b>Enables the control bit for IRQ generation if the alarm condition has been met</b><br>0: Disable IRQ generations<br>1: Enable the alarm time match interrupt. Clear the interrupt when ONESHOT is high upon generation of the corresponding IRQ. |

| A071000C | RTC_CII_EN |    |    |    | Counter increment IRQ enable |    |           |           |           |         |         |         | 0000    |         |         |         |
|----------|------------|----|----|----|------------------------------|----|-----------|-----------|-----------|---------|---------|---------|---------|---------|---------|---------|
| Bit      | 15         | 14 | 13 | 12 | 11                           | 10 | 9         | 8         | 7         | 6       | 5       | 4       | 3       | 2       | 1       | 0       |
| Mne      |            |    |    |    |                              |    | 1/8S ECCI | 1/4S ECCI | 1/2S ECCI | YEA CII | MTH CII | DOW CII | DOM CII | HOU CII | MIN CII | SEC CII |

|       |  |  |  |  |  |  |    |    |    |    |    |    |    |    |    |    |
|-------|--|--|--|--|--|--|----|----|----|----|----|----|----|----|----|----|
|       |  |  |  |  |  |  | I  | I  | I  |    |    |    |    |    |    |    |
| Type  |  |  |  |  |  |  | RW | RW | RW | RW | RW | RW | RW | RW | RW | RW |
| Reset |  |  |  |  |  |  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |

**Overview** This register activates or de-activates the IRQ generation when the TC counter reaches its maximum value.

| Bit (s) | Mnemonic         | Name       | Description                                                          |
|---------|------------------|------------|----------------------------------------------------------------------|
| 9       | <b>1/8SECCII</b> | SECCII_1_8 | Sets the bit to 1 to activate the IRQ at each 1/8 of a second update |
| 8       | <b>1/4SECCII</b> | SECCII_1_4 | Sets the bit to 1 to activate the IRQ at each 1/4 of a second update |
| 7       | <b>1/2SECCII</b> | SECCII_1_2 | Sets the bit to 1 to activate the IRQ at each 1/2 of a second update |
| 6       | <b>YEACII</b>    | YEACII     | Sets the bit to 1 to activate the IRQ at each year update            |
| 5       | <b>MTHCII</b>    | MTHCII     | Sets the bit to 1 to activate the IRQ at each month update           |
| 4       | <b>DOWCII</b>    | DOWCII     | Sets the bit to 1 to activate the IRQ at each day-of-week update     |
| 3       | <b>DOMCII</b>    | DOMCII     | Sets the bit to 1 to activate the IRQ at each day-of-month update    |
| 2       | <b>HOUCII</b>    | HOUCII     | Sets the bit to 1 to activate the IRQ at each hour update            |
| 1       | <b>MINCII</b>    | MINCII     | Sets the bit to 1 to activate the IRQ at each minute update          |
| 0       | <b>SECCII</b>    | SECCII     | Sets this bit to 1 to activate the IRQ at each second update         |

| A0710010 | RTC_AL_MASK |    |    | RTC alarm mask |    |    |   |   |   |          | 0000     |          |          |          |         |          |
|----------|-------------|----|----|----------------|----|----|---|---|---|----------|----------|----------|----------|----------|---------|----------|
| Bit      | 15          | 14 | 13 | 12             | 11 | 10 | 9 | 8 | 7 | 6        | 5        | 4        | 3        | 2        | 1       | 0        |
| Mne      |             |    |    |                |    |    |   |   |   | YEA_MS_K | MTH_MS_K | DOW_MS_K | DOM_MS_K | HOU_MS_K | MIN_MSK | SEC_MS_K |
| Type     |             |    |    |                |    |    |   |   |   | RW       | RW       | RW       | RW       | RW       | RW      | RW       |
| Reset    |             |    |    |                |    |    |   |   |   | 0        | 0        | 0        | 0        | 0        | 0       | 0        |

**Overview** The alarm condition for alarm IRQ generation depends on whether or not the corresponding bit in this register is masked. *Note that if all bits 1 in RTC\_AL\_MASK are set (i.e. RTC\_AL\_MASK = 0x7f) and PWREN = 1 in RTC\_BBPU, it means the alarm will come every second, not disabled.*

| Bit (s) | Mnemonic       | Name    | Description                                                                                                                                                                                                 |
|---------|----------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6       | <b>YEA_MSK</b> | YEA_MSK | 0: Condition (RTC_TC_YEA = RTC_AL_YEA) is checked to generate the alarm signal.<br>1: Condition (RTC_TC_YEA = RTC_AL_YEA) is masked, i.e. the value of RTC_TC_YEA does not affect the alarm IRQ generation. |

| Bit (s) | Mnemonic       | Name    | Description                                                                                                                                                                                                 |
|---------|----------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5       | <b>MTH_MSK</b> | MTH_MSK | 0: Condition (RTC_TC_MTH = RTC_AL_MTH) is checked to generate the alarm signal.<br>1: Condition (RTC_TC_MTH = RTC_AL_MTH) is masked, i.e. the value of RTC_TC_MTH does not affect the alarm IRQ generation. |
| 4       | <b>DOW_MSK</b> | DOW_MSK | 0: Condition (RTC_TC_DOW = RTC_AL_DOW) is checked to generate the alarm signal.<br>1: Condition (RTC_TC_DOW = RTC_AL_DOW) is masked, i.e. the value of RTC_TC_DOW does not affect the alarm IRQ generation. |
| 3       | <b>DOM_MSK</b> | DOM_MSK | 0: Condition (RTC_TC_DOM = RTC_AL_DOM) is checked to generate the alarm signal.<br>1: Condition (RTC_TC_DOM = RTC_AL_DOM) is masked, i.e. the value of RTC_TC_DOM does not affect the alarm IRQ generation. |
| 2       | <b>HOU_MSK</b> | HOU_MSK | 0: Condition (RTC_TC_HOU = RTC_AL_HOU) is checked to generate the alarm signal.<br>1: Condition (RTC_TC_HOU = RTC_AL_HOU) is masked, i.e. the value of RTC_TC_HOU does not affect the alarm IRQ generation. |
| 1       | <b>MIN_MSK</b> | MIN_MSK | 0: Condition (RTC_TC_MIN = RTC_AL_MIN) is checked to generate the alarm signal.<br>1: Condition (RTC_TC_MIN = RTC_AL_MIN) is masked, i.e. the value of RTC_TC_MIN does not affect the alarm IRQ generation. |
| 0       | <b>SEC_MSK</b> | SEC_MSK | 0: Condition (RTC_TC_SEC = RTC_AL_SEC) is checked to generate the alarm signal.<br>1: Condition (RTC_TC_SEC = RTC_AL_SEC) is masked, i.e. the value of RTC_TC_SEC does not affect the alarm IRQ generation. |

**A0710014    RTC TC SEC    RTC seconds time counter register**

**0000**

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5                | 4 | 3 | 2 | 1 | 0 |
|-------|----|----|----|----|----|----|---|---|---|---|------------------|---|---|---|---|---|
| Mne   |    |    |    |    |    |    |   |   |   |   | <b>TC_SECOND</b> |   |   |   |   |   |
| Type  |    |    |    |    |    |    |   |   |   |   | RW               |   |   |   |   |   |
| Reset |    |    |    |    |    |    |   |   |   |   | 0                | 0 | 0 | 0 | 0 | 0 |

**Bit (s)    Mnemonic    Name    Description**

5:0    **TC\_SECOND**    TC\_SECOND    Second initial value for the time counter. Range: 0 ~ 59

**A0710018    RTC TC MIN    RTC minutes time counter register**

**0000**

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5                | 4 | 3 | 2 | 1 | 0 |
|-------|----|----|----|----|----|----|---|---|---|---|------------------|---|---|---|---|---|
| Mne   |    |    |    |    |    |    |   |   |   |   | <b>TC_MINUTE</b> |   |   |   |   |   |
| Type  |    |    |    |    |    |    |   |   |   |   | RW               |   |   |   |   |   |
| Reset |    |    |    |    |    |    |   |   |   |   | 0                | 0 | 0 | 0 | 0 | 0 |



| Bit (s) | Mnemonic         | Name      | Description                                              |
|---------|------------------|-----------|----------------------------------------------------------|
| 5:0     | <b>TC_MINUTE</b> | TC_MINUTE | Minute initial value for the time counter. Range: 0 ~ 59 |

**A071001C    RTC TC\_HOU    RTC hours time counter register    0000**

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4              | 3 | 2 | 1 | 0 |
|-------|----|----|----|----|----|----|---|---|---|---|---|----------------|---|---|---|---|
| Mne   |    |    |    |    |    |    |   |   |   |   |   | <b>TC_HOUR</b> |   |   |   |   |
| Type  |    |    |    |    |    |    |   |   |   |   |   | RW             |   |   |   |   |
| Reset |    |    |    |    |    |    |   |   |   |   |   | 0              | 0 | 0 | 0 | 0 |

| Bit (s) | Mnemonic       | Name    | Description                                            |
|---------|----------------|---------|--------------------------------------------------------|
| 4:0     | <b>TC_HOUR</b> | TC_HOUR | Hour initial value for the time counter. Range: 0 ~ 23 |

**A0710020    RTC TC\_DOM    RTC day-of-month time counter register    0000**

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4             | 3 | 2 | 1 | 0 |
|-------|----|----|----|----|----|----|---|---|---|---|---|---------------|---|---|---|---|
| Mne   |    |    |    |    |    |    |   |   |   |   |   | <b>TC_DOM</b> |   |   |   |   |
| Type  |    |    |    |    |    |    |   |   |   |   |   | RW            |   |   |   |   |
| Reset |    |    |    |    |    |    |   |   |   |   |   | 0             | 0 | 0 | 0 | 0 |

| Bit (s) | Mnemonic      | Name   | Description                                                                                                                                                    |
|---------|---------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4:0     | <b>TC_DOM</b> | TC_DOM | Day-of-month initial value for the time counter. The day-of-month maximum value depends on the leap year condition, i.e. 2 LSB of year time counter are zeros. |

**A0710024    RTC TC\_DOW    RTC day-of-week time counter register    0000**

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2             | 1 | 0 |
|-------|----|----|----|----|----|----|---|---|---|---|---|---|---|---------------|---|---|
| Mne   |    |    |    |    |    |    |   |   |   |   |   |   |   | <b>TC_DOW</b> |   |   |
| Type  |    |    |    |    |    |    |   |   |   |   |   |   |   | RW            |   |   |
| Reset |    |    |    |    |    |    |   |   |   |   |   |   |   | 0             | 0 | 0 |

| Bit (s) | Mnemonic      | Name   | Description                                                  |
|---------|---------------|--------|--------------------------------------------------------------|
| 2:0     | <b>TC_DOW</b> | TC_DOW | Day-of-week initial value for the time counter. Range: 1 ~ 7 |

**A0710028    RTC TC\_MTH    RTC month time counter register    0000**

| Bit  | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3               | 2 | 1 | 0 |
|------|----|----|----|----|----|----|---|---|---|---|---|---|-----------------|---|---|---|
| Mne  |    |    |    |    |    |    |   |   |   |   |   |   | <b>TC_MONTH</b> |   |   |   |
| Type |    |    |    |    |    |    |   |   |   |   |   |   | RW              |   |   |   |



|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |   |   |   |   |
|-------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|---|---|---|---|
| Reset |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 0 | 0 | 0 | 0 |
|-------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|---|---|---|---|

| Bit (s) | Mnemonic        | Name     | Description                                             |
|---------|-----------------|----------|---------------------------------------------------------|
| 3:0     | <b>TC_MONTH</b> | TC_MONTH | Month initial value for the time counter. Range: 1 ~ 12 |

| A071002C |    | RTC TC_YEA |    |    |    | RTC year time counter register |   |   |   |         |   |   |   |   |   | 0000 |  |
|----------|----|------------|----|----|----|--------------------------------|---|---|---|---------|---|---|---|---|---|------|--|
| Bit      | 15 | 14         | 13 | 12 | 11 | 10                             | 9 | 8 | 7 | 6       | 5 | 4 | 3 | 2 | 1 | 0    |  |
| Mne      |    |            |    |    |    |                                |   |   |   | TC_YEAR |   |   |   |   |   |      |  |
| Type     |    |            |    |    |    |                                |   |   |   | RW      |   |   |   |   |   |      |  |
| Reset    |    |            |    |    |    |                                |   |   |   | 0       | 0 | 0 | 0 | 0 | 0 | 0    |  |

| Bit (s) | Mnemonic       | Name    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------|----------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6:0     | <b>TC_YEAR</b> | TC_YEAR | <b>Year initial value for the time counter. Range: 0 ~ 127 (2000-2127).</b><br>The software can bias the year as multiples of 4 for the internal leap-year formula. Here are 3 examples: 2000 ~ 2127, 1972 ~ 2099 and 1904 ~ 2031. To simplify the process, the RTC hardware treats all 4-multiple as leap years. If the range you define includes non-leap 4-multiple year (e.g. 2100), please adjust it to the correct date by yourselves. (e.g. change Feb. 29 <sup>th</sup> , 2100 to Mar. 1 <sup>st</sup> , 2100).<br>It is suggested to bias the range to be bigger than 1900 and smaller than 2100 to evade the manual adjustment, i.e. the bias values are suggested to be in the range of [-28,-96], that are (1972~ 2099) ~ (1904~ 2031).<br>The formal leap formula:<br>If year modulo 400 is 0 then leap<br>Else if year modulo 100 is 0 then no_leap<br>Else if year modulo 4 is 0 then leap<br>Else no_leap |

| A0710030 |    | RTC_AL_SEC |    |    |    | RTC second alarm setting register |   |   |   |   |           |   |   |   |   | 0000 |  |
|----------|----|------------|----|----|----|-----------------------------------|---|---|---|---|-----------|---|---|---|---|------|--|
| Bit      | 15 | 14         | 13 | 12 | 11 | 10                                | 9 | 8 | 7 | 6 | 5         | 4 | 3 | 2 | 1 | 0    |  |
| Mne      |    |            |    |    |    |                                   |   |   |   |   | AL_SECOND |   |   |   |   |      |  |
| Type     |    |            |    |    |    |                                   |   |   |   |   | RW        |   |   |   |   |      |  |
| Reset    |    |            |    |    |    |                                   |   |   |   |   | 0         | 0 | 0 | 0 | 0 | 0    |  |

| Bit (s) | Mnemonic          | Name       | Description                                              |
|---------|-------------------|------------|----------------------------------------------------------|
| 5:0     | <b>AL_SECON D</b> | AL_SECON D | Second value of the alarm counter setting. Range: 0 ~ 59 |

|                 |                   |                                          |  |  |  |  |  |  |  |  |  |  |  |  |  |             |  |  |  |
|-----------------|-------------------|------------------------------------------|--|--|--|--|--|--|--|--|--|--|--|--|--|-------------|--|--|--|
| <b>A0710034</b> | <b>RTC_AL_MIN</b> | <b>RTC minute alarm setting register</b> |  |  |  |  |  |  |  |  |  |  |  |  |  | <b>0000</b> |  |  |  |
|-----------------|-------------------|------------------------------------------|--|--|--|--|--|--|--|--|--|--|--|--|--|-------------|--|--|--|

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5         | 4 | 3 | 2 | 1 | 0 |
|-------|----|----|----|----|----|----|---|---|---|---|-----------|---|---|---|---|---|
| Mne   |    |    |    |    |    |    |   |   |   |   | AL_MINUTE |   |   |   |   |   |
| Type  |    |    |    |    |    |    |   |   |   |   | RW        |   |   |   |   |   |
| Reset |    |    |    |    |    |    |   |   |   |   | 0         | 0 | 0 | 0 | 0 | 0 |

| Bit (s) | Mnemonic  | Name      | Description                                              |
|---------|-----------|-----------|----------------------------------------------------------|
| 5:0     | AL_MINUTE | AL_MINUTE | Minute value of the alarm counter setting. Range: 0 ~ 59 |

**A0710038    RTC\_AL\_HOU    RTC hour alarm setting register    0000**

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4       | 3 | 2 | 1 | 0 |
|-------|----|----|----|----|----|----|---|---|---|---|---|---------|---|---|---|---|
| Mne   |    |    |    |    |    |    |   |   |   |   |   | AL_HOUR |   |   |   |   |
| Type  |    |    |    |    |    |    |   |   |   |   |   | RW      |   |   |   |   |
| Reset |    |    |    |    |    |    |   |   |   |   |   | 0       | 0 | 0 | 0 | 0 |

| Bit (s) | Mnemonic | Name    | Description                                            |
|---------|----------|---------|--------------------------------------------------------|
| 4:0     | AL_HOUR  | AL_HOUR | Hour value of the alarm counter setting. Range: 0 ~ 23 |

**A071003C    RTC\_AL\_DOM    RTC day-of-month alarm setting register    0000**

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4      | 3 | 2 | 1 | 0 |
|-------|----|----|----|----|----|----|---|---|---|---|---|--------|---|---|---|---|
| Mne   |    |    |    |    |    |    |   |   |   |   |   | AL_DOM |   |   |   |   |
| Type  |    |    |    |    |    |    |   |   |   |   |   | RW     |   |   |   |   |
| Reset |    |    |    |    |    |    |   |   |   |   |   | 0      | 0 | 0 | 0 | 0 |

| Bit (s) | Mnemonic | Name   | Description                                                                                                                                                    |
|---------|----------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4:0     | AL_DOM   | AL_DOM | Day-of-month value of the alarm counter setting. The day-of-month maximum value depends on the leap year condition, i.e. 2 LSB of year time counter are zeros. |

**A0710040    RTC\_AL\_DOW    RTC day-of-week alarm setting register    0000**

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2      | 1 | 0 |
|-------|----|----|----|----|----|----|---|---|---|---|---|---|---|--------|---|---|
| Mne   |    |    |    |    |    |    |   |   |   |   |   |   |   | AL_DOW |   |   |
| Type  |    |    |    |    |    |    |   |   |   |   |   |   |   | RW     |   |   |
| Reset |    |    |    |    |    |    |   |   |   |   |   |   |   | 0      | 0 | 0 |

| Bit (s) | Mnemonic | Name   | Description                                                  |
|---------|----------|--------|--------------------------------------------------------------|
| 2:0     | AL_DOW   | AL_DOW | Day-of-week value of the alarm counter setting. Range: 1 ~ 7 |

**A0710044    RTC\_AL\_MTH    RTC month alarm setting register**

**0000**

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| Mne   |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| Type  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| Reset |    |    |    |    |    |    |   |   |   |   |   |   | 0 | 0 | 0 | 0 |

| Bit (s) | Mnemonic        | Name     | Description                                             |
|---------|-----------------|----------|---------------------------------------------------------|
| 3:0     | <b>AL_MONTH</b> | AL_MONTH | Month value of the alarm counter setting. Range: 1 ~ 12 |

**A0710048    RTC\_AL\_YEA    RTC year alarm setting register**

**0000**

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| Mne   |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| Type  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| Reset |    |    |    |    |    |    |   |   |   | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit (s) | Mnemonic       | Name    | Description                                                         |
|---------|----------------|---------|---------------------------------------------------------------------|
| 6:0     | <b>AL_YEAR</b> | AL_YEAR | Year value of the alarm counter setting. Range: 0 ~ 127 (2000-2127) |

**A071004C    RTC\_OSC32CON    OSC32 control**

**0000**

| Bit   | 15                      | 14              | 13 | 12 | 11 | 10                    | 9               | 8 | 7                | 6                     | 5                 | 4               | 3 | 2 | 1 | 0 |
|-------|-------------------------|-----------------|----|----|----|-----------------------|-----------------|---|------------------|-----------------------|-------------------|-----------------|---|---|---|---|
| Mne   | <b>EMBCK_SEL_OPTION</b> | <b>EOSC_RSV</b> |    |    |    | <b>EOS_C_C_HOP_EN</b> | <b>EOSC_OPT</b> |   | <b>EMBCK_SEL</b> | <b>EMBCK_SEL_MODE</b> | <b>XOSC32_ENB</b> | <b>XOSCCALI</b> |   |   |   |   |
| Type  | <b>RW</b>               | <b>RW</b>       |    |    |    | <b>RW</b>             | <b>RW</b>       |   | <b>RW</b>        | <b>RW</b>             | <b>RW</b>         | <b>RW</b>       |   |   |   |   |
| Reset | -                       | 0               | 0  | 0  | 0  | 1                     | 0               | 0 | 0                | 0                     | -                 | 0               | 1 | 1 | 1 | 1 |

**Overview**    The function is only active when RTC\_POWERKEY1 and RTC\_POWERKEY2 match the correct values.

| Bit (s) | Mnemonic                | Name             | Description                                                                        |
|---------|-------------------------|------------------|------------------------------------------------------------------------------------|
| 15      | <b>EMBCK_SEL_OPTION</b> | EMBCK_SEL_OPTION | <b>Reserved bit</b><br>Please keep it to 0x0.                                      |
| 14:11   | <b>EOSC_RSV</b>         | EOSC_RSV         | <b>EOSC32 reserved bits</b><br>Please keep it to 0x0.                              |
| 10      | <b>EOSC_CHOP_EN</b>     | EOSC_CHOP_EN     | <b>Enables EOSC32 chopper</b><br>0: Disable<br>1: Enable<br>Please keep it to 0x1. |
| 9:8     | <b>EOSC_OPT</b>         | EOSC_OPT         | <b>EOSC32 optional bits</b>                                                        |

| Bit (s) | Mnemonic          | Name       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------|-------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:6     | <b>EMB_MODE</b>   | EMBCK_SEL  | Please keep it to 0x0.<br><b>Mode setting for crystal removal case</b>                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5       | <b>XOSC32_ENB</b> | XOSC32_ENB | Please keep it to 0x0.<br><b>XOSC32_ENB</b><br>Can read pin XOSC32_ENB configuration to know the 32k crystal usage.<br>0: Use 32k crystal.<br>1: Does not use 32k crystal.                                                                                                                                                                                                                                                                                               |
| 4:0     | <b>XOSCCALI</b>   | XOSCCALI   | <b>Controls XOSC32/EOSC32 cali.</b><br>If XOSC32_ENB(RTC_OSC32CON[5]) = 0, XOSCCALI will control the bias current for 32k xtal in XOSC32. When powerkeys do not match, the default value will be 0xf.<br>If XOSC32_ENB(RTC_OSC32CON[5]) = 1, XOSCCALI will be the trimming value for EOSC32. The software needs to find the best trimming value for EOSC32 when it is the first power-on by frequency meter. When powerkeys do not match, the default value will be 0xf. |

OSC32CON is not protected by RTC\_PROT because RTC\_PROT needs 32.768 kHz clock to unlock. Similarly, to modify RTC\_OSC32CON, writing RTC\_WRTGR is not needed, neither. To protect the OSC32 control bits, follow the *update sequence* to update RTC\_OSC32CON. After the updating sequence is completed, reload to acquire the internal RTC\_OSC32CON. This register needs to be initialized **before** writing powerkeys match.

Update sequence:

- Step 1: Write RTC\_OSC32CON = 0x1a57 and wait until CBUSY = 0
- Step 2: Write RTC\_OSC32CON = 0x2b68 and wait until CBUSY = 0
- Step 3: Write the real value you would like to write RTC\_OSC32CON and wait until CBUSY = 0
- Step 4: Return to step 1 if you need to modify RTC\_OSC32CON again.

*Note: RTC\_OSC32CON should be set before writing correct values to POWERKEY1 and POWERKEY2.*

|                 |                      |    |    |    |    |    |   |   |   |   |   |   |   |   |   |             |
|-----------------|----------------------|----|----|----|----|----|---|---|---|---|---|---|---|---|---|-------------|
| <b>A0710050</b> | <b>RTC_POWERKEY1</b> |    |    |    |    |    |   |   |   |   |   |   |   |   |   | <b>0000</b> |
| Bit             | 15                   | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0           |
| Mne             | <b>RTC_POWERKEY1</b> |    |    |    |    |    |   |   |   |   |   |   |   |   |   |             |
| Type            | RW                   |    |    |    |    |    |   |   |   |   |   |   |   |   |   |             |
| Reset           | 0                    | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0           |

| Bit (s) | Mnemonic | Name | Description |
|---------|----------|------|-------------|
|---------|----------|------|-------------|

| Bit (s) | Mnemonic                  | Name              | Description |
|---------|---------------------------|-------------------|-------------|
| 15:0    | <b>RTC_POWE<br/>RKEY1</b> | RTC_POWE<br>RKEY1 |             |

|          |               |    |    |    |    |                        |   |   |   |   |   |   |   |   |   |      |  |
|----------|---------------|----|----|----|----|------------------------|---|---|---|---|---|---|---|---|---|------|--|
| A0710054 | RTC_POWERKEY2 |    |    |    |    | RTC_POWERKEY2 register |   |   |   |   |   |   |   |   |   | 0000 |  |
| Bit      | 15            | 14 | 13 | 12 | 11 | 10                     | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0    |  |
| Mne      | RTC_POWERKEY2 |    |    |    |    |                        |   |   |   |   |   |   |   |   |   |      |  |
| Type     | RW            |    |    |    |    |                        |   |   |   |   |   |   |   |   |   |      |  |
| Reset    | 0             | 0  | 0  | 0  | 0  | 0                      | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0    |  |

| Bit (s) | Mnemonic                  | Name              | Description |
|---------|---------------------------|-------------------|-------------|
| 15:0    | <b>RTC_POWE<br/>RKEY2</b> | RTC_POWE<br>RKEY2 |             |

These register sets are used to determine if the real time clock has been programmed by the software, i.e. the time value in real time clock is correct. When the real-time clock is first powered on, the register contents are all undefined, and therefore the time values shown are incorrect. The software needs to know if the real-time clock has been programmed. Hence, the two registers are defined to solve this power-on issue. After the software programs the correct value, the two register sets will not need to be updated. In addition to programming the correct time value, when the contents of the register sets are wrong, the interrupt will not be generated. Therefore, the real-time clock will not generate the interrupts before the software programs the registers, and unwanted interrupt due to wrong time value will not occur. The correct values of these two register sets are:

**RTC\_POWERKEY1**    A357h  
**RTC\_POWERKEY2**    67D2h

|                 |                        |    |    |    |                    |    |   |   |   |   |   |   |   |   |   |   |             |  |
|-----------------|------------------------|----|----|----|--------------------|----|---|---|---|---|---|---|---|---|---|---|-------------|--|
| <b>A0710058</b> | <b><u>RTC_PDN1</u></b> |    |    |    | <b><u>PDN1</u></b> |    |   |   |   |   |   |   |   |   |   |   | <b>0000</b> |  |
| Bit             | 15                     | 14 | 13 | 12 | 11                 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |             |  |
| Mne             | RTC_PDN1               |    |    |    |                    |    |   |   |   |   |   |   |   |   |   |   |             |  |
| Type            | RW                     |    |    |    |                    |    |   |   |   |   |   |   |   |   |   |   |             |  |
| Reset           | 0                      | 0  | 0  | 0  | 0                  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |             |  |

| Bit (s) | Mnemonic        | Name     | Description                                                                       |
|---------|-----------------|----------|-----------------------------------------------------------------------------------|
| 15:0    | <b>RTC_PDN1</b> | RTC_PDN1 | Spare registers for software to keep the power-on and power-off state information |

|          |          |    |    |    |      |    |   |   |   |   |   |   |      |   |   |   |
|----------|----------|----|----|----|------|----|---|---|---|---|---|---|------|---|---|---|
| A071005C | RTC_PDN2 |    |    |    | PDN2 |    |   |   |   |   |   |   | 0000 |   |   |   |
| Bit      | 15       | 14 | 13 | 12 | 11   | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3    | 2 | 1 | 0 |
| Mne      | RTC_PDN2 |    |    |    |      |    |   |   |   |   |   |   |      |   |   |   |

| Type  | RW |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|-------|----|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Reset | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit (s) | Mnemonic        | Name     | Description                                                                   |
|---------|-----------------|----------|-------------------------------------------------------------------------------|
| 15:0    | <b>RTC_PDN2</b> | RTC_PDN2 | Spare registers for software to keep power-on and power-off state information |

**A0710060    RTC\_SPAR0    Spare register for specific purpose    0000**

| Bit   | 15               | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------|------------------|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| Mne   | <b>RTC_SPAR0</b> |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| Type  | RW               |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| Reset | 0                | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit (s) | Mnemonic         | Name      | Description                    |
|---------|------------------|-----------|--------------------------------|
| 15:0    | <b>RTC_SPAR0</b> | RTC_SPAR0 | Reserved for specific purposes |

**A0710064    RTC\_SPAR1    Spare register for specific purpose    0000**

| Bit   | 15               | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------|------------------|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| Mne   | <b>RTC_SPAR1</b> |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| Type  | RW               |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| Reset | 0                | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit (s) | Mnemonic         | Name      | Description                    |
|---------|------------------|-----------|--------------------------------|
| 15:0    | <b>RTC_SPAR1</b> | RTC_SPAR1 | Reserved for specific purposes |

**A0710068    RTC\_PROT    Lock/unlock scheme to prevent RTC miswriting    0000**

| Bit   | 15              | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------|-----------------|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| Mne   | <b>RTC_PROT</b> |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| Type  | RW              |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| Reset | 0               | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit (s) | Mnemonic        | Name     | Description                                                                                                                                                                                                                                                                                                                                                 |
|---------|-----------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:0    | <b>RTC_PROT</b> | RTC_PROT | <b>Protects RTC write interface by RTC_PROT</b><br>Whether the RTC writing interface is enabled or not is decided by RTC_PROT contents. When RTC_POWERKEY1 and RTC_POWERKEY2 are not equal to the correct values, the RTC writing interface will always be enabled. However, when they match, the user has to perform the unlock flow to enable the writing |

| Bit (s) | Mnemonic | Name | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------|----------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |          |      | interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|         |          |      | Unlock flow:<br>Step1: *RTC_PROT=0x586a;<br>Step2: *RTC_WRTGR=1;<br>Step3: while(*RTC_BBPU & 0x40) {}; // Timeout period: 120usec<br>Step4: *RTC_PROT=0x9136;<br>Step5: *RTC_WRTGR=1;<br>Step6: while(*RTC_BBPU & 0x40) {}; // Timeout period: 120usec<br><br><i>Note: Always keep RTC in the unlock state in the power-on mode. Once the normal RTC content writing is completed, DO NOT modify the RTC_PROT content to lock RTC. The RTC_PROT contents will be cleared automatically when being powered off immediately.</i> |

| A071006C | RTC_DIFF |    |    |    | One-time calibration offset |    |   |   |   |   |   |   |   |   |   |   | 0000 |  |
|----------|----------|----|----|----|-----------------------------|----|---|---|---|---|---|---|---|---|---|---|------|--|
| Bit      | 15       | 14 | 13 | 12 | 11                          | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |      |  |
| Mne      |          |    |    |    | RTC_DIFF                    |    |   |   |   |   |   |   |   |   |   |   |      |  |
| Type     |          |    |    |    | RW                          |    |   |   |   |   |   |   |   |   |   |   |      |  |
| Reset    |          |    |    |    | 0                           | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |      |  |

**Overview** This register is fixed to 0 when RTC\_POWERKEY1 and RTC\_POWERKEY2 unmatch the correct values.

| Bit (s) | Mnemonic | Name     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------|----------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11:0    | RTC_DIFF | RTC_DIFF | <b>Adjusts the internal counter of RTC. It takes effect once and returns to zero when done.</b><br><br>In some cases, RTC is faster or slower than the standard. To change RTC_TC_SEC from being coarse may cause alarm problems. RTC_DIFF provides a finer time unit. An internal 15-bit counter accumulates in each 32,768-Hz clock. Entering a non-zero value to RTC_DIFF will cause the internal RTC counter to increase or decrease RTC_DIFF when RTC_DIFF changes to zero again. RTC_DIFF represents 2's complement.<br><br>For example, if you fill 0xfff into RTC_DIFF, the internal counter will decrease by 1 when RTC_DIFF returns to 0. In other words, you can only use RTC_DIFF continuously if RTC_DIFF is equal to 0 now.<br><br><i>Note: RTC_DIFF ranges from 0x800 (-2048) to 0x7fd (2045). 0x7ff &amp; 0x7fe are forbidden.</i> |



**A0710070    RTC\_CALI    Repeat calibration offset    0000**

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6        | 5 | 4 | 3 | 2 | 1 | 0 |
|-------|----|----|----|----|----|----|---|---|---|----------|---|---|---|---|---|---|
| Mne   |    |    |    |    |    |    |   |   |   | RTC_CALI |   |   |   |   |   |   |
| Type  |    |    |    |    |    |    |   |   |   | RW       |   |   |   |   |   |   |
| Reset |    |    |    |    |    |    |   |   |   | 0        | 0 | 0 | 0 | 0 | 0 | 0 |

**Overview**    This register is fixed to 0 when RTC\_POWERKEY1 and RTC\_POWERKEY2 unmatch the correct values.

| Bit (s) | Mnemonic | Name     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------|----------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6:0     | RTC_CALI | RTC_CALI | <p><b>Provides a repeated calibration scheme</b></p> <p>RTC_CALI provides 7-bit calibration capability in 8-second duration, i.e. 5-bit calibration capability in each second. RTC_CALI represents 2's complement for the user to adjust RTC increase or decrease</p> <p>Due to RTC_CALI is revealed in 8 seconds, the resolution is less than a 1/32768 clock.<br/>Avg. resolution: 1/32768/8 = 3.81us<br/>Avg. adjust range: -0.244~0.240ms/sec in 2's complement: -0x40 ~ 0x3f (-64 ~ 63)</p> |

**A0710074    RTC\_WRTGR    Enable the transfers from core to RTC in the queue    0000**

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0     |
|-------|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|-------|
| Mne   |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | WRTGR |
| Type  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | WO    |
| Reset |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | 0     |

| Bit (s) | Mnemonic | Name  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------|----------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0       | WRTGR    | WRTGR | <p><b>Enables the transfers from core to RTC</b></p> <p>After you modify all the RTC registers and would like to change, write 1 to RTC_WRTGR to trigger the transfer. The prior writing operation is queued in the core power domain. The pending data will not be transferred to the RTC domain until WRTGR = 1.</p> <p>After WRTGR = 1, the pending data will be transferred to the RTC domain sequentially in order of register addresses, from low to high. For example: RTC_BBPU -&gt; RTC_IRQ_EN -&gt; RTC_CII_EN -&gt; RTC_AL_MASK -&gt; RTC_TC_SEC -&gt; etc. CBUSY in RTC_BBPU is equal to 1 in the writing process. Observe CBUSY to determine when the transmission is completed.</p> |

**A0710078    RTC\_CON    Other RTC control register    0000**

| Bit   | 15        | 14 | 13       | 12       | 11       | 10       | 9        | 8   | 7        | 6        | 5        | 4        | 3     | 2    | 1        | 0              |
|-------|-----------|----|----------|----------|----------|----------|----------|-----|----------|----------|----------|----------|-------|------|----------|----------------|
| Mne   | LPSTA_RAW |    | RESERVED | RESERVED | RESERVED | RESERVED | RESERVED | GSR | RESERVED | RESERVED | RESERVED | RESERVED | LPRST | LPEN | RESERVED | VBAT_LPSTA_RAW |
| Type  | RO        |    | RW       | RW       | RW       | RW       | RW       | RW  | RW       | RW       | RW       | RW       | RW    | RW   | RO       | RW             |
| Reset |           |    |          |          |          |          |          |     |          |          |          |          |       |      |          |                |

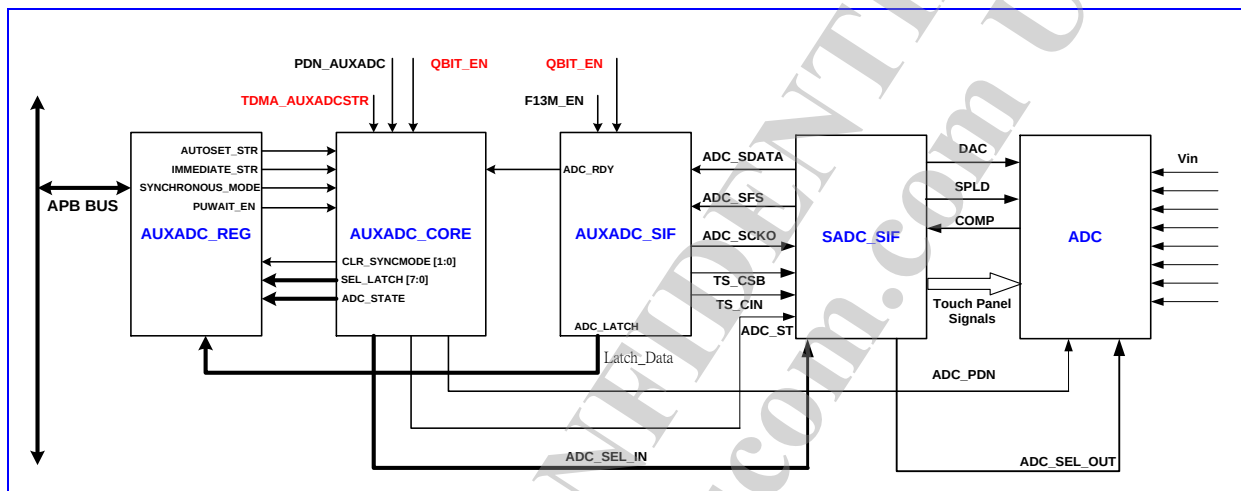
| Bit (s) | Mnemonic       | Name           | Description                                                                                                                                                                                                                                                               |
|---------|----------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15      | LPSTA_RAW      | LPSTA_RAW      | <b>Raw status of LP_STA</b><br>Re-initialize LPD to clear this bit. <i>Note that this bit is always high before LPD initialization sequence after the first power-on.</i>                                                                                                 |
| 8       | GSR            | GSR            | <b>SR of RTC_GPIO. Output slew rate control. High asserted. SR=0, slower slew. SR=1.</b><br>0: RTC_GPIO higher slew output.<br>1: RTC_GPIO slower slew output.                                                                                                            |
| 3       | LPRST          | LPRST          | <b>Resets LPDETB</b><br>This only takes effect when LPEN = 1.                                                                                                                                                                                                             |
| 2       | LPEN           | LPEN           | <b>Enables LPDETB</b><br>LP initialization sequence:<br>1. Write LPEN = 1, LPRST = 0. Write RTC_WRTGR = 1. wait cbusy down.<br>2. Write LPEN = 1, LPRST = 1. Write RTC_WRTGR = 1. wait cbusy down.<br>3. Write LPEN = 1, LPRST = 0. Write RTC_WRTGR = 1. wait cbusy down. |
| 0       | VBAT_LPSTA_RAW | VBAT_LPSTA_RAW | <b>Indicates the battery has been in LP state</b><br>The software needs to clear this bit for the next time use<br>0: VBAT has not been in LP state.<br>1: VBAT has been in LP state.<br>Note: VBAT LP state = VBAT < 2.5V                                                |

If Vcore always exists, the software can trust the registers and wait for the LP interrupt from RTC if the 32.768 kHz clock stops or has been stopped.

However, nothing can be trusted after the battery is off (Vrtc may drop). In every booting time, the software has to check if LPSTA\_RAW = 1. (LP\_STA = LPSTA\_RAW & LP\_IRQ\_EN) If true, then the RTC contents will no longer be trusted just like powerkeys do not match. You have to initialize the RTC contents in this case.

## 4.10 Auxiliary ADC Unit

The auxiliary ADC unit is used to monitor the status of the battery and charger, to identify the plugged peripheral, and to perform temperature measurement. There are 7 input channels allowing diverse applications in this unit.



**Figure 42. Auxadc Architecture**

Each channel operates in one of the two modes: immediate mode or timer-triggered mode. The mode of each channel can be individually selected through register AUXADC\_CON0. For example, if the flag SYN0 in register AUXADC\_CON0 is set, channel 0 will be set in the timer-triggered mode. Otherwise, the channel will operate in the immediate mode.

In the immediate mode, the A/D converter samples the value once only when the flag in the AUXADC\_CON1 register is set. For example, if the flag IMM0 in AUXADC\_CON1 is set, the A/D converter will sample the data for channel 0. The IMM flags must be cleared and set again to initialize another sampling.

The value sampled for channel 0 is stored in register AUXADC\_DAT0, and the value for channel 1 is stored in register AUXADC\_DAT1, and so on.

If the AUTOSSET flag in register AUXADC\_CON3 is set, the auto-sample function will be enabled. The A/D converter samples the data for the channel in which the corresponding data register is read. For example, in the case where the SYN1 flag is not set, the AUTOSSET flag is set. When the data register AUXADC\_DAT0 is read, the A/D converter will sample the next value for channel 1 immediately.

If multiple channels are selected at the same time, the task will be performed sequentially on every selected channel. For example, if AUXADC\_CON1 is set to 0x3f, i.e. 6 channels are selected, the state machine in the unit will start sampling from channel 6 to channel 0 and save the values of each input channel in respective registers. The same process also applies to the timer-triggered mode.

In the timer-triggered mode, the A/D converter samples the value for the channels in which the corresponding SYN flags are set when the TDMA timer counts to the value specified in register TDMA\_AUXEV1 placed in the TDMA timer. For example, if AUXADC\_CON0 is set to 0x3f, the 6 channels will be selected to be in the timer-triggered mode. The state machine will sample the 6 channels sequentially and save the values in registers from AUXADC\_DAT0 to AUXADC\_DAT5, as it does in the immediate mode.

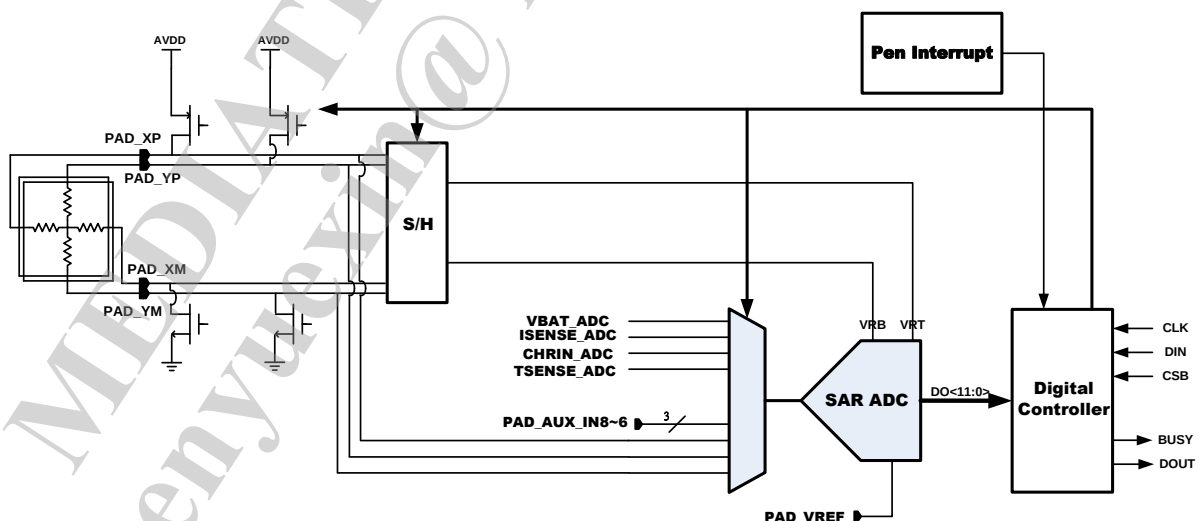
There is a dedicated timer-triggered scheme for channel 0. The scheme is enabled by setting up the SYN11 flag in register AUXADC\_CON2. The timing offset for this event is stored in register TDMA\_AUXEV0 in the TDMA timer. The sampled data triggered by this specific event are stored in register AUXADC\_DAT11. It is used to separate the results of two individual software routines that perform actions on the auxiliary ADC unit.

AUTOCLR<sub>n</sub> in register AUXADC\_CON3 is set when it is intended to sample only once after setting up the timer-triggered mode. If the AUTOCLR1 flag is set, after the data for the channels in the timer-triggered mode are stored, the SYN<sub>n</sub> flags in register AUXADC\_CON0 will be cleared. If the AUTOCLR0 flag is set, after the data for channel 0 are stored in register AUXADC\_DAT11, the SYN11 flag in register AUXADC\_CON2 will be cleared.

The uses of the immediate mode and timer-triggered mode are mutually exclusive in terms of individual channels.

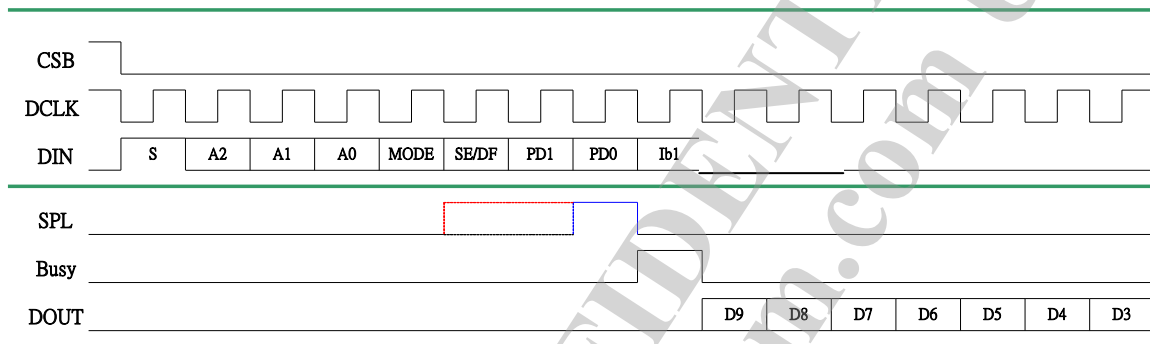
There are only two external pins (channel 4 ~ 5) for voltage detection. Other channels (0 ~ 3) are for battery voltage, battery current, charger and battery temperature respectively. Channel 9 is used for audio.

### Touch panel



**Figure 43. Touch Panel Circuit Structure**

Besides the normal sampling of external input voltage, the AUXADC includes the sampling of the touch panel function. For specified axis, the software should program AUX\_TS\_CMD first then trigger the sample of touch panel in register AUX\_TS\_CON. The touch panel sampling waveform is shown as the following. After the software polls the status bit in register AUXADC\_CON3 to know that the touch panel sample is finished, the software can read back the specified axis value from register AUX\_TS\_DAT0.



**Figure 44. Touch Panel Sampling Waveform**

S: Start bit  
A2 ~ A0: Addressing bits  
Mode: 10-bit or 8-bit  
SE/DF: Single end or differential mode  
PD1 ~ 0: Power down command

These values are defined in register AUX\_TS\_CMD. The table below shows the relationship between AUX\_TS\_CMD and touch panel control signals.

| A2 | A1 | A0 | SE/DFB | CHN_SEL | ADC In | X switches      | Y switches      | +REF | -REF |
|----|----|----|--------|---------|--------|-----------------|-----------------|------|------|
| 0  | 0  | 1  | 0      | 1100    | X+     | OFF             | ON              | Y+   | Y-   |
| 0  | 1  | 1  | 0      | 1100    | X+     | X+ OFF<br>X- ON | Y+ ON<br>Y- OFF | Y+   | X-   |
| 1  | 0  | 0  | 0      | 111 0   | Y-     | X+ OFF<br>X- ON | Y+ ON<br>Y- OFF | Y+   | X-   |
| 1  | 0  | 1  | 0      | 1101    | Y+     | ON              | OFF             | X+   | X-   |

**Table 63. Relationship between commands and touch panel control signals**

#### 4.10.1 Register Definition

| Register address | Register function                | Acronym     |
|------------------|----------------------------------|-------------|
| 0xA015_0000      | Auxiliary ADC control register 0 | AUXADC_CON0 |

| Register address | Register function                                       | Acronym      |
|------------------|---------------------------------------------------------|--------------|
| 0xA015_0004      | Auxiliary ADC control register 1                        | AUXADC_CON1  |
| 0xA015_0008      | Auxiliary ADC control register 2                        | AUXADC_CON2  |
| 0xA015_0010      | Auxiliary ADC channel 0 data register (VBAT)            | AUXADC_DAT0  |
| 0xA015_0014      | Auxiliary ADC channel 1 data register (ISENSE)          | AUXADC_DAT1  |
| 0xA015_0018      | Auxiliary ADC channel 2 data register (CHRIN)           | AUXADC_DAT2  |
| 0xA015_001C      | Auxiliary ADC channel 3 data register (VBATTMP)         | AUXADC_DAT3  |
| 0xA015_0020      | Auxiliary ADC channel 4 data register (External)        | AUXADC_DAT4  |
| 0xA015_0024      | Auxiliary ADC channel 5 data register (External/ACCDET) | AUXADC_DAT5  |
| 0xA015_0034      | Auxiliary ADC channel 9 data register (ClassAB)         | AUXADC_DAT9  |
| 0xA015_003C      | Auxiliary ADC channel 0 data register for TDMA event 0  | AUXADC_DAT11 |
| 0xA015_0050      | Touch screen de-bounce time                             | AUX_TS_DEBT  |
| 0xA015_0054      | Touch screen sample command                             | AUX_TS_CMD   |
| 0xA015_0058      | Touch screen control                                    | AUX_TS_CON   |

**Table 64. Auxadc Registers**

### 0xA015\_0000 Auxiliary ADC control register 0

**AUXADC\_CON  
0**

| Bit   | 15 | 14 | 13 | 12 | 11 | 10    | 9    | 8    | 7    | 6    | 5    | 4    | 3    | 2    | 1    | 0    |
|-------|----|----|----|----|----|-------|------|------|------|------|------|------|------|------|------|------|
| Name  |    |    |    |    |    | SYN10 | SYN9 | SYN8 | SYN7 | SYN6 | SYN5 | SYN4 | SYN3 | SYN2 | SYN1 | SYN0 |
| Type  |    |    |    |    |    | R/W   | R/W  | R/W  | R/W  | R/W  | R/W  | R/W  | R/W  | R/W  | R/W  | R/W  |
| Reset |    |    |    |    |    | 0     | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |

**SYN<sub>n</sub>** These 7 bits define whether the corresponding channel is sampled or not in the timer-triggered mode. It is associated with the timing offset register TDMA\_AUXEV1. It supports multiple flags. The flags can be automatically cleared after those channels are sampled if AUTOCLR1 in register AUXADC\_CON3 is set.

**0** The channel is not selected.

**1** The channel is selected.

### 0xA015\_0004 Auxiliary ADC control register 1

**AUXADC\_CON  
1**

| Bit   | 15 | 14 | 13 | 12 | 11 | 10    | 9    | 8    | 7    | 6    | 5    | 4    | 3    | 2    | 1    | 0    |
|-------|----|----|----|----|----|-------|------|------|------|------|------|------|------|------|------|------|
| Name  |    |    |    |    |    | IMM10 | IMM9 | IMM8 | IMM7 | IMM6 | IMM5 | IMM4 | IMM3 | IMM2 | IMM1 | IMM0 |
| Type  |    |    |    |    |    | R/W   | R/W  | R/W  | R/W  | R/W  | R/W  | R/W  | R/W  | R/W  | R/W  | R/W  |
| Reset |    |    |    |    |    | 0     | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |

**IMM<sub>n</sub>** These 7 bits are set individually to sample the data for the corresponding channel. It supports multiple flags.

- 0** The channel is not selected.
- 1** The channel is selected.

**0xA015\_0008 Auxiliary ADC control register 2**
**AUXADC\_CON**  
**2**

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0           |
|-------|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|-------------|
| Name  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | <b>SYN1</b> |
| Type  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | R/W         |
| Reset |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | 0           |

**SYN11** This bit is used only for channel 0 and is to be associated with timing offset register TDMA\_AUXEV0 in the TDMA timer in timer-triggered mode. The flag can be automatically cleared after channel 0 has been sampled if AUTOCLR0 in the register AUXADC\_CON3 is set.

- 0** The channel is not selected.
- 1** The channel is selected.

**0xA015\_000C Auxiliary ADC control register 3**
**AUXADC\_CON**  
**3**

| Bit   | 15              | 14 | 13 | 12 | 11         | 10 | 9                 | 8                 | 7                | 6 | 5 | 4 | 3 | 2 | 1 | 0          |
|-------|-----------------|----|----|----|------------|----|-------------------|-------------------|------------------|---|---|---|---|---|---|------------|
| Name  | <b>AUT OSET</b> |    |    |    | <b>RSV</b> |    | <b>AUT OCL R1</b> | <b>AUT OCL R0</b> | <b>SOFT _RST</b> |   |   |   |   |   |   | <b>STA</b> |
| Type  | R/W             |    |    |    | R/W        |    | R/W               | R/W               | R/W              |   |   |   |   |   |   | RO         |
| Reset | 0               |    |    |    | 0          |    | 0                 | 0                 | 0                |   |   |   |   |   |   | 0          |

**AUTOSET** Defines the auto-sample mode of the module. In auto-sample mode, each channel with its sample register being read can start sampling immediately without configuring the control register AUXADC\_CON1 again.

**PUWAIT\_EN** Enables the power warm-up period to ensure power stability before the SAR process takes place. It is recommended to activate this field.

- 0** The mode is not enabled.
- 1** The mode is enabled.

**SOFT\_RST** Software reset AUXADC state machine

- 0** Normal function
- 1** Reset AUXADC state machine

**AUTOCLR1** Defines the auto-clear mode of the module for event 1. In the auto-clear mode, each timer-triggered channel acquires samples of specified channels once the



SYN $n$  bit in register AUXADC\_CON0 is set. The SYN $n$  bits are automatically cleared and the channel is not enabled again by the timer event except when the SYN $n$  flags are set again.

- 0 The automatic clear mode is not enabled.
- 1 The automatic clear mode is enabled.

#### AUTOCLR0

Defines the auto-clear mode of the module for event 0. In the auto-clear mode, the timer-triggered channel 0 acquires the sample once the SYN11 bit in register AUXADC\_CON2 is set. The SYN11 bit is automatically cleared and the channel is not enabled again by the timer event 0 except when the SYN11 flag is set again.

- 0 The automatic clear mode is not enabled.
- 1 The automatic clear mode is enabled.

#### STA

Defines the state of the module

- 0 This module is idle.
- 1 This module is busy.

#### 0xA015\_0010 Auxiliary ADC channel 0 register

#### AUXADC\_DAT0

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0   |
|-------|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|-----|
| Name  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | DAT |
| Type  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | RO  |
| Reset |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | 0   |

The register stores the sampled data for channel 0. There are 8 registers of the same type for the corresponding channel. The overall register definition is listed in Table 65.

| Register address | Register function                                      | Acronym      |
|------------------|--------------------------------------------------------|--------------|
| 0xA015_0010      | Auxiliary ADC channel 0 data register                  | AUXADC_DAT0  |
| 0xA015_0014      | Auxiliary ADC channel 1 data register                  | AUXADC_DAT1  |
| 0xA015_0018      | Auxiliary ADC channel 2 data register                  | AUXADC_DAT2  |
| 0xA015_001C      | Auxiliary ADC channel 3 data register                  | AUXADC_DAT3  |
| 0xA015_0020      | Auxiliary ADC channel 4 data register                  | AUXADC_DAT4  |
| 0xA015_0024      | Auxiliary ADC channel 5 data register                  | AUXADC_DAT5  |
| 0xA015_0034      | Auxiliary ADC channel 9 data register                  | AUXADC_DAT9  |
| 0xA015_003C      | Auxiliary ADC channel 0 data register for TDMA event 0 | AUXADC_DAT11 |

**Table 65. Auxiliary ADC data register list**

### 0xA015\_0054 Touch screen sample command

### AUX\_TS\_CMD

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6       | 5   | 4   | 3        | 2         | 1   | 0   |
|-------|----|----|----|----|----|----|---|---|---|---------|-----|-----|----------|-----------|-----|-----|
| Name  |    |    |    |    |    |    |   |   |   | ADDRESS |     |     | MOD<br>E | SE/D<br>F | PD  |     |
| Type  |    |    |    |    |    |    |   |   |   | R/W     | R/W | R/W | R/W      | R/W       | R/W | R/W |
| Reset |    |    |    |    |    |    |   |   |   | 0       | 0   | 0   | 0        | 0         | 0   | 0   |

**ADDRESS** Defines which x or y or z data will be sampled

- 001** Y position
- 011** Z1 position
- 100** Z2 position
- 101** X position
- Others** Reserved

**MODE** Selects sample resolution

- 0** 10-bit
- 1** 8-bit

**SE/DF** Selects mode

- 0** Differential mode
- 1** Single-end mode

**PD** Power-down control for analog IRQ signal and touch screen sample control signal

- 00** Turn on Y\_drive signal and PDN\_sh\_ref
- 01** Turn on PDN\_IRQ and PDN\_sh\_ref
- 10** Reserved
- 11** Turn on PDN\_IRQ

### 0xA015\_0058 Touch screen control

### AUX\_TS\_CON

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1  | 0   |
|-------|----|----|----|----|----|----|---|---|---|---|---|---|---|---|----|-----|
| Name  |    |    |    |    |    |    |   |   |   |   |   |   |   |   | ST | SPL |
| Type  |    |    |    |    |    |    |   |   |   |   |   |   |   |   | R  | R/W |
| Reset |    |    |    |    |    |    |   |   |   |   |   |   |   |   | 0  | 0   |

**SPL** Touch screen sample trigger

- 0** No action
- 1** When software writes 1'b1, AUXADC will trigger the touch screen process. After the sampling process of touch screen is finished, this bit will be disserted.

### 0xA015\_005C Touch screen sample DATA

### AUX\_TS\_DAT0

|      |    |    |    |    |    |    |     |   |   |   |   |   |   |   |   |   |
|------|----|----|----|----|----|----|-----|---|---|---|---|---|---|---|---|---|
| Bit  | 15 | 14 | 13 | 12 | 11 | 10 | 9   | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| Name |    |    |    |    |    |    | DAT |   |   |   |   |   |   |   |   |   |

|       |  |  |  |  |  |  |  |  |    |
|-------|--|--|--|--|--|--|--|--|----|
| Type  |  |  |  |  |  |  |  |  | RO |
| Reset |  |  |  |  |  |  |  |  | 0  |

This register stores the touch screen sample data.

## 4.11 USB Device Controller

### 4.11.1 General Description

This chip provides a USB function interface which complies with Universal Serial Bus Specification Rev 1.1. The USB device controller supports only full-speed (12Mbps) operation. The chip can make use of this widely available USB interface to transmit/receive data with USB hosts, typically PC/laptop.

There are 6 endpoints in the USB device controller besides the mandatory control endpoint, 4 of which are for IN transactions and 2 are for OUT transactions. Word, half-word, and byte access are all allowed for loading and unloading the FIFO. The controller features 4 DMA channels to accelerate the data transfer for ACL and SCO data streams. The features of the endpoints are:

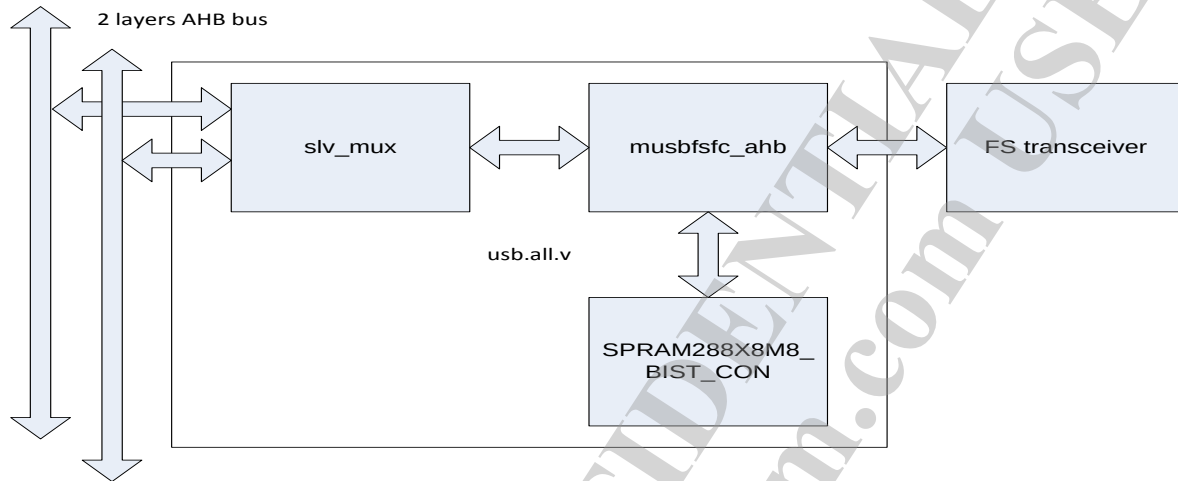
- Endpoint 0: The control endpoint feature 64 bytes FIFO and accommodates maximum packet size of up to 64 bytes. DMA transfer is not supported.
- IN endpoint 1: It features 64 bytes FIFO and accommodates maximum packet size of up to 64 bytes. GDMA Channel 4 Write Transfer is supported.
- IN endpoint 2: It features 64 bytes FIFO and accommodates maximum packet size of up to 64 bytes. GDMA Channel 6 Write Transfer is supported.
- IN endpoint 3: It features 16-byte FIFO and accommodates maximum packet size of 16 bytes. GDMA Transfer is not supported.
- IN endpoint 4: It features 16-byte FIFO and accommodates maximum packet size of 16 bytes. GDMA Transfer is not supported.
- OUT endpoint 1: It features 64 bytes FIFO and accommodates maximum packet size of 64 bytes. GDMA Channel 5 Read Transfer is supported.
- OUT endpoint 2: It features 64 bytes FIFO and accommodates maximum packet size of 64 bytes. GDMA Channel 7 Read Transfer is supported.

For each endpoint except for the control endpoint, when the packet size is smaller than half the size of the FIFO, at most 2 packets can be buffered.

This unit is highly software configurable. All endpoints except for the control endpoint can be configured to be a bulk, interrupt or isochronous endpoints. Composite devices are also supported. IN endpoint 1 and OUT endpoint 1 share the same endpoint number, but they can be used separately. So is the situation for the endpoint 2.

The USB device uses the cable-powered feature for the transceiver but only drains little current. An internal pull-up resistor is integrated across Vbus and D+ signal. The switch on/off of the pull-up resistor can be configured through the internal register. Two additional external serial resistors might

be needed to place on the output of D+ and D- signals to make the output impedance equivalent to 28 ~ 44Ohm.



**Figure 45. USB11 controller system diagram**

#### 4.11.2 Register Definition

| USB+0000h |     | USB function address register |       |   |   |   |   | USB_FADDR |  |
|-----------|-----|-------------------------------|-------|---|---|---|---|-----------|--|
| Bit       | 7   | 6                             | 5     | 4 | 3 | 2 | 1 | 0         |  |
| Name      | UPD |                               | FADDR |   |   |   |   |           |  |
| Type      | RO  |                               | R/W   |   |   |   |   |           |  |
| Reset     | 0   |                               | 0     |   |   |   |   |           |  |

This is an 8-bit register that should be written with the function's 7-bit address (received through a SET\_ADDRESS description). It is then used for decoding the function address in subsequent token packets.

**UPD** Set when FADDR is written. Cleared when the new address takes effect (at the end of the current transfer)

**FADDR** Function address of the device

| USB+0001h |        | USB power control register |           |   |       |        |          | USB POWER |  |
|-----------|--------|----------------------------|-----------|---|-------|--------|----------|-----------|--|
| Bit       | 7      | 6                          | 5         | 4 | 3     | 2      | 1        | 0         |  |
| Name      | ISO_UP |                            | SWRSTENAB |   | Reset | RESUME | SUSPMODE | SUSPENAB  |  |
| Type      | R/W    |                            | R/W       |   | RO    | R/W    | RO       | R/W       |  |
| Reset     | 0      |                            | 0         |   | 0     | 0      | 0        | 0         |  |

**ISO\_UP** When set by the MCU, the core will wait for an SOF token from the time INPKTRDY is set before sending the packet.

- SWRSTENAB** Set by the MCU to enable the mode in which the device can only be reset by the software after reset signals are detected on the bus. In case the software is delayed by other high-priority processes and cannot make it to read the command from the buffer before the hardware resets the device after the reset signal is detected on the bus, the command will be lost. That is why the software-reset mode is effective. When the flag is enabled, the hardware state machine cannot reset itself but by the software. In that sense, the software and hardware can keep synchronous detecting the reset signal.
- RESET** The read-only bit is set when Reset signaling is present on the bus.
- RESUME** Set by the MCU to generate Resume signaling when the function is in the suspend mode. The MCU should clear this bit after 10ms (maximum 15ms) to end Resume signaling.
- SUSPMODE** Set by the USB core when the Suspend mode is entered. Cleared when the Resume bit of this register is set.
- SUSPENAB** Set by the MCU to enable device into the Suspend mode when Suspend signaling is received on the bus.

**USB+0002h      USB IN endpoints interrupt register      USB\_INTRIN**

| Bit   | 7 | 6 | 5 | 4          | 3          | 2          | 1          | 0          |
|-------|---|---|---|------------|------------|------------|------------|------------|
| Name  |   |   |   | <b>EP4</b> | <b>EP3</b> | <b>EP2</b> | <b>EP1</b> | <b>EP0</b> |
| Type  |   |   |   | RC         | RC         | RC         | RC         | RC         |
| Reset |   |   |   | 0          | 0          | 0          | 0          | 0          |

This is a read-only register indicating which interrupts for IN endpoints 0 to 4 are currently active. It must be set after USB\_INTRINE is set. All active interrupts will be cleared when this register is read.

- EP4** IN endpoint #4 interrupt  
**EP3** IN endpoint #3 interrupt  
**EP2** IN endpoint #2 interrupt  
**EP1** IN endpoint #1 interrupt  
**EP0** IN endpoint #0 interrupt

**USB+0004h      USB OUT endpoints interrupt register      USB\_INTROUT**

| Bit   | 7 | 6 | 5 | 4 | 3 | 2          | 1          | 0 |
|-------|---|---|---|---|---|------------|------------|---|
| Name  |   |   |   |   |   | <b>EP2</b> | <b>EP1</b> |   |
| Type  |   |   |   |   |   | RC         | RC         |   |
| Reset |   |   |   |   |   | 0          | 0          |   |

This is a read-only register indicating which interrupts for OUT endpoints 1, and 2 are currently active. It must be set after USB\_INTRROUTE is set. All active interrupts will be cleared when this register is read.

**EP2**    OUT endpoint #2 interrupt  
**EP1**    OUT endpoint #1 interrupt

**USB+0006h      USB general interrupt register      USB\_INTRUSB**

| Bit   | 7 | 6 | 5 | 4               | 3          | 2            | 1             | 0           |
|-------|---|---|---|-----------------|------------|--------------|---------------|-------------|
| Name  |   |   |   | <b>POWERDWN</b> | <b>SOF</b> | <b>RESET</b> | <b>RESUME</b> | <b>SUSP</b> |
| Type  |   |   |   | RO              | RC         | RC           | RC            | RC          |
| Reset |   |   |   | 0               | 0          | 0            | 0             | 0           |

This is a read-only register indicating which USB interrupts are currently active. It must be set after USB\_INTRUSBE is set. All active interrupts will be cleared when this register is read.

**POWERDWN**    Set at SUSPMODE and LineState is JState. The programmer should have de-bounce scheme in SW code when using this interrupt.

**SOF**            Set at the start of each frame

**RESET**        Set when Reset signaling is detected on the bus

**RESUME**      Set when Resume signaling is detected on the bus while the USB core is in suspend mode

**SUSP**         Set when Suspend signaling is detected on the bus

**USB+0007h      USB IN endpoints interrupt enable register      USB\_INTRINE**

| Bit   | 7 | 6 | 5 | 4          | 3          | 2          | 1          | 0          |
|-------|---|---|---|------------|------------|------------|------------|------------|
| Name  |   |   |   | <b>EP4</b> | <b>EP3</b> | <b>EP2</b> | <b>EP1</b> | <b>EP0</b> |
| Type  |   |   |   | R/W        | R/W        | R/W        | R/W        | R/W        |
| Reset |   |   |   | 1          | 1          | 1          | 1          | 1          |

This register provides interrupt enable bits for the interrupts in USB\_INTRIN. On reset, the bits corresponding to endpoint 0 and all IN endpoints are set to 1.

**EP4**    Enables IN endpoint 4 interrupt

**EP3**    Enables IN endpoint 3 interrupt

**EP2**    Enables IN endpoint 2 interrupt

**EP1**    Enables IN endpoint 1 interrupt

**EP0**    Enables IN endpoint 0 interrupt

**USB+0009h      USB OUT endpoints interrupt enable register      USB\_INTRROUTE**

| Bit   | 7 | 6 | 5 | 4 | 3 | 2          | 1          | 0 |
|-------|---|---|---|---|---|------------|------------|---|
| Name  |   |   |   |   |   | <b>EP2</b> | <b>EP1</b> |   |
| Type  |   |   |   |   |   | R/W        | R/W        |   |
| Reset |   |   |   |   |   | 1          | 1          |   |

This register provides interrupt enable bits for the interrupts in USB\_INTROUT. On reset, the bits corresponding to all OUT endpoints are set to 1.

**EP2** Enables OUT endpoint 2 interrupt

**EP1** Enables OUT endpoint 1 interrupt

**USB+000Bh USB general interrupt enable register USB\_INTRUSBE**

| Bit   | 7 | 6 | 5 | 4               | 3          | 2            | 1             | 0           |
|-------|---|---|---|-----------------|------------|--------------|---------------|-------------|
| Name  |   |   |   | <b>POWERDWN</b> | <b>SOF</b> | <b>RESET</b> | <b>RESUME</b> | <b>SUSP</b> |
| Type  |   |   |   | R/W             | R/W        | R/W          | R/W           | R/W         |
| Reset |   |   |   | 0               | 0          | 1            | 1             | 0           |

This register provides interrupt enable bits for each of the interrupts for USB\_INTRUSB.

**POWERDWN** Enables PowerDown interrupt

**SOF** Enables SOF interrupt

**RESET** Enables Reset interrupt

**RESUME** Enables Resume interrupt

**SUSP** Enables Suspend interrupt

**USB+000Ch USB frame count #1 register usb\_frame1**

| Bit   | 7           | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------|-------------|---|---|---|---|---|---|---|
| Name  | <b>NUMI</b> |   |   |   |   |   |   |   |
| Type  | RO          |   |   |   |   |   |   |   |
| Reset | 0           |   |   |   |   |   |   |   |

The register holds the lower 8 bits of the last received frame number.

**NUML** The lower 8 bits of the frame number

**USB+000Dh USB frame count #2 register usb\_frame2**

| Bit   | 7           | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------|-------------|---|---|---|---|---|---|---|
| Name  | <b>NUMH</b> |   |   |   |   |   |   |   |
| Type  | RO          |   |   |   |   |   |   |   |
| Reset | 0           |   |   |   |   |   |   |   |

The register holds the upper 3 bits of the last received frame number.



**NUMH**     The upper 3 bits of the frame number

**USB+000Eh     USB endpoint register index     USB\_INDEX**

| Bit   | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0            |
|-------|---|---|---|---|---|---|---|--------------|
| Name  |   |   |   |   |   |   |   | <b>INDEX</b> |
| Type  |   |   |   |   |   |   |   | R/W          |
| Reset |   |   |   |   |   |   |   | 0            |

The register determines which endpoint control/status registers are to be accessed at addresses USB+10h to USB+17h. Each IN endpoint and OUT endpoint has its own set of control/status registers. Only one set of IN control/status and one set of OUT control/status registers appear in the memory map at a time. Before accessing the control/status registers of an endpoint, the endpoint number should be written to the USB\_INDEX register to ensure that the correct control/status registers appear in the memory map.

**INDEX**     Index of the endpoint.

**USB+000Fh     USB reset control     usb\_rstctrl**

| Bit   | 7            | 6 | 5 | 4 | 3 | 2 | 1 | 0              |
|-------|--------------|---|---|---|---|---|---|----------------|
| Name  | <b>SWRST</b> |   |   |   |   |   |   | <b>RSTCNTR</b> |
| Type  | R/W          |   |   |   |   |   |   | R/W            |
| Reset | 0            |   |   |   |   |   |   | 0              |

The register is used to control the reset process when the device detects the reset command issued from the host.

**SWRST**     If the flag SWRSTENAB in register USB\_POWER is set to 1, the software enable mode will be enabled, and the device can be reset by writing 1 to this flag.

**RSTCNTR**     Signifies the duration of the reset operation after reset signal is detected on the bus. It is only enabled when software reset is not enabled. If the value is 0, the duration will be 2.5us. Otherwise, the duration will be this value multiplied by 341 then added by 2.5 (unit: us). The range consequently starts from 2.5us to 5122.5us.

**USB+0011h     USB control/status register for endpoint 0     USB\_EP0\_CSR**

| Bit   | 7                     | 6                      | 5                     | 4               | 3              | 2                     | 1               | 0                     |
|-------|-----------------------|------------------------|-----------------------|-----------------|----------------|-----------------------|-----------------|-----------------------|
| Name  | <b>SSETUPEN<br/>D</b> | <b>SOUTPKTR<br/>DY</b> | <b>SENDSTAL<br/>L</b> | <b>SETUPEND</b> | <b>DATAEND</b> | <b>SENTSTAL<br/>L</b> | <b>INPKTRDY</b> | <b>OUTPKTRD<br/>Y</b> |
| Type  | R/WS                  | R/WS                   | R/WS                  | RO              | R/WS           | R/WC                  | R/WS            | RO                    |
| Reset | 0                     | 0                      | 0                     | 0               | 0              | 0                     | 0               | 0                     |

The register is used for all control/status of endpoint 0. The register is active when the USB\_INDEX register is set to 0.

- SSETUPEND**      The MCU writes 1 to this bit to clear the SETUPEND bit. It is cleared automatically. Only active when a transaction is started.
- SOUTPKTRDY**    The MCU writes 1 to this bit to clear the OUTPKTRDY bit. It is cleared automatically. Only active when an OUT transaction is started.
- SENDSTALL**      The MCU writes 1 to this bit to terminate the current transaction. The STALL handshake will be transmitted, and this bit will be cleared automatically.
- SETUPEND**        This bit will be set when a control transaction ends before the DATAEND bit is set. An interrupt will be generated and FIFO flushed at this time. The bit is cleared by the MCU writing 1 to the SSETUPEND bit.
- DATAEND**        The MCU sets up this bit:
1. When setting INPKTRDY for the last data packet
  2. When clearing OUTPKTRDY after unloading the last data packet
  3. When setting INPKTRDY for a zero length data packet
- It is cleared automatically.
- SENTSTALL**      This bit is set when a STALL handshake is transmitted. The MCU should clear this bit by writing 0 to it.
- INPKTRDY**        The MCU sets up this bit after loading a data packet into the FIFO. It is cleared automatically when the data packet is transmitted. An interrupt is generated when this bit is set.
- OUTPKTRDY**      This bit is set when a data packet is received. An interrupt is generated when this bit is set. The MCU clears this bit by setting up the SOUTPKTRDY bit.

| USB+0016h      USB byte count register |       | USB_ep0_COUNT |   |   |   |   |   |   |
|----------------------------------------|-------|---------------|---|---|---|---|---|---|
| Bit                                    | 7     | 6             | 5 | 4 | 3 | 2 | 1 | 0 |
| Name                                   | count |               |   |   |   |   |   |   |
| Type                                   | RO    |               |   |   |   |   |   |   |
| Reset                                  | 0     |               |   |   |   |   |   |   |

The register indicates the number of received data bytes in the endpoint 0. The value returned is valid while the OUTPKTRDY bit of the USB\_EP0\_CSR register is set. The register is active when the USB\_INDEX register is set to 0.

**COUNT**      Number of received data bytes in the endpoint 0

**USB+0010h      USB maximum packet size register for IN endpoint 1~4      USB\_EP\_INMAXP**

| Bit   | 7    | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------|------|---|---|---|---|---|---|---|
| Name  | MAXP |   |   |   |   |   |   |   |
| Type  | R/W  |   |   |   |   |   |   |   |
| Reset | 0    |   |   |   |   |   |   |   |

The register holds the maximum packet size for transactions through the currently selected IN endpoint – in units of one byte. When setting up the value, the programmer should note the constraints placed by the USB Specification on the packet size for bulk interrupt and isochronous transactions in full-speed operation. There is an INMAXP register for each IN endpoint except for endpoint 0. The registers are active when the USB\_INDEX register is set to 1, 2, 3, and 4 respectively.

The value written to this register should match the **wMaxPacketSize** field of the standard endpoint descriptor for the associated endpoint. A mismatch may cause unexpected results. If a value is bigger than the configured IN FIFO size for the endpoint written to the register, the value will be automatically changed to the IN FIFO size. If the value written to the register is smaller than or equal to half the IN FIFO size, two IN packets can be buffered. The configured IN FIFO size for the endpoint 1, 2, and 3 are 16 bytes, 64 bytes and 64 bytes respectively.

The register is reset to 0. If the register is changed after the packets are sent from the endpoint, the endpoint IN FIFO should be completely flushed after writing the new value to the register.

**MAXP**      Maximum packet size (unit: one byte)

**USB+0011h      USB control/status register #1 for IN endpoint 1~4      USB\_EP\_INCSR1**

| Bit   | 7           | 6           | 5          | 4          | 3         | 2        | 1             | 0        |
|-------|-------------|-------------|------------|------------|-----------|----------|---------------|----------|
| Name  | ABORTPKT_EN | CLRDATA TOG | SENTSTAL L | SENDSTAL L | FLUSHFIFO | UNDERRUN | FIFONOTE MPTY | INPKTRDY |
| Type  | R/W         | WO          | R/WC       | R/W        | WO        | R/WC     | RO            | R/WS     |
| Reset | 0           | 0           | 0          | 0          | 0         | 0        | 0             | 0        |

The register provides control and status bits for IN transactions through the currently selected endpoint. There is an INCSR1 register for each IN endpoint except endpoint 0. The registers are active when USB\_INDEX register is set to 1, 2, and 3, respectively.

**ABORTPKT\_EN**      When MCU writes 1 to ABORTPKT\_EN, FLUSHFIFO switches to abort the packet function. This bit should be enabled before FLUSHFIFO is set. If FLUSHFIFO is set and ABORTPKT\_EN is enabled, the data loaded into FIFO will be discarded. After the packet is aborted, EP will issue an interrupt. The programmer should wait for this interrupt to make sure the packet is aborted.

**CLRDATA TOG**      The MCU writes 1 to this bit to reset the endpoint IN data toggle to 0.

- SENTSTALL** The bit is set when a STALL handshake is transmitted. The FIFO is flushed and the INPKTRDY bit is cleared. The MCU should clear this bit by writing 0 to this bit.
- SENDSTALL** The MCU writes 1 to this bit to issue a STALL handshake to an IN token. The MCU clears this bit to terminate the stall condition.
- FLUSHFIFO** The MCU writes 1 to this bit to flush the next packet to be transmitted from the endpoint IN FIFO. The FIFO pointer is reset and the INPKTRDY bit is cleared. If the FIFO contains two packets, FLUSHFIFO will need to be set twice to completely clear the FIFO. FLUSHFIFO should only be used when INPKTRDY is set. At other times, it may cause data corruption.  
 If ABORTPKT\_EN is enabled and this bit is set, the function of this bit will become ABORTPKT to abort the next packet to be transmitted from the endpoint IN FIFO and does not need to set INPKTRDY. It is the same with the FLUSHFIFO function. This bit is only active when the endpoint is idle.
- UNDERRUN** In isochronous mode, this bit is set when a zero length data packet is sent after receiving an IN token with the INPKTRDY bit not set. In Bulk/Interrupt mode, this bit is set when a NAK is returned in response to an IN token. The MCU should clear this bit by writing a 0 to this bit.
- FIFONOTEMPTY** This bit is set when there is at least 1 packet in the IN FIFO.
- INPKTRDY** The MCU sets up this bit after loading a data packet into the FIFO. Only active when an IN transaction is started. It is cleared automatically when a data packet is transmitted. An interrupt will be generated (if enabled) when the bit is cleared.

| USB+0012h |                | USB control/status register #2 for IN endpoint 1~4 |             |                |                   |   | USB_EP_INCSR2 |   |
|-----------|----------------|----------------------------------------------------|-------------|----------------|-------------------|---|---------------|---|
| Bit       | 7              | 6                                                  | 5           | 4              | 3                 | 2 | 1             | 0 |
| Name      | <b>AUTOSET</b> | <b>ISO</b>                                         | <b>MODE</b> | <b>DMAENAB</b> | <b>RFCDATATOG</b> |   |               |   |
| Type      | R/W            | R/W                                                | R/W         | R/W            | R/W               |   |               |   |
| Reset     | 0              | 0                                                  | 0           | 0              | 0                 |   |               |   |

The register provides further control bits for IN transactions through the currently selected endpoint. There is an INCSR2 register for each IN endpoint except for endpoint 0. The registers are active when the USB\_INDEX register is set to 1, 2, and 3 respectively.

- AUTOSET** If the MCU sets up the bit, INPKTRDY will be automatically set when the data of the maximum packet size (value in INMAXP) is loaded into the IN FIFO. If a packet of smaller than the maximum packet size is loaded, INPKTRDY will have to be set manually. When 2 packets are in the IN FIFO, INPKTRDY will also be automatically set when the first packet is sent if the second packet is the maximum packet size.

- ISO** The MCU sets up this bit to enable the IN endpoint for isochronous transfer and clears it to enable the IN endpoint for bulk/interrupt transfers.
- MODE** The MCU sets up this bit to enable the endpoint direction as IN and clears it to enable the endpoint direction as OUT. It is valid only when the same endpoint FIFO is used for both IN and OUT transactions.
- DMAENAB** The MCU sets up this bit to enable the DMA request for the IN endpoint.
- FRCDATATOG** The MCU sets up this bit to force the endpoint's IN data toggle to switch after each data packet is sent regardless of whether an ACK has been received. This can be used by interrupt IN endpoints which are used to communicate rate feedback for isochronous endpoints.

**USB+0013h USB maximum packet size register for OUT endpoint 1~2 USB\_EP\_OUTMAXP**

| Bit   | 7           | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------|-------------|---|---|---|---|---|---|---|
| Name  | <b>MAXP</b> |   |   |   |   |   |   |   |
| Type  | R/W         |   |   |   |   |   |   |   |
| Reset | 0           |   |   |   |   |   |   |   |

This register holds the maximum packet size for transactions through the currently selected OUT endpoint (unit: one byte). When setting up this value, the programmer should note the constraints placed by the USB specification on packet sizes for bulk, interrupt, and isochronous transactions in full speed operation. There is an OUTMAXP register for each OUT endpoint except for endpoint 0. The registers are active when the USB\_INDEX register is set to 1 and 2 respectively.

The value written to this register should match the **wMaxPacketSize** field of the standard endpoint descriptor for the associated endpoint. A mismatch may cause unexpected results. The total amount of data represented by the value written to this register must not exceed the FIFO size for the OUT endpoint and should not exceed half the FIFO size if double buffering is required. If a value bigger than the configured OUT FIFO size for the endpoint is written to the register, the value will be automatically changed to the OUT FIFO size. If the value written to the register is smaller than or equal to half the OUT FIFO size, two OUT packets can be buffered. The configured IN FIFO size for the endpoint 1, 2, and 3 are both 16, 64, and 64 bytes respectively.

**MAXP** Maximum packet size (unit: one byte)

**USB+0014h USB control/status register #1 for OUT endpoint 1~2 USB\_EP\_OUTCSR 1**

| Bit  | 7                   | 6                 | 5                 | 4                 | 3                 | 2               | 1               | 0                 |
|------|---------------------|-------------------|-------------------|-------------------|-------------------|-----------------|-----------------|-------------------|
| Name | <b>CLRDATA TO G</b> | <b>SENTSTAL L</b> | <b>SENDSTAL L</b> | <b>FLUSHFIF O</b> | <b>DATAERRO R</b> | <b>OVERRU N</b> | <b>FIFOFULL</b> | <b>OUTPKTRD Y</b> |
| Type | WO                  | R/WC              | R/W               | WO                | RO                | R/WC            | RO              | R/WC              |

|       |   |   |   |   |   |   |   |   |
|-------|---|---|---|---|---|---|---|---|
| Reset | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
|-------|---|---|---|---|---|---|---|---|

The register provides control status bits for OUT transactions through the currently selected endpoint. The registers are active when the USB\_INDEX register is set to 1 and 2 respectively.

- CLRDATATOG**     The MCU writes 1 to this bit to reset the endpoint data toggle to 0.
- SENTSTALL**     The bit is set when a STALL handshake is transmitted. The MCU should clear this bit by writing 0 to it.
- SENDSTALL**     The MCU writes 1 to this bit to issue a STALL handshake. The MCU clears this bit to terminate the stall condition. This bit will have no effect if the OUT endpoint is in the isochronous mode.
- FLUSHFIFO**     The MCU writes 1 to this bit to flush the next packet to be read from the endpoint OUT FIFO. If the FIFO contains two packets, FLUSHFIFO will need to be set twice to completely clear the FIFO. FLUSHFIFO should only be used when OUTPKTRDY is set. At other times, it may cause data corruption.
- DATAERROR**     The bit is set when OUTPKTRDY is set if the data packet has a CRC or bit-stuff error. It is cleared when OUTPKTRDY is cleared. This bit is only valid in the isochronous mode.
- OVERRUN**     The bit will be set if an OUT packet cannot be loaded into the OUT FIFO. The MCU should clear the bit by writing 0 to it. This bit is only valid in the isochronous mode.
- FIFOFULL**     This bit is set when no more packets can be loaded into the OUT FIFO.
- OUTPKTRDY**     The bit is set when a data packet has been received. The MCU should clear (write 0 to it) the bit when the packet is unloaded from the OUT FIFO. An interrupt will be generated when the bit is set. When the null packet is received, OUTPKTRDY will be set after USB\_INTROUT1 is high.

**USB+0015h     USB control/status register #2 for OUT endpoint 1~2     USB\_EP\_OUTCSR2**

| Bit   | 7                | 6          | 5              | 4              | 3 | 2 | 1 | 0 |
|-------|------------------|------------|----------------|----------------|---|---|---|---|
| Name  | <b>AUTOCLEAR</b> | <b>ISO</b> | <b>DMAENAB</b> | <b>DMAMODE</b> |   |   |   |   |
| Type  | R/W              | R/W        | R/W            | R/W            |   |   |   |   |
| Reset | 0                | 0          | 0              | 0              |   |   |   |   |

The register provides further control bits for OUT transactions through the currently selected endpoint. The registers are active when the USB\_INDEX register is set to 1 and 2 respectively.

- AUTOCLEAR**     If the MCU sets up this bit, the OUTPKTRDY bit will be automatically cleared when a packet of OUTMAXP bytes is unloaded from the OUT FIFO. When packets of



smaller than the maximum packet size are unloaded, OUTPKTRDY will have to be cleared manually.

**ISO** The MCU sets up this bit to enable the OUT endpoint for isochronous transfers and clears it to enable the OUT endpoint for bulk/interrupt transfers.

**DMAENAB** The MCU sets up this bit to enable the DMA request for the OUT endpoint.

**DMA MODE** Two modes of DMA operation are supported: DMA mode 0 in which a DMA request is generated for all received packets, together with an interrupt (if enabled). DMA mode 1 in which a DMA request (but no interrupt) is generated for OUT packets of size OUTMAXP bytes and an interrupt (but no DMA request) is generated for OUT packets of any other size. The MCU sets up the bit to select DMA mode 1 and clears this bit to select DMA mode 0.

**USB+0016h**      **USB OUT endpoint byte counter register LSB part for endpoint 1~2**      **USB\_ep\_COUNT1**

| Bit   | 7    | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------|------|---|---|---|---|---|---|---|
| Name  | NUML |   |   |   |   |   |   |   |
| Type  | RO   |   |   |   |   |   |   |   |
| Reset | 0    |   |   |   |   |   |   |   |

The register holds the lower 8 bits of the number of received data bytes in the packet in the FIFO associated with the currently selected OUT endpoint. The value returned is valid while OUTPKTRDY in register USB\_OUTCSR1 is set. The registers are active when the USB\_INDEX register is set to 1 and 2 respectively.

**NUML** The lower 8 bits of the number of received data bytes for the OUT endpoint.

**USB+0017h**      **USB OUT endpoint byte counter register MSB part for endpoint 1~2**      **USB\_ep\_COUNT2**

| Bit   | 7    | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------|------|---|---|---|---|---|---|---|
| Name  | NUMH |   |   |   |   |   |   |   |
| Type  | RO   |   |   |   |   |   |   |   |
| Reset | 0    |   |   |   |   |   |   |   |

The register holds the upper 3 bits of the number of received data bytes in the packet in the FIFO associated with the currently selected OUT endpoint. The value returned is valid while OUTPKTRDY in register USB\_EP\_OUTCSR1 is set. The registers are active when the USB\_INDEX register is set to 1 and 2 respectively.

**NUMH** The upper 8 bits of the number of received data bytes for the OUT endpoint.



**USB+0020h      USB endpoint 0 FIFO access register      USB\_EP0\_FIFO**

| Bit  | 31         | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23         | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|------|------------|----|----|----|----|----|----|----|------------|----|----|----|----|----|----|----|
| Name | <b>DB3</b> |    |    |    |    |    |    |    | <b>DB2</b> |    |    |    |    |    |    |    |
| Type | R/W        |    |    |    |    |    |    |    | R/W        |    |    |    |    |    |    |    |
| Bit  | 15         | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7          | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Name | <b>DB1</b> |    |    |    |    |    |    |    | <b>DB0</b> |    |    |    |    |    |    |    |
| Type | R/W        |    |    |    |    |    |    |    | R/W        |    |    |    |    |    |    |    |

The register provides MCU access to the FIFO for the endpoint 0. Writing to this register will load data to the FIFO for the endpoint 0. Reading this register will unload data from the FIFO for the endpoint 0.

The register provides word, half-word, and byte mode accesses. If word or half-word accesses are performed, the less significant byte will correspond to the prior byte to load to or unload from the FIFO.

- DB0**      The first byte to be loaded to or unloaded from the FIFO
- DB1**      The second byte to be loaded to or unloaded from the FIFO
- DB2**      The third byte to be loaded to or unloaded from the FIFO
- DB3**      The forth byte to be loaded to or unloaded from the FIFO

**USB+0024h      USB endpoint 1 FIFO access register      USB\_EP1\_FIFO**

| Bit  | 31         | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23         | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|------|------------|----|----|----|----|----|----|----|------------|----|----|----|----|----|----|----|
| Name | <b>DB3</b> |    |    |    |    |    |    |    | <b>DB2</b> |    |    |    |    |    |    |    |
| Type | R/W        |    |    |    |    |    |    |    | R/W        |    |    |    |    |    |    |    |
| Bit  | 15         | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7          | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Name | <b>DB1</b> |    |    |    |    |    |    |    | <b>DB0</b> |    |    |    |    |    |    |    |
| Type | R/W        |    |    |    |    |    |    |    | R/W        |    |    |    |    |    |    |    |

The register provides MCU access to the IN FIFO and the OUT FIFO for the endpoint 1. Writing to the register will load data to the IN FIFO for the endpoint 1. Reading the register will unload data from the OUT FIFO for the endpoint 1.

The register provides word, half-word, and byte mode accesses. If word or half-word accesses are performed, the less significant byte will correspond to the prior byte to load to the IN FIFO or unload from the OUT FIFO.

- DB0**      The first byte to be loaded to the IN FIFO or unloaded from the OUT FIFO
- DB1**      The second byte to be loaded to the IN FIFO or unloaded from the OUT FIFO
- DB2**      The third byte to be loaded to the IN FIFO or unloaded from the OUT FIFO
- DB3**      The forth byte to be loaded to the IN FIFO or unloaded from the OUT FIFO

**USB+0028h      USB endpoint 2 FIFO access register      USB\_EP2\_FIFO**

| Bit  | 31  | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23  | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|------|-----|----|----|----|----|----|----|----|-----|----|----|----|----|----|----|----|
| Name | DB3 |    |    |    |    |    |    |    | DB2 |    |    |    |    |    |    |    |
| Type | R/W |    |    |    |    |    |    |    | R/W |    |    |    |    |    |    |    |
| Bit  | 15  | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7   | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Name | DB1 |    |    |    |    |    |    |    | DB0 |    |    |    |    |    |    |    |
| Type | R/W |    |    |    |    |    |    |    | R/W |    |    |    |    |    |    |    |

The register provides MCU access to the IN FIFO and the OUT FIFO for the endpoint 2. Writing to the register will load data to the IN FIFO for the endpoint 2. Reading the register will unload data from the OUT FIFO for the endpoint 2.

The register provides word, half-word, and byte mode accesses. If word or half-word accesses are performed, the less significant byte will correspond to the prior byte to load to the IN FIFO or unload from the OUT FIFO.

- DB0**     The first byte to be loaded to the IN FIFO or unloaded from the OUT FIFO
- DB1**     The second byte to be loaded to the IN FIFO or unloaded from the OUT FIFO
- DB2**     The third byte to be loaded to the IN FIFO or unloaded from the OUT FIFO
- DB3**     The forth byte to be loaded to the IN FIFO or unloaded from the OUT FIFO

| USB+002Ch |     |    |    |    |    |    |    |    | USB endpoint 3 FIFO access register |     |    |    |    |    |    |    |  | USB_EP3_FIFO |  |  |  |  |  |  |  |
|-----------|-----|----|----|----|----|----|----|----|-------------------------------------|-----|----|----|----|----|----|----|--|--------------|--|--|--|--|--|--|--|
| Bit       | 31  | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23                                  | 22  | 21 | 20 | 19 | 18 | 17 | 16 |  |              |  |  |  |  |  |  |  |
| Name      | DB3 |    |    |    |    |    |    |    |                                     | DB2 |    |    |    |    |    |    |  |              |  |  |  |  |  |  |  |
| Type      | R/W |    |    |    |    |    |    |    |                                     | R/W |    |    |    |    |    |    |  |              |  |  |  |  |  |  |  |
| Bit       | 15  | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7                                   | 6   | 5  | 4  | 3  | 2  | 1  | 0  |  |              |  |  |  |  |  |  |  |
| Name      | DB1 |    |    |    |    |    |    |    |                                     | DB0 |    |    |    |    |    |    |  |              |  |  |  |  |  |  |  |
| Type      | R/W |    |    |    |    |    |    |    |                                     | R/W |    |    |    |    |    |    |  |              |  |  |  |  |  |  |  |

The register provides MCU access to the IN FIFO and the OUT FIFO for the endpoint 3. Writing to the register will load data to the IN FIFO for the endpoint 3. Reading the register will unload data from the OUT FIFO for the endpoint 3.

The register provides word, half-word, and byte mode accesses. If word or half-word accesses are performed, the less significant byte will correspond to the prior byte to load to the IN FIFO or unload from the OUT FIFO.

- DB0**     The first byte to be loaded to the IN FIFO or unloaded from the OUT FIFO
- DB1**     The second byte to be loaded to the IN FIFO or unloaded from the OUT FIFO
- DB2**     The third byte to be loaded to the IN FIFO or unload from the OUT FIFO
- DB3**     The forth byte to be loaded to the IN FIFO or unload from the OUT FIFO

**USB+0240h     USB PHY control**

**USB\_CON**

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5           | 4 | 3 | 2 | 1        | 0        |
|-------|----|----|----|----|----|----|---|---|---|---|-------------|---|---|---|----------|----------|
| Name  |    |    |    |    |    |    |   |   |   |   | NULLPKT_FIX |   |   |   | DMPULLUP | DPPULLUP |
| Type  |    |    |    |    |    |    |   |   |   |   | R/W         |   |   |   | R/W      | R/W      |
| Reset |    |    |    |    |    |    |   |   |   |   | 1           | 0 | 0 | 0 | 0        | 0        |

**DPPULLUP** Pull-up enable pin. Enables the pull-up 1.5KOhm pull-up on D+ pin as a full-speed device by setting it to high

**DMPULLUP** Pull-up enable pin. Enables the pull-up 1.5KOhm pull-up on D- pin as a full-speed device by setting it to high

**nullPKT\_FIX** If NULLPKT\_FIX is set to "1", the USB controller will not issue a DMAreq when a null packet is received.

### 4.11.3 System Integration Guide

The target audience of this section is the software engineer.

The USB device controller features one control endpoint and 6 other endpoints. The configuration of interfaces and endpoints can be accommodated by software for specific functions, basically supporting Bluetooth HCI, and device firmware upgrade.

Bluetooth HCI transport layer defines the configuration of endpoints and interfaces.

✓ One voice channel with 16-bit encoding

| Endpoint             | Endpoint type  | Max. packet size (bytes) | Max. bandwidth (bytes/ms) | Min. bandwidth (bytes/ms) | Double buffer in controller | Generic DMA |
|----------------------|----------------|--------------------------|---------------------------|---------------------------|-----------------------------|-------------|
| Endpoint 0 (command) | Control        | 64                       |                           |                           | No                          | No          |
| Endpoint 1 IN        | Bulk (IN)      | 64                       |                           | 1,024                     | No                          | Yes         |
| Endpoint 2 OUT       | Bulk (OUT)     | 64                       |                           | 1,024                     | No                          | Yes         |
| Endpoint 2 IN        | Bulk (IN)      | 64                       | 1,024                     |                           | No                          | Yes         |
| Endpoint 2 OUT       | Bulk (OUT)     | 64                       | 1,024                     |                           | No                          | Yes         |
| Endpoint 3 IN        | Interrupt (IN) | 16                       |                           | 16                        | No                          | No          |
| Endpoint 4 IN        | Interrupt (IN) | 16                       |                           | 16                        | No                          | No          |

When the maximum packet size is smaller than one half of the device FIFO size (64 bytes), the double buffer will be automatically enabled by hardware.

The pull-up resistor on the USB transceiver is initially disconnected when boot-up. No external resistor is required. The software should enable it after performing the configuration of the USB device controller.

The clock, interrupt and DMA are defined as the system infrastructure. It requires several steps to bring up the USB. Those steps should be done in sequence to prevent from malfunction.

**Power-on**

1. Enable USB PLL.
2. Enable USB clock after USB PLL is settled.
3. Unmask the USB interrupt in the interrupt controller.
4. Enable the pull-up resistor.

The USB device controller can generate the interrupt when conditions are met as defined in USB\_INTRINE, USB\_INTROUTE, and USB\_INTRUSBE.

The generic DMA controller is used to move data from or to the USB device controller. The USB device controller will use at most 4 DMA channels for ACL and SCO. The user should use the half-channel DMA since only the half-channel DMA has the hardware flow control.

The USB FIFO provides byte and word accesses to the read/write port of USB\_EP0\_FIFO, USB\_EP1\_FIFO, USB\_EP2\_FIFO, USB\_EP3\_FIFO and USB\_EP4\_FIFO. If the data buffer allocated in memory is word aligned, the user can enable word transfer in the DMA controller. If the data buffer allocated in memory is not word aligned, the user should set to byte aligned and set B2W in DMAx\_CON to 1 to enable fast byte-to-word transfer. Please refer to the DMA section for more details.

**Power-on sequence after plug-in**

1. Turn on Vusb (PHY 3.3v power) – The control register is in PMIC document.
2. Turn on USB AHB clock (78MHz) – The control register is in config document.
3. Turn on internal 48MHz PLL – The control register is in config document.
4. Wait for 50 usec. (PHY 3.3v power stable time)
5. Turn on USB PHY BIAS current control → reg[USB+08C1h] bit3 = 1.  
(RG\_USB11\_FSLS\_ENBGRI).
6. Wait for 10 usec.
7. Set up D+ pull up register for connecting Host → reg[USB+0240h] bit 0 =1(PUB)

**Power-off sequence after plug-out**

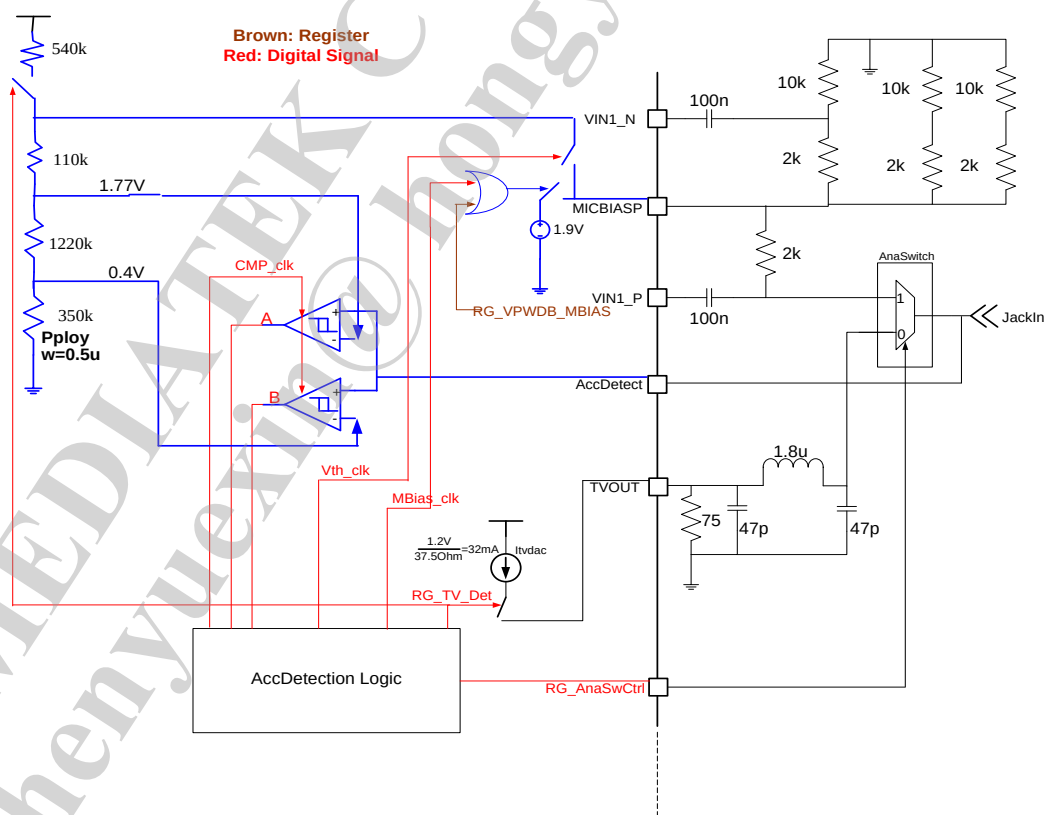
1. Release D+ pull up register for disconnecting host → Set reg[USB+0240h] bit 0 =0 (PUB)
2. Turn off USB PHY BIAS current control → reg[USB+08C1h] bit3 = 0.  
(RG\_USB11\_FSLS\_ENBGRI).
3. Turn off Vusb (PHY 3.3v power) – The control register is in PMIC document.

## 4.12 Accessory Detector

### 1.1.1 General Description

The hardware accessory detector (ACCDDET) detects plug-in/out of multiple types of external components. Based on the suggested circuit (see Figure 46), this design supports 3 types of external components, which are microphone, hook-switch and TV-OUT line. This design uses the internal 2-bit comparator to separate external components. The de-bounce scheme is also supported to resist uncertain input noises. When the plug-in/out state is stable, the PWM unit of ACCDET will enable the comparator, MBIAS and threshold voltage of the comparator periodically for the plugging detection. With suitable PWM settings, very low-power consumption can be achieved when the detection feature is enabled. In order to compensate the delay between the detection login and comparator, the delay enabling scheme is adopted. Given the suitable delay number compared to the rising edge of PWM high pulse, the stable plugging state can be prorogated to digital detection logic. Then the correct plugging state can be detected and reported.

Figure 47 shows the state machine without TV-OUT mode. The state machine is executed by the software to control the ACCDET design. The ACCDET design will send one interrupt to acknowledge the software after the ACCDET input state is changed and the duration of the state is longer than de-bounce time. The software needs to read out the memorized ACCDET input state and follow the recommend state machine to program the register in it.



**Figure 46. Suggested accessory detection circuit**

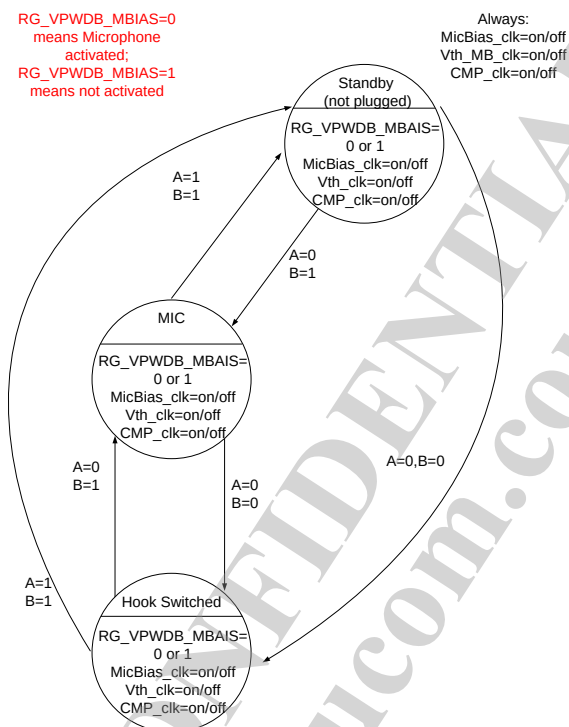


Figure 47. State machine between microphone and hook-switch plug-in/out change

## 1.1.2 Pulse Width Modulation

The ACCDET design also provides one Pulse-Width-Modulation (PWM) to enable the comparator, microphone's bias current and the threshold voltage of the comparator periodically. With suitable PWM and settings for delayed enabling, the ACCDET can achieve very low power consumption and accurate plug-in/out detection. Figure 48 shows a timing diagram example of such PWM design. The output from PWM keeps being at "0" until the value of the counter is smaller than the programmed threshold.

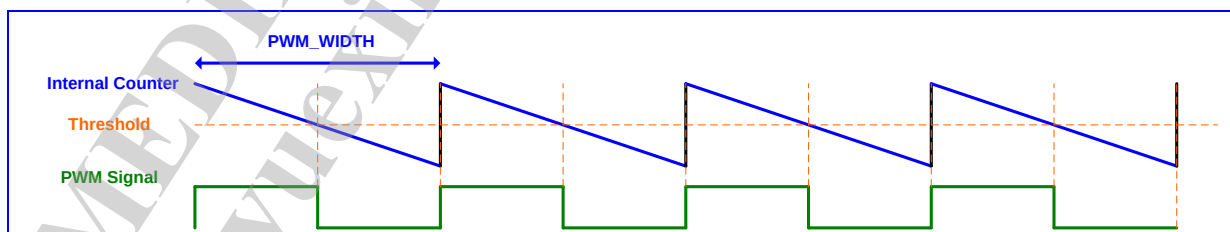


Figure 48. PWM waveform

### 1.1.3 Register Definition

Module name: ACCDET Base address: (+A0750000h)

| Address  | Name                                | Width | Register function                           |
|----------|-------------------------------------|-------|---------------------------------------------|
| A0750000 | <u>ACCDET_RSTB</u>                  | 32    | ACCDET software reset register              |
| A0750004 | <u>ACCDET_CTRL</u>                  | 32    | ACCDET control register                     |
| A0750008 | <u>ACCDET_STATE_SWCTRL</u>          | 32    | ACCDET state switch control register        |
| A075000C | <u>ACCDET_PWM_WIDTH</u>             | 32    | ACCDET PWM width register                   |
| A0750010 | <u>ACCDET_PWM_THRESH</u>            | 32    | ACCDET PWM threshold register               |
| A0750024 | <u>ACCDET_EN_DELAY_NUM</u>          | 32    | ACCDET enable delay number register         |
| A0750028 | <u>ACCDET_PWM_IDLE_VALU<br/>E</u>   | 32    | ACCDET PWM IDLE value register              |
| A075002C | <u>ACCDET_DEBOUNCE0</u>             | 32    | ACCDET debounce0 register                   |
| A0750030 | <u>ACCDET_DEBOUNCE1</u>             | 32    | ACCDET debounce1 register                   |
| A0750038 | <u>ACCDET_DEBOUNCE3</u>             | 32    | ACCDET debounce3 register                   |
| A075003C | <u>ACCDET_IRQ_STS</u>               | 32    | ACCDET interrupt status register            |
| A0750040 | <u>ACCDET_CURR_IN</u>               | 32    | ACCDET current input status register        |
| A0750044 | <u>ACCDET_SAMPLE_IN</u>             | 32    | ACCDET sampled input status register        |
| A0750048 | <u>ACCDET_MEMOIZED_IN</u>           | 32    | ACCDET memorized input status register      |
| A075004C | <u>ACCDET_LAST_MEMOIZED<br/>_IN</u> | 32    | ACCDET last memorized input status register |
| A0750050 | <u>ACCDET_FSM_STATE</u>             | 32    | ACCDET FSM status register                  |
| A0750054 | <u>ACCDET_CURR_DEBOUNC<br/>E</u>    | 32    | ACCDET current de-bounce status register    |
| A0750058 | <u>ACCDET_VERSION</u>               | 32    | ACCDET version code                         |
| A075005C | <u>ACCDET_IN_DEFAULT</u>            | 32    | default value of accdet_in                  |

**A0750000 ACCDET\_RSTB ACCDET software reset register 00000001**

| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16          |
|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-------------|
| Mne   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |             |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |             |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |             |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0           |
| Mne   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | <b>RSTB</b> |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | <b>RW</b>   |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | <b>1</b>    |

**Overview** After applying the setting to register, software reset is necessary for state initialization. Without this process, ACCDET may detect incorrect plug state.



| Bit(s) | Mnemonic Name    | Description                                                                                                                                                                                                                                                   |
|--------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0      | <b>RSTB</b> RSTB | <p><b>Set to 0 to reset the ACCDET unit and set to 1 after the reset process is finished.</b></p> <p>This software reset will clear ACCDET's enable signal but keep all ACCDET's settings. After the reset process, ACCDET will return to the IDLE state.</p> |

**A0750004 ACCDET\_CTRL ACCDET control register 00000001**

| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16               |
|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|------------------|
| Mne   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |                  |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |                  |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |                  |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0                |
| Mne   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | <b>ACCDET_EN</b> |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | RW               |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | 1                |

| Bit(s) | Mnemonic Name          | Description                         |
|--------|------------------------|-------------------------------------|
| 0      | <b>ACCDET_EEN</b><br>N | Set to 1 to enable the ACCDET unit. |

**A0750008 ACCDET\_STAT ACCDET state switch control register 00000001**  
**E\_SWCTRL**

| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20                  | 19                | 18                | 17 | 16 |
|-------|----|----|----|----|----|----|----|----|----|----|----|---------------------|-------------------|-------------------|----|----|
| Mne   |    |    |    |    |    |    |    |    |    |    |    |                     |                   |                   |    |    |
| Type  |    |    |    |    |    |    |    |    |    |    |    |                     |                   |                   |    |    |
| Reset |    |    |    |    |    |    |    |    |    |    |    |                     |                   |                   |    |    |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4                   | 3                 | 2                 | 1  | 0  |
| Mne   |    |    |    |    |    |    |    |    |    |    |    | <b>MBIAS_PWM_EN</b> | <b>VTH_PWM_EN</b> | <b>CMP_PWM_EN</b> |    |    |
| Type  |    |    |    |    |    |    |    |    |    |    |    | RW                  | RW                | RW                |    |    |
| Reset |    |    |    |    |    |    |    |    |    |    |    | 0                   | 0                 | 0                 |    |    |

| Bit(s) | Mnemonic Name             | Description                                  |
|--------|---------------------------|----------------------------------------------|
| 4      | <b>MBIAS_PWMEN</b>        | Enables PWM of ACCDET MBIAS unit             |
| 3      | <b>VTH_PWM</b> VTH_PWM_EN | Enables PWM of ACCDET voltage threshold unit |

| Bit(s) | Mnemonic Name     | Description                      |
|--------|-------------------|----------------------------------|
| 2      | <b>CMP_PWM_EN</b> | Enables PWM of ACCDET comparator |

**A075000C ACCDET\_PWM\_ACCDET PWM width register 00000000**  
**WIDTH**

| Bit   | 31               | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|-------|------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Mne   |                  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  |                  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset |                  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Bit   | 15               | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Mne   | <b>PWM_WIDTH</b> |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  | RW               |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset | 0                | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |

| Bit(s) | Mnemonic Name       | Description                                                                                                                                                                                                                                                                                                                   |
|--------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:0   | <b>PWM_WIDTH TH</b> | <b>ACCDET PWM width</b><br>It is PWM max. counter value. It will be the initial value for the internal counter. The PWM internal counter always counts down to zero to finish one complete period, and the value of the internal counter will return to the value of PWM_WIDTH.<br>PWM output frequency = (32k/PWM_WIDTH) Hz. |

**A0750010 ACCDET\_PWM\_ACCDET PWM threshold register 00000000**  
**THRESH**

| Bit   | 31                | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|-------|-------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Mne   |                   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  |                   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset |                   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Bit   | 15                | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Mne   | <b>PWM_THRESH</b> |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  | RW                |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset | 0                 | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |

| Bit(s) | Mnemonic Name      | Description                                                                                                                                                                                            |
|--------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:0   | <b>PWM_THR ESH</b> | <b>ACCDET PWM threshold</b><br>When the internal counter value is bigger than or equal to PWM_THRESH, the PWM output signal will be "0". When the internal counter is smaller than PWM_THRESH, the PWM |

| Bit(s) | Mnemonic Name | Description                                     |
|--------|---------------|-------------------------------------------------|
|        |               | output signal will be "1".                      |
|        |               | PWM output duty cycle = (PWM_THRESH)x(1/32) ms. |

Figure 49. PWM waveform with register value present

shows the PWM waveform with register value present.

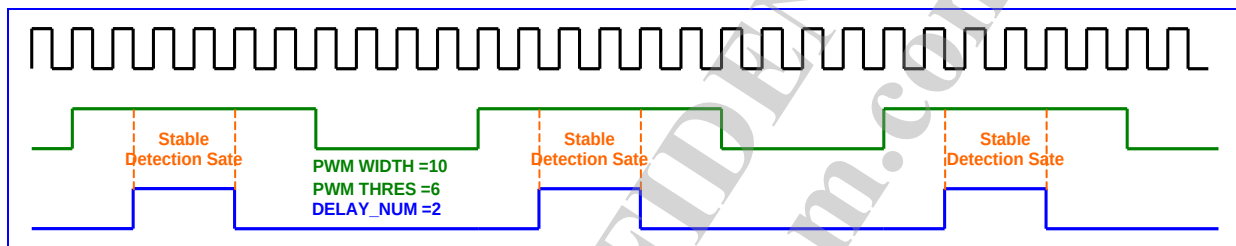


Figure 49. PWM waveform with register value present

**A0750024**    **ACCDT\_EN\_D**    **ACCDT enable delay number register**    **00000101**  
**ELAY\_NUM**

| Bit   | 31                 | 30             | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|-------|--------------------|----------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Mne   |                    |                |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  |                    |                |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset |                    |                |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Bit   | 15                 | 14             | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Mne   | FALL_DEL<br>AY_NUM | RISE_DELAY_NUM |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  | RW                 | RW             |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset | 0                  | 0              | 0  | 0  | 0  | 0  | 0  | 1  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1  |

| Bit(s) | Mnemonic Name                    | Description                                                                                                                                                                                                                                                                                                                                                              |
|--------|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15     | <b>FALL_DEL</b><br><b>AY_NUM</b> | <b>FALL_DELAY_N</b><br><b>UM</b>                                                                                                                                                                                                                                                                                                                                         |
|        |                                  | <b>Falling delay cycle compared to CMP PWM waveform</b><br>In order to make sure the plug state is stable after disabling ACCDET, the suitable delay cycle is necessary. This number indicates the clock cycle number between the point when the digital part of ACCDET stops receiving accdet_in and the point when the analog part of ACCDET stops working.            |
| 14:0   | <b>RISE_DEL</b><br><b>AY_NUM</b> | <b>RISE_DELAY_N</b><br><b>UM</b>                                                                                                                                                                                                                                                                                                                                         |
|        |                                  | <b>Rising delay cycle compared to PWM waveform</b><br>In order to make sure the plug state is stable before activating ACCDET, the suitable delay cycle is necessary. This number indicates the clock cycle number between the point when the analog part of ACCDET starts working and the point when the digital part of ACCDET starts receiving stable accdet_in. This |

| Bit(s) | Mnemonic Name | Description                                                              |
|--------|---------------|--------------------------------------------------------------------------|
|        |               | number should be fine tuned depending on different project requirements. |

**A0750028 ACCDET\_PWM\_ACCDET PWM IDLE value register IDLE\_VALUE** **00000001**

| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18    | 17  | 16  |
|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|-------|-----|-----|
| Mne   |    |    |    |    |    |    |    |    |    |    |    |    |    |       |     |     |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |       |     |     |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |       |     |     |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2     | 1   | 0   |
| Mne   |    |    |    |    |    |    |    |    |    |    |    |    |    | MBIAS | VTH | CMP |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    | RW    | RW  | RW  |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    | 0     | 0   | 1   |

| Bit(s) | Mnemonic Name | Description                   |
|--------|---------------|-------------------------------|
| 2      | <b>MBIAS</b>  | MBIAS IDLE value of MBIAS PWM |
| 1      | <b>VTH</b>    | VTH IDLE value of VTH PWM     |
| 0      | <b>CMP</b>    | CMP IDLE value of CMP PWM     |

**A075002C ACCDET\_DEBO ACCDET debounce0 register UNCE0** **00000010**

| Bit   | 31               | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|-------|------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Mne   |                  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  |                  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset |                  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Bit   | 15               | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Mne   | <b>DEBOUNCE0</b> |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  | RW               |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset | 0                | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 0  | 0  | 0  | 0  |

**Overview** This register defines the waiting period before plug-in/out or release events are considered stable. If the de-bounce setting is too small, the plug-in/out will be too sensitive and detect too many unexpected plug-ins/outs. The suitable de-bounce time setting must be adjusted for the user's demand.

| Bit(s) | Mnemonic Name     | Description                                      |
|--------|-------------------|--------------------------------------------------|
| 15:0   | <b>DEBOUNC E0</b> | De-bounce time control of the next state = 2'b00 |

**Bit(s) Mnemonic Name**

**Description**

De-bounce time = DEBOUNCE/32 ms

**A0750030 ACCDET\_DEBO ACCDET debounce1 register UNCE1 00000010**

| Bit   | 31               | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|-------|------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Mne   |                  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  |                  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset |                  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Bit   | 15               | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Mne   | <b>DEBOUNCE1</b> |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  | RW               |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset | 0                | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 0  | 0  | 0  | 0  |

**Overview** This register defines the waiting period before plug-in/out or release events are considered stable. If the de-bounce setting is too small, the plug-in/out will be too sensitive and detect too many unexpected plug-ins/outs. The suitable de-bounce time setting must be adjusted for the user's demand.

**Bit(s) Mnemonic Name**

**Description**

15:0 **DEBOUNC E1** DEBOUNCE1

**De-bounce time control of the next state = 2'b01**  
De-bounce time = DEBOUNCE/32 ms

**A0750034 ACCDET\_DEBO ACCDET debounce2 register UNCE2 00000010**

| Bit   | 31               | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|-------|------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Mne   |                  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  |                  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset |                  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Bit   | 15               | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Mne   | <b>DEBOUNCE2</b> |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  | RW               |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset | 0                | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 0  | 0  | 0  | 0  |

**Overview** This register defines the waiting period before plug-in/out or release events are considered stable. If the de-bounce setting is too small, the plug-in/out will be too sensitive and detect too many unexpected plug-ins/outs. The suitable de-bounce time setting must be adjusted for the user's demand.

| Bit(s) | Mnemonic Name     | Description                                                                                  |
|--------|-------------------|----------------------------------------------------------------------------------------------|
| 15:0   | <b>DEBOUNC E2</b> | <b>De-bounce time control of the next state = 2'b10</b><br>De-bounce time = DEBOUNC E2/32 ms |

**A0750038 ACCDET\_DEBO UNCE3** **ACCDET debounce3 register** **00000010**

| Bit   | 31                | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|-------|-------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Mne   |                   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  |                   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset |                   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Bit   | 15                | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Mne   | <b>DEBOUNC E3</b> |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  | <b>RW</b>         |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset | 0                 | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 0  | 0  | 0  | 0  |

**Overview** This register defines the waiting period before plug-in/out or release events are considered stable. If the de-bounce setting is too small, the plug-in/out will be too sensitive and detect too many unexpected plug-ins/outs. The suitable de-bounce time setting must be adjusted for the user's demand.

| Bit(s) | Mnemonic Name     | Description                                                                                  |
|--------|-------------------|----------------------------------------------------------------------------------------------|
| 15:0   | <b>DEBOUNC E3</b> | <b>De-bounce time control of the next state = 2'b11</b><br>De-bounce time = DEBOUNC E3/32 ms |

**A075003C ACCDET\_IRQ\_SACCD ETS** **ACCDET interrupt status register** **00000000**

| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24             | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16         |
|-------|----|----|----|----|----|----|----|----------------|----|----|----|----|----|----|----|------------|
| Mne   |    |    |    |    |    |    |    |                |    |    |    |    |    |    |    |            |
| Type  |    |    |    |    |    |    |    |                |    |    |    |    |    |    |    |            |
| Reset |    |    |    |    |    |    |    |                |    |    |    |    |    |    |    |            |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8              | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0          |
| Mne   |    |    |    |    |    |    |    | <b>IRQ CLR</b> |    |    |    |    |    |    |    | <b>IRQ</b> |
| Type  |    |    |    |    |    |    |    | <b>RW</b>      |    |    |    |    |    |    |    | <b>RW</b>  |
| Reset |    |    |    |    |    |    |    | 0              |    |    |    |    |    |    |    | 0          |

**Overview** When the interrupt of ACCDET is asserted, IRQ\_CLR must be set to 1 to clear the interrupt status. This bit will pause all activities in the ACCDET design until both interrupt status and IRQ\_CLR are cleared. The software should write 1 to IRQ\_CLR first to clear the interrupt (IRQ). After that, if pclk gating is enabled, the software should read

ACCDDET\_IRQ\_STS again to make IRQ\_CLR self-reset to 0.

| Bit(s)                                                                                                                                                                                                               | Mnemonic Name | Description |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------|
| 8                                                                                                                                                                                                                    | IRQ_CLR       | IRQ_CLR     |
| 0                                                                                                                                                                                                                    | IRQ           | IRQ         |
| <b>Clears interrupt status of ACCDET unit</b><br><b>Interrupt status of ACCDET unit</b><br>Because this register will be cleared by hardware, the interrupt edge-sensitive scheme should be adopted for this design. |               |             |

**A0750040**    **ACCDDET\_CURR\_IN**    **ACCDDET current input status register**    **00000003**

| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Mne   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Mne   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    | 1  | 1  |

| Bit(s)                              | Mnemonic Name | Description |
|-------------------------------------|---------------|-------------|
| 1:0                                 | CURR_IN       | CURR_IN     |
| Current input status of ACCDET unit |               |             |

**A0750044**    **ACCDDET\_SAMP\_LE\_IN**    **ACCDDET sampled input status register**    **00000003**

| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Mne   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Mne   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    | 1  | 1  |

| Bit(s)                                                                                                                 | Mnemonic Name | Description |
|------------------------------------------------------------------------------------------------------------------------|---------------|-------------|
| 1:0                                                                                                                    | SAMPLE_IN     | SAMPLE_IN   |
| <b>Samples input status of ACCDET unit</b><br>When the plug-in/out state is changed, the ACCDET unit will do sampling. |               |             |



**A0750048 ACCDET\_MEM ACCDET memorized input status register**  
**OIZED\_IN**

**00000003**

| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17                   | 16       |
|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----------------------|----------|
| Mne   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |                      |          |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |                      |          |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    |                      |          |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1                    | 0        |
| Mne   |    |    |    |    |    |    |    |    |    |    |    |    |    |    | <b>MEMORIZE D_IN</b> |          |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    | <b>RO</b>            |          |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    | <b>1</b>             | <b>1</b> |

| Bit(s) | Mnemonic Name                          | Description                                                                                                                                                                                                                                                                |
|--------|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1:0    | <b>MEMORIZE</b><br><b>MEMORIZED_IN</b> | <b>Memorized input status of ACCDET unit</b><br>When the plug-in/out states is changed and held longer than the de-bounce time, the ACCDET unit will save the sampled input state to the memorized state. The interrupt will also be asserted to acknowledge the software. |

**A075004C ACCDET\_LAST ACCDET Last memorized input status register**  
**\_MEMOIZED\_IN**

**00000003**

| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17                        | 16       |
|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|---------------------------|----------|
| Mne   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |                           |          |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |                           |          |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    |                           |          |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1                         | 0        |
| Mne   |    |    |    |    |    |    |    |    |    |    |    |    |    |    | <b>LAST_MEM ORIZED_IN</b> |          |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    | <b>RO</b>                 |          |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    | <b>1</b>                  | <b>1</b> |

| Bit(s) | Mnemonic Name                                  | Description                                                                       |
|--------|------------------------------------------------|-----------------------------------------------------------------------------------|
| 1:0    | <b>LAST_ME</b><br><b>MORIZED_</b><br><b>IN</b> | <b>LAST_MEMORI</b><br><b>ZED_IN</b><br>Last memorized input status of ACCDET unit |

**A0750050 ACCDET\_FSM\_ ACCDET FSM status register**  
**STATE**

**00000000**

| Bit  | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Mne  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |

|       |    |    |    |    |    |    |   |   |   |   |   |   |   |           |   |   |
|-------|----|----|----|----|----|----|---|---|---|---|---|---|---|-----------|---|---|
| Reset |    |    |    |    |    |    |   |   |   |   |   |   |   |           |   |   |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2         | 1 | 0 |
| Mne   |    |    |    |    |    |    |   |   |   |   |   |   |   | FSM_STATE |   |   |
| Type  |    |    |    |    |    |    |   |   |   |   |   |   |   | RO        |   |   |
| Reset |    |    |    |    |    |    |   |   |   |   |   |   |   | 0         | 0 | 0 |

| Bit(s) | Mnemonic Name                 | Description                                                                                                                                                             |
|--------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2:0    | <b>FSM_STAT</b> FSM_SATE<br>E | <b>State of ACCDET unit finite-state-machine</b><br>0: ACCDET_IDLE<br>1: ACCDET_SAMPLE<br>2: ACCDET_DEBOUNCE<br>3: ACCDET_CHECK<br>4: ACCDET_MEMORIZED<br>5: ACCDET_IRQ |

**A0750054**    **ACCDET\_CURR ACCDET current de-bounce status register**    **00000004**  
**\_DEBOUNCE**

|       |                      |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|-------|----------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Bit   | 31                   | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
| Mne   |                      |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  |                      |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset |                      |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Bit   | 15                   | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Mne   | <b>CURR_DEBOUNCE</b> |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  | RO                   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset | 0                    | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 0  |

| Bit(s) | Mnemonic Name                          | Description                           |
|--------|----------------------------------------|---------------------------------------|
| 15:0   | <b>CURR_DEBOUNCE</b> CURR_DEBOUN<br>CE | Currently used de-bounce time setting |

**A0750058**    **ACCDET\_VERSION**    **ACCDET version code**    **000000 03**

|       |    |    |    |    |    |    |    |    |    |    |    |    |    |    |                       |    |
|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----------------------|----|
| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17                    | 16 |
| Name  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |                       |    |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |                       |    |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    |                       |    |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1                     | 0  |
| Name  |    |    |    |    |    |    |    |    |    |    |    |    |    |    | <b>ACCDET_VERSION</b> |    |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    | RO                    |    |

|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |   |   |
|-------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|---|---|
| Reset |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 1 | 1 |
|-------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|---|---|

| Bit(s) | Mnemonic               | Name            | Description             |
|--------|------------------------|-----------------|-------------------------|
| 1:0    | <b>ACCDDET_VERSION</b> | ACCDDET_VERSION | Version code for ACCDET |

**A075005C**     **ACCDDET\_IN\_DEFAULT**     Default value of accdet\_in     **00000000**

|       |    |    |    |    |    |    |    |    |    |    |    |                               |    |    |                    |    |
|-------|----|----|----|----|----|----|----|----|----|----|----|-------------------------------|----|----|--------------------|----|
| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20                            | 19 | 18 | 17                 | 16 |
| Name  |    |    |    |    |    |    |    |    |    |    |    |                               |    |    |                    |    |
| Type  |    |    |    |    |    |    |    |    |    |    |    |                               |    |    |                    |    |
| Reset |    |    |    |    |    |    |    |    |    |    |    |                               |    |    |                    |    |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4                             | 3  | 2  | 1                  | 0  |
| Name  |    |    |    |    |    |    |    |    |    |    |    | ACCDDET_IN_DEFAULT_REFRESH_EN |    |    | ACCDDET_IN_DEFAULT |    |
| Type  |    |    |    |    |    |    |    |    |    |    |    | RW                            |    |    | RW                 |    |
| Reset |    |    |    |    |    |    |    |    |    |    |    | 0                             |    |    | 0                  | 0  |

**Overview**     The default value of sample\_accdet\_in and memorised\_accdet\_in can be set by software instead of using the default value set by hardware(i.e. 3).  
 ACCDET\_DEFAULT\_REFRESH\_EN is the enable bit controlling whether to use this additional function. The value of sample\_accdet\_in and memorized\_accdet\_in will change when accdet\_en rises from low to high. *Note that if software reset is applied when accdet\_en is high, the default value of sample\_accdet\_in and memorized\_accdet\_in will also be loaded when the software reset is de-asserted.*

| Bit(s) | Mnemonic                             | Name                          | Description                                                                                                                                    |
|--------|--------------------------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| 4      | <b>ACCDDET_IN_DEFAULT_REFRESH_EN</b> | ACCDDET_IN_DEFAULT_REFRESH_EN | <b>Enable signal for whether to load accdet_in_default</b><br>0: accdet_in_default will not be loaded.<br>1: accdet_in_default will be loaded. |
| 1:0    | <b>ACCDDET_IN_DEFAULT</b>            | ACCDDET_IN_DEFAULT            | <b>Default value of accdet_in set by software</b>                                                                                              |

## 4.13 SD Memory Card Controller

### 4.13.1 Introduction

The controller fully supports the SD memory card bus protocol as defined in SD Memory Card Specification Part 1 Physical Layer Specification version 2.0.

Furthermore, the controller also partially supports the SDIO card specification version 2.0. However, the controller can only be configured as the host of the SD memory card. Hereafter, the controller is also abbreviated as the SD controller. The following are the main features of the controller.

- Interface with MCU by APB bus
- 16/32-bit access on APB bus
- 16/32-bit access for control registers
- 32-bit access for FIFO
- Built-in 32 bytes FIFO buffers for transmit and receive, FIFO is shared for transmit and receive
- Built-in CRC circuit
- CRC generation can be disabled
- DMA supported
- Interrupt capabilities
- Data rate up to 48 Mbps in serial mode, 48x4 Mbps in parallel model, the module is targeted at 48 MHz operating clock
- Serial clock rate on SD bus is programmable
- Card detection capabilities during sleep mode
- Controllability of power for memory card
- Does not support SPI mode for SD memory card
- Does not support multiple SD memory cards

### 4.13.2 Overview

#### 4.13.2.1 Pin Assignment

The following lists pins required for the SD memory card. Table 66 shows how the pins are shared. Note that all I/O pads have embedded both pull-up and pull-down resistors because they are shared by the SD memory card. The pull-down resistors for these pins can be used for power saving. If optimal pull-up or pull-down resistors are required on the system board, all embedded pull-up and pull-down resistors can be disabled by programming the corresponding control registers. The VDDPD pin is used for power saving. Power for the SD memory card can be shut down by programming the corresponding control register. The WP (Write Protection) pin is used to detect the status of the Write Protection Switch on the SD memory card.

| No. | Name     | Type   | MMC  | SD      | Description                   |
|-----|----------|--------|------|---------|-------------------------------|
| 1   | SD_CLK   | O      | CLK  | CLK     | Clock                         |
| 2   | SD_DAT3  | I/O/PP |      | CD/DAT3 | Data Line [Bit 3]             |
| 3   | SD_DAT0  | I/O/PP | DAT0 | DAT0    | Data Line [Bit 0]             |
| 4   | SD_DAT1  | I/O/PP |      | DAT1    | Data Line [Bit 1]             |
| 5   | SD_DAT2  | I/O/PP |      | DAT2    | Data Line [Bit 2]             |
| 6   | SD_CMD   | I/O/PP | CMD  | CMD     | Command Or Bus State          |
| 7   | SD_PWRON | O      |      |         | VDD ON/OFF                    |
| 8   | SD_WP    | I      |      |         | Write Protection Switch in SD |
| 9   | SD_INS   | I      | VSS2 | VSS2    | Card Detection                |

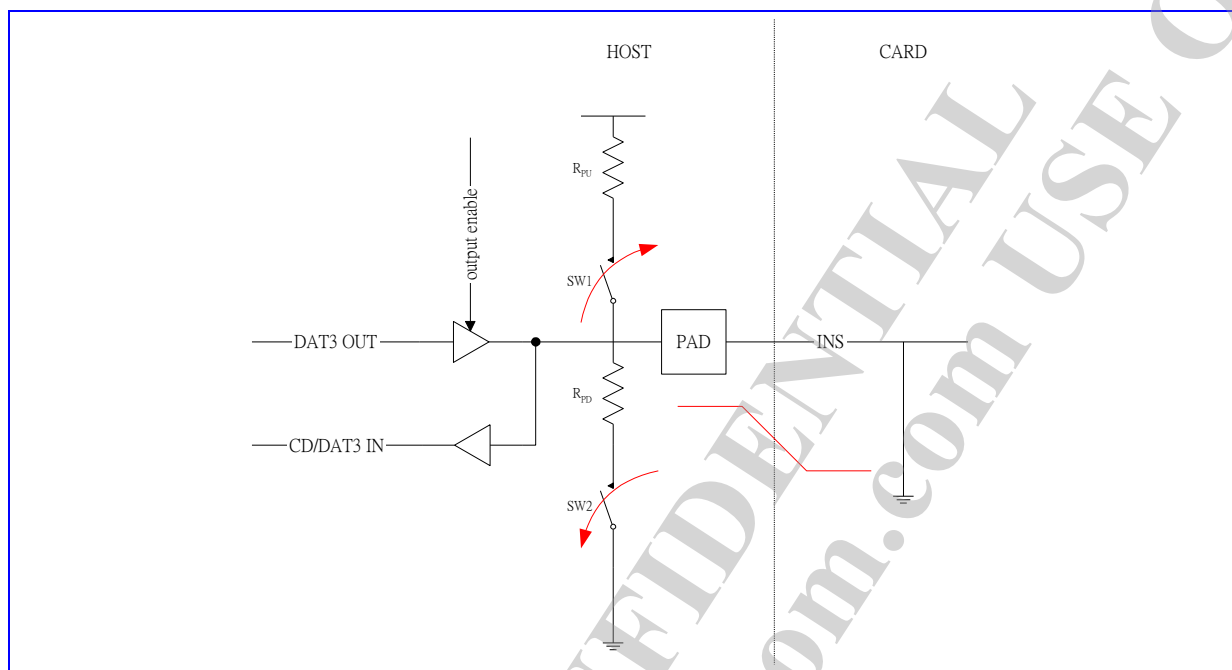
**Table 66. Sharing of pins for SD memory card controller**

#### 4.13.2.2 Card Detection

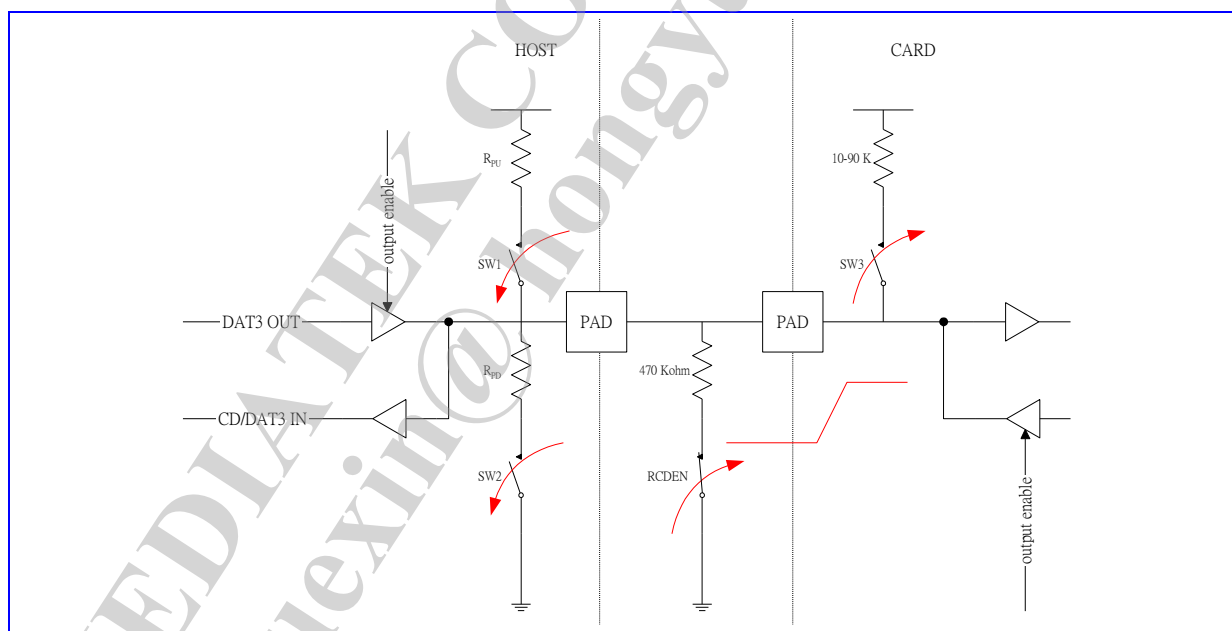
For SD memory card, detection of card insertion/removal by hardware is supported. Because a pull-down resistor with about 470 K $\Omega$  resistance which is impractical to embed in an I/O pad is needed on the signal CD/DAT3, and it has to be capable of being connected or disconnected dynamically onto the signal CD during initialization period, an additional I/O pad is needed to switch on/off the pull-down resistor on the system board. The card detection for SD memory card is shown in Figure 51. Before SD memory card is inserted or powered on, SW1 and SW2 must be opened for card detection on the host side. Meanwhile, the pull-down resistor R<sub>CD</sub> on the system board must attach onto the signal CD/DAT3 by the output signal RCDEN. In addition, the default position of SW3 on the card is closed.

Upon insertion of the SD memory card, the signal CD/DAT3 will transit from low to high. If the SD memory card is removed, the CD/DAT3 signal will return to logic low state. After the card identification process, the pull-down resistor R<sub>CD</sub> on the system board shall be disconnected with the CD/DAT3 signal, and SW3 on the card must be opened for normal operation.

Since the scheme above needs a mechanical switch such as a relay on the system board, it is not an ideal scheme. Thus, a dedicated pin "INS" is used to perform card insertion and removal for SD. The pin "INS" will be connected to the pin "VSS2" of a SD connector (see Figure 50).



**Figure 50. Card detection for SD memory card (Scheme 1)**



**Figure 51. Card detection for SD memory card (Scheme 2)**

### 4.13.3 Register Definition

For MT6250, MSDC base address is 0xA0130000.

| Register address | Register name                                     | Synonym      |
|------------------|---------------------------------------------------|--------------|
| MSDC + 0000h     | SD memory card controller configuration register  | MSDC_CFG     |
| MSDC + 0004h     | SD memory card controller status register         | MSDC_STA     |
| MSDC + 0008h     | SD memory card controller interrupt register      | MSDC_INT     |
| MSDC + 0000Ch    | SD memory card pin status register                | MSDC_PS      |
| MSDC + 0010h     | SD memory card controller data register           | MSDC_DAT     |
| MSDC + 00014h    | SD memory card controller IO control register 0   | MSDC_IOCON0  |
| MSDC + 00018h    | SD memory card controller IO control register 1   | MSDC_IOCON1  |
| MSDC + 0020h     | SD memory card controller configuration register  | SDC_CFG      |
| MSDC + 0024h     | SD memory card controller command register        | SDC_CMD      |
| MSDC + 0028h     | SD memory card controller argument register       | SDC_ARG      |
| MSDC + 002Ch     | SD memory card controller status register         | SDC_STA      |
| MSDC + 0030h     | SD memory card controller response register 0     | SDC_RESP0    |
| MSDC + 0034h     | SD memory card controller response register 1     | SDC_RESP1    |
| MSDC + 0038h     | SD memory card controller response register 2     | SDC_RESP2    |
| MSDC + 003Ch     | SD memory card controller response register 3     | SDC_RESP3    |
| MSDC + 0040h     | SD memory card controller command status register | SDC_CMDSTA   |
| MSDC + 0044h     | SD memory card controller data status register    | SDC_DATSTA   |
| MSDC + 0048h     | SD memory card status register                    | SDC_CSTA     |
| MSDC + 004Ch     | SD memory card IRQ mask register 0                | SDC_IRQMASK0 |
| MSDC + 0050h     | SD memory card IRQ mask register 1                | SDC_IRQMASK1 |
| MSDC + 0054h     | SDIO configuration register                       | SDIO_CFG     |
| MSDC + 0058h     | SDIO status register                              | SDIO_STA     |
| MSDC + 0080h     | CLK latching edge configuration register          | CLKPAD_RED   |

**Table 67. SD controller register map**

#### 4.13.3.1 Global Register Definition

##### MSDC+0000h SD memory card controller configuration register MSDC\_CFG

| Bit   | 31 | 30 | 29 | 28 | 27      | 26 | 25 | 24 | 23 | 22 | 21        | 20        | 19         | 18        | 17        | 16        |
|-------|----|----|----|----|---------|----|----|----|----|----|-----------|-----------|------------|-----------|-----------|-----------|
| Name  |    |    |    |    | FIFOTHD |    |    |    |    |    | VDDP<br>D | RCD<br>EN | DIRQ<br>EN | PINE<br>N | DMA<br>EN | INTE<br>N |
| Type  |    |    |    |    | R/W     |    |    |    |    |    | R/W       | R/W       | R/W        | R/W       | R/W       | R/W       |
| Reset |    |    |    |    | 0100    |    |    |    |    |    | 0         | 0         | 0          | 0         | 0         | 0         |
| Bit   | 15 | 14 | 13 | 12 | 11      | 10 | 9  | 8  | 7  | 6  | 5         | 4         | 3          | 2         | 1         | 0         |



| Name  | SCLKF    | SCLK ON | CRE D | STDB Y | CLKSRC | NOC RC | RST | MSD C |
|-------|----------|---------|-------|--------|--------|--------|-----|-------|
| Type  | R/W      | R/W     | R/W   | R/W    | R/W    | R/W    | W   | R/W   |
| Reset | 00000000 | 0       | 0     | 1      | 0      | 0      | 0   | 0     |

The register is for general configuration of the SD controller. *Note that MSDC\_CFG[31:16] can be accessed by the 16-bit APB bus.*

**MSDC** Configures the controller for SD memory card mode operation. The CLK/CMD/DAT line is pulled low when the SD memory card mode is disabled.

- 0** Disable SD memory card
- 1** Enable SD memory card

**NOCRC** Disables CRC. '1' indicates that data transfer without CRC is desired. For the write data block, the data will be transmitted without CRC. For the read data block, the CRC will not be checked. because it is for testing purposes.

- 0** Data transfer with CRC is desired.
- 1** Data transfer without CRC is desired.

**RST** Software seset. Writing '1' to this register bit will cause an internal synchronous reset of the SD controller. However, this does not reset register settings.

- 0** Otherwise
- 1** Reset SD controller

**CLKSRC** Specifies which clock is used as the source clock of the memory card. Use MPLL (780MHz) hopping clock as the source clock of the memory card. The fastest clock rate for the memory card is  $97.5/2 = 48.75\text{MHz}$  when frequency hopping is enabled.

- 00** Use MPLL clock divide by 8 as the source clock of the memory card. The source clock is 97.5MHz when MPLL is used.
- 01** Use MPLL clock divide by 9 as the source clock of the memory card. The source clock is 86.6MHz when MPLL is used.
- 10** Use MPLL clock divide by 10 as the source clock of the memory card. The source clock is 78MHz when MPLL is used.
- 11** Use MPLL clock divide by 12 as the source clock of the memory card. The source clock is 65MHz when MPLL is used.

**STDBY** Standby mode. If the module is powered down, the clock operation to the module will be stopped. At the same time, the clock to card detection circuitry will also be stopped. If detection of the memory card insertion and removal is desired, write '1' to the register bit. If the interrupt for detection of the memory card insertion and removal is enabled, an interrupt will take place whenever the memory is inserted or removed.

- 0** Standby mode is disabled.
- 1** Standby mode is enabled.

- CRED** Rising edge data. This register bit is used to determine the serial data input is latched at the falling edge or the rising edge of the serial clock. The default setting is at the rising edge. If the serial data have worse timing, set this register bit to '1'. When the memory card has worse timing on the return read data, set this register bit to '1'.
- 0** Serial data input is latched at the rising edge of the serial clock.
- 1** Serial data input is latched at the falling edge of the serial clock.
- SCLKON** Serial clock always on. Used for debugging purposes.
- 0** Serial clock not always on
- 1** Serial clock always on
- SCLKF** Controls clock frequency of serial clock on the SD bus. Denote the clock frequency of SD bus serial clock as  $f_{slave}$  and clock frequency of the SD controller as  $f_{host}$  which is 97.5 MHz. The value of this register field is as the following. *Note that the allowable maximum frequency of  $f_{slave}$  is 48.75MHz. While changing the clock rate, it needs "1T clock period before change + 1T clock period after change" for hardware signal to re-synchronize.*
- 00000000b**  $f_{slave} = (1/2) * f_{host}$
- 00000001b**  $f_{slave} = [1/(4*1)] * f_{host}$
- 00000010b**  $f_{slave} = [1/(4*2)] * f_{host}$
- 00000011b**  $f_{slave} = [1/(4*3)] * f_{host}$
- ...
- 00010000b**  $f_{slave} = [1/(4*16)] * f_{host}$
- ...
- 11111111b**  $f_{slave} = [1/(4*255)] * f_{host}$
- INTEN** Enables interrupt. *Note that if the interrupt capability is disabled, the application software must poll the status of the MSDC\_STA register to check for any interrupt requests.*
- 0** Disable interrupt induced by various conditions
- 1** Enables interrupt induced by various conditions
- DMAEN** Enables DMA. *Note that if DMA capability is disabled, the application software must poll the status of the MSDC\_STA register for checking any data transfer requests.* If DMA is desired, the register bit must be set before the command register is written.
- 0** Disable DMA request induced by various conditions
- 1** Enable DMA request induced by various conditions
- PINEN** Enables pin interrupt. This register bit is used to determine if the pin for card detection is used as an interrupt source.
- 0** The pin for card detection is not used as an interrupt source.
- 1** The pin for card detection is used as an interrupt source.
- DIRQEN** Enables data request interrupt. The register bit is used to determine if the data request is used as an interrupt source.
- 0** Data request is not used as an interrupt source.
- 1** Data request is used as an interrupt source.

- RCDEN** Controls the output pin RCDEN used for the card identification process when the controller is for SD memory card. Its output will control the pull down resistor on the system board to connect or disconnect with the CD/DAT3 signal.  
**0** The output pin RCDEN will output logic low.  
**1** The output pin RCDEN will output logic high.
- VDDPD** Controls the output pin VDDPD is used for power saving. The output pin VDDPD controls power for the memory card.  
**0** The output pin VDDPD will output logic low. The power for the memory card will be turned off.  
**1** The output pin VDDPD will output logic high. The power for the memory card will be turned on.
- FIFOTHD** FIFO threshold. Determines when to issue a DMA request. For write transactions, DMA requests will be asserted if the number of free entries in FIFO are larger than or equal to the value in the register field. For read transactions, DMA requests will be asserted if the number of valid entries in FIFO are larger than or equal to the value in the register field. The register field must be set according to the setting of the data transfer count in DMA burst mode. If the single mode for DMA transfer is used, the register field shall be set to 0b0001.  
**0000** Invalid  
**0001** Threshold value is 1.  
**0010** Threshold value is 2.  
...  
**1000** Threshold value is 8.  
**Others** Invalid

#### MSDC+0004h SD memory card controller status register

#### MSDC\_STA

| Bit   | 15          | 14              | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6              | 5 | 4 | 3          | 2          | 1         | 0         |
|-------|-------------|-----------------|----|----|----|----|---|---|---|----------------|---|---|------------|------------|-----------|-----------|
| Name  | <b>BUSY</b> | <b>FIFO CLR</b> |    |    |    |    |   |   |   | <b>FIFOCNT</b> |   |   | <b>INT</b> | <b>DRQ</b> | <b>BE</b> | <b>BF</b> |
| Type  | R           | W               |    |    |    |    |   |   |   | RO             |   |   | RO         | RO         | RO        | RO        |
| Reset | 0           | -               |    |    |    |    |   |   |   | 0000           |   |   | 0          | 0          | 0         | 0         |

The register contains the status of FIFO, interrupts and data requests.

- BF** Indicates if FIFO in the SD controller is full  
**0** FIFO in SD controller is not full.  
**1** FIFO in SD controller is full.
- BE** Indicates if FIFO in the SD controller is empty  
**0** FIFO in SD controller is not empty.

**1** FIFO in SD controller is empty.

**DRQ** Indicates if any data transfer is required. When any data transfer is required, this register bit will still be active even if register bit DIRQEN in register MSDC\_CFG is disabled. Data transfer can be achieved by the DMA channel, thus alleviating MCU loading, or by polling the register bit to check if any data transfer is requested. When register bit DIRQEN in register MSDC\_CFG is disabled, the second method is used.

**0** No DMA request exists.

**1** DMA request exists.

**INT** Indicates if any interrupt exists. When any interrupt exists, the register bit will still be active even if register bit INTEN in register MSDC\_CFG is disabled. The SD controller can interrupt MCU by issuing an interrupt request to the interrupt controller, or by the software/application endlessly polling the register to check for an interrupt request in the SD controller. While register bit INTEN in register MSDC\_CFG is disabled, the latter method is used. For read commands, it is possible that a time-out may occur. The software can read the status register to check and see if the time-out error takes place without OS time tick support or an asserted data request. *Note that the register bit will be cleared when reading register MSDC\_INT.*

**0** No interrupt request exists.

**1** Interrupt request exists.

**FIFOCNT** FIFO count. Shows how many valid entries are in FIFO

**0000** There is 0 valid entry in FIFO.

**0001** There is 1 valid entry in FIFO.

**0001** There are 2 valid entries in FIFO.

...

**1000** There are 8 valid entries in FIFO.

**Others** Invalid

**FIFOCLR** Clears FIFO. Writing '1' to the register bit will cause the content of FIFO to clear and reset the status of the FIFO controller.

**0** No effect on FIFO

**1** Clear the content of FIFO and reset the status of FIFO controller.

**BUSY** Status of the controller. If the controller is in busy state, the register bit will be '1'. Otherwise it will be '0'.

**0** The controller is in idle state.

**1** The controller is in busy state.

### MSDC+0008h SD memory card controller interrupt register

### MSDC\_INT

| Bit | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-----|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
|-----|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|

| Name  |  |  |  |  |  |  |  |  | SDIOI<br>RQ | SDR1<br>BIRQ |  | SDM<br>CIRQ | SDD<br>ATIR<br>Q | SDC<br>MDIR<br>Q | PINIR<br>Q | DIRQ |
|-------|--|--|--|--|--|--|--|--|-------------|--------------|--|-------------|------------------|------------------|------------|------|
| Type  |  |  |  |  |  |  |  |  | RC          | RC           |  | RC          | RC               | RC               | RC         | RC   |
| Reset |  |  |  |  |  |  |  |  | 0           | 0            |  | 0           | 0                | 0                | 0          | 0    |

The register contains the status of interrupts. *Note that the register still shows the status of an interrupt even though the interrupt is disabled, that is, register bit INTEN of register MSDC\_CFG is set to '0'. It implies that a software interrupt can be implemented by polling register bit INT of register MSDC\_STA and this register. However, if a hardware interrupt is desired, be sure to clear the register before setting register bit INTEN of register MSDC\_CFG to '1', or undesired hardware interrupt may occur from the previous interrupt status.*

**DIRQ** Data request interrupt. Indicates if an interrupt for data request exists. Whenever a data request exists, and the data request as an interrupt source is enabled, i.e. register bit DIRQEN in register MSDC\_CFG is set to '1', the register bit will be active. It will be reset upon being read. For software, the data requests can be recognized by polling register bit DRQ or by a data request interrupt. Data request interrupts will be generated with all FIFOTHD data transfers.

**0** No data request interrupt  
**1** Data request interrupt occurs.

**PINIRQ** Pin change interrupt. Indicates if an interrupt for the memory card insertion/removal exists. Whenever a memory card is inserted or removed and the card detection interrupt is enabled, i.e. register bit PINEN in register MSDC\_CFG is set to '1', the register bit will be set to '1'. It will be reset when the register is read.

**0** Otherwise  
**1** Card is inserted or removed.

**SDCMDIRQ** SD bus CMD interrupt. Indicates if an interrupt for the SD CMD line exists. Whenever an interrupt for the SD CMD line exists, i.e. any bit in register SDC\_CMDSTA is active, the register bit will be set to '1' if the interrupt is enabled. It will be reset when the register is read.

**0** No SD CMD line interrupt  
**1** SD CMD line interrupt exists.

**SDDATIRQ** SD bus DAT interrupt. Indicates if an interrupt for the SD DAT line exists. Whenever an interrupt for the SD DAT line exists, i.e. any bit in register SDC\_DATSTA is active, the register bit will be set to '1' if the interrupt is enabled. It will be reset when the register is read.

**0** No SD DAT line interrupt  
**1** SD DAT line interrupt exists.

**SDMCIRQ** SD memory card interrupt. Indicates if an interrupt for the SD memory card exists. Whenever an interrupt for the SD memory card exists, i.e. any bit in register

SDC\_CSTA is active, the register bit will be set to '1' if the interrupt is enabled. It will be reset when the register is read. *Note that this interrupt will not call interrupt process program.*

- 0 No SD memory card interrupt
- 1 SD memory card interrupt exists.

## SDR1BIRQ

SD R1b response interrupt. Active when a SD command with R1b response finishes and the DAT0 line has transited from busy to idle state. Single block write commands with R1b response will cause the interrupt when the command completes no matter whether it is successful or a CRC error has occurred. However, multi-block write commands with R1b response do not cause the interrupt because multi-block write commands are always stopped by STOP\_TRANS commands. STOP\_TRANS commands (with R1b response) behind multi-block write commands will cause the interrupt. Single block read command with R1b response will cause the interrupt when the command completes but multi-block read commands will not. *Note that STOP\_TRANS commands (with R1b response) behind multi-block read commands will cause the interrupt.*

- 0 No interrupt for SD R1b response
- 1 Interrupt for SD R1b response exists.

## SDIOIRQ

SDIO interrupt. Indicates if there is an SDIO interrupt on data line 1/5 (selected by SDIO\_CFG.INTSEL). When in the 1-bit data line mode, DAT1 is dedicated for SDIO interrupt, so in this mode when DAT1 goes low, SDIOIRQ will be active unless this interrupt is masked till read-clear. When in the 4-bit data line mode, if DAT1 goes low during the interrupt period, SDIOIRQ will be active unless this interrupt is masked till read-clear.

- 0 No interrupt for SDIO interrupt
- 1 Interrupt for SDIO interrupt

## MSDC+000Ch

### SD memory card pin status register

## MSDC\_PS

| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24  | 23 | 22 | 21 | 20         | 19   | 18        | 17        | 16       |
|-------|----|----|----|----|----|----|----|-----|----|----|----|------------|------|-----------|-----------|----------|
| Name  |    |    |    |    |    |    |    | CMD |    |    |    |            |      |           |           | DAT      |
| Type  |    |    |    |    |    |    |    | RO  |    |    |    |            |      |           |           | RO       |
| Reset |    |    |    |    |    |    |    | -   |    |    |    |            |      |           |           | -        |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8   | 7  | 6  | 5  | 4          | 3    | 2         | 1         | 0        |
| Name  |    |    |    |    |    |    |    |     |    |    |    | PINC<br>HG | PIN0 | POE<br>N0 | PIEN<br>0 | CDE<br>N |
| Type  |    |    |    |    |    |    |    |     |    |    |    | RC         | RO   | R/W       | R/W       | R/W      |
| Reset |    |    |    |    |    |    |    |     |    |    |    | 0          | 1    | 0         | 0         | 0        |

The register is used for card detection. When the memory card controller is powered on, and the system is powered on, the power for the memory card remains off unless the power has been supplied



by the PMIC. Meanwhile, the pad for card detection defaults to pull down when the system is powered on.

For detecting card insertion, first pull up the INS pin, then enable the card detection and input pin at the same time. After 32 cycles of controller clock, the status of pin changes will emerge. For detecting card removal, simply keep enabling card detection and the input pin.

- CDEN**      Enables card detection. Enables or disables card detection.  
**0**      Disable card detection  
**1**      Enable card detection
- PIEN0**      Controls input pin for card detection  
**0**      Disable input pin for card detection  
**1**      Enable input pin for card detection
- POEN0**      Controls output of input pin for card detection.  
**0**      Disable output of input pin for card detection  
**1**      Enable output of input pin for card detection
- PIN0**      Value of input pin for card detection  
**0**      The value of input pin for card detection is logic low.  
**1**      The value of input pin for card detection is logic high.
- PINCHG**      Pin change. Indicates the status of card insertion/removal. If the memory card is inserted or removed, the register bit will be set to '1' no matter if the pin change interrupt is enabled or not. It will be cleared when the register is read.  
**0**      Otherwise  
**1**      Card is inserted or removed.
- CDDEBOUNCE**      Specifies the time interval for card detection de-bounce. Its default value is 0. It means that the de-bounce interval is 32 cycle times of 32kHz. The interval will extend by one 32kHz cycle time by increasing the counter by 1.
- DAT**      Memory card data lines  
**CMD**      Memory card command lines

### MSDC+0010h SD memory card controller data register

### MSDC\_DAT

| Bit  | 31          | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|------|-------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Name | DATA[31:16] |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type | RW          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Bit  | 15          | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Name | DATA[15:0]  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |



|      |     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|------|-----|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
| Type | R/W |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|------|-----|--|--|--|--|--|--|--|--|--|--|--|--|--|--|

The register is used to read/write data from/to FIFO inside the SD controller. Data access unit: 32 bits.

### MSDC+0014h SD memory card controller IO control register MSDC\_IOCON

| Bit   | 31        | 30 | 29 | 28 | 27 | 26 | 25                  | 24 | 23       | 22         | 21             | 20         | 19     | 18      | 17 | 16  |
|-------|-----------|----|----|----|----|----|---------------------|----|----------|------------|----------------|------------|--------|---------|----|-----|
| Name  |           |    |    |    |    |    | SAMPLEDL<br>Y       |    | FIXDLY   |            | SA<br>MP<br>ON | CRCDS<br>S | CMDSEL | INTLH   |    | DSW |
| Type  |           |    |    |    |    |    | R/W                 |    | R/W      |            | R/<br>W        | R/W        | R/W    | R/W     |    | R/W |
| Reset |           |    |    |    |    |    | 01                  |    | 00       |            | 0              | 0          | 0      | 0       |    | 0   |
| Bit   | 15        | 14 | 13 | 12 | 11 | 10 | 9                   | 8  | 7        | 6          | 5              | 4          | 3      | 2       | 1  | 0   |
| Name  | CMD<br>RE |    |    |    |    |    | HIGH<br>_SPE<br>_ED |    | DMABURST | SRCF<br>G1 | SRCF<br>G0     | ODCCFG1    |        | ODCCFG0 |    |     |
| Type  | R/W       |    |    |    |    |    | R/W                 |    | R/W      | R/W        |                | R/W        |        | R/W     |    |     |
| Reset | 0         |    |    |    |    |    | 00                  |    | 1        | 1          |                | 000        |        | 011     |    |     |

The register specifies the output driving capability and slew rate of IO pads for MSDC. The reset value is the suggested setting. If the output driving capability of pins DAT0, DAT1, DAT2 and DAT3 is too large, it is possible to raise the ground bounce and thus result in glitch on SCLK. The actual driving current will depend on the PAD type selected for the chip.

**ODCCFG0**      Output driving capability of pins CMD/BS and SCLK

- 000**      4mA
- 001**      8mA
- 010**      12mA
- 011**      16mA

**ODCCFG1**      Output driving capability of pins DAT0, DAT1, DAT2 and DAT3

- 000**      4mA
- 001**      8mA
- 010**      12mA
- 011**      16mA

**SRCFG0**      Output driving capability of pins CMD/BS and SCLK

- 0**      Fast slew rate
- 1**      Slow slew rate

**SRCFG1**      Output driving capability of pins DAT0, DAT1, DAT2 and DAT3

- 0**      Fast slew rate
- 1**      Slow slew rate

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CMDRE</b>      | <p>Determines whether or not the host should latch a response token (sent from the card on the CMD line ) at the rising edge or falling edge of the serial clock.</p> <p><b>0</b> Host latches response at the rising edge of the serial clock</p> <p><b>1</b> Host latches response at the falling edge of the serial clock</p>                                                                                                                                                                                                                                                                                                                     |
| <b>DSW</b>        | <p>Determines whether or not the host should latch data with 1-T delay. For the SD card, this bit is suggest to be 0.</p> <p><b>0</b> Host latches the data with 1-T delay</p> <p><b>1</b> Host latches the data without 1-T delay</p>                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>INTLH</b>      | <p>Selects the latch timing for SDIO multi-block read interrupt.</p> <p><b>00</b> Host latches INT at the second backend clock after the end bit of the current data block from the card is received. (This is the default setting)</p> <p><b>01</b> Host latches INT at the first backend clock after the end bit of the current data block from the card is received.</p> <p><b>10</b> Host latches INT at the second backend clock after the end bit of the current data block from the card is received.</p> <p><b>11</b> Host latches INT at the third backend clock after the end bit of the current data block from the card is received.</p> |
| <b>CMDSEL</b>     | <p>Determines whether or not the host should delay 1-T to latch response from the card.</p> <p><b>0</b> Host latches response without 1-T delay</p> <p><b>1</b> Host latches response with 1-T delay</p>                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>CRCDIS</b>     | <p>Switches off the data CRC check for SD read data.</p> <p><b>0</b> CRC check is on.</p> <p><b>1</b> CRC check is off.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>HIGH_SPEED</b> | <p>For high-speed mode when multiple phase clock is used. High-speed mode means that the SD bus serial clock rate is bigger than 25MHz. The default speed mode means that the SD bus serial clock rate bigger than 25MHz.</p> <p><b>0</b> Default speed</p> <p><b>1</b> High speed</p>                                                                                                                                                                                                                                                                                                                                                               |
| <b>DMABURST</b>   | <p>Used for software to select the burst type when the data is transferred by DMA. <i>Note that only single mode operation can support non-4N bytes data transfer in read operation.</i></p> <p><b>00</b> Single mode</p> <p><b>01</b> 4-beat incrementing burst</p> <p><b>10</b> 8-beat incrementing burst</p> <p><b>11</b> Reserved</p>                                                                                                                                                                                                                                                                                                            |
| <b>SAMPON</b>     | <p>Data sample enable always on. This bit is suggested to be set to “1” when feedback clock is used and “0” when multiple phase clock is used.</p> <p><b>0</b> Data sample enable not always on</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

1 Data sample enable always on

**FIXDLY** Used for software to select the delay cycle after the clock fixes high for the host controller to SD card.

- 00** 0-T delay
- 01** 1-T delay
- 10** 2-T delay
- 11** 3-T delay

**SAMPLEDLY** Used for software to select the turn around delay cycle between the write data end bit and CRC status for SD card.

- 00** 0-T delay
- 01** 1-T delay
- 10** 2-T delay
- 11** 3-T delay

### MSDC+0018h SD memory card controller IO control register1 MSDC\_IOCON 1

| Bit   | 31 | 30         | 29       | 28 | 27 | 26          | 25       | 24 | 23 | 22         | 21       | 20 | 19 | 18           | 17           | 16 |
|-------|----|------------|----------|----|----|-------------|----------|----|----|------------|----------|----|----|--------------|--------------|----|
| Name  |    |            |          |    |    |             |          |    |    |            |          |    |    | PRCFG_RST/WP | PRVAL_RST/WP |    |
| Type  |    |            |          |    |    |             |          |    |    |            |          |    |    | R/W          | R/W          |    |
| Reset |    |            |          |    |    |             |          |    |    |            |          |    |    | 0            | 10           |    |
| Bit   | 15 | 14         | 13       | 12 | 11 | 10          | 9        | 8  | 7  | 6          | 5        | 4  | 3  | 2            | 1            | 0  |
| Name  |    | PRCFG_G_CK | PRVAL_CK |    |    | PRCFG_G_C_M | PRVAL_CM |    |    | PRCFG_G_DA | PRVAL_DA |    |    | PRCFG_G_IN_S | PRVAL_INS    |    |
| Type  |    | R/W        | R/W      |    |    | R/W         | R/W      |    |    | R/W        | R/W      |    |    | R/W          | R/W          |    |
| Reset |    | 0          | 10       |    |    | 0           | 00       |    |    | 0          | 10       |    |    | 0            | 10           |    |

**PRCFG\_RST/WP** Pull-up/down register configuration for pin RST/WP. Default value: **0**

- 0** Enable pull-up resistor in the I/O pad of pin WP
- 1** Enable pull-down resistor in the I/O pad of pin WP

**PRVAL\_RST/WP** Pull-up/down register value for pin RST/WP. Default value: **10**

- 00** Disable all pull-up and pull-down resistors in the I/O pad of pin WP
- 01** Pull-up/down resistor in the I/O pad of pin WP value is 47k.
- 10** Pull-up/down resistor in the I/O pad of pin WP value is 47k.
- 11** Pull-up/down resistor in the I/O pad of pin WP value is 23.5k.

**PRCFG\_CK** Pull-up/down register configuration for pin CK. Default value: **0**

- 0** Enable pull-up resistor in the I/O pad of pin CK
- 1** Enable pull-down resistor in the I/O pad of pin CK.

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PRVAL_CLK</b> | <p>Pull-up/down register value for pin CLK. Default value: <b>10</b></p> <p><b>00</b> Disable all pull-up and pull-down resistors in the I/O pad of pin CLK</p> <p><b>01</b> Pull-up/down resistor in the I/O pad of pin CLK value is 47k.</p> <p><b>10</b> Pull-up/down resistor in the I/O pad of pin CLK value is 47k.</p> <p><b>11</b> Pull-up/down resistor in the I/O pad of pin CLK value is 23.5k.</p>                       |
| <b>PRCFG_CM</b>  | <p>Pull-up/down register configuration for pin CM. Default value: <b>0</b>.</p> <p><b>0</b> Enable pull-up resistor in the I/O pad of pin CM</p> <p><b>1</b> Enable pull-down resistor in the I/O pad of pin CM</p>                                                                                                                                                                                                                  |
| <b>PRVAL_CM</b>  | <p>Pull-up/down register value for pin CMD/BS. Default value: <b>00</b></p> <p><b>00</b> Disable all pull-up and pull-down resistors in the I/O pad of pin CMD/BS</p> <p><b>01</b> Pull-up/down resistor in the I/O pad of pin CMD/BS value is 47k.</p> <p><b>10</b> Pull-up/down resistor in the I/O pad of pin CMD/BS value is 47k.</p> <p><b>11</b> Pull-up/down resistor in the I/O pad of pin CMD/BS value is 23.5k.</p>        |
| <b>PRCFG_DA</b>  | <p>Pull-up/down register configuration for pin DAT0, DAT1, DAT2 and DAT3. Default value: <b>0</b></p> <p><b>0</b> Enable pull-up resistor in the I/O pad of pin DAT</p> <p><b>1</b> Enable pull-down resistor in the I/O pad of pin DAT</p>                                                                                                                                                                                          |
| <b>PRVAL_DA</b>  | <p>Pull-up/down register value for pin DAT0, DAT1, DAT2 and DAT3. Default value: <b>10</b></p> <p><b>00</b> Disable all pull-up and pull-down resistors in the I/O pad of pin DAT</p> <p><b>01</b> Pull-up/down resistor in the I/O pad of pin DAT value is 47k.</p> <p><b>10</b> Pull-up/down resistor in the I/O pad of pin DAT value is 47k.</p> <p><b>11</b> Pull-up/down resistor in the I/O pad of pin DAT value is 23.5k.</p> |
| <b>PRCFG_INS</b> | <p>Pull-up/down register configuration for pin INS. Default value: <b>0</b></p> <p><b>0</b> Enable pull-up resistor in the I/O pad of pin INS</p> <p><b>1</b> Enable pull-down resistor in the I/O pad of pin INS</p>                                                                                                                                                                                                                |
| <b>PRVAL_INS</b> | <p>Pull-up/down register value for pin INS. Default value: <b>10</b></p> <p><b>00</b> Disable all pull-up and pull-down resistors in the I/O pad of pin INS</p> <p><b>01</b> Pull-up/down resistor in the I/O pad of pin INS value is 47k.</p> <p><b>10</b> Pull-up/down resistor in the I/O pad of pin INS value is 47k.</p> <p><b>11</b> Pull-up/down resistor in the I/O pad of pin INS value is 23.5k.</p>                       |

#### 4.13.3.2 SD Memory Card Controller Register Definition

##### MSDC+0020h SD memory card controller configuration register SDC\_CFG

| Bit   | 31       | 30 | 29 | 28 | 27           | 26 | 25 | 24 | 23   | 22 | 21 | 20 | 19   | 18 | 17        | 16   |
|-------|----------|----|----|----|--------------|----|----|----|------|----|----|----|------|----|-----------|------|
| Name  | DIOC     |    |    |    |              |    |    |    | WDOD |    |    |    | SDIO |    | MDL<br>EN | SIEN |
| Type  | R/W      |    |    |    |              |    |    |    | R/W  |    |    |    | R/W  |    | R/W       | R/W  |
| Reset | 00000000 |    |    |    |              |    |    |    | 0000 |    |    |    | 0    |    | 0         | 0    |
| Bit   | 15       | 14 | 13 | 12 | 11           | 10 | 9  | 8  | 7    | 6  | 5  | 4  | 3    | 2  | 1         | 0    |
| Name  | BSYDLY   |    |    |    | BLKLEN       |    |    |    |      |    |    |    |      |    |           |      |
| Type  | R/W      |    |    |    | R/W          |    |    |    |      |    |    |    |      |    |           |      |
| Reset | 1000     |    |    |    | 000000000000 |    |    |    |      |    |    |    |      |    |           |      |

The register is used for configuring the SD memory card controller when it is configured as the host of SD memory card. *Note that SDC\_CFG[31:16] can be accessed by the 16-bit APB bus.*

**BLKLEN** Block length. Defines the length (in bytes) of one block of a data transaction. Maximum value of block length: 2,048 bytes

000000000000 Reserved

000000000001 1 byte

000000000010 2 bytes

...

011111111111 2,047 bytes

100000000000 2,048 bytes

**BSYDLY** Only valid for commands with R1b response. If the command has a response of R1b type, the SD controller must monitor the data line 0 for card busy status from the bit time that is two serial clock cycles after the command end bit to check if the operation in the SD memory card have finished. This register field is used to expand the time between the command end bit and end of detection period to detect card busy status. If the time is up, and there is no card busy status on data line 0, the controller will abandon the detection.

0000 No extension

0001 Extend 1 more serial clock cycle

0010 Extend 2 more serial clock cycles

...

1111 Extend 15 more serial clock cycles

**SIEN** Enables serial interface. Should be enabled as soon as possible before any command.

0 Disable serial interface for SD

1 Enable serial interface for SD

**SDIO** Enables SDIO

0 Disable SDIO mode

1 Enable SDIO mode

- MDLEN** Enables multiple data line. Can be enabled only when the SD memory card is applied and detected by software application.  
**0** Disable 4-bit data line  
**1** Enable 4-bit data line
- WDOD** Write data output delay. The period from finish of the response for the initial host write command or the last write data block in a multiple block write operation to the start bit of the next write data block requires at least two serial clock cycles. The register field is used to extend the period (Write Data Output Delay) Unit: One serial clock  
**0000** No extension  
**0001** Extend 1 more serial clock cycle  
**0010** Extend 2 more serial clock cycles  
 ...  
**1111** Extend 15 more serial clock cycles
- DTOC** Data time-out counter. The period from finish of the initial host read command or the last read data block in a multiple block read operation to the start bit of the next read data block requires at least two serial clock cycles. The counter is used to extend the period (Read Data Access Time) Unit: 65,536 serial clock. Refer to descriptions of register bit RDINT.  
**00000000** Extend 65,536 more serial clock cycles  
**00000001** Extend 65,536x2 more serial clock cycles  
**00000010** Extend 65,536x3 more serial clock cycles  
 ...  
**11111111** Extend 65,536x 256 more serial clock cycles

#### MSDC+0024h SD memory card controller command register

#### SDC\_CMD

| Bit   | 31          | 30 | 29        | 28           | 27 | 26          | 25            | 24 | 23 | 22 | 21         | 20 | 19 | 18 | 17 | 16              |
|-------|-------------|----|-----------|--------------|----|-------------|---------------|----|----|----|------------|----|----|----|----|-----------------|
| Name  |             |    |           |              |    |             |               |    |    |    |            |    |    |    |    | <b>CMD FAIL</b> |
| Type  |             |    |           |              |    |             |               |    |    |    |            |    |    |    |    | R/W             |
| Reset |             |    |           |              |    |             |               |    |    |    |            |    |    |    |    | 0               |
| Bit   | 15          | 14 | 13        | 12           | 11 | 10          | 9             | 8  | 7  | 6  | 5          | 4  | 3  | 2  | 1  | 0               |
| Name  | <b>STOP</b> |    | <b>RW</b> | <b>DTYPE</b> |    | <b>IDRT</b> | <b>RSPTYP</b> |    |    |    | <b>CMD</b> |    |    |    |    |                 |
| Type  | R/W         |    | R/W       | R/W          |    | R/W         | R/W           |    |    |    | R/W        |    |    |    |    |                 |
| Reset | 0           |    | 0         | 00           |    | 0           | 000           |    |    |    | 000000     |    |    |    |    |                 |

The register defines a SD memory card command and its attribute. Before the SD controller issues a transaction onto SD bus, application shall specify other relative settings such as argument for command. After the application writes the register, the SD controller will issue the corresponding transaction onto SD serial bus. If the command is GO\_IDLE\_STATE, the controller will have serial clock on the SD bus run 128 cycles before issuing the command.



**CMD** SD memory card command. Total 6 bits.

**RSPTYP** Defines response type for the command. For commands with R1 and R1b response, register SDC\_CSTA (not SDC\_STA) will update after response token is received. Register SDC\_CSTA contains the status of the SD and it will be used as response interrupt sources. *Note that if CMD7 is used with all 0's RCA then RSPTYP must be "000".* Command "GO\_TO\_IDLE" also have RSPTYP = '000'.

- 000** No response for the command. For instance, broadcast command without response and GO\_INACTIVE\_STATE command.
- 001** The command has R1 response. R1 response token is 48-bit.
- 010** The command has R2 response. R2 response token is 136-bit.
- 011** The command has R3 response. Even though R3 is 48-bit response, but it does not contain CRC checksum.
- 100** Reserved.
- 101** Reserved.
- 110** The command has R6 response. R6 response token is 48-bit.
- 111** The command has R1b response. If the command has a response of R1b type, SD controller must monitor the data line 0 for card busy status from the bit time that is two or four serial clock cycles after the command end bit to check if the operation in the SD memory card is finished. There are two cases for detecting the card busy status. The first case is that the host stops the data transmission during an active write data transfer. The card will assert busy signal after the stop transmission command end bit followed by four serial clock cycles. The second case is that the card is in the idle state or under a scenario of receiving a stop transmission command between data blocks when multiple block write command is in progress. The register bit is valid only when the command has a response token.

**IDRT** Identification response time. The register bit indicates if the command has a response with  $N_{ID}$  (i.e. 5 serial clock cycles as defined in SD Memory Card Specification Part 1 Physical Layer Specification version 1.0) response time. The register bit is valid only when the command has a response token. Thus the register bit must be set to '1' for CMD2 (ALL\_SEND\_CID) and ACMD41 (SD\_APP\_OP\_CMD).

- 0** Otherwise
- 1** The command has a response with  $N_{ID}$  response time.

**DTYPE** Defines data token type for the command

- 00** No data token for the command
- 01** Single block transaction
- 10** Multiple block transaction, i.e. the command is a multiple-block read or write command.
- 11** Reserved.



**RW** Defines the command is a read command or write command. The register bit is valid only when the command will cause a transaction with data token.

- 0** The command is a read command.
- 1** The command is a write command.

**STOP** Indicates if the command is a stop transmission command.

- 0** The command is not a stop transmission command.
- 1** The command is a stop transmission command.

### MSDC+0028h SD memory card controller argument register

**SDC\_ARG**

| Bit  | 31          | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|------|-------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Name | ARG [31:16] |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type | R/W         |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Bit  | 15          | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Name | ARG [15:0]  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type | R/W         |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |

The register contains the argument of the SD memory card command.

### MSDC+002C h

### SD memory card controller status register

**SDC\_STA**

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4                | 3                 | 2                 | 1                 | 0                 |
|-------|----|----|----|----|----|----|---|---|---|---|---|------------------|-------------------|-------------------|-------------------|-------------------|
| Name  | WP |    |    |    |    |    |   |   |   |   |   | FED<br>TBUS<br>Y | FEC<br>MDB<br>USY | BED<br>ATBU<br>SY | BEC<br>MDB<br>USY | BES<br>DCB<br>USY |
| Type  | R  |    |    |    |    |    |   |   |   |   |   | RO               | RO                | RO                | RO                | RO                |
| Reset | -  |    |    |    |    |    |   |   |   |   |   | 0                | 0                 | 0                 | 0                 | 0                 |

The register contains various status of the SD controller as the controller is configured as the host of the SD memory card.

**BESDCBUSY** Indicates if the SD controller is busy, i.e. any transmission is going on CMD or DAT line on the SD bus. This bit shows backend controller's SDC busy state. The busy state is sync from card clock domain to bus clock domain.

- 0** Backend SD controller is idle.
- 1** Backend SD controller is busy.

**BECMDBUSY** Indicates if any transmission is going on CMD line on the SD bus. This bit shows backend controller's CMD busy state. The busy state is sync from the card clock domain to bus clock domain.

- 0** Backend SDC controller gets the info that no transmission is going on CMD line on the SD bus.

- 1 Backend SDC controller gets the info that there exists transmission going on CMD line on the SD bus.

**BEDATBUSY**

Indicates if any transmission is going on DAT line on the SD bus

- 0 Backend SDC controller gets the info that no transmission is going on DAT line on SD the bus.
- 1 Backend SDC controller gets the info that there exists transmission going on DAT line on the SD bus.

**FECMDBUSY**

Indicates if any transmission is going on CMD line on the SD bus. This bit indicates directly the CMD line at the card clock domain.

- 0 No transmission is going on CMD line on the SD bus.
- 1 There exists transmission going on CMD line on the SD bus.

**FEDATBUSY**

Indicates if any transmission is going on DAT line on the SD bus. This bit indicates directly the CMD line at the card clock domain. For those commands without data but still involving DAT line, the register bit is useless. For example, if an Erase command is issued, checking if the register bit is '0' before issuing the next command with data will not guarantee the controller is idle. In this situation, use register bit BESDCBUSY.

- 0 No transmission is going on DAT line on the SD bus.
- 1 There exists transmission going on DAT line on the SD bus.

**WP**

Detects the status of Write Protection Switch on the SD memory card. The register bit shows the status of Write Protection Switch on the SD memory card. There is no default reset value. The WP (Write Protection) pin is also only useful when the controller is configured for the SD memory card.

- 0 Write Protection Switch off. It means the memory card is writable.
- 1 Write Protection Switch on. It means the memory card is desired to be write-protected.

**MSDC+0030h SD memory card controller response register 0** **SDC\_RESP0**

| Bit  | 31                  | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|------|---------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Name | <b>RESP [31:16]</b> |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type | RO                  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Bit  | 15                  | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Name | <b>RESP [15:0]</b>  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type | RO                  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |

The register contains parts of the last SD memory card bus response. See the descriptions of register field SDC\_RESP3.

**MSDC+0034h SD memory card controller response register 1** **SDC\_RESP1**

| Bit  | 31           | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|------|--------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Name | RESP [63:48] |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type | RO           |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Bit  | 15           | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Name | RESP [47:32] |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type | RO           |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |

The register contains parts of the last SD memory card bus response. See the descriptions of register field SDC\_RESP3.

**MSDC+0038h SD memory card controller response register 2** **SDC\_RESP2**

| Bit  | 31           | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|------|--------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Name | RESP [95:80] |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type | RO           |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Bit  | 15           | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Name | RESP [79:64] |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type | RO           |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |

The register contains parts of the last SD memory card bus response. See the descriptions of register field SDC\_RESP3.

**MSDC+003Ch SD memory card controller response register 3** **SDC\_RESP3**

| Bit  | 31             | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|------|----------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Name | RESP [127:112] |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type | RO             |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Bit  | 15             | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Name | RESP [111:96]  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type | RO             |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |

The register contains parts of the last SD memory card bus response. Register fields SDC\_RESP0, SDC\_RESP1, SDC\_RESP2 and SDC\_RESP3 compose the last SD memory card bus response. For response of type R2, i.e. response of the command ALL\_SEND\_CID, SEND\_CSD and SEND\_CID, only bit 127 to 0 of response token are stored in register field SDC\_RESP0, SDC\_RESP1, SDC\_RESP2 and SDC\_RESP3. For response of other types, only bit 39 to 8 of response token are stored in register field SDC\_RESP0.

**MSDC+0040h SD memory card controller command status register**

**SDC\_CMDSTA**

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2                 | 1         | 0          |
|-------|----|----|----|----|----|----|---|---|---|---|---|---|---|-------------------|-----------|------------|
| Name  |    |    |    |    |    |    |   |   |   |   |   |   |   | RSPC<br>RCE<br>RR | CMD<br>TO | CMD<br>RDY |
| Type  |    |    |    |    |    |    |   |   |   |   |   |   |   | RC                | RC        | RC         |
| Reset |    |    |    |    |    |    |   |   |   |   |   |   |   | 0                 | 0         | 0          |

The register contains the status of SD controller during command execution and that of SD bus protocol after command execution when SD controller is configured as the host of SD memory card. The register will also be used as interrupt sources. The register will be cleared when being read. Meanwhile, if the interrupt is enabled and thus interrupt caused by the register is generated, reading the register will deassert the interrupt.

**CMDRDY** For command without response, the register bit will be '1' once the command is completed on the SD bus. For command with response, the register bit will be '1' whenever the command is issued onto SD bus and its corresponding response is received without CRC error.

**0** Otherwise

**1** Command with/without response finish successfully without CRC error.

**CMDTO** Time-out on CMD detected. '1' indicates that SD controller detected a timeout condition while waiting for a response on the CMD line.

**0** Otherwise

**1** SD controller detected a timeout condition while waiting for a response on the CMD line.

**RSPCRCERR** CRC error on CMD detected. '1' indicates that the SD controller detected a CRC error after reading a response from the CMD line. The register will also update in the clock auto calibration mode. After clock auto calibration, the software should read and clear the register.

**0** Otherwise

**1** SD controller detected a CRC error after reading a response from the CMD line.

**MSDC+0044h SD memory card controller data status register**

**SDC\_DATSTA**

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9         | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1         | 0               |
|-------|----|----|----|----|----|----|-----------|---|---|---|---|---|---|---|-----------|-----------------|
| Name  |    |    |    |    |    |    | DATCRCERR |   |   |   |   |   |   |   | DAT<br>TO | BLK<br>DON<br>E |
| Type  |    |    |    |    |    |    | RC        |   |   |   |   |   |   |   | RC        | RC              |
| Reset |    |    |    |    |    |    | 0         |   |   |   |   |   |   |   | 0         | 0               |

The register contains the status of SD controller during data transfer on DAT line(s) when the SD controller is configured as the host of SD memory card. The register also will be used as interrupt sources. The register will be cleared when being read. Meanwhile, if interrupt is enabled and thus interrupt caused by the register is generated, reading the register will deassert the interrupt.

- BLKDONE** Indicates the status of data block transfer  
**0** Otherwise.  
**1** A data block is successfully transferred.
- DATTO** Time-out on DAT detected. '1' indicates that SD controller detected a timeout condition while waiting for data token on the DAT line.  
**0** Otherwise  
**1** SD controller detected a timeout condition while waiting for data token on the DAT line.
- DATCRCERR[n]** CRC error on DAT detected. '1' indicates that SD controller detected a CRC error for bit n after reading a block of data from the DAT line or SD signaled a CRC error after writing a block of data to the DAT line. The register will also update in the clock auto calibration mode. After clock auto calibration, the software should read and clear the register.  
**0** Otherwise  
**1** SD controller detected a CRC error after reading a block of data from the DAT line or SD signaled a CRC error after writing a block of data to the DAT line.

*Note: n is 7~0 for 8-bit mode. Each bit is read and cleared individually.*

### MSDC+0048h SD memory card status register

### SDC\_CSTA

| Bit   | 31               | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|-------|------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Name  | CSTA [31:16]     |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  | RC               |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset | 0000000000000000 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Bit   | 15               | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Name  | CSTA [15:0]      |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  | RC               |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset | 0000000000000000 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |

After commands with R1 and R1b response this register contains the status of the SD card, and it will be used as response interrupt sources. In all register fields, logic high indicates error, and logic low indicates there is no error. The register will be cleared when being read. Meanwhile, if the interrupt is enabled and thus interrupt caused by the register is generated, reading the register will deassert the interrupt.

- CSTA31** OUT\_OF\_RANGE. The command's argument is out of the allowed range for this card.

- CSTA30** ADDRESS\_ERROR. A misaligned address that does not match the block length is used in the command.
- CSTA29** BLOCK\_LEN\_ERROR. The transferred block length is not allowed for this card, or the number of transferred bytes does not match the block length.
- CSTA28** ERASE\_SEQ\_ERROR. An error in the sequence of erase commands occurs.
- CSTA27** ERASE\_PARAM. An invalid selection of write-blocks for erase occurs.
- CSTA26** WP\_VIOLATION. Attempt to program a write-protected block.
- CSTA25** Reserved. Returns to zero.
- CSTA24** LOCK\_UNLOCK\_FAILED. Set when a sequence or password error has been detected in lock/unlock card command or if there is an attempt to access a locked card.
- CSTA23** COM\_CRC\_ERROR. CRC check of the previous failed command.
- CSTA22** ILLEGAL\_COMMAND. Command not legal for the card state.
- CSTA21** CARD\_ECC\_FAILED. Card internal ECC is applied but fails to correct the data.
- CSTA20** CC\_ERROR. Internal card controller error.
- CSTA19** ERROR. A general or unknown error occurs during the operation.
- CSTA18** UNDERRUN. The card cannot sustain data transfer in stream read mode.
- CSTA17** OVERRUN. The card cannot sustain data programming in stream write mode.
- CSTA16** CID/CSD\_OVERWRITE. It can be either one of the following errors: 1) The CID register has already been written and cannot be overwritten. 2) The read-only section of CSD does not match the card. 3) An attempt to reverse the copy (set as original) or permanent WP (unprotected) bits is made.
- CSTA[15:4]** Reserved. Returns to zero.
- CSTA3** AKE\_SEQ\_ERROR. Error in the sequence of authentication process
- CSTA[2:0]** Reserved. Returns to zero.

**MSDC+004C**  
**h SD memory card IRQ mask register 0**
**SDC\_IRQMAS**  
**K0**

| Bit   | 31               | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|-------|------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Name  | IRQMASK [31:16]  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  | R/W              |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset | 0000000000000000 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Bit   | 15               | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Name  | IRQMASK [15:0]   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  | R/W              |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset | 0000000000000000 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |

The register contains parts of the SD memory card interrupt mask register. Refer to the register descriptions of register SDC\_IRQMASK1. The register will mask interrupt sources from register SDC\_CMDSTA and SDC\_DATSTA. IRQMASK[3:0] is for SDC\_CMDSTA and IRQMASK[6:4] for SDC\_DATSTA, *Note that IRQMASK0[6] will mask the SDC\_DATSTA[9:2] together.* '1' in some bits of the register will mask the corresponding interrupt source with the same bit position. For example, if IRQMASK[0] is '1' then interrupt source from register field CMDRDY of register SDC\_CMDSTA will be masked. '0' in some bits will not cause interrupt mask on the corresponding interrupt source from register SDC\_CMDSTA and SDC\_DATSTA.

### MSDC+0050h SD memory card IRQ mask register 1

### SDC\_IRQMASK1

| Bit   | 31               | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|-------|------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Name  | IRQMASK [63:48]  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  | R/W              |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset | 0000000000000000 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Bit   | 15               | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Name  | IRQMASK [47:32]  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  | R/W              |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset | 0000000000000000 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |

The register contains parts of the SD memory card interrupt mask register. Registers SDC\_IRQMASK1 and SDC\_IRQMASK0 compose the SD memory card interrupt mask register. The register will mask interrupt sources from register SDC\_CSTA. '1' in some bits of the register will mask the corresponding interrupt source with the same bit position. For example, if IRQMASK[63] is '1' then interrupt source from register field OUT\_OF\_RANGE of register SDC\_CSTA will be masked. '0' in some bits will not cause interrupt mask on the corresponding interrupt source from register SDC\_CSTA.

### MSDC+0054h SDIO configuration register

### SDIO\_CFG

| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18         | 17         | 16    |
|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|------------|------------|-------|
| Name  |    |    |    |    |    |    |    |    |    |    |    |    |    |            |            |       |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |            |            |       |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |            |            |       |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2          | 1          | 0     |
| Name  |    |    |    |    |    |    |    |    |    |    |    |    |    | DSBS<br>EL | INTS<br>EL | INTEN |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    | R/W        | R/W        | R/W   |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    | 0          | 0          | 0     |

The register is used to configure functionality for SDIO.



**INTEN**      Enables interrupt for SDIO  
               **0**    Disable  
               **1**    Enable

**INTSEL**    Selects interrupt signal  
               **0**    Use data line 1 as interrupt signal  
               **1**    Use data line 5 as interrupt signal

**DSBSEL**    Selects data block start bit just for SDIO enable mode. If 1-bit mode is used, set it to 0.  
               **0**    Use data line 0 as start bit of data block and other data lines are ignored.  
               **1**    Start bit of a data block is received only when data line 0-3 all become low.

### MSDC+0058h SDIO status register

### SDIO\_STA

| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16         |
|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|------------|
| Name  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |            |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |            |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |            |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0          |
| Name  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | <b>IRQ</b> |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | RO         |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | 0          |

This register is used to identify if there is SDIO interrupt during the interrupt period on data line.

**IRQ**      SDIO interrupt exists on the data line. For example, during the interrupt period or in the 1-bit data line mode and DAT1/5 goes low from high, this bit will become 1 from 0. If DAT1/5 goes high from low, this bit will become 0 from 1.  
               **0**    There is no SDIO interrupt existing on the data line.  
               **1**    There is SDIO interrupt existing on the data line.

### MSDC+0080h CLK latching edge configuration register

### CLKPAD\_RED

| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23                          | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|-------|----|----|----|----|----|----|----|----|-----------------------------|----|----|----|----|----|----|----|
| Name  |    |    |    |    |    |    |    |    |                             |    |    |    |    |    |    |    |
| Type  |    |    |    |    |    |    |    |    |                             |    |    |    |    |    |    |    |
| Reset |    |    |    |    |    |    |    |    |                             |    |    |    |    |    |    |    |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7                           | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Name  |    |    |    |    |    |    |    |    | <b>CLKP<br/>AD_R<br/>ED</b> |    |    |    |    |    |    |    |
| Type  |    |    |    |    |    |    |    |    | R/W                         |    |    |    |    |    |    |    |
| Reset |    |    |    |    |    |    |    |    | 0                           |    |    |    |    |    |    |    |

**CLKPAD\_RED** Determines the input data from card is latched at the falling edge or rising edge of the feedback clock from pad. The suggested setting is “1” for the SD serial clock less than 25MHz and “0” for the SD serial clock higher than 25MHz. The suggested setting is “0” for MMC card no matter the serial clock rate is high speed or default speed.

**0** Rising edge to latch data  
**1** Falling edge to latch data

#### 4.13.4 Application Notes

##### 4.13.4.1 Initialization Procedures After Power-on

Disable power down control for MSDC module. Be sure to power on the MSDC module before starting any operation to it.

##### 4.13.4.2 Card Detection Procedures

The pseudo code is as the followinh:

```
MSDC_CFG.PRCFG0 = 2'b10
MSDC_PS = 2'b11
MSDC_CFG.VDDPD = 1
if(MSDC_PS.PINCHG) { // card is inserted
 . . .
}
```

The pseudo code segment performs the following tasks:

1. First pull up the CD/DAT3 (INS) pin.
2. Enable card detection and input pin at the same time.
3. Turn on the power for memory card.
4. Detect insertion of memory card.

##### 4.13.4.3 Notes On Commands

For SD, if the command desired to be issued involves data line, for example, commands with data transfer or R1b response, check if SDC\_STA.SDCBUSY is '0' before issuing. If the command desired to be issued does not involve data line, only check if SDC\_STA.CMDBUSY is '0' before issuing.

##### 4.13.4.4 Notes On Data Transfer

- For SD, if multiple-block-write command is issued then only issue the STOP\_TRANS command inter-blocks instead of intra-blocks.
- Once the software decides to issue STOP\_TRANS commands, there should be no more data transferred from or to the controller.

#### 4.13.4.5 Notes On Frequency Change

Before changing the frequency of serial clock on SD bus, it is necessary to disable the serial interface of the controller. That is, set register bit SIEN of register SDC\_CFG to '0' for SD controller. The serial interface of the controller needs to be enabled again before starting any operation to the memory card.

#### 4.13.4.6 Notes On Response Time-out

If a read command does not receive response, that is, it is terminated with a time-out, then register SDC\_DATSTA needs to be cleared by reading it. Register bit "DATTO" should be active. However, it may take a while before the register bit becomes active. The alternative is to send the STOP\_TRANS command. However, this method will receive a response with illegal-command information. Moreover, be sure to check if register bit SDC\_STA.CMDBUSY is active before issuing the STOP\_TRANS command. The procedure is as the following:

1. Read command => Response time-out
2. Issue STOP\_TRANS command => Get response
3. Read register SDC\_DATSTA to clear it.

#### 4.13.4.7 Source or Destination Address Not Word-aligned

It is possible that the source address is not word-aligned when the data are moved from the memory to MSDC. Similarly, the destination address may be not word-aligned when the data are moved from MSDC to the memory. This can be solved by setting up the DMA byte-to-word function.

1. DMA<sub>n</sub>\_CON.SIZE = 0
2. DMA<sub>n</sub>\_CON.BTW = 1
3. DMA<sub>n</sub>\_CON.BURST = 2 (or 4)
4. DMA<sub>n</sub>\_COUNT = byte number instead of word number
5. FIFO threshold setting must be 1 (or 2), depending on DMA<sub>n</sub>\_CON.BURST

Note:  $n = 4 \sim 11$

#### 4.13.4.8 Miscellaneous Notes

##### Siemens MMC card

When a write command is issued and followed by a STOP\_TRANS command, Siemens MMC card will de-assert busy status even though the flash programming has not yet been finished. The software must use "Get Status" command to make sure the flash programming is finished.

## 4.14 BTIF

### 4.14.1 General Description

Bluetooth Interface (BTIF) is designed in SOC (BT+GSM) in order to be instead of the UART interface between the BT chip and the baseband chip. As in the UART design, BTIF is an APB slave which transmits or receives data by MCU access or through DMA/VFIFO.

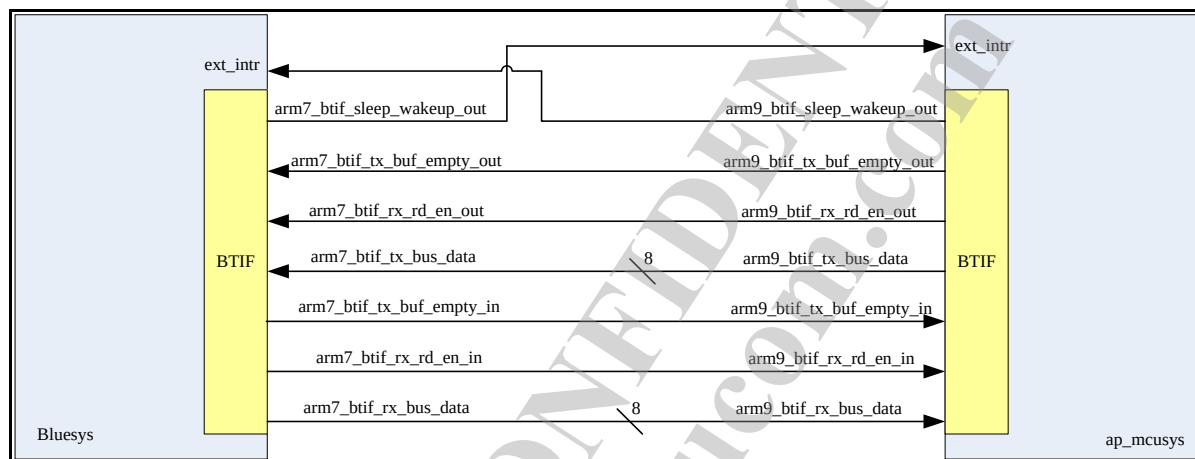


Figure 52. Interface connection between BT and Baseband system

### 4.14.2 Register Definition

#### BTIF+0000h Rx buffer register

#### BTIF\_RBR

| Bit  | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0        |
|------|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|----------|
| Name |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | RBR[7:0] |
| Type |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | RO       |

**RBR** Rx buffer register. A read-only register. The received data can be read by accessing this register. This register is valid only when BTIF\_FAKELCR[7] (0x0C) is 0.

#### BTIF+0000h Tx holding register

#### BTIF\_THR

| Bit  | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0        |
|------|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|----------|
| Name |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | THR[7:0] |
| Type |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | WO       |

**THR** Tx holding register. A write-only register. The data to be transmitted are written to this register and sent to the Bluetooth via BTIF. This register is valid only when BTIF\_FAKELCR[7] (0x0C) is 0.

### BTIF+0004h Interrupt enable register

### BTIF\_IER

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1     | 0     |
|-------|----|----|----|----|----|----|---|---|---|---|---|---|---|---|-------|-------|
| Name  |    |    |    |    |    |    |   |   |   |   |   |   |   |   | TXEEN | RXFEN |
| Type  |    |    |    |    |    |    |   |   |   |   |   |   |   |   | W/R   | W/R   |
| Reset |    |    |    |    |    |    |   |   |   |   |   |   |   |   | 0     | 0     |

This register is valid only when BTIF\_FAKELCR[7] is 0.

**TXEEN** Enables Tx empty interrupt. When set to 1, an interrupt will be generated if the Tx holding register is empty.

**0** No interrupt will be generated if the Tx holding register is empty.

**1** An interrupt will be generated if the Tx holding register is empty

**RXFEN** Enables Rx full interrupt. When set to 1, an interrupt will be generated if the Rx buffer contains data.

**0** No interrupt will be generated if the Rx buffer contains data.

**1** An interrupt will be generated if the Rx buffer contains data.

### BTIF+0008h Interrupt identification register

### BTIF\_IIR

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3   | 2   | 1   | 0    |
|-------|----|----|----|----|----|----|---|---|---|---|---|---|-----|-----|-----|------|
| Name  |    |    |    |    |    |    |   |   |   |   |   |   | ID2 | ID1 | ID0 | NINT |
| Type  |    |    |    |    |    |    |   |   |   |   |   |   | RO  | RO  | RO  | RO   |
| Reset |    |    |    |    |    |    |   |   |   |   |   |   | 0   | 0   | 0   | 1    |

This register is valid only when BTIF\_FAKELCR (0x0C) is not 0xBF.

**IIR** Identifies if there are pending interrupts. The following table lists the IIR[5:0] codes associated with the possible interrupts:

| IIR[3:0] | Priority level | Interrupt                 | Source                             |
|----------|----------------|---------------------------|------------------------------------|
| 0001     | -              | No pending interrupt      |                                    |
| 0100     | 1              | Rx data received          | Rx data received                   |
| 1100     | 2              | Rx data time-out          | Time-out on character in Rx buffer |
| 0010     | 3              | Tx holding register empty | Tx holding register empty.         |

**Table 68. IIR[5:0] codes associated with the possible interrupts**

### Rx data received interrupt

A Rx received interrupt (IIR[3:0] = 0x04) is generated when RXFEN (IER[0]) is set and Rx data are placed in the Rx buffer register. The interrupt is cleared by reading the Rx buffer register.

#### Rx data time-out interrupt

The Rx data time-out interrupt will be generated if all of the following conditions are applied:

1. Rx buffer is empty.
2. The most recent character is received longer than (RTOCNT\*bclk period\*4).
3. RXFEN (IER[0]) is set to 1.

The time-out timer is restarted upon receipt of a new byte from the Rx shift register. This interrupt is only valid while VFIFO is used. This register is cleared by reading the VFIFO status register (0x4C).

#### Tx holding register empty

A Tx holding register empty interrupt (IIR[3:0] = 0x02) is generated when TXEEN(IER[1]) is set and no data are placed in the Tx holding register. This interrupt is cleared by writing data into BTIF\_THR (0x00).

**BTIF+0008h      FIFO\_CTRL** **BTIF\_FIFOCR**  
**RL**

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2    | 1    | 0 |
|-------|----|----|----|----|----|----|---|---|---|---|---|---|---|------|------|---|
| Name  |    |    |    |    |    |    |   |   |   |   |   |   |   | CLRT | CLRR |   |
| Type  |    |    |    |    |    |    |   |   |   |   |   |   |   | WO   | WO   |   |
| Reset |    |    |    |    |    |    |   |   |   |   |   |   |   | 0    | 0    |   |

This register is valid only when BTIF\_FAKELCR (0x0C) is not 0xBF.

**CLRT**      Clears transmit FIFO. This bit is self-clearing.

- 0**      Leave Tx FIFO intact.  
**1**      Clear all the bytes in Tx FIFO.

**CLRR**      Clears receive FIFO. This bit is self-clearing.

- 0**      Leave Rx FIFO intact.  
**1**      Clear all the bytes in Rx FIFO.

**BTIF+000Ch      FAKE LCR** **BTIF\_FAKELCR**  
**CR**

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7            | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------|----|----|----|----|----|----|---|---|--------------|---|---|---|---|---|---|---|
| Name  |    |    |    |    |    |    |   |   | FAKELCR[7:0] |   |   |   |   |   |   |   |
| Type  |    |    |    |    |    |    |   |   | R/W          |   |   |   |   |   |   |   |
| Reset |    |    |    |    |    |    |   |   | 0            | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

**FAKELCR** This register is added to synchronize the software control method of UART. When FAKELCR[7] is 1, RBR(0x00), THR(0x00) and IER(0x04) will not be readable/writable. When FAKELCR is 0xBF, RBR(0x00), THR(0x00), IER(0x04), IIR(0x08) and LSR(0x14) will not be readable/writable.

### BTIF+0014h Line status register

### BTIF\_LSR

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6    | 5    | 4 | 3 | 2 | 1 | 0  |
|-------|----|----|----|----|----|----|---|---|---|------|------|---|---|---|---|----|
| Name  |    |    |    |    |    |    |   |   |   | TEMT | THRE |   |   |   |   | DR |
| Type  |    |    |    |    |    |    |   |   |   | RO   | RO   |   |   |   |   | RO |
| Reset |    |    |    |    |    |    |   |   |   | 1    | 1    |   |   |   |   | 0  |

**LSR** Line status register. Readable when LCR != 0xBF.

**TEMT** Tx holding register is empty.  
**0** Empty conditions are not met.  
**1** This bit is set when the Tx holding register is empty.

**THRE** Indicates if Tx FIFO is reduced to its trigger level  
**0** Reset whenever the contents of Tx FIFO are more than its trigger level (FIFOs are enabled)  
**1** Set whenever the contents of Tx FIFO are reduced to its trigger level (FIFOs are enabled)

**DR** Data Ready  
**0** Cleared by reading the Rx buffer.  
**1** Set by the Rx buffer becoming full.

### BTIF+0048h Sleep enable register

### BTIF\_SLEEP\_EN

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0        |
|-------|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|----------|
| Name  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | SLEEP_EN |
| Type  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | R/W      |
| Reset |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | 0        |

**SLEEP\_EN** For sleep mode issue  
**0** Does not deal with sleep mode indication signal  
**1** Activate flow control according to software initial settings when the chip enters the sleep mode. Release hardware flow when the chip wakes up.



## BTIFn+004Ch DMA enable register

### BTIF\_DMA\_EN

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2                  | 1         | 0         |
|-------|----|----|----|----|----|----|---|---|---|---|---|---|---|--------------------|-----------|-----------|
| Name  |    |    |    |    |    |    |   |   |   |   |   |   |   | TO_CNT_A<br>UTORST | TX_DMA_EN | RX_DMA_EN |
| Type  |    |    |    |    |    |    |   |   |   |   |   |   |   | R/W                | R/W       | R/W       |
| Reset |    |    |    |    |    |    |   |   |   |   |   |   |   | 0                  | 0         | 0         |

**RX\_DMA\_EN** RX\_DMA mechanism enable signal  
**0** Does not use DMA in Rx.  
**1** Use DMA in Rx. When this register is enabled, the flow control is based on the DMA threshold and generates a time-out interrupt

**TX\_DMA\_EN** TX\_DMA mechanism enable signal  
**0** Does not use DMA in Tx.  
**1** Use DMA in Tx. When this register is enabled, the flow control is based on the DMA threshold and generates a time-out interrupt for DMA.

**TO\_CNT\_AUTORST** Time-out counter auto reset register  
**0** After Rx time-out takes place, the software shall reset the interrupt by reading BTIF 0x4C.  
**1** The time-out counter will be auto reset.

## BTIF+0054h Rx time-out count

### BTIF\_RTOCNT

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0           |
|-------|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|-------------|
| Name  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | RTOCNT[7:0] |
| Type  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | R/W         |
| Reset |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | 0x40        |

**RTOCNT** Used for Rx time-out interrupt. The Rx time-out interrupt will be generated when:  
 1. RXFEN (0x04[0]) is set to 1.  
 2. Rx buffer is empty.  
 3. The most recent character is received longer than (RTOCNT\*bclk period\*4).

## BTIF+0060h TRX\_TRIGGER\_LEVEL

### BTIF\_TRI\_LVL

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7         | 6 | 5          | 4 | 3 | 2 | 1          | 0 |
|-------|----|----|----|----|----|----|---|---|-----------|---|------------|---|---|---|------------|---|
| Name  |    |    |    |    |    |    |   |   | BTIF_LOOP |   | RX_TRI_LVL |   |   |   | TX_TRI_LVL |   |
| Type  |    |    |    |    |    |    |   |   | R/W       |   | R/W        |   |   |   | R/W        |   |
| Reset |    |    |    |    |    |    |   |   | 0x0       |   | 0x5        |   |   |   | 0x0a       |   |

- TX\_TRI\_LVL**      Used for Tx FIFO trigger threshold. THRE(0x14[5]) will be set if the data in the TXFIFO are less than TX\_TRI\_LVL.
- RX\_TRI\_LVL**      Used for Rx FIFO trigger threshold. A Rx trigger interrupt (IIR(0x08)) = 4) might be set if the data in the RXFIFO are more than RX\_TRI\_LVL. The output flow control signal will also be set if the data in the RXFIFO are more than RX\_TRI\_LVL.
- BTIF\_LOOP**      This register is used to enable BTIF loop back mode. The data output from Tx will be received by Rx.

#### BTIF+0064h SLEEP\_WAKEUP

#### BTIF\_WAK

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0          |
|-------|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|------------|
| Name  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | SLEEP_WAKE |
| Type  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | WO         |
| Reset |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   | 1          |

- SLEEP\_WAKE**      ARM9 side btif\_sleep\_wakeup\_in\_b is connected to eint[16] (ARM9 has eint[19:0]).  
 ARM7 side btif\_sleep\_wakeup\_in\_b is connected to eint[0] (ARM7 has eint[3:0])

## 4.15 FM FSPI Master

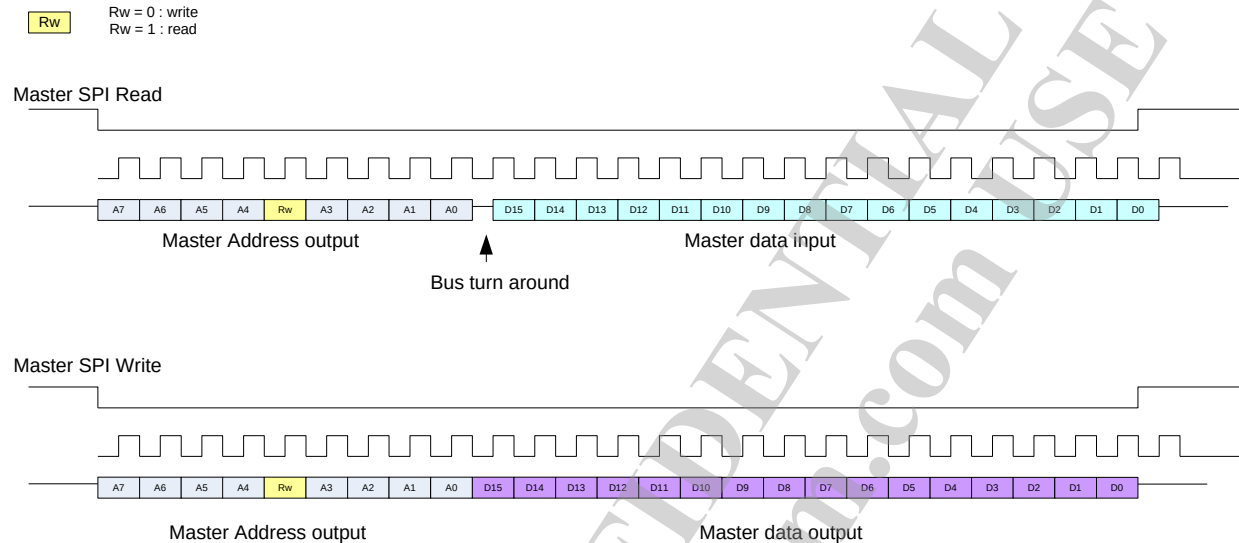
### 4.15.1 General Description

FSPI is an internal SPI master that provides access to the FMSYS register files. The protocol is proprietary and previously defined between the FSPI master controller and the FMSYS slave controller. All FMSYS register reads/writes are operated based on this controller which provides the following features:

- Basic single read or write access
- Supports multiple reads of up to 8 data reads within 1 SPI read transfer

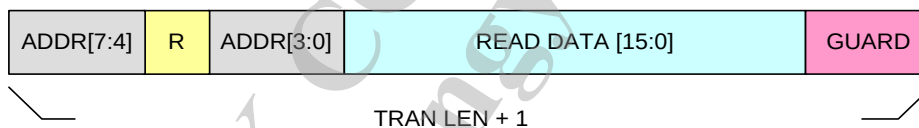
The multiple-read accesses can be utilized to off-load the MCU's control effort when accessing large data buffers such as RDS content from FMSYS. The internal read buffer is a FIFO with maximum depth of 5 entries.

## 4.15.2 Single RD/WR SPI Timing Diagram



## 4.15.3 Multiple Read Configuration

For the multiple-read access mode, the parameters can be set as the following:



## 4.15.4 Register Map

| Address  | Name            | Width | Register function     |
|----------|-----------------|-------|-----------------------|
| A0770000 | <b>CTRL</b>     | 32    | Main control          |
| A0770004 | <b>ADDR</b>     | 32    | Address + command     |
| A0770008 | <b>WR DATA</b>  | 32    | Write data port       |
| A077000C | <b>RD DATA</b>  | 32    | Read data port        |
| A0770010 | <b>CONFIG1</b>  | 32    | Config det 1          |
| A0770014 | <b>CONFIG2</b>  | 32    | Config det 2          |
| A0770020 | <b>MODE SEL</b> | 32    | FMSYS HCI mode select |
| A0770030 | <b>RESET</b>    | 32    | Soft reset            |
| A0770034 | <b>DEBUG1</b>   | 32    | Debug 1               |
| A0770038 | <b>DEBUG2</b>   | 32    | Debug 2               |
| A077003C | <b>DEBUG3</b>   | 32    | Debug 3               |

| Address  | Name          | Width | Register function              |
|----------|---------------|-------|--------------------------------|
| A0770040 | <b>DEBUG4</b> | 32    | Debug 4                        |
| A0770048 | <b>DEBUG5</b> | 32    | Debug 5                        |
| A0770050 | <b>RRCCAL</b> | 32    | R/RC calibration value from BT |

#### 4.15.5 Register Definition

**A0770000 CTRL Main control 00000000**

| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16           |
|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|--------------|
| Mne   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |              |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |              |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |              |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0            |
| Mne   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | <b>START</b> |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | R/W          |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | 0            |

| Bit(s) | Mnemonic     | Name  | Description                                                                                                                                                                                     |
|--------|--------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0      | <b>START</b> | START | <b>Set up this bit to start the SPI command sequence.</b><br>The bit will be auto-cleared upon completion of the SPI command. Therefore, this bit can also be polled as a completion indicator. |

**A0770004 ADDR Address + command 00000000**

| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24           | 23          | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|-------|----|----|----|----|----|----|----|--------------|-------------|----|----|----|----|----|----|----|
| Mne   |    |    |    |    |    |    |    |              |             |    |    |    |    |    |    |    |
| Type  |    |    |    |    |    |    |    |              |             |    |    |    |    |    |    |    |
| Reset |    |    |    |    |    |    |    |              |             |    |    |    |    |    |    |    |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8            | 7           | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Mne   |    |    |    |    |    |    |    | <b>RD_WR</b> | <b>ADDR</b> |    |    |    |    |    |    |    |
| Type  |    |    |    |    |    |    |    | R/W          | R/W         |    |    |    |    |    |    |    |
| Reset |    |    |    |    |    |    |    | 0            | 0           | 0  | 0  | 0  | 0  | 0  | 0  | 0  |

| Bit(s) | Mnemonic     | Name       | Description                                                                                         |
|--------|--------------|------------|-----------------------------------------------------------------------------------------------------|
| 7:0    | <b>ADDR</b>  | FMSYS ADDR | <b>Sets up the FMSYS register address.</b>                                                          |
| 8      | <b>RD_WR</b> | RD_WR CMD  | <b>Indicates if the transaction is a master read or master write transaction</b><br>0: Master write |

| Bit(s) | Mnemonic | Name | Description    |
|--------|----------|------|----------------|
|        |          |      | 1: Master read |

**A0770008 WR DATA Write data port 00000000**

| Bit   | 31     | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|-------|--------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Mne   |        |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  |        |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset |        |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Bit   | 15     | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Mne   | WRDATA |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  | R/W    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |

| Bit(s) | Mnemonic | Name    | Description |
|--------|----------|---------|-------------|
| 15:0   | WRDATA   | WR_DATA | Write data  |

**A077000C RD DATA Read data port 00000000**

| Bit   | 31     | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|-------|--------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Mne   |        |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  |        |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset |        |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Bit   | 15     | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Mne   | RDDATA |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  | R      |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |

| Bit(s) | Mnemonic | Name    | Description                                       |
|--------|----------|---------|---------------------------------------------------|
| 15:0   | RDDATA   | RD_DATA | This is the read port of the 8-entry read buffer. |

**A0770010 CONFIG1 Config set 1 00000000**

| Bit   | 31  | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23         | 22 | 21 | 20 | 19 | 18 | 17 | 16   |
|-------|-----|----|----|----|----|----|----|----|------------|----|----|----|----|----|----|------|
| Mne   |     |    |    |    |    |    |    |    |            |    |    |    |    |    |    |      |
| Type  |     |    |    |    |    |    |    |    |            |    |    |    |    |    |    |      |
| Reset |     |    |    |    |    |    |    |    |            |    |    |    |    |    |    |      |
| Bit   | 15  | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7          | 6  | 5  | 4  | 3  | 2  | 1  | 0    |
| Mne   | DIV |    |    |    |    |    |    |    | GUARD TIME |    |    |    |    |    |    | EXTR |

[illegible]

| Bit(s) | Mnemonic            | Name             | Description                                                                                                                                                                                     |
|--------|---------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0      | <b>EXTRA CLK EN</b> | RGF_EXTRA_CLK_EN | <b>For debugging only</b><br>This enables extra spi clks to be generated even after cs is de-asserted.<br>0: Disable<br>1: Enable<br><i>Note: this bit is effective only if EN_MAINCON = 0.</i> |
| 7:4    | <b>GUARD TIME</b>   | RGF_GUARD_TIME   | <b>Configures the amount of guard steps to insert between consecutive commands</b>                                                                                                              |
| 15:8   | <b>DIV</b>          | RGF_DIV          | <b>Sets up the final output SPI clk period</b><br>The clock period is calculated as the following:<br>$1/(\text{bclk\_ck freq}/\text{div}/2)$                                                   |

| A0770014 |    | CONFIG2 |    |    | Config set 2 |    |    |    |    |    |    |    | 00000000 |          |    |    |
|----------|----|---------|----|----|--------------|----|----|----|----|----|----|----|----------|----------|----|----|
| Bit      | 31 | 30      | 29 | 28 | 27           | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19       | 18       | 17 | 16 |
| Mne      |    |         |    |    |              |    |    |    |    |    |    |    |          |          |    |    |
| Type     |    |         |    |    |              |    |    |    |    |    |    |    |          |          |    |    |
| Reset    |    |         |    |    |              |    |    |    |    |    |    |    |          |          |    |    |
| Bit      | 15 | 14      | 13 | 12 | 11           | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3        | 2        | 1  | 0  |
| Mne      |    |         |    |    |              |    |    |    |    |    |    |    |          | TRAN LEN |    |    |
| Type     |    |         |    |    |              |    |    |    |    |    |    |    |          | R/W      |    |    |
| Reset    |    |         |    |    |              |    |    |    |    |    |    |    |          | 0        | 0  | 0  |

| Bit(s) | Mnemonic | Name      | Description                                                                                                                                                    |
|--------|----------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2:0    | TRAN LEN | TRANS LEN | <p><b>Allows the MCU to set up the number of data reads within 1 spi transfer to execute automatically</b></p> <p>Actual data reads = (TRAN LEN + 1) x 16b</p> |

| A0770020 |    | MODE SEL |    |    |    | FMSYS HCI mode select |    |    |    |    |    |    |    |    |    | 00000000 |  |
|----------|----|----------|----|----|----|-----------------------|----|----|----|----|----|----|----|----|----|----------|--|
| Bit      | 31 | 30       | 29 | 28 | 27 | 26                    | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16       |  |
| Mne      |    |          |    |    |    |                       |    |    |    |    |    |    |    |    |    |          |  |
| Type     |    |          |    |    |    |                       |    |    |    |    |    |    |    |    |    |          |  |
| Reset    |    |          |    |    |    |                       |    |    |    |    |    |    |    |    |    |          |  |
| Bit      | 15 | 14       | 13 | 12 | 11 | 10                    | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0        |  |

[illegible]

| Bit(s) | Mnemonic | Name         | Description                                                                                                                                                                        |
|--------|----------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0      | MODE_SEL | HCI_MODE_SEL | <b>Controls the top FMSYS HCI signal mux</b><br>The allowed selections are as the following:<br>1'b0: Select internal fspi_mas control<br>1'b1: Select external master SPI control |

[illegible]

| Bit(s) | Mnemonic             | Name          | Description                                                                                                                                  |
|--------|----------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 0      | <b>SOFT RESET</b>    | SOFT_RESET    | <b>For debugging only</b><br>This is used to clear the fspi_mas state to reset the state.<br>0: De-assert soft reset<br>1: Assert soft reset |
| 1      | <b>FIFO ADDR CLR</b> | FIFO_ADDR_CLR | <b>For debugging only</b><br>0: Normal operation.<br>1. Clear FIFO addresses.                                                                |

| A0770034 |    | DEBUG1 |    |    | Debug 1 |    |    |    |    |    |    |    |    |    |    | 00000000 |  |
|----------|----|--------|----|----|---------|----|----|----|----|----|----|----|----|----|----|----------|--|
| Bit      | 31 | 30     | 29 | 28 | 27      | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16       |  |
| Mne      |    |        |    |    |         |    |    |    |    |    |    |    |    |    |    |          |  |
| Type     |    |        |    |    |         |    |    |    |    |    |    |    |    |    |    |          |  |
| Reset    |    |        |    |    |         |    |    |    |    |    |    |    |    |    |    |          |  |
| Bit      | 15 | 14     | 13 | 12 | 11      | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0        |  |



|       |  |         |   |   |  |         |   |   |           |   |         |          |  |        |   |   |
|-------|--|---------|---|---|--|---------|---|---|-----------|---|---------|----------|--|--------|---|---|
| Mne   |  | RD_ADDR |   |   |  | WR_ADDR |   |   | FSPI_STAT |   | WR_FULL | RD_EMPTY |  | OFFSET |   |   |
| Type  |  | R       |   |   |  | R       |   |   | R         |   | R       | R        |  | R      |   |   |
| Reset |  | 0       | 0 | 0 |  | 0       | 0 | 0 | 0         | 0 | 0       | 0        |  | 0      | 0 | 0 |

| Bit(s) | Mnemonic         | Name        | Description |
|--------|------------------|-------------|-------------|
| 2:0    | <b>OFFSET</b>    | FIFO OFFSET |             |
| 4      | <b>RD_EMPTY</b>  | READ EMPTY  |             |
| 5      | <b>WR_FULL</b>   | WRITE FULL  |             |
| 7:6    | <b>FSPI_STAT</b> | FSPI STATE  |             |
| 10:8   | <b>WR_ADDR</b>   | WR ADDR     |             |
| 14:12  | <b>RD_ADDR</b>   | RD ADDR     |             |

**A0770038      DEBUG2      Debug 2      00000000**

|       |                   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|-------|-------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Bit   | 31                | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
| Mne   |                   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  |                   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset |                   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Bit   | 15                | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Mne   | <b>CUR_RDDATA</b> |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  | R                 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset | 0                 | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |

| Bit(s) | Mnemonic          | Name   | Description                       |
|--------|-------------------|--------|-----------------------------------|
| 15:0   | <b>CUR_RDDATA</b> | RDDATA | Reads data shift register content |

**A077003C      DEBUG3      Debug 3      00000000**

|       |    |    |    |    |    |    |    |    |    |                  |    |    |    |    |    |    |
|-------|----|----|----|----|----|----|----|----|----|------------------|----|----|----|----|----|----|
| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22               | 21 | 20 | 19 | 18 | 17 | 16 |
| Mne   |    |    |    |    |    |    |    |    |    |                  |    |    |    |    |    |    |
| Type  |    |    |    |    |    |    |    |    |    |                  |    |    |    |    |    |    |
| Reset |    |    |    |    |    |    |    |    |    |                  |    |    |    |    |    |    |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6                | 5  | 4  | 3  | 2  | 1  | 0  |
| Mne   |    |    |    |    |    |    |    |    |    | <b>AUTO_ADDR</b> |    |    |    |    |    |    |
| Type  |    |    |    |    |    |    |    |    |    | R                |    |    |    |    |    |    |
| Reset |    |    |    |    |    |    |    |    |    | 0                | 0  | 0  | 0  | 0  | 0  | 0  |

| Bit(s) | Mnemonic         | Name      | Description                      |
|--------|------------------|-----------|----------------------------------|
| 7:0    | <b>AUTO_ADDR</b> | AUTO_ADDR | Auto address increment read-back |

**A0770040      DEBUG4      Debug 4      00000000**

|       |    |    |    |    |    |    |    |    |    |                |    |    |    |                |    |    |
|-------|----|----|----|----|----|----|----|----|----|----------------|----|----|----|----------------|----|----|
| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22             | 21 | 20 | 19 | 18             | 17 | 16 |
| Mne   |    |    |    |    |    |    |    |    |    |                |    |    |    |                |    |    |
| Type  |    |    |    |    |    |    |    |    |    |                |    |    |    |                |    |    |
| Reset |    |    |    |    |    |    |    |    |    |                |    |    |    |                |    |    |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6              | 5  | 4  | 3  | 2              | 1  | 0  |
| Mne   |    |    |    |    |    |    |    |    |    | <b>RD_ADDR</b> |    |    |    | <b>WR_ADDR</b> |    |    |
| Type  |    |    |    |    |    |    |    |    |    | R              |    |    |    | R              |    |    |
| Reset |    |    |    |    |    |    |    |    |    | 0              | 0  | 0  |    | 0              | 0  | 0  |

| Bit(s) | Mnemonic       | Name    | Description |
|--------|----------------|---------|-------------|
| 2:0    | <b>WR_ADDR</b> | WR_ADDR |             |
| 6:4    | <b>RD_ADDR</b> | RD_ADDR |             |

**A0770048      DEBUG5      Debug 5      00000000**

|       |    |    |    |    |    |    |    |    |    |    |    |    |    |    |                      |                       |
|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----------------------|-----------------------|
| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17                   | 16                    |
| Mne   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |                      |                       |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |                      |                       |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    |                      |                       |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1                    | 0                     |
| Mne   |    |    |    |    |    |    |    |    |    |    |    |    |    |    | <b>FSPI_RGF_RSTB</b> | <b>FSPI_CORE_RSTB</b> |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    | R/W                  | R/W                   |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    | 0                    | 0                     |

| Bit(s) | Mnemonic              | Name           | Description                                                                            |
|--------|-----------------------|----------------|----------------------------------------------------------------------------------------|
| 0      | <b>FSPI_CORE_RSTB</b> | FSPI CORE_RSTB | <b>When set to 1, it resets the core logic inside FMSYS.</b><br>Set to 0 to de-assert. |
| 1      | <b>FSPI_RGF_RSTB</b>  | FSPI RGF_RSTB  | <b>When set to 1, it resets the RGF inside FMSYS.</b><br>Set to 0 to de-assert.        |

**A0770050      RRCCAL      R/RC calibration value from BT      00000000**

|       |    |    |    |    |    |                   |              |    |    |    |    |                  |             |    |    |    |
|-------|----|----|----|----|----|-------------------|--------------|----|----|----|----|------------------|-------------|----|----|----|
| Bit   | 31 | 30 | 29 | 28 | 27 | 26                | 25           | 24 | 23 | 22 | 21 | 20               | 19          | 18 | 17 | 16 |
| Mne   |    |    |    |    |    |                   |              |    |    |    |    |                  |             |    |    |    |
| Type  |    |    |    |    |    |                   |              |    |    |    |    |                  |             |    |    |    |
| Reset |    |    |    |    |    |                   |              |    |    |    |    |                  |             |    |    |    |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10                | 9            | 8  | 7  | 6  | 5  | 4                | 3           | 2  | 1  | 0  |
| Mne   |    |    |    |    |    | RCC<br>AL_V<br>AL | RCCAL NOBIAS |    |    |    |    | RCA<br>L_VA<br>L | RCAL NOBIAS |    |    |    |
| Type  |    |    |    |    |    | R/W               | R/W          |    |    |    |    | R/W              | R/W         |    |    |    |
| Reset |    |    |    |    |    | 0                 | 0            | 0  | 0  | 0  | 0  | 0                | 0           | 0  | 0  | 0  |

| Bit(s) | Mnemonic            | Name         | Description              |
|--------|---------------------|--------------|--------------------------|
| 3:0    | <b>RCAL_NOBIAS</b>  | RCAL_NOBIAS  | BT R calibration result  |
| 4      | <b>RCAL_VAL</b>     | RCAL_VAL     | BT R calibration valid   |
| 9:5    | <b>RCCAL_NOBIAS</b> | RCCAL_NOBIAS | BT RC calibration result |
| 4      | <b>RCCAL_VAL</b>    | RCCAL_VAL    | BT RC calibration valid  |

## 4.16 HIF

### 4.16.1 General Description

This module is used to help peripheral DMA to access the external device which is connected to the host interface (NLD[15:0], LWRB, LRDB, LPA0, LPCE1B, LPCE2B). There are two HIF modules in MT6250, and they are named HIF0 and HIF1. HIF0 and HIF1 are identical designs, the only difference is the base address. The base address of HIF0 is 0xA0190000, and the base address of HIF1 is 0xA01A0000.

The data path of writing to the external device is:

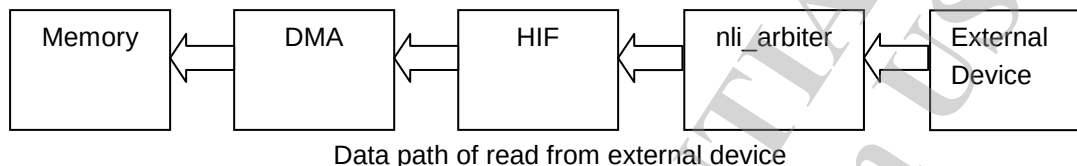
1. DMA moves data from memory to HIF internal buffer through dedicated HIF data port for DMA (HIF\_DMA\_DATA).
2. HIF moves data from its internal buffer to the external device through the host interface. nli\_arbiter is used to arbitrate the ownership of shared host interface, which is shared among LCM, NAND, Wi-Fi and TD-modem.



Data path of write to external device

The data path of reading from external device is

1. HIF reads data from external device, and puts data into HIF internal buffer.
2. DMA moves data from HIF internal buffer to memory through dedicated HIF data port for DMA (HIF\_DMA\_DATA).



#### 4.16.2 Programming Guide

HIF provides two access modes to transfer data from/to the external device. One is MCU mode, and the other is DMA mode.

In the MCU mode, MCU transfers data from/to the external device by reading/writing the two ports, HIF\_MCU\_DATA0 or HIF\_MCU\_DATA1. If MCU accesses HIF\_MCU\_DATA0, the LPA0 pin will be 0 during transferring. If MCU accesses HIF\_MCU\_DATA1, the LPA0 pin will be 1 during the transfer. Note that the two ports are only for MCU, and DMA is forbidden to access them.

##### Programming guides of DMA mode

1. Enable HIF interrupt
2. Set up HIF for transfer width: 8 bits or 16bits, direction = write or read, LPA0 value during transfer and transfer priority on the host interface. All settings are in HIF\_CON.
3. Chip selects that the external device is connected: LPCE1B or LPCE2B. It is set by NLI\_ARB\_CS (0x811A\_0014).
4. Setup HIF timing in HIF\_TIMING0 and HIF\_TIMING1. They only need to be set once, because for an external device, its access timing should be always the same.
5. Setup transfer data amount in HIF\_DAMOUNT.
6. Setup peripheral DMA required setting for HIF. Refer to the functional specifications of PDMA.
7. Start HIF and peripheral DMA.
8. After the transfer is completed, HIF issues an interrupt.

#### 4.16.3 Register Definition

| Register address | Register function             | Acronym    |
|------------------|-------------------------------|------------|
| A019_0000h       | HIF status register           | HIF_STA    |
| A019_0004h       | HIF interrupt enable register | HIF_INTEN  |
| A019_0008h       | HIF interrupt status register | HIF_INTSTA |
| A019_000Ch       | HIF start register            | HIF_START  |
| A019_0010h       | HIF software reset register   | HIF_SWRST  |
| A019_0014h       | HIF timing control register 0 | HIF_TIME0  |

| Register address | Register function                       | Acronym       |
|------------------|-----------------------------------------|---------------|
| A019_0018h       | HIF timing control register 1           | HIF_TIME1     |
| A019_0020h       | HIF control register                    | HIF_CON       |
| A019_0024h       | HIF transfer data amount register       | HIF_DAMOUNT   |
| A019_0030h       | HIF MCU/DMA access arbitration register | HIF_ACS_ARB   |
| A019_0040h       | HIF DMA side data counter register      | HIF_DMA_CNT   |
| A019_0044h       | HIF PIF side data counter register      | HIF_PIF_CNT   |
| A019_0048h       | HIF FIFO pointer register               | HIF_FIFOPTR   |
| A019_0200h       | HIF data port for DMA                   | HIF_DMA_DATA  |
| A019_0300h       | HIF data port 0 for MCU                 | HIF_MCU_DATA0 |
| A019_0310h       | HIF data port 1 for MCU                 | HIF_MCU_DATA1 |

**Table 69. HIF register map**

**A019\_0000h HIF status register**

**HIF\_STA**

| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16           |
|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|--------------|
| Name  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |              |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |              |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |              |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0            |
| Name  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | <b>BUS Y</b> |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | R            |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | 0            |

**BUS Y** HIF busy flag

**0** HIF is idle

**1** HIF is transferring data

**A019\_0004h HIF interrupt enable register**

**HIF\_INTEN**

| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16         |
|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|------------|
| Name  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |            |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |            |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |            |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0          |
| Name  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | <b>CPL</b> |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | R/W        |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | 0          |

**CPL** Enables HIF transfer completion interrupt

**0** Disable this interrupt

**1** Enable this interrupt. HIF will issue an interrupt when a transfer is completed.

### A019\_0008h HIF interrupt status register

### HIF\_INTSTA

| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16         |
|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|------------|
| Name  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |            |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |            |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |            |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0          |
| Name  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | <b>CPL</b> |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | WOC<br>LR  |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | 0          |

**CPL** HIF transfer completion interrupt status

- 0** This interrupt is not issued.
- 1** This interrupt is issued.

### A019\_000Ch HIF start register

### HIF\_START

| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16           |
|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|--------------|
| Name  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |              |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |              |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |              |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0            |
| Name  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | <b>START</b> |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | WO           |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | 0            |

**START** Starts HIF transfer

- 0** Stop HIF transfer. If this bit is cleared when HIF is still transferring data, HIF will continue to complete all transfers of data amount specified in HIF\_DAMOUNT.
- 1** Start HIF transfer. HIF will start transfer when this bit has a transition from 0 to 1.

### A019\_0010h HIF software reset register

### HIF\_SWRST

| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Name  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |

|       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |              |
|-------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--------------|
| Name  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | <b>SWRST</b> |
| Type  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | R/W          |
| Reset |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 0            |

**SWRST**     Software reset  
**0**     Does not reset HIF  
**1**     Reset HIF

### A019\_0014h HIF timing register 0

### HIF\_TIME0

|       |             |     |     |     |             |     |     |     |            |    |     |     |     |     |     |     |
|-------|-------------|-----|-----|-----|-------------|-----|-----|-----|------------|----|-----|-----|-----|-----|-----|-----|
| Bit   | 31          | 30  | 29  | 28  | 27          | 26  | 25  | 24  | 23         | 22 | 21  | 20  | 19  | 18  | 17  | 16  |
| Name  | <b>C2RH</b> |     |     |     | <b>C2RS</b> |     |     |     | <b>RLT</b> |    |     |     |     |     |     |     |
| Type  | R/W         | R/W | R/W | R/W | R/W         | R/W | R/W | R/W |            |    | R/W | R/W | R/W | R/W | R/W | R/W |
| Reset | 0           | 0   | 0   | 0   | 0           | 0   | 0   | 0   |            |    | 0   | 0   | 0   | 0   | 0   | 0   |
| Bit   | 15          | 14  | 13  | 12  | 11          | 10  | 9   | 8   | 7          | 6  | 5   | 4   | 3   | 2   | 1   | 0   |
| Name  | <b>C2WH</b> |     |     |     | <b>C2WS</b> |     |     |     | <b>WST</b> |    |     |     |     |     |     |     |
| Type  | R/W         | R/W | R/W | R/W | R/W         | R/W | R/W | R/W |            |    | R/W | R/W | R/W | R/W | R/W | R/W |
| Reset | 0           | 0   | 0   | 0   | 0           | 0   | 0   | 0   |            |    | 0   | 0   | 0   | 0   | 0   | 0   |

**WST**     Write wait state time. See Figure 53.  
**C2WS**     Chip select (LPCEB0) to write strobe (LWRB) setup time. See Figure 53. **C2WS ≤ WST.**  
**C2WH**     Chip select (LPCEB0) to write strobe (LWRB) hold time. See Figure 53.  
**RLT**     Read latency time. See Figure 54.  
**C2RS**     Chip select (LPCEB0) to read strobe (LRDB) setup time. See Figure 54. **C2RS ≤ RLT.**  
**C2RH**     Chip select (LPCEB0) to read strobe (LRDB) hold time. See Figure 54.

### A019\_0018h HIF timing register 1

### HIF\_TIME1

|       |    |    |    |    |    |    |    |    |    |    |    |    |            |     |     |     |
|-------|----|----|----|----|----|----|----|----|----|----|----|----|------------|-----|-----|-----|
| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19         | 18  | 17  | 16  |
| Name  |    |    |    |    |    |    |    |    |    |    |    |    |            |     |     |     |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |            |     |     |     |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |            |     |     |     |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3          | 2   | 1   | 0   |
| Name  |    |    |    |    |    |    |    |    |    |    |    |    | <b>CHW</b> |     |     |     |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    | R/W        | R/W | R/W | R/W |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    | 0          | 0   | 0   | 0   |

**CHW**     Chip select high width



## Parallel Interface Write timing

C2WS = 2, WST = 3, C2WH = 0, PERIOD = 0, **C2WS must <= WST**

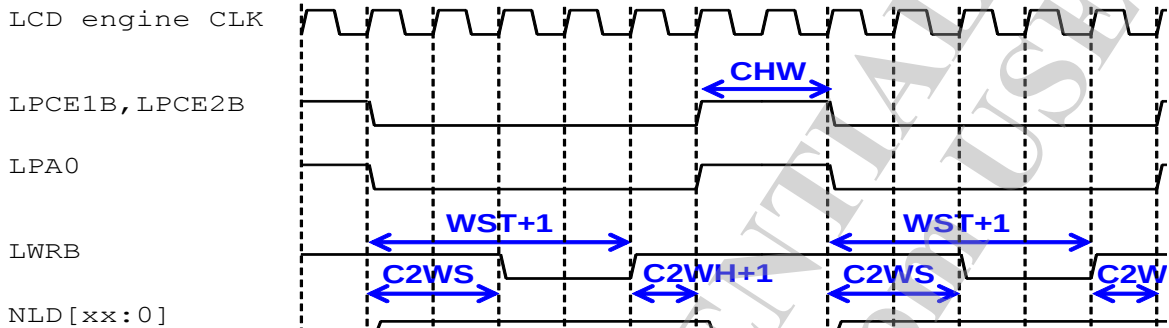


Figure 53. Interface write timing diagram

## Parallel Interface Read timing

C2RS=1, RLT=3, C2RH=0, PERIOD=0, **C2RS must <= RLT**

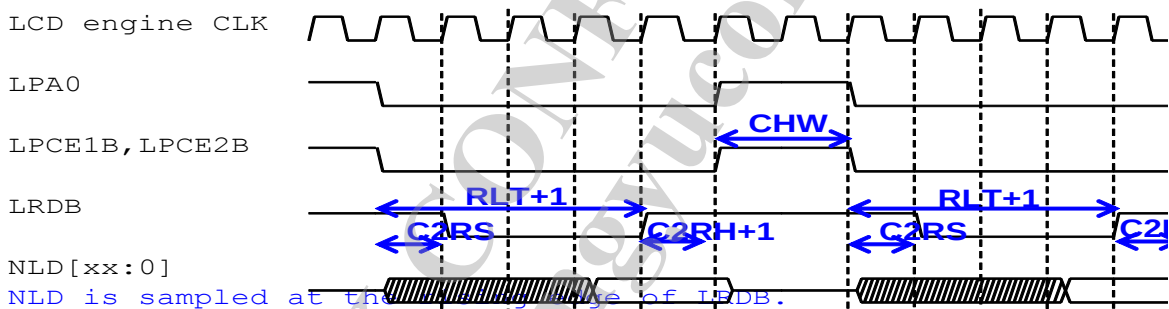


Figure 54. Interface read timing diagram

## A019\_0020h HIF control register

## HIF\_CON

| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20        | 19  | 18        | 17   | 16  |
|-------|----|----|----|----|----|----|----|----|----|----|----|-----------|-----|-----------|------|-----|
| Name  |    |    |    |    |    |    |    |    |    |    |    |           |     |           |      |     |
| Type  |    |    |    |    |    |    |    |    |    |    |    |           |     |           |      |     |
| Reset |    |    |    |    |    |    |    |    |    |    |    |           |     |           |      |     |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4         | 3   | 2         | 1    | 0   |
| Name  |    |    |    |    |    |    |    |    |    |    |    | ULTR<br>A | A0  | WRIT<br>E | HIFW |     |
| Type  |    |    |    |    |    |    |    |    |    |    |    | R/W       | R/W | R/W       | R/W  | R/W |
| Reset |    |    |    |    |    |    |    |    |    |    |    | 0         | 0   | 0         | 0    | 0   |

**HIFW** Host interface width  
**0** 8 bits  
**1** 16 bits  
**Others** Reserved

- WRITE** Write transfer flag  
**0** Read transfer  
**1** Write transfer
- A0** Sets up the value of LPA0 pin during transfer  
**0** LPA0=0 during transfer  
**1** LPA0=1 during transfer
- ULTRA** Enables ultra-high signal to nli\_arbiter  
**0** Does not issue ultra-high signal  
**1** Issue ultra high signal

**A019\_0024h HIF data amount register**
**HIF\_DAMOUNT**

| Bit  | 31             | 30  | 29  | 28  | 27  | 26  | 25  | 24  | 23  | 22  | 21  | 20  | 19  | 18  | 17  | 16  |
|------|----------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Name |                |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Type |                |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Bit  | 15             | 14  | 13  | 12  | 11  | 10  | 9   | 8   | 7   | 6   | 5   | 4   | 3   | 2   | 1   | 0   |
| Name | <b>DAMOUNT</b> |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Type | R/W            | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W |

**DAMOUNT** Specifies the data amount of this transfer. Unit: bytes. 0 means transferring 1 byte, 1 means transferring 2 bytes, and N means transferring N+1 bytes. DAMOUNT+1 must be multiple of HIF width, e.g., if HIF width is 16 bits, DAMOUNT+1 must be multiple of 2 bytes.

**A019\_0030h HIF MCU/DMA access arbitration register**
**HIF\_ACS\_ARB**

| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18                 | 17              | 16                 |
|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|--------------------|-----------------|--------------------|
| Name  |    |    |    |    |    |    |    |    |    |    |    |    |    |                    |                 |                    |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |                    |                 |                    |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |                    |                 |                    |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2                  | 1               | 0                  |
| Name  |    |    |    |    |    |    |    |    |    |    |    |    |    | <b>MCU_ACS_STR</b> | <b>PIF_BUSY</b> | <b>MCU_ACS_REQ</b> |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    | W                  | R               | R/W                |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    | 0                  | 0               | 0                  |

**MCU\_ACS\_REQ** MCU access request. MCU sets this bit to be 1'b1 to request for MCU data access via the NLD pin. This bit will block the next 8-bit/16-bit DMA data transfer that is intended to be transferred after the current one is completed.

Thus, MCU shall poll the PIF\_BUSY bit and wait until PIF\_BUSY = 0 before proceeding the following MCU access steps (set up MCU\_ACS\_STR and initiate MCU access).

- 0 MCU access request de-asserted
- 1 MCU access request asserted

**PIF\_BUSY**

Busy/idle status of the NLD pin data-transfer control FSM.

- 0 Idle
- 1 Busy

**MCU\_ACS\_STR (RO)**

Starts MCU access. The MCU sets this bit to be 1'b1 to switch data path for MCU access. MCU must wait PIF\_BUSY = 0 before setting up this bit. The MCU should clear the two bits MCU\_ACS\_STR and MCU\_ACS\_REQ after the MCU accesses. DMA transfer will auto resume after the two bits are cleared if DMA is originally running.

- 0 DMA access path is selected
- 1 MCU access path is selected

**A019\_0040h HIF DMA side data counter register**

**HIF\_DMA\_CNT**

| Bit   | 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|-------|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Name  | DATA_CNT |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  | R        | R  | R  | R  | R  | R  | R  | R  | R  | R  | R  | R  | R  | R  | R  | R  |
| Reset | 0        | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| Bit   | 15       | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Name  | CMD_CNT  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  | R        | R  | R  | R  | R  | R  | R  | R  | R  | R  | R  | R  | R  | R  | R  | R  |
| Reset | 0        | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |

**CMD\_CNT** This is a down counter in unit of byte to indicate the amount of data (command phase) waiting to be accessed by DMA. It counts down from {HIF\_DAMOUNT[15:2],2'b0} to 0. This value is always the multiple of 4 bytes due to one DMA access is 4 bytes. For debugging.

**DATA\_CNT** This is a down counter in unit of byte to indicate the amount of data (data phase) waiting to be accessed by DMA. It counts down from {HIF\_DAMOUNT[15:2],2'b0} to 0. This value is always the multiple of 4 bytes due to one DMA access is 4 bytes. For debugging.

**A019\_0044h HIF PIF side data counter register**

**HIF\_PIF\_CNT**

| Bit  | 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|------|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Name | DATA_CNT |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |

|       |                |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|-------|----------------|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| Type  | R              | R  | R  | R  | R  | R  | R | R | R | R | R | R | R | R | R | R |
| Reset | 0              | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Bit   | 15             | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| Name  | <b>CMD_CNT</b> |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| Type  | R              | R  | R  | R  | R  | R  | R | R | R | R | R | R | R | R | R | R |
| Reset | 0              | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

**CMD\_CNT** This is a down counter in unit of byte to indicate the amount of data (command phase) waiting to be transferred through the host interface. It counts down from HIF\_DAMOUNT to 0. For debugging.

**DATA\_CNT** This is a down counter in unit of byte to indicate the amount of data (data phase) waiting to be transferred through the host interface. It counts down from HIF\_DAMOUNT to 0. For debugging.

#### **A019\_0044h HIF FIFO pointer register**

#### **HIF\_FIFOPTR**

|       |    |    |    |    |    |    |    |    |    |                  |    |    |    |                  |    |    |
|-------|----|----|----|----|----|----|----|----|----|------------------|----|----|----|------------------|----|----|
| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22               | 21 | 20 | 19 | 18               | 17 | 16 |
| Name  |    |    |    |    |    |    |    |    |    |                  |    |    |    |                  |    |    |
| Type  |    |    |    |    |    |    |    |    |    |                  |    |    |    |                  |    |    |
| Reset |    |    |    |    |    |    |    |    |    |                  |    |    |    |                  |    |    |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6                | 5  | 4  | 3  | 2                | 1  | 0  |
| Name  |    |    |    |    |    |    |    |    |    | <b>FIFO_RPTR</b> |    |    |    | <b>FIFO_WPTR</b> |    |    |
| Type  |    |    |    |    |    |    |    |    |    | R                | R  | R  |    | R                | R  | R  |
| Reset |    |    |    |    |    |    |    |    |    | 0                | 0  | 0  |    | 0                | 0  | 0  |

**FIFO\_WPTR** FIFO write pointer. For debugging.

**FIFO\_RPTR** FIFO read pointer. For debugging.

#### **A019\_0200h HIF data port for DMA**

#### **HIF\_DMA\_DATA**

|      |                    |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|------|--------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Bit  | 31                 | 30  | 29  | 28  | 27  | 26  | 25  | 24  | 23  | 22  | 21  | 20  | 19  | 18  | 17  | 16  |
| Name | <b>DATA[31:16]</b> |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Type | R/W                | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W |
| Bit  | 15                 | 14  | 13  | 12  | 11  | 10  | 9   | 8   | 7   | 6   | 5   | 4   | 3   | 2   | 1   | 0   |
| Name | <b>DATA[15:0]</b>  |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Type | R/W                | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W |

**DATA** Data port of HIF for DMA mode. This port is only for DMA, and MCU is forbidden to access it.

**A019\_0300h HIF data port 0 for MCU**
**HIF\_MCU\_DAT  
A0**

| Bit  | 31          | 30  | 29  | 28  | 27  | 26  | 25  | 24  | 23  | 22  | 21  | 20  | 19  | 18  | 17  | 16  |
|------|-------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Name | DATA[31:16] |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Type | R/W         | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W |
| Bit  | 15          | 14  | 13  | 12  | 11  | 10  | 9   | 8   | 7   | 6   | 5   | 4   | 3   | 2   | 1   | 0   |
| Name | DATA[15:0]  |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Type | R/W         | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W |

**DATA** Data port of HIF for MCU mode. When MCU accesses this port, the LPA0 pin is 0 when transferring data. This port is only for MCU, and DMA is forbidden to access it.

**A019\_0310h HIF data port 1 for MCU**
**HIF\_MCU\_DAT  
A1**

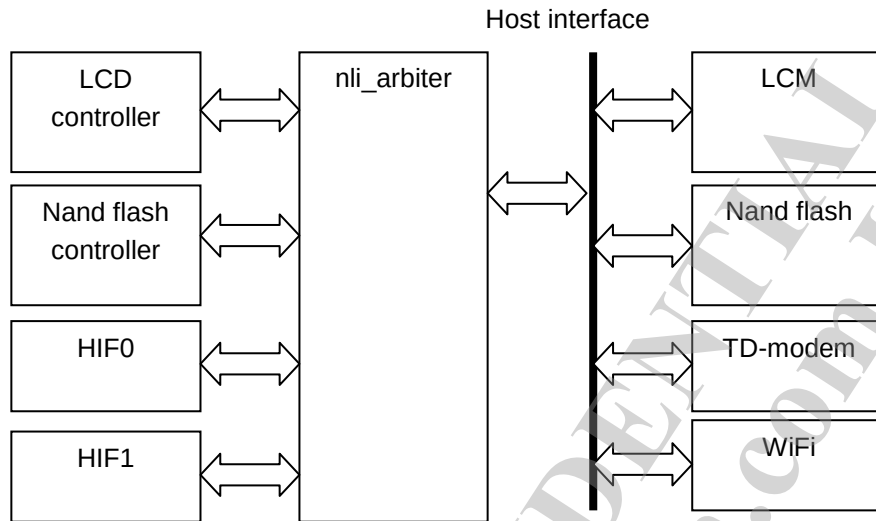
| Bit  | 31          | 30  | 29  | 28  | 27  | 26  | 25  | 24  | 23  | 22  | 21  | 20  | 19  | 18  | 17  | 16  |
|------|-------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Name | DATA[31:16] |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Type | R/W         | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W |
| Bit  | 15          | 14  | 13  | 12  | 11  | 10  | 9   | 8   | 7   | 6   | 5   | 4   | 3   | 2   | 1   | 0   |
| Name | DATA[15:0]  |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Type | R/W         | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W |

**DATA** Data port of HIF for MCU mode. When MCU accesses this port, the LPA0 pin is 1 when transferring data. This port is only for MCU, and DMA is forbidden to access it.

## 4.17 NLI\_ARBITER

### 4.17.1 General Description

This module is used to arbitrate the ownership of the host interface. There are maximal four external devices hooked on the host interface, which are LCM, NAND flash, TD-modem and WiFi. Each external device has its own controller. The LCD controller controls LCM, NAND flash controller controls NAND flash, HIF0 controls TD-modem and HIF1 controls WiFi. These controllers issue requests to nli\_arbiter, and nli\_arbiter will give grant according to the priority and then send data and control signals to the chip IO pins.



**Figure 55. Block diagram of NLI\_ARBTER**

| Register address | Register name                       | Synonym            |
|------------------|-------------------------------------|--------------------|
| A01B_0000h       | NLI_ARB status                      | NLI_ARB_STA        |
| A01B_0010h       | NLI_ARB software reset register     | NLI_ARB_SWRST      |
| A01B_0014h       | NLI_ARB chip select                 | NLI_ARB_CS         |
| A01B_0018h       | NLI_ARB continuous grant count      | NLI_ARB_CONT_GRANT |
| A01B_001Ch       | NLI_ARB handover count              | NLI_ARB_HANDOVER   |
| A01B_0020h       | NLI_ARB share-pin control           | NLI_ARB_SHRPIN_CON |
| A01B_0040h       | NLI_ARB monitor source select       | NLI_ARB_MON_SRC    |
| A01B_0044h       | NLI_ARB monitor clear               | NLI_ARB_MON_CLR    |
| A01B_0048h       | NLI_ARB monitor start               | NLI_ARB_MON_START  |
| A01B_0050h       | NLI_ARB monitor read counter        | NLI_ARB_MON_RD     |
| A01B_0054h       | NLI_ARB monitor write counter       | NLI_ARB_MON_WR     |
| A01B_0058h       | NLI_ARB monitor total cycle counter | NLI_ARB_MON_CYC    |

**Table 70. NLI\_ARBTER register map**

#### 4.17.2 Register Definition

##### A01B\_0000h NLI\_ARB status

##### NLI\_ARB\_STA

| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20        | 19 | 18 | 17 | 16 |
|-------|----|----|----|----|----|----|----|----|----|----|----|-----------|----|----|----|----|
| Name  |    |    |    |    |    |    |    |    |    |    |    | ULTRA_ARB |    |    |    |    |
| Type  |    |    |    |    |    |    |    |    |    |    |    | R         | R  | R  | R  | R  |
| Reset |    |    |    |    |    |    |    |    |    |    |    | 0         | 0  | 0  | 0  | 1  |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4         | 3  | 2  | 1  | 0  |

|       |  |  |  |                 |   |   |   |   |  |  |  |                 |               |                |                 |  |
|-------|--|--|--|-----------------|---|---|---|---|--|--|--|-----------------|---------------|----------------|-----------------|--|
| Name  |  |  |  | <b>NORM_ARB</b> |   |   |   |   |  |  |  | <b>HIF1_REQ</b> | <b>NF_REQ</b> | <b>LCD_REQ</b> | <b>HIF0_REQ</b> |  |
| Type  |  |  |  | R               | R | R | R | R |  |  |  | R               | R             | R              | R               |  |
| Reset |  |  |  | 0               | 0 | 0 | 0 | 1 |  |  |  | 0               | 0             | 0              | 0               |  |

This register is for debugging.

**HIF0\_REQ**      HIF0 request flag  
**LCD\_REQ**      LCD request flag  
**NF\_REQ**        NAND flash controller request flag  
**HIF1\_REQ**      HIF1 request flag

**NORM\_ARB**    Normal priority arbiter state  
**Bit 0** Idle state  
**Bit 1** HIF0 state  
**Bit 2** LCD state  
**Bit 3** NF state  
**Bit 4** HIF1 state

**ULTRA\_ARB**    Ultra-high priority arbiter state  
**Bit 0** Idle state  
**Bit 1** HIF0 state  
**Bit 2** LCD state  
**Bit 3** NF state  
**Bit 4** HIF1 state

**A01B\_0010h NLI\_ARB software reset register** **NLI\_ARB\_SWRST**

|       |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |              |
|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|--------------|
| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16           |
| Name  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |              |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |              |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |              |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0            |
| Name  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | <b>SWRST</b> |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | R/W          |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | 0            |

**SWRST**      Software reset  
**0**      Does not reset NLI\_ARB  
**1**      Reset NLI\_ARB



### A01B\_0014h NLI\_ARB chip select

### NLI\_ARB\_CS

| Bit   | 31 | 30         | 29  | 28  | 27 | 26         | 25  | 24  | 23 | 22         | 21  | 20  | 19 | 18         | 17          | 16           |
|-------|----|------------|-----|-----|----|------------|-----|-----|----|------------|-----|-----|----|------------|-------------|--------------|
| Name  |    |            |     |     |    |            |     |     |    |            |     |     |    |            | FAVOR_ULTRA | FAVOR_NORMAL |
| Type  |    |            |     |     |    |            |     |     |    |            |     |     |    |            | R/W         | R/W          |
| Reset |    |            |     |     |    |            |     |     |    |            |     |     |    |            | 0           | 1            |
| Bit   | 15 | 14         | 13  | 12  | 11 | 10         | 9   | 8   | 7  | 6          | 5   | 4   | 3  | 2          | 1           | 0            |
| Name  |    | LPCE3B_SEL |     |     |    | LPCE2B_SEL |     |     |    | LPCE1B_SEL |     |     |    | LPCE0B_SEL |             |              |
| Type  |    | R/W        | R/W | R/W |    | R/W        | R/W | R/W |    | R/W        | R/W | R/W |    | R/W        | R/W         | R/W          |
| Reset |    | 0          | 0   | 1   |    | 1          | 0   | 0   |    | 0          | 1   | 1   |    | 0          | 0           | 0            |

#### LPCE0B\_SEL

Selects pin LPCE0B source

- 0 Pin LPCE0B is controlled by LCD chip select 0
- 1 Pin LPCE0B is controlled by LCD chip select 1
- 2 Pin LPCE0B is controlled by LCD chip select 2
- 3 Pin LPCE0B is controlled by HIF0
- 4 Pin LPCE0B is controlled by HIF1

**Others** Reserved, LPCE0B is always high.

#### LPCE1B\_SEL

Selects pin LPCE1B source

- 0 Pin LPCE1B is controlled by LCD chip select 0
- 1 Pin LPCE1B is controlled by LCD chip select 1
- 2 Pin LPCE1B is controlled by LCD chip select 2
- 3 Pin LPCE1B is controlled by HIF0
- 4 Pin LPCE1B is controlled by HIF1

**Others** Reserved, LPCE1B is always high.

#### LPCE2B\_SEL

Selects pin LPCE2B source

- 0 Pin LPCE2B is controlled by LCD chip select 0
- 1 Pin LPCE2B is controlled by LCD chip select 1
- 2 Pin LPCE2B is controlled by LCD chip select 2
- 3 Pin LPCE2B is controlled by HIF0
- 4 Pin LPCE2B is controlled by HIF1

**Others** Reserved, LPCE2B is always high.

#### LPCE3B\_SEL

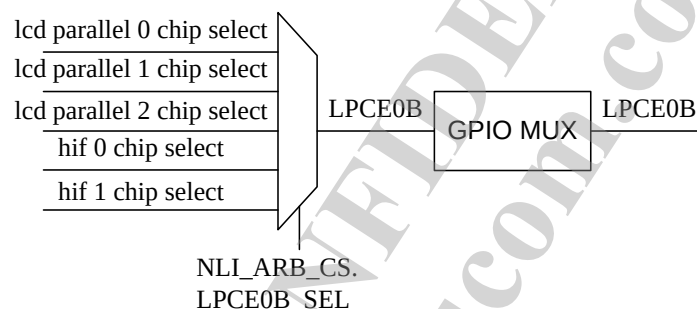
Selects pin LPCE3B source

- 0 Pin LPCE3B is controlled by LCD chip select 0
- 1 Pin LPCE3B is controlled by LCD chip select 1
- 2 Pin LPCE3B is controlled by LCD chip select 2
- 3 Pin LPCE3B is controlled by HIF0
- 4 Pin LPCE3B is controlled by HIF1

**Others** Reserved, LPCE3B is always high.

- FAVOR\_NORMAL** Makes nli\_arbiter favor normal requests
- 0** Make nli\_arbiter serves only ultra requests if ultra requests are issued. It serves normal requests only when there is no ultra request.
  - 1** Make nli\_arbiter serves normal requests when ultra requests are issued. Its behavior depends on FAVOR\_ULTRA.
- FAVOR\_ULTRA** Makes nli\_arbiter favor ultra requests when FAVOR\_NORMAL = 1.

The chip select pin selection circuit is described in Figure 56. The user must set LPCE\*B\_SEL and GPIO MUX properly.

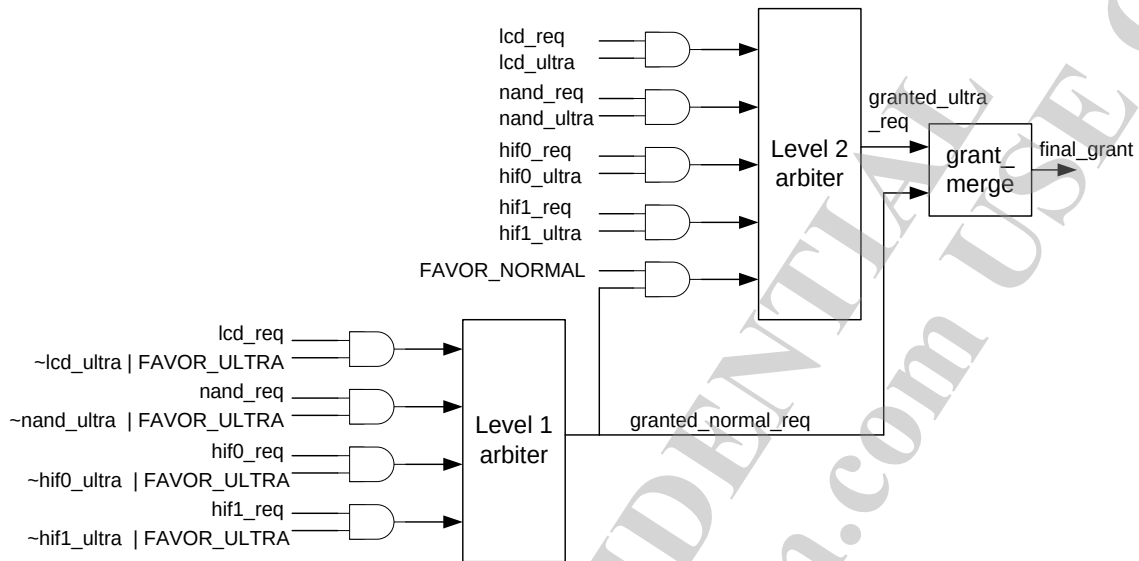


**Figure 56. Chip selection circuit**

The block diagram of nli\_arbiter is described in **Figure 57**. There are two levels of arbiters. Level 1 arbiter is for normal requests and level 2 arbiter is for ultra requests.

If FAVOR\_NORMAL = 0, only ultra requests are served when ultra requests are issued. If FAVOR\_NORMAL = 1, the winner of level 1 arbiter will be regarded as an ultra request and be arbitrated in the level 2 arbiter. Therefore, normal requests can also be served if FAVOR\_NORMAL = 1.

When FAVOR\_NORMAL = 1, FAVOR\_ULTRA can change the behavior of level 1 arbiter. If FAVOR\_ULTRA = 0, only normal requests will be arbitrated in the level 1 arbiter. If FAVOR\_ULTRA = 1, all normal requests and ultra requests will be arbitrated in the level 1 arbiter.



**Figure 57. NLI\_ARBITER block diagram**

An example is given in Table 71 to show how FAVOR\_NORMAL and FAVOR\_ULTRA affect nli\_arbiter's behavior.

Assume all the four masters issue requests, HIF0 and LCD controller issue ultra high requests, and HIF1 and NAND flash controller issue normal requests. The grant sequence is shown in Table 71. Red words mean ultra requests.

| FAVOR_NORMAL | FAVOR_ULTRA | Grant sequence                                                                                                                           |
|--------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 0            | 0           | HIF0, LCD, HIF0, LCD, HIF0, LCD, HIF0, LCD...                                                                                            |
| 0            | 1           | Same as FAVOR_NORMAL = 0 and FAVOR_ULTRA = 0                                                                                             |
| 1            | 0           | HIF0, LCD, NAND, HIF0, LCD, HIF1, HIF0, LCD, NAND, HIF0, LCD, HIF1...                                                                    |
| 1            | 1           | HIF0, LCD, NAND, HIF0, LCD, HIF1, HIF0, LCD, HIF0, HIF0, LCD, LCD, HIF0, LCD, NAND, HIF0, LCD, HIF1, HIF0, LCD, HIF0, HIF0, LCD, LCD,... |

**Table 71. NLI\_ARBITER grant example**

**A01B\_0018h NLI\_ARB continuous grant count** **NLI\_ARB\_CONT\_GRANT**

| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Name  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |

|       |                |     |     |     |     |     |     |     |  |  |  |  |  |  |  |  |               |
|-------|----------------|-----|-----|-----|-----|-----|-----|-----|--|--|--|--|--|--|--|--|---------------|
| Name  | CONT_GRANT_CNT |     |     |     |     |     |     |     |  |  |  |  |  |  |  |  | CONT_GRANT_EN |
| Type  | R/W            | R/W | R/W | R/W | R/W | R/W | R/W | R/W |  |  |  |  |  |  |  |  | R/W           |
| Reset | 0              | 0   | 0   | 0   | 0   | 0   | 0   | 0   |  |  |  |  |  |  |  |  | 0             |

**CONT\_GRANT\_EN** Enables continuous grant. When this function is enabled, nli\_arbiter will grant each master continuous transfer of CONT\_GRANT\_CNT times.

- 0** Disable continuous grant
- 1** Enable continuous grant

**CONT\_GRANT\_CNT** Continuous grant count

**A01B\_001Ch NLI\_ARB handover count**

**NLI\_ARB\_HANDOVER**

|       |    |    |    |    |              |     |     |     |                |                |             |              |                |                |             |              |
|-------|----|----|----|----|--------------|-----|-----|-----|----------------|----------------|-------------|--------------|----------------|----------------|-------------|--------------|
| Bit   | 31 | 30 | 29 | 28 | 27           | 26  | 25  | 24  | 23             | 22             | 21          | 20           | 19             | 18             | 17          | 16           |
| Name  |    |    |    |    |              |     |     |     |                |                |             |              |                |                |             |              |
| Type  |    |    |    |    |              |     |     |     |                |                |             |              |                |                |             |              |
| Reset |    |    |    |    |              |     |     |     |                |                |             |              |                |                |             |              |
| Bit   | 15 | 14 | 13 | 12 | 11           | 10  | 9   | 8   | 7              | 6              | 5           | 4            | 3              | 2              | 1           | 0            |
| Name  |    |    |    |    | HANDOVER_CYC |     |     |     | LEAVE_HI_F1_EN | LEAVE_HI_F0_EN | LEAVE_NF_EN | LEAVE_LCD_EN | ENTER_HI_F1_EN | ENTER_HI_F0_EN | ENTER_NF_EN | ENTER_LCD_EN |
| Type  |    |    |    |    | R/W          | R/W | R/W | R/W | R/W            | R/W            | R/W         | R/W          | R/W            | R/W            | R/W         | R/W          |
| Reset |    |    |    |    | 0            | 0   | 0   | 0   | 0              | 0              | 0           | 0            | 0              | 0              | 0           | 0            |

**ENTER\_LCD\_EN** Adds handover cycles when the bus grant is switched to LCD from another master

- 0** Does not add handover cycles
- 1** Add handover cycles

**ENTER\_NF\_EN** Adds handover cycles when the bus grant is switched to NFI from another master

- 0** Does not add handover cycles
- 1** Add handover cycles

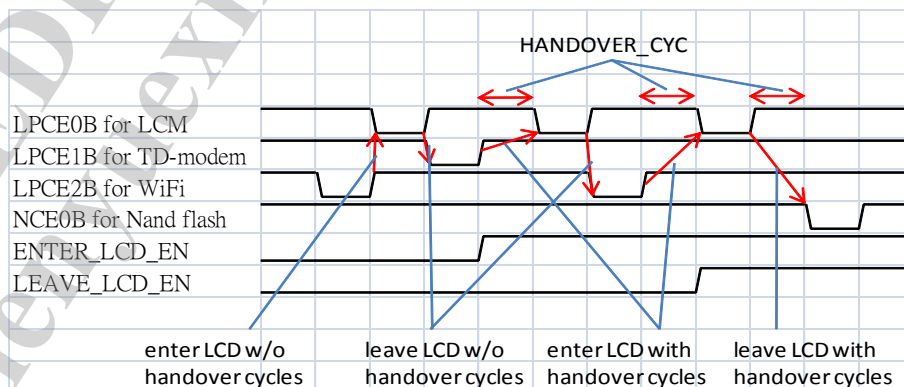
**ENTER\_HIF0\_EN** Adds handover cycles when the bus grant is switched to HIF0 from another master

- 0** Does not add handover cycles
- 1** Add handover cycles

- ENTER\_HIF1\_EN** Adds handover cycles when the bus grant is switched to HIF1 from another master  
**0** Does not add handover cycles  
**1** Add handover cycles
- LEAVE\_LCD\_EN** Adds handover cycles when the bus grant is switched from LCD to another master  
**0** Does not add handover cycles  
**1** Add handover cycles
- LEAVE\_NF\_EN** Adds handover cycles when the bus grant is switched from NFI to another master  
**0** Does not add handover cycles  
**1** Add handover cycles
- LEAVE\_HIF0\_EN** Adds handover cycles when the bus grant is switched from HIF0 to another master  
**0** Does not add handover cycles  
**1** Add handover cycles
- LEAVE\_HIF1\_EN** Adds handover cycles when the bus grant is switched from HIF1 to another master  
**0** Does not add handover cycles  
**1** Add handover cycles
- HANDOVER\_CYC** Idle cycles added when switching bus grant

The following waveform gives an example of how handover\_cyc works.

Note that the work clock frequency of nli\_arbiter in MT6250 is 130MHz or 104MHz, depending on the PSRAM operation frequency. The actual handover cycle number will be HANDOVER\_CYC + 1 once the handover is enabled. If not enabled, the handover cycle will be 0.



### A01B\_0020h NLI\_ARB share-pin control

### NLI\_ARB\_SHRPIN\_CON

| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19                               | 18                               | 17                          | 16                              |
|-------|----|----|----|----|----|----|----|----|----|----|----|----|----------------------------------|----------------------------------|-----------------------------|---------------------------------|
| Name  |    |    |    |    |    |    |    |    |    |    |    |    |                                  |                                  |                             |                                 |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |                                  |                                  |                             |                                 |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |                                  |                                  |                             |                                 |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3                                | 2                                | 1                           | 0                               |
| Name  |    |    |    |    |    |    |    |    |    |    |    |    | NRE<br>B_LP<br>A0_S<br>HR_E<br>N | NWE<br>B_LP<br>A0_S<br>HR_E<br>N | NCLE<br>LRD<br>B_SH<br>R_EN | NALE<br>LW<br>RB_S<br>HR_E<br>N |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    | R/W                              | R/W                              | R/W                         | R/W                             |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    | 0                                | 0                                | 0                           | 0                               |

**NALE\_LWRB\_SHR\_EN** Enables control of NF ALE pin shared with LWRB signal  
**0** Disabled. NF ALE pin is exclusively for NFI  
**1** Enabled. NF ALE pin is shared with LWRB signal. NLI\_ARB will arbitrate ownership of this pin.

**NCLE\_LRDB\_SHR\_EN** Enables control of NF CLE pin shared with LRDB signal  
**0** Disabled. NF CLE pin is exclusively for NFI  
**1** Enabled. NF CLE pin is shared with LRDB signal. NLI\_ARB will arbitrate ownership of this pin.

**NWEB\_LPA0\_SHR\_EN** Enables control of NF WEB pin shared with LPA0 signal  
**0** Disabled. NF WEB pin is exclusively for NFI  
**1** Enabled. NF WEB pin is shared with LPA0 signal. NLI\_ARB will arbitrate ownership of this pin.

**NREB\_LPA0\_SHR\_EN** Enables control of NF RDB pin shared with LPA0 signal  
**0** Disabled. NF RDB pin is exclusively for NFI  
**1** Enabled. NF RDB pin is shared with LPA0 signal. NLI\_ARB will arbitrate ownership of this pin.

### A01B\_0040h NLI\_ARB monitor source select

### NLI\_ARB\_MON\_SRC

| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Name  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |

| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1   | 0                  |
|-------|----|----|----|----|----|----|---|---|---|---|---|---|---|---|-----|--------------------|
| Name  |    |    |    |    |    |    |   |   |   |   |   |   |   |   |     | <b>MON_SRC_SEL</b> |
| Type  |    |    |    |    |    |    |   |   |   |   |   |   |   |   | R/W | R/W                |
| Reset |    |    |    |    |    |    |   |   |   |   |   |   |   |   | 0   | 0                  |

**MON\_SRC\_SEL** Selects NLI\_ARB monitor source

- 00** LCD
- 01** NAND flash
- 10** HIF0
- 11** HIF1

#### **A01B\_0044h NLI\_ARB monitor clear**

#### **NLI\_ARB\_MON\_CLR**

| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16         |
|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|------------|
| Name  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |            |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |            |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |            |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0          |
| Name  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | <b>CLR</b> |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | R/W        |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | 0          |

**CLR** Clears NLI\_ARB monitor counters, including NLI\_ARB\_MON\_RD, NLI\_ARB\_MON\_WR, and NLI\_ARB\_MON\_CYC. Before starting monitor counters, NLI\_ARB\_MON\_CLR should be set to 1 then cleared to 0, in order to clear monitor counters.

- 0** Does not clear
- 1** Clear

#### **A01B\_0048h NLI\_ARB monitor start**

#### **NLI\_ARB\_MON\_START**

| Bit   | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16           |
|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|--------------|
| Name  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |              |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |              |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |              |
| Bit   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0            |
| Name  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | <b>START</b> |
| Type  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | R/W          |
| Reset |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | 0            |



**START** Starts NLI\_ARB monitor to collect information  
**0** Stop collecting information  
**1** Start collecting information

### A01B\_0050h NLI\_ARB monitor read counter

**NLI\_ARB\_MON\_RD**

| Bit  | 31            | 30  | 29  | 28  | 27  | 26  | 25  | 24  | 23  | 22  | 21  | 20  | 19  | 18  | 17  | 16  |
|------|---------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Name | RD_CNT[31:16] |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Type | R/W           | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W |
| Bit  | 15            | 14  | 13  | 12  | 11  | 10  | 9   | 8   | 7   | 6   | 5   | 4   | 3   | 2   | 1   | 0   |
| Name | RD_CNT[15:0]  |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Type | R/W           | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W |

**RD\_CNT** Accumulated read counts when NLI\_ARB\_MON\_START = 1

### A01B\_0054h NLI\_ARB monitor write counter

**NLI\_ARB\_MON\_WR**

| Bit  | 31            | 30  | 29  | 28  | 27  | 26  | 25  | 24  | 23  | 22  | 21  | 20  | 19  | 18  | 17  | 16  |
|------|---------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Name | WR_CNT[31:16] |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Type | R/W           | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W |
| Bit  | 15            | 14  | 13  | 12  | 11  | 10  | 9   | 8   | 7   | 6   | 5   | 4   | 3   | 2   | 1   | 0   |
| Name | WR_CNT[15:0]  |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Type | R/W           | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W |

**WR\_CNT** Accumulated write counts when NLI\_ARB\_MON\_START = 1

### A01B\_0058h NLI\_ARB monitor total cycle counter

**NLI\_ARB\_MON\_CYC**

| Bit  | 31             | 30  | 29  | 28  | 27  | 26  | 25  | 24  | 23  | 22  | 21  | 20  | 19  | 18  | 17  | 16  |
|------|----------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Name | CYC_CNT[31:16] |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Type | R/W            | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W |
| Bit  | 15             | 14  | 13  | 12  | 11  | 10  | 9   | 8   | 7   | 6   | 5   | 4   | 3   | 2   | 1   | 0   |
| Name | CYC_CNT[15:0]  |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Type | R/W            | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W | R/W |

**CYC\_CNT** Accumulated cycles when NLI\_ARB\_MON\_START = 1